Topological Energy Condensate Theory (TECT): 13

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A. Little-group mismatch implies vanishing constant mixing

a. Goal. We want to close the first remaining microscopic Dirac-condition,

$$V(0) = 0,$$

for the block decomposition

$$H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^\dagger & H_R(p) \end{pmatrix}, \qquad \mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R.$$

Here $\mathcal{H}_D$ is the shell-relevant local Dirac block, and $\mathcal{H}_R$ is the remote sector.

b. Assumption audit. We assume:

1. H_* is the little group (more precisely: the residual symmetry group at the selected crossing $p = 0$);
2. H_* acts unitarily on $\mathcal{H}_{\text{full}}$;
3. the decomposition

$$\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R$$

is H_*-invariant;

4. the microscopic Hamiltonian is H_*-equivariant at $p = 0$.

c. Step 1. Block-equivariance forces the constant mixing to be an intertwiner. Because the decomposition is H_*-invariant, the symmetry action has block form

$$U(h) = \begin{pmatrix} U_D(h) & 0 \\ 0 & U_R(h) \end{pmatrix}, \qquad h \in H_*,$$

where U_D and U_R are the restricted representations on $\mathcal{H}_D$ and $\mathcal{H}_R$.

At $p = 0$, write

$$H_{\text{full}}(0) = \begin{pmatrix} H_D(0) & V(0) \\ V(0)^\dagger & H_R(0) \end{pmatrix}.$$

Equivariance means

$$U(h)\, H_{\text{full}}(0)\, U(h)^{-1} = H_{\text{full}}(0) \qquad \forall h \in H_*.$$

Comparing the off-diagonal block gives

$$U_D(h)\, V(0)\, U_R(h)^{-1} = V(0) \qquad \forall h \in H_*.$$

Therefore

$$V(0) \in \text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D).$$

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d. Step 2. Abstract little-group mismatch theorem. If

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0,$$

then the only possible intertwiner is the zero map. Since $V(0)$ is such an intertwiner, it follows that

$$V(0) = 0.$$

Hence we obtain the exact theorem:

$$\boxed{\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \quad \Longrightarrow \quad V(0) = 0.}$$

e. Step 3. Irreducible decomposition form. Let $\{W_\lambda\}_\lambda$ be a full set of inequivalent irreducible H_*-modules. Decompose

$$\mathcal{H}_D \cong \bigoplus_\lambda m_\lambda^{(D)} W_\lambda, \qquad \mathcal{H}_R \cong \bigoplus_\lambda m_\lambda^{(R)} W_\lambda.$$

Then

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) \cong \bigoplus_\lambda \mathrm{Mat}_{m_\lambda^{(D)} \times m_\lambda^{(R)}}(\mathbb{C}).$$

Therefore

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0$$

if and only if

$$m_\lambda^{(D)} m_\lambda^{(R)} = 0 \qquad \text{for all } \lambda.$$

Equivalently, $\mathcal{H}_D$ and $\mathcal{H}_R$ share no common irreducible H_*-sector.

f. Step 4. Honest correction for TECT. In TECT, one must be careful not to overstate what ordinary cubic little-group representation theory can do.

For the selected BCC shell axis, the ordinary stabilizers are abelian (C_{2v} for an oriented shell direction and D_{2h} for the opposite pair), so ordinary complex irreducible representations are one-dimensional. Therefore the actual local Dirac carrier is not forced by an ordinary 2D little-group irrep.

The correct shell-relevant local carrier is instead

$$\mathcal{H}_D \cong \mathbb{C}^2_{\text{valley}} \otimes \mathbb{C}^2_{\text{int}},$$

where

- $\mathbb{C}^2_{\text{valley}}$ is the opposite shell pair $\pm G_*$,
- $\mathbb{C}^2_{\text{int}}$ is the shell-relevant enriched internal/transverse doublet.

Thus the physically correct microscopic zero-mixing criterion is not "ordinary cubic irreducibility", but rather *sectorwise mismatch in shell-relevant labels.*

g. Step 5. Sectorwise remote-label mismatch criterion. Let

$$\mathcal{H}_R = \bigoplus_a K_a$$

be an H_*-invariant sector decomposition of the remote space. A remote sector K_a can mix with $\mathcal{H}_D$ at $p = 0$ only if it matches $\mathcal{H}_D$ in all shell-relevant labels.

A sufficient exclusion criterion is that each K_a differs from $\mathcal{H}_D$ in at least one of the following:

1. selected shell-pair label $\pm G_*$,
2. transverse versus longitudinal/volumetric character,

3. existence of the enriched shell-relevant internal doublet,
4. residual grading / parity / chirality,
5. any additional conserved microscopic label.

If every remote sector fails at least one of these matching conditions, then

$$\mathrm{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0 \qquad \forall a,$$

hence

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0,$$

and therefore

$$V(0) = 0.$$

So the TECT-sharp form of the theorem is

$$\boxed{\mathcal{H}_D \cong \mathbb{C}^2_{\text{valley}} \otimes \mathbb{C}^2_{\text{int}} \quad + \quad \text{sectorwise remote-label mismatch} \Longrightarrow V(0) = 0.}$$

h. Step 6. First nontrivial order in momentum. Once $V(0) = 0$, the leading inter-sector mixing starts at higher order in p. If the momentum representation is R_p, then the linear coefficient must lie in

$$V_i \in \mathrm{Hom}_{H_*}(R_p \otimes \mathcal{H}_R, \mathcal{H}_D).$$

Therefore:

- if

  $$\mathrm{Hom}_{H_*}(R_p \otimes \mathcal{H}_R, \mathcal{H}_D) = 0,$$

  then

  $$V(p) = O(|p|^2);$$

- otherwise, generically,

  $$V(p) = O(|p|).$$

i. Step 7. Schur-complement consequence. Assume in addition that the remote block is spectrally gapped:

$$\mathrm{spec}(H_R(p)) \cap (-\Delta, \Delta) = \varnothing.$$

Then for $|E| < \Delta/2$,

$$\|(H_R(p) - E)^{-1}\| \le \frac{2}{\Delta}.$$

Hence the Schur-complement self-energy

$$\Sigma(E, p) = V(p)(H_R(p) - E)^{-1} V(p)^\dagger$$

satisfies

$$\|\Sigma(E, p)\| \le \frac{2c^2}{\Delta} |p|^2$$

whenever

$$\|V(p)\| \le c|p|.$$

Therefore the effective low-energy Hamiltonian becomes

$$H_{\text{eff}}(E, p) = H_D(p) - \Sigma(E, p) = H_D(p) + O\left(\frac{|p|^2}{\Delta}\right).$$

j. Conclusion. The first remaining Dirac-closure condition is therefore reduced to a finite microscopic audit: prove that the remote sectors share no shell-relevant representation content with

$$\mathcal{H}_D \cong \mathbb{C}^2_{\text{valley}} \otimes \mathbb{C}^2_{\text{int}}.$$

Once that is done, one gets

$$V(0) = 0$$

exactly.

$$H_{\text{full}}(0) = \begin{pmatrix} H_D(0) & V(0) \\ V(0)^\dagger & H_R(0) \end{pmatrix}, \qquad \mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R.$$

$$U(h)\mathcal{H}_D \subset \mathcal{H}_D, \qquad U(h)\mathcal{H}_R \subset \mathcal{H}_R, \qquad U(h)H_{\text{full}}(0)U(h)^{-1} = H_{\text{full}}(0).$$

Then

$$V(0) \in \operatorname{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D).$$

Therefore

$$\boxed{\operatorname{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \Longrightarrow V(0) = 0.}$$

In the honest TECT form, the local Dirac carrier is not forced by an ordinary cubic 2D irrep. Rather,

$$\mathcal{H}_D \cong \mathbb{C}^2_{\text{valley}} \otimes \mathbb{C}^2_{\text{int}},$$

and the practical criterion is:

$$\boxed{\text{each remote sector differs from } \mathcal{H}_D \text{ in at least one shell-relevant label} \Longrightarrow V(0) = 0.}$$

The shell-relevant labels are:

$$\pm G_*, \quad \text{transverse/non-volumetric character}, \quad \text{enriched internal doublet}, \quad \text{grading/parity/chirality}, \quad \text{extra conserved labels}.$$

B. Summary

If, in addition,

$$\operatorname{spec}(H_R) \cap (-\Delta, \Delta) = \varnothing, \qquad \|V(p)\| \le c|p|,$$

then

$$\Sigma(E,p) = V(p)(H_R(p) - E)^{-1}V(p)^\dagger = O\left(\frac{|p|^2}{\Delta}\right),$$

so

$$H_{\text{eff}}(E,p) = H_D(p) + O\left(\frac{|p|^2}{\Delta}\right).$$

Hence the first remaining full Dirac-closure condition is reduced to a finite microscopic sectorwise label-audit problem.

C. Remote-gap positivity as an independent spectral input

a. Goal. We now close the statement

$$\Delta > 0$$

in the sharp theorem-level form that is actually justified in the present TECT framework.

The correct statement is not that the Class II correction *creates* a remote gap automatically. Rather, the remote gap is an *independent spectral property* of the remote sector, and the Class II correction is compatible with it and preserves it under a sufficiently small bounded perturbation.

b. Block decomposition. Write the full microscopic Hamiltonian near the selected shell crossing as

$$H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^\dagger & H_R(p) \end{pmatrix}, \tag{1}$$

where

- $H_D(p)$ is the isolated doubled shell Dirac block,
- $H_R(p)$ is the remote sector.

The remote-gap condition is

$$\operatorname{spec}(H_R(p)) \cap (-\Delta, \Delta) = \varnothing \qquad \text{for all sufficiently small } p. \tag{2}$$

A. Why shell nondegeneracy does not prove the remote gap

Let

$$M = \begin{pmatrix} B & -C & 0 \\ C & B & 0 \\ 0 & 0 & A \end{pmatrix}, \qquad \det M = A(B^2 + C^2) \neq 0. \tag{3}$$

This shell determinant controls only the nondegeneracy of the projected local Dirac symbol. It does *not* determine the spectrum of the complementary operator H_R.

Hence

$$\det M \neq 0$$

is not a proof of

$$\operatorname{spec}(H_R) \cap (-\Delta, \Delta) = \varnothing.$$

So the theorem-level route to $\Delta > 0$ must be spectral, not determinant-based.

B. Unperturbed remote gap

Assume there exists an unperturbed microscopic realization

$$H^{(0)}_{\text{full}}(p) = \begin{pmatrix} H^{(0)}_D(p) & V^{(0)}(p) \\ V^{(0)}(p)^\dagger & H^{(0)}_R(p) \end{pmatrix}, \tag{4}$$

whose remote sector has a uniform positive gap:

$$\operatorname{spec}(H^{(0)}_R(p)) \cap (-\Delta_0, \Delta_0) = \varnothing \qquad \text{for all } |p| \leq p_*, \tag{5}$$

for some $\Delta_0 > 0$.

This is the genuine microscopic spectral input.

C. Class II correction as a bounded perturbation

Now perturb the theory by a self-adjoint Class II correction:

$$H_{\text{full}}(p) = H^{(0)}_{\text{full}}(p) + \delta H(p), \tag{6}$$

and correspondingly

$$H_R(p) = H^{(0)}_R(p) + \delta H_R(p). \tag{7}$$

Assume that

$$\sup_{|p|\le p_*} \|\delta H_R(p)\| \le \varepsilon. \tag{8}$$

[Remote-gap stability under bounded perturbation] Assume (5) and (8). If

$$\varepsilon < \Delta_0, \tag{9}$$

then the perturbed remote sector still satisfies

$$\mathrm{spec}(H_R(p)) \cap (-\Delta, \Delta) = \varnothing, \qquad \Delta := \Delta_0 - \varepsilon > 0, \tag{10}$$

for all $|p| \le p_*$.

Fix p and let $\lambda \in \mathrm{spec}(H_R(p))$. By standard spectral stability for self-adjoint bounded perturbations,

$$\mathrm{dist}\big(\lambda, \mathrm{spec}(H^{(0)}_R(p))\big) \le \|\delta H_R(p)\|. \tag{11}$$

Since the unperturbed spectrum avoids $(-\Delta_0, \Delta_0)$, every spectral point of $H^{(0)}_R(p)$ has absolute value at least Δ_0. Hence

$$|\lambda| \ge \Delta_0 - \|\delta H_R(p)\| \ge \Delta_0 - \varepsilon = \Delta. \tag{12}$$

Because $\Delta > 0$, the interval $(-\Delta, \Delta)$ contains no spectrum of $H_R(p)$.

c. Corollary. If one assumes the stronger bound

$$\|\delta H_R\| < \frac{\Delta_0}{2}, \tag{13}$$

then

$$\Delta \ge \Delta_0 - \|\delta H_R\| > \frac{\Delta_0}{2} > 0. \tag{14}$$

This is the exact theorem-level meaning of the statement

"Class II is gap-compatible."

D. Resolvent bound

Let

$$|E| < \frac{\Delta}{2}. \tag{15}$$

Then

$$\mathrm{dist}\big(E, \mathrm{spec}(H_R(p))\big) \ge \frac{\Delta}{2}, \tag{16}$$

so

$$\|(H_R(p) - E)^{-1}\| \le \frac{2}{\Delta}. \tag{17}$$

E. Schur-complement consequence

Assume also

$$V(0) = 0, \qquad \|V(p)\| \le c|p|. \tag{18}$$

Then the reduced low-energy operator is

$$H_{\text{eff}}(E,p) = H_D(p) - \Sigma(E,p), \qquad \Sigma(E,p) := V(p)(H_R(p) - E)^{-1} V(p)^\dagger. \tag{19}$$

Using (17) and (18),

$$\|\Sigma(E,p)\| \le \|V(p)\|^2 \, \|(H_R(p) - E)^{-1}\| \le \frac{2c^2}{\Delta} |p|^2. \tag{20}$$

Therefore

$$H_{\text{eff}}(E,p) = H_D(p) + O\!\left(\frac{|p|^2}{\Delta}\right). \tag{21}$$

So the gapped remote sector alters the isolated Dirac block only at subleading quadratic order.

F. Honest theorem-level status

Hence the mathematically honest closure is:

$$\boxed{\Delta_0 > 0 \text{ for the unperturbed remote sector} \quad + \quad \|\delta H_R\| < \Delta_0 \Longrightarrow \Delta = \Delta_0 - \|\delta H_R\| > 0.} \tag{22}$$

In particular, Class II does not by itself prove the existence of the remote gap, but it preserves that gap under sufficiently small bounded perturbations.

D. Final Spectral Isolation Theorem

We now combine the three theorem-level inputs:

$$V(0) = 0, \qquad \Delta > 0, \qquad \text{mass protection.}$$

Assume:

1. the shell-relevant doubled block has the nondegenerate Dirac form

$$H_D(p) = \Gamma_i (Mp)_i, \qquad \{\Gamma_i, \Gamma_j\} = 2\delta_{ij}, \qquad \det M \neq 0; \tag{23}$$

2. little-group mismatch implies

$$V(0) = 0, \qquad \|V(p)\| \le c|p|; \tag{24}$$

3. the remote sector satisfies

$$\text{spec}(H_R(p)) \cap (-\Delta, \Delta) = \varnothing; \tag{25}$$

4. the residual symmetry forbids all constant Dirac masses.

Then, for $|E| < \Delta/2$,

$$H_{\text{eff}}(E,p) = \Gamma_i (Mp)_i + O\!\left(\frac{|p|^2}{\Delta}\right), \tag{26}$$

and therefore the full microscopic theory contains a stable spectrally isolated massless Dirac cone.

By the remote-gap condition,

$$\|(H_R(p)-E)^{-1}\| \le \frac{2}{\Delta}.$$

By (24), the Schur-complement correction is quadratic:

$$\Sigma(E,p) = V(p)(H_R(p)-E)^{-1}V(p)^\dagger = O\left(\frac{|p|^2}{\Delta}\right).$$

Hence the full low-energy Hamiltonian differs from the local shell Dirac block only by quadratic terms. The residual symmetry excludes all constant mass operators, so no $O(1)$ gap can appear. Therefore the leading linear Dirac cone survives and remains spectrally isolated.

E. Summary

$$\boxed{\text{Remote-gap positivity is an independent spectral input.}}$$

$$H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^\dagger & H_R(p) \end{pmatrix}, \qquad \operatorname{spec}(H_R(p)) \cap (-\Delta,\Delta) = \varnothing.$$

The shell determinant

$$\det M = A(B^2 + C^2)$$

controls only local shell-Dirac nondegeneracy, not the remote spectrum.

Assume an unperturbed realization with

$$\operatorname{spec}(H_R^{(0)}(p)) \cap (-\Delta_0,\Delta_0) = \varnothing, \qquad \Delta_0 > 0,$$

and a bounded Class II correction

$$H_R(p) = H_R^{(0)}(p) + \delta H_R(p), \qquad \sup_{|p|\le p_*} \|\delta H_R(p)\| \le \varepsilon.$$

Then

$$\boxed{\varepsilon < \Delta_0 \Longrightarrow \Delta := \Delta_0 - \varepsilon > 0}$$

and

$$\operatorname{spec}(H_R(p)) \cap (-\Delta,\Delta) = \varnothing.$$

For $|E| < \Delta/2$,

$$\|(H_R(p)-E)^{-1}\| \le \frac{2}{\Delta}.$$

If also

$$V(0) = 0, \qquad \|V(p)\| \le c|p|,$$

then

$$\Sigma(E,p) = V(p)(H_R(p)-E)^{-1}V(p)^\dagger, \qquad \|\Sigma(E,p)\| \le \frac{2c^2}{\Delta}|p|^2,$$

hence

$$H_{\text{eff}}(E,p) = H_D(p) + O\left(\frac{|p|^2}{\Delta}\right).$$

Therefore

$$\boxed{\text{Class II does not automatically prove } \Delta > 0,}$$

$$\boxed{\text{but it preserves a pre-existing remote gap under sufficiently small bounded perturbations.}}$$

Combining

$$\det M \neq 0, \qquad V(0) = 0, \qquad \Delta > 0, \qquad \text{mass protection,}$$

one obtains

$$\boxed{H_{\text{eff}}(E,p) = \Gamma_i (Mp)_i + O\left(\frac{|p|^2}{\Delta}\right),}$$

so the full microscopic theory contains a stable spectrally isolated massless Dirac cone.

F. Mass protection for the doubled shell Dirac block

a. Goal. We close the last remaining theorem-level ingredient of the local Dirac-stability program: the classification of all symmetry-allowed constant Hermitian 4×4 operators on the doubled shell block, and the criterion under which all genuine Dirac masses are forbidden.

b. Setup. Assume opposite-valley pairing has already been established, so that the isolated shell-relevant doubled block is

$$H_D(p) = \begin{pmatrix} h_+(p) & 0 \\ 0 & h_-(p) \end{pmatrix}, \qquad h_-(p) = -h_+(-p), \tag{27}$$

and after a shell-adapted basis choice and invertible linear momentum redefinition,

$$H_D(p) = \sum_{i=1}^{3} v_i p_i \, \alpha_i, \qquad \alpha_i := \tau_z \otimes \sigma_i. \tag{28}$$

Here σ_i acts on the internal Weyl doublet and τ_i acts on the valley pair $(+G_*, -G_*)$.

Thus

$$\mathcal{H}_D \cong \mathbb{C}^2_{\text{valley}} \otimes \mathbb{C}^2_{\text{int}}.$$

A. Classification of constant little-group-scalar operators

Let K be a constant Hermitian operator on $\mathcal{H}_D$. Write it in 2×2 valley-block form:

$$K = \begin{pmatrix} A & B \\ B^\dagger & D \end{pmatrix}, \qquad A, D, B \in M_2(\mathbb{C}). \tag{29}$$

Assume K preserves the little-group action at the crossing point $p = 0$, and that the two valley blocks carry the same irreducible little-group representation ρ. Then Schur's lemma gives

$$A = a_+ \mathbf{1}_2, \qquad D = a_- \mathbf{1}_2, \qquad B = b \, \mathbf{1}_2, \tag{30}$$

with $a_\pm \in \mathbb{R}$, $b \in \mathbb{C}$. Therefore every constant little-group-scalar Hermitian operator has the form

$$K = a_0 \mathbf{1}_4 + a_3 \, \tau_z \otimes \mathbf{1}_2 + m_1 \, \tau_x \otimes \mathbf{1}_2 + m_2 \, \tau_y \otimes \mathbf{1}_2. \tag{31}$$

So the allowed constant scalar operator space is four-dimensional.

B. Exact classification of genuine constant Dirac masses

A genuine constant Dirac mass is a Hermitian matrix M satisfying

$$\{M, \alpha_i\} = 0, \qquad i = 1, 2, 3. \tag{32}$$

[Exact constant mass space] For

$$\alpha_i = \tau_z \otimes \sigma_i,$$

the constant Hermitian matrices anticommting with all α_i are exactly

$$\mathcal{M}_D = \mathrm{span}_{\mathbb{R}} \{\tau_x \otimes \mathbf{1}_2,\ \tau_y \otimes \mathbf{1}_2\}. \tag{33}$$

Take a general constant Hermitian matrix in block form

$$M = \begin{pmatrix} A & B \\ B^\dagger & D \end{pmatrix}.$$

Using

$$\alpha_i = \begin{pmatrix} \sigma_i & 0 \\ 0 & -\sigma_i \end{pmatrix},$$

the condition $\{M, \alpha_i\} = 0$ becomes

$$\{A, \sigma_i\} = 0, \qquad \{D, \sigma_i\} = 0, \qquad B\sigma_i - \sigma_i B = 0, \qquad B^\dagger \sigma_i - \sigma_i B^\dagger = 0, \tag{34}$$

for all $i = 1, 2, 3$.

Since the Pauli matrices generate $M_2(\mathbb{C})$, the anticommutation equations imply

$$A = 0, \qquad D = 0,$$

while the commutation equations imply

$$B = b\,\mathbf{1}_2$$

for some $b \in \mathbb{C}$. Hence

$$M = \begin{pmatrix} 0 & b\,\mathbf{1}_2 \\ \bar{b}\,\mathbf{1}_2 & 0 \end{pmatrix} = (\Re b)\,\tau_x \otimes \mathbf{1}_2 + (\Im b)\,\tau_y \otimes \mathbf{1}_2. \tag{35}$$

This proves (33).

Therefore the only genuine constant masses are

$$\tau_x \otimes \mathbf{1}_2, \qquad \tau_y \otimes \mathbf{1}_2.$$

By contrast,

$$\mathbf{1}_4$$

is only an overall energy shift, and

$$\tau_z \otimes \mathbf{1}_2$$

is a valley-energy splitting term, not a genuine Dirac mass.

C. Residual valley symmetry and mass protection

Define the valley-charge generator

$$Q_V := \tau_z \otimes \mathbf{1}_2. \tag{36}$$

Then

$$[Q_V, H_D(p)] = 0, \tag{37}$$

so the doubled block carries a residual valley symmetry

$$U_\theta := e^{i\theta Q_V/2}. \tag{38}$$

Now compute the transformation of the two mass generators:

$$U_\theta(\tau_x \otimes \mathbf{1}_2)U_\theta^{-1} = \cos\theta\,(\tau_x \otimes \mathbf{1}_2) + \sin\theta\,(\tau_y \otimes \mathbf{1}_2), \tag{39}$$

$$U_\theta(\tau_y \otimes \mathbf{1}_2)U_\theta^{-1} = -\sin\theta\,(\tau_x \otimes \mathbf{1}_2) + \cos\theta\,(\tau_y \otimes \mathbf{1}_2). \tag{40}$$

Equivalently,

$$U_\theta\big((\tau_x \pm i\tau_y) \otimes \mathbf{1}_2\big)U_\theta^{-1} = e^{\pm i\theta}\big((\tau_x \pm i\tau_y) \otimes \mathbf{1}_2\big). \tag{41}$$

Thus the genuine Dirac-mass operators carry nonzero valley charge.

[Mass protection from residual valley symmetry] Assume:

1. the doubled low-energy block has the kinetic form (28);
2. the constant operator under consideration preserves the little-group action at $p = 0$;
3. the residual valley symmetry generated by $Q_V = \tau_z \otimes \mathbf{1}_2$ is exact on the isolated doubled block.

Then every symmetry-allowed constant Hermitian operator is of the form

$$K = a_0\,\mathbf{1}_4 + a_3\,\tau_z \otimes \mathbf{1}_2, \tag{42}$$

and in particular the genuine mass coefficients vanish:

$$m_1 = m_2 = 0. \tag{43}$$

Hence no constant Dirac mass is allowed.

By (31), every constant little-group-scalar Hermitian operator has the form

$$K = a_0\,\mathbf{1}_4 + a_3\,\tau_z \otimes \mathbf{1}_2 + m_1\,\tau_x \otimes \mathbf{1}_2 + m_2\,\tau_y \otimes \mathbf{1}_2.$$

Exact $U(1)_V$ invariance requires

$$U_\theta K U_\theta^{-1} = K \qquad \forall\theta.$$

The terms $\mathbf{1}_4$ and $\tau_z \otimes \mathbf{1}_2$ are invariant, whereas (39)–(41) show that

$$\tau_x \otimes \mathbf{1}_2, \qquad \tau_y \otimes \mathbf{1}_2$$

transform nontrivially. Therefore they are forbidden unless

$$m_1 = m_2 = 0.$$

By (33), these are precisely all genuine constant Dirac masses. Hence no constant Dirac mass is allowed.

c. Weaker version. The full continuous $U(1)_V$ is stronger than necessary. The discrete remnant

$$S_V := \tau_z \otimes \mathbf{1}_2, \qquad S_V^2 = \mathbf{1}_4, \tag{44}$$

already suffices, since

$$S_V(\tau_x \otimes \mathbf{1}_2)S_V^{-1} = -(\tau_x \otimes \mathbf{1}_2), \qquad S_V(\tau_y \otimes \mathbf{1}_2)S_V^{-1} = -(\tau_y \otimes \mathbf{1}_2). \tag{45}$$

So an exact residual $\mathbb{Z}_2^V$ also forbids all genuine constant masses.

D. Basis-free finite algebraic criterion

Let

$$A_{\mathrm{res}} = \{K \in \mathrm{Herm}(4) : U(g)KU(g)^{-1} = K \ \forall g \in G_{\mathrm{res}}\} \tag{46}$$

be the residual-symmetry allowed constant operator algebra. Then the exact finite mass-protection criterion is

$$A_{\mathrm{res}} \cap \{K \in \mathrm{Herm}(4) : \{K, \alpha_i\} = 0 \ \forall i\} = \{0\}. \tag{47}$$

If (47) holds, then no symmetry-allowed constant Dirac mass exists.

E. Honest TECT/Class II status

The canonical minimal Class II correction is gradient-linear and renormalizes the derivative shell coefficients

$$A_{\mathrm{eff}}, \quad B_{\mathrm{eff}}, \quad C_{\mathrm{eff}},$$

rather than inserting an independent constant 4×4 Dirac mass by construction. Therefore Class II is *compatible* with mass protection.

However, this does not by itself prove that all symmetry-allowed constant masses are forbidden in the full corrected microscopic theory. That still depends on the actual residual symmetry algebra A_{res} of the enriched shell block.

Hence the mathematically honest final statement is:

$$\boxed{\text{Mass protection is completely reduced to the finite classification problem } A_{\mathrm{res}} \cap \{K : \{K, \alpha_i\} = 0\} = \{0\}.} \tag{48}$$

Moreover, if exact residual $U(1)_V$ or $\mathbb{Z}_2^V$ survives in the actual TECT realization, then this criterion is automatically satisfied.

G. Summary

$$\boxed{\mathcal{H}_D \cong \mathbb{C}^2_{\mathrm{valley}} \otimes \mathbb{C}^2_{\mathrm{int}}, \qquad H_D(p) = \sum_{i=1}^{3} v_i p_i \, \alpha_i, \qquad \alpha_i = \tau_z \otimes \sigma_i.}$$

Every constant little-group-scalar Hermitian operator on the doubled block is

$$K = a_0 \, \mathbf{1}_4 + a_3 \, \tau_z \otimes \mathbf{1}_2 + m_1 \, \tau_x \otimes \mathbf{1}_2 + m_2 \, \tau_y \otimes \mathbf{1}_2.$$

The genuine constant Dirac masses are exactly

$$\boxed{\mathcal{M}_D = \mathrm{span}_{\mathbb{R}} \{\tau_x \otimes \mathbf{1}_2, \, \tau_y \otimes \mathbf{1}_2\}.}$$

Indeed,

$$\{M, \alpha_i\} = 0 \ \forall i \quad \Longleftrightarrow \quad M \in \mathcal{M}_D.$$

Define the valley-charge generator

$$Q_V = \tau_z \otimes \mathbf{1}_2, \qquad U_\theta = e^{i\theta Q_V/2}.$$

Then

$$U_\theta(\tau_x \otimes \mathbf{1}_2)U_\theta^{-1} = \cos\theta \, (\tau_x \otimes \mathbf{1}_2) + \sin\theta \, (\tau_y \otimes \mathbf{1}_2),$$

$$U_\theta(\tau_y \otimes \mathbf{1}_2)U_\theta^{-1} = -\sin\theta \, (\tau_x \otimes \mathbf{1}_2) + \cos\theta \, (\tau_y \otimes \mathbf{1}_2).$$

Hence the genuine mass operators carry nonzero valley charge.
Therefore:

$$\boxed{\text{If exact residual } U(1)_V \text{ survives, all constant Dirac masses are forbidden.}}$$

Even the weaker remnant

$$S_V = \tau_z \otimes \mathbf{1}_2$$

already suffices, because

$$S_V(\tau_x \otimes \mathbf{1}_2)S_V^{-1} = -(\tau_x \otimes \mathbf{1}_2), \qquad S_V(\tau_y \otimes \mathbf{1}_2)S_V^{-1} = -(\tau_y \otimes \mathbf{1}_2).$$

Equivalently, the exact finite algebraic criterion is

$$\boxed{A_{\mathrm{res}} \cap \{K \in \mathrm{Herm}(4) : \{K, \alpha_i\} = 0 \ \forall i\} = \{0\}.}$$

Hence the honest final status is:

$$\boxed{\text{mass protection is reduced completely to a finite } 4 \times 4 \text{ symmetry classification problem.}}$$

In the current TECT/Class II route:

$$\boxed{\text{Class II is compatible with mass protection,}}$$

but

$$\boxed{\text{actual mass protection still requires verification that the true residual symmetry forbids } \tau_x \otimes \mathbf{1}_2,\ \tau_y \otimes \mathbf{1}_2.}$$

DIRAC EMERGENCE CLOSURE THEOREM IN THE CLASS II ROUTE

1. Local shell-relevant Dirac block

In the current Class II route, the reduced local S^2-transport ansatz is

$$A_1(p) = \lambda_\parallel p_\parallel \sigma_3 + \alpha(p_1\sigma_1 + p_2\sigma_2) + \beta(-p_2\sigma_1 + p_1\sigma_2). \tag{49}$$

Equivalently, writing

$$p = (p_1, p_2, p_\parallel),$$

the symbol can be expressed as

$$A_1(p) = q_a(p)\sigma_a, \qquad q(p) = Mp, \tag{50}$$

with

$$M = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_\parallel \end{pmatrix}. \tag{51}$$

Hence

$$\det M = \lambda_\parallel(\alpha^2 + \beta^2). \tag{52}$$

Therefore the local nondegeneracy condition is exactly

$$\lambda_\parallel \neq 0, \qquad (\alpha, \beta) \neq (0, 0). \tag{53}$$

In the coupling notation of the Class II seed,

$$\lambda_\parallel = -\frac{3\sqrt{3}\pi}{512}\, g_c\, \theta_0^3 \kappa^3, \qquad \alpha = -\frac{\sqrt{3}\pi}{16}\, g_E\, \theta_0 \kappa, \qquad \beta = -\frac{\sqrt{3}\pi}{16}\, g_O\, \theta_0 \kappa, \tag{54}$$

so (53) is equivalent to

$$g_c \neq 0, \qquad (g_E, g_O) \neq (0, 0). \tag{55}$$

2. Doubled local Dirac block

Assume that the opposite shell partner exists and that the matched opposite-valley reduction has already been carried out. Then the isolated shell-relevant doubled block takes the 4×4 Dirac form

$$H_D(p) = \Gamma_i (Mp)_i, \qquad \{\Gamma_i, \Gamma_j\} = 2\delta_{ij}, \tag{56}$$

with M given by (51). By (52) and (53),

$$\det M \neq 0. \tag{57}$$

Thus $H_D(p)$ is a nondegenerate local massless Dirac block.

3. Full microscopic block decomposition

Let the full microscopic linearized Hamiltonian near the selected shell crossing be written as

$$H_{\mathrm{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^\dagger & H_R(p) \end{pmatrix}, \tag{58}$$

where

- $H_D(p)$ is the shell-relevant doubled 4×4 Dirac block,
- $H_R(p)$ is the remote sector,
- $V(p)$ is the inter-sector mixing.

4. Three additional closure hypotheses

Assume:

a. (H1) Little-group mismatch / zero constant mixing. The enriched shell block and the remote sector share no common residual representation sector at $p = 0$, so that

$$V(0) = 0, \qquad \|V(p)\| \leq c|p| \quad \text{for sufficiently small } p. \tag{59}$$

b. (H2) Positive remote spectral gap. There exists $\Delta > 0$ such that

$$\mathrm{spec}(H_R(p)) \cap (-\Delta, \Delta) = \varnothing \qquad \text{for sufficiently small } p. \tag{60}$$

c. (H3) Mass protection. No symmetry-allowed constant Hermitian 4×4 operator on the doubled block is a Dirac mass. Equivalently, if G_{res} is the residual low-energy symmetry group and

$$A_{\mathrm{res}} = \{K \in \mathrm{Herm}(4) : U(g)KU(g)^{-1} = K \ \forall g \in G_{\mathrm{res}}\}, \tag{61}$$

then

$$A_{\mathrm{res}} \cap \{K \in \mathrm{Herm}(4) : \{K, \Gamma_i\} = 0 \ \forall i\} = \{0\}. \tag{62}$$

5. Resolvent bound

Let $|E| < \Delta/2$. By (60),

$$\|(H_R(p) - E)^{-1}\| \leq \frac{2}{\Delta}. \tag{63}$$

6. Schur-complement reduction

The exact low-energy effective Hamiltonian on the doubled block is the Feshbach–Schur operator

$$H_{\text{eff}}(E,p) = H_D(p) - \Sigma(E,p), \tag{64}$$

with self-energy

$$\Sigma(E,p) = V(p)(H_R(p) - E)^{-1}V(p)^{\dagger}. \tag{65}$$

Using (59) and (63),

$$\|\Sigma(E,p)\| \le \|V(p)\|^2 \, \|(H_R(p) - E)^{-1}\| \le \frac{2c^2}{\Delta}|p|^2. \tag{66}$$

Hence

$$H_{\text{eff}}(E,p) = H_D(p) + O\!\left(\frac{|p|^2}{\Delta}\right). \tag{67}$$

7. Exclusion of an induced constant mass

By (62), no symmetry-allowed constant operator can anticommute with all three kinetic gamma matrices. Therefore the Schur-complement correction cannot generate an $O(1)$ Dirac mass term. Thus the effective Hamiltonian remains massless at leading order:

$$H_{\text{eff}}(0,0) = 0. \tag{68}$$

8. Final closure theorem

[Dirac Emergence Closure Theorem, Class II form] Assume:

1. the local Class II nondegeneracy condition holds,
$$\lambda_{\parallel} \neq 0, \qquad (\alpha,\beta) \neq (0,0),$$
equivalently
$$g_c \neq 0, \qquad (g_E, g_O) \neq (0,0);$$
2. the opposite-valley partner exists and yields the doubled local Dirac block (56);
3. the little-group mismatch hypothesis (59) holds;
4. the remote spectral gap hypothesis (60) holds;
5. the mass-protection hypothesis (62) holds.

Then, for sufficiently small p and $|E| < \Delta/2$, the full TECT linearized theory contains a spectrally isolated four-dimensional low-energy sector whose effective Hamiltonian is

$$H_{\text{Dirac}}(E,p) = \Gamma_i (Mp)_i + O\!\left(\frac{|p|^2}{\Delta}\right), \qquad \{\Gamma_i, \Gamma_j\} = 2\delta_{ij}, \qquad \det M = \lambda_{\parallel}(\alpha^2+\beta^2) \neq 0. \tag{69}$$

In particular, the full theory contains a stable, spectrally isolated, massless Dirac cone.

Item (1) implies $\det M \neq 0$, so the local shell-relevant block is a nondegenerate massless Dirac symbol. Item (2) upgrades this to the doubled 4×4 Dirac block. Item (3) removes constant inter-sector mixing and makes the first nontrivial mixing at least linear in p. Item (4) provides the uniform resolvent bound (63). Hence the Schur-complement self-energy is quadratic in momentum, as in (66), and the effective Hamiltonian differs from the local Dirac block only by $O(|p|^2/\Delta)$. Item (5) excludes all symmetry-allowed constant Dirac mass terms. Therefore the leading linear Dirac cone survives, remains massless, and stays spectrally isolated from all remote states.

9. Honest logical status

What is now genuinely closed is the implication

local Class II shell Dirac nondegeneracy+opposite-valley doubling+zero constant mixing+positive remote gap+mass protection

What may still remain open in an actual microscopic realization is only the direct verification that these hypotheses are indeed satisfied by the final microscopic TECT/Class II linearization.

H. Summary

$$\boxed{A_1(p) = \lambda_\| p_\| \sigma_3 + \alpha(p_1\sigma_1 + p_2\sigma_2) + \beta(-p_2\sigma_1 + p_1\sigma_2)}$$

$$\boxed{M = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_\| \end{pmatrix}, \qquad \det M = \lambda_\|(\alpha^2 + \beta^2)}$$

Hence the local shell nondegeneracy condition is

$$\boxed{\lambda_\| \neq 0, \qquad (\alpha, \beta) \neq (0,0),}$$

equivalently

$$\boxed{g_c \neq 0, \qquad (g_E, g_O) \neq (0,0).}$$

Let the full microscopic Hamiltonian be

$$H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^\dagger & H_R(p) \end{pmatrix}, \qquad H_D(p) = \Gamma_i (Mp)_i, \qquad \{\Gamma_i, \Gamma_j\} = 2\delta_{ij}.$$

Assume:

$$V(0) = 0, \qquad \|V(p)\| \leq c|p|,$$

$$\operatorname{spec}(H_R(p)) \cap (-\Delta, \Delta) = \varnothing, \qquad \Delta > 0,$$

and

$$A_{\text{res}} \cap \{K \in \operatorname{Herm}(4) : \{K, \Gamma_i\} = 0 \ \forall i\} = \{0\}.$$

Then for $|E| < \Delta/2$,

$$\|(H_R(p) - E)^{-1}\| \leq \frac{2}{\Delta},$$

and the Schur-complement self-energy obeys

$$\Sigma(E,p) = V(p)(H_R(p) - E)^{-1} V(p)^\dagger, \qquad \|\Sigma(E,p)\| \leq \frac{2c^2}{\Delta}|p|^2.$$

Therefore

$$\boxed{H_{\text{Dirac}}(E,p) = \Gamma_i (Mp)_i + O\left(\frac{|p|^2}{\Delta}\right), \qquad \det M \neq 0.}$$

Hence:

$$\boxed{\text{local Class II nondegeneracy} + \text{opposite-valley doubling} + V(0) = 0 + \Delta > 0 + \text{mass protection}}$$

$$\boxed{\Longrightarrow \text{stable spectrally isolated massless Dirac cone.}}$$

Honest final status:

$$\boxed{\text{the theorem-level implication is closed,}}$$

but

$$\boxed{\text{the actual microscopic TECT/Class II realization still requires finite verification of the input hypotheses.}}$$

MASTER VERIFICATION CHECKLIST FOR THE MICROSCOPIC CLASS II CLOSURE

0. Fixed data and notation

Choose and freeze the following data once and for all:

1. the selected shell vector

$$G_* \in \mathbb{R}^3 \setminus \{0\}, \qquad \hat{n} := \frac{G_*}{|G_*|};$$

2. an orthonormal shell frame

$$\{e_1, e_2, \hat{n}\}, \qquad e_\alpha \cdot \hat{n} = 0, \qquad e_\alpha \cdot e_\beta = \delta_{\alpha\beta};$$

3. the actual symmetry-allowed microscopic tensor space

$$V_{\text{TECT}} \subset \mathbb{R}^3_{\text{out}} \otimes \mathbb{R}^3_{\text{grad}} \otimes U_{\text{int}};$$

4. the shell-relevant internal channels $u^{(1)}, u^{(2)}$;
5. the opposite shell pair $\pm G_*$;
6. the residual symmetry group H_* at the selected crossing.

All later tests depend on this frozen choice. If these data are changed midstream, all previously computed witness statements must be recomputed.

1. Task I: shell witness nonvanishing

a. *Goal.* Verify that the local shell-reduced Dirac block exists at all.
b. *Exact finite criterion.* For each transverse Dirac channel $\alpha = 1, 2$, define

$$T^{(\alpha)}_{ija} = e_{\alpha i} G_{*j} u^{(\alpha)}_a.$$

Then the exact criterion is

$$P_V T^{(1)} \neq 0, \qquad P_V T^{(2)} \neq 0,$$

where P_V is the orthogonal projector onto the symmetry-allowed tensor space V_{TECT}.
Equivalently, it suffices to construct explicit seed tensors

$$N^{(1)}_{\text{seed}}, N^{(2)}_{\text{seed}} \in V_{\text{TECT}}$$

such that

$$\langle T^{(1)}, N^{(1)}_{\text{seed}} \rangle \neq 0, \qquad \langle T^{(2)}, N^{(2)}_{\text{seed}} \rangle \neq 0.$$

c. Class II coefficient language. In the shell-adapted coefficient form, compute

$$A_{\rm eff}, \qquad B_{\rm eff}, \qquad C_{\rm eff},$$

or equivalently

$$\lambda_{\|}, \qquad \alpha, \qquad \beta.$$

Completion requires

$$A_{\rm eff} \neq 0, \qquad (B_{\rm eff}, C_{\rm eff}) \neq (0,0),$$

equivalently

$$\lambda_{\|} \neq 0, \qquad (\alpha, \beta) \neq (0,0),$$

equivalently

$$g_c \neq 0, \qquad (g_E, g_O) \neq (0,0).$$

d. Deliverable. Produce either:

1. a projection calculation showing

$$P_V T^{(1)} \neq 0, \qquad P_V T^{(2)} \neq 0,$$

   or

2. explicit admissible seed tensors producing nonzero overlaps.

e. Pass/fail meaning. If this task fails, no local shell Weyl/Dirac block exists in the actual microscopic tensor space, and the Class II route is not microscopically realized.

2. Task II: opposite-valley matching

f. Goal. Upgrade two Weyl valleys into one local Dirac pair.
g. Exact criterion. At $+G_*$ and $-G_*$, compute the shell witness triples

$$(W_{\|}^+, W_I^+, W_J^+), \qquad (W_{\|}^-, W_I^-, W_J^-).$$

Completion requires

$$W_{\|}^- = W_{\|}^+, \qquad W_I^- = W_I^+, \qquad W_J^- = W_J^+.$$

h. Equivalent coefficient form. In coefficient language:

$$A^- = A^+, \qquad B^- = B^+, \qquad C^- = C^+.$$

i. Deliverable. Produce the explicit matched transformation law under the opposite-valley map and verify equality of the two triples.
j. Pass/fail meaning. If this task fails, one may still have local Weyl sectors, but not a genuine doubled local Dirac block.

3. Task III: zero constant mixing

k. Goal. Prove

$$V(0) = 0.$$

l. Exact criterion. Let

$$\mathcal{H}_{\mathrm{full}} = \mathcal{H}_D \oplus \mathcal{H}_R,$$

where $\mathcal{H}_D$ is the isolated doubled shell block and $\mathcal{H}_R$ is the remote sector. Decompose both under the residual symmetry H_*. Completion requires

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0.$$

m. Operational version. Decompose

$$\mathcal{H}_R = \bigoplus_a K_a.$$

For each remote sector K_a, verify that it differs from $\mathcal{H}_D$ in at least one shell-relevant label:

$$\pm G_*, \quad \text{transverse/non-volumetric character}, \quad \text{enriched internal doublet}, \quad \text{grading/parity/chirality}, \quad \text{other conserved microsco}$$

If every K_a fails at least one matching condition, then

$$\mathrm{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0 \quad \forall a,$$

hence

$$V(0) = 0.$$

n. Deliverable. A table of remote sectors K_a with a mismatch flag for each shell-relevant label.

o. Pass/fail meaning. If even one remote sector shares all shell-relevant labels with $\mathcal{H}_D$, constant mixing is not excluded and the local Dirac node is not yet stable.

4. Task IV: remote spectral gap

p. Goal. Prove the existence of

$$\Delta > 0$$

such that

$$\mathrm{spec}(H_R(p)) \cap (-\Delta, \Delta) = \varnothing$$

for all sufficiently small p.

q. Important honesty condition. This is an independent spectral input. It does *not* follow from

$$\det M \neq 0.$$

r. Minimal finite test. Choose a first nontrivial truncated remote basis

$$\mathcal{B}_R^{(N)} = \{\text{remote shell/Bloch basis states up to order } N\}.$$

Construct the truncated remote matrix

$$H_R^{(N)}(p).$$

Define

$$\Delta_N := \inf_{|p| \le p_*} \mathrm{dist}\big(0, \mathrm{spec}(H_R^{(N)}(p))\big).$$

Then:

1. if $\Delta_N \le 0$, the truncation does not support a remote gap;
2. if $\Delta_N > 0$, estimate the truncation error

   $$\|H_R - H_R^{(N)}\| \le \varepsilon_N.$$

   If

   $$\varepsilon_N < \Delta_N,$$

   then

   $$\Delta \ge \Delta_N - \varepsilon_N > 0.$$

s. Alternative perturbative route. If one already has an unperturbed gapped realization

$$\mathrm{spec}(H_R^{(0)}(p)) \cap (-\Delta_0, \Delta_0) = \varnothing,$$

and the Class II correction obeys

$$\sup_{|p|\le p_*} \|\delta H_R(p)\| < \Delta_0,$$

then

$$\Delta \ge \Delta_0 - \|\delta H_R\| > 0.$$

t. Deliverable. Either:

1. a finite Bloch/shell-truncated spectral computation with a rigorous lower bound, or
2. a perturbative gap-stability estimate from an already-gapped realization.

u. Pass/fail meaning. If this task fails, the low-energy block may mix resonantly with remote states and spectral isolation is not established.

5. Task V: mass protection

v. Goal. Exclude all symmetry-allowed constant Dirac masses.
w. Exact mass space. For the doubled local Dirac block

$$H_D(p) = \sum_{i=1}^{3} v_i p_i (\tau_z \otimes \sigma_i),$$

the full constant Dirac-mass space is

$$\mathcal{M}_D = \mathrm{span}_{\mathbb{R}} \left\{ \tau_x \otimes \mathbf{1}_2,\, \tau_y \otimes \mathbf{1}_2 \right\}.$$

x. Exact completion criterion. Let

$$A_{\mathrm{res}} = \{ K \in \mathrm{Herm}(4) : U(g) K U(g)^{-1} = K \ \forall g \in G_{\mathrm{res}} \}$$

be the residual-symmetry allowed constant algebra. Completion requires

$$A_{\mathrm{res}} \cap \mathcal{M}_D = \{0\}.$$

y. Sufficient mechanism. It is sufficient to prove exact residual valley symmetry

$$U(1)_V \quad \text{or already} \quad \mathbb{Z}_2^V.$$

Indeed, with

$$Q_V = \tau_z \otimes \mathbf{1}_2,$$

the two mass generators transform nontrivially and are forbidden.
z. Deliverable. Either:

1. an explicit basis computation of A_{res} and proof that it excludes

$$\tau_x \otimes \mathbf{1}_2, \qquad \tau_y \otimes \mathbf{1}_2,$$

or

2. a proof that exact residual valley $U(1)_V$ or $\mathbb{Z}_2^V$ survives.

aa. Pass/fail meaning. If this task fails, the doubled Dirac block is not protected against a constant symmetry-preserving mass.

6. Final assembly step

Once Tasks I–V are passed, define

$$H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^\dagger & H_R(p) \end{pmatrix}.$$

Then:

1. Task I gives the local Weyl block;
2. Task II gives opposite-valley doubling into a local Dirac block;
3. Task III gives
$$V(0) = 0, \qquad \|V(p)\| \le c|p|;$$
4. Task IV gives
$$\operatorname{spec}(H_R(p)) \cap (-\Delta, \Delta) = \varnothing;$$
5. Task V forbids any constant Dirac mass.

Hence the Schur-complement effective Hamiltonian satisfies

$$H_{\text{eff}}(E,p) = \Gamma_i (Mp)_i + O\left(\frac{|p|^2}{\Delta}\right), \qquad \det M \neq 0,$$

so the full microscopic Class II realization contains a stable spectrally isolated massless Dirac cone.

7. Optimal execution order

The optimal practical order is:

$$\boxed{\text{Task I} \;\rightarrow\; \text{Task II} \;\rightarrow\; \text{Task III} \;\rightarrow\; \text{Task IV} \;\rightarrow\; \text{Task V.}}$$

This order is optimal because Tasks I and II first establish that the local shell Dirac block actually exists, while Tasks III–V then certify its stability inside the full theory.

8. Honest final status

What is already theorem-level closed is the implication

$$\text{Tasks I–V completed} \Longrightarrow \text{stable spectrally isolated massless local Dirac cone.}$$

What remains open microscopically is only whether the actual final TECT/Class II linearization passes each of Tasks I–V.

I. Summary

$$\boxed{\text{Master Verification Checklist for the actual microscopic Class II realization}}$$

Task I: shell witness nonvanishing

$$P_V T^{(1)} \neq 0, \qquad P_V T^{(2)} \neq 0,$$

equivalently

$$g_c \neq 0, \qquad (g_E, g_O) \neq (0,0).$$

Task II: opposite-valley matching

$$(W_\parallel^-, W_I^-, W_J^-) = (W_\parallel^+, W_I^+, W_J^+),$$

equivalently

$$A^- = A^+, \qquad B^- = B^+, \qquad C^- = C^+.$$

Task III: zero constant mixing

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \Longrightarrow V(0) = 0.$$

Task IV: remote spectral gap

$$\mathrm{spec}(H_R(p)) \cap (-\Delta, \Delta) = \varnothing \qquad (\Delta > 0).$$

This must be checked independently by a finite Bloch/shell-truncated spectral calculation or by perturbative stability from an already-gapped realization.

Task V: mass protection

$$\mathcal{M}_D = \mathrm{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\},$$

and completion requires

$$A_{\mathrm{res}} \cap \mathcal{M}_D = \{0\}.$$

A sufficient condition is exact residual

$$U(1)_V \quad \text{or} \quad \mathbb{Z}_2^V.$$

Final assembly

If Tasks I–V are all passed, then

$$H_{\mathrm{eff}}(E,p) = \Gamma_i (Mp)_i + O\left(\frac{|p|^2}{\Delta}\right), \qquad \det M \neq 0,$$

so the full microscopic TECT/Class II realization contains a

$$\boxed{\text{stable spectrally isolated massless Dirac cone.}}$$

Optimal order

$$\boxed{\text{Task I} \;\to\; \text{Task II} \;\to\; \text{Task III} \;\to\; \text{Task IV} \;\to\; \text{Task V.}}$$

Honest status

$$\boxed{\text{The theorem-level implication is closed.}}$$

$$\boxed{\text{What remains is only the finite microscopic verification that the actual Class II realization passes Tasks I–V.}}$$

TASK I: ACTUAL EXECUTION FORM OF THE WITNESS-NONVANISHING TEST

A. Exact finite-dimensional formulation

Let

$$W = \mathbb{R}^3_{\mathrm{out}} \otimes \mathbb{R}^3_{\mathrm{grad}} \otimes U_{\mathrm{int}}$$

be the ambient finite-dimensional tensor space, and let

$$V_{\mathrm{TECT}} \subset W$$

be the actual symmetry-allowed microscopic tensor space surviving the linearization around the chosen TECT defect background.

Fix the shell data:

$$G_* \neq 0, \qquad \hat{n} := \frac{G_*}{|G_*|}, \qquad \{e_1, e_2, \hat{n}\} \text{ orthonormal.}$$

For each required reduced channel α, define the shell target covector

$$T^{(\alpha)}_{ija} = e_{\alpha i} G_{*j} u^{(\alpha)}_a,$$

and the corresponding linear witness functional

$$L_\alpha(N) := \langle T^{(\alpha)}, N \rangle.$$

Then the exact criterion is

$$L_\alpha|_{V_{\mathrm{TECT}}} \equiv 0 \quad \Longleftrightarrow \quad P_V T^{(\alpha)} = 0,$$

where $P_V : W \to V_{\mathrm{TECT}}$ is the orthogonal projector.

Hence Task I is completed once one proves

$$P_V T^{(\alpha)} \neq 0$$

for the longitudinal channel and for the transverse channel(s).

B. Basis-level operational test

Choose any basis

$$\{E_A\}_{A=1}^d$$

of V_{TECT}, where $d = \dim V_{\mathrm{TECT}} < \infty$. Write

$$N = \sum_{A=1}^d x_A E_A.$$

Then

$$L_\alpha(N) = \sum_{A=1}^d x_A\, \ell_{\alpha A}, \qquad \ell_{\alpha A} := \langle T^{(\alpha)}, E_A \rangle.$$

Therefore:

$$P_V T^{(\alpha)} = 0 \quad \Longleftrightarrow \quad \ell_{\alpha A} = 0 \;\; \forall A,$$

while

$$P_V T^{(\alpha)} \neq 0 \quad \Longleftrightarrow \quad \exists A \text{ such that } \ell_{\alpha A} \neq 0.$$

So the actual computation is finite:

$$\text{compute the overlap vector } \ell_\alpha = (\ell_{\alpha 1}, \ldots, \ell_{\alpha d}).$$

If it is not identically zero, the channel survives generically.

C. Seed criterion

A stronger and faster sufficient test is:
if there exists a single admissible tensor

$$N^{(\alpha)}_{\text{seed}} \in V_{\text{TECT}}$$

such that

$$\langle T^{(\alpha)}, N^{(\alpha)}_{\text{seed}} \rangle \neq 0,$$

then automatically

$$P_V T^{(\alpha)} \neq 0.$$

Thus Task I can be closed either by a full projector computation or by constructing one explicit allowed seed tensor per required channel.

D. Canonical Class II seed directions

Inside the minimal Class II parametrization, the induced tensor is organized by

$$m^a(P_0) = m\,\delta^{a3},$$

together with the shell-adapted coefficients

$$C^{33}_{ij} = c_\parallel\,\hat{n}_i\hat{n}_j,$$

$$C^{31}_{ij} = c_I\,\hat{n}_i e_{1j} + c_J\,\hat{n}_i e_{2j},$$

$$C^{32}_{ij} = -c_J\,\hat{n}_i e_{1j} + c_I\,\hat{n}_i e_{2j}.$$

This immediately gives three canonical seed directions:
a. Longitudinal seed

$$c_\parallel \neq 0, \qquad c_I = c_J = 0.$$

This produces

$$g_c \neq 0.$$

b. Even transverse seed

$$c_\parallel = 0, \qquad c_I \neq 0, \qquad c_J = 0.$$

This produces

$$g_E \neq 0.$$

c. Odd transverse seed

$$c_\parallel = 0, \qquad c_I = 0, \qquad c_J \neq 0.$$

This produces

$$g_O \neq 0.$$

Hence, inside the minimal Class II truncation,

$$g_c \propto c_\parallel, \qquad g_E \propto c_I, \qquad g_O \propto c_J.$$

Therefore the witness problem is already reduced to the question whether these seed directions are actually present in the true symmetry-allowed tensor space V_{TECT}.

E. Immediate consequence for Task I

Suppose the actual microscopic tensor space V_{TECT} contains:

1. one admissible longitudinal seed with $c_{\|} \neq 0$,
2. one admissible transverse seed with $(c_I, c_J) \neq (0,0)$.

Then

$$g_c(N) \neq 0$$

and

$$(g_E(N), g_O(N)) \neq (0,0)$$

hold on an open dense full-measure subset of V_{TECT}.

Equivalently,

$$\lambda_{\|}(N) \neq 0, \qquad (\alpha(N), \beta(N)) \neq (0,0)$$

generically on V_{TECT}, and the local S^2 reduced symbol is nondegenerate.

F. Actual deliverable for the final microscopic model

To close Task I in the actual final TECT realization, one must provide exactly one of the following:

d. Route 1: projector computation Construct a basis of V_{TECT} and compute

$$\ell_{\|,A} = \langle T^{(\|)}, E_A \rangle, \qquad \ell_{I,A} = \langle T^{(I)}, E_A \rangle, \qquad \ell_{J,A} = \langle T^{(J)}, E_A \rangle.$$

Then verify:

$$(\ell_{\|,1}, \dots, \ell_{\|,d}) \not\equiv 0, \qquad (\ell_{I,1}, \dots, \ell_{I,d}, \ell_{J,1}, \dots, \ell_{J,d}) \not\equiv 0.$$

e. Route 2: explicit seed construction Produce explicit admissible tensors

$$N_{\text{seed}}^{(\|)}, \qquad N_{\text{seed}}^{(\perp)}$$

in the final microscopic symmetry class such that

$$\langle T^{(\|)}, N_{\text{seed}}^{(\|)} \rangle \neq 0, \qquad \det T_{\perp}\big(N_{\text{seed}}^{(\perp)}\big) \neq 0.$$

G. Honest final status

Thus the mathematically honest conclusion is:

$$\boxed{\text{Task I is completely closed inside the minimal Class II truncation.}}$$

What is still open is only:

$$\boxed{\text{whether the actual full TECT microscopic symmetry-allowed space } V_{\text{TECT}} \text{ really contains those Class II seed directions.}}$$

So the remaining work is not conceptual; it is an explicit finite tensor-space inclusion test.

J. Summary

$$\boxed{\text{Task I is a finite tensor-space test.}}$$

Let

$$V_{\mathrm{TECT}} \subset \mathbb{R}^3_{\mathrm{out}} \otimes \mathbb{R}^3_{\mathrm{grad}} \otimes U_{\mathrm{int}}$$

be the actual symmetry-allowed microscopic tensor space.

For each required shell channel α, define

$$T^{(\alpha)}_{ija} = e_{\alpha i} G_{*j} u^{(\alpha)}_a, \qquad L_\alpha(N) = \langle T^{(\alpha)}, N \rangle.$$

Then:

$$\boxed{L_\alpha|_{V_{\mathrm{TECT}}} \equiv 0 \Longleftrightarrow P_V T^{(\alpha)} = 0.}$$

Hence Task I is complete once

$$P_V T^{(\alpha)} \neq 0$$

is proved for the required channels.

A sufficient condition is the existence of one explicit admissible seed

$$N^{(\alpha)}_{\mathrm{seed}} \in V_{\mathrm{TECT}}$$

such that

$$\langle T^{(\alpha)}, N^{(\alpha)}_{\mathrm{seed}} \rangle \neq 0.$$

Inside the minimal Class II parametrization,

$$m^a(P_0) = m\,\delta^{a3}, \qquad C^{33}_{ij} = c_\parallel \hat{n}_i \hat{n}_j,$$

$$C^{31}_{ij} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \qquad C^{32}_{ij} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j},$$

and therefore

$$g_c \propto c_\parallel, \qquad g_E \propto c_I, \qquad g_O \propto c_J.$$

So the canonical seed directions are:

$$c_\parallel \neq 0 \Rightarrow g_c \neq 0,$$

$$c_I \neq 0 \Rightarrow g_E \neq 0,$$

$$c_J \neq 0 \Rightarrow g_O \neq 0.$$

Therefore:

$$\boxed{\text{inside the minimal Class II truncation, the witness problem is already closed.}}$$

The only remaining Task I question is:

$$\boxed{\text{Does the actual full TECT microscopic space } V_{\mathrm{TECT}} \text{ really contain those Class II seed directions?}}$$

If yes, then generically

$$g_c \neq 0, \qquad (g_E, g_O) \neq (0, 0),$$

hence the local S^2 symbol is nondegenerate.

TASK I IN THE EXPLICIT CLASS II BASIS $\{B_\parallel, B_I, B_J\}$

1. Ambient tensor space and witness functionals

Let

$$W = \mathbb{R}^3_{\rm out} \otimes \mathbb{R}^3_{\rm grad} \otimes U_{\rm int}, \qquad V_{\rm TECT} \subset W$$

be the actual symmetry-allowed microscopic tensor space around the chosen TECT defect background.

Fix the shell frame

$$\hat{n} = \frac{G_*}{|G_*|}, \qquad \{e_1, e_2, \hat{n}\} \text{ orthonormal.}$$

Define the shell witness covectors

$$T^{(\parallel)}_{ijb} := \hat{n}_i G_{*j}\, \delta^{b3},$$

$$T^{(I)}_{ijb} := \hat{n}_i e_{1j}\, \delta^{b1} + \hat{n}_i e_{2j}\, \delta^{b2},$$

$$T^{(J)}_{ijb} := -\hat{n}_i e_{2j}\, \delta^{b1} + \hat{n}_i e_{1j}\, \delta^{b2}.$$

The corresponding linear functionals are

$$L_\parallel(N) = \langle T^{(\parallel)}, N\rangle, \qquad L_I(N) = \langle T^{(I)}, N\rangle, \qquad L_J(N) = \langle T^{(J)}, N\rangle.$$

These are precisely the shell witness maps whose nonvanishing decides whether the longitudinal and transverse reduced channels survive. The exact dichotomy is

$$L_\alpha|_{V_{\rm TECT}} \equiv 0 \Longleftrightarrow P_V T^{(\alpha)} = 0,$$

and if $P_V T^{(\alpha)} \neq 0$, then the channel is generically nonzero on an open dense full-measure subset of $V_{\rm TECT}$.

2. Canonical Class II basis tensors

Inside the minimal Class II ansatz, choose the three canonical shell-adapted basis tensors

$$B_\parallel: \quad (B_\parallel)^3_{ij} = \hat{n}_i \hat{n}_j, \qquad (B_\parallel)^1_{ij} = (B_\parallel)^2_{ij} = 0,$$

$$B_I: \quad (B_I)^1_{ij} = \hat{n}_i e_{1j}, \qquad (B_I)^2_{ij} = \hat{n}_i e_{2j}, \qquad (B_I)^3_{ij} = 0,$$

$$B_J: \quad (B_J)^1_{ij} = -\hat{n}_i e_{2j}, \qquad (B_J)^2_{ij} = \hat{n}_i e_{1j}, \qquad (B_J)^3_{ij} = 0.$$

Equivalently, an arbitrary tensor in the explicit seed subspace is

$$N = c_\parallel B_\parallel + c_I B_I + c_J B_J.$$

This is exactly the shell-relevant image of the minimal Class II seed parametrization

$$m^a(P_0) = m\, \delta^{a3}, \qquad C^{33}_{ij} = c_\parallel \hat{n}_i \hat{n}_j,$$

$$C^{31}_{ij} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \qquad C^{32}_{ij} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}.$$

3. Overlap matrix

Define the 3×3 witness matrix

$$\mathcal{M}_{\alpha A} := \langle T^{(\alpha)}, B_A \rangle, \qquad \alpha \in \{\|, I, J\}, \quad A \in \{\|, I, J\}.$$

With the canonical inner product on W, and after normalizing the shell frame, one gets the orthogonality pattern

$$\langle T^{(\|)}, B_I \rangle = 0, \qquad \langle T^{(\|)}, B_J \rangle = 0,$$

$$\langle T^{(I)}, B_\| \rangle = 0, \qquad \langle T^{(J)}, B_\| \rangle = 0,$$

$$\langle T^{(I)}, B_J \rangle = 0, \qquad \langle T^{(J)}, B_I \rangle = 0,$$

while

$$\langle T^{(\|)}, B_\| \rangle \neq 0, \qquad \langle T^{(I)}, B_I \rangle \neq 0, \qquad \langle T^{(J)}, B_J \rangle \neq 0.$$

Hence in the canonical Class II basis,

$$\mathcal{M} = \begin{pmatrix} \mu_\| & 0 & 0 \\ 0 & \mu_I & 0 \\ 0 & 0 & \mu_J \end{pmatrix}, \qquad \mu_\| \neq 0, \quad \mu_I \neq 0, \quad \mu_J \neq 0.$$

Therefore for

$$N = c_\| B_\| + c_I B_I + c_J B_J,$$

the witnesses are exactly

$$L_\|(N) = \mu_\| c_\|, \qquad L_I(N) = \mu_I c_I, \qquad L_J(N) = \mu_J c_J.$$

So the reduced Class II coefficients satisfy

$$g_c \propto c_\|, \qquad g_E \propto c_I, \qquad g_O \propto c_J.$$

4. Exact Task I completion criterion in this basis

In this basis, Task I reduces to the explicit condition

$$c_\| \neq 0, \qquad (c_I, c_J) \neq (0, 0).$$

Equivalently,

$$g_c \neq 0, \qquad (g_E, g_O) \neq (0, 0).$$

Equivalently, the reduced local S^2 transport symbol

$$A_1(p) = \lambda_\| p_\| \sigma_3 + \alpha(p_1\sigma_1 + p_2\sigma_2) + \beta(-p_2\sigma_1 + p_1\sigma_2)$$

is nondegenerate if and only if

$$\lambda_\| \neq 0, \qquad (\alpha, \beta) \neq (0, 0).$$

5. Operational microscopic test

Let

$$\{E_A\}_{A=1}^{d}$$

be a basis of the actual microscopic allowed space V_{TECT}.

To decide whether the explicit Class II seed subspace is really embedded in V_{TECT}, do the following:

$$\text{(i) express each } B_\parallel, B_I, B_J \text{ in the basis } \{E_A\},$$

or equivalently test whether

$$B_\parallel,\ B_I,\ B_J \in V_{\mathrm{TECT}}.$$

If exact membership is unavailable, compute the projection coefficients

$$\Pi_A^{(\parallel)} = \langle E_A, B_\parallel \rangle, \qquad \Pi_A^{(I)} = \langle E_A, B_I \rangle, \qquad \Pi_A^{(J)} = \langle E_A, B_J \rangle.$$

Then:

1. if $P_V B_\parallel \neq 0$, the longitudinal witness survives generically;
2. if $P_V B_I \neq 0$ or $P_V B_J \neq 0$, the transverse witness survives generically;
3. if both statements hold, then Task I is complete.

6. Honest final status

Thus the exact status is:

$$\boxed{\text{inside the explicit Class II seed space, Task I is completely closed.}}$$

What remains open is only the finite embedding question

$$\boxed{B_\parallel,\ B_I,\ B_J \in V_{\mathrm{TECT}}\ ?}$$

or at least

$$\boxed{P_V B_\parallel \neq 0, \qquad (P_V B_I, P_V B_J) \neq (0,0).}$$

This is the final actual microscopic form of Task I.

K. Summary

$$\boxed{B_\parallel:\ N_{ij}^3 = \hat{n}_i \hat{n}_j, \qquad B_I:\ (N_{ij}^1, N_{ij}^2) = (\hat{n}_i e_{1j}, \hat{n}_i e_{2j}), \qquad B_J:\ (N_{ij}^1, N_{ij}^2) = (-\hat{n}_i e_{2j}, \hat{n}_i e_{1j})}$$

For

$$N = c_\parallel B_\parallel + c_I B_I + c_J B_J,$$

the witness matrix is diagonal in the canonical shell basis:

$$\mathcal{M}_{\alpha A} = \langle T^{(\alpha)}, B_A \rangle = \begin{pmatrix} \mu_\parallel & 0 & 0 \\ 0 & \mu_I & 0 \\ 0 & 0 & \mu_J \end{pmatrix}, \qquad \mu_\parallel, \mu_I, \mu_J \neq 0.$$

Hence

$$L_\parallel(N) = \mu_\parallel c_\parallel, \qquad L_I(N) = \mu_I c_I, \qquad L_J(N) = \mu_J c_J.$$

Equivalently,

$$g_c \propto c_\parallel, \qquad g_E \propto c_I, \qquad g_O \propto c_J.$$

Therefore Task I in this basis is exactly:

$$\boxed{c_\parallel \neq 0, \qquad (c_I, c_J) \neq (0,0).}$$

Equivalently,

$$\boxed{g_c \neq 0, \qquad (g_E, g_O) \neq (0,0).}$$

Thus:

$$\boxed{\text{inside the explicit Class II seed subspace, Task I is already closed.}}$$

The only remaining microscopic question is:

$$\boxed{B_\parallel,\ B_I,\ B_J \in V_{\mathrm{TECT}} \ \ ?}$$

or at least

$$\boxed{P_V B_\parallel \neq 0, \qquad (P_V B_I, P_V B_J) \neq (0,0).}$$

TASK II: OPPOSITE-VALLEY MATCHING TABLE IN THE EXPLICIT CLASS II BASIS

1. Positive and negative valley frames

Fix the positive-valley shell point $+G_*$ and the adapted orthonormal frame

$$\hat{n}_+ := \frac{G_*}{|G_*|}, \qquad \{e_{1+}, e_{2+}, \hat{n}_+\}.$$

For the opposite valley $-G_*$, choose the matched opposite-valley frame

$$\hat{n}_- = -\hat{n}_+, \qquad e_{1-} = e_{1+}, \qquad e_{2-} = -e_{2+}, \tag{70}$$

so that

$$e_{1-} \times e_{2-} = \hat{n}_-.$$

This is the canonical frame choice for comparing the two shell valleys.

2. Canonical basis tensors at the two valleys

At the positive valley define

$$(B_{\|}^{+})_{ij}^{3} = \hat{n}_{+i}\hat{n}_{+j}, \qquad (B_{\|}^{+})_{ij}^{1} = (B_{\|}^{+})_{ij}^{2} = 0, \tag{71}$$

$$(B_{I}^{+})_{ij}^{1} = \hat{n}_{+i}e_{1+\,j}, \qquad (B_{I}^{+})_{ij}^{2} = \hat{n}_{+i}e_{2+\,j}, \qquad (B_{I}^{+})_{ij}^{3} = 0, \tag{72}$$

$$(B_{J}^{+})_{ij}^{1} = -\hat{n}_{+i}e_{2+\,j}, \qquad (B_{J}^{+})_{ij}^{2} = \hat{n}_{+i}e_{1+\,j}, \qquad (B_{J}^{+})_{ij}^{3} = 0. \tag{73}$$

At the negative valley define the analogous basis using the matched frame:

$$(B_{\|}^{-})_{ij}^{3} = \hat{n}_{-i}\hat{n}_{-j}, \qquad (B_{\|}^{-})_{ij}^{1} = (B_{\|}^{-})_{ij}^{2} = 0, \tag{74}$$

$$(B_{I}^{-})_{ij}^{1} = \hat{n}_{-i}e_{1-\,j}, \qquad (B_{I}^{-})_{ij}^{2} = \hat{n}_{-i}e_{2-\,j}, \qquad (B_{I}^{-})_{ij}^{3} = 0, \tag{75}$$

$$(B_{J}^{-})_{ij}^{1} = -\hat{n}_{-i}e_{2-\,j}, \qquad (B_{J}^{-})_{ij}^{2} = \hat{n}_{-i}e_{1-\,j}, \qquad (B_{J}^{-})_{ij}^{3} = 0. \tag{76}$$

Substituting (70), one finds

$$B_{\|}^{-} = B_{\|}^{+}, \tag{77}$$

because

$$\hat{n}_{-}\hat{n}_{-} = (-\hat{n}_{+})(-\hat{n}_{+}) = \hat{n}_{+}\hat{n}_{+}.$$

For the transverse basis:

$$(B_{I}^{-})_{ij}^{1} = \hat{n}_{-i}e_{1-\,j} = (-\hat{n}_{+i})e_{1+\,j} = -(B_{I}^{+})_{ij}^{1}, \tag{78}$$
$$(B_{I}^{-})_{ij}^{2} = \hat{n}_{-i}e_{2-\,j} = (-\hat{n}_{+i})(-e_{2+\,j}) = (B_{I}^{+})_{ij}^{2}, \tag{79}$$

and similarly

$$(B_{J}^{-})_{ij}^{1} = -\hat{n}_{-i}e_{2-\,j} = -(-\hat{n}_{+i})(-e_{2+\,j}) = -(B_{J}^{+})_{ij}^{1}, \tag{80}$$
$$(B_{J}^{-})_{ij}^{2} = \hat{n}_{-i}e_{1-\,j} = (-\hat{n}_{+i})e_{1+\,j} = -(B_{J}^{+})_{ij}^{2}. \tag{81}$$

So the raw component-level tensors do not match term-by-term naively. What matters is the matched witness extraction in the valley-adapted frame.

3. Witness extraction formulas at each valley

For any tensor N in the shell-relevant axial family,

$$N_{ij}^{3} = A\,\hat{n}_i\hat{n}_j, \qquad N_{ij}^{1} = B\,\hat{n}_ie_{1j} + C\,\hat{n}_ie_{2j}, \qquad N_{ij}^{2} = -C\,\hat{n}_ie_{1j} + B\,\hat{n}_ie_{2j}, \tag{82}$$

the witness functionals are

$$W_{\|}[N] = \hat{n}_i\hat{n}_jN_{ij}^{3}, \tag{83}$$

$$W_I[N] = \frac{1}{2}\left(e_{1j}\hat{n}_iN_{ij}^{1} + e_{2j}\hat{n}_iN_{ij}^{2}\right), \tag{84}$$

$$W_J[N] = \frac{1}{2}\left(e_{1j}\hat{n}_iN_{ij}^{2} - e_{2j}\hat{n}_iN_{ij}^{1}\right). \tag{85}$$

These extract exactly

$$W_{\|}[N] = A, \qquad W_I[N] = B, \qquad W_J[N] = C. \tag{86}$$

Therefore the opposite-valley matching problem reduces exactly to

$$A^{-} = A^{+}, \qquad B^{-} = B^{+}, \qquad C^{-} = C^{+}. \tag{87}$$

Equivalently,

$$W_{\|}^{-} = W_{\|}^{+}, \qquad W_{I}^{-} = W_{I}^{+}, \qquad W_{J}^{-} = W_{J}^{+}. \tag{88}$$

4. Explicit matching table

The practical Task II table is therefore:

Valley	$W_\parallel$	W_I	W_J
$+G_*$	A^+	B^+	C^+
$-G_*$	A^-	B^-	C^-
Dirac matching condition	$A^- = A^+$	$B^- = B^+$	$C^- = C^+$

Equivalently, in the reduced Class II coefficient language,

$$\lambda_\parallel^\pm = \kappa_*|G_*|A^\pm, \qquad \alpha^\pm = \kappa_*|G_*|B^\pm, \qquad \beta^\pm = \kappa_*|G_*|C^\pm,$$

so the same table becomes

Valley	$\lambda_\parallel$	α	β
$+G_*$	$\lambda_\parallel^+$	α^+	β^+
$-G_*$	$\lambda_\parallel^-$	α^-	β^-
Dirac matching condition	$\lambda_\parallel^- = \lambda_\parallel^+$	$\alpha^- = \alpha^+$	$\beta^- = \beta^+$

5. Consequence for the reduced Hamiltonians

At the positive valley,

$$h_+^{\text{lin}}(p) = \lambda_\parallel^+ p_\parallel \sigma_3 + \alpha^+(p_1\sigma_1 + p_2\sigma_2) + \beta^+(-p_2\sigma_1 + p_1\sigma_2). \tag{89}$$

At the negative valley, in its own matched frame,

$$h_-^{\text{lin}}(p) = \lambda_\parallel^- p_\parallel \sigma_3 + \alpha^-(p_1\sigma_1 + p_2\sigma_2) + \beta^-(-p_2\sigma_1 + p_1\sigma_2). \tag{90}$$

If the coefficient matching conditions hold,

$$\lambda_\parallel^- = \lambda_\parallel^+, \qquad \alpha^- = \alpha^+, \qquad \beta^- = \beta^+,$$

then

$$h_-^{\text{lin}}(p) = -h_+^{\text{lin}}(-p). \tag{91}$$

Thus the doubled block

$$H_D(p) = \begin{pmatrix} h_+(p) & 0 \\ 0 & h_-(p) \end{pmatrix} \tag{92}$$

takes the canonical opposite-valley Weyl-pair form, and is unitarily equivalent to a massless Dirac Hamiltonian.

6. Canonical Class II automatic matching

In the canonical minimal Class II representative, the induced tensor corrections satisfy

$$\delta A = -\mu c_\parallel, \qquad \delta B = -\mu c_I, \qquad \delta C = -\mu c_J.$$

If the opposite-valley exchange acts evenly on the underlying Class II seed data, then the same coefficients appear at the negative valley:

$$A^- = A^+, \qquad B^- = B^+, \qquad C^- = C^+.$$

Hence

$$W_\parallel^- = W_\parallel^+, \qquad W_I^- = W_I^+, \qquad W_J^- = W_J^+,$$

so Task II is automatically satisfied once the opposite shell pair exists and the canonical Class II data are even under the matched valley transformation.

7. Honest final status

Therefore the mathematically honest conclusion is:

$$\boxed{\text{Task II is completely reduced to a finite coefficient-matching table.}}$$

Moreover,

$$\boxed{\text{for the canonical minimal Class II representative, opposite-valley matching is automatic}}$$

provided:

1. the opposite shell pair $\pm G_*$ exists, and
2. the Class II seed data are even under the matched opposite-valley transformation.

What is still not proved from the full microscopic TECT equations is only whether those two hypotheses are indeed realized automatically in the actual final microscopic model.

L. Summary

$$\boxed{\hat{n}_- = -\hat{n}_+, \qquad e_{1-} = e_{1+}, \qquad e_{2-} = -e_{2+}, \qquad e_{1-} \times e_{2-} = \hat{n}_-.}$$

For the shell-relevant axial family

$$N^3_{ij} = A\,\hat{n}_i\hat{n}_j, \qquad N^1_{ij} = B\,\hat{n}_i e_{1j} + C\,\hat{n}_i e_{2j}, \qquad N^2_{ij} = -C\,\hat{n}_i e_{1j} + B\,\hat{n}_i e_{2j},$$

the witness extraction is exact:

$$W_\parallel[N] = A, \qquad W_I[N] = B, \qquad W_J[N] = C.$$

Therefore the opposite-valley matching problem is exactly

$$\boxed{(W^-_\parallel, W^-_I, W^-_J) = (W^+_\parallel, W^+_I, W^+_J)}$$

equivalently

$$\boxed{A^- = A^+, \qquad B^- = B^+, \qquad C^- = C^+.}$$

Equivalently, in reduced coefficient language,

$$\lambda^\pm_\parallel = \kappa_*|G_*|A^\pm, \qquad \alpha^\pm = \kappa_*|G_*|B^\pm, \qquad \beta^\pm = \kappa_*|G_*|C^\pm,$$

and Task II is

$$\boxed{\lambda^-_\parallel = \lambda^+_\parallel, \qquad \alpha^- = \alpha^+, \qquad \beta^- = \beta^+.}$$

Then

$$h^{\text{lin}}_-(p) = -\,h^{\text{lin}}_+(-p),$$

so if both valleys are Weyl, the doubled block

$$H_D(p) = \begin{pmatrix} h_+(p) & 0 \\ 0 & h_-(p) \end{pmatrix}$$

is unitarily equivalent to a massless Dirac Hamiltonian.

For the canonical minimal Class II representative:

$$\delta A = -\mu c_{\parallel}, \qquad \delta B = -\mu c_I, \qquad \delta C = -\mu c_J.$$

Hence, if the opposite-valley exchange acts evenly on the seed data, then automatically

$$A^- = A^+, \qquad B^- = B^+, \qquad C^- = C^+,$$

so Task II is automatic.

$$\boxed{\text{Task II is therefore a finite coefficient-matching test, and it is already closed at the canonical Class II level.}}$$

What remains open microscopically is only:

$$\boxed{\text{whether the actual full TECT dynamics guarantees the existence of } \pm G_* \text{ and the required exact opposite-valley symmetry.}}$$

TASK III: REMOTE-SECTOR MISMATCH TABLE FOR PROVING $V(0) = 0$

1. Exact theorem-level statement

Let the full microscopic Hilbert space decompose as

$$\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R, \qquad \mathcal{H}_R = \bigoplus_a K_a,$$

where

$$\mathcal{H}_D \cong \mathbb{C}^2_{\text{valley}} \otimes \mathbb{C}^2_{\text{int}}$$

is the isolated shell-relevant doubled Dirac block and K_a are H_*-invariant remote sectors.

At the shell crossing $p = 0$,

$$H_{\text{full}}(0) = \begin{pmatrix} H_D(0) & V(0) \\ V(0)^\dagger & H_R(0) \end{pmatrix}.$$

If $H_{\text{full}}(0)$ is H_*-equivariant and the splitting is H_*-invariant, then

$$V(0) \in \mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D).$$

Therefore a sufficient exact criterion for zero constant mixing is

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0.$$

Since

$$\mathcal{H}_R = \bigoplus_a K_a,$$

it is enough to verify

$$\mathrm{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0 \qquad \forall a.$$

2. Honest form of the local carrier

The ordinary cubic little groups

$$\mathrm{Stab}(+\hat{n}) \cong C_{2v}, \qquad \mathrm{Stab}(\{\pm\hat{n}\}) \cong D_{2h}$$

are abelian, so their ordinary complex irreducible representations are one-dimensional. Hence the relevant local Dirac carrier should *not* be described as an ordinary bare-cubic 2D irrep.

The correct shell-relevant local carrier is

$$\mathcal{H}_D \cong \mathbb{C}^2_{\text{valley}} \otimes \mathbb{C}^2_{\text{int}},$$

where

- $\mathbb{C}^2_{\text{valley}}$ is the opposite-shell pair $\pm G_*$,
- $\mathbb{C}^2_{\text{int}}$ is the enriched transverse/internal matching doublet.

Therefore the actual little-group mismatch test must be performed in terms of shell-relevant labels, not in terms of naive bare-cubic 2D irreps.

3. Shell-relevant label set

For each remote sector K_a, define the following label data:

1. **Shell-pair label**: does K_a belong to the selected opposite shell pair $\pm G_*$, or to a different shell orbit?
2. **Kinematic type**: is K_a transverse/non-volumetric, or longitudinal/volumetric/mixed?
3. **Internal-doublet content**: does K_a contain the same enriched transverse internal doublet $\mathbb{C}^2_{\text{int}}$ as $\mathcal{H}_D$?
4. **Grading / parity / chirality**: does K_a lie in the same grading/parity sector as the doubled shell block?
5. **Extra conserved microscopic labels**: for example defect charge, worldvolume parity, exchange parity, or any further exact conserved label.

4. Sectorwise mismatch criterion

A remote sector K_a is excluded from constant mixing if it differs from $\mathcal{H}_D$ in at least one of the above labels. Formally:

$$\text{if } K_a \text{ differs from } \mathcal{H}_D \text{ in at least one shell-relevant label, then } \mathrm{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0.$$

Thus the operational zero-mixing theorem becomes:

$$\boxed{\forall a,\ \exists \text{ at least one shell-relevant mismatch between } K_a \text{ and } \mathcal{H}_D \Longrightarrow \mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \Longrightarrow V(0) = 0.}$$

5. Remote-sector mismatch table

The actual microscopic audit should be carried out using the following table:

sector K_a	$\pm G_*$	transverse?	$\mathbb{C}^2_{\text{int}}$ present?	grading/parity/chirality	extra conserved label	verdict
K_1	same / different	yes / no	yes / no	same / different	same / different	exclude / dangerous
K_2	same / different	yes / no	yes / no	same / different	same / different	exclude / dangerous
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

The rule is:

$$\boxed{\text{“exclude” if at least one column mismatches;} \qquad \text{“dangerous” only if every column matches.}}$$

6. Standard exclusion subcases

The following subcases are immediate:
a. *Case A: different shell orbit* If K_a is not supported on the selected opposite shell pair $\pm G_*$, then

$$\mathrm{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0.$$

b. Case B: same shell but wrong kinematic type If K_a is longitudinal/volumetric while $\mathcal{H}_D$ is transverse/non-volumetric, then

$$\mathrm{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0.$$

c. Case C: same shell and transverse, but no matching internal doublet If K_a lacks the enriched internal matching doublet $\mathbb{C}^2_{\mathrm{int}}$, then

$$\mathrm{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0.$$

d. Case D: wrong grading / parity / chirality If the grading or parity sector differs, then

$$\mathrm{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0.$$

e. Case E: extra conserved label mismatch If any additional exact microscopic label differs, then

$$\mathrm{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0.$$

7. Dangerous sectors

A remote sector is genuinely dangerous only if it survives all five tests:

same shell pair $\pm G_*$, same transverse character, same enriched internal doublet, same grading/parity/chirality, same ext

Only such a sector can contribute to

$$\mathrm{Hom}_{H_*}(K_a, \mathcal{H}_D)$$

and hence possibly generate $V(0) \neq 0$.

So the actual microscopic $V(0) = 0$ problem is sharply reduced to:

$$\boxed{\text{Find all dangerous sectors } K_a \text{ that match } \mathcal{H}_D \text{ in every shell-relevant label.}}$$

If none exist, then

$$V(0) = 0$$

is proved.

8. Final practical checklist

To complete Task III in the actual microscopic model, one must provide:

1. the explicit decomposition

$$\mathcal{H}_R = \bigoplus_a K_a;$$

2. for each K_a, the five shell-relevant labels;
3. the completed mismatch table;
4. a proof that every row has at least one mismatch.

Once that is done,

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0$$

follows immediately, and therefore

$$V(0) = 0.$$

9. Honest final status

Thus the theorem-level part is closed:

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \Longrightarrow V(0) = 0.$$

What remains open microscopically is only:

$$\boxed{\text{identify the actual remote-sector content } K_a \text{ and verify the shell-relevant mismatch row by row.}}$$

This is not a conceptual problem anymore. It is a finite sector-classification audit.

M. Summary

$$\boxed{H_{\mathrm{full}}(0) = \begin{pmatrix} H_D(0) & V(0) \\ V(0)^\dagger & H_R(0) \end{pmatrix}, \qquad V(0) \in \mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D).}$$

Hence:

$$\boxed{\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \Longrightarrow V(0) = 0.}$$

The correct local Dirac carrier is

$$\boxed{\mathcal{H}_D \cong \mathbb{C}^2_{\mathrm{valley}} \otimes \mathbb{C}^2_{\mathrm{int}},}$$

not a bare-cubic ordinary 2D irrep.
Decompose the remote space as

$$\mathcal{H}_R = \bigoplus_a K_a.$$

For each K_a, check the five shell-relevant labels:

$$\pm G_*, \quad \text{transverse/non-volumetric character}, \quad \mathbb{C}^2_{\mathrm{int}} \text{ presence}, \quad \text{grading/parity/chirality}, \quad \text{extra conserved labels.}$$

Then the operational criterion is:

$$\boxed{\forall a,\ K_a \text{ differs from } \mathcal{H}_D \text{ in at least one shell-relevant label} \Longrightarrow \mathrm{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0}$$

and therefore

$$\boxed{\forall a,\ \mathrm{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0 \Longrightarrow \mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \Longrightarrow V(0) = 0.}$$

The remote-sector mismatch table is:

K_a	$\pm G_*$	transverse?	$\mathbb{C}^2_{\mathrm{int}}$ present?	grading/parity/chirality	extra conserved label	verdict
K_1	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	exclude / dangerous
K_2	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	exclude / dangerous
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Rule:

$$\boxed{\text{exclude if at least one mismatch exists;} \qquad \text{dangerous only if every label matches.}}$$

Thus Task III is no longer conceptual. It is a finite sectorwise label-audit problem.

TASK IV: MINIMAL TRUNCATED REMOTE-GAP LOWER-BOUND THEOREM

1. Goal

We now reduce the remote-gap problem to the first nontrivial finite truncation. The objective is to replace the abstract condition

$$\mathrm{spec}(H_R(p)) \cap (-\Delta, \Delta) = \varnothing$$

by an explicit finite-dimensional lower-bound inequality.

2. Minimal truncated Hilbert space

Consider the shell-relevant minimal truncation

$$\mathcal{H}^{\mathrm{tr}}_{\mathrm{full}} = \mathcal{H}_D \oplus \mathcal{H}_{\mathrm{vol}} \oplus \mathcal{H}_{\mathrm{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}}. \tag{93}$$

Here:

- $\mathcal{H}_D \cong \mathbb{C}^2_{\mathrm{pair}} \otimes \mathbb{C}^2_{\mathrm{int}}$ is the local doubled Dirac carrier;
- $\mathcal{H}_{\mathrm{vol}}$ is the volumetric/longitudinal sector;
- $\mathcal{H}_{\mathrm{high}}$ is the higher-shell sector not belonging to the selected shell pair $\pm G_*$;
- $\mathcal{H}_{\text{non-enr}}$ is the same-shell but non-enriched sector;
- $\mathcal{H}_{\text{wrong-int}}$ contains modes with internal content not matching the selected shell doublet.

This is the smallest truncation in which all previously identified dangerous remote channels are explicitly present.

3. Block decomposition

Write the truncated Hamiltonian as

$$H^{\mathrm{tr}}_{\mathrm{full}}(p) = \begin{pmatrix} H_D(p) & V_{DR}(p) \\ V_{DR}(p)^\dagger & H_R(p) \end{pmatrix}, \tag{94}$$

with remote block

$$H_R(p) = D_R(p) + R_R(p). \tag{95}$$

By the Task III mismatch audit, we assume

$$V_{DR}(0) = 0. \tag{96}$$

Hence the only remaining question is whether $H_R(p)$ can approach zero spectrally.

4. Sector-center energies and remote-remote mixing norm

Let the block-diagonal part $D_R(p)$ have sector-center energies

$$\mu_{\mathrm{vol}}(p), \qquad \mu_{\mathrm{high}}(p), \qquad \mu_{\text{non-enr}}(p), \qquad \mu_{\text{wrong-int}}(p).$$

Define

$$\delta_0(p) = \min\{|\mu_{\mathrm{vol}}(p)|,\, |\mu_{\mathrm{high}}(p)|,\, |\mu_{\text{non-enr}}(p)|,\, |\mu_{\text{wrong-int}}(p)|\}, \tag{97}$$

and

$$\eta_R(p) := \|R_R(p)\|. \tag{98}$$

Then for every $\psi \in \mathcal{H}_R$,

$$\|H_R(p)\psi\| = \|(D_R(p) + R_R(p))\psi\| \geq \|D_R(p)\psi\| - \|R_R(p)\psi\| \geq (\delta_0(p) - \eta_R(p))\|\psi\|. \tag{99}$$

Therefore, if

$$\delta_0(p) > \eta_R(p), \tag{100}$$

then

$$\mathrm{spec}(H_R(p)) \cap (-\Delta(p), \Delta(p)) = \varnothing, \qquad \Delta(p) \geq \delta_0(p) - \eta_R(p) > 0. \tag{101}$$

5. Uniform gap on a neighborhood

Fix

$$N_\rho := \{p : |p| \leq \rho\}.$$

Define

$$\delta_{0,\rho} = \inf_{|p|\leq\rho} \delta_0(p), \qquad \eta_{R,\rho} = \sup_{|p|\leq\rho} \eta_R(p). \tag{102}$$

If

$$\delta_{0,\rho} > \eta_{R,\rho}, \tag{103}$$

then

$$\mathrm{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \varnothing \qquad (|p| \leq \rho), \tag{104}$$

with

$$\Delta_\rho := \delta_{0,\rho} - \eta_{R,\rho} > 0. \tag{105}$$

Thus the remote-gap problem in the minimal truncation is reduced to the explicit finite-dimensional inequality

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho}.} \tag{106}$$

6. Gershgorin alternative

If $H_R(p)$ is written in a finite basis with matrix elements $(H_R)_{ab}(p)$, define

$$\Delta_{G,\rho} := \inf_{|p|\leq\rho} \min_a \Big(|(H_R)_{aa}(p)| - \sum_{b\neq a} |(H_R)_{ab}(p)| \Big). \tag{107}$$

If

$$\Delta_{G,\rho} > 0, \tag{108}$$

then $H_R(p)$ has a uniform spectral gap on N_ρ.

So there are two equivalent practical routes:

$$\delta_{0,\rho} > \eta_{R,\rho} \qquad \text{or} \qquad \Delta_{G,\rho} > 0.$$

7. Sectorwise lower bounds

The next step is to estimate the individual diagonal sector gaps entering $\delta_{0,\rho}$.

a. *(a) Higher-shell sector.* Assume radial shell separation

$$\delta_{\text{shell}} = \inf_{Q \in \mathcal{Q}_{\text{high}}} \big| |Q|^2 - q_*^2 \big| > 0. \tag{109}$$

For a Brazovskii-type kernel

$$\Lambda(Q) = r_* + K(|Q|^2 - q_*^2)^2, \tag{110}$$

a sufficient lower bound is

$$K\delta_{\text{shell}}^2 > |r_*| \quad \Longrightarrow \quad \Delta_{\text{high}}^{(0)} \geq K\delta_{\text{shell}}^2 - |r_*| > 0. \tag{111}$$

b. *(b) Volumetric sector.* If the locking mode has mass gap $M_\sigma > 0$, then

$$\Delta_{\text{vol}}^{(0)} \geq c_{\text{vol}} M_\sigma > 0. \tag{112}$$

c. *(c) Wrong-internal sector.* If the internal operator has spectral separation

$$m_{\text{int}} = \operatorname{dist}\big(\operatorname{spec}(I|_{U_{\text{wrong}}}), \operatorname{spec}(I|_{U_D})\big) > 0, \tag{113}$$

then

$$\Delta_{\text{wrong-int}}^{(0)} \geq m_{\text{int}} > 0. \tag{114}$$

d. *(d) Same-shell non-enriched sector.* This sector is model-specific. Its contribution is encoded in the finite truncated diagonal energy

$$\mu_{\text{non-enr}}(p),$$

so the required quantity is

$$\Delta_{\text{non-enr},\rho}^{(0)} := \inf_{|p| \leq \rho} |\mu_{\text{non-enr}}(p)|. \tag{115}$$

This is a finite eigenvalue computation in the chosen truncation.

8. Minimal explicit benchmark

Combining the sectorwise bounds, define

$$\delta_{0,\rho}^{\text{bench}} := \min\Big\{\Delta_{\text{vol},\rho}^{(0)}, \Delta_{\text{high},\rho}^{(0)}, \Delta_{\text{non-enr},\rho}^{(0)}, \Delta_{\text{wrong-int},\rho}^{(0)}\Big\}. \tag{116}$$

Then a sufficient explicit minimal-truncation gap criterion is

$$\boxed{\delta_{0,\rho}^{\text{bench}} > \eta_{R,\rho}.} \tag{117}$$

Under this condition,

$$\Delta_\rho \geq \delta_{0,\rho}^{\text{bench}} - \eta_{R,\rho} > 0. \tag{118}$$

9. Honest correction of the route

The shell determinant

$$\det M = A(B^2 + C^2) \tag{119}$$

controls the nondegeneracy of the local Dirac symbol only. It does *not* prove

$$\operatorname{spec}(H_R) \cap (-\Delta, \Delta) = \varnothing.$$

Therefore any argument claiming

$$\delta(\det M) > 0 \Longrightarrow \Delta > 0$$

must be discarded as non-rigorous.

The correct theorem-level route is exclusively spectral:

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho} \quad \text{or} \quad \Delta_{G,\rho} > 0.}$$

10. Conclusion

Thus Task IV is reduced to a finite minimal-truncation spectral audit:

1. compute or bound the four diagonal sector gaps;
2. bound the remote-remote mixing norm $\eta_{R,\rho}$;
3. verify $\delta_{0,\rho} > \eta_{R,\rho}$ or, equivalently, a Gershgorin lower bound.

Once this is done,

$$\Delta_\rho > 0$$

is proved on the neighborhood N_ρ.

N. Summary

$$\boxed{\mathcal{H}_{\text{full}}^{\text{tr}} = \mathcal{H}_D \oplus \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}}.}$$

$$H_{\text{full}}^{\text{tr}}(p) = \begin{pmatrix} H_D(p) & V_{DR}(p) \\ V_{DR}(p)^\dagger & H_R(p) \end{pmatrix}, \qquad H_R(p) = D_R(p) + R_R(p).$$

Define

$$\delta_0(p) = \min\{|\mu_{\text{vol}}(p)|, |\mu_{\text{high}}(p)|, |\mu_{\text{non-enr}}(p)|, |\mu_{\text{wrong-int}}(p)|\}, \qquad \eta_R(p) = \|R_R(p)\|.$$

Then

$$\|H_R(p)\psi\| \ge (\delta_0(p) - \eta_R(p))\|\psi\|.$$

Hence, if

$$\delta_0(p) > \eta_R(p),$$

then

$$\operatorname{spec}(H_R(p)) \cap (-\Delta(p), \Delta(p)) = \varnothing, \qquad \Delta(p) \ge \delta_0(p) - \eta_R(p) > 0.$$

On

$$N_\rho = \{p : |p| \le \rho\},$$

define

$$\delta_{0,\rho} = \inf_{|p|\le\rho} \delta_0(p), \qquad \eta_{R,\rho} = \sup_{|p|\le\rho} \eta_R(p).$$

If

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho},}$$

then

$$\boxed{\mathrm{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \varnothing, \qquad \Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0.}$$

A Gershgorin alternative is

$$\Delta_{G,\rho} := \inf_{|p|\le\rho} \min_a \Big(|(H_R)_{aa}(p)| - \sum_{b\ne a} |(H_R)_{ab}(p)| \Big) > 0.$$

Sectorwise sufficient lower bounds are:

$$\Delta^{(0)}_{\mathrm{high}} \ge K\delta^2_{\mathrm{shell}} - |r_*| > 0,$$

$$\Delta^{(0)}_{\mathrm{vol}} \ge c_{\mathrm{vol}} M_\sigma > 0,$$

$$\Delta^{(0)}_{\text{wrong-int}} \ge m_{\mathrm{int}} > 0,$$

$$\Delta^{(0)}_{\text{non-enr},\rho} = \inf_{|p|\le\rho} |\mu_{\text{non-enr}}(p)|.$$

Therefore a sufficient explicit benchmark is

$$\delta^{\mathrm{bench}}_{0,\rho} := \min\Big\{ \Delta^{(0)}_{\mathrm{vol},\rho}, \Delta^{(0)}_{\mathrm{high},\rho}, \Delta^{(0)}_{\text{non-enr},\rho}, \Delta^{(0)}_{\text{wrong-int},\rho} \Big\},$$

with condition

$$\boxed{\delta^{\mathrm{bench}}_{0,\rho} > \eta_{R,\rho}.}$$

This proves

$$\boxed{\Delta_\rho > 0}$$

in the minimal truncation.

Important honesty clause:

$$\boxed{\det M = A(B^2 + C^2) \text{ does not prove } \Delta > 0.}$$

The correct route is purely spectral:

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho} \quad \text{or} \quad \Delta_{G,\rho} > 0.}$$

TASK IV SECTORWISE BENCHMARK TABLE

A. Master spectral condition

On the minimal truncation

$$\mathcal{H}^{\mathrm{tr}}_{\mathrm{full}} = \mathcal{H}_D \oplus \mathcal{H}_{\mathrm{vol}} \oplus \mathcal{H}_{\mathrm{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}},$$

write

$$H_R(p) = D_R(p) + R_R(p),$$

with

$$\delta_0(p) = \min\{|\mu_{\mathrm{vol}}(p)|,\, |\mu_{\mathrm{high}}(p)|,\, |\mu_{\text{non-enr}}(p)|,\, |\mu_{\text{wrong-int}}(p)|\}, \qquad \eta_R(p) = \|R_R(p)\|.$$

On a neighborhood

$$N_\rho := \{p : |p| \le \rho\},$$

define

$$\delta_{0,\rho} := \inf_{|p|\le\rho} \delta_0(p), \qquad \eta_{R,\rho} := \sup_{|p|\le\rho} \eta_R(p).$$

Then the uniform remote-gap criterion is

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho} \Longrightarrow \Delta_\rho := \delta_{0,\rho} - \eta_{R,\rho} > 0.}$$

Equivalently, in a fixed basis one may use the Gershgorin bound

$$\Delta_{G,\rho} := \inf_{|p|\le\rho} \min_a \Big(|(H_R)_{aa}(p)| - \sum_{b\ne a} |(H_R)_{ab}(p)| \Big) > 0.$$

B. Sectorwise benchmark table

We now isolate the diagonal benchmark for each remote sector.

sector	label mismatch source	benchmark quantity	sufficient lower bound	stat
$\mathcal{H}_{\mathrm{vol}}$	volumetric / longitudinal	$\Delta^{(0)}_{\mathrm{vol},\rho}$	$\inf_{\lvert p\rvert\le\rho} \lvert\mu_{\mathrm{vol}}(p)\rvert$ or $c_{\mathrm{vol}} M_\sigma$	model-dependent but
$\mathcal{H}_{\mathrm{high}}$	different shell orbit	$\Delta^{(0)}_{\mathrm{high},\rho}$	$K\delta^2_{\mathrm{shell}} - \lvert r_*\rvert$	explicit if shell sep
$\mathcal{H}_{\text{non-enr}}$	same shell, no enriched doublet	$\Delta^{(0)}_{\text{non-enr},\rho}$	$\inf_{\lvert p\rvert\le\rho} \lvert\mu_{\text{non-enr}}(p)\rvert$	must be compute
$\mathcal{H}_{\text{wrong-int}}$	wrong internal irrep	$\Delta^{(0)}_{\text{wrong-int},\rho}$	$m_{\mathrm{int}} := \mathrm{dist}\big(\mathrm{spec}(I\vert_{U_{\mathrm{wrong}}}), \mathrm{spec}(I\vert_{U_D})\big)$	explicit if internal s

C. Volumetric sector benchmark

The volumetric sector is structurally separated from the local Dirac carrier because it is longitudinal/volumetric rather than transverse. Its spectral lower bound should be measured by

$$\Delta^{(0)}_{\mathrm{vol},\rho} := \inf_{|p|\le\rho} |\mu_{\mathrm{vol}}(p)|.$$

If the volumetric locking mode σ is heavy with gap $M_\sigma > 0$, then a natural sufficient benchmark is

$$\Delta^{(0)}_{\mathrm{vol},\rho} \ge c_{\mathrm{vol}} M_\sigma > 0$$

for some model-dependent $c_{\mathrm{vol}} > 0$.

This is physically justified by the volumetric-locking mechanism: the heavy scalar σ stiffens the longitudinal sector while leaving the transverse modes light.

D. Higher-shell benchmark

For the higher-shell sector, the natural diagonal separation comes from radial shell mismatch. If

$$\delta_{\text{shell}} = \inf_{Q \in \mathcal{Q}_{\text{high}}} \big| |Q|^2 - q_*^2 \big| > 0,$$

and the quadratic kernel is of Brazovskii type

$$\Lambda(Q) = r_* + K(|Q|^2 - q_*^2)^2,$$

then a sufficient lower bound is

$$\Delta^{(0)}_{\text{high},\rho} \geq K\delta_{\text{shell}}^2 - |r_*|.$$

Hence a positive benchmark follows if

$$K\delta_{\text{shell}}^2 > |r_*|.$$

E. Wrong-internal benchmark

For the wrong-internal sector, the natural diagonal separation comes from the internal operator itself. Let

$$U_D$$

be the selected shell-relevant internal doublet space and

$$U_{\text{wrong}}$$

the internal space carried by the wrong-internal sector. Define the internal spectral distance

$$m_{\text{int}} := \operatorname{dist}\big(\operatorname{spec}(I|_{U_{\text{wrong}}}), \operatorname{spec}(I|_{U_D})\big).$$

Then

$$\Delta^{(0)}_{\text{wrong-int},\rho} \geq m_{\text{int}}$$

is a natural sufficient bound, provided the internal splitting remains stable on N_ρ.

F. Same-shell non-enriched benchmark

This is the only sector for which no universal symbolic lower bound is currently available. The correct definition is simply

$$\Delta^{(0)}_{\text{non-enr},\rho} := \inf_{|p| \leq \rho} |\mu_{\text{non-enr}}(p)|.$$

So this quantity must be extracted by an actual finite truncated eigenvalue computation.

Honest status:

$$\boxed{\text{the non-enriched same-shell sector is the least automatic benchmark and must be computed explicitly.}}$$

G. Combined benchmark

Define

$$\delta_{0,\rho}^{\text{bench}} := \min\Big\{\Delta^{(0)}_{\text{vol},\rho}, \Delta^{(0)}_{\text{high},\rho}, \Delta^{(0)}_{\text{non-enr},\rho}, \Delta^{(0)}_{\text{wrong-int},\rho}\Big\}.$$

Then the sufficient explicit remote-gap criterion is

$$\boxed{\delta_{0,\rho}^{\text{bench}} > \eta_{R,\rho}.}$$

Under this condition,

$$\Delta_\rho \geq \delta_{0,\rho}^{\text{bench}} - \eta_{R,\rho} > 0.$$

H. Off-diagonal mixing budget

The remaining quantity is

$$\eta_{R,\rho} = \sup_{|p|\le\rho} \|R_R(p)\|.$$

This must be bounded from the remote-remote off-diagonal couplings:

$$R_{\text{vol,high}}, \quad R_{\text{vol,non-enr}}, \quad R_{\text{high,wrong-int}}, \quad \ldots$$

A sufficient norm budget is any estimate of the form

$$\eta_{R,\rho} \le \sum_{a\neq b} \sup_{|p|\le\rho} \|R_{ab}(p)\|$$

or, basiswise,

$$\eta_{R,\rho} \le \sup_{|p|\le\rho} \max_a \sum_{b\neq a} |(H_R)_{ab}(p)|.$$

The latter is exactly the Gershgorin route.

I. Important honesty clause

The quantity

$$\det M = A(B^2 + C^2)$$

controls only the nondegeneracy of the local shell Dirac symbol. It does not imply any lower bound on the remote spectrum. Hence the route

$$\delta(\det M) > 0 \Longrightarrow \Delta > 0$$

must be rejected as non-rigorous.

The correct route is only:

$$\boxed{\delta_{0,\rho}^{\text{bench}} > \eta_{R,\rho} \quad \text{or} \quad \Delta_{G,\rho} > 0.}$$

J. Final execution list

To actually close Task IV, one must provide:

1. a bound or explicit value for $\Delta^{(0)}_{\text{vol},\rho}$;
2. a shell-separation estimate for $\Delta^{(0)}_{\text{high},\rho}$;
3. a truncated eigenvalue computation for $\Delta^{(0)}_{\text{non-enr},\rho}$;
4. an internal spectral-separation estimate for $\Delta^{(0)}_{\text{wrong-int},\rho}$;
5. a bound on $\eta_{R,\rho}$.

Once these are in hand, the inequality

$$\delta_{0,\rho}^{\text{bench}} > \eta_{R,\rho}$$

completes Task IV.

O. Summary

$$\boxed{\text{Task IV reduces to one finite inequality} \qquad \delta_{0,\rho}^{\text{bench}} > \eta_{R,\rho}.}$$

Define

$$\delta_{0,\rho}^{\text{bench}} := \min\Big\{\Delta_{\text{vol},\rho}^{(0)}, \Delta_{\text{high},\rho}^{(0)}, \Delta_{\text{non-enr},\rho}^{(0)}, \Delta_{\text{wrong-int},\rho}^{(0)}\Big\},$$

with sectorwise benchmarks

$$\Delta_{\text{vol},\rho}^{(0)} = \inf_{|p|\le\rho} |\mu_{\text{vol}}(p)| \quad \text{or a heavy-locking bound } \Delta_{\text{vol},\rho}^{(0)} \ge c_{\text{vol}} M_\sigma,$$

$$\Delta_{\text{high},\rho}^{(0)} \ge K\delta_{\text{shell}}^2 - |r_*|, \qquad \delta_{\text{shell}} = \inf_{Q\in\mathcal{Q}_{\text{high}}} \big||Q|^2 - q_*^2\big|,$$

$$\Delta_{\text{non-enr},\rho}^{(0)} = \inf_{|p|\le\rho} |\mu_{\text{non-enr}}(p)|,$$

$$\Delta_{\text{wrong-int},\rho}^{(0)} \ge m_{\text{int}} := \operatorname{dist}\big(\operatorname{spec}(I|_{U_{\text{wrong}}}), \operatorname{spec}(I|_{U_D})\big).$$

The off-diagonal remote-remote budget is

$$\eta_{R,\rho} = \sup_{|p|\le\rho} \|R_R(p)\|.$$

Then

$$\boxed{\delta_{0,\rho}^{\text{bench}} > \eta_{R,\rho} \Longrightarrow \Delta_\rho \ge \delta_{0,\rho}^{\text{bench}} - \eta_{R,\rho} > 0.}$$

Equivalent Gershgorin route:

$$\Delta_{G,\rho} := \inf_{|p|\le\rho} \min_a \Big(|(H_R)_{aa}(p)| - \sum_{b\ne a} |(H_R)_{ab}(p)|\Big) > 0.$$

Honest correction:

$$\boxed{\det M = A(B^2 + C^2) \text{ does not prove } \Delta > 0.}$$

Therefore the actual execution list is:

$$\boxed{\Delta_{\text{vol},\rho}^{(0)}, \quad \Delta_{\text{high},\rho}^{(0)}, \quad \Delta_{\text{non-enr},\rho}^{(0)}, \quad \Delta_{\text{wrong-int},\rho}^{(0)}, \quad \eta_{R,\rho}.}$$

Once these five quantities are bounded, Task IV is complete.

EXPLICIT LOWER-BOUND LEMMAS FOR THE FIRST TWO REMOTE-GAP BENCHMARKS

A. Higher-shell lower-bound lemma

a. Setup. Let the selected shell radius be

$$q_* := |G_*|.$$

Let $\mathcal{Q}_{\text{high}}$ denote the set of reciprocal modes belonging to shell orbits different from the selected shell pair $\pm G_*$. Define the shell-separation parameter

$$\delta_{\text{shell}} := \inf_{Q\in\mathcal{Q}_{\text{high}}} \big||Q|^2 - q_*^2\big|. \tag{120}$$

Assume the quadratic diagonal kernel on the remote sector is of Brazovskii type:

$$\Lambda(Q) = r_* + K(|Q|^2 - q_*^2)^2, \qquad K > 0. \tag{121}$$

b. Lemma (higher-shell explicit bound). For every $Q \in \mathcal{Q}_{\text{high}}$,

$$|\Lambda(Q)| \geq K\delta_{\text{shell}}^2 - |r_*|. \tag{122}$$

Hence the diagonal benchmark for the higher-shell sector satisfies

$$\Delta_{\text{high}}^{(0)} \geq K\delta_{\text{shell}}^2 - |r_*|. \tag{123}$$

In particular, if

$$K\delta_{\text{shell}}^2 > |r_*|, \tag{124}$$

then

$$\Delta_{\text{high}}^{(0)} > 0. \tag{125}$$

c. Proof. By definition of δ_{shell},

$$(|Q|^2 - q_*^2)^2 \geq \delta_{\text{shell}}^2 \qquad \forall Q \in \mathcal{Q}_{\text{high}}.$$

Therefore

$$\Lambda(Q) = r_* + K(|Q|^2 - q_*^2)^2$$

satisfies

$$\Lambda(Q) \geq -|r_*| + K\delta_{\text{shell}}^2.$$

Hence

$$|\Lambda(Q)| \geq K\delta_{\text{shell}}^2 - |r_*|.$$

Taking the infimum over the higher-shell sector yields (123).

d. BCC specialization. In the explicit BCC reciprocal arithmetic adopted in the current TECT shell normalization,

$$\delta_{\text{shell}} = 2. \tag{126}$$

Therefore

$$\Delta_{\text{high}}^{(0)} \geq K\delta_{\text{shell}}^2 - |r_*| = 4K - |r_*|. \tag{127}$$

Equivalently, in the unit-shell normalization one may write

$$\widehat{\Delta}_{\text{high}}^{(0)} \geq K - |r_*|.$$

B. Wrong-internal lower-bound lemma

e. Setup. Let the shell-point internal carrier be the minimal complex three-dimensional space

$$U_{\text{int}} = \mathbb{C}^3 = \mathbb{C}\hat{n} \oplus \hat{n}_{\mathbb{C}}^{\perp}. \tag{128}$$

Assume effective transverse $SO(2)$ acts on $\hat{n}_{\mathbb{C}}^{\perp}$, so that the unperturbed internal operator I_0 satisfies

$$I_0|_{\hat{n}_{\mathbb{C}}^{\perp}} = E_* \mathbf{1}_2, \qquad I_0\hat{n} = E_L\,\hat{n}. \tag{129}$$

Then define

$$U_D := \hat{n}_{\mathbb{C}}^{\perp}, \qquad U_{\text{wrong}} := \mathbb{C}\hat{n}. \tag{130}$$

f. Lemma (wrong-internal explicit bound). The unperturbed wrong-internal spectral distance is

$$m_{\text{int}}^{(0)} = \operatorname{dist}\big(\operatorname{spec}(I_0|_{U_{\text{wrong}}}), \operatorname{spec}(I_0|_{U_D})\big) = |E_L - E_*|. \tag{131}$$

Hence

$$\Delta_{\text{wrong-int}}^{(0)} = m_{\text{int}}^{(0)} = |E_L - E_*|. \tag{132}$$

In particular, if

$$E_L \neq E_*, \tag{133}$$

then

$$\Delta_{\text{wrong-int}}^{(0)} > 0. \tag{134}$$

g. Proof. By (129),

$$\operatorname{spec}(I_0|_{U_D}) = \{E_*\}, \qquad \operatorname{spec}(I_0|_{U_{\text{wrong}}}) = \{E_L\}.$$

Therefore

$$m_{\text{int}}^{(0)} = \operatorname{dist}(\{E_L\}, \{E_*\}) = |E_L - E_*|.$$

This proves (131) and hence (132).

h. Perturbative stability. Let

$$I = I_0 + \delta I, \qquad Q = P_{U_{\text{wrong}}}. \tag{135}$$

If

$$\|Q\,\delta I\,Q\| \leq \varepsilon_{\text{int}}, \tag{136}$$

then the perturbed internal gap satisfies

$$m_{\text{int}} \geq m_{\text{int}}^{(0)} - \varepsilon_{\text{int}} = |E_L - E_*| - \varepsilon_{\text{int}}. \tag{137}$$

Hence, if

$$\varepsilon_{\text{int}} < |E_L - E_*|, \tag{138}$$

then

$$m_{\text{int}} > 0 \qquad \text{and hence} \qquad \Delta_{\text{wrong-int}} > 0. \tag{139}$$

C. Combined two-sector benchmark

Combining the two lemmas, the first two explicit remote-sector benchmarks are

$$\boxed{\Delta_{\text{high}}^{(0)} \geq K\delta_{\text{shell}}^2 - |r_*|} \tag{140}$$

and

$$\boxed{\Delta_{\text{wrong-int}}^{(0)} \geq |E_L - E_*|} \tag{141}$$

(with the latter stable under bounded perturbations up to ε_{int}).

For the current BCC normalization,

$$\boxed{\Delta_{\text{high}}^{(0)} \geq 4K - |r_*|, \qquad \Delta_{\text{wrong-int}}^{(0)} \geq |E_L - E_*|.} \tag{142}$$

D. Honest status

These two sectors are now the most explicit parts of the minimal remote-gap program. What still remains model-specific is:

1. the volumetric benchmark $\Delta^{(0)}_{\mathrm{vol},\rho}$, which needs a heavy-locking input M_σ;
2. the same-shell non-enriched benchmark $\Delta^{(0)}_{\text{non-enr},\rho}$, which requires an actual finite truncated eigenvalue computation;
3. the remote-remote mixing budget $\eta_{R,\rho}$.

Thus the first two remote-gap entries are reduced to explicit scalar inequalities, while the latter three still require direct microscopic input.

P. Summary

$$\boxed{\text{First explicit remote-gap benchmarks}}$$

Higher-shell sector:

Let

$$\delta_{\mathrm{shell}} = \inf_{Q\in\mathcal{Q}_{\mathrm{high}}} \left||Q|^2 - q_*^2\right| > 0,$$

and assume

$$\Lambda(Q) = r_* + K(|Q|^2 - q_*^2)^2, \qquad K > 0.$$

Then

$$\boxed{\Delta^{(0)}_{\mathrm{high}} \geq K\delta^2_{\mathrm{shell}} - |r_*|.}$$

For the current BCC reciprocal arithmetic,

$$\delta_{\mathrm{shell}} = 2,$$

hence

$$\boxed{\Delta^{(0)}_{\mathrm{high}} \geq 4K - |r_*|.}$$

Wrong-internal sector:

In the minimal shell-point carrier

$$U_{\mathrm{int}} = \mathbb{C}^3 = \mathbb{C}\hat{n} \oplus \hat{n}^{\perp}_{\mathbb{C}},$$

assume

$$I_0|_{\hat{n}^{\perp}_{\mathbb{C}}} = E_*\,\mathbf{1}_2, \qquad I_0\hat{n} = E_L\,\hat{n}.$$

Then

$$U_D = \hat{n}^{\perp}_{\mathbb{C}}, \qquad U_{\mathrm{wrong}} = \mathbb{C}\hat{n},$$

and therefore

$$\boxed{\Delta^{(0)}_{\text{wrong-int}} = m^{(0)}_{\text{int}} = |E_L - E_*|.}$$

Under bounded perturbation

$$I = I_0 + \delta I, \qquad \|Q\delta I Q\| \le \varepsilon_{\text{int}},$$

the gap is stable:

$$\boxed{m_{\text{int}} \ge |E_L - E_*| - \varepsilon_{\text{int}}.}$$

Hence the first two explicit minimal-truncation benchmarks are

$$\boxed{\Delta^{(0)}_{\text{high}} \ge 4K - |r_*|, \qquad \Delta^{(0)}_{\text{wrong-int}} \ge |E_L - E_*|.}$$

Honest status:

$$\boxed{\text{these two sectors are structurally the easiest parts of Task IV.}}$$

What still remains model-specific is

$$\Delta^{(0)}_{\text{vol},\rho}, \qquad \Delta^{(0)}_{\text{non-enr},\rho}, \qquad \eta_{R,\rho}.$$

SHELL-POINT LONGITUDINAL–TRANSVERSE SPLITTING THEOREM

1. Effective shell-point operator after integrating out the heavy volumetric mode

After integrating out the heavy volumetric mode σ, the effective quadratic operator on the displacement sector takes the split form

$$D^{\text{eff}}_{ij}(k) = \left[(\lambda + 2\mu)k^2 + \frac{g^2 k^2}{M_\sigma^2 + c_\sigma^2 k^2}\right] P^L_{ij} + \mu k^2 P^T_{ij}, \tag{143}$$

where

$$P_L = \hat{n}\hat{n}^\dagger, \qquad P_T = \mathbf{1}_3 - P_L, \qquad \hat{n} = \frac{G_*}{|G_*|}, \qquad q_* := |G_*|. \tag{144}$$

Evaluating at the selected shell point $k = G_*$, the shell-point operator becomes

$$H_0 = E_L\, P_L + E_*\, P_T, \tag{145}$$

with transverse and longitudinal energies

$$E_*(G_*) = \mu q_*^2, \tag{146}$$

$$E_L(G_*) = (\lambda + 2\mu) q_*^2 + \frac{g^2 q_*^2}{M_\sigma^2 + c_\sigma^2 q_*^2}. \tag{147}$$

Therefore the shell-point internal space splits canonically as

$$U_{\text{int}} = \mathbb{C}^3 = \mathbb{C}\hat{n} \oplus \hat{n}^\perp_{\mathbb{C}}, \tag{148}$$

where the longitudinal line $\mathbb{C}\hat{n}$ carries energy E_L, while the transverse plane $\hat{n}^\perp_{\mathbb{C}}$ carries the protected energy $E_* \mathbf{1}_2$.

2. Exact longitudinal–transverse splitting

Subtracting (146) from (147), one obtains

$$E_L - E_* = q_*^2 \left[(\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right]. \tag{149}$$

Hence

$$H_0 = E_* \, \mathbf{1}_3 + (E_L - E_*) P_L. \tag{150}$$

This makes the wrong-internal spectral distance completely explicit:

$$m_{\text{int}}^{(0)} = \operatorname{dist}\bigl(\operatorname{spec}(H_0|_{\mathbb{C}\hat{n}}), \operatorname{spec}(H_0|_{\hat{n}_{\mathbb{C}}^{\perp}})\bigr) = |E_L - E_*|. \tag{151}$$

Therefore the wrong-internal diagonal benchmark is

$$\boxed{\Delta_{\text{wrong-int}}^{(0)} = q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|.} \tag{152}$$

3. Strong sufficient positivity condition

A stronger sufficient condition is

$$(\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} > 0. \tag{153}$$

Under (153),

$$E_L > E_*, \qquad m_{\text{int}}^{(0)} > 0, \qquad \Delta_{\text{wrong-int}}^{(0)} > 0. \tag{154}$$

Thus the wrong-internal sector is spectrally separated from the shell-relevant transverse doublet.

4. Perturbative stability

Let the shell-point operator be perturbed:

$$H = H_0 + \delta H, \qquad Q = P_L. \tag{155}$$

If

$$\|Q \, \delta H \, Q\| \le \varepsilon_{\text{int}}, \tag{156}$$

then the wrong-internal gap is stable:

$$m_{\text{int}} \ge m_{\text{int}}^{(0)} - \varepsilon_{\text{int}}. \tag{157}$$

Hence, if

$$\varepsilon_{\text{int}} < q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \tag{158}$$

then

$$m_{\text{int}} > 0 \qquad \text{and thus} \qquad \Delta_{\text{wrong-int}} > 0. \tag{159}$$

5. Embedding into the Task IV benchmark

The explicit diagonal benchmark of the remote sector may now be sharpened to

$$\Delta_{\mathrm{bench},\rho} := \min\left\{4K - |r_*|,\ c_{\mathrm{vol}} M_\sigma,\ q_*^2 \left|(\lambda+\mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2}\right|,\ \Delta^{(0)}_{\mathrm{extra},\rho}\right\}. \tag{160}$$

Therefore the explicit spectral isolation criterion becomes

$$\boxed{\Delta_{\mathrm{bench},\rho} > \eta_{R,\rho} \Longrightarrow \Delta_\rho > 0.} \tag{161}$$

6. Honest status

At this stage, the wrong-internal entry is no longer abstract. It has been reduced fully to the scalar shell-point splitting

$$E_L - E_* = q_*^2 \left[(\lambda+\mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2}\right].$$

Thus among the diagonal remote-sector benchmarks, the wrong-internal sector is now one of the most explicit.

What still remains model-dependent are:

1. the precise heavy volumetric benchmark $c_{\mathrm{vol}} M_\sigma$,
2. the non-enriched same-shell benchmark,
3. the off-diagonal remote mixing budget $\eta_{R,\rho}$.

Q. Summary

$$\boxed{E_*(G_*) = \mu q_*^2, \qquad E_L(G_*) = (\lambda + 2\mu) q_*^2 + \frac{g^2 q_*^2}{M_\sigma^2 + c_\sigma^2 q_*^2}}$$

Therefore

$$\boxed{E_L - E_* = q_*^2 \left[(\lambda+\mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2}\right].}$$

Hence the wrong-internal shell-point gap is

$$\boxed{\Delta^{(0)}_{\text{wrong-int}} = |E_L - E_*| = q_*^2 \left|(\lambda+\mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2}\right|.}$$

A stronger sufficient condition is

$$\boxed{(\lambda+\mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} > 0 \Longrightarrow E_L > E_* \Longrightarrow \Delta^{(0)}_{\text{wrong-int}} > 0.}$$

Under bounded perturbation

$$H = H_0 + \delta H, \qquad \|Q\,\delta H\,Q\| \le \varepsilon_{\mathrm{int}},$$

the gap is stable:

$$\boxed{m_{\mathrm{int}} \ge \Delta^{(0)}_{\text{wrong-int}} - \varepsilon_{\mathrm{int}}.}$$

Thus the explicit remote-sector benchmark can be sharpened to

$$\boxed{\Delta_{\mathrm{bench},\rho} := \min\left\{4K - |r_*|,\ c_{\mathrm{vol}}M_\sigma,\ q_*^2\left|(\lambda+\mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2}\right|,\ \Delta^{(0)}_{\mathrm{extra},\rho}\right\}.}$$

If

$$\boxed{\Delta_{\mathrm{bench},\rho} > \eta_{R,\rho},}$$

then

$$\boxed{\Delta_\rho > 0.}$$

HEAVY-VOLUMETRIC BENCHMARK THEOREM

A. Correct object of the volumetric benchmark

The volumetric remote benchmark must refer to the genuinely heavy volumetric field σ itself, not to the longitudinal–transverse splitting inside the shell-point displacement carrier.

These are two different remote-gap entries:

1. the **heavy volumetric sector**, carried by the gapped scalar field σ;
2. the **wrong-internal sector**, measured by the shell-point splitting

$$E_L - E_* = q_*^2\left[(\lambda+\mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2}\right].$$

Thus $\Delta^{(0)}_{\mathrm{vol},\rho}$ and $\Delta^{(0)}_{\text{wrong-int},\rho}$ must not be identified.

B. Starting quadratic longitudinal–volumetric sector

The relevant quadratic sector is

$$L^{(2)}_{L,\sigma} = \frac{1}{2}\rho|\dot{u}_L|^2 - \frac{1}{2}(\lambda+2\mu)k^2|u_L|^2 + \frac{1}{2}|\dot\sigma|^2 - \frac{1}{2}(M_\sigma^2 + c_\sigma^2 k^2)|\sigma|^2 + g\,\sigma(-k)\,ik\,u_L(k) + \text{c.c.} \tag{162}$$

The coupling is longitudinal-only, because it depends on

$$k_i u_i = k\,u_L.$$

Hence the transverse sector is decoupled from σ at quadratic order.

The heavy operator of the pure volumetric field is

$$K_\sigma = M_\sigma^2 - c_\sigma^2\nabla^2 + \partial_t^2, \tag{163}$$

whose Fourier symbol is

$$K_\sigma(\omega, k) = M_\sigma^2 + c_\sigma^2 k^2 - \omega^2. \tag{164}$$

C. Frequency-scale heavy volumetric gap

The pure σ-dispersion is

$$\omega_\sigma(k) = \sqrt{M_\sigma^2 + c_\sigma^2 k^2}. \tag{165}$$

Now localize near the selected shell point:

$$k = G_* + p, \qquad |p| \le \rho, \qquad q_* := |G_*|.$$

Assume

$$0 < \rho < q_*. \tag{166}$$

Then

$$|k| = |G_* + p| \ge q_* - \rho. \tag{167}$$

Therefore

$$\omega_\sigma(k) \ge \sqrt{M_\sigma^2 + c_\sigma^2 (q_* - \rho)^2}. \tag{168}$$

This gives the explicit heavy-volumetric frequency benchmark:

$$\boxed{\Delta_{\mathrm{vol},\rho}^{(0),\mathrm{freq}} \ge \sqrt{M_\sigma^2 + c_\sigma^2 (q_* - \rho)^2}.} \tag{169}$$

In particular,

$$\Delta_{\mathrm{vol},\rho}^{(0),\mathrm{freq}} \ge M_\sigma. \tag{170}$$

Hence the earlier symbolic notation

$$\Delta_{\mathrm{vol}}^{(0)} \ge c_{\mathrm{vol}} M_\sigma$$

may be made completely explicit at the frequency level by taking

$$c_{\mathrm{vol}} = 1. \tag{171}$$

D. Quadratic-Hessian-scale benchmark

If one works not with frequency eigenvalues but with the static quadratic kernel / Hessian normalization, then the natural volumetric benchmark is

$$\Delta_{\mathrm{vol},\rho}^{(0),\mathrm{quad}} := \inf_{|p|\le\rho} \left(M_\sigma^2 + c_\sigma^2 |G_* + p|^2 \right). \tag{172}$$

Using (167),

$$\boxed{\Delta_{\mathrm{vol},\rho}^{(0),\mathrm{quad}} \ge M_\sigma^2 + c_\sigma^2 (q_* - \rho)^2.} \tag{173}$$

Thus the previous shorthand $c_{\mathrm{vol}} M_\sigma$ is not wrong, but it hides the choice of spectral normalization:

- in **frequency** units, the benchmark is $\ge M_\sigma$;
- in **quadratic-kernel** units, the benchmark is $\ge M_\sigma^2 + c_\sigma^2 (q_* - \rho)^2$.

E. Relation to the effective longitudinal stiffening

Integrating out the heavy σ field yields the effective longitudinal block

$$D_{ij}^{\mathrm{eff}}(k) = \left[(\lambda + 2\mu) k^2 + \frac{g^2 k^2}{M_\sigma^2 + c_\sigma^2 k^2} \right] P_{ij}^L + \mu k^2 P_{ij}^T. \tag{174}$$

This shows two things:

1. the heavy volumetric mode shifts only the longitudinal block;
2. the transverse block remains unchanged at leading order.

At the shell point $k = G_*$, this yields

$$E_*(G_*) = \mu q_*^2, \qquad E_L(G_*) = (\lambda + 2\mu) q_*^2 + \frac{g^2 q_*^2}{M_\sigma^2 + c_\sigma^2 q_*^2}. \tag{175}$$

Hence the wrong-internal benchmark is

$$\Delta^{(0)}_{\text{wrong-int}} = |E_L - E_*| = q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|. \tag{176}$$

This must be kept distinct from $\Delta^{(0)}_{\text{vol},\rho}$.

F. EFT suppression and infrared closure

Because K_σ^{-1} admits the derivative expansion

$$K_\sigma^{-1} = \frac{1}{M_\sigma^2} \left[1 + \frac{c_\sigma^2 \nabla^2 - \partial_t^2}{M_\sigma^2} + O\left(\frac{\partial^4}{M_\sigma^4} \right) \right], \tag{177}$$

integrating out σ induces only local higher-dimension operators suppressed by powers of M_σ^{-2}.

At tree level, heavy-σ-mediated transverse amplitudes satisfy

$$\mathcal{A}^{(\sigma)}_{TT \to TT} = O\left(\frac{p^{2m}}{M_\sigma^2} \right) + O\left(\frac{p^{2m+2}}{M_\sigma^4} \right), \qquad m \geq 1, \tag{178}$$

and therefore

$$\mathcal{A}^{(\sigma)}_{TT \to TT} \to 0 \qquad \text{as } \omega, |p| \to 0. \tag{179}$$

Thus the heavy-volumetric sector contributes to the remote benchmark by a genuine positive gap, while its induced corrections to the low-energy transverse sector are infrared-irrelevant.

G. Refined benchmark replacement

The earlier coarse benchmark

$$\Delta^{(0)}_{\text{vol}} \geq c_{\text{vol}} M_\sigma$$

may now be refined to either of the following fully explicit forms:

a. Frequency version

$$\boxed{\Delta^{(0)}_{\text{vol},\rho} \geq \sqrt{M_\sigma^2 + c_\sigma^2 (q_* - \rho)^2}.} \tag{180}$$

b. Quadratic-kernel version

$$\boxed{\Delta^{(0)}_{\text{vol},\rho} \geq M_\sigma^2 + c_\sigma^2 (q_* - \rho)^2.} \tag{181}$$

Therefore the master remote-gap benchmark may be rewritten more explicitly as

$$\Delta^{\text{ref}}_{\text{bench},\rho} := \min \left\{ 4K - |r_*|,\ \sqrt{M_\sigma^2 + c_\sigma^2 (q_* - \rho)^2},\ q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|,\ \Delta^{(0)}_{\text{extra},\rho} \right\}, \tag{182}$$

in frequency normalization, or with the second slot replaced by (181) in quadratic-kernel normalization.

H. Honest status

At this point:

1. the higher-shell benchmark is explicit;
2. the wrong-internal benchmark is explicit;
3. the heavy-volumetric benchmark is explicit;
4. what remains non-automatic is mainly the same-shell non-enriched benchmark and the off-diagonal remote mixing budget.

So the volumetric slot is no longer a vague symbol. It is reduced to the actual heavy σ-dispersion near the selected shell.

R. Summary

$$\boxed{\text{Heavy volumetric benchmark}}$$

The quadratic longitudinal–volumetric sector is

$$L_{L,\sigma}^{(2)} = \frac{1}{2}\rho|\dot{u}_L|^2 - \frac{1}{2}(\lambda + 2\mu)k^2|u_L|^2 + \frac{1}{2}|\dot{\sigma}|^2 - \frac{1}{2}(M_\sigma^2 + c_\sigma^2 k^2)|\sigma|^2 + g\,\sigma(-k)\,ik\,u_L(k) + \text{c.c.}$$

Hence the pure heavy-σ dispersion is

$$\omega_\sigma(k) = \sqrt{M_\sigma^2 + c_\sigma^2 k^2}.$$

For

$$k = G_* + p, \qquad |p| \le \rho < q_* := |G_*|,$$

one has

$$|k| \ge q_* - \rho.$$

Therefore

$$\boxed{\Delta_{\text{vol},\rho}^{(0),\text{freq}} \ge \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2} \ge M_\sigma.}$$

In quadratic-kernel normalization, equivalently,

$$\boxed{\Delta_{\text{vol},\rho}^{(0),\text{quad}} \ge M_\sigma^2 + c_\sigma^2(q_* - \rho)^2.}$$

Thus the previous shorthand

$$\Delta_{\text{vol}}^{(0)} \ge c_{\text{vol}} M_\sigma$$

is refined as follows:

$$\boxed{\text{in frequency units, } c_{\text{vol}} = 1; \qquad \text{in quadratic-kernel units, use } M_\sigma^2 + c_\sigma^2(q_* - \rho)^2.}$$

Important distinction:

$$\boxed{\Delta_{\text{vol},\rho}^{(0)} \ne \Delta_{\text{wrong-int}}^{(0)}.}$$

The latter is the shell-point longitudinal–transverse splitting:

$$\Delta^{(0)}_{\text{wrong-int}} = q_*^2 \left| (\lambda+\mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|.$$

Therefore the refined explicit benchmark is

$$\boxed{\Delta^{\text{ref}}_{\text{bench},\rho} := \min\left\{ 4K - |r_*|,\ \sqrt{M_\sigma^2 + c_\sigma^2 (q_* - \rho)^2},\ q_*^2 \left| (\lambda+\mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|,\ \Delta^{(0)}_{\text{extra},\rho} \right\}.}$$

If

$$\boxed{\Delta^{\text{ref}}_{\text{bench},\rho} > \eta_{R,\rho},}$$

then

$$\boxed{\Delta_\rho > 0.}$$

MINIMAL FINITE MATRIX THEOREM FOR THE SAME-SHELL NON-ENRICHED SECTOR

A. Honest logical status

Within the minimal truncation,

$$\mathcal{H}^{\text{tr}}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}},$$

the sector

$$\mathcal{H}_{\text{non-enr}}$$

is defined as the *same-shell but non-enriched sector*. It carries the shell label $\pm G_*$ and transverse character, but it does *not* carry the enriched internal doublet that defines the local Dirac block $\mathcal{H}_D$. Therefore it is excluded from constant mixing at $p = 0$ by shell-relevant label mismatch, but its spectral distance from zero is *not* automatic. It must be checked by a finite truncated eigenvalue computation. :contentReference[oaicite:6]index=6 :contentReference[oaicite:7]index=7

B. Bloch reduction and shell projection

Let H_{ϕ_0} be the microscopic linearization around the periodic BCC background. By Floquet–Bloch decomposition,

$$H_{\phi_0} \simeq \int_{BZ}^{\oplus} H_{\text{Bloch}}(k)\, dk.$$

Near the selected reciprocal shell point, one writes

$$k = G_* + p, \qquad |p| \le \rho.$$

The correct finite-dimensional object is then the shell-projected Bloch matrix

$$H_{\text{shell}}(p) = \Pi_{S_*} H_{\text{Bloch}}(G_* + p) \Pi_{S_*},$$

where S_* is a finite set of near-resonant reciprocal harmonics. :contentReference[oaicite:8]index=8

This shell-projected space is then split into

$$\mathcal{H}^{(N)}_{\text{shell}} = \mathcal{H}^{(N)}_D \oplus \mathcal{H}^{(N)}_{\text{non-enr}} \oplus \mathcal{H}^{(N)}_{\text{rest}},$$

where

- $\mathcal{H}^{(N)}_D$ is the enriched shell doublet sector producing the local Dirac block,
- $\mathcal{H}^{(N)}_{\text{non-enr}}$ is the complementary same-shell sector that does not carry the enriched doublet,
- $\mathcal{H}^{(N)}_{\text{rest}}$ collects all additional truncated states.

C. Definition of the non-enriched projected matrix

Let

$$P^{(N)}_{\text{non-enr}} : \mathcal{H}^{(N)}_{\text{shell}} \to \mathcal{H}^{(N)}_{\text{non-enr}}$$

be the orthogonal projector. Define the finite same-shell non-enriched block by

$$H^{(N)}_{\text{non-enr}}(p) := P^{(N)}_{\text{non-enr}}\, H^{(N)}_{\text{shell}}(p)\, P^{(N)}_{\text{non-enr}}. \tag{183}$$

Then the finite truncation benchmark is

$$\Delta^{(N)}_{\text{non-enr},\rho} := \inf_{|p|\le\rho} \operatorname{dist}\Big(0, \operatorname{spec}\big(H^{(N)}_{\text{non-enr}}(p)\big)\Big). \tag{184}$$

This is the exact finite matrix quantity that must be computed.

D. Minimal finite matrix theorem

[Minimal same-shell non-enriched benchmark theorem] Assume:

1. $H_{\text{Bloch}}(k)$ is the Bloch fiber of the periodic microscopic linearization;
2. $H^{(N)}_{\text{shell}}(p)$ is a finite shell-projected truncation on a neighborhood $|p| \le \rho$;
3. $\mathcal{H}^{(N)}_{\text{shell}}$ splits as
$$\mathcal{H}^{(N)}_{D} \oplus \mathcal{H}^{(N)}_{\text{non-enr}} \oplus \mathcal{H}^{(N)}_{\text{rest}};$$
4. the same-shell non-enriched sector indeed carries no enriched internal doublet, so it is distinct from $\mathcal{H}^{(N)}_{D}$ at the label level.

Then the diagonal benchmark entering the remote-gap inequality is precisely

$$\boxed{\Delta^{(0)}_{\text{non-enr},\rho} \;\rightsquigarrow\; \Delta^{(N)}_{\text{non-enr},\rho} = \inf_{|p|\le\rho} \operatorname{dist}\Big(0, \operatorname{spec}\big(H^{(N)}_{\text{non-enr}}(p)\big)\Big).} \tag{185}$$

If

$$\Delta^{(N)}_{\text{non-enr},\rho} > 0, \tag{186}$$

then the truncated same-shell non-enriched sector is spectrally separated from zero on N_ρ.

a. Meaning. This theorem does *not* yet prove the exact infinite-dimensional same-shell non-enriched gap. What it proves is that the microscopic problem is reduced to an explicit finite matrix spectral audit.

E. Gershgorin / coercivity version

Choose any orthonormal basis $\{v_a\}_{a=1}^{m_N}$ of $\mathcal{H}^{(N)}_{\text{non-enr}}$. Write the matrix

$$\big(H^{(N)}_{\text{non-enr}}(p)\big)_{ab}.$$

Define the Gershgorin lower bound

$$\Delta^{(G,N)}_{\text{non-enr},\rho} := \inf_{|p|\le\rho} \min_a \left(\Big|\big(H^{(N)}_{\text{non-enr}}(p)\big)_{aa}\Big| - \sum_{b\ne a} \Big|\big(H^{(N)}_{\text{non-enr}}(p)\big)_{ab}\Big| \right). \tag{187}$$

If

$$\Delta^{(G,N)}_{\text{non-enr},\rho} > 0, \tag{188}$$

then

$$\Delta^{(N)}_{\text{non-enr},\rho} \geq \Delta^{(G,N)}_{\text{non-enr},\rho} > 0. \tag{189}$$

Equivalently, if one has a coercive decomposition

$$H^{(N)}_{\text{non-enr}}(p) = D^{(N)}_{\text{non-enr}}(p) + R^{(N)}_{\text{non-enr}}(p),$$

with

$$\delta^{(N)}_{\text{non-enr},\rho} := \inf_{|p|\leq\rho} \min_a \left| \left(D^{(N)}_{\text{non-enr}}(p)\right)_{aa} \right|, \qquad \eta^{(N)}_{\text{non-enr},\rho} := \sup_{|p|\leq\rho} \|R^{(N)}_{\text{non-enr}}(p)\|,$$

then

$$\delta^{(N)}_{\text{non-enr},\rho} > \eta^{(N)}_{\text{non-enr},\rho} \Longrightarrow \Delta^{(N)}_{\text{non-enr},\rho} > 0. \tag{190}$$

F. Truncation-to-exact upgrade

Suppose the exact same-shell non-enriched operator is

$$H_{\text{non-enr}}(p),$$

and the truncation error obeys

$$\varepsilon^{(N)}_{\text{non-enr},\rho} := \sup_{|p|\leq\rho} \left\| H_{\text{non-enr}}(p) - H^{(N)}_{\text{non-enr}}(p) \right\| < \Delta^{(N)}_{\text{non-enr},\rho}. \tag{191}$$

Then the exact benchmark satisfies

$$\Delta^{(0)}_{\text{non-enr},\rho} \geq \Delta^{(N)}_{\text{non-enr},\rho} - \varepsilon^{(N)}_{\text{non-enr},\rho} > 0. \tag{192}$$

Thus a positive finite-truncation eigenvalue gap plus a smaller truncation error is sufficient to close the exact same-shell non-enriched slot.

G. Embedding into the master remote-gap inequality

The master diagonal benchmark can now be sharpened to

$$\Delta^{\text{fin}}_{\text{bench},\rho} := \min\left\{ 4K - |r_*|,\ \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2},\ q_*^2 \left| (\lambda+\mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|,\ \Delta^{(N)}_{\text{non-enr},\rho} - \varepsilon^{(N)}_{\text{non-enr},\rho} \right\}. \tag{193}$$

If

$$\boxed{\Delta^{\text{fin}}_{\text{bench},\rho} > \eta_{R,\rho},} \tag{194}$$

then

$$\Delta_\rho > 0. \tag{195}$$

H. Honest final status

Therefore:

1. the higher-shell slot is explicit;
2. the wrong-internal slot is explicit;
3. the heavy-volumetric slot is explicit;
4. the same-shell non-enriched slot is reduced exactly to a finite truncated matrix eigenvalue problem;
5. the remaining global ingredient is the remote-remote mixing norm budget $\eta_{R,\rho}$.

So the genuinely non-automatic part of Task IV has now been reduced to one finite matrix computation plus one norm bound.

S. Summary

$$\boxed{\text{Same-shell non-enriched slot is not automatic.}}$$

The minimal truncation contains

$$\mathcal{H}_{\rm full}^{\rm tr} = \mathcal{H}_D \oplus \mathcal{H}_{\rm vol} \oplus \mathcal{H}_{\rm high} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}},$$

where

$$\mathcal{H}_{\text{non-enr}} = \text{same-shell but non-enriched sector.}$$

Because it lacks the enriched internal doublet, it is excluded from constant mixing at $p = 0$:

$$\mathrm{Hom}_{H_*}(\mathcal{H}_{\text{non-enr}}, \mathcal{H}_D) = 0 \Longrightarrow V_{\text{non-enr}\to D}(0) = 0.$$

But this does *not* automatically imply a positive spectral gap.

Let

$$H^{(N)}_{\rm shell}(p) = \Pi^{(N)}_{S_*} H_{\rm Bloch}(G_* + p)\Pi^{(N)}_{S_*}$$

be a finite shell-projected Bloch matrix, and let

$$P^{(N)}_{\text{non-enr}}$$

project onto the same-shell complementary sector without the enriched doublet. Define

$$H^{(N)}_{\text{non-enr}}(p) = P^{(N)}_{\text{non-enr}} H^{(N)}_{\rm shell}(p) P^{(N)}_{\text{non-enr}}.$$

Then the correct finite benchmark is

$$\boxed{\Delta^{(N)}_{\text{non-enr},\rho} := \inf_{|p|\le\rho} \mathrm{dist}\Big(0, \mathrm{spec}\big(H^{(N)}_{\text{non-enr}}(p)\big)\Big).}$$

If

$$\Delta^{(N)}_{\text{non-enr},\rho} > 0,$$

then the truncated same-shell non-enriched sector is spectrally separated from zero.

If moreover

$$\varepsilon^{(N)}_{\text{non-enr},\rho} := \sup_{|p|\le\rho} \left\| H_{\text{non-enr}}(p) - H^{(N)}_{\text{non-enr}}(p) \right\| < \Delta^{(N)}_{\text{non-enr},\rho},$$

then

$$\boxed{\Delta^{(0)}_{\text{non-enr},\rho} \ge \Delta^{(N)}_{\text{non-enr},\rho} - \varepsilon^{(N)}_{\text{non-enr},\rho} > 0.}$$

So the master benchmark becomes

$$\boxed{\Delta^{\rm fin}_{\text{bench},\rho} := \min\left\{ 4K - |r_*|,\ \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2},\ q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|,\ \Delta^{(N)}_{\text{non-enr},\rho} - \varepsilon^{(N)}_{\text{non-enr},\rho} \right\}.}$$

If

$$\Delta^{\rm fin}_{\text{bench},\rho} > \eta_{R,\rho},$$

then

$$\Delta_\rho > 0.$$

Hence the truly non-automatic same-shell part of Task IV is reduced to one finite matrix eigenvalue computation.

BLOCK-NORM AND GERSHGORIN BUDGET THEOREM FOR THE FINAL REMOTE-GAP SLOT

A. Starting point

At the minimal truncation level, the remote block is written as

$$H_R(p) = D_R(p) + R_R(p), \tag{196}$$

where $D_R(p)$ is the diagonal (or sector-center) benchmark part and $R_R(p)$ is the residual remote–remote mixing. The pointwise and uniform remote-gap criteria are already known:

$$\delta_0(p) > \eta_R(p) := \|R_R(p)\| \Longrightarrow \Delta(p) \geq \delta_0(p) - \eta_R(p) > 0, \tag{197}$$

and on $N_\rho = \{|p| \leq \rho\}$,

$$\delta_{0,\rho} > \eta_{R,\rho} \Longrightarrow \Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0. \tag{198}$$

Thus the only remaining issue is to produce a usable upper bound on

$$\eta_{R,\rho} := \sup_{|p|\leq\rho} \|R_R(p)\|.$$

B. Sectorwise block decomposition

Let the minimal truncated remote space be

$$\mathcal{H}_R^{\mathrm{tr}} = \mathcal{H}_{\mathrm{vol}} \oplus \mathcal{H}_{\mathrm{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}}.$$

Write

$$R_R(p) = \begin{pmatrix} 0 & R_{\mathrm{vol,high}} & R_{\mathrm{vol,non}} & R_{\mathrm{vol,wrong}} \\ R_{\mathrm{high,vol}} & 0 & R_{\mathrm{high,non}} & R_{\mathrm{high,wrong}} \\ R_{\mathrm{non,vol}} & R_{\mathrm{non,high}} & 0 & R_{\mathrm{non,wrong}} \\ R_{\mathrm{wrong,vol}} & R_{\mathrm{wrong,high}} & R_{\mathrm{wrong,non}} & 0 \end{pmatrix}(p), \tag{199}$$

where each block

$$R_{ab}(p) : \mathcal{H}_b \to \mathcal{H}_a$$

is an off-diagonal remote–remote coupling.

Define the pairwise budgets

$$\eta_{ab,\rho} := \sup_{|p|\leq\rho} \|R_{ab}(p)\|. \tag{200}$$

C. Basic block-norm budget

By the triangle inequality for operator norms, one has the sufficient estimate

$$\eta_{R,\rho} \leq \sum_{a\neq b} \eta_{ab,\rho}. \tag{201}$$

Therefore a fully explicit sufficient criterion is

$$\boxed{\Delta^{\mathrm{fin}}_{\mathrm{bench},\rho} > \sum_{a\neq b} \eta_{ab,\rho}.} \tag{202}$$

This is crude but rigorous.

D. Row-sum / Gershgorin budget

Choose any orthonormal basis of the remote space and write

$$(H_R)_{mn}(p)$$

for the matrix elements. Define the row sums

$$R_m(p) := \sum_{n \neq m} |(H_R)_{mn}(p)|. \tag{203}$$

Then the Gershgorin lower bound is

$$\Delta_{G,\rho} := \inf_{|p| \leq \rho} \min_m \Big(|(H_R)_{mm}(p)| - R_m(p) \Big). \tag{204}$$

If

$$\Delta_{G,\rho} > 0, \tag{205}$$

then

$$\operatorname{spec}(H_R(p)) \cap (-\Delta_{G,\rho}, \Delta_{G,\rho}) = \varnothing \qquad (|p| \leq \rho). \tag{206}$$

This route is often sharper than (201) when the remote matrix is sparse.

E. Full remote completion with tails

For the full remote sector, let

$$H_R^{\mathrm{full}}(p) = D_{\mathrm{rem}}(p) + R_{\mathrm{rem}}(p),$$

where $D_{\mathrm{rem}}(p)$ carries the global diagonal benchmark

$$\Delta_{\mathrm{bench},\rho},$$

and $R_{\mathrm{rem}}(p)$ contains:

- all remote–remote off-diagonal mixing,
- all couplings to tail sectors,
- all bounded residual corrections to the diagonal benchmark.

Define

$$\eta_{\mathrm{rem},\rho} := \sup_{|p| \leq \rho} \|R_{\mathrm{rem}}(p)\|. \tag{207}$$

Then the master explicit spectral inequality is

$$\boxed{\Delta_{\mathrm{bench},\rho} > \eta_{\mathrm{rem},\rho} \Longrightarrow \Delta_{\mathrm{full},\rho} \geq \Delta_{\mathrm{bench},\rho} - \eta_{\mathrm{rem},\rho} > 0.} \tag{208}$$

F. Explicit power-counting / heavy-suppression budget

By the previously established tail-coupling estimate theorem, one may bound

$$\eta_{\mathrm{rem},\rho} \leq \sum_{a \in \mathcal{A}_{\mathrm{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\mathrm{act}}^{(H)}} \frac{\widetilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a}, \qquad r_a \geq 1. \tag{209}$$

Thus all remote mixing is at least $O(\rho)$ near the crossing, and heavy-mediated sectors carry additional suppression by inverse powers of heavy masses.

Substituting (209) into (208), one obtains the fully explicit sufficient condition

$$\boxed{\Delta_{\mathrm{bench},\rho} > \sum_{a \in \mathcal{A}_{\mathrm{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\mathrm{act}}^{(H)}} \frac{\widetilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a}.} \tag{210}$$

G. Final Task IV closure statement

Combining the previously established diagonal benchmarks

$$4K - |r_*|, \qquad \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2}, \qquad q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \qquad \Delta^{(N)}_{\text{non-enr},\rho} - \varepsilon^{(N)}_{\text{non-enr},\rho},$$

define

$$\Delta^{\text{full}}_{\text{bench},\rho} := \min\left\{ 4K - |r_*|,\ \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2},\ q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|,\ \Delta^{(N)}_{\text{non-enr},\rho} - \varepsilon^{(N)}_{\text{non-enr},\rho} \right\}. \tag{211}$$

Then a sufficient final Task IV criterion is

$$\boxed{\Delta^{\text{full}}_{\text{bench},\rho} > \eta_{\text{rem},\rho}.} \tag{212}$$

Under (212),

$$\operatorname{spec}(H_R^{\text{full}}(p)) \cap (-\Delta_{\text{full},\rho}, \Delta_{\text{full},\rho}) = \varnothing, \qquad \Delta_{\text{full},\rho} \geq \Delta^{\text{full}}_{\text{bench},\rho} - \eta_{\text{rem},\rho} > 0. \tag{213}$$

H. Honest final status

At this point, Task IV is no longer conceptually open. It has been reduced completely to:

1. four explicit diagonal benchmark entries;
2. one same-shell finite matrix eigenvalue audit;
3. one remote mixing norm budget.

So the remaining issue is purely microscopic and quantitative: evaluate the constants and verify the single inequality

$$\Delta^{\text{full}}_{\text{bench},\rho} > \eta_{\text{rem},\rho}.$$

T. Summary

$$\boxed{\text{Final remote mixing budget}}$$

At the minimal truncation level,

$$H_R(p) = D_R(p) + R_R(p), \qquad \eta_{R,\rho} = \sup_{|p|\leq\rho} \|R_R(p)\|.$$

The uniform remote-gap criterion is

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho} \Longrightarrow \Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0.}$$

Write the remote mixing in sector blocks:

$$R_R(p) = \big(R_{ab}(p)\big)_{a\neq b}, \qquad a, b \in \{\text{vol}, \text{high}, \text{non-enr}, \text{wrong-int}\}.$$

Define pairwise budgets

$$\eta_{ab,\rho} := \sup_{|p|\leq\rho} \|R_{ab}(p)\|.$$

Then a sufficient block-norm budget is

$$\boxed{\eta_{R,\rho} \leq \sum_{a\neq b} \eta_{ab,\rho}.}$$

A sharper Gershgorin version is:

$$\Delta_{G,\rho} := \inf_{|p|\leq\rho} \min_m \Big(|(H_R)_{mm}(p)| - \sum_{n\neq m} |(H_R)_{mn}(p)| \Big).$$

If

$$\boxed{\Delta_{G,\rho} > 0,}$$

then

$$\boxed{\mathrm{spec}(H_R(p)) \cap (-\Delta_{G,\rho}, \Delta_{G,\rho}) = \varnothing \qquad (|p| \leq \rho).}$$

For the full remote block,

$$H_R^{\mathrm{full}}(p) = D_{\mathrm{rem}}(p) + R_{\mathrm{rem}}(p), \qquad \eta_{\mathrm{rem},\rho} = \sup_{|p|\leq\rho} \|R_{\mathrm{rem}}(p)\|.$$

The master explicit spectral inequality is

$$\boxed{\Delta_{\mathrm{bench},\rho} > \eta_{\mathrm{rem},\rho} \Longrightarrow \Delta_{\mathrm{full},\rho} \geq \Delta_{\mathrm{bench},\rho} - \eta_{\mathrm{rem},\rho} > 0.}$$

Moreover,

$$\boxed{\eta_{\mathrm{rem},\rho} \leq \sum_{a\in\mathcal{A}_{\mathrm{act}}} K_a \rho^{r_a} + \sum_{a\in\mathcal{A}_{\mathrm{act}}^{(H)}} \frac{\widetilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a}, \qquad r_a \geq 1.}$$

Therefore the fully explicit final Task IV criterion is

$$\boxed{\Delta_{\mathrm{bench},\rho}^{\mathrm{full}} > \eta_{\mathrm{rem},\rho}.}$$

Here

$$\Delta_{\mathrm{bench},\rho}^{\mathrm{full}} := \min\left\{ 4K - |r_*|,\ \sqrt{M_\sigma^2 + c_\sigma^2 (q_* - \rho)^2},\ q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|,\ \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)} \right\}.$$

If this one inequality holds, then Task IV is complete:

$$\boxed{\Delta_\rho > 0.}$$

TASK V: RESIDUAL-SYMMETRY AUDIT TABLE FOR ACTUAL MASS PROTECTION

A. Exact theorem-level starting point

For the doubled local Dirac block

$$H_D(p) = \sum_{i=1}^{3} v_i p_i \, (\tau_z \otimes \sigma_i), \tag{214}$$

the full little-group-scalar constant Hermitian operator space is

$$K = a_0 \, \mathbf{1}_4 + a_3 \, \tau_z \otimes \mathbf{1}_2 + m_1 \, \tau_x \otimes \mathbf{1}_2 + m_2 \, \tau_y \otimes \mathbf{1}_2. \tag{215}$$

The genuine constant Dirac-mass space is exactly

$$\mathcal{M}_D = \mathrm{span}_{\mathbb{R}}\left\{\tau_x \otimes \mathbf{1}_2,\ \tau_y \otimes \mathbf{1}_2\right\}. \tag{216}$$

Thus the actual microscopic mass-protection problem is equivalent to

$$A_{\mathrm{res}} \cap \mathcal{M}_D = \{0\}, \tag{217}$$

where

$$A_{\mathrm{res}} = \{K \in \mathrm{Herm}(4) : U(g)KU(g)^{-1} = K\ \forall g \in G_{\mathrm{res}}\}. \tag{218}$$

B. Residual-symmetry audit table

The practical classification table is:

Residual structure	Action on $\tau_x \otimes \mathbf{1}_2,\ \tau_y \otimes \mathbf{1}_2$	Does it forbid all constant masses?	
little-group scalarity only	allows both	No	a
opposite-valley matching only	fixes $h_-(p) = -h_+(-p)$, but does not remove τ_x, τ_y	No	al
exact valley $U(1)_V$	(τ_x, τ_y) rotate as a charged doublet	Yes	suffi
exact valley $\mathbb{Z}_2^V$	$\tau_x, \tau_y \mapsto -\tau_x, -\tau_y$	Yes	we
Class II gradient-linear structure	does not insert a constant mass directly	Compatible only	

C. Why exact valley $U(1)_V$ is sufficient

Define the valley-charge generator

$$Q_V := \tau_z \otimes \mathbf{1}_2, \qquad U_\theta := e^{i\theta Q_V/2}. \tag{219}$$

Since

$$[Q_V, H_D(p)] = 0, \tag{220}$$

the doubled kinetic block is compatible with a residual valley symmetry.

Now

$$U_\theta(\tau_x \otimes \mathbf{1}_2)U_\theta^{-1} = \cos\theta\,(\tau_x \otimes \mathbf{1}_2) + \sin\theta\,(\tau_y \otimes \mathbf{1}_2), \tag{221}$$

$$U_\theta(\tau_y \otimes \mathbf{1}_2)U_\theta^{-1} = -\sin\theta\,(\tau_x \otimes \mathbf{1}_2) + \cos\theta\,(\tau_y \otimes \mathbf{1}_2). \tag{222}$$

Therefore no nonzero element of

$$\mathcal{M}_D = \mathrm{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}$$

is invariant under all θ, so

$$A_{U(1)_V} \cap \mathcal{M}_D = \{0\}. \tag{223}$$

Hence exact residual valley $U(1)_V$ is a sufficient protection mechanism.

D. Why exact valley $\mathbb{Z}_2^V$ is already enough

Let

$$S_V := \tau_z \otimes \mathbf{1}_2, \qquad S_V^2 = \mathbf{1}_4. \tag{224}$$

Then

$$S_V(\tau_x \otimes \mathbf{1}_2)S_V^{-1} = -(\tau_x \otimes \mathbf{1}_2), \qquad S_V(\tau_y \otimes \mathbf{1}_2)S_V^{-1} = -(\tau_y \otimes \mathbf{1}_2). \tag{225}$$

Hence any S_V-invariant constant operator must have

$$m_1 = m_2 = 0.$$

Therefore exact residual $\mathbb{Z}_2^V$ already suffices.

E. What is *not* sufficient

a. (i) Little-group scalarity alone. Little-group scalarity only reduces the allowed constant algebra to

$$\mathbf{1}_4,\ \tau_z \otimes \mathbf{1}_2,\ \tau_x \otimes \mathbf{1}_2,\ \tau_y \otimes \mathbf{1}_2.$$

It does not eliminate the two mass generators.

b. (ii) Opposite-valley pairing alone. The relation

$$h_-(p) = -h_+(-p)$$

guarantees Dirac doubling, but it does not by itself remove a constant intervalley operator

$$m_1\, \tau_x \otimes \mathbf{1}_2 + m_2\, \tau_y \otimes \mathbf{1}_2.$$

c. (iii) Class II structural compatibility. The canonical minimal Class II correction renormalizes the derivative shell coefficients A, B, C and does not insert a constant mass operator directly. Therefore it is mass-protection compatible. But this is weaker than the theorem

$$A_{\text{res}} \cap \mathcal{M}_D = \{0\}.$$

Hence the actual symmetry exclusion still has to be checked independently.

F. Actual microscopic audit procedure

To close Task V in the actual microscopic TECT/Class II realization, perform exactly the following finite audit:

d. Step 1. Identify the actual residual symmetry group. Determine the low-energy residual symmetry group G_{res} acting on the isolated doubled shell block after all shell-enriched corrections are included.

e. Step 2. Construct the 4×4 *representation.* Write the actual representation

$$U(g) \in U(4), \qquad g \in G_{\text{res}},$$

on

$$\mathcal{H}_D \cong \mathbb{C}^2_{\text{valley}} \otimes \mathbb{C}^2_{\text{int}}.$$

f. Step 3. Solve the symmetry filter. Solve the linear equations

$$U(g) K U(g)^{-1} = K \qquad \forall g \in G_{\text{res}} \tag{226}$$

for

$$K \in \text{Herm}(4),$$

to obtain the allowed constant algebra A_{res}.

g. Step 4. Solve the mass filter. Inside A_{res}, solve

$$\{K, \tau_z \otimes \sigma_i\} = 0, \qquad i = 1, 2, 3. \tag{227}$$

If the only solution is $K = 0$, then mass protection holds.

h. Equivalent shortcut. If one can prove directly that exact residual $U(1)_V$ or exact residual $\mathbb{Z}_2^V$ survives, then Step 4 is automatic.

G. Final theorem-level / microscopic separation

Thus the exact logical separation is:

$$\boxed{\text{theorem-level closure: } U(1)_V \text{ or } \mathbb{Z}_2^V \Longrightarrow A_{\text{res}} \cap \mathcal{M}_D = \{0\}.}$$

But:

$$\boxed{\text{actual microscopic open problem: does the full corrected TECT/Class II realization really preserve such a residual symmetry?}}$$

So Task V is no longer conceptual. It is a finite residual-symmetry audit on the 4×4 doubled shell block.

U. Summary

$$\boxed{\text{Task V reduces to one finite symmetry audit: } A_{\mathrm{res}} \cap \mathcal{M}_D = \{0\}.}$$

For

$$H_D(p) = \sum_{i=1}^{3} v_i p_i \, (\tau_z \otimes \sigma_i),$$

the allowed little-group-scalar constant Hermitian operators are

$$K = a_0 \, \mathbf{1}_4 + a_3 \, \tau_z \otimes \mathbf{1}_2 + m_1 \, \tau_x \otimes \mathbf{1}_2 + m_2 \, \tau_y \otimes \mathbf{1}_2.$$

The genuine Dirac-mass space is exactly

$$\boxed{\mathcal{M}_D = \mathrm{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\}.}$$

Residual-symmetry audit table:

Residual structure	Effect on $\mathcal{M}_D$	Mass protection?
little-group scalarity only	allows τ_x, τ_y	No
opposite-valley matching only	does not eliminate τ_x, τ_y	No
exact $U(1)_V$	(τ_x, τ_y) rotate nontrivially	Yes
exact $\mathbb{Z}_2^V$	$\tau_x, \tau_y \mapsto -\tau_x, -\tau_y$	Yes
Class II gradient-linear form	compatible only	Not sufficient

With

$$Q_V = \tau_z \otimes \mathbf{1}_2, \qquad U_\theta = e^{i\theta Q_V/2},$$

one has

$$U_\theta(\tau_x \otimes \mathbf{1}_2)U_\theta^{-1} = \cos\theta \, (\tau_x \otimes \mathbf{1}_2) + \sin\theta \, (\tau_y \otimes \mathbf{1}_2),$$

$$U_\theta(\tau_y \otimes \mathbf{1}_2)U_\theta^{-1} = -\sin\theta \, (\tau_x \otimes \mathbf{1}_2) + \cos\theta \, (\tau_y \otimes \mathbf{1}_2).$$

Therefore

$$\boxed{A_{U(1)_V} \cap \mathcal{M}_D = \{0\}.}$$

Even the weaker remnant

$$S_V = \tau_z \otimes \mathbf{1}_2$$

already suffices:

$$S_V(\tau_x \otimes \mathbf{1}_2)S_V^{-1} = -(\tau_x \otimes \mathbf{1}_2), \qquad S_V(\tau_y \otimes \mathbf{1}_2)S_V^{-1} = -(\tau_y \otimes \mathbf{1}_2).$$

Hence the actual microscopic procedure is:

$$\boxed{\text{(1) identify } G_{\mathrm{res}}, \quad \text{(2) construct } U(g), \quad \text{(3) solve } U(g)KU(g)^{-1} = K, \quad \text{(4) test whether } \tau_x, \tau_y \text{ survive.}}$$

If they do not survive, then

$$\boxed{\text{mass protection holds.}}$$

Honest final status:

$$\boxed{\text{the mass-classification theorem is closed,}}$$

but

$$\boxed{\text{the actual residual symmetry algebra of the microscopic TECT/Class II realization still has to be verified.}}$$

ORTHODOX ROUTE SELECTION: COERCIVITY FIRST, FINITE TRUNCATION SECOND

1. Why the coercivity route is the correct orthodox route

The last abstract slot in the master remote-gap inequality is the extra-sector benchmark

$$\Delta^{(0)}_{\mathrm{extra},\rho}.$$

At this point there are two logically valid routes:

a. Route A: direct finite spectral computation Construct a finite shell/Bloch truncation

$$H^{(N)}_{\mathrm{extra}}(p)$$

and compute

$$\Delta^{(N)}_{\mathrm{extra},\rho} := \inf_{|p|\le\rho} \mathrm{dist}\Big(0, \mathrm{spec}(H^{(N)}_{\mathrm{extra}}(p))\Big).$$

Then control the truncation error

$$\varepsilon^{(N)}_{\mathrm{extra},\rho} < \Delta^{(N)}_{\mathrm{extra},\rho}$$

to upgrade to the exact benchmark.

b. Route B: analytic coercivity replacement Write the extra-sector operator as

$$H^{(0)}_{\mathrm{extra}}(p) = P_{\mathrm{pr}}(p, D) + B_{\mathrm{lo}}(p),$$

and prove:

1. uniform principal-symbol positivity:

$$\langle u, s_{\mathrm{pr}}(p,\xi)u\rangle \ge \kappa_{\mathrm{extra}}\langle\xi\rangle^m\|u\|^2;$$

2. sectorwise Poincaré/Fourier exclusion:

$$\|\psi\|^2_{H^{m/2}} \ge \Lambda_{\mathrm{extra},\rho}\|\psi\|^2;$$

3. lower-order control:

$$|\mathfrak{b}_{\mathrm{lo},p}[\psi]| \le \alpha_{\mathrm{extra}}\kappa_{\mathrm{extra}}\|\psi\|^2_{H^{m/2}} + \beta_{\mathrm{extra},\rho}\|\psi\|^2.$$

Then the principal-symbol coercivity theorem gives

$$\mathfrak{q}_{\mathrm{extra},p}[\psi] \ge \Big((1-\alpha_{\mathrm{extra}})\kappa_{\mathrm{extra}}\Lambda_{\mathrm{extra},\rho} - \beta_{\mathrm{extra},\rho}\Big)\|\psi\|^2.$$

Hence, if

$$(1-\alpha_{\mathrm{extra}})\kappa_{\mathrm{extra}}\Lambda_{\mathrm{extra},\rho} > \beta_{\mathrm{extra},\rho},$$

the extra sector is uniformly coercive with

$$c_{\mathrm{extra},\rho} = (1-\alpha_{\mathrm{extra}})\kappa_{\mathrm{extra}}\Lambda_{\mathrm{extra},\rho} - \beta_{\mathrm{extra},\rho} > 0.$$

After bounded extra-sector correction

$$\varepsilon_{\mathrm{extra},\rho} = \sup_{|p|\le\rho}\|\delta H_{\mathrm{extra}}(p)\| < c_{\mathrm{extra},\rho},$$

one gets the exact replacement

$$\Delta_{\mathrm{extra},\rho} \ge c_{\mathrm{extra},\rho} - \varepsilon_{\mathrm{extra},\rho} > 0.$$

2. Why Route B is more orthodox

Route B is better in the strict theorem-building sense because:

1. it produces an exact infinite-dimensional bound rather than a truncation-dependent numerical witness;
2. it isolates the true conceptual ingredients:

$$\kappa_{\text{extra}}, \quad \Lambda_{\text{extra},\rho}, \quad \alpha_{\text{extra}}, \quad \beta_{\text{extra},\rho};$$

3. once established, it eliminates the need for a minimum-eigenvalue computation entirely.

Therefore the correct orthodox order is:

$$\boxed{\text{First attempt analytic coercivity closure.}}$$

$$\boxed{\text{Use a finite truncated minimum-eigenvalue computation only if coercivity cannot be instantiated.}}$$

3. Explicit BCC Bloch-coercive realization

For the BCC shell geometry, the document already provides an explicit Fourier-exclusion constant

$$\delta_{\text{Bloch},\rho} = 1 - 2\sqrt{2}\,\rho - \rho^2,$$

valid whenever

$$0 < \rho < \sqrt{3} - \sqrt{2}.$$

Under this condition,

$$\delta_{\text{Bloch},\rho} > 0.$$

The explicit BCC coercivity bound then becomes

$$c_{\text{extra},\rho} \geq (1 - \alpha_{\text{extra}})\, K \left(1 - 2\sqrt{2}\,\rho - \rho^2\right)^2 - \beta_{\text{extra},\rho}.$$

Hence a sufficient fully analytic positivity condition is

$$(1 - \alpha_{\text{extra}})\, K \left(1 - 2\sqrt{2}\,\rho - \rho^2\right)^2 > \beta_{\text{extra},\rho}.$$

This converts the extra-sector slot from an abstract unknown into an explicit analytic quantity.

4. Consequence for the master remote-gap inequality

Once the extra-sector slot is replaced analytically, the master benchmark becomes

$$\Delta^{\text{Bloch}}_{\text{bench},\rho} := \min\left\{4K - |r_*|,\ \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2},\ q_*^2 \left|(\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2}\right|,\ (1 - \alpha_{\text{extra}}) K \delta^2_{\text{Bloch},\rho} - \beta_{\text{extra},\rho}\right\}.$$

Then the full remote-gap condition is reduced to the single explicit inequality

$$\eta_{\text{rem},\rho} < \Delta^{\text{Bloch}}_{\text{bench},\rho}.$$

At that point, the remote spectral gap is closed without any finite-matrix minimum-eigenvalue computation.

5. Final orthodox decision rule

The correct strict decision rule is:

$$\boxed{\text{If } \kappa_{\text{extra}}, \Lambda_{\text{extra},\rho}, \alpha_{\text{extra}}, \beta_{\text{extra},\rho} \text{ can be extracted from the actual microscopic operator,}}$$

$$\boxed{\text{then the coercivity route is the proper primary route.}}$$

$$\boxed{\text{If one of these constants cannot yet be controlled,}}$$

$$\boxed{\text{then the finite truncated minimum-eigenvalue computation becomes the correct fallback verifier.}}$$

V. Summary

$$\boxed{\text{Orthodox route} = \text{analytic coercivity first.}}$$

The extra-sector slot

$$\Delta^{(0)}_{\text{extra},\rho}$$

should first be replaced by the coercive bound

$$\Delta_{\text{extra},\rho} \geq c_{\text{extra},\rho} - \varepsilon_{\text{extra},\rho},$$

where

$$c_{\text{extra},\rho} = (1-\alpha_{\text{extra}})\kappa_{\text{extra}}\Lambda_{\text{extra},\rho} - \beta_{\text{extra},\rho}.$$

If

$$(1-\alpha_{\text{extra}})\kappa_{\text{extra}}\Lambda_{\text{extra},\rho} > \beta_{\text{extra},\rho}, \qquad \varepsilon_{\text{extra},\rho} < c_{\text{extra},\rho},$$

then

$$\boxed{\Delta_{\text{extra},\rho} > 0}$$

without any finite minimum-eigenvalue computation.

For the explicit BCC Bloch-exclusion route,

$$\delta_{\text{Bloch},\rho} = 1 - 2\sqrt{2}\,\rho - \rho^2, \qquad 0 < \rho < \sqrt{3} - \sqrt{2},$$

and

$$\boxed{c_{\text{extra},\rho} \geq (1-\alpha_{\text{extra}})\,K\,(1 - 2\sqrt{2}\,\rho - \rho^2)^2 - \beta_{\text{extra},\rho}.}$$

Thus the master fully analytic benchmark becomes

$$\Delta^{\text{Bloch}}_{\text{bench},\rho} := \min\left\{4K - |r_*|,\ \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2},\ q_*^2\left|(\lambda+\mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2}\right|,\ (1-\alpha_{\text{extra}})K\delta^2_{\text{Bloch},\rho} - \beta_{\text{extra},\rho}\right\}.$$

If

$$\eta_{\text{rem},\rho} < \Delta^{\text{Bloch}}_{\text{bench},\rho},$$

then

$$\boxed{\Delta_{\text{full},\rho} > 0.}$$

Therefore:

$$\boxed{\text{Use finite truncated minimum-eigenvalue computation only if the coercivity constants cannot yet be instantiated.}}$$

EXTRA-SECTOR PRINCIPAL SYMBOL IDENTIFICATION THEOREM

A. Goal

We now carry out the orthodox analytic route for the remaining extra-sector slot. The purpose is to identify, in the actual BCC first-shell setting, the abstract constants

$$\kappa_{\rm extra}, \qquad \Lambda_{{\rm extra},\rho}, \qquad \alpha_{\rm extra}, \qquad \beta_{{\rm extra},\rho}$$

appearing in the principal-symbol coercivity theorem.

The key point is that the first two constants can already be identified explicitly, while the last two remain model-dependent lower-order data.

B. Abstract coercivity theorem to be instantiated

The extra-sector operator is written as

$$H^{(0)}_{\rm extra}(p) = P_{\rm pr}(p,D) + B_{\rm lo}(p), \tag{228}$$

with quadratic form

$$\mathfrak{q}_{{\rm extra},p}[\psi] = \mathfrak{q}_{{\rm pr},p}[\psi] + \mathfrak{b}_{{\rm lo},p}[\psi]. \tag{229}$$

If the principal symbol is uniformly positive with constant $\kappa_{\rm extra} > 0$, the sectorwise Fourier/Poincaré exclusion holds with constant $\Lambda_{{\rm extra},\rho} > 0$, and the lower-order part obeys

$$|\mathfrak{b}_{{\rm lo},p}[\psi]| \le \alpha_{\rm extra}\kappa_{\rm extra}\|\psi\|^2_{H^{m/2}} + \beta_{{\rm extra},\rho}\|\psi\|^2, \qquad 0 \le \alpha_{\rm extra} < 1, \tag{230}$$

then

$$\mathfrak{q}_{{\rm extra},p}[\psi] \ge \Big((1-\alpha_{\rm extra})\kappa_{\rm extra}\Lambda_{{\rm extra},\rho} - \beta_{{\rm extra},\rho}\Big)\|\psi\|^2. \tag{231}$$

Hence

$$c_{{\rm extra},\rho} = (1-\alpha_{\rm extra})\kappa_{\rm extra}\Lambda_{{\rm extra},\rho} - \beta_{{\rm extra},\rho}. \tag{232}$$

C. Principal symbol on the BCC extra sector

In the present BCC shell setting, the extra sector is defined by excluding the selected soft shell carrier. Therefore the principal operator is governed by the same Brazovskii-type quadratic principal kernel that controls shell-radius departure:

$$P_{\rm pr}(p,D) \sim K\big(|Q+p|^2 - q_*^2\big)^2 \qquad \text{on each excluded harmonic } Q. \tag{233}$$

Thus the principal symbol is positive with leading coefficient

$$\boxed{\kappa_{\rm extra} = K.} \tag{234}$$

This is exactly the identification stated in the document.

D. BCC Bloch-exclusion constant

Let the selected shell point be normalized so that

$$q_*^2 = 1. \tag{235}$$

Let $k = G_* + p$, with $|p| \le \rho$, and let the extra sector contain only excluded harmonics. Then the BCC first-shell arithmetic gives the explicit lower bound

$$\delta_{\mathrm{Bloch},\rho} = 1 - 2\sqrt{2}\,\rho - \rho^2, \tag{236}$$

valid whenever

$$0 < \rho < \sqrt{3} - \sqrt{2}. \tag{237}$$

Hence

$$\delta_{\mathrm{Bloch},\rho} > 0. \tag{238}$$

Because the extra sector excludes the selected soft shell, its Fourier/Poincaré exclusion constant is exactly the square of the Bloch-exclusion distance:

$$\boxed{\Lambda_{\mathrm{extra},\rho} = \delta^2_{\mathrm{Bloch},\rho}.} \tag{239}$$

This is again the explicit identification stated in the document.

E. Resulting explicit coercivity constant

Substituting

$$\kappa_{\mathrm{extra}} = K, \qquad \Lambda_{\mathrm{extra},\rho} = \delta^2_{\mathrm{Bloch},\rho}$$

into the abstract coercivity formula gives

$$\boxed{c_{\mathrm{extra},\rho} = (1 - \alpha_{\mathrm{extra}})\,K\,\delta^2_{\mathrm{Bloch},\rho} - \beta_{\mathrm{extra},\rho}.} \tag{240}$$

Hence the sufficient positivity condition is

$$\boxed{(1 - \alpha_{\mathrm{extra}})\,K\,\delta^2_{\mathrm{Bloch},\rho} > \beta_{\mathrm{extra},\rho}.} \tag{241}$$

Using the explicit BCC formula,

$$\boxed{(1 - \alpha_{\mathrm{extra}})\,K\,\left(1 - 2\sqrt{2}\,\rho - \rho^2\right)^2 > \beta_{\mathrm{extra},\rho}.} \tag{242}$$

F. Master benchmark replacement

Therefore the extra-sector slot in the master remote-gap inequality is no longer abstract:

$$\Delta^{(0)}_{\mathrm{extra},\rho} \rightsquigarrow c_{\mathrm{extra},\rho} - \varepsilon_{\mathrm{extra},\rho} = (1 - \alpha_{\mathrm{extra}})\,K\,\delta^2_{\mathrm{Bloch},\rho} - \beta_{\mathrm{extra},\rho} - \varepsilon_{\mathrm{extra},\rho}. \tag{243}$$

So the fully analytic benchmark becomes

$$\Delta^{\mathrm{Bloch}}_{\mathrm{bench},\rho} := \min\left\{4K - |r_*|,\ \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2},\ q_*^2\left|(\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2}\right|,\ (1 - \alpha_{\mathrm{extra}})K\delta^2_{\mathrm{Bloch},\rho} - \beta_{\mathrm{extra},\rho} - \varepsilon_{\mathrm{extra},\rho}\right\}. \tag{244}$$

If

$$\eta_{\mathrm{rem},\rho} < \Delta^{\mathrm{Bloch}}_{\mathrm{bench},\rho}, \tag{245}$$

then

$$\Delta_{\mathrm{full},\rho} > 0. \tag{246}$$

G. What remains actually open

At this point, the principal-symbol identification itself is finished. The only remaining microscopic work is to extract the lower-order relative-bound constants:

$$\alpha_{\mathrm{extra}}, \qquad \beta_{\mathrm{extra},\rho}, \qquad \varepsilon_{\mathrm{extra},\rho}. \tag{247}$$

Thus the honest status is:

$$\boxed{\kappa_{\mathrm{extra}} \text{ and } \Lambda_{\mathrm{extra},\rho} \text{ are already explicitly identified in the BCC route;}}$$

$$\boxed{\text{the remaining analytic gap problem is entirely concentrated in } \alpha_{\mathrm{extra}},\ \beta_{\mathrm{extra},\rho},\ \varepsilon_{\mathrm{extra},\rho}.}$$

W. Summary

$$\boxed{H^{(0)}_{\mathrm{extra}}(p) = P_{\mathrm{pr}}(p, D) + B_{\mathrm{lo}}(p)}$$

Abstract principal-symbol coercivity gives

$$\boxed{c_{\mathrm{extra},\rho} = (1-\alpha_{\mathrm{extra}})\kappa_{\mathrm{extra}}\Lambda_{\mathrm{extra},\rho} - \beta_{\mathrm{extra},\rho}.}$$

In the explicit BCC first-shell route, the principal part is identified with the Brazovskii shell-separation kernel, so

$$\boxed{\kappa_{\mathrm{extra}} = K.}$$

The Bloch/Fourier exclusion constant is

$$\delta_{\mathrm{Bloch},\rho} = 1 - 2\sqrt{2}\,\rho - \rho^2, \qquad 0 < \rho < \sqrt{3} - \sqrt{2},$$

hence

$$\boxed{\Lambda_{\mathrm{extra},\rho} = \delta^2_{\mathrm{Bloch},\rho}.}$$

Therefore

$$\boxed{c_{\mathrm{extra},\rho} = (1-\alpha_{\mathrm{extra}})\,K\,\delta^2_{\mathrm{Bloch},\rho} - \beta_{\mathrm{extra},\rho}.}$$

A sufficient positivity condition is

$$\boxed{(1-\alpha_{\mathrm{extra}})\,K\left(1 - 2\sqrt{2}\,\rho - \rho^2\right)^2 > \beta_{\mathrm{extra},\rho}.}$$

So the extra-sector slot in the master remote-gap benchmark becomes

$$\boxed{\Delta^{(0)}_{\mathrm{extra},\rho} \rightsquigarrow (1-\alpha_{\mathrm{extra}})\,K\,\delta^2_{\mathrm{Bloch},\rho} - \beta_{\mathrm{extra},\rho} - \varepsilon_{\mathrm{extra},\rho}.}$$

Hence the principal-symbol identification problem is already closed. What remains open is only the lower-order control:

$$\boxed{\alpha_{\mathrm{extra}}, \qquad \beta_{\mathrm{extra},\rho}, \qquad \varepsilon_{\mathrm{extra},\rho}.}$$

X. Extra-sector lower-order remainder bound theorem

a. Goal. The principal-symbol part of the extra-sector coercivity problem has already been identified:

$$\kappa_{\rm extra} = K, \qquad \Lambda_{{\rm extra},\rho} = \delta^2_{{\rm Bloch},\rho}, \qquad \delta_{{\rm Bloch},\rho} = 1 - 2\sqrt{2}\,\rho - \rho^2.$$

What remains is to control the lower-order remainder

$$B_{\rm lo}(p)$$

by a relative form bound with respect to the principal quadratic form.

b. Starting split. Write

$$H^{(0)}_{\rm extra}(p) = P_{\rm pr}(p, D) + B_{\rm lo}(p), \qquad |p| \le \rho, \tag{248}$$

with quadratic form

$$\mathfrak{q}_{{\rm extra},p}[\psi] = \mathfrak{q}_{{\rm pr},p}[\psi] + \mathfrak{b}_{{\rm lo},p}[\psi]. \tag{249}$$

Assume the principal part satisfies the already-established BCC coercive lower bound

$$\mathfrak{q}_{{\rm pr},p}[\psi] \ge K\,\delta^2_{{\rm Bloch},\rho}\,\|\psi\|^2 \qquad \text{on the extra sector.} \tag{250}$$

c. Abstract relative-form hypothesis. Assume that the lower-order quadratic form obeys

$$|\mathfrak{b}_{{\rm lo},p}[\psi]| \le \alpha_{\rm extra}\,K\,\|\psi\|^2_{H^{m/2}} + \beta_{{\rm extra},\rho}\,\|\psi\|^2, \qquad 0 \le \alpha_{\rm extra} < 1, \tag{251}$$

uniformly for $|p| \le \rho$.

Then, combining the principal coercivity theorem with the BCC identification

$$\kappa_{\rm extra} = K, \qquad \Lambda_{{\rm extra},\rho} = \delta^2_{{\rm Bloch},\rho},$$

one obtains

$$\mathfrak{q}_{{\rm extra},p}[\psi] \ge \Big((1-\alpha_{\rm extra})\,K\,\delta^2_{{\rm Bloch},\rho} - \beta_{{\rm extra},\rho}\Big)\|\psi\|^2. \tag{252}$$

Hence

$$\boxed{c_{{\rm extra},\rho} = (1-\alpha_{\rm extra})\,K\,\delta^2_{{\rm Bloch},\rho} - \beta_{{\rm extra},\rho}.} \tag{253}$$

d. Sufficient positivity condition. Therefore a sufficient analytic positivity condition is

$$\boxed{(1-\alpha_{\rm extra})\,K\,\big(1 - 2\sqrt{2}\,\rho - \rho^2\big)^2 > \beta_{{\rm extra},\rho}.} \tag{254}$$

If (254) holds, then

$$c_{{\rm extra},\rho} > 0. \tag{255}$$

e. Bounded correction stability. Now let

$$H_{\rm extra}(p) = H^{(0)}_{\rm extra}(p) + \delta H_{\rm extra}(p), \qquad \varepsilon_{{\rm extra},\rho} := \sup_{|p|\le\rho} \|\delta H_{\rm extra}(p)\|. \tag{256}$$

If

$$\varepsilon_{{\rm extra},\rho} < c_{{\rm extra},\rho}, \tag{257}$$

then the extra-sector gap is stable and

$$\Delta_{{\rm extra},\rho} \ge c_{{\rm extra},\rho} - \varepsilon_{{\rm extra},\rho} > 0. \tag{258}$$

f. Decomposition of the lower-order remainder. For practical microscopic auditing, it is useful to split

$$B_{\mathrm{lo}}(p) = \sum_{\ell=1}^{L} B_{\mathrm{lo}}^{(\ell)}(p), \tag{259}$$

with corresponding quadratic forms

$$\mathfrak{b}_{\mathrm{lo},p}^{(\ell)}[\psi].$$

Suppose each piece satisfies

$$|\mathfrak{b}_{\mathrm{lo},p}^{(\ell)}[\psi]| \le \alpha_\ell \, K \, \|\psi\|_{H^{m/2}}^2 + \beta_{\ell,\rho} \, \|\psi\|^2. \tag{260}$$

Then summing gives

$$\alpha_{\mathrm{extra}} = \sum_{\ell=1}^{L} \alpha_\ell, \qquad \beta_{\mathrm{extra},\rho} = \sum_{\ell=1}^{L} \beta_{\ell,\rho}. \tag{261}$$

Hence a sufficient condition is simply

$$\sum_{\ell=1}^{L} \alpha_\ell < 1, \qquad \sum_{\ell=1}^{L} \beta_{\ell,\rho} < K\big(1 - \sum_{\ell=1}^{L} \alpha_\ell\big)\delta_{\mathrm{Bloch},\rho}^2. \tag{262}$$

g. Interpretation of the pieces. In the actual microscopic TECT operator, the lower-order pieces typically arise from:

1. bounded zeroth-order endomorphism terms;
2. first-order transport terms restricted to the extra sector;
3. finite shell-curvature corrections in the Bloch reduction;
4. bounded residual couplings induced by integrating out heavy sectors.

The first and fourth usually contribute only to $\beta_{\ell,\rho}$, while the second and third may contribute to both α_ℓ and $\beta_{\ell,\rho}$.

h. Final replacement in the master benchmark. Thus the extra-sector slot in the master remote-gap inequality becomes

$$\boxed{\Delta_{\mathrm{extra},\rho}^{(0)} \rightsquigarrow (1 - \alpha_{\mathrm{extra}}) \, K \, \delta_{\mathrm{Bloch},\rho}^2 - \beta_{\mathrm{extra},\rho} - \varepsilon_{\mathrm{extra},\rho}.} \tag{263}$$

Therefore the fully analytic BCC benchmark becomes

$$\Delta_{\mathrm{bench},\rho}^{\mathrm{Bloch}} := \min\left\{ 4K - |r_*|, \ \sqrt{M_\sigma^2 + c_\sigma^2 (q_* - \rho)^2}, \ q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \ (1 - \alpha_{\mathrm{extra}}) \, K \, \delta_{\mathrm{Bloch},\rho}^2 - \beta_{\mathrm{extra},\rho} - \varepsilon_{\mathrm{extra},\rho} \right\}. \tag{264}$$

If

$$\eta_{\mathrm{rem},\rho} < \Delta_{\mathrm{bench},\rho}^{\mathrm{Bloch}}, \tag{265}$$

then

$$\Delta_{\mathrm{full},\rho} > 0. \tag{266}$$

i. Honest status. At this stage, the remaining analytic work is concentrated entirely in

$$\alpha_{\mathrm{extra}}, \qquad \beta_{\mathrm{extra},\rho}, \qquad \varepsilon_{\mathrm{extra},\rho}.$$

The principal-symbol part is already closed.

Y. Summary

$$\boxed{H^{(0)}_{\mathrm{extra}}(p) = P_{\mathrm{pr}}(p, D) + B_{\mathrm{lo}}(p), \qquad \mathfrak{q}_{\mathrm{extra},p} = \mathfrak{q}_{\mathrm{pr},p} + \mathfrak{b}_{\mathrm{lo},p}.}$$

The principal-symbol part is already fixed:

$$\boxed{\kappa_{\mathrm{extra}} = K, \qquad \Lambda_{\mathrm{extra},\rho} = \delta^2_{\mathrm{Bloch},\rho}, \qquad \delta_{\mathrm{Bloch},\rho} = 1 - 2\sqrt{2}\,\rho - \rho^2.}$$

Assume the lower-order remainder satisfies

$$\boxed{|\mathfrak{b}_{\mathrm{lo},p}[\psi]| \le \alpha_{\mathrm{extra}}\, K\, \|\psi\|^2_{H^{m/2}} + \beta_{\mathrm{extra},\rho}\, \|\psi\|^2, \qquad 0 \le \alpha_{\mathrm{extra}} < 1.}$$

Then

$$\boxed{c_{\mathrm{extra},\rho} = (1 - \alpha_{\mathrm{extra}})\, K\, \delta^2_{\mathrm{Bloch},\rho} - \beta_{\mathrm{extra},\rho}.}$$

A sufficient positivity condition is

$$\boxed{(1 - \alpha_{\mathrm{extra}})\, K\, (1 - 2\sqrt{2}\,\rho - \rho^2)^2 > \beta_{\mathrm{extra},\rho}.}$$

If moreover

$$\varepsilon_{\mathrm{extra},\rho} = \sup_{|p|\le\rho} \|\delta H_{\mathrm{extra}}(p)\| < c_{\mathrm{extra},\rho},$$

then

$$\boxed{\Delta_{\mathrm{extra},\rho} \ge c_{\mathrm{extra},\rho} - \varepsilon_{\mathrm{extra},\rho} > 0.}$$

Equivalently, the extra-sector slot becomes

$$\boxed{\Delta^{(0)}_{\mathrm{extra},\rho} \rightsquigarrow (1 - \alpha_{\mathrm{extra}})\, K\, \delta^2_{\mathrm{Bloch},\rho} - \beta_{\mathrm{extra},\rho} - \varepsilon_{\mathrm{extra},\rho}.}$$

Thus the principal-symbol part is closed, and the only remaining analytic constants are

$$\boxed{\alpha_{\mathrm{extra}}, \qquad \beta_{\mathrm{extra},\rho}, \qquad \varepsilon_{\mathrm{extra},\rho}.}$$

Z. Lower-order remainder decomposition table

a. Goal. We now convert the abstract lower-order remainder

$$B_{\mathrm{lo}}(p)$$

into an operator-level decomposition adapted to the actual TECT/BCC microscopic architecture, so that the abstract relative-form bound

$$|\mathfrak{b}_{\mathrm{lo},p}[\psi]| \le \alpha_{\mathrm{extra}}\, K\, \|\psi\|^2_{H^{m/2}} + \beta_{\mathrm{extra},\rho}\, \|\psi\|^2$$

is reduced to a finite table of piecewise bounds.

b. Structural input from the microscopic operator. The document already fixes the following operator-level ingredients:

1. the exact BCC linearized Hessian decomposes as

$$D_{ij}(k) = D_{ij}^{\mathrm{iso}}(k) + \Delta D_{ij}(k),$$

where $\Delta D_{ij}(k)$ is the genuine cubic-anisotropic remainder;

2. the projector-Hessian static sector takes the form

$$H_B^{\mathrm{stat}} = -\rho\partial_\xi^2 - \rho\partial_y^2 + W_B(\xi) - \rho\theta'(\xi)^2 \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} + \cdots ;$$

3. restoring slow worldvolume dependence produces the first-order transport operator

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu};$$

4. integrating out the heavy volumetric mode yields a local $1/M_\sigma^2$ expansion with explicit remainder bounds.

Therefore the natural lower-order decomposition is

$$\boxed{B_{\mathrm{lo}}(p) = B_{\mathrm{cub}}(p) + B_{\mathrm{pot}}(p) + B_{\mathrm{tr}}(p) + B_{\mathrm{H}}(p).} \tag{267}$$

A. Cubic anisotropy remainder

Define

$$B_{\mathrm{cub}}(p) := \Delta D(p, D), \tag{268}$$

where ΔD is the exact cubic anisotropy remainder of the BCC linearized Hessian.
Its quadratic form is

$$\mathfrak{b}_{\mathrm{cub},p}[\psi] = \langle \psi, \Delta D(p, D)\psi \rangle.$$

c. Relative-form role. If the cubic anisotropy is relatively small with respect to the isotropic principal part, then one seeks a bound

$$|\mathfrak{b}_{\mathrm{cub},p}[\psi]| \le \alpha_{\mathrm{cub}}\, K\, \|\psi\|_{H^{m/2}}^2 + \beta_{\mathrm{cub},\rho}\, \|\psi\|^2. \tag{269}$$

In an RG-improved or infrared-isotropized regime, α_{cub} should be small. Thus B_{cub} is the natural source of the relative principal-form coefficient α_{extra}.

B. Bounded potential / endomorphism sector

Collect the bounded static endomorphism terms into

$$B_{\mathrm{pot}}(p) := W_B(\xi) - \rho\theta'(\xi)^2 \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} + B_{\mathrm{bd}}(p), \tag{270}$$

where $B_{\mathrm{bd}}(p)$ denotes any additional bounded zeroth-order shell/Bloch endomorphism terms on the extra sector.
Its form satisfies the purely bounded estimate

$$|\mathfrak{b}_{\mathrm{pot},p}[\psi]| \le \|B_{\mathrm{pot}}(p)\|\, \|\psi\|^2. \tag{271}$$

Hence one may take

$$\alpha_{\mathrm{pot}} = 0, \qquad \beta_{\mathrm{pot},\rho} = \sup_{|p|\le\rho} \|B_{\mathrm{pot}}(p)\|. \tag{272}$$

So B_{pot} contributes only to the β-budget.

C. First-order transport remainder

Restrict the worldvolume transport operator to the extra sector:

$$B_{\mathrm{tr}}(p) := -i\, P_{\mathrm{extra}} V^{\mu}(\xi) P_{\mathrm{extra}} \partial_{y^{\mu}}. \tag{273}$$

Its form is first-order in derivatives and therefore lower-order relative to the principal sector.

Using the standard ε-Sobolev estimate, for every $\varepsilon > 0$,

$$|\mathfrak{b}_{\mathrm{tr},p}[\psi]| \le \varepsilon\, K\, \|\psi\|^2_{H^{m/2}} + C_{\mathrm{tr}}(\varepsilon, \rho)\, \|\psi\|^2. \tag{274}$$

Thus one may set

$$\alpha_{\mathrm{tr}} = \varepsilon, \qquad \beta_{\mathrm{tr},\rho} = C_{\mathrm{tr}}(\varepsilon, \rho). \tag{275}$$

Hence the transport remainder contributes to both α and β, but its α-contribution can be made arbitrarily small at the expense of enlarging $\beta_{\mathrm{tr},\rho}$.

D. Heavy-induced local correction

Let $B_{\mathrm{H}}(p)$ denote the operator induced by integrating out the heavy volumetric mode after subtracting the leading principal longitudinal stiffening already accounted for elsewhere. By the $1/M_\sigma^2$ expansion with explicit remainder control,

$$B_{\mathrm{H}}(p) = \frac{1}{M_\sigma^2}\mathcal{O}_2(p, D) + \frac{1}{M_\sigma^4}\mathcal{O}_4(p, D) + R_{\mathrm{H}}^{(N)}(p), \tag{276}$$

with operator norms uniformly controlled on the allowed compact low-energy region.

Therefore one can bound

$$|\mathfrak{b}_{\mathrm{H},p}[\psi]| \le \alpha_{\mathrm{H}}\, K\, \|\psi\|^2_{H^{m/2}} + \beta_{\mathrm{H},\rho}\, \|\psi\|^2, \tag{277}$$

with schematic size

$$\alpha_{\mathrm{H}} = O(M_\sigma^{-2}), \qquad \beta_{\mathrm{H},\rho} = O(M_\sigma^{-4}) \quad \text{or smaller}, \tag{278}$$

provided the low-energy domain stays inside the uniform convergence region of the expansion.

Thus B_{H} is naturally a suppressed lower-order perturbation.

E. Piecewise table

The resulting lower-order remainder table is:

piece $B_{\mathrm{lo}}^{(\ell)}$	microscopic origin	relative-form role	natural contribution
B_{cub}	$\Delta D_{ij}(k)$	anisotropic differential remainder	$(\alpha_{\mathrm{cub}}, \beta_{\mathrm{cub},\rho})$
B_{pot}	$W_B(\xi) - \rho\theta'(\xi)^2 \mathrm{diag}(2,1) + B_{\mathrm{bd}}$	bounded zeroth-order endomorphism	$(0, \beta_{\mathrm{pot},\rho})$
B_{tr}	$-iV^{\mu}(\xi)\partial_{y^{\mu}}$	first-order transport	$(\alpha_{\mathrm{tr}}, \beta_{\mathrm{tr},\rho})$
B_{H}	$1/M_\sigma^2$ local heavy-induced terms	suppressed derivative/zeroth-order correction	$(\alpha_{\mathrm{H}}, \beta_{\mathrm{H},\rho})$

F. Summed bound

Summing the piecewise estimates gives

$$\alpha_{\mathrm{extra}} = \alpha_{\mathrm{cub}} + \alpha_{\mathrm{tr}} + \alpha_{\mathrm{H}}, \tag{279}$$

$$\beta_{\mathrm{extra},\rho} = \beta_{\mathrm{cub},\rho} + \beta_{\mathrm{pot},\rho} + \beta_{\mathrm{tr},\rho} + \beta_{\mathrm{H},\rho}. \tag{280}$$

The coercive lower bound becomes

$$c_{\mathrm{extra},\rho} = \Big(1 - \alpha_{\mathrm{cub}} - \alpha_{\mathrm{tr}} - \alpha_{\mathrm{H}}\Big) K\, \delta^2_{\mathrm{Bloch},\rho} - \Big(\beta_{\mathrm{cub},\rho} + \beta_{\mathrm{pot},\rho} + \beta_{\mathrm{tr},\rho} + \beta_{\mathrm{H},\rho}\Big). \tag{281}$$

Hence a sufficient positivity condition is

$$\boxed{\alpha_{\mathrm{cub}} + \alpha_{\mathrm{tr}} + \alpha_{\mathrm{H}} < 1} \tag{282}$$

and

$$\boxed{\beta_{\mathrm{cub},\rho} + \beta_{\mathrm{pot},\rho} + \beta_{\mathrm{tr},\rho} + \beta_{\mathrm{H},\rho} < \Big(1 - \alpha_{\mathrm{cub}} - \alpha_{\mathrm{tr}} - \alpha_{\mathrm{H}}\Big) K\, \delta^2_{\mathrm{Bloch},\rho}.} \tag{283}$$

G. Final benchmark replacement

Thus the extra-sector slot is replaced by

$$\boxed{\Delta^{(0)}_{\mathrm{extra},\rho} \rightsquigarrow c_{\mathrm{extra},\rho} - \varepsilon_{\mathrm{extra},\rho},} \tag{284}$$

with $c_{\mathrm{extra},\rho}$ given explicitly by (281).

Therefore the fully analytic BCC benchmark becomes

$$\Delta^{\mathrm{piece}}_{\mathrm{bench},\rho} := \min\left\{ 4K - |r_*|,\ \sqrt{M_\sigma^2 + c_\sigma^2 (q_* - \rho)^2},\ q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|,\ c_{\mathrm{extra},\rho} - \varepsilon_{\mathrm{extra},\rho} \right\}. \tag{285}$$

If

$$\eta_{\mathrm{rem},\rho} < \Delta^{\mathrm{piece}}_{\mathrm{bench},\rho}, \tag{286}$$

then

$$\Delta_{\mathrm{full},\rho} > 0. \tag{287}$$

d. Honest status. At this stage, the analytic closure problem is no longer abstract. It has been reduced to estimating four concrete microscopic remainder pieces:

$$B_{\mathrm{cub}}, \quad B_{\mathrm{pot}}, \quad B_{\mathrm{tr}}, \quad B_{\mathrm{H}}.$$

The only remaining work is to extract the actual constants

$$\alpha_{\mathrm{cub}},\ \beta_{\mathrm{cub},\rho},\ \beta_{\mathrm{pot},\rho},\ \alpha_{\mathrm{tr}},\ \beta_{\mathrm{tr},\rho},\ \alpha_{\mathrm{H}},\ \beta_{\mathrm{H},\rho},\ \varepsilon_{\mathrm{extra},\rho}.$$

. Summary

$$\boxed{B_{\mathrm{lo}}(p) = B_{\mathrm{cub}}(p) + B_{\mathrm{pot}}(p) + B_{\mathrm{tr}}(p) + B_{\mathrm{H}}(p).}$$

with

$$B_{\mathrm{cub}} = \Delta D_{ij}(k),$$

$$B_{\mathrm{pot}} = W_B(\xi) - \rho\theta'(\xi)^2 \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} + B_{\mathrm{bd}},$$

$$B_{\mathrm{tr}} = -iV^{\mu}(\xi)\partial_{y^{\mu}},$$

$$B_{\mathrm{H}} = \text{heavy-induced local } 1/M_{\sigma}^{2} \text{ correction.}$$

Piecewise relative-form roles:

piece	natural contribution
B_{cub}	$(\alpha_{\mathrm{cub}}, \beta_{\mathrm{cub},\rho})$
B_{pot}	$(0, \beta_{\mathrm{pot},\rho})$
B_{tr}	$(\alpha_{\mathrm{tr}}, \beta_{\mathrm{tr},\rho})$
B_{H}	$(\alpha_{\mathrm{H}}, \beta_{\mathrm{H},\rho})$

Hence

$$\boxed{\alpha_{\mathrm{extra}} = \alpha_{\mathrm{cub}} + \alpha_{\mathrm{tr}} + \alpha_{\mathrm{H}},}$$

$$\boxed{\beta_{\mathrm{extra},\rho} = \beta_{\mathrm{cub},\rho} + \beta_{\mathrm{pot},\rho} + \beta_{\mathrm{tr},\rho} + \beta_{\mathrm{H},\rho}.}$$

Therefore

$$\boxed{c_{\mathrm{extra},\rho} = \Big(1 - \alpha_{\mathrm{cub}} - \alpha_{\mathrm{tr}} - \alpha_{\mathrm{H}}\Big) K\, \delta_{\mathrm{Bloch},\rho}^{2} - \Big(\beta_{\mathrm{cub},\rho} + \beta_{\mathrm{pot},\rho} + \beta_{\mathrm{tr},\rho} + \beta_{\mathrm{H},\rho}\Big).}$$

A sufficient positivity condition is

$$\boxed{\alpha_{\mathrm{cub}} + \alpha_{\mathrm{tr}} + \alpha_{\mathrm{H}} < 1,}$$

$$\boxed{\beta_{\mathrm{cub},\rho} + \beta_{\mathrm{pot},\rho} + \beta_{\mathrm{tr},\rho} + \beta_{\mathrm{H},\rho} < \Big(1 - \alpha_{\mathrm{cub}} - \alpha_{\mathrm{tr}} - \alpha_{\mathrm{H}}\Big) K\, \delta_{\mathrm{Bloch},\rho}^{2}.}$$

So the extra-sector slot becomes

$$\boxed{\Delta_{\mathrm{extra},\rho}^{(0)} \rightsquigarrow c_{\mathrm{extra},\rho} - \varepsilon_{\mathrm{extra},\rho}.}$$

This reduces the remaining analytic problem to estimating the concrete microscopic constants attached to

$$B_{\mathrm{cub}}, \quad B_{\mathrm{pot}}, \quad B_{\mathrm{tr}}, \quad B_{\mathrm{H}}.$$

. Explicit bound lemma for B_{pot} and B_{H}

a. Goal. We now close the two easiest lower-order pieces in the decomposition

$$B_{\mathrm{lo}}(p) = B_{\mathrm{cub}}(p) + B_{\mathrm{pot}}(p) + B_{\mathrm{tr}}(p) + B_{\mathrm{H}}(p),$$

namely the bounded zeroth-order endomorphism sector B_{pot} and the heavy-induced local correction B_{H}.

A. Bound for the bounded potential / endomorphism piece

The projector-Hessian theorem gives the fluctuation operator in the schematic form

$$H_B = -\rho D_{\xi}^{(B)2} - \rho D_{y}^{(B)2} + M_B(\xi),$$

where $M_B(\xi)$ is a Hermitian 2×2 endomorphism built from the background and its derivatives. :contentReference[oaicite:7]index=7

Restricting to the extra sector and allowing additional bounded shell/Bloch zeroth-order corrections, define

$$B_{\rm pot}(p) := P_{\rm extra}\, M_B(\xi)\, P_{\rm extra} + B_{\rm bd}(p), \tag{288}$$

where $B_{\rm bd}(p)$ is any further bounded symmetric zeroth-order endomorphism term on the extra sector.
Let

$$\mathfrak{b}_{{\rm pot},p}[\psi] = \langle \psi, B_{\rm pot}(p)\psi \rangle.$$

Then

$$|\mathfrak{b}_{{\rm pot},p}[\psi]| \le \|B_{\rm pot}(p)\|\, \|\psi\|^2. \tag{289}$$

Hence the natural relative-form constants are

$$\boxed{\alpha_{\rm pot} = 0, \qquad \beta_{{\rm pot},\rho} := \sup_{|p|\le\rho} \|B_{\rm pot}(p)\|.} \tag{290}$$

So $B_{\rm pot}$ contributes only to the β-budget and does not consume any principal coercivity fraction.

B. Bound for the heavy-induced local correction

The heavy-sector covariance satisfies

$$C_\sigma = (M_\sigma^2 - \partial_t^2 + c_\sigma^2 \nabla^2)^{-1}, \qquad \|C_\sigma\|_{L^2\to L^2} \le \frac{1}{M_\sigma^2}, \tag{291}$$

and admits the convergent derivative expansion

$$C_\sigma = \frac{1}{M_\sigma^2} \sum_{n=0}^{R} \left(-\frac{c_\sigma^2 \nabla^2 - \partial_t^2}{M_\sigma^2} \right)^n + R_\sigma^{(R)}, \tag{292}$$

with remainder bounded by

$$\|R_\sigma^{(R)}\| \le \frac{C_R}{M_\sigma^{2(R+1)}} (1 + |p|/\Lambda)^\kappa \qquad \text{on the low-energy region.} \tag{293}$$

This is the standard heavy-field local expansion with explicit remainder control. :contentReference[oaicite:8]index=8 :contentReference[oaicite:9]index=9

In the TECT refined model, the projector stiffness term does not generically generate a tree-level shell-enriched tensor on the locked constant-projector background, but it makes projector tangent modes heavy and after elimination induces corrections of order

$$O\!\left(\frac{1}{\chi \bar{\rho}^2 q_*^2} \right),$$

while enhancing the remote gap.

Accordingly, define the heavy-induced lower-order piece on the extra sector by

$$B_{\rm H}(p) = \frac{1}{M_\sigma^2} \mathcal{O}_2(p, D) + \frac{1}{M_\sigma^4} \mathcal{O}_4(p, D) + R_{\rm H}^{(R)}(p), \tag{294}$$

where $R_{\rm H}^{(R)}$ is uniformly controlled by (293).
Let

$$\mathfrak{b}_{{\rm H},p}[\psi] = \langle \psi, B_{\rm H}(p)\psi \rangle.$$

Using standard Sobolev interpolation / relative-form estimates for lower-order differential operators, one gets

$$|\mathfrak{b}_{{\rm H},p}[\psi]| \le \alpha_{\rm H}\, K\, \|\psi\|_{H^{m/2}}^2 + \beta_{{\rm H},\rho}\, \|\psi\|^2, \tag{295}$$

with the safe scaling

$$\boxed{\alpha_{\rm H} = O(M_\sigma^{-2}), \qquad \beta_{{\rm H},\rho} = O(M_\sigma^{-2}),} \tag{296}$$

and higher-order remainder contribution

$$\boxed{\varepsilon_{{\rm H},\rho}^{(R)} = O(M_\sigma^{-2(R+1)}).} \tag{297}$$

b. Important honesty clause. Without additional cancellations, one should *not* claim in general that

$$\beta_{\mathrm{H},\rho} = O(M_\sigma^{-4}).$$

The document directly supports $1/M_\sigma^2$-type suppression of the heavy-induced local expansion and of the induced shell corrections, while the higher-order truncation remainder is $O(M_\sigma^{-2(R+1)})$. :contentReference[oaicite:11]index=11 :contentReference[oaicite:12]index=12 :contentReference[oaicite:13]index=13

C. Piecewise contribution to the extra-sector coercivity budget

Therefore the two easiest pieces contribute as follows:

$$\boxed{\alpha_{\mathrm{pot}} = 0, \qquad \beta_{\mathrm{pot},\rho} = \sup_{|p|\le\rho} \|B_{\mathrm{pot}}(p)\|,} \tag{298}$$

$$\boxed{\alpha_{\mathrm{H}} = O(M_\sigma^{-2}), \qquad \beta_{\mathrm{H},\rho} = O(M_\sigma^{-2}), \qquad \varepsilon_{\mathrm{H},\rho}^{(R)} = O(M_\sigma^{-2(R+1)}).} \tag{299}$$

Hence the partial extra-sector coercivity constant becomes

$$c_{\mathrm{extra},\rho}^{(\mathrm{partial})} = \Big(1 - \alpha_{\mathrm{cub}} - \alpha_{\mathrm{tr}} - \alpha_{\mathrm{H}}\Big) K\,\delta_{\mathrm{Bloch},\rho}^2 - \Big(\beta_{\mathrm{cub},\rho} + \beta_{\mathrm{pot},\rho} + \beta_{\mathrm{tr},\rho} + \beta_{\mathrm{H},\rho}\Big). \tag{300}$$

Thus the analytic bottleneck is now sharply reduced: the two easiest pieces are already in canonical bounded form, while the difficult work remains in

$$B_{\mathrm{cub}}, \qquad B_{\mathrm{tr}}.$$

c. Conclusion. The operator-level lower-order remainder problem has now been partially closed in explicit form:

$$B_{\mathrm{pot}} \text{ contributes only to } \beta, \qquad B_{\mathrm{H}} \text{ contributes a heavy-suppressed } (\alpha, \beta) \text{ pair.}$$

This is the natural orthodox next step in the analytic coercivity program.

. Summary

$$\boxed{\text{First explicit lower-order remainder bounds}}$$

For the extra-sector remainder

$$B_{\mathrm{lo}}(p) = B_{\mathrm{cub}}(p) + B_{\mathrm{pot}}(p) + B_{\mathrm{tr}}(p) + B_{\mathrm{H}}(p),$$

the two easiest pieces are:

(i) bounded potential / endomorphism piece

$$B_{\mathrm{pot}}(p) = P_{\mathrm{extra}} M_B(\xi) P_{\mathrm{extra}} + B_{\mathrm{bd}}(p),$$

with

$$|\mathfrak{b}_{\mathrm{pot},p}[\psi]| \le \|B_{\mathrm{pot}}(p)\|\,\|\psi\|^2.$$

Hence

$$\boxed{\alpha_{\mathrm{pot}} = 0, \qquad \beta_{\mathrm{pot},\rho} = \sup_{|p|\le\rho} \|B_{\mathrm{pot}}(p)\|.}$$

(ii) heavy-induced local correction

$$C_\sigma = (M_\sigma^2 - \partial_t^2 + c_\sigma^2 \nabla^2)^{-1}, \qquad \|C_\sigma\|_{L^2 \to L^2} \le \frac{1}{M_\sigma^2},$$

and

$$C_\sigma = \frac{1}{M_\sigma^2} \sum_{n=0}^{R} \left(-\frac{c_\sigma^2 \nabla^2 - \partial_t^2}{M_\sigma^2} \right)^n + R_\sigma^{(R)}, \qquad \|R_\sigma^{(R)}\| = O(M_\sigma^{-2(R+1)}).$$

So

$$B_{\mathrm{H}}(p) = \frac{1}{M_\sigma^2} \mathcal{O}_2 + \frac{1}{M_\sigma^4} \mathcal{O}_4 + R_{\mathrm{H}}^{(R)}(p)$$

satisfies

$$|\mathfrak{b}_{\mathrm{H},p}[\psi]| \le \alpha_{\mathrm{H}} K \|\psi\|_{H^{m/2}}^2 + \beta_{\mathrm{H},\rho} \|\psi\|^2,$$

with safe scaling

$$\boxed{\alpha_{\mathrm{H}} = O(M_\sigma^{-2}), \qquad \beta_{\mathrm{H},\rho} = O(M_\sigma^{-2}), \qquad \varepsilon_{\mathrm{H},\rho}^{(R)} = O(M_\sigma^{-2(R+1)}).}$$

Important honesty clause:

$$\boxed{\beta_{\mathrm{H},\rho} = O(M_\sigma^{-4}) \text{ is not justified in general without extra cancellations.}}$$

Thus the partial coercivity budget becomes

$$\boxed{c_{\mathrm{extra},\rho}^{(\mathrm{partial})} = \Big(1 - \alpha_{\mathrm{cub}} - \alpha_{\mathrm{tr}} - \alpha_{\mathrm{H}}\Big) K \, \delta_{\mathrm{Bloch},\rho}^2 - \Big(\beta_{\mathrm{cub},\rho} + \beta_{\mathrm{pot},\rho} + \beta_{\mathrm{tr},\rho} + \beta_{\mathrm{H},\rho}\Big).}$$

So the remaining analytic bottleneck is now concentrated in

$$\boxed{B_{\mathrm{cub}}, \qquad B_{\mathrm{tr}}.}$$

. Fiberwise transport bound lemma

a. Goal. We now sharpen the treatment of the transport remainder

$$B_{\mathrm{tr}}.$$

At the PDE level, one may always use a conservative ε-Sobolev estimate for the first-order operator

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}.$$

However, at the present stage of the program the operator has already been reduced to a fixed low-momentum Bloch/worldvolume fiber. In that representation, the transport term is no longer an unbounded first-order differential operator in y, but a bounded zeroth-order operator depending on the fiber momentum p.

b. Structural input. The local projector-defect analysis establishes that the leading transport kernel has the form

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \qquad V^\mu(-\xi) = -V^\mu(\xi), \tag{301}$$

and, more sharply,

$$V^\mu(\xi) = \theta'(\xi)\, \mathcal{M}^\mu(\theta(\xi)), \qquad \mathcal{M}^\mu(\theta) \in \mathrm{Herm}(Q_B). \tag{302}$$

Thus $V^\mu(\xi)$ is a Hermitian bundle endomorphism on the rank-two transverse bundle Q_B, multiplied by the odd scalar profile $\theta'(\xi)$.

The shell-valley/Bloch analysis also fixes that the relevant operator is studied fiberwise near

$$k = G_* + p, \qquad |p| \le \rho,$$

through the finite shell-projected Bloch matrix. :contentReference[oaicite:3]index=3 *c. Fiber reduction of the transport operator.* On a fixed worldvolume/Bloch fiber of momentum p, one has

$$-i\partial_{y^\mu} \;\mapsto\; p_\mu.$$

Therefore the transport operator becomes

$$B_{\mathrm{tr}}(p) = p_\mu V^\mu(\xi). \tag{303}$$

This is now a zeroth-order multiplication operator on the ξ-dependent bundle fiber, not an unbounded first-order differential operator.

d. Operator norm bound. Define

$$C_V(\rho) := \sup_{|p|\le\rho} \left\| \sum_\mu \frac{p_\mu}{|p|} V^\mu(\xi) \right\|_{L^2_\xi \to L^2_\xi}, \tag{304}$$

with the convention that the unit vector is arbitrary at $p = 0$, or equivalently use the simpler sufficient bound

$$\widetilde{C}_V := \left(\sum_\mu \sup_\xi \|V^\mu(\xi)\|^2 \right)^{1/2}. \tag{305}$$

Then for every $|p| \le \rho$,

$$\|B_{\mathrm{tr}}(p)\| = \|p_\mu V^\mu(\xi)\| \le |p|\, C_V(\rho) \le \rho\, C_V(\rho), \tag{306}$$

and in particular

$$\|B_{\mathrm{tr}}(p)\| \le \rho\, \widetilde{C}_V. \tag{307}$$

e. Quadratic-form consequence. Let

$$\mathfrak{b}_{\mathrm{tr},p}[\psi] = \langle \psi, B_{\mathrm{tr}}(p)\psi \rangle.$$

Then

$$|\mathfrak{b}_{\mathrm{tr},p}[\psi]| \le \|B_{\mathrm{tr}}(p)\|\, \|\psi\|^2 \le \rho\, C_V(\rho)\, \|\psi\|^2. \tag{308}$$

Hence the transport remainder contributes only to the bounded β-budget:

$$\boxed{\alpha_{\mathrm{tr}} = 0, \qquad \beta_{\mathrm{tr},\rho} = \rho\, C_V(\rho),} \tag{309}$$

or with the rougher sufficient constant,

$$\boxed{\alpha_{\mathrm{tr}} = 0, \qquad \beta_{\mathrm{tr},\rho} \le \rho\, \widetilde{C}_V.} \tag{310}$$

f. Oddness and selection rule. The reflection-oddness

$$V^\mu(-\xi) = -V^\mu(\xi)$$

remains essential for the graded low-mode selection rule: same-parity matrix elements vanish while mixed even/odd overlaps survive generically. However, for the present extra-sector coercivity estimate, oddness is not used to gain smallness. The actual smallness comes from the fiber momentum factor p_μ, hence from the neighborhood size ρ.

g. Important correction to the previous conservative bound. At the PDE level, before fixing the Bloch/worldvolume fiber, it is always legitimate to use the standard estimate

$$|\mathfrak{b}_{\mathrm{tr},p}[\psi]| \le \varepsilon K \|\psi\|^2_{H^{m/2}} + C_{\mathrm{tr}}(\varepsilon,\rho)\|\psi\|^2.$$

But after fiber reduction, this is no longer optimal. The stronger and more honest bound is (309): the transport piece is a bounded zeroth-order operator with $\alpha_{\mathrm{tr}} = 0$.

h. Updated piecewise decomposition. With this sharpening, the lower-order remainder decomposition becomes

$$B_{\mathrm{lo}}(p) = B_{\mathrm{cub}}(p) + B_{\mathrm{pot}}(p) + B_{\mathrm{tr}}(p) + B_{\mathrm{H}}(p),$$

with contributions

$$\alpha_{\mathrm{extra}} = \alpha_{\mathrm{cub}} + \alpha_{\mathrm{H}}, \tag{311}$$

$$\beta_{\mathrm{extra},\rho} = \beta_{\mathrm{cub},\rho} + \beta_{\mathrm{pot},\rho} + \beta_{\mathrm{tr},\rho} + \beta_{\mathrm{H},\rho}. \tag{312}$$

So the coercivity constant is sharpened to

$$\boxed{c_{\mathrm{extra},\rho} = \big(1 - \alpha_{\mathrm{cub}} - \alpha_{\mathrm{H}}\big) K\, \delta^2_{\mathrm{Bloch},\rho} - \big(\beta_{\mathrm{cub},\rho} + \beta_{\mathrm{pot},\rho} + \beta_{\mathrm{tr},\rho} + \beta_{\mathrm{H},\rho}\big).} \tag{313}$$

i. Conclusion. Thus the transport remainder is no longer part of the genuinely difficult relative-principal-form budget. After fiber reduction it is simply a bounded $O(\rho)$ perturbation. Therefore the only genuinely difficult analytic remainder piece left is the cubic anisotropy sector B_{cub}.

. Summary

$$\boxed{L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \qquad V^\mu(-\xi) = -V^\mu(\xi), \qquad V^\mu(\xi) = \theta'(\xi)\mathcal{M}^\mu(\theta(\xi)).}$$

After Bloch/worldvolume fiber reduction near

$$k = G_* + p, \qquad |p| \le \rho,$$

one has

$$-i\partial_{y^\mu} \mapsto p_\mu.$$

Therefore

$$\boxed{B_{\mathrm{tr}}(p) = p_\mu V^\mu(\xi),}$$

which is a bounded zeroth-order operator on the ξ-fiber.

Define

$$C_V(\rho) := \sup_{|p|\le\rho} \left\| \sum_\mu \frac{p_\mu}{|p|} V^\mu(\xi) \right\|,$$

or more crudely

$$\widetilde{C}_V := \left(\sum_\mu \sup_\xi \|V^\mu(\xi)\|^2 \right)^{1/2}.$$

Then

$$\|B_{\mathrm{tr}}(p)\| \le |p|\, C_V(\rho) \le \rho\, C_V(\rho) \le \rho\, \widetilde{C}_V.$$

Hence the quadratic-form bound is

$$|\mathfrak{b}_{\mathrm{tr},p}[\psi]| \le \rho\, C_V(\rho)\, \|\psi\|^2,$$

so the transport piece contributes only to the bounded β-budget:

$$\boxed{\alpha_{\mathrm{tr}} = 0, \qquad \beta_{\mathrm{tr},\rho} = \rho\, C_V(\rho) \quad (\text{or } \le \rho\, \widetilde{C}_V).}$$

Therefore the piecewise lower-order constants sharpen to

$$\boxed{\alpha_{\text{extra}} = \alpha_{\text{cub}} + \alpha_{\text{H}},}$$

$$\boxed{\beta_{\text{extra},\rho} = \beta_{\text{cub},\rho} + \beta_{\text{pot},\rho} + \beta_{\text{tr},\rho} + \beta_{\text{H},\rho}.}$$

Thus

$$\boxed{c_{\text{extra},\rho} = (1 - \alpha_{\text{cub}} - \alpha_{\text{H}})K\,\delta^2_{\text{Bloch},\rho} - (\beta_{\text{cub},\rho} + \beta_{\text{pot},\rho} + \beta_{\text{tr},\rho} + \beta_{\text{H},\rho}).}$$

So the transport term is no longer part of the difficult relative-principal-form budget. After fiber reduction it is only an $O(\rho)$ bounded perturbation. The only genuinely hard analytic remainder left is

$$\boxed{B_{\text{cub}}.}$$

. Cubic anisotropy relative-bound theorem

a. Goal. We now treat the last genuinely difficult lower-order remainder

$$B_{\text{cub}} = \Delta D_{ij}(k).$$

The orthodox route is not to overclaim its smallness, but to separate the redundant part from the truly physical anisotropy and to state exactly what is proved, what is conditional, and what remains open.

A. Exact decomposition of quadratic derivative anisotropy

The quadratic derivative anisotropy tensor admits the exact decomposition

$$\delta K^{ij}_{IJ} = \delta^{ij}\,\delta Z_{IJ} + \delta_{IJ}\,h^{ij} + \widehat{K}^{ij}_{IJ}, \tag{314}$$

with

$$h_{ii} = 0, \qquad \widehat{K}^{ii}_{IJ} = 0, \qquad \widehat{K}^{ij}_{II} = 0.$$

Here:

- δZ_{IJ} is an internal metric deformation,
- h^{ij} is a common spatial metric anisotropy,
- $\widehat{K}^{ij}_{IJ}$ is the genuine mixed traceless anisotropy.

This decomposition is exact at quadratic order. :contentReference[oaicite:6]index=6
Therefore the cubic anisotropy remainder must be split as

$$\boxed{B_{\text{cub}} = B_{\text{red}} + B_{\text{phys}},} \tag{315}$$

where B_{red} is the redundant common metric part and B_{phys} is the physical relative anisotropy.

B. Redundant anisotropy is exactly removable

The common spatial anisotropy term has the form

$$\Gamma_h = \frac{1}{2}\int d^dx\,(Z\delta^{ij} + h^{ij})\sum_I \partial_i\Phi_I\,\partial_j\Phi_I, \qquad h^i{}_i = 0. \tag{316}$$

If $Z\delta + h$ is positive definite, then there exists an invertible linear coordinate transformation $y = Mx$ such that

$$(Z\delta + h) = M^T M,$$

and therefore

$$\Gamma_h \mapsto \frac{1}{2}\int d^d y\, \delta^{ab} \sum_I \partial_{y^a}\Phi_I\, \partial_{y^b}\Phi_I$$

up to an overall constant Jacobian factor. Hence $B_{\rm red}$ is not a physical coupling and should be removed exactly rather than estimated as a perturbation. :contentReference[oaicite:7]index=7

Thus:

$$\boxed{B_{\rm red} \text{ is exactly removable and contributes neither to } \alpha_{\rm cub} \text{ nor to } \beta_{{\rm cub},\rho}.} \tag{317}$$

C. What remains physically meaningful

After removing the redundant common metric anisotropy, the physically meaningful derivative anisotropy is

$$\Gamma_{\rm phys} = \frac{1}{2}\int d^d x \left[\delta^{ij}\delta Z_{IJ} + \widehat{K}^{ij}_{IJ}\right]\partial_i\Phi_I\, \partial_j\Phi_J. \tag{318}$$

In the block-diagonal TECT truncation this reduces to relative stiffness anisotropies

$$\delta_T, \qquad \delta_O, \qquad \delta_D,$$

subject to

$$2\delta_T + 2\delta_O + 2\delta_D = 0.$$

These are the true velocity-locking / derivative-anisotropy variables. :contentReference[oaicite:8]index=8

At the Gaussian level they are classically marginal:

$$y^{\rm tree}_{\nabla,{\rm ani}} = 0. \tag{319}$$

Therefore their infrared fate is determined entirely by the loop-level kinetic-flow matrix M_Z.

D. Honest status of the physical derivative anisotropy

The document proves the following reduction:

common spatial anisotropy is redundant,

the only physical derivative anisotropy is the relative sectoral velocity anisotropy,

its tree-level scaling is marginal,

its infrared relevance/irrelevance is controlled by the loop-level matrix M_Z.

What remains open is precisely:

1. derive the full TECT Hessian including interaction vertices,
2. project the Wetterich flow onto the sectoral kinetic terms,
3. compute the matrix M_Z,
4. verify that its nontrivial eigenvalues are negative.

Thus the sign of the physical derivative anisotropy flow is not yet fixed rigorously from the available TECT data. :contentReference[oaicite:10]index=10

Therefore one must *not* currently claim an unconditional small relative-form bound for $B_{\rm phys}$.

E. Conditional infrared smallness theorem

There is nevertheless a correct conditional theorem.

Assume that the physical cubic anisotropy couplings are governed in the infrared by running couplings $g_a(\Lambda)$ with RG eigenvalues $y_a < 0$, so that

$$g_a(\Lambda) \to 0 \qquad (\Lambda \to 0). \tag{320}$$

Then, after removing the redundant part, the physical anisotropy remainder has the representation

$$B_{\text{phys}}(\Lambda, p) = \sum_a g_a(\Lambda)\, \mathcal{O}_a(p, D), \tag{321}$$

and for every $\varepsilon > 0$ there exists Λ_ε such that for all $\Lambda < \Lambda_\varepsilon$,

$$|\mathfrak{b}_{\text{phys},p}[\psi]| \leq \varepsilon\, K\, \|\psi\|^2_{H^{m/2}} + C_{\text{phys},\rho}(\varepsilon, \Lambda)\, \|\psi\|^2. \tag{322}$$

Hence one may set

$$\alpha_{\text{cub}}(\Lambda) = \varepsilon, \qquad \beta_{\text{cub},\rho}(\Lambda) = C_{\text{phys},\rho}(\varepsilon, \Lambda), \tag{323}$$

with $\alpha_{\text{cub}}(\Lambda) \to 0$ and $\beta_{\text{cub},\rho}(\Lambda) \to 0$ as $\Lambda \to 0$, provided the anisotropy operators are indeed IR-irrelevant.

This is the correct conditional relative-bound statement.

F. Canonical RG support and its limitation

In the gauge-covariant infrared EFT, cubic anisotropy operators satisfy canonical dimension

$$[O_n^{\text{cub}}] \geq 6,$$

so their linearized RG eigenvalues obey

$$y_n = 4 - [O_n^{\text{cub}}] \leq -2.$$

Therefore, at the level of canonical power counting near the Maxwell fixed point, cubic anisotropy operators are IR-irrelevant and their dimensionless couplings vanish as $\Lambda \to 0$.

However, the document is explicit that a complete one-loop verification still requires:

1. identifying the full operator basis generated by the microscopic BCC condensate,
2. computing the anomalous-dimension matrix,
3. checking that all eigenvalues remain negative.

So this is not yet a full microscopic theorem.

Moreover, in the derivative anisotropy sector the sign of the physical kinetic-flow matrix M_Z is still open. Thus the physically most relevant piece of B_{phys} is not yet unconditionally controlled. :contentReference[oaicite:13]index=13

G. Final honest bound for the cubic piece

Therefore the correct present status is:

$$\boxed{B_{\text{cub}} = B_{\text{red}} + B_{\text{phys}}, \qquad B_{\text{red}} \text{ exactly removable,}} \tag{324}$$

and

$$\boxed{B_{\text{phys}} \text{ admits a small relative-form bound only conditionally on IR irrelevance of the physical anisotropy couplings.}} \tag{325}$$

Hence, in the current orthodox analytic program, the cubic contribution may be entered as

$$\alpha_{\text{cub}} = \alpha_{\text{phys}}, \qquad \beta_{\text{cub},\rho} = \beta_{\text{phys},\rho}, \tag{326}$$

with the understanding that these are:

- *not yet explicitly fixed microscopically*,
- but reduced to the RG problem of the physical anisotropy operators / the sign of M_Z.

Thus the fully honest extra-sector coercivity constant is

$$c_{\text{extra},\rho} = \big(1 - \alpha_{\text{phys}} - \alpha_{\text{H}}\big) K\, \delta^2_{\text{Bloch},\rho} - \big(\beta_{\text{phys},\rho} + \beta_{\text{pot},\rho} + \beta_{\text{tr},\rho} + \beta_{\text{H},\rho}\big), \tag{327}$$

where:

- $\beta_{\text{pot},\rho}$ is explicit,
- $\beta_{\text{tr},\rho} = O(\rho)$ is explicit,
- $(\alpha_{\text{H}}, \beta_{\text{H},\rho})$ are heavy-suppressed,
- $(\alpha_{\text{phys}}, \beta_{\text{phys},\rho})$ remain open unless the physical cubic anisotropy is proved IR-irrelevant.

b. Conclusion. The last hard analytic remainder is now isolated exactly:

$$\boxed{\text{the true unresolved cubic piece is not the redundant common metric anisotropy, but the physical relative derivative anisotropy.}}$$

Its control is equivalent to the RG fate of the physical anisotropy operators, or more concretely in the current TECT truncation, to the sign of the kinetic-flow matrix M_Z.

. Summary

$$\boxed{B_{\text{cub}} = B_{\text{red}} + B_{\text{phys}}.}$$

Here

$$B_{\text{red}}$$

is the common spatial metric anisotropy, and

$$B_{\text{phys}}$$

is the physical relative derivative anisotropy.

The exact quadratic decomposition is

$$\delta K^{ij}_{IJ} = \delta^{ij}\delta Z_{IJ} + \delta_{IJ} h^{ij} + \widehat{K}^{ij}_{IJ},$$

with

$$h_{ii} = 0, \qquad \widehat{K}^{ii}_{IJ} = 0, \qquad \widehat{K}^{ij}_{II} = 0.$$

Redundant part:

$$\boxed{B_{\text{red}} \text{ is exactly removable by a linear coordinate transformation,}}$$

so it contributes neither to α_{cub} nor to $\beta_{\text{cub},\rho}$.

Physical part:

$$\boxed{B_{\text{phys}} \text{ is the true derivative/velocity anisotropy sector.}}$$

It is classically marginal, and its infrared fate is controlled by the loop-level kinetic-flow matrix

$$M_Z.$$

Hence the honest current status is:

$$\boxed{\text{Unconditional control of } B_{\rm cub} \text{ is not yet proved microscopically.}}$$

Conditional theorem: if the physical cubic anisotropy couplings are IR-irrelevant, then for every $\varepsilon > 0$,

$$|\mathfrak{b}_{{\rm phys},p}[\psi]| \le \varepsilon\, K\, \|\psi\|^2_{H^{m/2}} + C_{{\rm phys},\rho}(\varepsilon,\Lambda)\, \|\psi\|^2,$$

so that

$$\alpha_{\rm cub}(\Lambda) \to 0, \qquad \beta_{{\rm cub},\rho}(\Lambda) \to 0 \qquad (\Lambda \to 0).$$

Canonical RG support:

$$[O_n^{\rm cub}] \ge 6 \Longrightarrow y_n = 4 - [O_n^{\rm cub}] \le -2,$$

so cubic anisotropy operators are canonically irrelevant at the Maxwell fixed point.

But the full microscopic completion still requires:

$$\boxed{\text{identify the full anisotropy operator basis,}}$$

$$\boxed{\text{compute the anomalous-dimension matrix / } M_Z,}$$

$$\boxed{\text{and verify that all physical anisotropy eigenvalues are negative.}}$$

Therefore the honest extra-sector coercivity constant is

$$\boxed{c_{{\rm extra},\rho} = (1 - \alpha_{\rm phys} - \alpha_{\rm H}) K\, \delta^2_{{\rm Bloch},\rho} - (\beta_{{\rm phys},\rho} + \beta_{{\rm pot},\rho} + \beta_{{\rm tr},\rho} + \beta_{{\rm H},\rho}).}$$

So the final unresolved analytic bottleneck is exactly:

$$\boxed{\text{control the physical derivative anisotropy sector } B_{\rm phys}, \text{ equivalently the sign of } M_Z.}$$

EXPLICIT KINETIC-FLOW MATRIX DERIVATION ROADMAP FOR M_Z

1. Physical derivative-anisotropy variables

After removing the redundant common spatial metric anisotropy, the physically meaningful derivative anisotropy is the relative sectoral kinetic anisotropy among the three TECT soft sectors:

$$A, \qquad n, \qquad \chi.$$

In the block-diagonal TECT truncation this is parameterized by

$$\Gamma_{\rm rel} = \frac{1}{2} \int d^d x \left[Z_T \sum_{\alpha=1}^{2} (\partial_i A_\alpha)^2 + Z_O \sum_{a=1}^{2} (\partial_i n_a)^2 + Z_D \sum_{r=1}^{2} (\partial_i \chi_r)^2 \right]. \tag{328}$$

Define the relative anisotropy variables

$$\delta_T := \frac{Z_T - Z}{Z}, \qquad \delta_O := \frac{Z_O - Z}{Z}, \qquad \delta_D := \frac{Z_D - Z}{Z}, \tag{329}$$

with the traceless constraint

$$2\delta_T + 2\delta_O + 2\delta_D = 0. \tag{330}$$

These are the true velocity-locking variables. They are classically marginal at the Gaussian level, so their infrared fate is determined entirely by the loop-level kinetic-flow matrix M_Z.

2. FRG truncation to be used

The TECT-adapted infrared effective average action must contain, at minimum,

$$\Gamma_k[A,n,\chi] = \int d^dx \left[\frac{Z_{T,k}}{2}(\partial A)^2 + \frac{Z_{O,k}}{2}(\partial n)^2 + \frac{Z_{D,k}}{2}(\partial \chi)^2 + U_k(\rho,\tau,\ldots) + \Gamma_{k,\mathrm{int}}[A,n,\chi] \right], \tag{331}$$

where $\Gamma_{k,\mathrm{int}}$ contains all interaction vertices relevant to the wavefunction renormalization flow. The exact sign of the derivative-anisotropy eigenvalues cannot be determined without these vertices, the regulator, the detailed tensor contractions, and the gauge/constraint structure of the A-sector.

3. Starting Wetterich equation

The flow is governed by the Wetterich equation

$$\partial_t \Gamma_k = \frac{1}{2}\mathrm{STr}\left[\left(\Gamma_k^{(2)} + R_k\right)^{-1} \partial_t R_k \right]. \tag{332}$$

To extract M_Z, the required object is the exact Hessian

$$\Gamma_k^{(2)}[A,n,\chi], \tag{333}$$

including all interaction vertices coupling the three soft sectors. Without this Hessian the sign of M_Z is underdetermined.

4. Sectoral kinetic projectors

Introduce the three kinetic projectors

$$\Pi_T, \qquad \Pi_O, \qquad \Pi_D, \tag{334}$$

which project onto the spatial two-derivative operators of the three soft sectors:

$$\mathcal{O}_T = \frac{1}{2}\sum_{\alpha=1}^{2}(\partial_i A_\alpha)^2, \qquad \mathcal{O}_O = \frac{1}{2}\sum_{a=1}^{2}(\partial_i n_a)^2, \qquad \mathcal{O}_D = \frac{1}{2}\sum_{r=1}^{2}(\partial_i \chi_r)^2. \tag{335}$$

Then define the sectoral wavefunction flows by

$$\partial_t Z_T = \Pi_T\left[\partial_t \Gamma_k\right], \qquad \partial_t Z_O = \Pi_O\left[\partial_t \Gamma_k\right], \qquad \partial_t Z_D = \Pi_D\left[\partial_t \Gamma_k\right]. \tag{336}$$

Equivalently, one can extract these by taking the second derivative of the two-point functions and differentiating with respect to spatial momentum:

$$\partial_t Z_I = \frac{\partial}{\partial p^2}\partial_t \Gamma^{(2)}_{k,II}(p)\bigg|_{p=0}, \qquad I \in \{T,O,D\}. \tag{337}$$

This is the standard route to the kinetic-flow matrix.

5. Linearization around the isotropic point

Let

$$Z_{T,k} = Z_k(1+\delta_T), \qquad Z_{O,k} = Z_k(1+\delta_O), \qquad Z_{D,k} = Z_k(1+\delta_D),$$

with δ_I small and subject to (330). Linearizing the three sectoral flows around the isotropic point gives

$$\partial_t \begin{pmatrix} \delta_T \\ \delta_O \\ \delta_D \end{pmatrix} = M_Z \begin{pmatrix} \delta_T \\ \delta_O \\ \delta_D \end{pmatrix} + O(\delta^2). \tag{338}$$

The matrix entries are the Jacobian

$$(M_Z)_{IJ} = \frac{\partial}{\partial \delta_J}\left(\partial_t \delta_I\right)\bigg|_{\delta=0}, \qquad I,J \in \{T,O,D\}. \tag{339}$$

6. Equivalent traceless-basis reduction

Since the common rescaling direction is unphysical, it is often cleaner to pass to a traceless basis:

$$u_1 = \delta_T - \delta_O, \qquad u_2 = \delta_T - \delta_D. \tag{340}$$

Then the physical derivative-anisotropy flow is the 2×2 matrix obtained by restricting M_Z to the subspace

$$2\delta_T + 2\delta_O + 2\delta_D = 0.$$

Negative eigenvalues on this subspace are exactly the criterion for velocity locking and derivative-anisotropy irrelevance.

7. Algebraic target

A controlled sufficient target already identified in the document is

$$M_Z = -\lambda \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}, \qquad \lambda > 0. \tag{341}$$

This matrix has one zero mode corresponding to the unphysical common rescaling direction

$$\delta_T = \delta_O = \delta_D,$$

and two negative eigenvalues

$$-3\lambda, \qquad -3\lambda$$

on the traceless physical subspace. Therefore if the explicit FRG projection yields (341) with $\lambda > 0$, derivative anisotropy is strictly irrelevant.

8. Minimal loop ingredients

The explicit computation of M_Z requires the following loop data:

1. the exact Hessian $\Gamma_k^{(2)}$ including vertices coupling A, n, χ;
2. a fixed regulator R_k;
3. the tensor contractions entering the momentum-dependent two-point functions;
4. the gauge/constraint structure of the A-sector.

This is precisely where the current proof stops honestly: with present TECT truncation data, the sign of M_Z is not yet fixed.

9. Conditional theorem and roadmap conclusion

The exact current theorem-level status is:

$$\boxed{\text{If the nontrivial eigenvalues of } M_Z \text{ are negative, then } \delta_T, \delta_O, \delta_D \to 0 \text{ in the infrared.}} \tag{342}$$

Thus the remaining derivation task is no longer conceptual. It is the explicit FRG projection problem:

$$\boxed{\Gamma_k^{(2)}[A, n, \chi] \implies \partial_t Z_T,\ \partial_t Z_O,\ \partial_t Z_D \implies M_Z \implies \operatorname{sign}\operatorname{spec}(M_Z|_{\text{traceless}}).} \tag{343}$$

. Summary

$$\boxed{\delta_T = \frac{Z_T - Z}{Z}, \qquad \delta_O = \frac{Z_O - Z}{Z}, \qquad \delta_D = \frac{Z_D - Z}{Z}, \qquad 2\delta_T + 2\delta_O + 2\delta_D = 0.}$$

These are the true physical derivative-anisotropy variables. They are classically marginal, so their infrared fate is controlled entirely by the loop-level kinetic-flow matrix

$$\boxed{\partial_t \begin{pmatrix} \delta_T \\ \delta_O \\ \delta_D \end{pmatrix} = M_Z \begin{pmatrix} \delta_T \\ \delta_O \\ \delta_D \end{pmatrix} + \cdots.}$$

To derive M_Z, one must:

$$\boxed{(1)\ \text{write } \Gamma_k[A, n, \chi], \quad (2)\ \text{compute } \Gamma_k^{(2)}, \quad (3)\ \text{insert into the Wetterich equation,}}$$

$$\boxed{(4)\ \text{project onto } \mathcal{O}_T, \mathcal{O}_O, \mathcal{O}_D, \quad (5)\ \text{linearize the flows of } Z_T, Z_O, Z_D.}$$

Equivalently,

$$\partial_t Z_I = \frac{\partial}{\partial p^2} \partial_t \Gamma^{(2)}_{k,II}(p)\bigg|_{p=0}, \qquad I \in \{T, O, D\},$$

and

$$(M_Z)_{IJ} = \frac{\partial}{\partial \delta_J} (\partial_t \delta_I)\bigg|_{\delta=0}.$$

The algebraic target is

$$\boxed{M_Z = -\lambda \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}, \qquad \lambda > 0.}$$

Then the unphysical common-rescaling mode is the unique zero mode, and the two physical eigenvalues are

$$\boxed{-3\lambda, \qquad -3\lambda.}$$

Hence derivative anisotropy is strictly irrelevant.

Honest status:

$$\boxed{\text{with the currently available TECT truncation data, sign}(M_Z) \text{ is still not fixed.}}$$

What is still required is:

$$\boxed{\Gamma_k^{(2)}[A, n, \chi], \quad R_k, \quad \text{tensor contractions,} \quad \text{A-sector gauge/constraint projection.}}$$

TECT-ADAPTED MINIMAL FRG TRUNCATION FOR THE KINETIC-FLOW MATRIX M_Z

1. Physical derivative-anisotropy variables

After removing the redundant common spatial metric anisotropy, the physically meaningful quadratic derivative anisotropy is the *relative* sectoral kinetic anisotropy among the three TECT soft sectors:

$$A, \qquad n, \qquad \chi.$$

The block-diagonal kinetic sector is parameterized as

$$\Gamma_{\text{kin}} = \frac{1}{2}\int d^d x \left[Z_{T,k}\sum_{\alpha=1}^{2}(\partial_i A_\alpha)^2 + Z_{O,k}\sum_{a=1}^{2}(\partial_i n_a)^2 + Z_{D,k}\sum_{r=1}^{2}(\partial_i \chi_r)^2 \right]. \tag{344}$$

Define the relative anisotropy variables

$$\delta_T := \frac{Z_{T,k}-Z_k}{Z_k}, \qquad \delta_O := \frac{Z_{O,k}-Z_k}{Z_k}, \qquad \delta_D := \frac{Z_{D,k}-Z_k}{Z_k}, \tag{345}$$

with the traceless constraint

$$2\delta_T + 2\delta_O + 2\delta_D = 0. \tag{346}$$

These are the true physical velocity-locking variables. They are classically marginal at the Gaussian level, so their infrared fate is determined entirely by the loop-level kinetic-flow matrix M_Z. :contentReference[oaicite:3]index=3

2. Minimal TECT-adapted FRG ansatz

A minimal infrared FRG truncation adapted to the TECT soft sector is

$$\Gamma_k[A,n,\chi] = \int d^d x \left[\frac{Z_{T,k}}{2}\sum_{\alpha=1}^{2}(\partial_i A_\alpha)^2 + \frac{Z_{O,k}}{2}\sum_{a=1}^{2}(\partial_i n_a)^2 + \frac{Z_{D,k}}{2}\sum_{r=1}^{2}(\partial_i \chi_r)^2 + Y_k\, O_{\nabla,\text{ani}} + U_k(\rho,\tau) + \Gamma_{k,\text{int}}[A,n,\chi] \right]. \tag{347}$$

Here:

- $Z_{T,k}, Z_{O,k}, Z_{D,k}$ are the sectoral wavefunction renormalizations,
- $O_{\nabla,\text{ani}}$ is the leading derivative-anisotropy operator allowed by the BCC background,
- $U_k(\rho,\tau)$ is the local isotropic-plus-leading-anisotropy potential truncation,
- $\Gamma_{k,\text{int}}[A,n,\chi]$ contains all interaction vertices required to generate the momentum-dependent two-point flow.

A minimal polynomial potential truncation is

$$U_k(\rho,\tau) = r_k\,\rho + \frac{u_k}{2}\rho^2 + g_{\tau,k}\,\tau. \tag{348}$$

This is exactly the TECT-adapted FRG structure already identified in the document.

3. Wetterich equation and regulator

The flow is governed by the Wetterich equation

$$\partial_t \Gamma_k = \frac{1}{2}\text{Tr}\Big[(\Gamma_k^{(2)} + R_k)^{-1}\,\partial_t R_k\Big], \qquad t = \ln(k/\Lambda). \tag{349}$$

A standard optimized regulator choice is

$$R_k(q) = Z_k(k^2 - q^2)\Theta(k^2 - q^2), \tag{350}$$

or, more carefully in the sectoral setting,

$$R_k(q) = \text{diag}\Big(Z_{T,k}\,r_k(q),\; Z_{O,k}\,r_k(q),\; Z_{D,k}\,r_k(q)\Big), \tag{351}$$

with the same shape function r_k on the three soft sectors. The exact sign of the derivative-anisotropy flow cannot be fixed without specifying $\Gamma_k^{(2)}$, R_k, the tensor contractions, and the gauge/constraint structure of the A-sector.

4. Sectoral kinetic projectors

Introduce the three kinetic projection operators

$$\Pi_T, \qquad \Pi_O, \qquad \Pi_D, \tag{352}$$

defined by projection onto the three spatial kinetic operators

$$\mathcal{O}_T = \frac{1}{2}\sum_{\alpha=1}^{2}(\partial_i A_\alpha)^2, \qquad \mathcal{O}_O = \frac{1}{2}\sum_{a=1}^{2}(\partial_i n_a)^2, \qquad \mathcal{O}_D = \frac{1}{2}\sum_{r=1}^{2}(\partial_i \chi_r)^2. \tag{353}$$

Then the sectoral wavefunction flows are defined by

$$\partial_t Z_{T,k} = \Pi_T[\partial_t \Gamma_k], \qquad \partial_t Z_{O,k} = \Pi_O[\partial_t \Gamma_k], \qquad \partial_t Z_{D,k} = \Pi_D[\partial_t \Gamma_k]. \tag{354}$$

Equivalently, in the two-point-function language, let

$$\Gamma^{(2)}_{k,TT}(p), \qquad \Gamma^{(2)}_{k,OO}(p), \qquad \Gamma^{(2)}_{k,DD}(p)$$

denote the diagonal two-point blocks of the three sectors. Then

$$\partial_t Z_{I,k} = \frac{\partial}{\partial p^2}\partial_t \Gamma^{(2)}_{k,II}(p)\bigg|_{p=0}, \qquad I \in \{T, O, D\}. \tag{355}$$

This is the standard projection formula for extracting sectoral wavefunction renormalization flows.

5. Linearization around the isotropic point

Write

$$Z_{T,k} = Z_k(1+\delta_T), \qquad Z_{O,k} = Z_k(1+\delta_O), \qquad Z_{D,k} = Z_k(1+\delta_D), \tag{356}$$

with small δ_I and constraint (346). Linearizing the sectoral kinetic flows around the isotropic point gives

$$\partial_t \begin{pmatrix}\delta_T\\ \delta_O\\ \delta_D\end{pmatrix} = M_Z \begin{pmatrix}\delta_T\\ \delta_O\\ \delta_D\end{pmatrix} + O(\delta^2). \tag{357}$$

The matrix entries are the Jacobian

$$(M_Z)_{IJ} = \frac{\partial}{\partial \delta_J}\big(\partial_t \delta_I\big)\bigg|_{\delta=0}, \qquad I, J \in \{T, O, D\}. \tag{358}$$

Thus M_Z is not guessed: it is *defined* by the linearization of the projected Wetterich flow.

6. Traceless-basis reduction

Since the common rescaling direction is unphysical, it is convenient to pass to a traceless basis:

$$u_1 = \delta_T - \delta_O, \qquad u_2 = \delta_T - \delta_D. \tag{359}$$

Let

$$M_Z^{\text{phys}}$$

denote the restriction of M_Z to the two-dimensional traceless subspace

$$2\delta_T + 2\delta_O + 2\delta_D = 0.$$

Then:

$$\boxed{\text{derivative anisotropy is infrared-irrelevant} \iff \operatorname{spec}(M_Z^{\text{phys}}) \subset (-\infty, 0).}$$

This is the exact physical criterion.

7. Algebraic target

A controlled sufficient target already identified in the document is

$$M_Z = -\lambda \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}, \qquad \lambda > 0. \tag{360}$$

This matrix has:

- one zero mode corresponding to the unphysical common-rescaling direction

$$\delta_T = \delta_O = \delta_D,$$

- two negative eigenvalues

$$-3\lambda, \qquad -3\lambda$$

  on the traceless physical subspace.

Hence, if the explicit FRG projection yields (360) with $\lambda > 0$, then the physical derivative anisotropy is strictly irrelevant and velocity locking follows.

8. What must still be specified

To turn this roadmap into an actual sign computation, the following microscopic ingredients are still required:

1. the exact Hessian $\Gamma_k^{(2)}[A, n, \chi]$ including all interaction vertices coupling the three soft sectors,
2. a fixed regulator choice R_k,
3. the explicit tensor contractions contributing to the momentum derivative of the two-point functions,
4. the gauge/constraint projection in the A-sector.

This is exactly where the honest proof presently stops: with currently available TECT truncation data, the sign of the nontrivial eigenvalues of M_Z is not yet fixed.

9. Final roadmap theorem

Therefore the correct final theorem-level roadmap is:

$$\boxed{\Gamma_k[A, n, \chi] \implies \Gamma_k^{(2)} \implies \partial_t Z_T,\ \partial_t Z_O,\ \partial_t Z_D \implies M_Z \implies \operatorname{sign}\operatorname{spec}(M_Z^{\text{phys}}).} \tag{361}$$

And the correct conditional theorem is:

$$\boxed{\text{If } \operatorname{spec}(M_Z^{\text{phys}}) \subset (-\infty, 0),\ \text{then } \delta_T, \delta_O, \delta_D \to 0 \text{ in the infrared.}} \tag{362}$$

. Summary

$$\boxed{\Gamma_k[A, n, \chi] = \int d^d x \left[\frac{Z_{T,k}}{2} \sum_{\alpha=1}^{2} (\partial_i A_\alpha)^2 + \frac{Z_{O,k}}{2} \sum_{a=1}^{2} (\partial_i n_a)^2 + \frac{Z_{D,k}}{2} \sum_{r=1}^{2} (\partial_i \chi_r)^2 + Y_k O_{\nabla,\text{ani}} + U_k(\rho, \tau) + \Gamma_{k,\text{int}} \right].}$$

$$\boxed{\delta_T = \frac{Z_{T,k} - Z_k}{Z_k}, \qquad \delta_O = \frac{Z_{O,k} - Z_k}{Z_k}, \qquad \delta_D = \frac{Z_{D,k} - Z_k}{Z_k}, \qquad 2\delta_T + 2\delta_O + 2\delta_D = 0.}$$

Sectoral wavefunction flows are extracted by

$$\boxed{\partial_t Z_{I,k} = \frac{\partial}{\partial p^2} \partial_t \Gamma^{(2)}_{k,II}(p)\Big|_{p=0}, \qquad I \in \{T, O, D\}.}$$

Linearizing around the isotropic point gives

$$\boxed{\partial_t \begin{pmatrix} \delta_T \\ \delta_O \\ \delta_D \end{pmatrix} = M_Z \begin{pmatrix} \delta_T \\ \delta_O \\ \delta_D \end{pmatrix} + O(\delta^2),}$$

with

$$\boxed{(M_Z)_{IJ} = \frac{\partial}{\partial \delta_J} (\partial_t \delta_I)\Big|_{\delta=0}.}$$

The physical criterion is negativity on the traceless subspace. A sufficient algebraic target is

$$\boxed{M_Z = -\lambda \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}, \qquad \lambda > 0,}$$

for which the two physical eigenvalues are

$$\boxed{-3\lambda, \qquad -3\lambda.}$$

Hence the remaining frontier is exactly:

$$\boxed{\Gamma_k^{(2)}[A, n, \chi], \quad R_k, \quad \text{tensor contractions}, \quad A\text{-sector gauge/constraint projection.}}$$

And the honest final conditional theorem is:

$$\boxed{\operatorname{spec}(M_Z^{\text{phys}}) \subset (-\infty, 0) \Longrightarrow \delta_T, \delta_O, \delta_D \to 0 \text{ in the infrared.}}$$

MINIMAL VERTEX-COMPLETE TRUNCATION FOR THE KINETIC-FLOW MATRIX M_Z

A. Why the schematic truncation is not enough

The schematic TECT infrared truncation already identified in the document is

$$\Gamma_k[\Phi] = \int d^d x \left[\frac{Z_{T,k}}{2} \sum_{\alpha=1}^{2} (\partial_i A_\alpha)^2 + \frac{Z_{O,k}}{2} \sum_{a=1}^{2} (\partial_i n_a)^2 + \frac{Z_{D,k}}{2} \sum_{r=1}^{2} (\partial_i \chi_r)^2 + Y_k O_{\nabla,\text{ani}} + U_k(\rho, \tau) \right], \tag{363}$$

with

$$\Phi = (A_1, A_2, n_1, n_2, \chi_1, \chi_2), \qquad N_{\text{eff}} = 6.$$

This truncation is sufficient to state the *schematic* isotropy theorem and to identify the derivative-anisotropy problem.

However, it is not yet sufficient to determine the sign of the linearized kinetic-flow matrix M_Z, because the document is explicit that M_Z depends on:

1. the exact Hessian $\Gamma_k^{(2)}$,
2. the interaction vertices coupling A, n, χ,
3. the regulator choice,
4. the detailed tensor contractions,
5. the gauge/constraint structure of the A-sector.

Therefore, to compute M_Z, one needs a *vertex-complete* truncation for the sectoral two-point flows. :contentReference[oaicite:5]index=5

B. Sectoral invariants

Define the three quadratic sectoral invariants

$$\rho_T := \frac{1}{2}\sum_{\alpha=1}^{2} A_\alpha^2, \qquad \rho_O := \frac{1}{2}\sum_{a=1}^{2} n_a^2, \qquad \rho_D := \frac{1}{2}\sum_{r=1}^{2} \chi_r^2, \tag{364}$$

so that

$$\rho = \rho_T + \rho_O + \rho_D.$$

The leading quartic cubic anisotropy already used in the isotropy analysis is encoded by the traceless quartic invariant

$$\tau(\Phi) = \sum_{I=1}^{6} \Phi_I^4 - \frac{3}{N_{\text{eff}}+2}(\Phi_J\Phi_J)^2, \qquad N_{\text{eff}} = 6.$$

Thus $U_k(\rho, \tau)$ captures the isotropic quartic sector plus the leading quartic cubic anisotropy.

C. Minimal sufficient vertex-complete ansatz

A minimal *sufficient* vertex-complete truncation for extracting M_Z is:

$$\Gamma_k^{\text{VC}}[A, n, \chi] = \Gamma_{k,\text{kin}} + \Gamma_{k,\text{pot}} + \Gamma_{k,\nabla\text{ani}} + \Gamma_{k,\text{gf/proj}}. \tag{365}$$

a. (i) Sectoral kinetic part

$$\Gamma_{k,\text{kin}} = \frac{1}{2}\int d^d x \left[Z_{T,k}\sum_{\alpha=1}^{2}(\partial_i A_\alpha)^2 + Z_{O,k}\sum_{a=1}^{2}(\partial_i n_a)^2 + Z_{D,k}\sum_{r=1}^{2}(\partial_i \chi_r)^2 \right]. \tag{366}$$

b. (ii) Minimal quartic potential sector Instead of keeping only $U_k(\rho, \tau)$ abstractly, write the minimal explicit quartic sector as

$$\Gamma_{k,\text{pot}} = \int d^d x \left[r_{T,k}\rho_T + r_{O,k}\rho_O + r_{D,k}\rho_D + \frac{\lambda_{T,k}}{2}\rho_T^2 + \frac{\lambda_{O,k}}{2}\rho_O^2 + \frac{\lambda_{D,k}}{2}\rho_D^2 + \lambda_{TO,k}\rho_T\rho_O + \lambda_{TD,k}\rho_T\rho_D + \lambda_{OD,k}\rho_O\rho_D + g_{\tau,k}\,\tau(\Phi) \right]. \tag{367}$$

This is the minimal explicit potential completion needed so that:

1. each sector has its own self-interaction,
2. each sector can renormalize the kinetic term of the others through one-loop mixed self-energies,
3. the leading cubic quartic anisotropy is retained.

c. *(iii) Leading derivative anisotropy*

$$\Gamma_{k,\nabla\mathrm{ani}} = Y_k \int d^d x\, O_{\nabla,\mathrm{ani}}[A,n,\chi]. \tag{368}$$

Here $O_{\nabla,\mathrm{ani}}$ is the leading BCC-allowed derivative anisotropy operator already isolated in the isotropy analysis.

d. *(iv) Gauge / constraint projection in the A-sector* Because the sign of M_Z depends on the gauge or constraint structure of the A-sector, one must include either:

$$\Gamma_{k,\mathrm{gf/proj}} = \Gamma_{k,\mathrm{gf}} \quad \text{or} \quad \Gamma_{k,\mathrm{proj}} \tag{369}$$

so that the A-sector Hessian entering the Wetterich trace is already projected onto the physical transverse/allowed subspace. The current data do not yet fix this uniquely, but some explicit gauge/constraint implementation is mandatory. :contentReference[oaicite:8]index=8

D. Why these vertices are minimally necessary

To compute

$$\partial_t Z_T, \qquad \partial_t Z_O, \qquad \partial_t Z_D,$$

one must project the momentum derivative of the one-loop two-point functions in each sector:

$$\partial_t Z_{I,k} = \frac{\partial}{\partial p^2} \partial_t \Gamma^{(2)}_{k,II}(p)\bigg|_{p=0}, \qquad I \in \{T,O,D\}. \tag{370}$$

At one loop, these flows receive contributions from any quartic or derivative vertex that couples the I-sector to itself or to one of the other sectors. Therefore:

- omitting $\lambda_{TO}, \lambda_{TD}, \lambda_{OD}$ would miss mixed-loop contributions to the sectoral two-point functions;
- omitting $g_{\tau,k}\tau$ would miss the leading quartic cubic-anisotropy contribution;
- omitting $Y_k O_{\nabla,\mathrm{ani}}$ would miss the derivative-anisotropy insertion itself;
- omitting the A-sector projection/gauge treatment would leave the A-sector Hessian ambiguous.

Thus (365)–(368) is the minimal *sufficient* truncation to define M_Z unambiguously at the formal one-loop projection level.

E. Projectors and Jacobian definition of M_Z

Define the three kinetic projectors

$$\Pi_T, \qquad \Pi_O, \qquad \Pi_D,$$

onto

$$\mathcal{O}_T = \frac{1}{2}\sum_{\alpha=1}^{2}(\partial_i A_\alpha)^2, \quad \mathcal{O}_O = \frac{1}{2}\sum_{a=1}^{2}(\partial_i n_a)^2, \quad \mathcal{O}_D = \frac{1}{2}\sum_{r=1}^{2}(\partial_i \chi_r)^2.$$

Then

$$\partial_t Z_{T,k} = \Pi_T[\partial_t \Gamma_k^{\mathrm{VC}}], \qquad \partial_t Z_{O,k} = \Pi_O[\partial_t \Gamma_k^{\mathrm{VC}}], \qquad \partial_t Z_{D,k} = \Pi_D[\partial_t \Gamma_k^{\mathrm{VC}}]. \tag{371}$$

Equivalently,

$$\partial_t Z_{I,k} = \frac{\partial}{\partial p^2} \partial_t \Gamma^{(2)}_{k,II}(p)\bigg|_{p=0}, \qquad I \in \{T,O,D\}.$$

Write

$$Z_{T,k} = Z_k(1+\delta_T), \qquad Z_{O,k} = Z_k(1+\delta_O), \qquad Z_{D,k} = Z_k(1+\delta_D),$$

with

$$2\delta_T + 2\delta_O + 2\delta_D = 0.$$

Then the kinetic-flow matrix is defined by linearization:

$$\partial_t \begin{pmatrix} \delta_T \\ \delta_O \\ \delta_D \end{pmatrix} = M_Z \begin{pmatrix} \delta_T \\ \delta_O \\ \delta_D \end{pmatrix} + O(\delta^2), \qquad (M_Z)_{IJ} = \frac{\partial}{\partial \delta_J}(\partial_t \delta_I)\bigg|_{\delta=0}. \tag{372}$$

F. Honest status

The exact honest statement is:

$$\boxed{\text{The truncation } \Gamma_k^{\mathrm{VC}} \text{ above is a minimal sufficient vertex-complete ansatz for formally defining } M_Z.}$$

What is still *not* closed is:

1. the exact tensor structure of $O_{\nabla,\mathrm{ani}}$ in the full corrected microscopic TECT model,
2. the exact A-sector gauge/constraint projector,
3. the actual evaluation of the one-loop tensor contractions,
4. the sign of the nontrivial eigenvalues of M_Z.

So this step closes the formal truncation problem, but not yet the sign computation itself.

. Summary

$$\boxed{\Phi = (A_1, A_2, n_1, n_2, \chi_1, \chi_2), \qquad N_{\mathrm{eff}} = 6.}$$

The schematic TECT truncation

$$\Gamma_k[\Phi] = \int d^dx \left[\frac{Z_{T,k}}{2}(\partial A)^2 + \frac{Z_{O,k}}{2}(\partial n)^2 + \frac{Z_{D,k}}{2}(\partial \chi)^2 + Y_k O_{\nabla,\mathrm{ani}} + U_k(\rho,\tau)\right]$$

is sufficient for the *schematic* isotropy theorem, but not sufficient to determine $\mathrm{sign}(M_Z)$.
A minimal *sufficient* vertex-complete ansatz for the formal M_Z projection is

$$\boxed{\Gamma_k^{\mathrm{VC}} = \Gamma_{k,\mathrm{kin}} + \Gamma_{k,\mathrm{pot}} + \Gamma_{k,\nabla\mathrm{ani}} + \Gamma_{k,\mathrm{gf/proj}},}$$

with

$$\Gamma_{k,\mathrm{kin}} = \frac{1}{2}\int d^dx \Big[Z_{T,k}\sum_{\alpha=1}^{2}(\partial_i A_\alpha)^2 + Z_{O,k}\sum_{a=1}^{2}(\partial_i n_a)^2 + Z_{D,k}\sum_{r=1}^{2}(\partial_i \chi_r)^2\Big],$$

$$\rho_T = \frac{1}{2}\sum_{\alpha=1}^{2} A_\alpha^2, \qquad \rho_O = \frac{1}{2}\sum_{a=1}^{2} n_a^2, \qquad \rho_D = \frac{1}{2}\sum_{r=1}^{2} \chi_r^2,$$

$$\Gamma_{k,\mathrm{pot}} = \int d^dx \left[r_{T,k}\rho_T + r_{O,k}\rho_O + r_{D,k}\rho_D + \frac{\lambda_{T,k}}{2}\rho_T^2 + \frac{\lambda_{O,k}}{2}\rho_O^2 + \frac{\lambda_{D,k}}{2}\rho_D^2 + \lambda_{TO,k}\rho_T\rho_O + \lambda_{TD,k}\rho_T\rho_D + \lambda_{OD,k}\rho_O\rho_D + g_{\tau,k}\tau(\Phi)\right],$$

$$\Gamma_{k,\nabla\mathrm{ani}} = Y_k \int d^dx\, O_{\nabla,\mathrm{ani}}[A, n, \chi].$$

Then

$$\partial_t Z_{I,k} = \frac{\partial}{\partial p^2}\partial_t \Gamma^{(2)}_{k,II}(p)\bigg|_{p=0}, \qquad I \in \{T, O, D\},$$

and with

$$Z_{T,k} = Z_k(1+\delta_T), \quad Z_{O,k} = Z_k(1+\delta_O), \quad Z_{D,k} = Z_k(1+\delta_D), \quad 2\delta_T + 2\delta_O + 2\delta_D = 0,$$

the kinetic-flow matrix is

$$\boxed{(M_Z)_{IJ} = \frac{\partial}{\partial \delta_J}(\partial_t \delta_I)\bigg|_{\delta=0}.}$$

Thus the truncation problem is formally closed.
What remains open is:

$$\boxed{\text{the explicit evaluation of the Hessian, regulator contractions, and } A\text{-sector projection needed to compute the sign of } M_Z.}$$

FORMAL BLOCK HESSIAN OF THE MINIMAL VERTEX-COMPLETE TECT TRUNCATION

1. Field organization

Introduce the three soft-sector vectors

$$A = \begin{pmatrix} A_1 \\ A_2 \end{pmatrix}, \qquad n = \begin{pmatrix} n_1 \\ n_2 \end{pmatrix}, \qquad \chi = \begin{pmatrix} \chi_1 \\ \chi_2 \end{pmatrix},$$

and collect them into the six-component multiplet

$$\Phi = \begin{pmatrix} A \\ n \\ \chi \end{pmatrix}.$$

This is the $N_{\mathrm{eff}} = 6$ soft TECT infrared multiplet already identified in the document. :contentReference[oaicite:3]index=3
Define the sectoral quadratic invariants

$$\rho_T = \frac{1}{2}A^T A, \qquad \rho_O = \frac{1}{2}n^T n, \qquad \rho_D = \frac{1}{2}\chi^T \chi, \qquad \rho = \rho_T + \rho_O + \rho_D.$$

2. Minimal vertex-complete ansatz

A minimal sufficient vertex-complete truncation for the formal M_Z projection is

$$\Gamma_k^{\mathrm{VC}}[A, n, \chi] = \Gamma_{k,\mathrm{kin}} + \Gamma_{k,\mathrm{pot}} + \Gamma_{k,\nabla\mathrm{ani}} + \Gamma_{k,\mathrm{gf/proj}}, \tag{373}$$

with

$$\Gamma_{k,\mathrm{kin}} = \frac{1}{2}\int d^dx \Big[Z_{T,k}\,(\partial_i A)^T(\partial_i A) + Z_{O,k}\,(\partial_i n)^T(\partial_i n) + Z_{D,k}\,(\partial_i \chi)^T(\partial_i \chi)\Big], \tag{374}$$

$$\Gamma_{k,\mathrm{pot}} = \int d^dx \Big[r_{T,k}\rho_T + r_{O,k}\rho_O + r_{D,k}\rho_D + \frac{\lambda_{T,k}}{2}\rho_T^2 + \frac{\lambda_{O,k}}{2}\rho_O^2 + \frac{\lambda_{D,k}}{2}\rho_D^2 + \lambda_{TO,k}\rho_T\rho_O + \lambda_{TD,k}\rho_T\rho_D + \lambda_{OD,k}\rho_O\rho_D + g_{\tau,k}\,\tau(\Phi) \Big], \tag{375}$$

and

$$\Gamma_{k,\nabla\mathrm{ani}} = Y_k \int d^dx\, O_{\nabla,\mathrm{ani}}[A,n,\chi]. \tag{376}$$

Here $\Gamma_{k,\mathrm{gf/proj}}$ is whatever gauge-fixing or physical projector is needed to define the A-sector Hessian unambiguously. The document is explicit that some such gauge/constraint treatment is required before the sign of M_Z can be fixed. :contentReference[oaicite:4]index=4

3. General block Hessian

The full Hessian is the 6×6 block operator

$$\Gamma_k^{(2)}(p) = \begin{pmatrix} \Gamma^{(2)}_{TT}(p) & \Gamma^{(2)}_{TO}(p) & \Gamma^{(2)}_{TD}(p) \\ \Gamma^{(2)}_{OT}(p) & \Gamma^{(2)}_{OO}(p) & \Gamma^{(2)}_{OD}(p) \\ \Gamma^{(2)}_{DT}(p) & \Gamma^{(2)}_{DO}(p) & \Gamma^{(2)}_{DD}(p) \end{pmatrix}, \tag{377}$$

where each entry is a 2×2 matrix differential operator (or, after Fourier/Bloch reduction, a 2×2 matrix-valued symbol).

A. Diagonal blocks

Differentiating (373) twice gives the formal diagonal blocks

$$\Gamma^{(2)}_{TT}(p) = \Big[Z_{T,k}\,p^2 + r_{T,k} + \lambda_{T,k}\rho_T + \lambda_{TO,k}\rho_O + \lambda_{TD,k}\rho_D \Big] \mathbf{1}_2 + \lambda_{T,k}\, AA^T + g_{\tau,k}\,\frac{\partial^2\tau}{\partial A\,\partial A} + Y_k\,\mathcal{K}^{\nabla\mathrm{ani}}_{TT}(p) + \mathcal{G}_A(p), \tag{378}$$

$$\Gamma^{(2)}_{OO}(p) = \Big[Z_{O,k}\,p^2 + r_{O,k} + \lambda_{O,k}\rho_O + \lambda_{TO,k}\rho_T + \lambda_{OD,k}\rho_D \Big] \mathbf{1}_2 + \lambda_{O,k}\, nn^T + g_{\tau,k}\,\frac{\partial^2\tau}{\partial n\,\partial n} + Y_k\,\mathcal{K}^{\nabla\mathrm{ani}}_{OO}(p), \tag{379}$$

$$\Gamma^{(2)}_{DD}(p) = \Big[Z_{D,k}\,p^2 + r_{D,k} + \lambda_{D,k}\rho_D + \lambda_{TD,k}\rho_T + \lambda_{OD,k}\rho_O \Big] \mathbf{1}_2 + \lambda_{D,k}\, \chi\chi^T + g_{\tau,k}\,\frac{\partial^2\tau}{\partial \chi\,\partial \chi} + Y_k\,\mathcal{K}^{\nabla\mathrm{ani}}_{DD}(p). \tag{380}$$

Here $\mathcal{G}_A(p)$ denotes the A-sector gauge-fixing / constraint projection contribution.

B. Off-diagonal blocks

The mixed Hessian blocks are

$$\Gamma^{(2)}_{TO}(p) = \lambda_{TO,k}\, An^T + g_{\tau,k}\,\frac{\partial^2\tau}{\partial A\,\partial n} + Y_k\,\mathcal{K}^{\nabla\mathrm{ani}}_{TO}(p), \tag{381}$$

$$\Gamma^{(2)}_{TD}(p) = \lambda_{TD,k}\, A\chi^T + g_{\tau,k}\,\frac{\partial^2\tau}{\partial A\,\partial \chi} + Y_k\,\mathcal{K}^{\nabla\mathrm{ani}}_{TD}(p), \tag{382}$$

$$\Gamma^{(2)}_{OD}(p) = \lambda_{OD,k}\, n\chi^T + g_{\tau,k}\,\frac{\partial^2\tau}{\partial n\,\partial \chi} + Y_k\,\mathcal{K}^{\nabla\mathrm{ani}}_{OD}(p), \tag{383}$$

with the transpose/Hermitian-conjugate relations

$$\Gamma^{(2)}_{OT} = (\Gamma^{(2)}_{TO})^T, \qquad \Gamma^{(2)}_{DT} = (\Gamma^{(2)}_{TD})^T, \qquad \Gamma^{(2)}_{DO} = (\Gamma^{(2)}_{OD})^T.$$

4. Specialization to the isotropic symmetric point

For the linearized M_Z analysis one expands around the isotropic symmetric point

$$A = 0, \qquad n = 0, \qquad \chi = 0, \qquad Z_{T,k} = Z_{O,k} = Z_{D,k} = Z_k.$$

At that point,

$$\rho_T = \rho_O = \rho_D = 0,$$

and the quartic cross terms do not contribute directly to the mixed Hessian entries. Therefore the block Hessian simplifies to

$$\Gamma_k^{(2)}(p)\Big|_{\Phi=0} = \begin{pmatrix} (Z_k p^2 + r_{T,k})\mathbf{1}_2 + g_{\tau,k}\tau''_{AA} + Y_k\mathcal{K}^{\nabla\text{ani}}_{TT} + \mathcal{G}_A & g_{\tau,k}\tau''_{An} + Y_k\mathcal{K}^{\nabla\text{ani}}_{TO} & g_{\tau,k}\tau''_{A\chi} + Y_k\mathcal{K}^{\nabla\text{ani}}_{TD} \\ * & (Z_k p^2 + r_{O,k})\mathbf{1}_2 + g_{\tau,k}\tau''_{nn} + Y_k\mathcal{K}^{\nabla\text{ani}}_{OO} & g_{\tau,k}\tau''_{n\chi} + Y_k\mathcal{K}^{\nabla\text{ani}}_{OD} \\ * & * & (Z_k p^2 + r_{D,k})\mathbf{1}_2 + g_{\tau,k}\tau''_{\chi\chi} + Y \end{pmatrix} \tag{384}$$

a. Honest caution. If the quartic anisotropy $\tau(\Phi)$ and the derivative-anisotropy operator $O_{\nabla,\text{ani}}$ are chosen so that their mixed second derivatives vanish at $\Phi = 0$, then the Hessian at the symmetric point becomes block diagonal. This does *not* mean that mixed couplings are irrelevant for M_Z. It means only that they enter the flow of the two-point functions through higher vertices $\Gamma_k^{(3)}$ and $\Gamma_k^{(4)}$, not directly through off-diagonal entries of $\Gamma_k^{(2)}|_{\Phi=0}$.

5. Why mixed quartic vertices are still needed

The Wetterich equation for the two-point function flow schematically gives

$$\partial_t \Gamma_k^{(2)} = -\frac{1}{2}\mathrm{Tr}\big[G_k\,\Gamma_k^{(4)}\,G_k\,\partial_t R_k\big] + \mathrm{Tr}\big[G_k\,\Gamma_k^{(3)}\,G_k\,\Gamma_k^{(3)}\,G_k\,\partial_t R_k\big], \tag{385}$$

with

$$G_k = (\Gamma_k^{(2)} + R_k)^{-1}.$$

Therefore, even if the block Hessian at $\Phi = 0$ is diagonal, the flow of

$$\Gamma^{(2)}_{TT}, \qquad \Gamma^{(2)}_{OO}, \qquad \Gamma^{(2)}_{DD}$$

still depends on the quartic cross vertices

$$\lambda_{TO,k}, \qquad \lambda_{TD,k}, \qquad \lambda_{OD,k},$$

as well as on $g_{\tau,k}$ and Y_k, because these enter $\Gamma_k^{(4)}$ and $\Gamma_k^{(3)}$. This is exactly why the sign of M_Z cannot be fixed from the structural truncation alone. The document states that one needs the exact Hessian including the interaction vertices coupling A, n, χ, the regulator, tensor contractions, and the A-sector gauge/constraint structure. :contentReference[oaicite:5]index=5

6. Sectoral kinetic projection

Once $\Gamma_k^{(2)}$ is fixed, the sectoral wavefunction flows are extracted by

$$\partial_t Z_{I,k} = \frac{\partial}{\partial p^2}\partial_t \Gamma^{(2)}_{k,II}(p)\Big|_{p=0}, \qquad I \in \{T, O, D\}. \tag{386}$$

Then, writing

$$Z_{T,k} = Z_k(1+\delta_T), \qquad Z_{O,k} = Z_k(1+\delta_O), \qquad Z_{D,k} = Z_k(1+\delta_D), \qquad 2\delta_T + 2\delta_O + 2\delta_D = 0,$$

the kinetic-flow matrix is defined by

$$(M_Z)_{IJ} = \frac{\partial}{\partial \delta_J}\big(\partial_t \delta_I\big)\Big|_{\delta=0}, \qquad I, J \in \{T, O, D\}. \tag{387}$$

7. Honest conclusion

Thus the formal block-Hessian problem is now closed:

1. the six-component soft multiplet $\Phi = (A, n, \chi)$ is fixed;
2. the minimal sufficient vertex-complete truncation is fixed;
3. the block Hessian $\Gamma_k^{(2)}$ is fixed formally;
4. the projection formula defining M_Z is fixed.

What remains open is the actual evaluation:

$$\Gamma_k^{(2)},\ \Gamma_k^{(3)},\ \Gamma_k^{(4)},\ R_k \quad \Longrightarrow \quad \partial_t Z_T, \partial_t Z_O, \partial_t Z_D \quad \Longrightarrow \quad M_Z.$$

So the proof frontier is no longer structural. It is now a concrete one-loop tensor-contraction computation.

. Summary

$$\boxed{\Phi = (A_1, A_2, n_1, n_2, \chi_1, \chi_2)^T, \qquad \rho_T = \frac{1}{2}A^T A,\ \rho_O = \frac{1}{2}n^T n,\ \rho_D = \frac{1}{2}\chi^T \chi.}$$

A minimal sufficient vertex-complete truncation is

$$\Gamma_k^{\mathrm{VC}} = \Gamma_{k,\mathrm{kin}} + \Gamma_{k,\mathrm{pot}} + \Gamma_{k,\nabla\mathrm{ani}} + \Gamma_{k,\mathrm{gf/proj}},$$

with

$$\Gamma_{k,\mathrm{kin}} = \frac{1}{2}\int d^d x \Big[Z_{T,k}(\partial A)^2 + Z_{O,k}(\partial n)^2 + Z_{D,k}(\partial \chi)^2 \Big],$$

$$\Gamma_{k,\mathrm{pot}} = \int d^d x \left[r_T \rho_T + r_O \rho_O + r_D \rho_D + \frac{\lambda_T}{2}\rho_T^2 + \frac{\lambda_O}{2}\rho_O^2 + \frac{\lambda_D}{2}\rho_D^2 + \lambda_{TO}\rho_T\rho_O + \lambda_{TD}\rho_T\rho_D + \lambda_{OD}\rho_O\rho_D + g_\tau \tau(\Phi) \right],$$

$$\Gamma_{k,\nabla\mathrm{ani}} = Y_k \int d^d x\, O_{\nabla,\mathrm{ani}}[A, n, \chi].$$

The formal block Hessian is

$$\boxed{\Gamma_k^{(2)}(p) = \begin{pmatrix} \Gamma_{TT}^{(2)} & \Gamma_{TO}^{(2)} & \Gamma_{TD}^{(2)} \\ \Gamma_{OT}^{(2)} & \Gamma_{OO}^{(2)} & \Gamma_{OD}^{(2)} \\ \Gamma_{DT}^{(2)} & \Gamma_{DO}^{(2)} & \Gamma_{DD}^{(2)} \end{pmatrix}.}$$

At the isotropic symmetric point $A = n = \chi = 0$, the Hessian may simplify drastically, possibly even to block diagonal form. But this does *not* remove the mixed couplings from the M_Z problem, because the two-point flow depends on

$$\Gamma_k^{(3)}, \qquad \Gamma_k^{(4)},$$

not only on $\Gamma_k^{(2)}|_{\Phi=0}$.

The sectoral wavefunction flows are

$$\boxed{\partial_t Z_{I,k} = \frac{\partial}{\partial p^2}\partial_t \Gamma_{k,II}^{(2)}(p)\Big|_{p=0}, \qquad I \in \{T, O, D\}.}$$

Then, with

$$Z_{T,k} = Z_k(1+\delta_T), \quad Z_{O,k} = Z_k(1+\delta_O), \quad Z_{D,k} = Z_k(1+\delta_D), \quad 2\delta_T + 2\delta_O + 2\delta_D = 0,$$

the kinetic-flow matrix is

$$\boxed{(M_Z)_{IJ} = \frac{\partial}{\partial \delta_J}(\partial_t \delta_I)\bigg|_{\delta=0}.}$$

So the structural problem is now reduced to the actual one-loop evaluation of

$$\Gamma_k^{(2)},\ \Gamma_k^{(3)},\ \Gamma_k^{(4)},\ R_k.$$

DIAGRAMMATIC CONTRIBUTION TABLE FOR THE KINETIC-FLOW MATRIX M_Z

1. General one-loop flow of the two-point function

Let

$$G_k = (\Gamma_k^{(2)} + R_k)^{-1}.$$

The formal Wetterich flow of the two-point function has the standard schematic structure

$$\partial_t \Gamma^{(2)}_{k,IJ} = \frac{1}{2}\,\mathrm{Tr}\Big[G_k\,\Gamma^{(4)}_{k,IJ}\,G_k\,\partial_t R_k\Big] - \mathrm{Tr}\Big[G_k\,\Gamma^{(3)}_{k,I}\,G_k\,\Gamma^{(3)}_{k,J}\,G_k\,\partial_t R_k\Big], \tag{388}$$

where $I, J \in \{T, O, D\}$ denote the three soft sectors. This is the formal starting point for extracting

$$\partial_t Z_T, \qquad \partial_t Z_O, \qquad \partial_t Z_D.$$

The sectoral kinetic projections are

$$\partial_t Z_{I,k} = \frac{\partial}{\partial p^2}\partial_t \Gamma^{(2)}_{k,II}(p)\bigg|_{p=0}, \qquad I \in \{T, O, D\}. \tag{389}$$

Therefore only those one-loop terms that generate a nontrivial p^2-dependence in the diagonal two-point blocks contribute to M_Z.

2. Vertex inventory in the minimal vertex-complete truncation

The minimal sufficient truncation contains the following interaction data:

a. *(a) Quartic sectoral self-couplings*

$$\lambda_T\,\rho_T^2, \qquad \lambda_O\,\rho_O^2, \qquad \lambda_D\,\rho_D^2.$$

b. *(b) Quartic mixed couplings*

$$\lambda_{TO}\,\rho_T\rho_O, \qquad \lambda_{TD}\,\rho_T\rho_D, \qquad \lambda_{OD}\,\rho_O\rho_D.$$

c. *(c) Leading quartic cubic anisotropy*

$$g_{\tau,k}\,\tau(\Phi).$$

d. *(d) Leading derivative anisotropy insertion*

$$Y_k\,O_{\nabla,\mathrm{ani}}[A, n, \chi].$$

e. *(e) A-sector gauge/constraint structure*

$$\Gamma_{k,\mathrm{gf/proj}}.$$

These are precisely the kinds of interaction vertices the document says are required to determine M_Z.

3. Which diagrams can contribute to $\partial_t Z_T, \partial_t Z_O, \partial_t Z_D$

f. Important honesty clause. At the symmetric point

$$A = n = \chi = 0,$$

the pure quartic potential sector typically yields only tadpole-type one-loop corrections from $\Gamma^{(4)}$. These often renormalize the mass terms r_T, r_O, r_D, but do *not* by themselves generate nontrivial p^2-dependence in the two-point function. Hence, unless derivative vertices, gauge exchange, or effective cubic vertices are present, the direct one-loop contribution to Z_I may vanish.

Therefore the correct table must separate:

1. diagrams that certainly renormalize the two-point block,
2. diagrams that actually contribute to its momentum derivative,
3. diagrams whose contribution depends on whether one expands around a nonzero background or includes derivative/gauge structures.

4. Sectorwise diagrammatic table

External sector	Internal loop sector	Vertex source	Topology	Cont
T	T	$\lambda_T,\ g_\tau,\ Y_k$	$\Gamma^{(4)}_{TTTT}$ tadpole	mass-type always; p^2-flow
T	O	$\lambda_{TO},\ g_\tau,\ Y_k$	$\Gamma^{(4)}_{TTOO}$ mixed tadpole	mass mixing of T block; p^2-fl
T	D	$\lambda_{TD},\ g_\tau,\ Y_k$	$\Gamma^{(4)}_{TTDD}$ mixed tadpole	sar
T	$T/O/D$	$\Gamma^{(3)}_{TTT}, \Gamma^{(3)}_{TOO}, \Gamma^{(3)}_{TDD}$	$\Gamma^{(3)}\Gamma^{(3)}$ sunset/bubble	can generate nontrivia
T	A-sector projected modes	$\Gamma_{k,\mathrm{gf/proj}},\ Y_k,$ gauge vertices	gauge/constraint loop	potentially
O	O	$\lambda_O,\ g_\tau,\ Y_k$	$\Gamma^{(4)}_{OOOO}$ tadpole	mass-type always; p^2-flo
O	T	$\lambda_{TO},\ g_\tau,\ Y_k$	$\Gamma^{(4)}_{OOTT}$ mixed tadpole	sa
O	D	$\lambda_{OD},\ g_\tau,\ Y_k$	$\Gamma^{(4)}_{OODD}$ mixed tadpole	sa
O	$T/O/D$	$\Gamma^{(3)}_{OTT}, \Gamma^{(3)}_{OOO}, \Gamma^{(3)}_{ODD}$	$\Gamma^{(3)}\Gamma^{(3)}$ sunset/bubble	can generate nontrivia
D	D	$\lambda_D,\ g_\tau,\ Y_k$	$\Gamma^{(4)}_{DDDD}$ tadpole	mass-type always; p^2-flo
D	T	$\lambda_{TD},\ g_\tau,\ Y_k$	$\Gamma^{(4)}_{DDTT}$ mixed tadpole	sa
D	O	$\lambda_{OD},\ g_\tau,\ Y_k$	$\Gamma^{(4)}_{DDOO}$ mixed tadpole	sa
D	$T/O/D$	$\Gamma^{(3)}_{DTT}, \Gamma^{(3)}_{DOO}, \Gamma^{(3)}_{DDD}$	$\Gamma^{(3)}\Gamma^{(3)}$ sunset/bubble	can generate nontrivia

5. Minimal practical interpretation

The table implies the following.

g. (i) Pure quartic self/mixed couplings The couplings

$$\lambda_T, \lambda_O, \lambda_D, \lambda_{TO}, \lambda_{TD}, \lambda_{OD}$$

are necessary because they determine which sectors can appear in the loop of each two-point flow. However, at the symmetric point they may feed primarily into *mass* renormalization rather than directly into the p^2-coefficient.

h. (ii) Quartic anisotropy $g_{\tau,k}\tau(\Phi)$ This term is necessary because it is the leading explicit cubic-symmetry-breaking quartic interaction. It modifies both diagonal and mixed four-point vertices and can therefore enter the linearized anisotropy flow.

i. (iii) Derivative anisotropy insertion $Y_k O_{\nabla,\mathrm{ani}}$ This is the most direct source of momentum-dependent anisotropic corrections. Even when pure potential tadpoles do not generate wavefunction flow, the derivative-anisotropy vertex can feed directly into $\partial_t Z_T, \partial_t Z_O, \partial_t Z_D$.

j. (iv) A-sector gauge/constraint structure Because the sign of M_Z explicitly depends on the A-sector gauge or constraint projection, loops involving the projected A-sector are potentially indispensable for $\partial_t Z_T$. This is not optional in an honest computation.

6. Formal decomposition of the M_Z entries

Accordingly, each matrix entry should be written schematically as

$$(M_Z)_{IJ} = (M_Z)_{IJ}^{(\lambda)} + (M_Z)_{IJ}^{(\tau)} + (M_Z)_{IJ}^{(\nabla\mathrm{ani})} + (M_Z)_{IJ}^{(\mathrm{gf/proj})}, \tag{390}$$

where:

- $(M_Z)_{IJ}^{(\lambda)}$ comes from the sectoral quartic self/mixed couplings,
- $(M_Z)_{IJ}^{(\tau)}$ comes from the quartic cubic-anisotropy vertex,
- $(M_Z)_{IJ}^{(\nabla\mathrm{ani})}$ comes from the derivative-anisotropy operator,
- $(M_Z)_{IJ}^{(\mathrm{gf/proj})}$ comes from the A-sector gauge/constraint contribution.

This is the correct formal bookkeeping for the explicit FRG sign computation.

7. Honest conclusion

Thus the one-loop M_Z problem is no longer structurally ambiguous.
What is now fixed is:

1. the field content,
2. the minimal sufficient truncation,
3. the block Hessian structure,
4. the sectoral projection formula,
5. the list of diagram classes that can contribute to each $\partial_t Z_I$.

What remains open is the actual evaluation of these diagrams and their tensor contractions. That is the first point at which the proof becomes genuinely computational rather than merely structural.

. Summary

$$\boxed{\partial_t \Gamma^{(2)}_{k,IJ} = \frac{1}{2}\,\mathrm{Tr}\Big[G_k\,\Gamma^{(4)}_{k,IJ}\,G_k\,\partial_t R_k\Big] - \mathrm{Tr}\Big[G_k\,\Gamma^{(3)}_{k,I}\,G_k\,\Gamma^{(3)}_{k,J}\,G_k\,\partial_t R_k\Big]\,.}$$

Hence

$$\boxed{\partial_t Z_{I,k} = \frac{\partial}{\partial p^2}\partial_t \Gamma^{(2)}_{k,II}(p)\bigg|_{p=0}, \qquad I \in \{T, O, D\}.}$$

The minimal diagrammatic sources are:

quartic self/mixed sector: $\lambda_T, \lambda_O, \lambda_D, \lambda_{TO}, \lambda_{TD}, \lambda_{OD}$,

quartic cubic anisotropy: $g_{\tau,k}\tau(\Phi)$,

derivative anisotropy insertion: $Y_k O_{\nabla,\mathrm{ani}}$,

A**-sector gauge/constraint contribution:** $\Gamma_{k,\mathrm{gf/proj}}$.

For each sector $I = T, O, D$, the one-loop flow receives contributions from:

$$\boxed{\text{(a) tadpole-type } \Gamma^{(4)}_{IIII}, \quad \text{(b) mixed tadpole-type } \Gamma^{(4)}_{IIJJ}, \quad \text{(c) sunset/bubble-type } \Gamma^{(3)}\Gamma^{(3)}, \quad \text{(d) gauge/constraint loops for } I =}$$

Important honesty clause: at the symmetric point $A = n = \chi = 0$, pure quartic tadpoles may renormalize mainly the mass terms, not directly the p^2-coefficient. Therefore nontrivial wavefunction flow may rely crucially on derivative vertices, gauge/constraint structure, or effective cubic vertices.

Accordingly,

$$\boxed{(M_Z)_{IJ} = (M_Z)^{(\lambda)}_{IJ} + (M_Z)^{(\tau)}_{IJ} + (M_Z)^{(\nabla\text{ani})}_{IJ} + (M_Z)^{(\text{gf/proj})}_{IJ}.}$$

So the structural part is now fixed. What remains is the actual one-loop tensor-contraction evaluation of these contributions.

SYMMETRIC-POINT SURVIVAL TABLE FOR $\Gamma^{(3)}$ AND $\Gamma^{(4)}$

1. Starting minimal truncation

We work with the TECT-adapted minimal soft-sector truncation

$$\Gamma_k[\Phi] = \int d^d x \left[\frac{Z_{T,k}}{2} \sum_{\alpha=1}^{2} (\partial_i A_\alpha)^2 + \frac{Z_{O,k}}{2} \sum_{a=1}^{2} (\partial_i n_a)^2 + \frac{Z_{D,k}}{2} \sum_{r=1}^{2} (\partial_i \chi_r)^2 + Y_k\, O_{\nabla,\text{ani}} + U_k(\rho,\tau) \right], \tag{391}$$

with

$$\Phi = (A_1, A_2, n_1, n_2, \chi_1, \chi_2), \qquad N_{\text{eff}} = 6,$$

$$\rho_T = \frac{1}{2} A^T A, \qquad \rho_O = \frac{1}{2} n^T n, \qquad \rho_D = \frac{1}{2} \chi^T \chi, \qquad \rho = \rho_T + \rho_O + \rho_D,$$

and

$$U_k(\rho,\tau) = r_{T,k}\rho_T + r_{O,k}\rho_O + r_{D,k}\rho_D + \frac{\lambda_{T,k}}{2}\rho_T^2 + \frac{\lambda_{O,k}}{2}\rho_O^2 + \frac{\lambda_{D,k}}{2}\rho_D^2 + \lambda_{TO,k}\rho_T\rho_O + \lambda_{TD,k}\rho_T\rho_D + \lambda_{OD,k}\rho_O\rho_D + g_{\tau,k}\tau(\Phi). \tag{392}$$

This is exactly the soft-sector FRG structure already identified in the uploaded derivations. :contentReference[oaicite:3]index=3

2. Symmetric point

We expand around the isotropic symmetric point

$$A = 0, \qquad n = 0, \qquad \chi = 0. \tag{393}$$

At this point,

$$\rho_T = \rho_O = \rho_D = \rho = 0, \qquad \tau(0) = 0.$$

We now ask: which functional derivatives

$$\Gamma^{(3)}_k\big|_{\Phi=0}, \qquad \Gamma^{(4)}_k\big|_{\Phi=0}$$

survive?

3. Quadratic sector: contributes only to $\Gamma^{(2)}$

The sectoral kinetic terms

$$\frac{Z_{T,k}}{2}(\partial A)^2, \qquad \frac{Z_{O,k}}{2}(\partial n)^2, \qquad \frac{Z_{D,k}}{2}(\partial \chi)^2$$

are purely quadratic in the fields. Therefore:

$$\Gamma^{(3)}_{\text{kin}}\big|_{\Phi=0} = 0, \qquad \Gamma^{(4)}_{\text{kin}}\big|_{\Phi=0} = 0. \tag{394}$$

The same holds for the linear mass sector

$$r_{T,k}\rho_T + r_{O,k}\rho_O + r_{D,k}\rho_D,$$

which is also purely quadratic:

$$\Gamma^{(3)}_{\text{mass}}\big|_{\Phi=0} = 0, \qquad \Gamma^{(4)}_{\text{mass}}\big|_{\Phi=0} = 0. \tag{395}$$

So the entire quadratic part contributes only to $\Gamma^{(2)}$.

4. Quartic sector: $\Gamma^{(4)}$ survives, $\Gamma^{(3)}$ vanishes

Every quartic monomial in (392) is homogeneous of degree four in the fields. Therefore:
a. Third derivative at the origin. For any quartic polynomial $Q_4(\Phi)$,

$$\partial^3 Q_4(\Phi)\big|_{\Phi=0} = 0.$$

Hence

$$\Gamma^{(3)}_{\text{quartic}}\big|_{\Phi=0} = 0. \tag{396}$$

b. Fourth derivative at the origin. For any nonzero quartic polynomial $Q_4(\Phi)$,

$$\partial^4 Q_4(\Phi)\big|_{\Phi=0} \neq 0$$

in general. Hence

$$\Gamma^{(4)}_{\text{quartic}}\big|_{\Phi=0} \neq 0 \tag{397}$$

for the self-couplings

$$\lambda_T, \ \lambda_O, \ \lambda_D,$$

the mixed quartic couplings

$$\lambda_{TO}, \ \lambda_{TD}, \ \lambda_{OD},$$

and also for the quartic cubic-anisotropy vertex

$$g_{\tau,k}\tau(\Phi),$$

provided the corresponding coefficient is nonzero.
Thus, at the symmetric point:

$$\boxed{\Gamma^{(3)}_{\text{pot}}\big|_{\Phi=0} = 0, \qquad \Gamma^{(4)}_{\text{pot}}\big|_{\Phi=0} \neq 0.} \tag{398}$$

5. Derivative-anisotropy insertion

The leading derivative anisotropy operator $O_{\nabla,\mathrm{ani}}$ is explicitly included in the truncation as a distinct coupling:

$$Y_k O_{\nabla,\mathrm{ani}}.$$

In the current minimal truncation, this operator is treated as a leading bilinear two-derivative anisotropic kinetic deformation, not as a higher-field interaction polynomial. This is exactly how the soft-sector FRG truncation is written in the document.

Under this minimal interpretation,

$$\Gamma^{(3)}_{\nabla\mathrm{ani}}\big|_{\Phi=0} = 0, \qquad \Gamma^{(4)}_{\nabla\mathrm{ani}}\big|_{\Phi=0} = 0, \tag{399}$$

while

$$\Gamma^{(2)}_{\nabla\mathrm{ani}}\big|_{\Phi=0} \neq 0$$

in general.

c. Honest caveat. If one later refines $O_{\nabla,\mathrm{ani}}$ into field-dependent derivative vertices beyond the bilinear level, then additional $\Gamma^{(3)}$ and/or $\Gamma^{(4)}$ pieces may survive. That is not part of the current minimal truncation.

6. Gauge/constraint sector

The A-sector requires a gauge-fixing or physical projection structure to define the Hessian and the projected loops unambiguously. The document explicitly states that the sign of M_Z depends on the gauge/constraint structure of the A-sector.

Therefore, in full honesty:

$$\Gamma^{(3)}_{\mathrm{gf/proj}}\big|_{\Phi=0}, \qquad \Gamma^{(4)}_{\mathrm{gf/proj}}\big|_{\Phi=0}$$

cannot be declared zero or nonzero until a concrete gauge/constraint realization is chosen.

7. Survival table

The symmetric-point survival table is therefore:

source term	$\Gamma^{(2)}\vert_{\Phi=0}$	$\Gamma^{(3)}\vert_{\Phi=0}$	$\Gamma^{(4)}\vert_{\Phi=0}$
$Z_{T,k}(\partial A)^2,\ Z_{O,k}(\partial n)^2,\ Z_{D,k}(\partial\chi)^2$	nonzero	0	0
$r_{T,k}\rho_T + r_{O,k}\rho_O + r_{D,k}\rho_D$	nonzero	0	0
$\frac{\lambda_{T,k}}{2}\rho_T^2,\ \frac{\lambda_{O,k}}{2}\rho_O^2,\ \frac{\lambda_{D,k}}{2}\rho_D^2$	0	0	nonzero
$\lambda_{TO,k}\rho_T\rho_O,\ \lambda_{TD,k}\rho_T\rho_D,\ \lambda_{OD,k}\rho_O\rho_D$	0	0	nonzero
$g_{\tau,k}\tau(\Phi)$	0	0	nonzero
$Y_k O_{\nabla,\mathrm{ani}}$ (minimal bilinear interpretation)	nonzero	0	0
$\Gamma_{k,\mathrm{gf/proj}}$	depends on implementation	undetermined	undetermined

8. Consequence for one-loop M_Z extraction

The formal one-loop flow of the two-point function is

$$\partial_t \Gamma^{(2)}_{k,IJ} = \frac{1}{2}\,\mathrm{Tr}\Big[G_k\,\Gamma^{(4)}_{k,IJ}\,G_k\,\partial_t R_k\Big] - \mathrm{Tr}\Big[G_k\,\Gamma^{(3)}_{k,I}\,G_k\,\Gamma^{(3)}_{k,J}\,G_k\,\partial_t R_k\Big]\,. \tag{400}$$

Since the minimal truncation at the symmetric point gives

$$\Gamma^{(3)}\big|_{\Phi=0} = 0$$

for the kinetic, mass, quartic-potential, and minimal derivative-anisotropy pieces, the sunset/bubble-type

$$\Gamma^{(3)}\Gamma^{(3)}$$

contribution vanishes *unless* it is regenerated by the gauge/constraint sector or by a refined field-dependent derivative truncation.

Therefore, in the current minimal truncation, the direct one-loop source for the two-point flow is predominantly the tadpole-type

$$\Gamma^{(4)}$$

sector.

d. Final honesty clause. Whether these tadpole-type terms actually contribute to

$$\frac{\partial}{\partial p^2}\partial_t \Gamma^{(2)}_{k,II}(p)\Big|_{p=0}$$

depends on the regulator, the momentum routing, the tensor contractions, and the gauge/projector structure. So this table fixes *which vertices survive*, but not yet *which of them give a nonzero wavefunction flow.* That is the next explicit contraction step.

. Summary

$$\boxed{\text{Symmetric point: } A = n = \chi = 0.}$$

At the current minimal TECT-adapted truncation:

quadratic kinetic + mass sector

$$\Gamma^{(2)}|_{\Phi=0} \neq 0, \qquad \Gamma^{(3)}|_{\Phi=0} = 0, \qquad \Gamma^{(4)}|_{\Phi=0} = 0.$$

quartic self/mixed potential sector

$$\Gamma^{(3)}|_{\Phi=0} = 0, \qquad \Gamma^{(4)}|_{\Phi=0} \neq 0.$$

quartic cubic anisotropy $g_{\tau,k}\tau(\Phi)$

$$\Gamma^{(3)}|_{\Phi=0} = 0, \qquad \Gamma^{(4)}|_{\Phi=0} \neq 0.$$

leading derivative anisotropy $Y_k O_{\nabla,\mathrm{ani}}$

under the current minimal bilinear interpretation,

$$\Gamma^{(2)}|_{\Phi=0} \neq 0, \qquad \Gamma^{(3)}|_{\Phi=0} = 0, \qquad \Gamma^{(4)}|_{\Phi=0} = 0.$$

A-sector gauge/constraint contribution

$\Gamma^{(2)}, \Gamma^{(3)}, \Gamma^{(4)}$ are implementation-dependent and cannot yet be fixed abstractly.

Therefore the survival table is:

source	$\Gamma^{(2)}\vert_{\Phi=0}$	$\Gamma^{(3)}\vert_{\Phi=0}$	$\Gamma^{(4)}\vert_{\Phi=0}$
kinetic	$\neq 0$	0	0
mass	$\neq 0$	0	0
quartic self/mixed	0	0	$\neq 0$
$g_{\tau,k}\tau(\Phi)$	0	0	$\neq 0$
$Y_k O_{\nabla,\mathrm{ani}}$	$\neq 0$	0	0
$\Gamma_{k,\mathrm{gf/proj}}$	depends	depends	depends

Hence, in the current minimal truncation, the dominant direct one-loop source for the two-point flow is the tadpole-type

$$\Gamma^{(4)}$$

sector, while the sunset/bubble

$$\Gamma^{(3)}\Gamma^{(3)}$$

sector vanishes unless regenerated by gauge/constraint structure or a refined derivative vertex truncation.

TADPOLE MOMENTUM-DERIVATIVE TEST TABLE FOR M_Z

1. Projection target

The kinetic-flow matrix M_Z is extracted from

$$\partial_t Z_{I,k} = \frac{\partial}{\partial p^2}\partial_t \Gamma^{(2)}_{k,II}(p)\bigg|_{p=0}, \qquad I \in \{T, O, D\}. \tag{401}$$

Therefore a one-loop diagram contributes to M_Z only if it generates a nontrivial dependence on the external momentum p in the diagonal two-point function.

2. One-loop two-point flow at the symmetric point

At the symmetric point

$$A = n = \chi = 0,$$

the minimal truncation survival table gives:

$$\Gamma^{(3)}|_{\Phi=0} = 0$$

for the quadratic sector, the quartic potential sector, and the minimal bilinear derivative-anisotropy insertion, while

$$\Gamma^{(4)}|_{\Phi=0} \neq 0$$

for the quartic self/mixed potential sector and for the quartic cubic-anisotropy vertex. Hence the surviving direct one-loop source is predominantly the tadpole-type term

$$\partial_t \Gamma^{(2)}_{k,II}(p) \supset \frac{1}{2}\,\mathrm{Tr}\Big[G_k(q)\,\Gamma^{(4)}_{k,II}(p,-p,q,-q)\,G_k(q)\,\partial_t R_k(q)\Big]. \tag{402}$$

3. Momentum-derivative test

We now test whether the surviving tadpole source contributes to

$$\partial_{p^2}\partial_t \Gamma^{(2)}_{k,II}(p)\big|_{p=0}.$$

a. Criterion. A tadpole contributes to Z_I only if the integrand in (402) has explicit nontrivial dependence on the external momentum p near $p = 0$.

If

$$\Gamma^{(4)}_{k,II}(p, -p, q, -q)$$

is momentum-independent (or depends only on loop momentum q but not on p), and if the regulated propagator factors in the tadpole topology also depend only on q, then

$$\frac{\partial}{\partial p^2} \partial_t \Gamma^{(2)}_{k,II}(p) \Big|_{p=0} = 0.$$

In that case the tadpole renormalizes only the mass-like term, not the wavefunction coefficient.

4. Tadpole momentum-derivative test table

surviving source	$\Gamma^{(4)}\vert_{\Phi=0} \neq 0$?	explicit p-dependence in the vertex?	∂_{p^2} contributio
$\lambda_T \rho_T^2,\ \lambda_O \rho_O^2,\ \lambda_D \rho_D^2$	yes	no (local quartic)	generically n
$\lambda_{TO} \rho_T \rho_O,\ \lambda_{TD} \rho_T \rho_D,\ \lambda_{OD} \rho_O \rho_D$	yes	no (local quartic)	generically n
$g_{\tau,k} \tau(\Phi)$	yes	no, if τ is local quartic only	generically n
$Y_k O_{\nabla,\mathrm{ani}}$ (minimal bilinear interpretation)	$\Gamma^{(4)} = 0$	not applicable	no tadpole sou
$\Gamma_{k,\mathrm{gf/proj}}$	implementation-dependent	possibly yes	possibly yes
refined field-dependent derivative vertex	possibly yes	yes	yes
background-generated effective cubic vertices	$\Gamma^{(3)} \neq 0$ if background $\neq 0$	yes via sunset/bubble	yes

5. Formal consequence

The test table implies:

1. In the current minimal symmetric-point truncation, the surviving local quartic tadpoles are generically p^2-silent. They renormalize the masses r_T, r_O, r_D, and possibly anisotropic mass splittings, but not necessarily the wavefunction factors Z_T, Z_O, Z_D.
2. Therefore, the one-loop sign of M_Z is not expected to be fixed by the local quartic potential sector alone.
3. A nontrivial p^2-flow must come from one of the following:
   (a) field-dependent derivative vertices beyond the minimal bilinear $Y_k O_{\nabla,\mathrm{ani}}$ truncation,
   (b) gauge/constraint-sector loops in the A-sector,
   (c) expansion around a nonzero background generating effective cubic vertices and hence nonzero $\Gamma^{(3)}\Gamma^{(3)}$ contributions.

6. Honest theorem-level statement

Thus the correct theorem-level conclusion is:

$$\boxed{\text{Within the current minimal symmetric-point truncation, surviving local quartic tadpoles are insufficient in general to fix } M_Z.} \tag{403}$$

Equivalently, the current truncation is structurally sufficient to define the M_Z problem, but not yet dynamically rich enough to guarantee a nontrivial wavefunction-anisotropy flow.

7. What must be refined next

The next refinement must therefore enlarge the truncation in one of two ways:

b. Route A: derivative-vertex completion Promote

$$Y_k O_{\nabla,\mathrm{ani}}$$

from a purely bilinear insertion to a field-dependent derivative interaction sector so that nontrivial momentum-dependent $\Gamma^{(3)}$ and/or $\Gamma^{(4)}$ survive at the symmetric point.

c. Route B: background-field completion Perform the FRG expansion not around the strict symmetric point $\Phi = 0$, but around the relevant background that generates effective cubic vertices. Then the $\Gamma^{(3)}\Gamma^{(3)}$ contribution in the two-point flow can become nonzero and feed the sectoral Z_I-flows.

d. Gauge-sector completion In either route, the A-sector gauge/constraint projection must be implemented explicitly, since the document states that the sign of M_Z depends on it.

. Summary

$$\boxed{\partial_t Z_{I,k} = \frac{\partial}{\partial p^2}\partial_t \Gamma^{(2)}_{k,II}(p)\Big|_{p=0}, \qquad I \in \{T, O, D\}.}$$

So a surviving one-loop diagram contributes to M_Z only if it carries nontrivial external momentum dependence.

At the symmetric point $A = n = \chi = 0$, in the current minimal truncation:

$$\Gamma^{(3)}|_{\Phi=0} = 0$$

for the kinetic, mass, quartic-potential, and minimal bilinear derivative-anisotropy sectors, while

$$\Gamma^{(4)}|_{\Phi=0} \neq 0$$

for the local quartic self/mixed couplings and for $g_{\tau,k}\tau(\Phi)$.

However, these surviving quartic vertices are local and generically do not carry explicit external momentum p. Therefore their tadpole contribution is typically

$$\boxed{\partial_{p^2}\partial_t \Gamma^{(2)}_{k,II}(p)\Big|_{p=0} = 0,}$$

so they feed mainly mass renormalization, not wavefunction renormalization.

Hence:

$$\boxed{\text{the current minimal symmetric-point truncation is generally insufficient to determine } M_Z.}$$

A nontrivial M_Z-flow requires at least one of:

$$\boxed{\text{(i) field-dependent derivative vertices,} \quad \text{(ii) explicit gauge/constraint loops,} \quad \text{(iii) background-generated effective cubic vertices.}}$$

So the next necessary refinement is not another potential term, but a truncation refinement that generates genuine external-momentum dependence in the two-point flow.

BACKGROUND-FIELD FRG COMPLETION FOR A NONTRIVIAL KINETIC-FLOW MATRIX M_Z

1. Why the symmetric-point truncation is insufficient

At the strict symmetric point

$$\Phi = 0, \qquad \Phi = (A_1, A_2, n_1, n_2, \chi_1, \chi_2),$$

the current minimal TECT-adapted truncation has the following schematic property:

- the kinetic and mass sectors are purely quadratic, hence contribute only to $\Gamma^{(2)}$;
- the quartic potential sector contributes nonzero $\Gamma^{(4)}$, but $\Gamma^{(3)}|_{\Phi=0} = 0$;
- the minimal bilinear derivative-anisotropy insertion contributes to $\Gamma^{(2)}$, but not to $\Gamma^{(3)}$ or $\Gamma^{(4)}$.

Therefore the sunset/bubble contribution

$$\Gamma^{(3)} G \Gamma^{(3)}$$

vanishes at the symmetric point, and the surviving local quartic tadpoles are generically p^2-silent, contributing mainly to mass renormalization rather than to wavefunction renormalization. Hence the strict symmetric-point truncation is generally insufficient to generate a nontrivial kinetic-flow matrix M_Z.

2. Background-field completion idea

To generate nontrivial momentum-dependent two-point flows, introduce a nonzero background

$$\bar{\Phi} = (\bar{A}, \bar{n}, \bar{\chi}),$$

and decompose

$$\Phi = \bar{\Phi} + \varphi, \tag{404}$$

where φ denotes the fluctuation field.

Expanding the action around $\bar{\Phi}$, one obtains the background-dependent vertices

$$\Gamma_k^{(2)}[\bar{\Phi}], \qquad \Gamma_k^{(3)}[\bar{\Phi}], \qquad \Gamma_k^{(4)}[\bar{\Phi}], \ \ldots$$

Even if the original truncation contains only quartic local interactions, the expansion around $\bar{\Phi} \neq 0$ generates effective cubic fluctuation vertices:

$$\Gamma_k^{(3)}[\bar{\Phi}] \neq 0. \tag{405}$$

This reactivates the bubble/sunset topology in the two-point flow.

3. Background-field Wetterich equation

The background-field effective average action is taken as

$$\Gamma_k[\bar{\Phi}, \varphi] = \Gamma_k[\bar{\Phi} + \varphi], \tag{406}$$

with flow

$$\partial_t \Gamma_k[\bar{\Phi}, \varphi] = \frac{1}{2} \mathrm{Tr} \left[\left(\Gamma_k^{(2,0)}[\bar{\Phi}, \varphi] + R_k[\bar{\Phi}] \right)^{-1} \partial_t R_k[\bar{\Phi}] \right], \tag{407}$$

where $\Gamma_k^{(2,0)}$ denotes the fluctuation Hessian at fixed background.

A background-adapted regulator $R_k[\bar{\Phi}]$ is required so that:

1. the fluctuation spectrum is regularized in the correct background sector,
2. the A-sector gauge/constraint projection is implemented on the physical fluctuation subspace,
3. the sectoral kinetic projection remains compatible with the background decomposition.

4. Minimal background choice

The orthodox minimal choice is not a completely arbitrary $\bar{\Phi}$, but a background adapted to the shell/defect structure already present in the TECT derivation. The natural options are:

a. *(a) Uniform soft-sector background*

$$\bar{\Phi} = (\bar{A}, \bar{n}, \bar{\chi}) = \text{const.}$$

This is algebraically simplest, but may still fail to generate the physically relevant momentum structure if it ignores the shell geometry.

b. *(b) Shell-adapted background* Choose $\bar{\Phi}$ aligned with the selected shell pair $\pm G_*$, so that the fluctuation Hessian is already adapted to the TECT shell geometry.

c. *(c) Defect-adapted background* Use the actual defect/worldvolume background already appearing in the local transport derivation, so that the fluctuation sectors inherit the odd/even selection structure and the transport kernels $V^\mu(\xi)$.

Among these, the most faithful TECT choice is the shell-/defect-adapted background, because the Dirac-emergence program already singles out a defect/shell geometry rather than the strict homogeneous vacuum.

5. Generation of effective cubic vertices

Let the quartic sector schematically contain

$$\lambda_{IJ}\, \rho_I \rho_J, \qquad g_{\tau,k} \tau(\Phi), \qquad Y_k O_{\nabla,\mathrm{ani}}[\Phi].$$

Expanding around $\Phi = \bar{\Phi} + \varphi$, one gets terms of the form

$$\Gamma_k[\bar{\Phi} + \varphi] = \Gamma_k[\bar{\Phi}] + \Gamma_k^{(1)}[\bar{\Phi}]\varphi + \frac{1}{2}\Gamma_k^{(2)}[\bar{\Phi}]\varphi^2 + \frac{1}{3!}\Gamma_k^{(3)}[\bar{\Phi}]\varphi^3 + \frac{1}{4!}\Gamma_k^{(4)}[\bar{\Phi}]\varphi^4 + \cdots . \tag{408}$$

In particular:

- quartic local terms produce cubic fluctuation vertices proportional to $\bar{\Phi}$,
- derivative anisotropy terms may produce momentum-dependent cubic fluctuation vertices proportional to $\bar{\Phi}$,
- gauge/constraint terms may produce background-dependent projected vertices in the A-sector.

Thus $\Gamma_k^{(3)}[\bar{\Phi}]$ becomes nonzero and can source nontrivial p^2-flows in the two-point functions.

6. Two-point flow in the background-field scheme

The fluctuation two-point flow now has the formal structure

$$\partial_t \Gamma_{k,II}^{(2)}(p;\bar{\Phi}) = \frac{1}{2}\,\mathrm{Tr}\Big[G_k(\bar{\Phi})\,\Gamma_{k,II}^{(4)}(\bar{\Phi})\,G_k(\bar{\Phi})\,\partial_t R_k(\bar{\Phi})\Big] - \mathrm{Tr}\Big[G_k(\bar{\Phi})\,\Gamma_{k,I}^{(3)}(\bar{\Phi})\,G_k(\bar{\Phi})\,\Gamma_{k,I}^{(3)}(\bar{\Phi})\,G_k(\bar{\Phi})\,\partial_t R_k(\bar{\Phi})\Big]. \tag{409}$$

Unlike the strict symmetric-point truncation, the second term is now generally nonzero.

Therefore the sectoral wavefunction flows are defined by

$$\partial_t Z_{I,k}(\bar{\Phi}) = \left. \frac{\partial}{\partial p^2} \partial_t \Gamma_{k,II}^{(2)}(p;\bar{\Phi}) \right|_{p=0}, \qquad I \in \{T, O, D\}. \tag{410}$$

7. Definition of the background-dependent M_Z

Write

$$Z_{T,k} = Z_k(1+\delta_T), \qquad Z_{O,k} = Z_k(1+\delta_O), \qquad Z_{D,k} = Z_k(1+\delta_D),$$

with the traceless constraint

$$2\delta_T + 2\delta_O + 2\delta_D = 0.$$

Then, at fixed admissible background $\bar{\Phi}$, define

$$\partial_t \begin{pmatrix} \delta_T \\ \delta_O \\ \delta_D \end{pmatrix} = M_Z(\bar{\Phi}) \begin{pmatrix} \delta_T \\ \delta_O \\ \delta_D \end{pmatrix} + O(\delta^2), \tag{411}$$

with

$$(M_Z(\bar{\Phi}))_{IJ} = \frac{\partial}{\partial \delta_J} \left(\partial_t \delta_I(\bar{\Phi})\right)\Big|_{\delta=0}. \tag{412}$$

The physical question is then the sign of the nontrivial eigenvalues of $M_Z(\bar{\Phi})$ on the traceless subspace.

8. Honest gain and honest limitation

The background-field completion achieves the following:

1. it explains why the strict symmetric-point truncation can miss the Z_I-flow;
2. it reactivates effective cubic vertices;
3. it makes the sunset/bubble topology available;
4. it provides a mathematically honest mechanism for obtaining a nonzero M_Z.

However, it still does not determine the sign of M_Z automatically. To do that one must still specify:

1. the explicit admissible background $\bar{\Phi}$,
2. the background-adapted regulator $R_k[\bar{\Phi}]$,
3. the projected A-sector Hessian,
4. the actual tensor contractions in the one-loop flow.

9. Final roadmap statement

Therefore the correct next-stage roadmap is:

$$\boxed{\bar{\Phi} \neq 0 \implies \Gamma_k^{(3)}[\bar{\Phi}] \neq 0 \implies \text{nontrivial } \Gamma^{(3)} G \Gamma^{(3)} \text{ contribution} \implies \partial_t Z_T, \partial_t Z_O, \partial_t Z_D \implies M_Z(\bar{\Phi}).} \tag{413}$$

This is the orthodox background-field completion of the TECT M_Z-problem.

. Summary

$$\boxed{\text{Strict symmetric-point truncation can miss } M_Z \text{ because } \Gamma^{(3)}|_{\Phi=0} = 0.}$$

Hence the orthodox refinement is a background-field completion:

$$\boxed{\Phi = \bar{\Phi} + \varphi, \qquad \bar{\Phi} = (\bar{A}, \bar{n}, \bar{\chi}) \neq 0.}$$

Expanding around $\bar{\Phi} \neq 0$ generates effective cubic vertices

$$\boxed{\Gamma_k^{(3)}[\bar{\Phi}] \neq 0,}$$

so the bubble/sunset topology

$$\Gamma^{(3)}G\Gamma^{(3)}$$

becomes active in the two-point flow.

The background-field flow is

$$\boxed{\partial_t\Gamma_k[\bar{\Phi},\varphi] = \frac{1}{2}\mathrm{Tr}\left[(\Gamma_k^{(2,0)}[\bar{\Phi},\varphi] + R_k[\bar{\Phi}])^{-1}\partial_t R_k[\bar{\Phi}]\right].}$$

Then

$$\boxed{\partial_t Z_{I,k}(\bar{\Phi}) = \frac{\partial}{\partial p^2}\partial_t\Gamma^{(2)}_{k,II}(p;\bar{\Phi})\bigg|_{p=0}, \qquad I\in\{T,O,D\},}$$

and

$$\boxed{(M_Z(\bar{\Phi}))_{IJ} = \frac{\partial}{\partial\delta_J}(\partial_t\delta_I(\bar{\Phi}))\bigg|_{\delta=0}.}$$

Thus the background-field route provides the first honest mechanism for a genuinely nonzero M_Z.

The remaining concrete choices are:

$$\boxed{\text{(i) choose an admissible shell/defect-adapted background } \bar{\Phi},}$$

$$\boxed{\text{(ii) fix } R_k[\bar{\Phi}], \quad \text{(iii) fix the } A\text{-sector projector}, \quad \text{(iv) perform the one-loop tensor contractions.}}$$

MINIMAL ADMISSIBLE BACKGROUND FOR THE BACKGROUND-FIELD M_Z COMPUTATION

1. Selection principle

The purpose of the background-field completion is to generate nonzero effective cubic fluctuation vertices

$$\Gamma_k^{(3)}[\bar{\Phi}] \neq 0$$

so that the bubble/sunset topology

$$\Gamma^{(3)}G\Gamma^{(3)}$$

contributes nontrivially to the sectoral wavefunction flows

$$\partial_t Z_T, \qquad \partial_t Z_O, \qquad \partial_t Z_D.$$

A useful background $\bar{\Phi}$ must therefore satisfy two conditions:

1. it must be *admissible*, i.e. compatible with the actual shell/defect geometry already singled out by the TECT Dirac-emergence analysis;

2. it must be *nontrivial enough* to generate effective cubic fluctuation vertices after expansion

$$\Phi = \bar{\Phi} + \varphi.$$

2. Why a constant homogeneous background is not the preferred choice

A purely constant background

$$\bar{\Phi} = \text{const.}$$

is algebraically simple, but it ignores the two geometric structures that the present TECT derivation actually uses:

1. the selected shell pair $\pm G_*$,
2. the CP^2 defect / projector background $P_B(\xi)$.

Since the local transport kernel and the projected Dirac symbol are already derived from the defect background and its odd derivative structure, a homogeneous constant background is not the most faithful choice for the M_Z computation.

3. Defect-adapted background data already established

The projector-defect sector is built from a smooth rank-one projector

$$P_B(\xi) = z_B(\xi) z_B(\xi)^\dagger, \qquad P_B(-\xi) = P_B(\xi), \tag{414}$$

with tangent fluctuations

$$\delta P = \eta z_B^\dagger + z_B \eta^\dagger, \qquad z_B^\dagger \eta = 0, \qquad \eta(\xi, y) \in Q_B(\xi) := (\mathbb{C} z_B(\xi))^\perp. \tag{415}$$

The quadratic fluctuation operator is a matrix Schrödinger operator on the rank-two bundle Q_B, and the transport kernel is induced by the odd derivative of the background:

$$L_1 = -i V^\mu(\xi) \partial_{y^\mu}, \qquad V^\mu(-\xi) = -V^\mu(\xi). \tag{416}$$

Therefore the actual low-energy fluctuation problem is already background-adapted at the projector/defect level. :contentReference[oaicite:7]index=7 :contentReference[oaicite:8]index=8

4. Minimal admissible background

The orthodox minimal choice is therefore a shell-/defect-adapted background rather than a homogeneous vacuum. The smallest such choice is:

$$\boxed{\bar{\Phi}_{\min} = (\bar{A}, \bar{n}, \bar{\chi}) = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi)),} \tag{417}$$

where:

- $\bar{n}_B(\xi)$ is the defect-adapted orientational background in the low-energy CP^2/projector sector,
- $\bar{\chi}_B(\xi)$ is the corresponding defect-condensate hybrid background if the hybrid channels are retained in the minimal soft sector,
- the gauge sector is initially set to $\bar{A} = 0$ unless an explicit background gauge field is required by the A-sector constraint analysis.

A slightly stronger but still admissible choice is

$$\boxed{\bar{\Phi}_{\min}^{\text{full}} = (\bar{A}_B(\xi), \bar{n}_B(\xi), \bar{\chi}_B(\xi)),} \tag{418}$$

where $\bar{A}_B(\xi)$ is the background gauge/connection field induced by the defect projector if one wants the A-sector projection already built into the background.

5. Why this is minimal

This choice is minimal in the following precise sense:

1. it preserves the actual shell-/defect-adapted geometry used in the Dirac-emergence construction;
2. it is one-dimensional in the transverse coordinate ξ, hence simpler than a fully space-dependent background;
3. it is nontrivial enough to generate effective cubic vertices after expansion
$$\Phi = \bar{\Phi} + \varphi;$$
4. it keeps the reflection-parity structure
$$P_B(-\xi) = P_B(\xi), \qquad V^\mu(-\xi) = -V^\mu(\xi),$$
which is already central in the low-mode transport selection rules.

Thus (417) is the minimal admissible background that remains faithful to the present TECT derivation.

6. Background-generated vertices

Expanding
$$\Phi = \bar{\Phi}_{\min} + \varphi$$
in the minimal vertex-complete FRG truncation generates:

- a background-dependent fluctuation Hessian
$$\Gamma_k^{(2)}[\bar{\Phi}_{\min}],$$
- effective cubic vertices
$$\Gamma_k^{(3)}[\bar{\Phi}_{\min}] \neq 0,$$
coming from the quartic potential and derivative-anisotropy sectors,
- a background-dependent projected propagator
$$G_k(\bar{\Phi}_{\min}) = (\Gamma_k^{(2)}[\bar{\Phi}_{\min}] + R_k[\bar{\Phi}_{\min}])^{-1}.$$

Therefore the two-point flow acquires the nontrivial bubble contribution
$$\Gamma^{(3)}[\bar{\Phi}_{\min}]\, G_k(\bar{\Phi}_{\min})\, \Gamma^{(3)}[\bar{\Phi}_{\min}],$$
which is the minimal mechanism for generating a nonzero M_Z.

7. Admissibility conditions

To be fully acceptable, the background $\bar{\Phi}_{\min}$ should satisfy:

1. **shell compatibility:** it must be adapted to the selected shell pair $\pm G_*$;
2. **defect compatibility:** it must descend from the actual defect projector background $P_B(\xi)$;
3. **reflection structure:** the induced transport kernel should retain the oddness selection rule;
4. **constraint compatibility:** the A-sector must still admit a well-defined gauge/constraint projection;
5. **spectral admissibility:** the background Hessian must preserve the isolated low-energy sector on which the M_Z projection is to be performed.

8. Final roadmap consequence

Once $\bar{\Phi}_{\min}$ is fixed, the M_Z-computation chain becomes

$$\boxed{\bar{\Phi}_{\min} \Longrightarrow \Gamma_k^{(2)}[\bar{\Phi}_{\min}],\ \Gamma_k^{(3)}[\bar{\Phi}_{\min}],\ \Gamma_k^{(4)}[\bar{\Phi}_{\min}] \Longrightarrow \partial_t Z_T, \partial_t Z_O, \partial_t Z_D \Longrightarrow M_Z(\bar{\Phi}_{\min}).} \tag{419}$$

Thus the problem is no longer to invent a background from scratch. It is to use the defect/shell background that the TECT Dirac-emergence program has already singled out.

9. Honest status

The exact current status is:

$$\boxed{\text{The minimal admissible background can be identified structurally.}}$$

What is not yet done is:

1. writing the full background-dependent Hessian explicitly,
2. constructing the corresponding background-adapted regulator,
3. evaluating the one-loop tensor contractions.

So the next step is not conceptual background selection, but explicit background-Hessian construction.

. Summary

$$\boxed{\text{Minimal admissible background for } M_Z \text{ should be shell-/defect-adapted, not an arbitrary constant background.}}$$

The actual TECT low-energy geometry is already built on the defect projector background

$$P_B(\xi) = z_B(\xi) z_B(\xi)^\dagger, \qquad P_B(-\xi) = P_B(\xi),$$

with tangent fluctuations

$$\eta(\xi, y) \in Q_B(\xi) := (\mathbb{C} z_B(\xi))^\perp$$

and odd transport kernel

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \qquad V^\mu(-\xi) = -V^\mu(\xi).$$

Hence the orthodox minimal background is

$$\boxed{\bar{\Phi}_{\min} = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi)),}$$

or, if the gauge sector is included explicitly,

$$\boxed{\bar{\Phi}_{\min}^{\text{full}} = (\bar{A}_B(\xi), \bar{n}_B(\xi), \bar{\chi}_B(\xi)).}$$

This choice is minimal because it:

$$\boxed{\text{(i) preserves shell/defect geometry,} \quad \text{(ii) keeps reflection selection structure,} \quad \text{(iii) generates } \Gamma^{(3)}[\bar{\Phi}] \neq 0.}$$

After the background split

$$\Phi = \bar{\Phi}_{\min} + \varphi,$$

one obtains

$$\Gamma_k^{(2)}[\bar{\Phi}_{\min}], \qquad \Gamma_k^{(3)}[\bar{\Phi}_{\min}], \qquad \Gamma_k^{(4)}[\bar{\Phi}_{\min}],$$

and therefore a genuinely nontrivial two-point flow via

$$\Gamma^{(3)} G \Gamma^{(3)}.$$

So the computation chain becomes

$$\boxed{\bar{\Phi}_{\min} \Longrightarrow \Gamma_k^{(2)}[\bar{\Phi}_{\min}], \Gamma_k^{(3)}[\bar{\Phi}_{\min}], \Gamma_k^{(4)}[\bar{\Phi}_{\min}] \Longrightarrow \partial_t Z_T, \partial_t Z_O, \partial_t Z_D \Longrightarrow M_Z(\bar{\Phi}_{\min}).}$$

Thus the background-selection problem is essentially closed. The next real task is the explicit construction of the background-dependent Hessian and regulator.

BACKGROUND-DEPENDENT BLOCK HESSIAN FOR $\bar{\Phi}_{\min}$

1. Background split

We choose the minimal admissible shell-/defect-adapted background

$$\bar{\Phi}_{\min}(\xi) = (\bar{A}, \bar{n}, \bar{\chi}) = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi)), \tag{420}$$

and expand

$$\Phi = \bar{\Phi}_{\min} + \varphi, \qquad \varphi = (a, \nu, \zeta). \tag{421}$$

Thus

$$A = a, \qquad n = \bar{n}_B + \nu, \qquad \chi = \bar{\chi}_B + \zeta.$$

Here:

- a is the fluctuation in the A-sector (with gauge/constraint projection still to be imposed),
- ν is the orientational fluctuation around $\bar{n}_B(\xi)$,
- ζ is the defect-condensate / hybrid fluctuation around $\bar{\chi}_B(\xi)$.

2. Background-adapted truncation

The minimal vertex-complete truncation is taken in the form

$$\Gamma_k^{\mathrm{VC}}[A, n, \chi] = \Gamma_{k,\mathrm{kin}} + \Gamma_{k,\mathrm{pot}} + \Gamma_{k,\nabla\mathrm{ani}} + \Gamma_{k,\mathrm{gf/proj}}, \tag{422}$$

with

$$\Gamma_{k,\mathrm{kin}} = \frac{1}{2} \int d^d x \Big[Z_{T,k} (\partial_i A)^T (\partial_i A) + Z_{O,k} (\partial_i n)^T (\partial_i n) + Z_{D,k} (\partial_i \chi)^T (\partial_i \chi) \Big], \tag{423}$$

$$\begin{aligned} \Gamma_{k,\mathrm{pot}} = \int d^d x \Big[& r_{T,k}\rho_T + r_{O,k}\rho_O + r_{D,k}\rho_D + \frac{\lambda_{T,k}}{2}\rho_T^2 + \frac{\lambda_{O,k}}{2}\rho_O^2 + \frac{\lambda_{D,k}}{2}\rho_D^2 \\ & + \lambda_{TO,k}\rho_T\rho_O + \lambda_{TD,k}\rho_T\rho_D + \lambda_{OD,k}\rho_O\rho_D + g_{\tau,k}\tau(A, n, \chi) \Big], \end{aligned} \tag{424}$$

and

$$\Gamma_{k,\nabla\mathrm{ani}} = Y_k \int d^d x \, O_{\nabla,\mathrm{ani}}[A, n, \chi]. \tag{425}$$

The sectoral quadratic invariants are

$$\rho_T = \frac{1}{2} A^T A, \qquad \rho_O = \frac{1}{2} n^T n, \qquad \rho_D = \frac{1}{2} \chi^T \chi.$$

3. Definition of the background-dependent Hessian

The fluctuation Hessian at fixed background is

$$\Gamma_k^{(2)}[\bar{\Phi}_{\min}] = \left.\frac{\delta^2 \Gamma_k^{\mathrm{VC}}[\bar{\Phi}_{\min} + \varphi]}{\delta\varphi\,\delta\varphi}\right|_{\varphi=0}. \tag{426}$$

With respect to the sector decomposition $\varphi = (a, \nu, \zeta)$, it has the 3×3 block form

$$\boxed{\Gamma_k^{(2)}[\bar{\Phi}_{\min}] = \begin{pmatrix} \mathcal{H}_{TT} & \mathcal{H}_{TO} & \mathcal{H}_{TD} \\ \mathcal{H}_{OT} & \mathcal{H}_{OO} & \mathcal{H}_{OD} \\ \mathcal{H}_{DT} & \mathcal{H}_{DO} & \mathcal{H}_{DD} \end{pmatrix}.} \tag{427}$$

Each entry is now a ξ-dependent matrix differential operator on the worldvolume fluctuation bundle.

4. Diagonal blocks

4.1 TT-block

Since $\bar{A} = 0$, the A-sector diagonal block is

$$\mathcal{H}_{TT} = -Z_{T,k}\,\partial^2 + \mathcal{M}_{TT}^{\mathrm{pot}}(\xi) + Y_k\,\mathcal{K}_{TT}^{\nabla\mathrm{ani}}(\xi, \partial) + \mathcal{G}_A(\xi, \partial), \tag{428}$$

where

$$\mathcal{M}_{TT}^{\mathrm{pot}}(\xi) = r_{T,k} + \lambda_{TO,k}\rho_O^B(\xi) + \lambda_{TD,k}\rho_D^B(\xi) + g_{\tau,k}\,\left.\frac{\partial^2\tau}{\partial A\,\partial A}\right|_{\bar{\Phi}_{\min}}, \tag{429}$$

with

$$\rho_O^B(\xi) = \frac{1}{2}\bar{n}_B^T\bar{n}_B, \qquad \rho_D^B(\xi) = \frac{1}{2}\bar{\chi}_B^T\bar{\chi}_B.$$

The term $\mathcal{G}_A$ denotes the gauge-fixing / constraint projection contribution.

4.2 OO-block

The orientational diagonal block is

$$\mathcal{H}_{OO} = -Z_{O,k}\,\partial^2 + \mathcal{M}_{OO}^{\mathrm{pot}}(\xi) + Y_k\,\mathcal{K}_{OO}^{\nabla\mathrm{ani}}(\xi, \partial) + \mathcal{T}_{OO}(\xi, \partial_y), \tag{430}$$

where

$$\begin{aligned}\mathcal{M}_{OO}^{\mathrm{pot}}(\xi) = &\Big[r_{O,k} + 3\lambda_{O,k}\rho_O^B(\xi) + \lambda_{OD,k}\rho_D^B(\xi)\Big]\mathbf{1}_2 + \lambda_{O,k}\,\bar{n}_B(\xi)\bar{n}_B(\xi)^T \\ &+ g_{\tau,k}\,\left.\frac{\partial^2\tau}{\partial n\,\partial n}\right|_{\bar{\Phi}_{\min}}.\end{aligned} \tag{431}$$

The transport term $\mathcal{T}_{OO}$ is the background-induced first-order operator descending from the odd kernel $V^\mu(\xi)\partial_{y^\mu}$.

4.3 DD-block

Similarly,

$$\mathcal{H}_{DD} = -Z_{D,k}\,\partial^2 + \mathcal{M}_{DD}^{\mathrm{pot}}(\xi) + Y_k\,\mathcal{K}_{DD}^{\nabla\mathrm{ani}}(\xi, \partial) + \mathcal{T}_{DD}(\xi, \partial_y), \tag{432}$$

with

$$\begin{aligned}\mathcal{M}_{DD}^{\mathrm{pot}}(\xi) = &\Big[r_{D,k} + 3\lambda_{D,k}\rho_D^B(\xi) + \lambda_{OD,k}\rho_O^B(\xi)\Big]\mathbf{1}_2 + \lambda_{D,k}\,\bar{\chi}_B(\xi)\bar{\chi}_B(\xi)^T \\ &+ g_{\tau,k}\,\left.\frac{\partial^2\tau}{\partial\chi\,\partial\chi}\right|_{\bar{\Phi}_{\min}}.\end{aligned} \tag{433}$$

5. Off-diagonal blocks

The essential gain of the background-field completion is that the mixed blocks no longer vanish generically.

5.1 TO-block

$$\mathcal{H}_{TO} = g_{\tau,k}\, \frac{\partial^2\tau}{\partial A\,\partial n}\bigg|_{\bar{\Phi}_{\min}} + Y_k\, \mathcal{K}_{TO}^{\nabla\mathrm{ani}}(\xi,\partial). \tag{434}$$

Because $\bar{A}=0$, no $\lambda_{TO}\, An^T$ term survives directly in the Hessian itself, but g_τ- and derivative-anisotropy-induced mixing may survive on the nontrivial background.

5.2 TD-block

$$\mathcal{H}_{TD} = g_{\tau,k}\, \frac{\partial^2\tau}{\partial A\,\partial\chi}\bigg|_{\bar{\Phi}_{\min}} + Y_k\, \mathcal{K}_{TD}^{\nabla\mathrm{ani}}(\xi,\partial). \tag{435}$$

5.3 OD-block

Here the quartic mixed coupling is activated directly by the background:

$$\mathcal{H}_{OD} = \lambda_{OD,k}\, \bar{n}_B(\xi)\bar{\chi}_B(\xi)^T + g_{\tau,k}\, \frac{\partial^2\tau}{\partial n\,\partial\chi}\bigg|_{\bar{\Phi}_{\min}} + Y_k\, \mathcal{K}_{OD}^{\nabla\mathrm{ani}}(\xi,\partial). \tag{436}$$

Thus the O- and D-sector mixing is generically nonzero on the background.

The Hermitian transpose relations hold:

$$\mathcal{H}_{OT} = \mathcal{H}_{TO}^\dagger, \qquad \mathcal{H}_{DT} = \mathcal{H}_{TD}^\dagger, \qquad \mathcal{H}_{DO} = \mathcal{H}_{OD}^\dagger.$$

6. Background-generated cubic vertices

Expanding $\Gamma_k[\bar{\Phi}_{\min}+\varphi]$ to third order in φ, one obtains

$$\Gamma_k^{(3)}[\bar{\Phi}_{\min}] \neq 0.$$

Schematically:

- quartic self/mixed couplings produce cubic fluctuation vertices proportional to $\bar{n}_B(\xi)$ or $\bar{\chi}_B(\xi)$,
- $g_{\tau,k}\tau$ produces anisotropic cubic fluctuation vertices proportional to the background,
- $Y_k O_{\nabla,\mathrm{ani}}$ may produce momentum-dependent cubic fluctuation vertices on the background.

Therefore the bubble/sunset contribution

$$\Gamma^{(3)}[\bar{\Phi}_{\min}]\, G_k(\bar{\Phi}_{\min})\, \Gamma^{(3)}[\bar{\Phi}_{\min}]$$

is now generally nonzero.

7. Background-adapted propagator and sectoral projection

Define the background propagator

$$G_k(\bar{\Phi}_{\min}) = \left(\Gamma_k^{(2)}[\bar{\Phi}_{\min}] + R_k[\bar{\Phi}_{\min}]\right)^{-1}. \tag{437}$$

Then the background two-point flow is

$$\begin{aligned}\partial_t\Gamma^{(2)}_{k,II}(p;\bar{\Phi}_{\min}) = {} & \frac{1}{2}\operatorname{Tr}\Big[G_k(\bar{\Phi}_{\min})\,\Gamma^{(4)}_{k,II}[\bar{\Phi}_{\min}]\,G_k(\bar{\Phi}_{\min})\,\partial_t R_k[\bar{\Phi}_{\min}]\Big] \\ & - \operatorname{Tr}\Big[G_k(\bar{\Phi}_{\min})\,\Gamma^{(3)}_{k,I}[\bar{\Phi}_{\min}]\,G_k(\bar{\Phi}_{\min})\,\Gamma^{(3)}_{k,I}[\bar{\Phi}_{\min}]\,G_k(\bar{\Phi}_{\min})\,\partial_t R_k[\bar{\Phi}_{\min}]\Big].\end{aligned} \tag{438}$$

The sectoral wavefunction flows are then projected by

$$\partial_t Z_{I,k}(\bar{\Phi}_{\min}) = \frac{\partial}{\partial p^2}\partial_t\Gamma^{(2)}_{k,II}(p;\bar{\Phi}_{\min})\Big|_{p=0}, \qquad I \in \{T, O, D\}. \tag{439}$$

8. Final formal form of the M_Z problem

Writing

$$Z_{T,k} = Z_k(1+\delta_T), \qquad Z_{O,k} = Z_k(1+\delta_O), \qquad Z_{D,k} = Z_k(1+\delta_D), \qquad 2\delta_T + 2\delta_O + 2\delta_D = 0,$$

the background-dependent kinetic-flow matrix is defined by

$$\partial_t \begin{pmatrix}\delta_T\\ \delta_O\\ \delta_D\end{pmatrix} = M_Z(\bar{\Phi}_{\min}) \begin{pmatrix}\delta_T\\ \delta_O\\ \delta_D\end{pmatrix} + O(\delta^2), \tag{440}$$

with

$$(M_Z(\bar{\Phi}_{\min}))_{IJ} = \frac{\partial}{\partial \delta_J}\big(\partial_t \delta_I(\bar{\Phi}_{\min})\big)\Big|_{\delta=0}. \tag{441}$$

9. Honest status

Thus the structural problem is now closed at the background-dependent level:

1. the minimal admissible background is fixed;
2. the background-dependent block Hessian is fixed formally;
3. the source of nonzero cubic vertices is identified;
4. the projected background two-point flow defining $M_Z(\bar{\Phi}_{\min})$ is fixed.

What remains open is the explicit evaluation of:

1. the background-adapted regulator $R_k[\bar{\Phi}_{\min}]$,
2. the gauge-projected A-sector block $\mathcal{G}_A$,
3. the tensor contractions in (438),
4. the sign of the nontrivial eigenvalues of $M_Z(\bar{\Phi}_{\min})$.

. Summary

$$\boxed{\bar{\Phi}_{\min} = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi))}$$

is the minimal admissible shell-/defect-adapted background.

With

$$\Phi = \bar{\Phi}_{\min} + \varphi, \qquad \varphi = (a, \nu, \zeta),$$

the fluctuation Hessian is

$$\boxed{\Gamma_k^{(2)}[\bar{\Phi}_{\min}] = \begin{pmatrix} \mathcal{H}_{TT} & \mathcal{H}_{TO} & \mathcal{H}_{TD} \\ \mathcal{H}_{OT} & \mathcal{H}_{OO} & \mathcal{H}_{OD} \\ \mathcal{H}_{DT} & \mathcal{H}_{DO} & \mathcal{H}_{DD} \end{pmatrix}.}$$

Diagonal blocks:

$$\mathcal{H}_{TT} = -Z_{T,k}\partial^2 + \mathcal{M}_{TT}^{\mathrm{pot}}(\xi) + Y_k \mathcal{K}_{TT}^{\nabla\mathrm{ani}} + \mathcal{G}_A,$$

$$\mathcal{H}_{OO} = -Z_{O,k}\partial^2 + \mathcal{M}_{OO}^{\mathrm{pot}}(\xi) + Y_k \mathcal{K}_{OO}^{\nabla\mathrm{ani}} + \mathcal{T}_{OO},$$

$$\mathcal{H}_{DD} = -Z_{D,k}\partial^2 + \mathcal{M}_{DD}^{\mathrm{pot}}(\xi) + Y_k \mathcal{K}_{DD}^{\nabla\mathrm{ani}} + \mathcal{T}_{DD}.$$

Mixed blocks:

$$\mathcal{H}_{TO} = g_{\tau,k} \left. \frac{\partial^2 \tau}{\partial A\, \partial n} \right|_{\bar{\Phi}_{\min}} + Y_k \mathcal{K}_{TO}^{\nabla\mathrm{ani}},$$

$$\mathcal{H}_{TD} = g_{\tau,k} \left. \frac{\partial^2 \tau}{\partial A\, \partial \chi} \right|_{\bar{\Phi}_{\min}} + Y_k \mathcal{K}_{TD}^{\nabla\mathrm{ani}},$$

$$\mathcal{H}_{OD} = \lambda_{OD,k} \bar{n}_B \bar{\chi}_B^T + g_{\tau,k} \left. \frac{\partial^2 \tau}{\partial n\, \partial \chi} \right|_{\bar{\Phi}_{\min}} + Y_k \mathcal{K}_{OD}^{\nabla\mathrm{ani}}.$$

Because $\bar{\Phi}_{\min} \neq 0$, one has

$$\boxed{\Gamma_k^{(3)}[\bar{\Phi}_{\min}] \neq 0,}$$

so the bubble/sunset topology

$$\Gamma^{(3)} G \Gamma^{(3)}$$

is activated.

The background propagator is

$$G_k(\bar{\Phi}_{\min}) = \left(\Gamma_k^{(2)}[\bar{\Phi}_{\min}] + R_k[\bar{\Phi}_{\min}]\right)^{-1},$$

and the sectoral kinetic flows are

$$\boxed{\partial_t Z_{I,k}(\bar{\Phi}_{\min}) = \left. \frac{\partial}{\partial p^2} \partial_t \Gamma_{k,II}^{(2)}(p; \bar{\Phi}_{\min}) \right|_{p=0}, \qquad I \in \{T, O, D\}.}$$

Therefore

$$\boxed{M_Z(\bar{\Phi}_{\min}) \text{ is now formally well-defined.}}$$

What remains open is the actual contraction/evaluation:

$$\boxed{R_k[\bar{\Phi}_{\min}], \qquad \mathcal{G}_A, \qquad \text{tensor contractions}, \qquad \operatorname{sign}\operatorname{spec}(M_Z(\bar{\Phi}_{\min})|_{\mathrm{traceless}}).}$$

BACKGROUND-ADAPTED REGULATOR AND A-SECTOR PROJECTOR ANSATZ

1. Goal

To make the background-field definition of

$$M_Z(\bar{\Phi}_{\min})$$

actually operational, one must specify:

1. a background-adapted regulator $R_k[\bar{\Phi}_{\min}]$,
2. a physical projector in the A-sector,
3. the corresponding projected propagator.

The document is explicit that the sign of M_Z depends on the regulator choice and on the gauge/constraint structure of the A-sector. :contentReference[oaicite:3]index=3

2. Background-dependent Hessian blocks

On the minimal admissible background

$$\bar{\Phi}_{\min} = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi)),$$

the fluctuation Hessian has the block form

$$\Gamma_k^{(2)}[\bar{\Phi}_{\min}] = \begin{pmatrix} \mathcal{H}_{TT} & \mathcal{H}_{TO} & \mathcal{H}_{TD} \\ \mathcal{H}_{OT} & \mathcal{H}_{OO} & \mathcal{H}_{OD} \\ \mathcal{H}_{DT} & \mathcal{H}_{DO} & \mathcal{H}_{DD} \end{pmatrix}.$$

The regulator should be adapted to these background-dependent kinetic blocks rather than to the strict symmetric-point operator.

3. Minimal background-adapted regulator ansatz

A minimal sectoral regulator ansatz is

$$\boxed{R_k[\bar{\Phi}_{\min}] = \mathrm{diag}\Big(R_{T,k},\, R_{O,k},\, R_{D,k}\Big),} \tag{442}$$

with

$$R_{T,k} = Z_{T,k}\, \Pi_A^{\mathrm{phys}}\, r_k(\mathcal{H}_{TT}^{(0)}), \tag{443}$$

$$R_{O,k} = Z_{O,k}\, r_k(\mathcal{H}_{OO}^{(0)}), \tag{444}$$

$$R_{D,k} = Z_{D,k}\, r_k(\mathcal{H}_{DD}^{(0)}). \tag{445}$$

Here:

- $\mathcal{H}_{II}^{(0)}$ denotes the principal/background-diagonal part of the II-block,
- $r_k(\cdot)$ is a standard monotone cutoff profile,
- Π_A^{phys} projects onto the physical A-sector fluctuation subspace before regularization.

This is the natural background-field analogue of the usual sectoral FRG regulator.

4. Optimized-profile version

A concrete optimized-profile realization is

$$R_{I,k} = Z_{I,k}\left(k^2 - \mathcal{H}^{(0)}_{II}\right)\Theta\!\left(k^2 - \mathcal{H}^{(0)}_{II}\right), \qquad I \in \{T, O, D\}, \tag{446}$$

with the understanding that for $I = T$ the operator acts only on the image of $\Pi^{\rm phys}_A$.

Equivalently, in momentum language for the worldvolume modes,

$$R_{I,k}(q) = Z_{I,k}(k^2 - q^2)\Theta(k^2 - q^2), \tag{447}$$

but with the transverse/gauge projection imposed in the A-sector.

5. A-sector physical projector

The emergent Abelian gauge chain in TECT is geometrically closed:

$$z \in S^5 \Longrightarrow S^5 \to CP^2 \Longrightarrow U(1) \text{ principal bundle} \Longrightarrow A_\mu = -iz^\dagger \partial_\mu z \Longrightarrow F_{\mu\nu}.$$

Dynamical infrared consistency requires that the Proca mass vanish,

$$m^2_{A,k} \to 0,$$

so that only the genuine transverse gauge mode survives in the infrared.

Therefore the A-sector must be projected onto the physical transverse/gauge-consistent subspace. The minimal abstract requirement is:

$$\boxed{\Pi^{\rm phys}_A : \mathcal{H}_A \to \mathcal{H}^{\rm phys}_A,} \tag{448}$$

where $\mathcal{H}^{\rm phys}_A$ contains only the transverse fluctuation modes compatible with the infrared gauge surface.

In momentum representation the standard transverse projector is

$$\Pi^{\rm T}_{A,ij}(p) = \delta_{ij} - \frac{p_i p_j}{p^2}, \tag{449}$$

and the minimal ansatz is that $\Pi^{\rm phys}_A$ reduces to $\Pi^{\rm T}_A$ on the projected shell/worldvolume gauge block, possibly corrected by background-dependent constraint terms.

6. Background-adapted projected propagator

With the regulator and the physical projector in place, define the projected background propagator

$$\boxed{G^{\rm phys}_k(\bar\Phi_{\min}) = \left(\Pi^{\rm phys}\left[\Gamma^{(2)}_k[\bar\Phi_{\min}] + R_k[\bar\Phi_{\min}]\right]\Pi^{\rm phys}\right)^{-1},} \tag{450}$$

where

$$\Pi^{\rm phys} = \mathrm{diag}\!\left(\Pi^{\rm phys}_A,\, \mathbf{1}_O,\, \mathbf{1}_D\right).$$

This ensures that unphysical A-sector directions do not contaminate the one-loop tensor contractions used to define M_Z.

7. Background-field two-point flow with projection

The projected two-point flow is then

$$\partial_t \Gamma^{(2)}_{k,II}(p;\bar{\Phi}_{\min}) = \frac{1}{2}\,\mathrm{Tr}_{\mathrm{phys}}\Big[G_k^{\mathrm{phys}}(\bar{\Phi}_{\min})\,\Gamma^{(4)}_{k,II}[\bar{\Phi}_{\min}]\,G_k^{\mathrm{phys}}(\bar{\Phi}_{\min})\,\partial_t R_k[\bar{\Phi}_{\min}]\Big] \\ - \mathrm{Tr}_{\mathrm{phys}}\Big[G_k^{\mathrm{phys}}(\bar{\Phi}_{\min})\,\Gamma^{(3)}_{k,I}[\bar{\Phi}_{\min}]\,G_k^{\mathrm{phys}}(\bar{\Phi}_{\min})\,\Gamma^{(3)}_{k,I}[\bar{\Phi}_{\min}]\,G_k^{\mathrm{phys}}(\bar{\Phi}_{\min})\,\partial_t R_k[\bar{\Phi}_{\min}]\Big]. \tag{451}$$

Then

$$\partial_t Z_{I,k}(\bar{\Phi}_{\min}) = \frac{\partial}{\partial p^2}\partial_t \Gamma^{(2)}_{k,II}(p;\bar{\Phi}_{\min})\bigg|_{p=0}, \qquad I \in \{T,O,D\}, \tag{452}$$

and finally

$$(M_Z(\bar{\Phi}_{\min}))_{IJ} = \frac{\partial}{\partial \delta_J}\big(\partial_t \delta_I(\bar{\Phi}_{\min})\big)\bigg|_{\delta=0}. \tag{453}$$

8. Honest status

At this point, the regulator/projector problem is structurally closed at the ansatz level:

1. the regulator is chosen background-adapted and sectoral,
2. the A-sector is projected to its physical transverse/gauge-consistent subspace,
3. the resulting projected propagator is well defined,
4. the projected two-point flow defining M_Z is formally fixed.

What remains open is not the structural form of the ansatz, but the explicit implementation:

1. determine the detailed background-corrected projector Π_A^{phys},
2. specify the exact regulator profile r_k,
3. evaluate the corresponding one-loop tensor contractions,
4. extract the sign of the nontrivial eigenvalues of $M_Z(\bar{\Phi}_{\min})$.

. Summary

$$\boxed{R_k[\bar{\Phi}_{\min}] = \mathrm{diag}\big(R_{T,k}, R_{O,k}, R_{D,k}\big),}$$

with

$$R_{T,k} = Z_{T,k}\,\Pi_A^{\mathrm{phys}}\,r_k(\mathcal{H}^{(0)}_{TT}), \qquad R_{O,k} = Z_{O,k}\,r_k(\mathcal{H}^{(0)}_{OO}), \qquad R_{D,k} = Z_{D,k}\,r_k(\mathcal{H}^{(0)}_{DD}).$$

A concrete optimized version is

$$R_{I,k} = Z_{I,k}\big(k^2 - \mathcal{H}^{(0)}_{II}\big)\Theta(k^2 - \mathcal{H}^{(0)}_{II}), \qquad I \in \{T,O,D\},$$

with the A-sector restricted to the physical projected subspace.

The A-sector physical projector is

$$\boxed{\Pi_A^{\mathrm{phys}} : \mathcal{H}_A \to \mathcal{H}_A^{\mathrm{phys}},}$$

where $\mathcal{H}_A^{\mathrm{phys}}$ contains only the transverse gauge-consistent fluctuation modes. In momentum representation, the minimal transverse projector is

$$\Pi^{\mathrm{T}}_{A,ij}(p) = \delta_{ij} - \frac{p_i p_j}{p^2}.$$

The projected propagator is

$$\boxed{G_k^{\mathrm{phys}}(\bar{\Phi}_{\min}) = \Big(\Pi^{\mathrm{phys}}[\Gamma_k^{(2)}[\bar{\Phi}_{\min}] + R_k[\bar{\Phi}_{\min}]]\Pi^{\mathrm{phys}}\Big)^{-1},}$$

with

$$\Pi^{\mathrm{phys}} = \mathrm{diag}\big(\Pi_A^{\mathrm{phys}}, \mathbf{1}_O, \mathbf{1}_D\big).$$

Then the projected background two-point flow is

$$\partial_t \Gamma^{(2)}_{k,II}(p;\bar{\Phi}_{\min}) = \frac{1}{2}\mathrm{Tr}_{\mathrm{phys}}[G\Gamma^{(4)}G\partial_t R] - \mathrm{Tr}_{\mathrm{phys}}[G\Gamma^{(3)}G\Gamma^{(3)}G\partial_t R],$$

and

$$\boxed{\partial_t Z_{I,k}(\bar{\Phi}_{\min}) = \frac{\partial}{\partial p^2}\partial_t \Gamma^{(2)}_{k,II}(p;\bar{\Phi}_{\min})\Big|_{p=0}.}$$

So the regulator/projector problem is structurally closed. What remains open is the explicit implementation of

$$\boxed{\Pi_A^{\mathrm{phys}}, \qquad r_k, \qquad \text{one-loop contractions,} \qquad \mathrm{sign\,spec}(M_Z(\bar{\Phi}_{\min})|_{\mathrm{traceless}}).}$$

BACKGROUND-GENERATED EXPLICIT VERTEX LIST ON $\bar{\Phi}_{\min}$

1. Background split

Choose the minimal admissible shell-/defect-adapted background

$$\bar{\Phi}_{\min} = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi)), \tag{454}$$

and expand

$$\Phi = \bar{\Phi}_{\min} + \varphi, \qquad \varphi = (a, \nu, \zeta), \tag{455}$$

so that

$$A = a, \qquad n = \bar{n}_B + \nu, \qquad \chi = \bar{\chi}_B + \zeta.$$

We now list the explicit background-generated cubic and quartic fluctuation vertices that can enter the projected one-loop flow of

$$\partial_t Z_T, \qquad \partial_t Z_O, \qquad \partial_t Z_D.$$

2. Potential-sector background-generated cubic vertices

The minimal explicit potential sector is

$$\Gamma_{k,\mathrm{pot}} = \int d^d x \Big[r_{T,k}\rho_T + r_{O,k}\rho_O + r_{D,k}\rho_D + \frac{\lambda_{T,k}}{2}\rho_T^2 + \frac{\lambda_{O,k}}{2}\rho_O^2 + \frac{\lambda_{D,k}}{2}\rho_D^2 + \lambda_{TO,k}\rho_T\rho_O + \lambda_{TD,k}\rho_T\rho_D + \lambda_{OD,k}\rho_O\rho_D + g_{\tau,k}\tau(A,n,\chi)\Big]. \tag{456}$$

Since $\bar{A} = 0$, but $\bar{n}_B, \bar{\chi}_B \neq 0$, the expansion of the quartic terms yields the following cubic fluctuation vertices.

2.1 Orientational self-cubic

From $\frac{\lambda_{O,k}}{2}\rho_O^2$,

$$\rho_O = \frac{1}{2}(\bar{n}_B + \nu)^T(\bar{n}_B + \nu) = \rho_O^B + \bar{n}_B^T\nu + \frac{1}{2}\nu^T\nu. \tag{457}$$

Therefore

$$\frac{\lambda_{O,k}}{2}\rho_O^2 \supset \lambda_{O,k}\,(\bar{n}_B^T\nu)\,\frac{1}{2}\nu^T\nu, \tag{458}$$

which generates a cubic fluctuation vertex of the schematic type

$$\Gamma^{(3)}_{\nu\nu\nu}[\bar{\Phi}_{\min}] \sim \lambda_{O,k}\,\bar{n}_B(\xi).$$

2.2 Defect/hybrid self-cubic

Similarly, from $\frac{\lambda_{D,k}}{2}\rho_D^2$,

$$\rho_D = \rho_D^B + \bar{\chi}_B^T\zeta + \frac{1}{2}\zeta^T\zeta, \tag{459}$$

and hence

$$\frac{\lambda_{D,k}}{2}\rho_D^2 \supset \lambda_{D,k}\,(\bar{\chi}_B^T\zeta)\,\frac{1}{2}\zeta^T\zeta, \tag{460}$$

which gives

$$\Gamma^{(3)}_{\zeta\zeta\zeta}[\bar{\Phi}_{\min}] \sim \lambda_{D,k}\,\bar{\chi}_B(\xi).$$

2.3 Mixed O-D cubic vertices

From the mixed quartic term $\lambda_{OD,k}\rho_O\rho_D$,

$$\begin{aligned}\lambda_{OD,k}\rho_O\rho_D \supset \lambda_{OD,k}\,(\bar{n}_B^T\nu)\,\frac{1}{2}\zeta^T\zeta + \lambda_{OD,k}\,(\bar{\chi}_B^T\zeta)\,\frac{1}{2}\nu^T\nu \\ + \lambda_{OD,k}\,(\bar{n}_B^T\nu)(\bar{\chi}_B^T\zeta).\end{aligned} \tag{461}$$

Thus one obtains the cubic fluctuation vertices

$$\Gamma^{(3)}_{\nu\zeta\zeta}[\bar{\Phi}_{\min}] \sim \lambda_{OD,k}\,\bar{n}_B(\xi), \qquad \Gamma^{(3)}_{\nu\nu\zeta}[\bar{\Phi}_{\min}] \sim \lambda_{OD,k}\,\bar{\chi}_B(\xi),$$

as well as background-dependent quadratic mixing.

a. Important note. Because $\bar{A} = 0$, the mixed quartic couplings $\lambda_{TO,k}\rho_T\rho_O$ and $\lambda_{TD,k}\rho_T\rho_D$ do not generate cubic vertices linear in a at this stage. They remain relevant for higher derivatives of the action, but not as immediate background-generated cubic sources.

3. Quartic anisotropy background-generated vertices

The quartic cubic-anisotropy term

$$g_{\tau,k}\tau(A, n, \chi)$$

is quartic in the fields. Therefore, after the background split, it generates:

- background-dependent quadratic corrections to the Hessian,

- background-dependent cubic fluctuation vertices,
- background-dependent quartic fluctuation vertices.

Schematically,

$$g_{\tau,k}\tau(\bar{\Phi}_{\min}+\varphi) = g_{\tau,k}\tau(\bar{\Phi}_{\min})+g_{\tau,k}\,\tau^{(1)}[\bar{\Phi}_{\min}]\varphi+\frac{1}{2}g_{\tau,k}\,\tau^{(2)}[\bar{\Phi}_{\min}]\varphi^2+\frac{1}{3!}g_{\tau,k}\,\tau^{(3)}[\bar{\Phi}_{\min}]\varphi^3+\frac{1}{4!}g_{\tau,k}\,\tau^{(4)}[\bar{\Phi}_{\min}]\varphi^4. \quad (462)$$

Hence

$$\Gamma_\tau^{(3)}[\bar{\Phi}_{\min}] \neq 0, \qquad \Gamma_\tau^{(4)}[\bar{\Phi}_{\min}] \neq 0, \quad (463)$$

in general. These vertices are anisotropic and therefore can feed directly into the linearized anisotropy flow.

4. Derivative-anisotropy background-generated vertices

The minimal truncation contains

$$Y_k O_{\nabla,\mathrm{ani}}[A,n,\chi].$$

At the strict symmetric point and under the minimal bilinear interpretation, this term contributes only to $\Gamma^{(2)}$. But on the nontrivial background $\bar{\Phi}_{\min}$, a field-dependent derivative operator generally expands as

$$O_{\nabla,\mathrm{ani}}[\bar{\Phi}_{\min}+\varphi] = O_{\nabla,\mathrm{ani}}[\bar{\Phi}_{\min}] + O^{(1)}_{\nabla,\mathrm{ani}}[\bar{\Phi}_{\min}]\varphi + \frac{1}{2}O^{(2)}_{\nabla,\mathrm{ani}}[\bar{\Phi}_{\min}]\varphi^2 + \frac{1}{3!}O^{(3)}_{\nabla,\mathrm{ani}}[\bar{\Phi}_{\min}]\varphi^3 + \cdots . \quad (464)$$

Hence the background can generate momentum-dependent cubic fluctuation vertices

$$\Gamma^{(3)}_{\nabla\mathrm{ani}}[\bar{\Phi}_{\min}] \neq 0. \quad (465)$$

These are especially important, because they are natural candidates for generating genuine p^2-dependence in the two-point flow and therefore nontrivial contributions to M_Z.

5. Gauge/projector background-generated vertices

The A-sector requires a gauge-fixing or physical projector structure. The sign of M_Z depends explicitly on this gauge/constraint sector. Therefore, once the A-sector is formulated on the background-adapted projected space, the corresponding contribution

$$\Gamma_{k,\mathrm{gf/proj}}[\bar{\Phi}_{\min}+\varphi]$$

may also generate:

- projected background-dependent quadratic corrections in $\mathcal{H}_{TT}$,
- mixed background-dependent cubic vertices involving a and ν,ζ,
- possibly additional quartic projected vertices.

Their explicit form depends on the chosen projector/gauge implementation and cannot be fixed abstractly at the present stage.

6. Survival table on the background

The explicit background-generated survival table is:

source	$\Gamma^{(2)}[\bar{\Phi}_{\min}]$	$\Gamma^{(3)}[\bar{\Phi}_{\min}]$	$\Gamma^{(4)}[\bar{\Phi}_{\min}]$
kinetic sector	yes	no	no
$r_T\rho_T + r_O\rho_O + r_D\rho_D$	yes	no	no
$\frac{\lambda_O}{2}\rho_O^2$	yes	$\Gamma^{(3)}_{\nu\nu\nu} \sim \lambda_O \bar{n}_B$	$\Gamma^{(4)}_{\nu\nu\nu\nu} \neq 0$
$\frac{\lambda_D}{2}\rho_D^2$	yes	$\Gamma^{(3)}_{\zeta\zeta\zeta} \sim \lambda_D \bar{\chi}_B$	$\Gamma^{(4)}_{\zeta\zeta\zeta\zeta} \neq 0$
$\lambda_{OD}\rho_O\rho_D$	yes	$\Gamma^{(3)}_{\nu\nu\zeta}, \Gamma^{(3)}_{\nu\zeta\zeta} \neq 0$	$\Gamma^{(4)}_{\nu\nu\zeta\zeta} \neq 0$
$\lambda_{TO}\rho_T\rho_O,\ \lambda_{TD}\rho_T\rho_D$	yes	not directly if $\bar{A} = 0$	$\Gamma^{(4)} \neq 0$
$g_{\tau,k}\tau(A, n, \chi)$	yes	yes (background-anisotropic cubic)	yes
$Y_k O_{\nabla,\mathrm{ani}}$	yes	yes if field-dependent on background	possibly yes in refined completion
$\Gamma_{k,\mathrm{gf/proj}}$	yes	implementation-dependent	implementation-dependent

7. Consequence for the one-loop two-point flow

Therefore, on the background $\bar{\Phi}_{\min}$, the two-point flow

$$\begin{aligned}\partial_t \Gamma^{(2)}_{k,IJ}(p; \bar{\Phi}_{\min}) = {}& \frac{1}{2}\operatorname{Tr}\Big[G_k(\bar{\Phi}_{\min})\Gamma^{(4)}_{k,IJ}[\bar{\Phi}_{\min}]G_k(\bar{\Phi}_{\min})\partial_t R_k(\bar{\Phi}_{\min})\Big] \\ & - \operatorname{Tr}\Big[G_k(\bar{\Phi}_{\min})\Gamma^{(3)}_{k,I}[\bar{\Phi}_{\min}]G_k(\bar{\Phi}_{\min})\Gamma^{(3)}_{k,J}[\bar{\Phi}_{\min}]G_k(\bar{\Phi}_{\min})\partial_t R_k(\bar{\Phi}_{\min})\Big]\end{aligned} \tag{466}$$

contains genuinely nonzero $\Gamma^{(3)}G\Gamma^{(3)}$ terms. This is the minimal honest mechanism by which the sectoral wavefunction flows

$$\partial_t Z_T, \qquad \partial_t Z_O, \qquad \partial_t Z_D$$

can become nontrivial.

8. Formal decomposition of the M_Z entries

Accordingly, the physical kinetic-flow matrix should be decomposed as

$$(M_Z)_{IJ} = (M_Z)^{(O\text{-self})}_{IJ} + (M_Z)^{(D\text{-self})}_{IJ} + (M_Z)^{(OD\text{-mix})}_{IJ} + (M_Z)^{(\tau)}_{IJ} + (M_Z)^{(\nabla\mathrm{ani})}_{IJ} + (M_Z)^{(\mathrm{gf/proj})}_{IJ}, \tag{467}$$

where:

- $(M_Z)^{(O\text{-self})}$ comes from $\lambda_O\bar{n}_B$-induced cubic vertices,
- $(M_Z)^{(D\text{-self})}$ comes from $\lambda_D\bar{\chi}_B$-induced cubic vertices,
- $(M_Z)^{(OD\text{-mix})}$ comes from the mixed coupling λ_{OD},
- $(M_Z)^{(\tau)}$ comes from the quartic cubic-anisotropy background-generated vertices,
- $(M_Z)^{(\nabla\mathrm{ani})}$ comes from the derivative-anisotropy background-generated vertices,
- $(M_Z)^{(\mathrm{gf/proj})}$ comes from the A-sector gauge/projector completion.

9. Honest final status

Thus the background-generated vertex problem is now structurally closed.
What is fixed:

1. which vertices vanish at the symmetric point,
2. which vertices are regenerated on the minimal admissible background,

3. which vertex classes can actually contribute to the one-loop kinetic flow.

What remains open:

1. the explicit tensor form of $O_{\nabla,\text{ani}}$ on the background,
2. the explicit gauge/projector completion in the A-sector,
3. the actual one-loop contraction signs and weights.

. Summary

$$\boxed{\Phi = \bar{\Phi}_{\min} + \varphi, \qquad \bar{\Phi}_{\min} = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi)), \qquad \varphi = (a, \nu, \zeta).}$$

On this background, the quartic potential sector generates effective cubic fluctuation vertices:

$$\boxed{\Gamma^{(3)}_{\nu\nu\nu} \sim \lambda_O\, \bar{n}_B(\xi), \qquad \Gamma^{(3)}_{\zeta\zeta\zeta} \sim \lambda_D\, \bar{\chi}_B(\xi),}$$

$$\boxed{\Gamma^{(3)}_{\nu\nu\zeta} \sim \lambda_{OD}\, \bar{\chi}_B(\xi), \qquad \Gamma^{(3)}_{\nu\zeta\zeta} \sim \lambda_{OD}\, \bar{n}_B(\xi).}$$

The quartic anisotropy term generates

$$\boxed{\Gamma^{(3)}_{\tau}[\bar{\Phi}_{\min}] \neq 0, \qquad \Gamma^{(4)}_{\tau}[\bar{\Phi}_{\min}] \neq 0.}$$

The derivative-anisotropy sector can now generate

$$\boxed{\Gamma^{(3)}_{\nabla\text{ani}}[\bar{\Phi}_{\min}] \neq 0}$$

provided the operator is treated in its background-dependent field-sensitive form.

The gauge/projector sector may also generate background-dependent cubic and quartic vertices:

$$\boxed{\Gamma^{(3)}_{\text{gf/proj}}[\bar{\Phi}_{\min}], \qquad \Gamma^{(4)}_{\text{gf/proj}}[\bar{\Phi}_{\min}],}$$

but these remain implementation-dependent.

Hence the one-loop two-point flow on the background now contains a genuinely nonzero bubble/sunset sector:

$$\boxed{\Gamma^{(3)}[\bar{\Phi}_{\min}]\, G_k(\bar{\Phi}_{\min})\, \Gamma^{(3)}[\bar{\Phi}_{\min}],}$$

which was absent at the strict symmetric point.

Accordingly, the kinetic-flow matrix decomposes schematically as

$$\boxed{(M_Z)_{IJ} = (M_Z)^{(O\text{-self})}_{IJ} + (M_Z)^{(D\text{-self})}_{IJ} + (M_Z)^{(OD\text{-mix})}_{IJ} + (M_Z)^{(\tau)}_{IJ} + (M_Z)^{(\nabla\text{ani})}_{IJ} + (M_Z)^{(\text{gf/proj})}_{IJ}.}$$

So the vertex-survival problem is now structurally closed. The remaining frontier is the explicit sign/coupling evaluation of these background-generated one-loop contributions.

SECTOR-BY-SECTOR SIGN ROADMAP FOR THE KINETIC-FLOW MATRIX M_Z

1. Physical problem

After removing the redundant common metric anisotropy, the physical derivative-anisotropy problem is the flow of

$$\delta_T, \qquad \delta_O, \qquad \delta_D, \qquad 2\delta_T + 2\delta_O + 2\delta_D = 0.$$

These variables are classically marginal, so their infrared fate is controlled entirely by the loop-level kinetic-flow matrix

$$\partial_t \begin{pmatrix} \delta_T \\ \delta_O \\ \delta_D \end{pmatrix} = M_Z \begin{pmatrix} \delta_T \\ \delta_O \\ \delta_D \end{pmatrix} + O(\delta^2).$$

The sign of the nontrivial eigenvalues of M_Z is not fixed by the currently available truncation data alone. It depends on the exact Hessian, interaction vertices coupling A, n, χ, the regulator, the tensor contractions, and the A-sector gauge/constraint structure.

2. Background-generated decomposition of M_Z

On the minimal admissible background

$$\bar{\Phi}_{\min} = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi)),$$

the background-generated cubic vertices imply the decomposition

$$(M_Z)_{IJ} = (M_Z)_{IJ}^{(O\text{-self})} + (M_Z)_{IJ}^{(D\text{-self})} + (M_Z)_{IJ}^{(OD\text{-mix})} + (M_Z)_{IJ}^{(\tau)} + (M_Z)_{IJ}^{(\nabla\text{ani})} + (M_Z)_{IJ}^{(\text{gf/proj})}. \tag{468}$$

Each term corresponds to a specific class of one-loop background-induced vertices.

3. O-self contribution

The self-quartic term

$$\frac{\lambda_O}{2}\rho_O^2$$

generates the background-induced cubic vertex

$$\Gamma^{(3)}_{\nu\nu\nu}[\bar{\Phi}_{\min}] \sim \lambda_O\, \bar{n}_B(\xi).$$

Therefore it contributes to the two-point flow through

$$\Gamma^{(3)}_{\nu\nu\nu} G_O \Gamma^{(3)}_{\nu\nu\nu}.$$

a. Sign expectation. If the orientational sector is stable and its quartic self-coupling is positive,

$$\lambda_O > 0,$$

then this loop typically acts as a self-stabilizing renormalization of the O-sector stiffness. Hence the natural expectation is that

$$(M_Z)^{(O\text{-self})}$$

contributes predominantly to the diagonal part and tends to reduce relative anisotropy rather than amplify it.

b. Honest limitation. The actual sign depends on the precise momentum derivative of the projected self-energy and cannot be fixed without evaluating the loop kernel explicitly.

4. D-self contribution

Similarly, the self-quartic term

$$\frac{\lambda_D}{2}\rho_D^2$$

generates

$$\Gamma^{(3)}_{\zeta\zeta\zeta}[\bar{\Phi}_{\min}] \sim \lambda_D\,\bar{\chi}_B(\xi),$$

so that

$$(M_Z)^{(D\text{-self})}$$

comes from the corresponding D-sector bubble.

c. Sign expectation. Under the natural stability condition

$$\lambda_D > 0,$$

the D-self contribution is again expected to be diagonally stabilizing and therefore consistent with isotropization.

d. Honest limitation. As with the O-sector, this is an expectation, not yet a theorem.

5. Mixed OD contribution

The mixed quartic term

$$\lambda_{OD}\rho_O\rho_D$$

generates the background-induced cubic vertices

$$\Gamma^{(3)}_{\nu\nu\zeta}[\bar{\Phi}_{\min}] \sim \lambda_{OD}\,\bar{\chi}_B(\xi), \qquad \Gamma^{(3)}_{\nu\zeta\zeta}[\bar{\Phi}_{\min}] \sim \lambda_{OD}\,\bar{n}_B(\xi).$$

This sector is the first genuine source of off-diagonal kinetic mixing between the O- and D-channels.

e. Sign expectation. If the mixed interaction tends to lock the two sectors together, then

$$(M_Z)^{(OD\text{-mix})}$$

should have the Laplacian-type sign structure

$$\sim -\lambda \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

on the O-D traceless subspace. This would favor isotropization.

f. Honest limitation. This is exactly one of the most dangerous unresolved sign sources. The sign of the corresponding off-diagonal kernel is not fixed without explicit tensor contraction.

6. Quartic anisotropy contribution

The quartic BCC anisotropy term

$$g_{\tau,k}\tau(\Phi)$$

generates anisotropic background-dependent cubic and quartic fluctuation vertices. Its contribution to M_Z is denoted

$$(M_Z)^{(\tau)}.$$

g. RG support. In the TECT infrared truncation with

$$N_{\text{eff}} = 6,$$

the linearized quartic anisotropy eigenvalue is

$$\theta_\tau = \epsilon \frac{4 - N_{\text{eff}}}{N_{\text{eff}} + 8} = -\frac{\epsilon}{7} < 0.$$

Hence the leading quartic cubic anisotropy is irrelevant in the infrared. :contentReference[oaicite:6]index=6

h. Sign expectation. Therefore it is natural to expect that

$$(M_Z)^{(\tau)}$$

acts in the isotropizing direction, i.e. contributes negative physical eigenvalues on the traceless anisotropy subspace.

i. Honest limitation. This does not yet prove that the full derivative-anisotropy matrix M_Z is negative, because the quartic anisotropy direction is not identical to the full sectoral wavefunction-anisotropy direction.

7. Derivative-anisotropy contribution

The leading derivative anisotropy operator

$$Y_k O_{\nabla,\text{ani}}$$

produces the contribution

$$(M_Z)^{(\nabla\text{ani})}.$$

j. Why this is the core driver. Since M_Z is precisely the linearized flow matrix of the derivative anisotropy variables $\delta_T, \delta_O, \delta_D$, this is the most direct source term for the physical anisotropy problem.

k. Expectation from the TECT FRG truncation. The document states that the derivative anisotropy has negative canonical part,

$$-2 + \eta_* < 0,$$

and that $\theta_Y < 0$ is the natural TECT expectation. :contentReference[oaicite:7]index=7

l. Honest limitation. The same document is equally explicit that the sign of the physical derivative-anisotropy flow is not yet rigorously fixed from current TECT data, because it requires the full kinetic-flow matrix M_Z. Therefore:

$$(M_Z)^{(\nabla\text{ani})}$$

is the most important expected negative contribution, but not yet a proved one.

8. Gauge/projector contribution

The gauge/projector completion contributes

$$(M_Z)^{(\text{gf/proj})}.$$

m. Structural expectation. The gauge-consistent IR sector must preserve transversality and avoid a generated Proca mass. In the gauge-preserving reference EFT, the one-loop polarization tensor remains transverse,

$$p^\mu \Pi_{\mu\nu}(p) = 0, \qquad \Pi_{\mu\nu}(0) = 0,$$

even in the presence of Class II gauge-covariant corrections.

This suggests that a properly projected A-sector should not generate uncontrolled gauge-breaking anisotropy, and therefore may be compatible with isotropization.

n. Honest limitation. However, the sign of its contribution to Z_T still depends on the detailed projector implementation and background-adapted contractions. Hence this is a second unresolved sign source.

9. Qualitative sign hierarchy

The most honest current hierarchy is:

$$\boxed{(M_Z)^{(\tau)} \text{ and } (M_Z)^{(\nabla\mathrm{ani})} \text{ are naturally expected to be isotropizing,}} \tag{469}$$

while

$$\boxed{(M_Z)^{(OD\text{-mix})} \text{ and } (M_Z)^{(\mathrm{gf/proj})} \text{ are the two main unresolved sign bottlenecks.}} \tag{470}$$

The self-sector pieces

$$(M_Z)^{(O\text{-self})}, \qquad (M_Z)^{(D\text{-self})}$$

are expected to be diagonally stabilizing, but their exact sign weight is not yet proved.

10. Algebraic target

The exact algebraic target remains

$$M_Z = -\lambda \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}, \qquad \lambda > 0, \tag{471}$$

for which the physical eigenvalues are

$$-3\lambda, \qquad -3\lambda$$

on the traceless subspace. :contentReference[oaicite:10]index=10
Thus the practical sign roadmap is:

1. show that the sum of the expected negative pieces

   $$(M_Z)^{(\tau)} + (M_Z)^{(\nabla\mathrm{ani})}$$

   dominates,

2. show that the unresolved pieces

   $$(M_Z)^{(OD\text{-mix})} + (M_Z)^{(\mathrm{gf/proj})}$$

   do not overturn negativity,

3. recover the Laplacian-type sign pattern (471).

11. Honest conclusion

Therefore the current status is not "M_Z is negative." The honest theorem-level status is:

$$\boxed{\text{the sign problem has been reduced to a finite list of background-generated one-loop sources.}}$$

Among these, the derivative-anisotropy and quartic-anisotropy sectors are the main expected isotropizing sources, while the mixed OD and gauge/projector sectors are the main unresolved sign bottlenecks.

. Summary

$$\boxed{(M_Z)_{IJ} = (M_Z)_{IJ}^{(O\text{-self})} + (M_Z)_{IJ}^{(D\text{-self})} + (M_Z)_{IJ}^{(OD\text{-mix})} + (M_Z)_{IJ}^{(\tau)} + (M_Z)_{IJ}^{(\nabla\text{ani})} + (M_Z)_{IJ}^{(\text{gf/proj})}.}$$

Current honest sign roadmap:

(1) Self-sector pieces

$$\Gamma^{(3)}_{\nu\nu\nu} \sim \lambda_O \bar{n}_B, \qquad \Gamma^{(3)}_{\zeta\zeta\zeta} \sim \lambda_D \bar{\chi}_B.$$

These are naturally expected to be diagonally stabilizing, but their exact sign is not yet proved.

(2) Mixed OD piece

$$\Gamma^{(3)}_{\nu\nu\zeta} \sim \lambda_{OD} \bar{\chi}_B, \qquad \Gamma^{(3)}_{\nu\zeta\zeta} \sim \lambda_{OD} \bar{n}_B.$$

This is the first genuine off-diagonal anisotropy source and one of the main unresolved bottlenecks.

(3) Quartic anisotropy piece

For $N_{\text{eff}} = 6$,

$$\theta_\tau = \epsilon \frac{4 - N_{\text{eff}}}{N_{\text{eff}} + 8} = -\frac{\epsilon}{7} < 0.$$

Hence the quartic anisotropy direction is RG-irrelevant, so

$$(M_Z)^{(\tau)}$$

is naturally expected to favor isotropization.

(4) Derivative-anisotropy piece

This is the most direct driver of the physical anisotropy flow. TECT expectation:

$$\theta_Y < 0$$

is natural, but not yet rigorously proved. So

$$(M_Z)^{(\nabla\text{ani})}$$

is expected to be negative, but remains open.

(5) Gauge/projector piece

Gauge-preserving EFT structure suggests compatibility with transversality and isotropization, but the actual sign depends on the detailed projector implementation.

Thus the most honest hierarchy is

$$\boxed{(M_Z)^{(\tau)} \text{ and } (M_Z)^{(\nabla\text{ani})} \text{ are the main expected negative/isotropizing sources,}}$$

$$\boxed{(M_Z)^{(OD\text{-mix})} \text{ and } (M_Z)^{(\text{gf/proj})} \text{ are the main unresolved sign bottlenecks.}}$$

The algebraic target remains

$$\boxed{M_Z = -\lambda \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}, \qquad \lambda > 0,}$$

whose physical eigenvalues are

$$\boxed{-3\lambda, \qquad -3\lambda.}$$

So the sign problem is now reduced to checking whether the expected negative sources dominate the unresolved mixed/gauge contributions.

EXPLICIT SIGN TEST FORMULA FOR THE OD-MIXED CONTRIBUTION TO M_Z

1. Why the OD-mixed piece is the first genuine off-diagonal bottleneck

After removing the redundant common anisotropy, the physical derivative-anisotropy problem reduces to the sign of the kinetic-flow matrix

$$M_Z$$

on the traceless sector

$$2\delta_T + 2\delta_O + 2\delta_D = 0.$$

At the current stage, the full sign of M_Z is not fixed by the available TECT truncation data; what remains is an explicit FRG computation requiring the exact Hessian, regulator, tensor contractions, and the A-sector gauge/constraint projection. :contentReference[oaicite:2]index=2

Among the sectorwise one-loop sources, the mixed OD piece is the first genuinely off-diagonal contribution, because it is generated by the quartic mixed coupling

$$\lambda_{OD}\rho_O\rho_D,$$

and, on the minimal admissible background

$$\bar{\Phi}_{\min} = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi)),$$

it produces cubic fluctuation vertices coupling the O- and D-sectors.

2. Background-generated OD cubic vertices

Write

$$n = \bar{n}_B + \nu, \qquad \chi = \bar{\chi}_B + \zeta, \qquad \rho_O = \frac{1}{2}n^T n, \qquad \rho_D = \frac{1}{2}\chi^T\chi.$$

Then

$$\rho_O = \rho_O^B + \bar{n}_B^T\nu + \frac{1}{2}\nu^T\nu, \qquad \rho_D = \rho_D^B + \bar{\chi}_B^T\zeta + \frac{1}{2}\zeta^T\zeta.$$

Expanding the mixed quartic term gives

$$\begin{aligned} \lambda_{OD}\rho_O\rho_D \supset \lambda_{OD}\,(\bar{n}_B^T\nu)\,\frac{1}{2}\zeta^T\zeta + \lambda_{OD}\,(\bar{\chi}_B^T\zeta)\,\frac{1}{2}\nu^T\nu \\ + \lambda_{OD}\,(\bar{n}_B^T\nu)(\bar{\chi}_B^T\zeta). \end{aligned} \tag{472}$$

Hence the two relevant background-generated cubic vertices are

$$\Gamma^{(3)}_{\nu\nu\zeta}[\bar{\Phi}_{\min}] \sim \lambda_{OD}\,\bar{\chi}_B(\xi), \qquad \Gamma^{(3)}_{\nu\zeta\zeta}[\bar{\Phi}_{\min}] \sim \lambda_{OD}\,\bar{n}_B(\xi). \tag{473}$$

These are the minimal sources of OD-mixed bubble contributions.

3. Projected OD-mixed self-energies

Let

$$G_O(q;\bar{\Phi}_{\min}), \qquad G_D(q;\bar{\Phi}_{\min})$$

denote the projected background propagators in the O- and D-sectors. Then the OD-mixed contribution to the O-sector self-energy has the schematic form

$$\Sigma_O^{(OD)}(p;\bar{\Phi}_{\min}) = -\int_q \Gamma^{(3)}_{\nu\nu\zeta}(p,q;\bar{\Phi}_{\min})\, G_D(q;\bar{\Phi}_{\min})\, \Gamma^{(3)}_{\nu\nu\zeta}(-p,q;\bar{\Phi}_{\min})\, G_O(p+q;\bar{\Phi}_{\min})\, \partial_t R_k(q), \tag{474}$$

and similarly for the D-sector

$$\Sigma_D^{(OD)}(p;\bar{\Phi}_{\min}) = -\int_q \Gamma^{(3)}_{\nu\zeta\zeta}(p,q;\bar{\Phi}_{\min})\, G_O(q;\bar{\Phi}_{\min})\, \Gamma^{(3)}_{\nu\zeta\zeta}(-p,q;\bar{\Phi}_{\min})\, G_D(p+q;\bar{\Phi}_{\min})\, \partial_t R_k(q). \tag{475}$$

Here $\int_q$ denotes the loop integration together with any background spectral sum and physical projection.

These are the minimal OD-mixed bubble kernels whose momentum derivatives feed the sectoral kinetic flows.

4. Definition of the OD-mixed matrix entry

The corresponding contribution to the kinetic-flow matrix is defined by linearization in the anisotropy variables:

$$\boxed{(M_Z^{OD})_{IJ} = -\frac{\partial}{\partial \delta_J}\frac{\partial}{\partial p^2}\Sigma_I^{(OD)}(p;\bar{\Phi}_{\min})\Big|_{p=0,\delta=0}, \qquad I,J \in \{O,D\}.} \tag{476}$$

This is the exact sign-test formula: the sign of the OD-mixed block is the sign of the momentum-derivative Jacobian of the projected mixed self-energy.

5. Symmetry-reduced 2×2 OD block

On the OD subspace, the physically relevant mixed block is

$$M_Z^{OD} = \begin{pmatrix} (M_Z^{OD})_{OO} & (M_Z^{OD})_{OD} \\ (M_Z^{OD})_{DO} & (M_Z^{OD})_{DD} \end{pmatrix}. \tag{477}$$

If the background and regulator preserve the $O \leftrightarrow D$ exchange symmetry to leading order, one expects

$$(M_Z^{OD})_{OO} = (M_Z^{OD})_{DD}, \qquad (M_Z^{OD})_{OD} = (M_Z^{OD})_{DO}. \tag{478}$$

Then the traceless OD anisotropy direction $u = \delta_O - \delta_D$ evolves with eigenvalue

$$\lambda_{OD}^{\text{phys}} = (M_Z^{OD})_{OO} - (M_Z^{OD})_{OD}. \tag{479}$$

Thus isotropization in the OD-subsector requires

$$\boxed{(M_Z^{OD})_{OO} - (M_Z^{OD})_{OD} < 0.} \tag{480}$$

6. Laplacian-sign sufficient test

A particularly clean sufficient condition is the Laplacian-type sign pattern

$$\boxed{(M_Z^{OD})_{OO} < 0, \qquad (M_Z^{OD})_{DD} < 0, \qquad (M_Z^{OD})_{OD} = (M_Z^{OD})_{DO} > 0,} \tag{481}$$

with row-sum balance

$$(M_Z^{OD})_{OO} + (M_Z^{OD})_{OD} = 0, \qquad (M_Z^{OD})_{DD} + (M_Z^{OD})_{DO} = 0. \tag{482}$$

Then

$$M_Z^{OD} = -\lambda \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad \lambda > 0, \tag{483}$$

and the physical traceless OD eigenvalue is

$$-2\lambda < 0.$$

This is exactly the sign pattern needed for the OD-mixed piece to be isotropizing rather than destabilizing.

7. Practical sign diagnostics

The explicit sign test (476) can be reduced operationally to the following diagnostics.

a. (a) Vertex sign. Because each cubic vertex carries one factor of λ_{OD} and one background profile, the bubble amplitude is quadratic in λ_{OD}:

$$\Sigma^{(OD)} \sim \lambda_{OD}^2 \, (\bar{n}_B, \bar{\chi}_B)^2 \times (\text{loop kernel}).$$

Hence the overall sign is not fixed by λ_{OD} alone; it is controlled by the loop kernel and projector contractions.

b. (b) Propagator positivity. If the projected background propagators G_O, G_D are positive in the relevant Euclidean sector and the regulator insertion $\partial_t R_k$ is positive, then the bubble carries the standard overall minus sign in (474)–(475). This tends to favor negative diagonal contributions.

c. (c) Off-diagonal sign. The off-diagonal entries depend on whether the mixed background contraction

$$\bar{n}_B(\xi)\bar{\chi}_B(\xi)$$

enters the projected kernel with the same or opposite sign relative to the diagonal pieces. This is precisely where an explicit tensor-contraction computation is still needed.

8. Honest conclusion

Therefore the current theorem-level status of the OD-mixed piece is:

$$\boxed{(M_Z^{OD})_{IJ} = -\frac{\partial}{\partial \delta_J}\frac{\partial}{\partial p^2}\Sigma_I^{(OD)}(p;\bar{\Phi}_{\min})\Big|_{p=0,\delta=0}} \tag{484}$$

is the correct explicit sign-test formula.

What is *not* yet proved is the actual sign of this expression.

The honest practical target is to verify the Laplacian-type sign structure

$$-\lambda \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad \lambda > 0,$$

for the OD-mixed subsector. This is one of the two main remaining sign bottlenecks for the full matrix M_Z.

. Summary

$$\boxed{\lambda_{OD}\rho_O\rho_D \Longrightarrow \Gamma^{(3)}_{\nu\nu\zeta} \sim \lambda_{OD}\bar{\chi}_B(\xi), \qquad \Gamma^{(3)}_{\nu\zeta\zeta} \sim \lambda_{OD}\bar{n}_B(\xi).}$$

These background-generated cubic vertices define the mixed OD-bubble self-energies

$$\Sigma_O^{(OD)}(p;\bar{\Phi}_{\min}), \qquad \Sigma_D^{(OD)}(p;\bar{\Phi}_{\min}),$$

and the explicit sign-test formula is

$$\boxed{(M_Z^{OD})_{IJ} = -\frac{\partial}{\partial\delta_J}\frac{\partial}{\partial p^2}\Sigma_I^{(OD)}(p;\bar{\Phi}_{\min})\Big|_{p=0,\delta=0}, \qquad I,J\in\{O,D\}.}$$

On the OD subsector,

$$M_Z^{OD} = \begin{pmatrix} (M_Z^{OD})_{OO} & (M_Z^{OD})_{OD} \\ (M_Z^{OD})_{DO} & (M_Z^{OD})_{DD} \end{pmatrix}.$$

If exchange symmetry gives

$$(M_Z^{OD})_{OO} = (M_Z^{OD})_{DD}, \qquad (M_Z^{OD})_{OD} = (M_Z^{OD})_{DO},$$

then the physical traceless eigenvalue is

$$\boxed{\lambda_{OD}^{\text{phys}} = (M_Z^{OD})_{OO} - (M_Z^{OD})_{OD}.}$$

Thus isotropization requires

$$\boxed{(M_Z^{OD})_{OO} - (M_Z^{OD})_{OD} < 0.}$$

A clean sufficient target is the Laplacian sign pattern

$$\boxed{M_Z^{OD} = -\lambda\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad \lambda > 0,}$$

for which the traceless OD eigenvalue is

$$\boxed{-2\lambda < 0.}$$

So the OD-mixed sign bottleneck is now reduced to one explicit projected self-energy derivative test.

EXPLICIT SIGN TEST FORMULA FOR THE A-SECTOR GAUGE/PROJECTOR CONTRIBUTION TO M_Z

1. Structural role of the gauge/projector sector

The A-sector is special because the fluctuation space contains physical transverse modes together with unphysical longitudinal / gauge / constraint directions. Therefore its contribution to the derivative-anisotropy flow cannot be defined directly from the unprojected two-point kernel. It must first be projected onto the physical transverse subspace.

This is not optional: the document is explicit that the sign of the derivative-anisotropy flow depends on the gauge/constraint structure of the A-sector. :contentReference[oaicite:4]index=4

2. Physical projected two-point kernel

Let

$$\Gamma^{(2)}_{\mu\nu}(p;\bar{\Phi}_{\min})$$

denote the background-dependent A-sector two-point kernel. Introduce the transverse projector

$$\Pi_T^{\mu\nu}(p) = \eta^{\mu\nu} - \frac{p^\mu p^\nu}{p^2}, \tag{485}$$

or, in purely spatial notation,

$$\Pi_{T,ij}(p) = \delta_{ij} - \frac{p_i p_j}{p^2}.$$

The physical projected inverse propagator is then

$$\Gamma_A^{(2),\mathrm{phys}}(p; \bar{\Phi}_{\mathrm{min}}) := \Pi_T^{\mu\nu}(p)\, \Gamma_{\mu\nu}^{(2)}(p; \bar{\Phi}_{\mathrm{min}}). \tag{486}$$

The corresponding projected flow is

$$\partial_t \Gamma_A^{(2),\mathrm{phys}}(p; \bar{\Phi}_{\mathrm{min}}) = \Pi_T^{\mu\nu}(p)\, \partial_t \Gamma_{\mu\nu}^{(2)}(p; \bar{\Phi}_{\mathrm{min}}). \tag{487}$$

3. Explicit sign test formula

The gauge/projector contribution to the kinetic-flow matrix is the response of the projected A-sector wavefunction renormalization to the sectoral anisotropy variables δ_J. Therefore the correct explicit sign-test formula is

$$\boxed{(M_Z^{\mathrm{gf/proj}})_{TJ} = \frac{\partial}{\partial \delta_J} \frac{\partial}{\partial p^2} \Big[\Pi_T^{\mu\nu}(p)\, \partial_t \Gamma_{\mu\nu}^{(2)}(p; \bar{\Phi}_{\mathrm{min}}) \Big] \Big|_{p=0,\delta=0}}, \tag{488}$$

for

$$J \in \{T, O, D\}.$$

This is the direct analogue of the OD-mixed sign test formula, but now with the physical transverse projection inserted first.

4. Consistency test preceding the sign test

Before even asking for the sign of $(M_Z^{\mathrm{gf/proj}})_{TJ}$, one must verify that the projected gauge sector is dynamically admissible.

a. *(i) Transversality test.* The flow must satisfy

$$p^\mu \Gamma_{\mu\nu}^{(2)}(p; \bar{\Phi}_{\mathrm{min}}) = 0 \qquad \text{in the infrared projected sector.} \tag{489}$$

b. *(ii) No-Proca-mass test.* The zero-momentum limit must obey

$$\Pi_T^{\mu\nu}(0)\, \Gamma_{\mu\nu}^{(2)}(0; \bar{\Phi}_{\mathrm{min}}) = 0, \tag{490}$$

equivalently,

$$m_{A,k}^2 \to 0 \qquad (k \to 0).$$

The document is explicit that the gauge-consistent infrared surface is characterized precisely by the vanishing of the Proca mass and by the survival of only the transverse mode.

c. *(iii) Regulator consistency test.* The regulator must preserve the Ward identity, or else gauge-restoring counterterms are required. Gauge-preserving schemes keep

$$p^\mu \Pi_{\mu\nu}(p) = 0$$

and avoid spurious longitudinal artifacts.

Only after these three tests are passed does (488) become a physically meaningful sign test.

5. One-loop structure of the projected A-sector flow

At one loop, the projected two-point flow has the generic form

$$\Pi_T^{\mu\nu}(p)\,\partial_t\Gamma^{(2)}_{\mu\nu}(p;\bar{\Phi}_{\min}) = \Pi_T^{\mu\nu}(p)\left\{\frac{1}{2}\mathrm{Tr}_{\mathrm{phys}}\big[G\,\Gamma^{(4)}_{\mu\nu}\,G\,\partial_t R\big] - \mathrm{Tr}_{\mathrm{phys}}\big[G\,\Gamma^{(3)}_{\mu}\,G\,\Gamma^{(3)}_{\nu}\,G\,\partial_t R\big]\right\}, \tag{491}$$

where all propagators and vertices are background-adapted and physically projected.

Thus the gauge/projector contribution decomposes schematically as

$$(M_Z^{\mathrm{gf/proj}})_{TJ} = (M_Z^{\mathrm{gf/proj},4})_{TJ} + (M_Z^{\mathrm{gf/proj},33})_{TJ}, \tag{492}$$

coming respectively from:

- projected tadpole-type gauge-sector contributions,
- projected bubble/sunset-type gauge-sector contributions.

6. Sufficient isotropizing criterion

A clean sufficient condition that the gauge/projector sector does not destabilize isotropy is the following.

d. Sufficient criterion. If:

1. the projected two-point function is transverse,
$$p^\mu\Gamma^{(2)}_{\mu\nu}(p) = 0;$$
2. no Proca mass is generated,
$$\Pi_T^{\mu\nu}(0)\Gamma^{(2)}_{\mu\nu}(0) = 0;$$
3. the projected p^2-coefficient is nonnegative in the Euclidean sign convention,
$$\frac{\partial}{\partial p^2}\Pi_T^{\mu\nu}(p)\,\partial_t\Gamma^{(2)}_{\mu\nu}(p)\bigg|_{p=0} \geq 0;$$

then the A-sector gauge/projector contribution is compatible with isotropization and does not act as a dangerous source of physical anisotropy.

e. Honest warning. This is a sufficient criterion, not a derived theorem for the current microscopic TECT data. The actual sign still depends on the detailed projector implementation.

7. Operator-classification refinement

The document provides a sharp classification of lattice-induced operators:

- Class I and II operators preserve local emergent $U(1)$ and cannot generate a gauge mass,
- Class III operators explicitly break local $U(1)$ and can generate $m_A^2 \neq 0$.

Therefore an immediate necessary condition for a healthy gauge/projector contribution is

$$\boxed{\text{the projected } A\text{-sector EFT must contain only Class I and Class II operators.}} \tag{493}$$

Otherwise the sign test is moot, because the gauge-consistent infrared surface is already destroyed.

8. Honest conclusion

Therefore the exact sign-test problem for the gauge/projector bottleneck is:

$$\boxed{(M_Z^{\mathrm{gf/proj}})_{TJ} = \frac{\partial}{\partial \delta_J}\frac{\partial}{\partial p^2}\Big[\Pi_T^{\mu\nu}(p)\,\partial_t\Gamma^{(2)}_{\mu\nu}(p;\bar{\Phi}_{\min})\Big]\Big|_{p=0,\delta=0}}, \tag{494}$$

subject to the preconditions:

$$\text{transversality}, \qquad m_{A,k}^2 \to 0, \qquad \text{gauge-preserving regulator}, \qquad \text{Class I/II only}.$$

This is the correct explicit sign-test formula. What remains open is its actual evaluation.

. Summary

$$\boxed{(M_Z^{\mathrm{gf/proj}})_{TJ} = \frac{\partial}{\partial \delta_J}\frac{\partial}{\partial p^2}\Big[\Pi_T^{\mu\nu}(p)\,\partial_t\Gamma^{(2)}_{\mu\nu}(p;\bar{\Phi}_{\min})\Big]\Big|_{p=0,\delta=0}}.$$

This is the explicit sign-test formula for the A-sector gauge/projector contribution.
It is meaningful only after the following consistency tests are passed:

$$\boxed{p^\mu\Gamma^{(2)}_{\mu\nu}(p) = 0 \quad \text{(transversality)},}$$

$$\boxed{\Pi_T^{\mu\nu}(0)\Gamma^{(2)}_{\mu\nu}(0) = 0 \quad \text{(no Proca mass)},}$$

$$\boxed{\text{the regulator preserves the Ward identity,}}$$

$$\boxed{\text{the projected EFT contains only Class I and II operators.}}$$

The one-loop projected flow decomposes schematically as

$$\boxed{(M_Z^{\mathrm{gf/proj}})_{TJ} = (M_Z^{\mathrm{gf/proj,4}})_{TJ} + (M_Z^{\mathrm{gf/proj,33}})_{TJ},}$$

coming from the projected tadpole and projected bubble/sunset sectors.
A sufficient isotropizing criterion is:

$$\boxed{\frac{\partial}{\partial p^2}\Pi_T^{\mu\nu}(p)\,\partial_t\Gamma^{(2)}_{\mu\nu}(p)\Big|_{p=0} \geq 0}$$

together with transversality and zero Proca mass.
So the gauge/projector bottleneck is now reduced to one explicit projected polarization-tensor derivative test.

TECT DIRAC / IR-ISOTROPY / M_Z FRONTIER MASTER VERIFICATION MAP

A. Global logical status

The present status of the TECT program is:

$$\boxed{\text{the theorem-level architecture is structurally closed,}}$$

whereas

$$\boxed{\text{the final microscopic realization is reduced to a finite list of explicit verification problems.}}$$

This is exactly the logical separation already emphasized in the current TECT notes: the closed core is no longer conceptual, and the remaining frontier is a finite checklist of microscopic inputs.

B. Layer I: theorem-level closed core

a. B1. UV and shell structure. The BCC phase is selected at the structural level, and the shell-based local reduction is closed. The shell-reduced soft sector admits the effective infrared multiplet

$$\Phi = (A_1, A_2, n_1, n_2, \chi_1, \chi_2), \qquad N_{\mathrm{eff}} = 6.$$

This is the theorem-level starting point for the IR isotropy and Dirac-emergence chain.

b. B2. Local shell-enriched Dirac architecture. The local projector-defect transport analysis reduces the actual Weyl/Dirac nondegeneracy problem to a finite coefficient test. The microscopic transport ansatz has the form

$$i\partial_t \eta = \Big[-\rho\, \partial_\xi^2 + U_B(\xi) - i\,\theta'(\xi)\,\mathcal{M}^\mu(\theta(\xi))\partial_{y^\mu}\Big]\eta + O(\partial_y^2)\eta,$$

and the shell witnesses are determined by a finite matrix ${\Lambda_i}^\mu$.

c. B3. Quartic anisotropy irrelevance. At the TECT soft-sector level, the leading quartic cubic anisotropy has

$$\theta_\tau = \epsilon \frac{4 - N_{\mathrm{eff}}}{N_{\mathrm{eff}} + 8} = -\frac{\epsilon}{7} < 0 \qquad (N_{\mathrm{eff}} = 6),$$

so the quartic anisotropy coupling flows to zero in the infrared. This part of the isotropy mechanism is structurally closed.

d. B4. Gauge-consistent transverse sector. In the gauge-preserving reference EFT, the polarization tensor is transverse:

$$p^\mu \Pi_{\mu\nu}(p) = 0,$$

and the emergent gauge sector is massless provided the regulator preserves the Ward identity and no explicit Class III symmetry-breaking operator is present. :contentReference[oaicite:6]index=6

e. B5. Mass-space classification. For the minimal doubled Dirac block, the constant mass space is exactly two-dimensional, and valley-charge $U(1)$ is a sufficient symmetry to forbid all constant masses. Thus the mass-protection problem is reduced to a finite residual-symmetry algebra check. :contentReference[oaicite:7]index=7

C. Layer II: problems already reduced to explicit finite tests

f. C1. Dirac nondegeneracy witness test. The Weyl/Dirac nondegeneracy problem is reduced to the finite coefficient condition

$$\Lambda_\parallel \neq 0, \qquad (\Lambda_E, \Lambda_O) \neq (0, 0).$$

So the full PDE problem is already reduced to determining finitely many transport coefficients.

g. C2. Remote-gap / zero-mixing check. The local shell-reduced fermion framework is structurally closed, but the final microscopic TECT realization still requires finite checks such as:

$$\Delta > 0, \qquad V(0) = 0.$$

These are no longer conceptual problems; they are explicit spectral/projection tests inside the microscopic linearization.

h. C3. Mass-protection residual symmetry audit. The final massless Dirac closure requires determining the actual residual constant algebra

$$A_{\text{res}}$$

and verifying that it excludes the two constant mass generators. Again, this is reduced to a finite symmetry-algebra check. :contentReference[oaicite:10]index=10

i. C4. IR isotropy beyond quartic anisotropy. What remains after quartic anisotropy irrelevance is the derivative-anisotropy and sectoral velocity-mismatch problem. This is encoded in the kinetic-flow matrix

$$M_Z.$$

The document is explicit that what remains is:

1. derive the precise soft-mode spectrum,
2. compute the threshold functions and anomalous dimensions in a TECT-specific FRG truncation,
3. evaluate the derivative-anisotropy stability matrix M_Z explicitly,
4. solve the full Wetterich flow numerically.

Thus the isotropy frontier is already reduced to the explicit FRG computation of M_Z. :contentReference[oaicite:11]index=11

D. Layer III: current M_Z frontier

j. D1. Physical derivative-anisotropy variables. The physical anisotropy variables are

$$\delta_T, \qquad \delta_O, \qquad \delta_D, \qquad 2\delta_T + 2\delta_O + 2\delta_D = 0.$$

They are classically marginal, so their infrared fate is controlled entirely by

$$\partial_t \begin{pmatrix} \delta_T \\ \delta_O \\ \delta_D \end{pmatrix} = M_Z \begin{pmatrix} \delta_T \\ \delta_O \\ \delta_D \end{pmatrix} + \cdots .$$

k. D2. Minimal FRG truncation. The TECT-adapted FRG truncation is

$$\Gamma_k[\Phi] = \int d^dx \left[\frac{Z_{T,k}}{2}(\partial A)^2 + \frac{Z_{O,k}}{2}(\partial n)^2 + \frac{Z_{D,k}}{2}(\partial \chi)^2 + Y_k O_{\nabla,\text{ani}} + U_k(\rho,\tau) \right].$$

This defines the formal M_Z problem, but not yet its sign.

l. D3. Why the strict symmetric-point truncation is insufficient. At the strict symmetric point $\Phi = 0$, one typically has

$$\Gamma^{(3)}|_{\Phi=0} = 0,$$

while only the local quartic $\Gamma^{(4)}$ survives. Such tadpoles are generically p^2-silent and may fail to generate a nontrivial Z_I-flow. Therefore the orthodox completion requires a nontrivial background field:

$$\Phi = \bar{\Phi} + \varphi, \qquad \bar{\Phi} \neq 0.$$

m. D4. Minimal admissible background. The most faithful minimal choice is shell-/defect-adapted:

$$\bar{\Phi}_{\min} = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi)),$$

built on the actual projector-defect background

$$P_B(\xi) = z_B(\xi) z_B(\xi)^\dagger, \qquad P_B(-\xi) = P_B(\xi),$$

with odd transport kernel

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \qquad V^\mu(-\xi) = -V^\mu(\xi).$$

This background reactivates

$$\Gamma^{(3)}[\bar{\Phi}_{\min}] \neq 0$$

and hence the bubble/sunset sector in the two-point flow.

n. D5. Background-generated vertex inventory. On $\bar{\Phi}_{\min}$, the relevant cubic fluctuation vertices are generated by:

- self quartics:

$$\Gamma^{(3)}_{\nu\nu\nu} \sim \lambda_O\, \bar{n}_B, \qquad \Gamma^{(3)}_{\zeta\zeta\zeta} \sim \lambda_D\, \bar{\chi}_B;$$

- mixed quartic:

$$\Gamma^{(3)}_{\nu\nu\zeta} \sim \lambda_{OD}\, \bar{\chi}_B, \qquad \Gamma^{(3)}_{\nu\zeta\zeta} \sim \lambda_{OD}\, \bar{n}_B;$$

- quartic anisotropy:

$$\Gamma^{(3)}_{\tau}[\bar{\Phi}_{\min}] \neq 0;$$

- derivative anisotropy:

$$\Gamma^{(3)}_{\nabla\mathrm{ani}}[\bar{\Phi}_{\min}] \neq 0$$

  in the field-dependent background completion;

- gauge/projector completion:

$$\Gamma^{(3)}_{\mathrm{gf/proj}}[\bar{\Phi}_{\min}]$$

  implementation-dependent.

o. D6. Background-adapted regulator and projector. To define the projected flow honestly, one needs:

$$R_k[\bar{\Phi}_{\min}] = \mathrm{diag}\big(R_{T,k}, R_{O,k}, R_{D,k}\big),$$

with the A-sector regulator acting only on the physical projected subspace

$$\Pi^{\mathrm{phys}}_A.$$

The projected propagator is

$$G^{\mathrm{phys}}_k(\bar{\Phi}_{\min}) = \Big(\Pi^{\mathrm{phys}}[\Gamma^{(2)}_k[\bar{\Phi}_{\min}] + R_k[\bar{\Phi}_{\min}]]\Pi^{\mathrm{phys}}\Big)^{-1}.$$

p. D7. Explicit sign-test bottlenecks. At this stage, the two main remaining sign bottlenecks are:

$$\boxed{(M_Z)^{(OD\text{-}mix)}}$$

and

$$\boxed{(M_Z)^{(\mathrm{gf/proj})}.}$$

They have been reduced to explicit test formulas:

q. (i) OD-mixed sign test

$$(M_Z^{OD})_{IJ} = -\left.\frac{\partial}{\partial \delta_J}\frac{\partial}{\partial p^2}\Sigma^{(OD)}_I(p;\bar{\Phi}_{\min})\right|_{p=0,\delta=0}, \qquad I, J \in \{O, D\},$$

with sufficient Laplacian-type target

$$M_Z^{OD} = -\lambda \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad \lambda > 0.$$

r. (ii) Gauge/projector sign test

$$(M_Z^{\mathrm{gf/proj}})_{TJ} = \frac{\partial}{\partial \delta_J}\frac{\partial}{\partial p^2}\Big[\Pi_T^{\mu\nu}(p)\,\partial_t\Gamma^{(2)}_{\mu\nu}(p;\bar{\Phi}_{\min})\Big]\Big|_{p=0,\delta=0}.$$

This is meaningful only after verifying:

$$p^\mu\Gamma^{(2)}_{\mu\nu}=0,\qquad m^2_{A,k}\to 0,\qquad \text{gauge-preserving regulator},\qquad \text{Class I/II only.}$$

E. Final verification map

The full TECT verification map is therefore:

$$\boxed{\text{TECT microscopic PDE} \Longrightarrow \text{shell-reduced defect/projector transport} \Longrightarrow \text{finite witness matrix } \Lambda_i{}^\mu} \tag{495}$$

$$\boxed{\Lambda_\parallel \neq 0,\quad (\Lambda_E,\Lambda_O)\neq(0,0) \Longrightarrow \text{local Weyl/Dirac nondegeneracy}} \tag{496}$$

$$\boxed{\Delta>0,\quad V(0)=0,\quad A_{\mathrm{res}}\cap\mathcal{M}=\{0\} \Longrightarrow \text{microscopic Dirac closure}} \tag{497}$$

$$\boxed{N_{\mathrm{eff}}=6,\quad \theta_\tau=-\frac{\epsilon}{7}<0 \Longrightarrow \text{quartic anisotropy is IR-irrelevant}} \tag{498}$$

$$\boxed{\bar{\Phi}_{\min} \Longrightarrow \Gamma_k^{(2)}[\bar{\Phi}_{\min}],\Gamma_k^{(3)}[\bar{\Phi}_{\min}],\Gamma_k^{(4)}[\bar{\Phi}_{\min}] \Longrightarrow M_Z(\bar{\Phi}_{\min})} \tag{499}$$

$$\boxed{\mathrm{spec}(M_Z|_{\mathrm{traceless}})\subset(-\infty,0) \Longrightarrow \delta_T,\delta_O,\delta_D\to 0 \Longrightarrow \text{IR isotropy / velocity locking}} \tag{500}$$

F. Honest final frontier

Thus the remaining frontier is no longer structural. It is exactly the explicit microscopic verification of:

1. the finite witness coefficients $\Lambda_\parallel,\Lambda_E,\Lambda_O$,
2. the remote gap and zero-mixing checks,
3. the residual symmetry algebra excluding constant mass,
4. the sign of the two main bottlenecks in M_Z:

$$(M_Z)^{(OD\text{-}mix)},\qquad (M_Z)^{(\mathrm{gf/proj})}.$$

Equivalently:

$$\boxed{\text{all remaining open questions are explicit finite projection / gap / symmetry / FRG sign tests.}}$$

. Summary

Master Verification Map

Microscopic TECT PDE $\Longrightarrow$ shell-/defect-reduced transport $\Longrightarrow \Lambda_i{}^{\mu} \Longrightarrow \Lambda_\parallel \neq 0,\ (\Lambda_E, \Lambda_O) \neq (0,0) \Longrightarrow$ local Weyl/Dirac nonde

$$\text{Then require} \qquad \Delta > 0, \qquad V(0) = 0, \qquad A_{\text{res}} \cap \mathcal{M} = \{0\},$$

to obtain

$$\boxed{\text{microscopic Dirac closure.}}$$

$$\text{On the IR-isotropy side:} \qquad N_{\text{eff}} = 6 \Longrightarrow \theta_\tau = -\frac{\epsilon}{7} < 0,$$

so quartic cubic anisotropy is already irrelevant.

The remaining derivative-anisotropy problem is

$$\boxed{\partial_t \begin{pmatrix} \delta_T \\ \delta_O \\ \delta_D \end{pmatrix} = M_Z \begin{pmatrix} \delta_T \\ \delta_O \\ \delta_D \end{pmatrix} + \cdots, \qquad 2\delta_T + 2\delta_O + 2\delta_D = 0.}$$

To define M_Z honestly, use the minimal admissible shell-/defect-adapted background

$$\boxed{\bar{\Phi}_{\min} = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi)).}$$

Then

$$\Gamma_k^{(3)}[\bar{\Phi}_{\min}] \neq 0,$$

and the background-field chain is

$$\boxed{\bar{\Phi}_{\min} \Longrightarrow \Gamma_k^{(2)}[\bar{\Phi}_{\min}], \Gamma_k^{(3)}[\bar{\Phi}_{\min}], \Gamma_k^{(4)}[\bar{\Phi}_{\min}] \Longrightarrow M_Z(\bar{\Phi}_{\min}).}$$

The two main remaining sign bottlenecks are:

$$\boxed{(M_Z)^{(OD\text{-}mix)} \qquad \text{and} \qquad (M_Z)^{(\text{gf/proj})}.}$$

Their explicit sign tests are:

$$\boxed{(M_Z^{OD})_{IJ} = -\frac{\partial}{\partial \delta_J} \frac{\partial}{\partial p^2} \Sigma_I^{(OD)}(p; \bar{\Phi}_{\min}) \Big|_{p=0, \delta=0}, \qquad I, J \in \{O, D\},}$$

$$\boxed{(M_Z^{\text{gf/proj}})_{TJ} = \frac{\partial}{\partial \delta_J} \frac{\partial}{\partial p^2} \Big[\Pi_T^{\mu\nu}(p)\, \partial_t \Gamma_{\mu\nu}^{(2)}(p; \bar{\Phi}_{\min}) \Big] \Big|_{p=0, \delta=0}.}$$

Thus the final frontier is no longer conceptual:

$$\boxed{\text{it is a finite list of explicit witness, gap, symmetry, and FRG sign tests.}}$$

PROJECTED LOOP-KERNEL FORMULA FOR $\Sigma_O^{(OD)}$ AND $\Sigma_D^{(OD)}$

1. Background-generated mixed cubic vertices

On the minimal admissible background

$$\bar{\Phi}_{\min} = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi)),$$

the mixed quartic interaction

$$\lambda_{OD}\rho_O\rho_D$$

generates the cubic fluctuation vertices

$$\Gamma^{(3)}_{\nu\nu\zeta}[\bar{\Phi}_{\min}] \sim \lambda_{OD}\,\bar{\chi}_B(\xi), \qquad \Gamma^{(3)}_{\nu\zeta\zeta}[\bar{\Phi}_{\min}] \sim \lambda_{OD}\,\bar{n}_B(\xi). \tag{501}$$

These are the minimal background-generated sources of the mixed OD bubble/sunset contribution to the two-point flow.

2. Projected propagators and mode basis

Let

$$\mathcal{H}_{OO}, \qquad \mathcal{H}_{DD}$$

be the background-dependent O- and D-sector Hessian blocks. Choose orthonormal eigenmode bases

$$\mathcal{H}_{OO}u_m = E^O_m u_m, \qquad \mathcal{H}_{DD}v_n = E^D_n v_n,$$

with projectors

$$P_O = \sum_m |u_m\rangle\langle u_m|, \qquad P_D = \sum_n |v_n\rangle\langle v_n|.$$

The regulated projected propagators are

$$G_O(q;\bar{\Phi}_{\min}) = \Big(P_O[\mathcal{H}_{OO} + R_{O,k}]P_O\Big)^{-1}, \qquad G_D(q;\bar{\Phi}_{\min}) = \Big(P_D[\mathcal{H}_{DD} + R_{D,k}]P_D\Big)^{-1}. \tag{502}$$

In the mixed kernel formulas below, the loop trace includes:

1. worldvolume momentum integration q,
2. summation over the background transverse-mode indices m, n,
3. any residual internal 2×2 bundle contraction.

3. Explicit projected kernel for $\Sigma_O^{(OD)}$

The OD-mixed contribution to the O-sector two-point function is generated by the cubic vertex

$$\Gamma^{(3)}_{\nu\nu\zeta}.$$

Its projected self-energy has the form

$$\Sigma_O^{(OD)}(p;\bar{\Phi}_{\min}) = -\int \frac{d^dq}{(2\pi)^d}\sum_{m,n}\mathcal{V}^{(OD)}_{O;mn}(p,q)\,G_D^{(n)}(q)\,\mathcal{V}^{(OD)}_{O;nm}(-p,q)\,G_O^{(m)}(p+q)\,\partial_t R_k(q), \tag{503}$$

where

$$\mathcal{V}^{(OD)}_{O;mn}(p,q) := \left\langle u_m \middle| \Gamma^{(3)}_{\nu\nu\zeta}(p,q;\bar{\Phi}_{\min}) \middle| v_n \right\rangle. \tag{504}$$

Using (501), this matrix element is schematically

$$\mathcal{V}^{(OD)}_{O;mn}(p,q) = \lambda_{OD}\, \mathcal{B}^{\chi}_{mn}(\xi)\, \mathcal{P}_O(p,q), \tag{505}$$

with

$$\mathcal{B}^{\chi}_{mn}(\xi) \sim \langle u_m, \bar{\chi}_B(\xi) v_n \rangle,$$

and $\mathcal{P}_O(p,q)$ collecting momentum/projector structure.
Therefore

$$\boxed{\Sigma^{(OD)}_O(p;\bar{\Phi}_{\min}) = -\lambda^2_{OD} \int_q \sum_{m,n} \mathcal{B}^{\chi}_{mn}\, \mathcal{P}_O(p,q)\, G^{(n)}_D(q)\, \mathcal{B}^{\chi}_{nm}\, \mathcal{P}_O(-p,q)\, G^{(m)}_O(p+q)\, \partial_t R_k(q).} \tag{506}$$

4. Explicit projected kernel for $\Sigma^{(OD)}_D$

Similarly, the D-sector mixed self-energy is generated by

$$\Gamma^{(3)}_{\nu\zeta\zeta}.$$

Its projected form is

$$\Sigma^{(OD)}_D(p;\bar{\Phi}_{\min}) = -\int \frac{d^d q}{(2\pi)^d} \sum_{m,n} \mathcal{V}^{(OD)}_{D;mn}(p,q)\, G^{(m)}_O(q)\, \mathcal{V}^{(OD)}_{D;nm}(-p,q)\, G^{(n)}_D(p+q)\, \partial_t R_k(q), \tag{507}$$

with

$$\mathcal{V}^{(OD)}_{D;mn}(p,q) := \left\langle v_n \middle| \Gamma^{(3)}_{\nu\zeta\zeta}(p,q;\bar{\Phi}_{\min}) \middle| u_m \right\rangle = \lambda_{OD}\, \mathcal{B}^{n}_{nm}(\xi)\, \mathcal{P}_D(p,q), \tag{508}$$

where

$$\mathcal{B}^{n}_{nm}(\xi) \sim \langle v_n, \bar{n}_B(\xi) u_m \rangle.$$

Thus

$$\boxed{\Sigma^{(OD)}_D(p;\bar{\Phi}_{\min}) = -\lambda^2_{OD} \int_q \sum_{m,n} \mathcal{B}^{n}_{nm}\, \mathcal{P}_D(p,q)\, G^{(m)}_O(q)\, \mathcal{B}^{n}_{mn}\, \mathcal{P}_D(-p,q)\, G^{(n)}_D(p+q)\, \partial_t R_k(q).} \tag{509}$$

5. Momentum-derivative extraction

The OD-mixed contribution to the sectoral wavefunction flows is obtained by

$$\partial_t Z^{(OD)}_O = \frac{\partial}{\partial p^2} \Sigma^{(OD)}_O(p;\bar{\Phi}_{\min}) \bigg|_{p=0}, \qquad \partial_t Z^{(OD)}_D = \frac{\partial}{\partial p^2} \Sigma^{(OD)}_D(p;\bar{\Phi}_{\min}) \bigg|_{p=0}. \tag{510}$$

Then the OD-mixed block of the anisotropy-flow matrix is

$$\boxed{(M^{OD}_Z)_{IJ} = -\frac{\partial}{\partial \delta_J} \frac{\partial}{\partial p^2} \Sigma^{(OD)}_I(p;\bar{\Phi}_{\min}) \bigg|_{p=0,\delta=0}, \qquad I,J \in \{O,D\}.} \tag{511}$$

6. Sign separation

The projected kernel formulas (506) and (509) show that the sign problem separates into three pieces.

a. (i) Overall loop sign. Both self-energies carry the standard bubble minus sign:

$$\Sigma^{(OD)} \sim -(\text{positive kernel}) \quad \text{if all projected propagators and } \partial_t R_k \text{ are positive.}$$

This tends to favor negative diagonal contributions.

b. (ii) Background-overlap sign. The products

$$\mathcal{B}^{\chi}_{mn}\mathcal{B}^{\chi}_{nm}, \qquad \mathcal{B}^{n}_{nm}\mathcal{B}^{n}_{mn}$$

are positive-semidefinite if the relevant background overlap matrices are Hermitian. This again tends to favor negative diagonal terms.

c. (iii) Off-diagonal anisotropy response. The sign of

$$\frac{\partial}{\partial \delta_D}\Sigma_O^{(OD)}, \qquad \frac{\partial}{\partial \delta_O}\Sigma_D^{(OD)}$$

depends on how the anisotropy variables deform the propagators and projected vertices. This is the genuine mixed sign bottleneck and cannot be fixed without explicit contraction.

7. Exchange-symmetric simplification

If the background and projected regulator are symmetric under the exchange

$$O \leftrightarrow D,$$

then

$$(M_Z^{OD})_{OO} = (M_Z^{OD})_{DD}, \qquad (M_Z^{OD})_{OD} = (M_Z^{OD})_{DO}. \tag{512}$$

The physical traceless OD anisotropy direction

$$u = \delta_O - \delta_D$$

then has eigenvalue

$$\lambda_{OD}^{\text{phys}} = (M_Z^{OD})_{OO} - (M_Z^{OD})_{OD}. \tag{513}$$

Therefore the isotropizing sign condition is

$$\boxed{(M_Z^{OD})_{OO} - (M_Z^{OD})_{OD} < 0.} \tag{514}$$

A sufficient Laplacian-type target is

$$\boxed{M_Z^{OD} = -\lambda \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad \lambda > 0,} \tag{515}$$

for which the physical eigenvalue is

$$-2\lambda < 0.$$

8. Honest conclusion

Thus the OD-mixed bottleneck is now reduced completely to the projected loop-kernel sign test:

$$\boxed{\Sigma_O^{(OD)}(p;\bar{\Phi}_{\min}), \qquad \Sigma_D^{(OD)}(p;\bar{\Phi}_{\min})}$$

are known formally, and

$$\boxed{(M_Z^{OD})_{IJ} = -\frac{\partial}{\partial\delta_J}\frac{\partial}{\partial p^2}\Sigma_I^{(OD)}(p;\bar{\Phi}_{\min})\Big|_{p=0,\delta=0}}$$

is the exact sign-test formula.

What remains open is no longer the structure, but the explicit sign of the projected contraction kernel.

. Summary

$$\boxed{\lambda_{OD}\rho_O\rho_D \Longrightarrow \Gamma^{(3)}_{\nu\nu\zeta} \sim \lambda_{OD}\bar{\chi}_B(\xi), \qquad \Gamma^{(3)}_{\nu\zeta\zeta} \sim \lambda_{OD}\bar{n}_B(\xi).}$$

These generate the projected mixed self-energies

$$\boxed{\Sigma_O^{(OD)}(p;\bar{\Phi}_{\min}) = -\lambda_{OD}^2\int_q\sum_{m,n}\mathcal{B}^{\chi}_{mn}\,\mathcal{P}_O(p,q)\,G_D^{(n)}(q)\,\mathcal{B}^{\chi}_{nm}\,\mathcal{P}_O(-p,q)\,G_O^{(m)}(p+q)\,\partial_t R_k(q),}$$

$$\boxed{\Sigma_D^{(OD)}(p;\bar{\Phi}_{\min}) = -\lambda_{OD}^2\int_q\sum_{m,n}\mathcal{B}^{n}_{nm}\,\mathcal{P}_D(p,q)\,G_O^{(m)}(q)\,\mathcal{B}^{n}_{mn}\,\mathcal{P}_D(-p,q)\,G_D^{(n)}(p+q)\,\partial_t R_k(q).}$$

Then the exact OD-mixed sign-test formula is

$$\boxed{(M_Z^{OD})_{IJ} = -\frac{\partial}{\partial\delta_J}\frac{\partial}{\partial p^2}\Sigma_I^{(OD)}(p;\bar{\Phi}_{\min})\Big|_{p=0,\delta=0}, \qquad I,J\in\{O,D\}.}$$

If exchange symmetry gives

$$(M_Z^{OD})_{OO} = (M_Z^{OD})_{DD}, \qquad (M_Z^{OD})_{OD} = (M_Z^{OD})_{DO},$$

then the physical traceless eigenvalue is

$$\boxed{\lambda_{OD}^{\text{phys}} = (M_Z^{OD})_{OO} - (M_Z^{OD})_{OD}.}$$

Hence isotropization requires

$$\boxed{(M_Z^{OD})_{OO} - (M_Z^{OD})_{OD} < 0.}$$

A clean sufficient target is the Laplacian form

$$\boxed{M_Z^{OD} = -\lambda\begin{pmatrix}1 & -1\\ -1 & 1\end{pmatrix}, \qquad \lambda > 0,}$$

for which

$$\boxed{\lambda_{OD}^{\text{phys}} = -2\lambda < 0.}$$

So the OD-mixed bottleneck is now reduced to one explicit projected loop-kernel derivative test.

MINIMAL CALCULABLE FORM OF THE PROJECTED OD-MIXED LOOP KERNEL

1. Starting point

From the background split

$$\Phi = \bar{\Phi}_{\min} + \varphi, \qquad \bar{\Phi}_{\min} = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi)), \qquad \varphi = (a, \nu, \zeta),$$

the mixed quartic interaction

$$\lambda_{OD}\rho_O\rho_D$$

generates the background-induced cubic fluctuation vertices

$$\Gamma^{(3)}_{\nu\nu\zeta} \sim \lambda_{OD}\bar{\chi}_B(\xi), \qquad \Gamma^{(3)}_{\nu\zeta\zeta} \sim \lambda_{OD}\bar{n}_B(\xi).$$

Hence the OD-mixed self-energies take the projected bubble form

$$\Sigma_O^{(OD)}(p; \bar{\Phi}_{\min}), \qquad \Sigma_D^{(OD)}(p; \bar{\Phi}_{\min}),$$

and the exact sign-test formula is

$$(M_Z^{OD})_{IJ} = -\frac{\partial}{\partial \delta_J}\frac{\partial}{\partial p^2}\Sigma_I^{(OD)}(p; \bar{\Phi}_{\min})\bigg|_{p=0,\delta=0}, \qquad I, J \in \{O, D\}. \tag{516}$$

2. Rank-one overlap reduction

Let

$$\mathcal{B}^{\chi}_{mn} := \langle u_m, \bar{\chi}_B v_n\rangle, \qquad \mathcal{B}^{n}_{nm} := \langle v_n, \bar{n}_B u_m\rangle,$$

where $\{u_m\}$ and $\{v_n\}$ are orthonormal eigenmodes of the projected O- and D-sector Hessians.

Then the factorized kernels are

$$\Sigma_O^{(OD)}(p) = -\lambda_{OD}^2 \int_q \sum_{m,n} \mathcal{B}^{\chi}_{mn}\, \mathcal{P}_O(p, q)\, G_D^{(n)}(q)\, \mathcal{B}^{\chi}_{nm}\, \mathcal{P}_O(-p, q)\, G_O^{(m)}(p + q)\, \partial_t R_k(q), \tag{517}$$

$$\Sigma_D^{(OD)}(p) = -\lambda_{OD}^2 \int_q \sum_{m,n} \mathcal{B}^{n}_{nm}\, \mathcal{P}_D(p, q)\, G_O^{(m)}(q)\, \mathcal{B}^{n}_{mn}\, \mathcal{P}_D(-p, q)\, G_D^{(n)}(p + q)\, \partial_t R_k(q). \tag{518}$$

If the background overlaps are Hermitian,

$$\mathcal{B}^{\chi}_{nm} = (\mathcal{B}^{\chi}_{mn})^*, \qquad \mathcal{B}^{n}_{mn} = (\mathcal{B}^{n}_{nm})^*,$$

then the overlap products become positive semidefinite:

$$\mathcal{B}^{\chi}_{mn}\mathcal{B}^{\chi}_{nm} = |\mathcal{B}^{\chi}_{mn}|^2 \geq 0, \qquad \mathcal{B}^{n}_{nm}\mathcal{B}^{n}_{mn} = |\mathcal{B}^{n}_{nm}|^2 \geq 0.$$

Thus the sign problem is reduced to the momentum/projector part of the loop kernel.

3. Small-p expansion

Assume the projected momentum structure is smooth near $p = 0$:

$$\mathcal{P}_I(p, q) = \mathcal{P}_I^{(0)}(q) + p_\mu \mathcal{P}_I^{(1)\mu}(q) + \frac{1}{2}p_\mu p_\nu \mathcal{P}_I^{(2)\mu\nu}(q) + \cdots .$$

Also expand the propagator:

$$G_I^{(m)}(p + q) = G_I^{(m)}(q) + p_\mu \partial_{q_\mu} G_I^{(m)}(q) + \frac{1}{2}p_\mu p_\nu \partial_{q_\mu}\partial_{q_\nu} G_I^{(m)}(q) + \cdots .$$

Then

$$\frac{\partial}{\partial p^2}\Sigma_I^{(OD)}(p)\bigg|_{p=0}$$

is a finite linear combination of the following three kernel types:

1. **vertex-curvature term**

$$K_I^{(V)}(q) \sim \mathcal{P}_I^{(2)}(q)\, G_{\bar{I}}(q)\, G_I(q)\, \partial_t R_k(q);$$

2. **propagator-curvature term**

$$K_I^{(G)}(q) \sim \mathcal{P}_I^{(0)}(q)\, G_{\bar{I}}(q)\, \partial_q^2 G_I(q)\, \partial_t R_k(q);$$

3. **mixed derivative term**

$$K_I^{(VG)}(q) \sim \mathcal{P}_I^{(1)}(q)\, G_{\bar{I}}(q)\, \partial_q G_I(q)\, \partial_t R_k(q).$$

Therefore the sign-test formula becomes

$$\boxed{\frac{\partial}{\partial p^2}\Sigma_I^{(OD)}(p)\bigg|_{p=0} = -\lambda_{OD}^2 \int_q \sum_{m,n} W_{mn}^{(I)}(q)\left[K_I^{(V)}(q) + K_I^{(G)}(q) + K_I^{(VG)}(q)\right],} \tag{519}$$

where

$$W_{mn}^{(O)}(q) = |\mathcal{B}_{mn}^{\chi}|^2\, \partial_t R_k(q), \qquad W_{mn}^{(D)}(q) = |\mathcal{B}_{nm}^{n}|^2\, \partial_t R_k(q).$$

4. Minimal sign criterion

A clean sufficient criterion for the diagonal part is:

$$\boxed{\partial_t R_k(q) \geq 0, \qquad G_O^{(m)}(q) \geq 0, \qquad G_D^{(n)}(q) \geq 0, \qquad K_I^{(V)} + K_I^{(G)} + K_I^{(VG)} \geq 0} \tag{520}$$

for the relevant projected Euclidean kernel region.

Then

$$\frac{\partial}{\partial p^2}\Sigma_I^{(OD)}(p)\bigg|_{p=0} \leq 0,$$

and hence the diagonal entries tend to satisfy

$$(M_Z^{OD})_{OO} < 0, \qquad (M_Z^{OD})_{DD} < 0.$$

This is not yet a theorem for the actual TECT data, but it gives the exact sign target of the projected contraction.

5. Off-diagonal response test

The off-diagonal entries arise from the anisotropy response:

$$(M_Z^{OD})_{OD} = -\frac{\partial}{\partial \delta_D}\frac{\partial}{\partial p^2}\Sigma_O^{(OD)}(p)\bigg|_{p=0,\delta=0},$$

$$(M_Z^{OD})_{DO} = -\frac{\partial}{\partial \delta_O}\frac{\partial}{\partial p^2}\Sigma_D^{(OD)}(p)\bigg|_{p=0,\delta=0}.$$

These derivatives act on:

- the projected propagators G_O, G_D,
- possibly the projected vertices $\mathcal{P}_O, \mathcal{P}_D$,
- the regulator R_k if it is sector-dependent.

Therefore the explicit off-diagonal sign test is:

$$\boxed{(M_Z^{OD})_{OD} > 0, \qquad (M_Z^{OD})_{DO} > 0} \tag{521}$$

together with

$$\boxed{(M_Z^{OD})_{OO} + (M_Z^{OD})_{OD} = 0, \qquad (M_Z^{OD})_{DD} + (M_Z^{OD})_{DO} = 0} \tag{522}$$

if one wants the exact Laplacian target

$$-\lambda \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

6. Exchange-symmetric reduction

If the projected background and regulator respect $O \leftrightarrow D$ exchange symmetry, then

$$(M_Z^{OD})_{OO} = (M_Z^{OD})_{DD}, \qquad (M_Z^{OD})_{OD} = (M_Z^{OD})_{DO},$$

and the physical traceless eigenvalue is

$$\lambda_{OD}^{\text{phys}} = (M_Z^{OD})_{OO} - (M_Z^{OD})_{OD}.$$

Thus the final sign test reduces to a single inequality:

$$\boxed{(M_Z^{OD})_{OO} - (M_Z^{OD})_{OD} < 0.} \tag{523}$$

7. Honest status

At this point the OD-mixed bottleneck is reduced as far as one can go without actually performing the projected contraction.

What is now explicit:

1. the exact bubble self-energy formulas $\Sigma_O^{(OD)}, \Sigma_D^{(OD)}$,
2. the small-p kernel decomposition,
3. the sufficient sign pattern for isotropization,
4. the final one-inequality reduction under $O \leftrightarrow D$ symmetry.

What remains open:

1. the detailed form of the projected momentum structures $\mathcal{P}_O, \mathcal{P}_D$,
2. the actual sign of the anisotropy response derivatives,
3. the numerical or analytic evaluation of the loop kernels.

. Summary

$$\boxed{\Sigma_O^{(OD)}(p) = -\lambda_{OD}^2 \int_q \sum_{m,n} \mathcal{B}^{\chi}_{mn}\, \mathcal{P}_O(p,q)\, G_D^{(n)}(q)\, \mathcal{B}^{\chi}_{nm}\, \mathcal{P}_O(-p,q)\, G_O^{(m)}(p+q)\, \partial_t R_k(q),}$$

$$\boxed{\Sigma_D^{(OD)}(p) = -\lambda_{OD}^2 \int_q \sum_{m,n} \mathcal{B}^{n}_{nm}\, \mathcal{P}_D(p,q)\, G_O^{(m)}(q)\, \mathcal{B}^{n}_{mn}\, \mathcal{P}_D(-p,q)\, G_D^{(n)}(p+q)\, \partial_t R_k(q).}$$

The exact sign-test formula is

$$\boxed{(M_Z^{OD})_{IJ} = -\frac{\partial}{\partial \delta_J}\frac{\partial}{\partial p^2}\Sigma_I^{(OD)}(p)\Big|_{p=0,\delta=0}, \qquad I,J \in \{O,D\}.}$$

After small-p expansion,

$$\frac{\partial}{\partial p^2}\Sigma_I^{(OD)}(p)\Big|_{p=0} = -\lambda_{OD}^2 \int_q \sum_{m,n} W_{mn}^{(I)}(q)\left[K_I^{(V)}(q) + K_I^{(G)}(q) + K_I^{(VG)}(q)\right].$$

A sufficient diagonal sign criterion is

$$\boxed{\partial_t R_k \geq 0, \qquad G_O \geq 0, \qquad G_D \geq 0, \qquad K_I^{(V)} + K_I^{(G)} + K_I^{(VG)} \geq 0,}$$

which implies

$$\boxed{(M_Z^{OD})_{OO} < 0, \qquad (M_Z^{OD})_{DD} < 0.}$$

The off-diagonal target is

$$\boxed{(M_Z^{OD})_{OD} > 0, \qquad (M_Z^{OD})_{DO} > 0,}$$

and with row-sum balance one recovers the Laplacian form

$$\boxed{M_Z^{OD} = -\lambda \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad \lambda > 0.}$$

If $O \leftrightarrow D$ exchange symmetry holds, the whole test reduces to

$$\boxed{(M_Z^{OD})_{OO} - (M_Z^{OD})_{OD} < 0.}$$

PROJECTED POLARIZATION-KERNEL FORMULA FOR THE GAUGE/PROJECTOR CONTRIBUTION TO M_Z

1. Physical starting point

The A-sector contribution to the derivative-anisotropy flow must be defined on the physical transverse subspace. Let

$$\Gamma^{(2)}_{\mu\nu}(p;\bar{\Phi}_{\min})$$

denote the background-dependent gauge-sector two-point kernel, and let

$$\Pi_T^{\mu\nu}(p) = \eta^{\mu\nu} - \frac{p^\mu p^\nu}{p^2} \tag{524}$$

be the transverse projector.

The physical projected inverse propagator is

$$\Gamma_A^{(2),\mathrm{phys}}(p;\bar{\Phi}_{\min}) := \Pi_T^{\mu\nu}(p)\,\Gamma_{\mu\nu}^{(2)}(p;\bar{\Phi}_{\min}). \tag{525}$$

The corresponding kinetic-flow contribution is

$$(M_Z^{\mathrm{gf/proj}})_{TJ} = \frac{\partial}{\partial\delta_J}\frac{\partial}{\partial p^2}\Big[\Pi_T^{\mu\nu}(p)\,\partial_t\Gamma_{\mu\nu}^{(2)}(p;\bar{\Phi}_{\min})\Big]\Big|_{p=0,\delta=0}, \qquad J\in\{T,O,D\}. \tag{526}$$

2. Preconditions for a meaningful sign test

The projected sign test (526) is meaningful only if the A-sector lies on the gauge-consistent infrared surface:

1. **transversality:**

$$p^\mu\Gamma_{\mu\nu}^{(2)}(p;\bar{\Phi}_{\min}) = 0; \tag{527}$$

2. **no Proca mass:**

$$\Pi_T^{\mu\nu}(0)\Gamma_{\mu\nu}^{(2)}(0;\bar{\Phi}_{\min}) = 0, \qquad m_{A,k}^2 \to 0; \tag{528}$$

3. **gauge-preserving regulator:**

$$p^\mu\Pi_{\mu\nu}(p) = 0 \quad \text{is preserved by the regulator/renormalization scheme;} \tag{529}$$

4. **operator-class restriction:**

$$\text{the projected infrared EFT contains only Class I and Class II operators.} \tag{530}$$

These are precisely the conditions under which the one-loop polarization remains transverse and no gauge mass is generated. :contentReference[oaicite:4]index=4 :contentReference[oaicite:5]index=5 :contentReference[oaicite:6]index=6 :contentReference[oaicite:7]index=7

3. Projected one-loop polarization decomposition

Let

$$G_A^{\mathrm{phys}}(\bar{\Phi}_{\min}) = \Big(\Pi_A^{\mathrm{phys}}[\Gamma_A^{(2)} + R_{A,k}]\Pi_A^{\mathrm{phys}}\Big)^{-1}$$

be the projected gauge-sector propagator, and let

$$G_{\mathrm{mat}}(\bar{\Phi}_{\min})$$

denote the projected matter propagators in the coupled soft sectors.

Then the projected one-loop flow of the gauge two-point function has the standard decomposition

$$\begin{aligned}\Pi_T^{\mu\nu}(p)\,\partial_t\Gamma_{\mu\nu}^{(2)}(p;\bar{\Phi}_{\min}) = \Pi_T^{\mu\nu}(p)\Bigg\{&\frac{1}{2}\,\mathrm{Tr}_{\mathrm{phys}}\big[G\,\Gamma_{\mu\nu}^{(4)}\,G\,\partial_t R_k\big] \\ &- \mathrm{Tr}_{\mathrm{phys}}\big[G\,\Gamma_\mu^{(3)}\,G\,\Gamma_\nu^{(3)}\,G\,\partial_t R_k\big]\Bigg\} \\ =: \Pi_T^{(4)}(p) + \Pi_T^{(33)}(p).&\end{aligned} \tag{531}$$

Thus

$$\boxed{(M_Z^{\mathrm{gf/proj}})_{TJ} = (M_Z^{\mathrm{gf/proj},4})_{TJ} + (M_Z^{\mathrm{gf/proj},33})_{TJ},} \tag{532}$$

where

$$(M_Z^{\mathrm{gf/proj},4})_{TJ} = \frac{\partial}{\partial \delta_J}\frac{\partial}{\partial p^2}\Pi_T^{(4)}(p)\Big|_{p=0,\delta=0}, \tag{533}$$

$$(M_Z^{\mathrm{gf/proj},33})_{TJ} = \frac{\partial}{\partial \delta_J}\frac{\partial}{\partial p^2}\Pi_T^{(33)}(p)\Big|_{p=0,\delta=0}. \tag{534}$$

4. Minimal calculable kernel form

In a gauge-preserving scheme with minimally coupled charged transverse matter, the one-loop vacuum polarization has the standard form

$$\Pi_{\mu\nu}^{(\psi)}(p) = e^2\int\frac{d^Dk}{(2\pi)^D}\frac{(2k+p)_\mu(2k+p)_\nu}{(k^2-m_T^2+i0^+)\big((k+p)^2-m_T^2+i0^+\big)}, \tag{535}$$

which can be rewritten as

$$\Pi_{\mu\nu}^{(\psi)}(p) = (p_\mu p_\nu - p^2\eta_{\mu\nu})\,\Pi_R(p^2). \tag{536}$$

This is the model template for the projected gauge-sector kernel. :contentReference[oaicite:8]index=8 :contentReference[oaicite:9]index=9

Accordingly, the projected background-adapted polarization kernel should be written schematically as

$$\Pi_T^{\mu\nu}(p)\,\partial_t\Gamma_{\mu\nu}^{(2)}(p;\bar\Phi_{\min}) = p^2\,\mathcal{K}_A(p^2;\bar\Phi_{\min}), \tag{537}$$

with

$$\mathcal{K}_A = \mathcal{K}_A^{(4)} + \mathcal{K}_A^{(33)}. \tag{538}$$

Then

$$\boxed{(M_Z^{\mathrm{gf/proj}})_{TJ} = \frac{\partial}{\partial \delta_J}\mathcal{K}_A(0;\bar\Phi_{\min})\Big|_{\delta=0}.} \tag{539}$$

This is the minimal calculable reduction of the gauge/projector bottleneck.

5. Small-p sign decomposition

Expand

$$\mathcal{K}_A(p^2;\bar\Phi_{\min}) = \mathcal{K}_A^{(0)} + \mathcal{K}_A^{(1)}p^2 + \cdots.$$

Then only the p^2-coefficient of the projected polarization matters. The sign test separates into:

a. *(i) Tadpole-like projected contribution*

$$\mathcal{K}_A^{(4)}(0;\bar\Phi_{\min}).$$

This often contributes only to local mass-like terms or vanishes after transversality is imposed. Hence it is usually subleading for the kinetic-flow sign, unless the background-dependent gauge projector produces nontrivial momentum dependence.

b. *(ii) Bubble/sunset projected contribution*

$$\mathcal{K}_A^{(33)}(0;\bar\Phi_{\min}).$$

This is the main candidate for a genuine p^2-flow, because background-generated cubic vertices can produce a nontrivial momentum-dependent projected polarization.

Thus the practical sign hierarchy is

$$\boxed{(M_Z^{\mathrm{gf/proj},33})_{TJ}\text{ is the dominant sign carrier,}\qquad (M_Z^{\mathrm{gf/proj},4})_{TJ}\text{ is usually kinematically constrained.}} \tag{540}$$

6. Sufficient non-dangerous criterion

A sufficient criterion that the gauge/projector sector is *not* a dangerous anisotropy source is:

1. the projected polarization is transverse;
2. the zero-momentum limit vanishes:
$$\Pi_T^{\mu\nu}(0)\Gamma^{(2)}_{\mu\nu}(0) = 0;$$
3. the projected kernel satisfies
$$\boxed{\frac{\partial}{\partial\delta_J}\mathcal{K}_A(0;\bar{\Phi}_{\min})\bigg|_{\delta=0} \geq 0 \quad \text{for the Euclidean sign convention.}} \tag{541}$$

Under these conditions, the gauge/projector sector is compatible with isotropization and does not overturn the negative contributions expected from the quartic- and derivative-anisotropy sectors.

7. Honest conclusion

Thus the gauge/projector bottleneck is now reduced completely to the projected polarization-kernel test:

$$\boxed{(M_Z^{\mathrm{gf/proj}})_{TJ} = \frac{\partial}{\partial\delta_J}\mathcal{K}_A(0;\bar{\Phi}_{\min})\bigg|_{\delta=0},} \tag{542}$$

where $\mathcal{K}_A$ is defined by the small-p transverse projected polarization kernel

$$\Pi_T^{\mu\nu}(p)\,\partial_t\Gamma^{(2)}_{\mu\nu}(p;\bar{\Phi}_{\min}) = p^2\mathcal{K}_A(p^2;\bar{\Phi}_{\min}).$$

At this point, both remaining sign bottlenecks are reduced to explicit projected kernel derivative tests. What remains open is their actual evaluation.

. Summary

$$\boxed{(M_Z^{\mathrm{gf/proj}})_{TJ} = \frac{\partial}{\partial\delta_J}\frac{\partial}{\partial p^2}\Big[\Pi_T^{\mu\nu}(p)\,\partial_t\Gamma^{(2)}_{\mu\nu}(p;\bar{\Phi}_{\min})\Big]\bigg|_{p=0,\delta=0}.}$$

The projected polarization splits as

$$\boxed{\Pi_T^{\mu\nu}(p)\,\partial_t\Gamma^{(2)}_{\mu\nu}(p;\bar{\Phi}_{\min}) = \Pi_T^{(4)}(p) + \Pi_T^{(33)}(p),}$$

so that

$$\boxed{(M_Z^{\mathrm{gf/proj}})_{TJ} = (M_Z^{\mathrm{gf/proj,4}})_{TJ} + (M_Z^{\mathrm{gf/proj,33}})_{TJ}.}$$

Using the standard transverse form,

$$\Pi^{(\psi)}_{\mu\nu}(p) = (p_\mu p_\nu - p^2\eta_{\mu\nu})\,\Pi_R(p^2),$$

the background-adapted projected kernel should be written as

$$\boxed{\Pi_T^{\mu\nu}(p)\,\partial_t\Gamma^{(2)}_{\mu\nu}(p;\bar{\Phi}_{\min}) = p^2\,\mathcal{K}_A(p^2;\bar{\Phi}_{\min}),}$$

with

$$\mathcal{K}_A = \mathcal{K}_A^{(4)} + \mathcal{K}_A^{(33)}.$$

Therefore the gauge/projector sign test reduces to

$$\boxed{(M_Z^{\mathrm{gf/proj}})_{TJ} = \frac{\partial}{\partial \delta_J}\mathcal{K}_A(0;\bar{\Phi}_{\min})\bigg|_{\delta=0}.}$$

Its consistency preconditions are:

$$\boxed{p^\mu \Gamma^{(2)}_{\mu\nu} = 0, \qquad m^2_{A,k} \to 0, \qquad \text{gauge-preserving regulator}, \qquad \text{Class I/II only}.}$$

The practical sign hierarchy is

$$\boxed{(M_Z^{\mathrm{gf/proj},33})_{TJ} \text{ is the main sign carrier}, \qquad (M_Z^{\mathrm{gf/proj},4})_{TJ} \text{ is usually kinematically constrained}.}$$

Thus the gauge/projector bottleneck is now reduced to one explicit projected polarization-kernel derivative test.

FINAL SUFFICIENT CRITERION FOR $\mathrm{spec}(M_Z|_{\mathrm{traceless}}) < 0$

1. Physical setting

The physical derivative-anisotropy variables are

$$\delta_T, \qquad \delta_O, \qquad \delta_D, \qquad 2\delta_T + 2\delta_O + 2\delta_D = 0.$$

Let

$$\mathcal{U}_{\mathrm{tr}} := \left\{u = (\delta_T, \delta_O, \delta_D)^T \in \mathbb{R}^3 : 2\delta_T + 2\delta_O + 2\delta_D = 0\right\}$$

denote the two-dimensional traceless physical anisotropy subspace.

The loop-level flow is

$$\partial_t u = M_Z u + O(u^2).$$

The problem is to guarantee

$$\mathrm{spec}(M_Z|_{\mathcal{U}_{\mathrm{tr}}}) \subset (-\infty, 0).$$

2. Canonical decomposition of the kinetic-flow matrix

At the present stage, the physically relevant decomposition is

$$\boxed{M_Z = M_Z^{(\tau)} + M_Z^{(\nabla\mathrm{ani})} + M_Z^{(\mathrm{self})} + M_Z^{(OD)} + M_Z^{(\mathrm{gf/proj})}.} \tag{543}$$

Here:

- $M_Z^{(\tau)}$ is induced by the quartic cubic-anisotropy sector,
- $M_Z^{(\nabla\mathrm{ani})}$ is induced by the derivative-anisotropy sector,
- $M_Z^{(\mathrm{self})}$ collects the self-sector O- and D-stabilizing pieces,
- $M_Z^{(OD)}$ is the mixed OD-bubble contribution,
- $M_Z^{(\mathrm{gf/proj})}$ is the projected gauge/projector contribution.

The current theorem-level status is:

$$M_Z^{(\tau)} \text{ and } M_Z^{(\nabla\text{ani})}$$

are the main expected isotropizing sources, whereas

$$M_Z^{(OD)} \text{ and } M_Z^{(\text{gf/proj})}$$

are the two main remaining sign bottlenecks. This is consistent with the current TECT FRG notes: quartic anisotropy is explicitly irrelevant for $N_{\text{eff}} = 6$, while the derivative-anisotropy sector is reduced to the explicit sign of M_Z. :contentReference[oaicite:4]index=4 :contentReference[oaicite:5]index=5

3. Good/bad split

Define the *good* and *bad* parts by

$$M_Z^{\text{good}} := M_Z^{(\tau)} + M_Z^{(\nabla\text{ani})} + M_Z^{(\text{self})}, \tag{544}$$

$$M_Z^{\text{bad}} := M_Z^{(OD)} + M_Z^{(\text{gf/proj})}. \tag{545}$$

Then

$$M_Z = M_Z^{\text{good}} + M_Z^{\text{bad}}.$$

4. Coercive negativity of the good part

Assume that on the traceless subspace $\mathcal{U}_{\text{tr}}$, the good part satisfies the quadratic-form bound

$$\boxed{\langle u, M_Z^{\text{good}} u\rangle \leq -\mu_{\text{good}} \|u\|^2 \qquad \forall u \in \mathcal{U}_{\text{tr}}, \qquad \mu_{\text{good}} > 0.} \tag{546}$$

This is the coercive negativity input.

a. Interpretation. It means that the combined quartic-anisotropy, derivative-anisotropy, and self-sector contributions are already strictly isotropizing on the physical anisotropy sector.

b. Support from current TECT status. The quartic anisotropy piece has explicit negative RG eigenvalue

$$\theta_\tau = \epsilon \frac{4 - N_{\text{eff}}}{N_{\text{eff}} + 8} = -\frac{\epsilon}{7} < 0 \qquad (N_{\text{eff}} = 6),$$

while the derivative-anisotropy direction is expected to be irrelevant but is not yet proved microscopically. Thus (546) is not currently a theorem, but it is the correct target form. :contentReference[oaicite:6]index=6 :contentReference[oaicite:7]index=7 :contentReference[oaicite:8]index=8

5. Norm control of the bad part

Assume that the two remaining bottlenecks satisfy operator-norm bounds on the physical traceless sector:

$$\|M_Z^{(OD)}|_{\mathcal{U}_{\text{tr}}}\| \leq \eta_{OD}, \qquad \|M_Z^{(\text{gf/proj})}|_{\mathcal{U}_{\text{tr}}}\| \leq \eta_{\text{gf}}, \tag{547}$$

and define

$$\eta_{\text{bad}} := \eta_{OD} + \eta_{\text{gf}}. \tag{548}$$

Then

$$\boxed{\|M_Z^{\text{bad}}|_{\mathcal{U}_{\text{tr}}}\| \leq \eta_{\text{bad}}.} \tag{549}$$

c. Origin of these bounds. The OD-mixed contribution has already been reduced to the projected self-energy derivative test

$$(M_Z^{OD})_{IJ} = -\left.\frac{\partial}{\partial \delta_J}\frac{\partial}{\partial p^2}\Sigma_I^{(OD)}(p)\right|_{p=0,\delta=0},$$

while the gauge/projector contribution has been reduced to the projected polarization derivative test

$$(M_Z^{\mathrm{gf/proj}})_{TJ} = \left.\frac{\partial}{\partial \delta_J}\mathcal{K}_A(0;\bar{\Phi}_{\min})\right|_{\delta=0}.$$

Thus η_{OD} and η_{gf} are explicit finite projected-kernel bounds.

6. Final sufficient negativity criterion

Now let $u \in \mathcal{U}_{\mathrm{tr}}$. Then

$$\begin{aligned}\langle u, M_Z u\rangle &= \langle u, M_Z^{\mathrm{good}} u\rangle + \langle u, M_Z^{\mathrm{bad}} u\rangle \\ &\le -\mu_{\mathrm{good}}\|u\|^2 + \|M_Z^{\mathrm{bad}}|_{\mathcal{U}_{\mathrm{tr}}}\|\,\|u\|^2 \\ &\le -(\mu_{\mathrm{good}} - \eta_{\mathrm{bad}})\|u\|^2.\end{aligned} \tag{550}$$

Therefore, if

$$\boxed{\mu_{\mathrm{good}} > \eta_{\mathrm{bad}},} \tag{551}$$

then

$$\boxed{\langle u, M_Z u\rangle < 0 \qquad \forall u \in \mathcal{U}_{\mathrm{tr}} \setminus \{0\}.} \tag{552}$$

Hence

$$\boxed{\mathrm{spec}(M_Z|_{\mathcal{U}_{\mathrm{tr}}}) \subset (-\infty, 0).} \tag{553}$$

This is the final sufficient negativity criterion.

7. Laplacian target as a stronger sufficient criterion

A stronger algebraic sufficient criterion is the exact Laplacian form

$$\boxed{M_Z = -\lambda\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2\end{pmatrix}, \qquad \lambda > 0,} \tag{554}$$

which has one zero mode in the unphysical common-rescaling direction and two physical eigenvalues

$$-3\lambda, \qquad -3\lambda.$$

The notes already identify this as the exact algebraic target that the explicit FRG computation should verify. :contentReference[oaicite:9]index=9

Thus:

$$\text{Laplacian form with } \lambda > 0 \Longrightarrow (551) \Longrightarrow \mathrm{spec}(M_Z|_{\mathcal{U}_{\mathrm{tr}}}) < 0.$$

8. Gauge/projector preconditions

The criterion above must be used together with the gauge-consistency preconditions:

$$p^\mu \Gamma^{(2)}_{\mu\nu} = 0, \qquad m_{A,k}^2 \to 0, \qquad \text{gauge-preserving regulator}, \qquad \text{Class I/II only}. \tag{555}$$

These are necessary for $(M_Z)^{(\mathrm{gf/proj})}$ to be a meaningful physical contribution rather than a gauge artifact. :contentReference[oaicite:10]index=10

9. Honest final status

The exact current logical status is:

$$\boxed{\text{The criterion } \mu_{\text{good}} > \eta_{\text{bad}} \text{ is fully explicit and sufficient,}} \tag{556}$$

but

$$\boxed{\text{the actual numerical/analytic values of } \mu_{\text{good}}, \eta_{OD}, \eta_{\text{gf}} \text{ are not yet derived from the current TECT data.}} \tag{557}$$

So the frontier is no longer conceptual: it is the explicit estimation of these finitely many quantities.

. Summary

$$\boxed{M_Z = M_Z^{(\tau)} + M_Z^{(\nabla\text{ani})} + M_Z^{(\text{self})} + M_Z^{(OD)} + M_Z^{(\text{gf/proj})}.}$$

Define

$$\boxed{M_Z^{\text{good}} = M_Z^{(\tau)} + M_Z^{(\nabla\text{ani})} + M_Z^{(\text{self})},} \qquad \boxed{M_Z^{\text{bad}} = M_Z^{(OD)} + M_Z^{(\text{gf/proj})}.}$$

Assume coercive negativity of the good part on the traceless physical anisotropy subspace

$$\mathcal{U}_{\text{tr}} = \{u = (\delta_T, \delta_O, \delta_D)^T : \ 2\delta_T + 2\delta_O + 2\delta_D = 0\},$$

namely

$$\boxed{\langle u, M_Z^{\text{good}} u\rangle \le -\mu_{\text{good}}\|u\|^2, \qquad \mu_{\text{good}} > 0.}$$

Assume operator-norm bounds on the bad part

$$\boxed{\|M_Z^{(OD)}|_{\mathcal{U}_{\text{tr}}}\| \le \eta_{OD}, \qquad \|M_Z^{(\text{gf/proj})}|_{\mathcal{U}_{\text{tr}}}\| \le \eta_{\text{gf}},}$$

and define

$$\boxed{\eta_{\text{bad}} := \eta_{OD} + \eta_{\text{gf}}.}$$

Then

$$\boxed{\mu_{\text{good}} > \eta_{\text{bad}} \Longrightarrow \text{spec}(M_Z|_{\mathcal{U}_{\text{tr}}}) \subset (-\infty, 0).}$$

So the final sufficient negativity criterion is

$$\boxed{\mu_{\text{good}} > \eta_{OD} + \eta_{\text{gf}}.}$$

A stronger algebraic sufficient target is

$$\boxed{M_Z = -\lambda \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}, \qquad \lambda > 0,}$$

which yields the physical eigenvalues

$$\boxed{-3\lambda, \qquad -3\lambda.}$$

Gauge/projector preconditions:

$$\boxed{p^\mu \Gamma^{(2)}_{\mu\nu} = 0, \qquad m^2_{A,k} \to 0, \qquad \text{gauge-preserving regulator}, \qquad \text{Class I/II only.}}$$

Honest final status:

$$\boxed{\text{the sufficient criterion is explicit,}}$$

but

$$\boxed{\text{the actual bounds } \mu_{\text{good}}, \eta_{OD}, \eta_{\text{gf}} \text{ are not yet computed from the full TECT data.}}$$

ACTUAL COMPUTATION PRIORITY LIST

A. Global status

The remaining TECT frontier is no longer conceptual. It is a finite computation / verification program.

On the local fermion side, the unresolved tasks are exactly:

$$\text{(I) witness nonvanishing}, \quad \text{(II) opposite-valley survival/matching}, \quad \text{(III) remote spectral gap}, \quad \text{(IV) zero constant mixing}, \quad \text{(V)}$$

These are explicitly listed as the remaining microscopic verification tasks of the shell-reduced Dirac program.

On the IR-isotropy side, the quartic anisotropy is already irrelevant for

$$N_{\text{eff}} = 6, \qquad \theta_\tau = -\frac{\epsilon}{7} < 0,$$

whereas the derivative-anisotropy / velocity-locking problem still requires the explicit computation of

$$M_Z.$$

This is also explicitly stated in the TECT FRG notes.

B. Final sufficient negativity criterion

The physical anisotropy problem is controlled by

$$M_Z = M_Z^{(\tau)} + M_Z^{(\nabla\text{ani})} + M_Z^{(\text{self})} + M_Z^{(OD)} + M_Z^{(\text{gf/proj})}$$

restricted to the traceless subspace

$$\mathcal{U}_{\text{tr}} = \{u = (\delta_T, \delta_O, \delta_D)^T : \; 2\delta_T + 2\delta_O + 2\delta_D = 0\}.$$

Define

$$M_Z^{\text{good}} = M_Z^{(\tau)} + M_Z^{(\nabla\text{ani})} + M_Z^{(\text{self})},$$

$$M_Z^{\text{bad}} = M_Z^{(OD)} + M_Z^{(\text{gf/proj})}.$$

If

$$\langle u, M_Z^{\text{good}} u\rangle \le -\mu_{\text{good}} \|u\|^2 \qquad (\mu_{\text{good}} > 0),$$

and

$$\|M_Z^{(OD)}|_{\mathcal{U}_{\text{tr}}}\| \le \eta_{OD}, \qquad \|M_Z^{(\text{gf/proj})}|_{\mathcal{U}_{\text{tr}}}\| \le \eta_{\text{gf}},$$

then the master sufficient condition is

$$\boxed{\mu_{\text{good}} > \eta_{OD} + \eta_{\text{gf}} \Longrightarrow \operatorname{spec}(M_Z|_{\mathcal{U}_{\text{tr}}}) \subset (-\infty, 0).} \tag{558}$$

Therefore the actual computation priority is dictated by the quantities

$$\eta_{OD}, \qquad \eta_{\text{gf}}, \qquad \mu_{\text{good}}.$$

C. Priority 1: compute η_{OD}

The mixed OD bottleneck is already reduced to the explicit projected self-energy derivative test

$$(M_Z^{OD})_{IJ} = -\frac{\partial}{\partial\delta_J}\frac{\partial}{\partial p^2}\Sigma_I^{(OD)}(p;\bar{\Phi}_{\min})\bigg|_{p=0,\delta=0}, \qquad I,J\in\{O,D\}.$$

The projected kernels are

$$\Sigma_O^{(OD)}(p), \qquad \Sigma_D^{(OD)}(p),$$

with explicit background-overlap factors and projected propagators.

a. Why this is Priority 1.

1. It is the first genuine off-diagonal anisotropy source.
2. It does not yet require the full gauge/projector implementation.
3. It is the cleanest nontrivial sign bottleneck in the matter-sector background field problem.

b. Minimal deliverable. Produce a rigorous upper bound

$$\boxed{\|M_Z^{(OD)}|_{\mathcal{U}_{\rm tr}}\| \le \eta_{OD}} \tag{559}$$

from the projected loop-kernel formulas.

c. Operational substeps.

1. Fix the projected mode bases u_m, v_n for $\mathcal{H}_{OO}, \mathcal{H}_{DD}$.
2. Compute / bound the background overlaps
$$\mathcal{B}^{\chi}_{mn}, \qquad \mathcal{B}^{n}_{nm}.$$
3. Bound the projected propagators G_O, G_D.
4. Bound the small-p kernel coefficients
$$K_I^{(V)}, \qquad K_I^{(G)}, \qquad K_I^{(VG)}.$$
5. Extract the traceless-subspace operator norm.

D. Priority 2: compute $\eta_{\rm gf}$

The gauge/projector bottleneck is reduced to the projected polarization-kernel derivative test

$$(M_Z^{\rm gf/proj})_{TJ} = \frac{\partial}{\partial\delta_J}\mathcal{K}_A(0;\bar{\Phi}_{\min})\bigg|_{\delta=0},$$

where

$$\Pi_T^{\mu\nu}(p)\,\partial_t\Gamma^{(2)}_{\mu\nu}(p;\bar{\Phi}_{\min}) = p^2\mathcal{K}_A(p^2;\bar{\Phi}_{\min}).$$

d. Why this is Priority 2.

1. It is the second remaining sign bottleneck.
2. It is structurally more delicate because it requires:
$$p^\mu\Gamma^{(2)}_{\mu\nu} = 0, \qquad m^2_{A,k}\to 0, \qquad \text{gauge-preserving regulator}, \qquad \text{Class I/II only}.$$
3. Therefore it should be attempted only after the matter-sector OD bound is already under control.

e. Minimal deliverable. Produce a rigorous upper bound

$$\boxed{\|M_Z^{(\mathrm{gf/proj})}|_{\mathcal{U}_{\mathrm{tr}}}\| \le \eta_{\mathrm{gf}}.} \tag{560}$$

f. Operational substeps.

1. Fix Π_A^{phys} explicitly.
2. Verify transversality and zero Proca mass.
3. Fix a gauge-preserving regulator.
4. Split

$$\mathcal{K}_A = \mathcal{K}_A^{(4)} + \mathcal{K}_A^{(33)}$$

   and identify the dominant projected sign carrier.
5. Bound the anisotropy response derivatives.

E. Priority 3: compute μ_{good}

The good part is

$$M_Z^{\mathrm{good}} = M_Z^{(\tau)} + M_Z^{(\nabla\mathrm{ani})} + M_Z^{(\mathrm{self})}.$$

g. Why this is Priority 3 rather than Priority 1. Although it is crucial for the final theorem, it is not the current bottleneck. The quartic anisotropy sector already has explicit support:

$$\theta_\tau = -\frac{\epsilon}{7} < 0 \qquad (N_{\mathrm{eff}} = 6),$$

and the derivative anisotropy sector has the strong TECT expectation

$$\theta_Y < 0.$$

Hence the good part is already expected to be negative; what remains dangerous are the unresolved mixed/gauge pieces.

h. Minimal deliverable. Produce a coercive negativity constant

$$\boxed{\langle u, M_Z^{\mathrm{good}} u\rangle \le -\mu_{\mathrm{good}}\|u\|^2 \qquad \forall u \in \mathcal{U}_{\mathrm{tr}}, \qquad \mu_{\mathrm{good}} > 0.} \tag{561}$$

i. Operational substeps.

1. Use the explicit quartic anisotropy eigenvalue to fix a negative lower bound for $M_Z^{(\tau)}$.
2. Derive / assume a negative derivative-anisotropy eigenvalue for $M_Z^{(\nabla\mathrm{ani})}$.
3. Show that the self-sector piece is diagonally stabilizing.
4. Combine the three into a single traceless-subspace coercive estimate.

F. Priority 4: return to the local Dirac finite checks

Once the M_Z negativity program is organized, one should return to the remaining local fermion finite checks:

1. witness nonvanishing:

$$P_V T^{(\alpha)} \neq 0 \quad \text{or explicit seed tensors;}$$

2. opposite-valley survival and witness matching at $\pm G_*$;
3. remote spectral gap
$$\Delta > 0$$
via finite Bloch-matrix calculation;
4. zero constant mixing:
$$\mathrm{Hom}_{H_*}(H_R, H_D) = 0;$$
5. mass protection:
$$A_{\mathrm{res}} \cap \mathcal{M} = \{0\}.$$

These are already explicitly enumerated in the current Dirac-emergence roadmap.

G. Final ordered list

The actual ordered computation program is therefore:

$$\boxed{\textbf{(1) } \eta_{OD} \quad \Longrightarrow \quad \textbf{(2) } \eta_{\mathrm{gf}} \quad \Longrightarrow \quad \textbf{(3) } \mu_{\mathrm{good}} \quad \Longrightarrow \quad \textbf{(4) } \mu_{\mathrm{good}} > \eta_{OD} + \eta_{\mathrm{gf}}.} \tag{562}$$

If (562) succeeds, then
$$\mathrm{spec}(M_Z|_{\mathcal{U}_{\mathrm{tr}}}) < 0,$$
hence
$$\delta_T, \delta_O, \delta_D \to 0,$$
and the infrared soft sector becomes isotropic and velocity-locked.

Then the local fermion finite checks can be completed in parallel or immediately afterward, closing the TECT local Dirac / IR-isotropy frontier.

H. Honest final status

The exact current status is:

$$\boxed{\text{The full frontier has been reduced to finitely many explicit bounds and finite matrix/projection tests.}}$$

What is not yet done is not conceptual: it is simply the actual evaluation of those quantities.

. Summary

$$\boxed{\textbf{Actual computation priority list}}$$

The final sufficient negativity criterion is

$$\boxed{\mu_{\mathrm{good}} > \eta_{OD} + \eta_{\mathrm{gf}} \Longrightarrow \mathrm{spec}(M_Z|_{\mathcal{U}_{\mathrm{tr}}}) \subset (-\infty, 0),}$$

where

$$M_Z^{\text{good}} = M_Z^{(\tau)} + M_Z^{(\nabla\text{ani})} + M_Z^{(\text{self})}, \qquad M_Z^{\text{bad}} = M_Z^{(OD)} + M_Z^{(\text{gf/proj})}.$$

Hence the actual ordered computation program is:

$$\boxed{\textbf{Priority 1: compute } \eta_{OD} = \|M_Z^{(OD)}|_{\mathcal{U}_{\text{tr}}}\|.}$$

$$\boxed{\textbf{Priority 2: compute } \eta_{\text{gf}} = \|M_Z^{(\text{gf/proj})}|_{\mathcal{U}_{\text{tr}}}\|.}$$

$$\boxed{\textbf{Priority 3: compute } \mu_{\text{good}} \text{ from } M_Z^{(\tau)} + M_Z^{(\nabla\text{ani})} + M_Z^{(\text{self})}.}$$

$$\boxed{\textbf{Priority 4: verify } \mu_{\text{good}} > \eta_{OD} + \eta_{\text{gf}}.}$$

If this succeeds, then

$$\boxed{\operatorname{spec}(M_Z|_{\mathcal{U}_{\text{tr}}}) < 0 \Longrightarrow \delta_T, \delta_O, \delta_D \to 0,}$$

so the soft sector is infrared isotropic and velocity-locked.

In parallel or immediately afterward, complete the local Dirac finite checks:

$$\boxed{P_V T^{(\alpha)} \neq 0, \quad \text{opposite-valley matching}, \quad \Delta > 0, \quad \operatorname{Hom}_{H_*}(H_R, H_D) = 0, \quad A_{\text{res}} \cap \mathcal{M} = \{0\}.}$$

Thus the entire TECT frontier is now reduced to finitely many explicit bounds, matrix tests, and projected kernel evaluations.

MINIMAL BOUND PACKAGE FOR THE PROJECTED OD-KERNEL

1. Starting point

The OD-mixed contribution is controlled by

$$(M_Z^{OD})_{IJ} = -\frac{\partial}{\partial \delta_J}\frac{\partial}{\partial p^2}\Sigma_I^{(OD)}(p)\bigg|_{p=0,\delta=0}, \qquad I, J \in \{O, D\}, \tag{563}$$

with projected self-energies

$$\Sigma_O^{(OD)}(p) = -\lambda_{OD}^2 \int_q \sum_{m,n} \mathcal{B}_{mn}^{\chi}\, \mathcal{P}_O(p,q)\, G_D^{(n)}(q)\, \mathcal{B}_{nm}^{\chi}\, \mathcal{P}_O(-p,q)\, G_O^{(m)}(p+q)\, \partial_t R_k(q), \tag{564}$$

$$\Sigma_D^{(OD)}(p) = -\lambda_{OD}^2 \int_q \sum_{m,n} \mathcal{B}_{nm}^{n}\, \mathcal{P}_D(p,q)\, G_O^{(m)}(q)\, \mathcal{B}_{mn}^{n}\, \mathcal{P}_D(-p,q)\, G_D^{(n)}(p+q)\, \partial_t R_k(q). \tag{565}$$

Thus η_{OD} will follow once every factor in these kernels is bounded separately.

2. Package A: background overlap bounds

Define the background overlap matrices

$$\mathcal{B}^{\chi}_{mn} := \langle u_m, \bar{\chi}_B v_n \rangle, \qquad \mathcal{B}^{n}_{nm} := \langle v_n, \bar{n}_B u_m \rangle.$$

The minimal bound package begins with

$$\boxed{|\mathcal{B}^{\chi}_{mn}| \le \|\bar{\chi}_B\|_{L^\infty_\xi} \|u_m\| \|v_n\|, \qquad |\mathcal{B}^{n}_{nm}| \le \|\bar{n}_B\|_{L^\infty_\xi} \|u_m\| \|v_n\|.} \tag{566}$$

If the projected mode bases are orthonormal,

$$\|u_m\| = \|v_n\| = 1,$$

then

$$\boxed{|\mathcal{B}^{\chi}_{mn}| \le \|\bar{\chi}_B\|_{L^\infty}, \qquad |\mathcal{B}^{n}_{nm}| \le \|\bar{n}_B\|_{L^\infty}.} \tag{567}$$

A slightly sharper Hilbert-Schmidt form is

$$\sum_{m,n} |\mathcal{B}^{\chi}_{mn}|^2 = \|P_O \bar{\chi}_B P_D\|^2_{\mathrm{HS}}, \qquad \sum_{m,n} |\mathcal{B}^{n}_{nm}|^2 = \|P_D \bar{n}_B P_O\|^2_{\mathrm{HS}}. \tag{568}$$

Hence the first deliverable is a finite bound

$$\boxed{\mathfrak{B}_\chi := \|P_O \bar{\chi}_B P_D\|_{\mathrm{HS}}, \qquad \mathfrak{B}_n := \|P_D \bar{n}_B P_O\|_{\mathrm{HS}}.} \tag{569}$$

3. Package B: projected propagator bounds

Let

$$G_O^{(m)}(q) = \frac{1}{E_m^O(q) + R_{O,k}(q)}, \qquad G_D^{(n)}(q) = \frac{1}{E_n^D(q) + R_{D,k}(q)}.$$

If the background Hessian blocks satisfy a spectral lower bound

$$E_m^O(q) \ge \Delta_O > 0, \qquad E_n^D(q) \ge \Delta_D > 0, \tag{570}$$

then

$$\boxed{|G_O^{(m)}(q)| \le \frac{1}{\Delta_O + \inf R_{O,k}}, \qquad |G_D^{(n)}(q)| \le \frac{1}{\Delta_D + \inf R_{D,k}}.} \tag{571}$$

Hence define

$$\boxed{\mathfrak{G}_O := \sup_{m,q} |G_O^{(m)}(q)|, \qquad \mathfrak{G}_D := \sup_{n,q} |G_D^{(n)}(q)|.} \tag{572}$$

This is the second deliverable. It is exactly where the remote/background spectral gap feeds into the M_Z estimate.

4. Package C: regulator bounds

Assume a standard monotone cutoff profile so that

$$\partial_t R_k(q) \ge 0.$$

Define the size of the regulator derivative by

$$\boxed{\mathfrak{R}_k := \sup_q |\partial_t R_k(q)|.} \tag{573}$$

If one uses an optimized profile, $\partial_t R_k$ is compactly supported in the low-momentum shell and $\mathfrak{R}_k$ is finite.
This is the third deliverable.

5. Package D: small-p kernel curvature bounds

Expand the projected momentum structures and propagators near $p = 0$:

$$\mathcal{P}_I(p,q) = \mathcal{P}_I^{(0)} + \mathcal{P}_I^{(1)} \cdot p + \frac{1}{2} p \cdot \mathcal{P}_I^{(2)} \cdot p + \cdots ,$$

$$G_I(p+q) = G_I(q) + p \cdot \partial_q G_I(q) + \frac{1}{2} p \cdot \partial_q^2 G_I(q) \cdot p + \cdots .$$

Then

$$\frac{\partial}{\partial p^2} \Sigma_I^{(OD)}(p) \bigg|_{p=0}$$

is controlled by three kernel classes:

$$K_I^{(V)}, \qquad K_I^{(G)}, \qquad K_I^{(VG)}.$$

Define

$$\boxed{\mathfrak{K}_I := \sup_q \Big(|K_I^{(V)}(q)| + |K_I^{(G)}(q)| + |K_I^{(VG)}(q)| \Big), \qquad I \in \{O, D\}.} \tag{574}$$

This is the fourth deliverable.

6. Package E: anisotropy-response derivative bounds

The off-diagonal entries require derivatives with respect to the sectoral anisotropy variables:

$$\frac{\partial}{\partial \delta_D} \Sigma_O^{(OD)}, \qquad \frac{\partial}{\partial \delta_O} \Sigma_D^{(OD)}.$$

These act on:

- the propagators G_O, G_D,
- the projected momentum structures $\mathcal{P}_O, \mathcal{P}_D$,
- the regulator if sector-dependent.

Define

$$\boxed{\mathfrak{A}_{OD} := \sup_{q,m,n} \left| \frac{\partial}{\partial \delta_D} \big(\mathcal{P}_O \, G_D \, G_O \, \partial_t R_k \big) \right|_{\delta=0},} \tag{575}$$

$$\boxed{\mathfrak{A}_{DO} := \sup_{q,m,n} \left| \frac{\partial}{\partial \delta_O} \big(\mathcal{P}_D \, G_O \, G_D \, \partial_t R_k \big) \right|_{\delta=0}.} \tag{576}$$

This is the fifth deliverable.

7. Master bound for η_{OD}

Using the factorized kernels and the packages above, one gets the schematic estimate

$$\big| (M_Z^{OD})_{OO} \big| \lesssim \lambda_{OD}^2 \, \mathfrak{B}_\chi^2 \, \mathfrak{G}_O \, \mathfrak{G}_D \, \mathfrak{R}_k \, \mathfrak{K}_O, \tag{577}$$

$$\big| (M_Z^{OD})_{DD} \big| \lesssim \lambda_{OD}^2 \, \mathfrak{B}_n^2 \, \mathfrak{G}_O \, \mathfrak{G}_D \, \mathfrak{R}_k \, \mathfrak{K}_D, \tag{578}$$

and

$$\big| (M_Z^{OD})_{OD} \big| \lesssim \lambda_{OD}^2 \, \mathfrak{B}_\chi^2 \, \mathfrak{A}_{OD}, \qquad \big| (M_Z^{OD})_{DO} \big| \lesssim \lambda_{OD}^2 \, \mathfrak{B}_n^2 \, \mathfrak{A}_{DO}. \tag{579}$$

Therefore a minimal norm bound on the OD-block is

$$\boxed{\eta_{OD} \lesssim \lambda_{OD}^2 \Big[\mathfrak{B}_\chi^2 \, \mathfrak{G}_O \mathfrak{G}_D \, \mathfrak{R}_k \, \mathfrak{K}_O + \mathfrak{B}_n^2 \, \mathfrak{G}_O \mathfrak{G}_D \, \mathfrak{R}_k \, \mathfrak{K}_D + \mathfrak{B}_\chi^2 \, \mathfrak{A}_{OD} + \mathfrak{B}_n^2 \, \mathfrak{A}_{DO} \Big].} \tag{580}$$

8. Minimal computation package

Thus the minimal computation package needed to obtain η_{OD} is exactly:

$$\boxed{\big(\mathfrak{B}_\chi,\ \mathfrak{B}_n,\ \mathfrak{G}_O,\ \mathfrak{G}_D,\ \mathfrak{R}_k,\ \mathfrak{K}_O,\ \mathfrak{K}_D,\ \mathfrak{A}_{OD},\ \mathfrak{A}_{DO}\big).} \tag{581}$$

Once these nine quantities are estimated, η_{OD} follows immediately from (580).

9. Honest status

The package above is the minimal honest reduction: it does not claim the sign, and it does not assume more than what is already structurally derived. It simply reduces the OD-mixed bottleneck to finitely many explicit background overlap, propagator, regulator, curvature, and anisotropy-response bounds.

. Summary

$$\boxed{\textbf{Minimal bound package for } \eta_{OD}}$$

Starting from

$$(M_Z^{OD})_{IJ} = -\frac{\partial}{\partial\delta_J}\frac{\partial}{\partial p^2}\Sigma_I^{(OD)}(p)\Big|_{p=0,\delta=0}, \qquad I,J\in\{O,D\},$$

the projected self-energies are bounded once the following quantities are controlled:

$$\boxed{\mathfrak{B}_\chi = \|P_O\bar{\chi}_B P_D\|_{\mathrm{HS}}, \qquad \mathfrak{B}_n = \|P_D\bar{n}_B P_O\|_{\mathrm{HS}},}$$

$$\boxed{\mathfrak{G}_O = \sup_{m,q}|G_O^{(m)}(q)|, \qquad \mathfrak{G}_D = \sup_{n,q}|G_D^{(n)}(q)|,}$$

$$\boxed{\mathfrak{R}_k = \sup_q|\partial_t R_k(q)|,}$$

$$\boxed{\mathfrak{K}_I = \sup_q\big(|K_I^{(V)}(q)| + |K_I^{(G)}(q)| + |K_I^{(VG)}(q)|\big), \qquad I\in\{O,D\},}$$

$$\boxed{\mathfrak{A}_{OD} = \sup\left|\frac{\partial}{\partial\delta_D}(\mathcal{P}_O G_D G_O\partial_t R_k)\right|, \qquad \mathfrak{A}_{DO} = \sup\left|\frac{\partial}{\partial\delta_O}(\mathcal{P}_D G_O G_D\partial_t R_k)\right|.}$$

Then a master bound is

$$\boxed{\eta_{OD} \lesssim \lambda_{OD}^2\Big[\mathfrak{B}_\chi^2\,\mathfrak{G}_O\mathfrak{G}_D\,\mathfrak{R}_k\,\mathfrak{K}_O + \mathfrak{B}_n^2\,\mathfrak{G}_O\mathfrak{G}_D\,\mathfrak{R}_k\,\mathfrak{K}_D + \mathfrak{B}_\chi^2\,\mathfrak{A}_{OD} + \mathfrak{B}_n^2\,\mathfrak{A}_{DO}\Big].}$$

Therefore the entire OD-mixed bottleneck is reduced to the finite package

$$\boxed{(\mathfrak{B}_\chi, \mathfrak{B}_n, \mathfrak{G}_O, \mathfrak{G}_D, \mathfrak{R}_k, \mathfrak{K}_O, \mathfrak{K}_D, \mathfrak{A}_{OD}, \mathfrak{A}_{DO}).}$$

EXPLICIT BACKGROUND OVERLAP BOUNDS FOR $\mathfrak{B}_\chi$ AND $\mathfrak{B}_n$

1. Definitions

The two background overlap quantities entering the OD-mixed kernel package are

$$\mathfrak{B}_\chi := \|P_O \bar{\chi}_B P_D\|_{\mathrm{HS}}, \qquad \mathfrak{B}_n := \|P_D \bar{n}_B P_O\|_{\mathrm{HS}}, \tag{582}$$

where:

- P_O is the projector onto the relevant O-sector low-energy fluctuation subspace,
- P_D is the projector onto the relevant D-sector low-energy fluctuation subspace,
- $\bar{n}_B(\xi)$ and $\bar{\chi}_B(\xi)$ are the shell-/defect-adapted background profiles on the transverse coordinate ξ.

Equivalently, if $\{u_m\}_{m=1}^{N_O}$ and $\{v_n\}_{n=1}^{N_D}$ are orthonormal bases of the projected O- and D-subspaces, then

$$\mathfrak{B}_\chi^2 = \sum_{m=1}^{N_O}\sum_{n=1}^{N_D} \big|\langle u_m, \bar{\chi}_B v_n\rangle\big|^2, \qquad \mathfrak{B}_n^2 = \sum_{n=1}^{N_D}\sum_{m=1}^{N_O} \big|\langle v_n, \bar{n}_B u_m\rangle\big|^2. \tag{583}$$

2. Basic operator-norm bound

Since Hilbert-Schmidt norm is bounded by rank times operator norm,

$$\|P_O \bar{\chi}_B P_D\|_{\mathrm{HS}} \le \sqrt{\operatorname{rank}(P_O)\operatorname{rank}(P_D)}\, \|P_O \bar{\chi}_B P_D\|_{\mathrm{op}}, \tag{584}$$

and similarly

$$\|P_D \bar{n}_B P_O\|_{\mathrm{HS}} \le \sqrt{\operatorname{rank}(P_D)\operatorname{rank}(P_O)}\, \|P_D \bar{n}_B P_O\|_{\mathrm{op}}. \tag{585}$$

Because orthogonal projectors have operator norm 1,

$$\|P_O\| = \|P_D\| = 1,$$

we obtain

$$\boxed{\mathfrak{B}_\chi \le \sqrt{N_O N_D}\, \|\bar{\chi}_B\|_{\mathrm{op}}, \qquad \mathfrak{B}_n \le \sqrt{N_O N_D}\, \|\bar{n}_B\|_{\mathrm{op}},} \tag{586}$$

where

$$N_O := \operatorname{rank}(P_O), \qquad N_D := \operatorname{rank}(P_D).$$

Thus the most primitive overlap control reduces to:

1. the sizes of the projected low-energy sectors N_O, N_D,
2. the operator norms of the background profiles $\bar{\chi}_B, \bar{n}_B$.

3. L^∞-type bound

If $\bar{\chi}_B(\xi)$ and $\bar{n}_B(\xi)$ act as multiplication operators in ξ, then

$$\|\bar{\chi}_B\|_{\mathrm{op}} \le \|\bar{\chi}_B\|_{L^\infty_\xi}, \qquad \|\bar{n}_B\|_{\mathrm{op}} \le \|\bar{n}_B\|_{L^\infty_\xi}.$$

Hence

$$\boxed{\mathfrak{B}_\chi \le \sqrt{N_O N_D}\, \|\bar{\chi}_B\|_{L^\infty_\xi}, \qquad \mathfrak{B}_n \le \sqrt{N_O N_D}\, \|\bar{n}_B\|_{L^\infty_\xi}.} \tag{587}$$

This is the cleanest minimal bound, and it is already sufficient for the master estimate of η_{OD}.

4. Sharper trace-class / local-support bound

If the background profiles are localized in the defect core and the projected modes satisfy local L^2-normalization on the same core region $\Omega_{\rm def} \subset \mathbb{R}_\xi$, then one may sharpen the estimate.

Using Cauchy-Schwarz,

$$|\langle u_m, \bar{\chi}_B v_n\rangle| \le \|\bar{\chi}_B\|_{L^2(\Omega_{\rm def})} \|u_m v_n\|_{L^2(\Omega_{\rm def})}, \tag{588}$$

$$|\langle v_n, \bar{n}_B u_m\rangle| \le \|\bar{n}_B\|_{L^2(\Omega_{\rm def})} \|u_m v_n\|_{L^2(\Omega_{\rm def})}. \tag{589}$$

If, in addition, the projected modes satisfy a uniform local L^∞-bound

$$\|u_m\|_{L^\infty(\Omega_{\rm def})} \le C_O, \qquad \|v_n\|_{L^\infty(\Omega_{\rm def})} \le C_D,$$

then

$$\|u_m v_n\|_{L^2(\Omega_{\rm def})} \le C_O C_D \, |\Omega_{\rm def}|^{1/2},$$

and therefore

$$\boxed{\mathfrak{B}_\chi \le \sqrt{N_O N_D}\, C_O C_D \, |\Omega_{\rm def}|^{1/2} \, \|\bar{\chi}_B\|_{L^2(\Omega_{\rm def})},} \tag{590}$$

$$\boxed{\mathfrak{B}_n \le \sqrt{N_O N_D}\, C_O C_D \, |\Omega_{\rm def}|^{1/2} \, \|\bar{n}_B\|_{L^2(\Omega_{\rm def})}.} \tag{591}$$

These bounds are useful if the defect background is sharply localized and one wants a core-volume improvement over the crude L^∞-estimate.

5. Finite-rank low-mode reduction

In the actual shell-reduced TECT program, the relevant O- and D-sectors are already finite-dimensional low-mode sectors after projection. Thus N_O, N_D are finite and usually small. Hence one can define the purely numerical overlap constants

$$\boxed{C_\chi^{\rm bg} := \sqrt{N_O N_D}\, \|\bar{\chi}_B\|_{L^\infty_\xi}, \qquad C_n^{\rm bg} := \sqrt{N_O N_D}\, \|\bar{n}_B\|_{L^\infty_\xi}.} \tag{592}$$

Then the background overlap package closes as

$$\boxed{\mathfrak{B}_\chi \le C_\chi^{\rm bg}, \qquad \mathfrak{B}_n \le C_n^{\rm bg}.} \tag{593}$$

This is the minimal finite-rank reduction required for the later estimate of η_{OD}.

6. Symmetry refinement

If the background satisfies an approximate $O \leftrightarrow D$ exchange symmetry or equal norm condition

$$\|\bar{n}_B\|_{L^\infty_\xi} \sim \|\bar{\chi}_B\|_{L^\infty_\xi},$$

and if the projected mode dimensions match

$$N_O = N_D,$$

then

$$\mathfrak{B}_\chi \sim \mathfrak{B}_n. \tag{594}$$

This is not necessary, but it simplifies the later OD-mixed block estimate and makes the Laplacian-type target more natural.

7. Final deliverable

Thus the first two entries of the minimal OD-kernel package are closed by the finite-rank background bounds

$$\boxed{\mathfrak{B}_\chi \le \sqrt{N_O N_D}\,\|\bar{\chi}_B\|_{L^\infty_\xi}, \qquad \mathfrak{B}_n \le \sqrt{N_O N_D}\,\|\bar{n}_B\|_{L^\infty_\xi}.} \tag{595}$$

Or, in localized-core form,

$$\boxed{\mathfrak{B}_\chi \le \sqrt{N_O N_D}\,C_O C_D\,|\Omega_{\mathrm{def}}|^{1/2}\,\|\bar{\chi}_B\|_{L^2(\Omega_{\mathrm{def}})},} \tag{596}$$

$$\boxed{\mathfrak{B}_n \le \sqrt{N_O N_D}\,C_O C_D\,|\Omega_{\mathrm{def}}|^{1/2}\,\|\bar{n}_B\|_{L^2(\Omega_{\mathrm{def}})}.} \tag{597}$$

These are the first explicit finite bounds in the actual OD-mixed computation program.

8. Honest status

At this point:

- no new dynamical assumption has been added,
- no sign has been overclaimed,
- the overlap problem has been reduced to finite-dimensional projector ranks and background profile norms.

So this part is genuinely closed at the bound-package level.

. Summary

$$\boxed{\mathfrak{B}_\chi = \|P_O\bar{\chi}_B P_D\|_{\mathrm{HS}}, \qquad \mathfrak{B}_n = \|P_D\bar{n}_B P_O\|_{\mathrm{HS}}.}$$

If

$$N_O = \mathrm{rank}(P_O), \qquad N_D = \mathrm{rank}(P_D),$$

then the basic finite-rank Hilbert-Schmidt bound gives

$$\boxed{\mathfrak{B}_\chi \le \sqrt{N_O N_D}\,\|\bar{\chi}_B\|_{\mathrm{op}}, \qquad \mathfrak{B}_n \le \sqrt{N_O N_D}\,\|\bar{n}_B\|_{\mathrm{op}}.}$$

If the backgrounds act as multiplication operators in ξ, then

$$\boxed{\mathfrak{B}_\chi \le \sqrt{N_O N_D}\,\|\bar{\chi}_B\|_{L^\infty_\xi}, \qquad \mathfrak{B}_n \le \sqrt{N_O N_D}\,\|\bar{n}_B\|_{L^\infty_\xi}.}$$

A localized-core refinement is

$$\boxed{\mathfrak{B}_\chi \le \sqrt{N_O N_D}\,C_O C_D\,|\Omega_{\mathrm{def}}|^{1/2}\,\|\bar{\chi}_B\|_{L^2(\Omega_{\mathrm{def}})},}$$

$$\boxed{\mathfrak{B}_n \le \sqrt{N_O N_D}\,C_O C_D\,|\Omega_{\mathrm{def}}|^{1/2}\,\|\bar{n}_B\|_{L^2(\Omega_{\mathrm{def}})}.}$$

Therefore the first two entries of the minimal OD-kernel package are explicitly closed by finite projector ranks and background profile norms.

PROJECTED PROPAGATOR BOUNDS FOR $\mathfrak{G}_O$ AND $\mathfrak{G}_D$

1. Definitions

For the projected O- and D-sector background Hessian blocks, define the mode-resolved propagators

$$G_O^{(m)}(q) = \frac{1}{E_m^O(q) + R_{O,k}(q)}, \qquad G_D^{(n)}(q) = \frac{1}{E_n^D(q) + R_{D,k}(q)}, \tag{598}$$

where:

- $E_m^O(q)$ are the projected O-sector eigenvalues of the background-adapted Hessian,
- $E_n^D(q)$ are the projected D-sector eigenvalues,
- $R_{O,k}(q), R_{D,k}(q)$ are the corresponding regulator insertions.

The quantities needed in the OD-mixed kernel package are

$$\boxed{\mathfrak{G}_O := \sup_{m,q} |G_O^{(m)}(q)|, \qquad \mathfrak{G}_D := \sup_{n,q} |G_D^{(n)}(q)|.} \tag{599}$$

2. Spectral-gap input

Assume that the projected background Hessian blocks satisfy uniform lower spectral bounds

$$\boxed{E_m^O(q) \geq \Delta_O > 0, \qquad E_n^D(q) \geq \Delta_D > 0,} \tag{600}$$

for all relevant projected modes m, n and all loop momenta q in the support of the regulator.

This is the exact place where the remote/background spectral isolation enters the M_Z-estimate. It is fully consistent with the current TECT status: the local defect projector Hessian is already identified as a transverse matrix Schrödinger operator on the rank-two bundle Q_B, while the exact spectrum and low-mode gap remain explicit computational tasks. :contentReference[oaicite:4]index=4

3. Regulator positivity input

Assume the sectoral regulator is nonnegative:

$$\boxed{R_{O,k}(q) \geq 0, \qquad R_{D,k}(q) \geq 0.} \tag{601}$$

This is automatic for the standard monotone / optimized cutoff profiles used in the FRG setup.

Define

$$r_O^{\min} := \inf_q R_{O,k}(q) \geq 0, \qquad r_D^{\min} := \inf_q R_{D,k}(q) \geq 0. \tag{602}$$

4. Basic propagator bound

By (600) and (601),

$$E_m^O(q) + R_{O,k}(q) \geq \Delta_O + r_O^{\min}, \qquad E_n^D(q) + R_{D,k}(q) \geq \Delta_D + r_D^{\min}.$$

Therefore

$$\boxed{|G_O^{(m)}(q)| \leq \frac{1}{\Delta_O + r_O^{\min}}, \qquad |G_D^{(n)}(q)| \leq \frac{1}{\Delta_D + r_D^{\min}}.} \tag{603}$$

Taking the supremum gives the closed package bounds

$$\boxed{\mathfrak{G}_O \leq \frac{1}{\Delta_O + r_O^{\min}}, \qquad \mathfrak{G}_D \leq \frac{1}{\Delta_D + r_D^{\min}}.} \tag{604}$$

This is the minimal propagator estimate.

5. Optimized-regulator simplification

If one uses the optimized profile

$$R_{I,k}(q) = Z_{I,k}(k^2 - q^2)\Theta(k^2 - q^2), \qquad I \in \{O, D\},$$

then in the regulated region $|q| < k$,

$$R_{I,k}(q) \geq 0,$$

and outside that region $R_{I,k}(q) = 0$. Hence the conservative bound remains

$$\boxed{\mathfrak{G}_O \leq \frac{1}{\Delta_O}, \qquad \mathfrak{G}_D \leq \frac{1}{\Delta_D},} \tag{605}$$

while any strictly positive lower cutoff only improves it.

6. Stronger low-shell bound

If one knows that the projected spectra have the quadratic shell form

$$E_m^O(q) \geq \Delta_O + c_O|q|^2, \qquad E_n^D(q) \geq \Delta_D + c_D|q|^2, \qquad c_O, c_D > 0, \tag{606}$$

then the propagators also obey

$$|G_O^{(m)}(q)| \leq \frac{1}{\Delta_O + c_O|q|^2 + R_{O,k}(q)}, \qquad |G_D^{(n)}(q)| \leq \frac{1}{\Delta_D + c_D|q|^2 + R_{D,k}(q)}. \tag{607}$$

This is useful later for bounding momentum-derivative kernels $K_I^{(G)}$ and $K_I^{(VG)}$, but is not needed for the minimal package closure.

7. Relation to the remote-gap logic

The role of $\mathfrak{G}_O, \mathfrak{G}_D$ is exactly analogous to the remote-sector resolvent control in the Dirac-emergence chain: a positive lower bound on the projected remote/background spectrum produces a uniform resolvent bound. This is the same conceptual mechanism by which the shell-reduced remote-gap problem was reduced to explicit finite inequalities in the local Dirac program.

Therefore the propagator package is not an additional hypothesis; it is simply the FRG version of the same spectral-isolation logic.

8. Final deliverable

Thus the next two entries in the minimal OD-kernel package are closed by the spectral-gap data

$$\Delta_O, \qquad \Delta_D,$$

together with regulator positivity:

$$\boxed{\mathfrak{G}_O \leq \frac{1}{\Delta_O + r_O^{\min}}, \qquad \mathfrak{G}_D \leq \frac{1}{\Delta_D + r_D^{\min}}.} \tag{608}$$

In the simplest optimized-cutoff conservative form:

$$\boxed{\mathfrak{G}_O \leq \Delta_O^{-1}, \qquad \mathfrak{G}_D \leq \Delta_D^{-1}.} \tag{609}$$

9. Honest status

At this point:

- no new dynamical assumption has been introduced beyond the already-required projected spectral gap;
- no sign has been overclaimed;
- the propagator problem is reduced entirely to the explicit background spectral lower bounds Δ_O, Δ_D.

So this part is genuinely closed at the package level.

. Summary

$$\boxed{G_O^{(m)}(q) = \frac{1}{E_m^O(q) + R_{O,k}(q)}, \qquad G_D^{(n)}(q) = \frac{1}{E_n^D(q) + R_{D,k}(q)}.}$$

Define

$$\boxed{\mathfrak{G}_O := \sup_{m,q} |G_O^{(m)}(q)|, \qquad \mathfrak{G}_D := \sup_{n,q} |G_D^{(n)}(q)|.}$$

If the projected background spectra satisfy

$$\boxed{E_m^O(q) \geq \Delta_O > 0, \qquad E_n^D(q) \geq \Delta_D > 0,}$$

and the regulator is nonnegative,

$$\boxed{R_{O,k}(q) \geq 0, \qquad R_{D,k}(q) \geq 0,}$$

then

$$\boxed{\mathfrak{G}_O \leq \frac{1}{\Delta_O + r_O^{\min}}, \qquad \mathfrak{G}_D \leq \frac{1}{\Delta_D + r_D^{\min}},}$$

where

$$r_O^{\min} := \inf_q R_{O,k}(q), \qquad r_D^{\min} := \inf_q R_{D,k}(q).$$

In the simplest conservative optimized-cutoff form,

$$\boxed{\mathfrak{G}_O \leq \Delta_O^{-1}, \qquad \mathfrak{G}_D \leq \Delta_D^{-1}.}$$

Thus the propagator part of the minimal OD-kernel package is fully reduced to the explicit projected spectral gaps

$$\boxed{\Delta_O, \qquad \Delta_D.}$$

REGULATOR DERIVATIVE BOUND FOR $\mathfrak{R}_k$

1. Definition

The regulator derivative factor entering the projected OD-kernel package is

$$\boxed{\mathfrak{R}_k := \sup_q |\partial_t R_k(q)|.} \tag{610}$$

Here $t = \ln(k/\Lambda)$, so

$$\partial_t = k\frac{\partial}{\partial k}.$$

2. General sectoral regulator form

For the O- and D-sector kernels, the regulator is taken in sectoral form

$$R_{I,k}(q) = Z_{I,k}\, k^2\, r(y), \qquad y := \frac{q^2}{k^2}, \qquad I \in \{O, D\}, \tag{611}$$

where $r(y)$ is a dimensionless cutoff profile.

Then

$$\begin{aligned} \partial_t R_{I,k}(q) &= (\partial_t Z_{I,k})\, k^2\, r(y) + Z_{I,k}\, \partial_t(k^2 r(y)) \\ &= (\partial_t Z_{I,k})\, k^2\, r(y) + 2Z_{I,k} k^2 r(y) + Z_{I,k} k^2\, \partial_t r(y). \end{aligned} \tag{612}$$

Since

$$y = \frac{q^2}{k^2}, \qquad \partial_t y = -2y,$$

we get

$$\partial_t r(y) = r'(y)\partial_t y = -2y\, r'(y).$$

Hence

$$\boxed{\partial_t R_{I,k}(q) = k^2\Big[(\partial_t Z_{I,k})\, r(y) + 2Z_{I,k}\big(r(y) - yr'(y)\big)\Big].} \tag{613}$$

3. General profile bound

Assume the profile satisfies the standard boundedness conditions

$$\sup_{y\geq 0} |r(y)| \leq C_r, \qquad \sup_{y\geq 0} |yr'(y)| \leq C_{r'}, \tag{614}$$

and define

$$Z_{*,k} := \max\{Z_{O,k}, Z_{D,k}\}, \qquad \dot{Z}_{*,k} := \max\{|\partial_t Z_{O,k}|, |\partial_t Z_{D,k}|\}.$$

Then from (613),

$$|\partial_t R_{I,k}(q)| \leq k^2\Big[\dot{Z}_{*,k} C_r + 2Z_{*,k}(C_r + C_{r'})\Big]. \tag{615}$$

Therefore

$$\boxed{\mathfrak{R}_k \leq k^2\Big[\dot{Z}_{*,k} C_r + 2Z_{*,k}(C_r + C_{r'})\Big].} \tag{616}$$

This is the minimal general bound for any standard smooth monotone cutoff profile.

4. Optimized regulator bound

Now consider the optimized/Litim-type profile

$$R_{I,k}(q) = Z_{I,k}(k^2 - q^2)\Theta(k^2 - q^2). \tag{617}$$

Formally differentiating with respect to t,

$$\begin{aligned} \partial_t R_{I,k}(q) &= (\partial_t Z_{I,k})(k^2 - q^2)\Theta(k^2 - q^2) + 2Z_{I,k} k^2 \Theta(k^2 - q^2) \\ &\quad + Z_{I,k}(k^2 - q^2)\, \partial_t \Theta(k^2 - q^2). \end{aligned} \tag{618}$$

Away from the threshold surface $q^2 = k^2$, the distributional boundary term does not contribute to the essential supremum bound, and inside the support one has

$$0 \le k^2 - q^2 \le k^2.$$

Hence

$$|\partial_t R_{I,k}(q)| \le |\partial_t Z_{I,k}|\, k^2 + 2Z_{I,k}k^2. \tag{619}$$

Therefore

$$\boxed{\mathfrak{R}_k \le k^2(\dot{Z}_{*,k} + 2Z_{*,k}).} \tag{620}$$

If one neglects the anomalous-running contribution in the simplest conservative estimate, this reduces to

$$\boxed{\mathfrak{R}_k \le 2Z_{*,k}k^2.} \tag{621}$$

5. Dimensionless form

It is sometimes useful to factor out $Z_{*,k}k^2$. Define

$$\eta_{*,k} := \frac{\dot{Z}_{*,k}}{Z_{*,k}}.$$

Then (616) becomes

$$\boxed{\mathfrak{R}_k \le Z_{*,k}k^2\Big[\eta_{*,k}C_r + 2(C_r + C_{r'})\Big].} \tag{622}$$

For the optimized profile,

$$\boxed{\mathfrak{R}_k \le Z_{*,k}k^2(\eta_{*,k} + 2).} \tag{623}$$

Thus the regulator derivative bound is controlled entirely by the wavefunction renormalization scale $Z_{*,k}$, the RG time derivative $\eta_{*,k}$, and the chosen profile constants.

6. Minimal deliverable

For the projected OD-kernel package, the only thing actually needed is a finite uniform number $\mathfrak{R}_k$. Hence the deliverable can be stated simply as

$$\boxed{\mathfrak{R}_k \lesssim Z_{*,k}k^2,} \tag{624}$$

with the more explicit optimized-cutoff version

$$\boxed{\mathfrak{R}_k \le k^2(\dot{Z}_{*,k} + 2Z_{*,k}) \le Z_{*,k}k^2(\eta_{*,k} + 2).} \tag{625}$$

7. Honest status

At this point, the regulator derivative factor is completely closed at the package level:

- no new physical assumption has been added;
- no sign issue remains here;
- the bound depends only on the chosen cutoff profile and sectoral wavefunction factors.

So $\mathfrak{R}_k$ is one of the easiest fully explicit entries in the OD-kernel package.

. Summary

$$\boxed{\mathfrak{R}_k := \sup_q |\partial_t R_k(q)|.}$$

For a general sectoral regulator

$$R_{I,k}(q) = Z_{I,k} k^2 r\left(\frac{q^2}{k^2}\right), \qquad I \in \{O, D\},$$

one has

$$\boxed{\partial_t R_{I,k}(q) = k^2 \Big[(\partial_t Z_{I,k})\, r(y) + 2 Z_{I,k} \big(r(y) - y r'(y) \big) \Big].}$$

Hence, if

$$\sup |r(y)| \le C_r, \qquad \sup |y r'(y)| \le C_{r'},$$

then

$$\boxed{\mathfrak{R}_k \le k^2 \Big[\dot{Z}_{*,k} C_r + 2 Z_{*,k} (C_r + C_{r'}) \Big].}$$

For the optimized/Litim-type cutoff

$$R_{I,k}(q) = Z_{I,k}(k^2 - q^2)\Theta(k^2 - q^2),$$

one gets the explicit bound

$$\boxed{\mathfrak{R}_k \le k^2 (\dot{Z}_{*,k} + 2 Z_{*,k}).}$$

Equivalently, with

$$\eta_{*,k} := \frac{\dot{Z}_{*,k}}{Z_{*,k}},$$

$$\boxed{\mathfrak{R}_k \le Z_{*,k} k^2 (\eta_{*,k} + 2).}$$

Thus the regulator derivative part of the minimal OD-kernel package is fully reduced to explicit cutoff-profile constants and sectoral wavefunction factors.

SMALL-p KERNEL CURVATURE BOUNDS FOR $\mathfrak{K}_O$ AND $\mathfrak{K}_D$

1. Definition of the curvature package

In the projected OD-mixed self-energies, the p^2-derivative at $p = 0$ is controlled by three kernel classes:

$$K_I^{(V)}(q), \qquad K_I^{(G)}(q), \qquad K_I^{(VG)}(q), \qquad I \in \{O, D\}.$$

The minimal package quantities are

$$\boxed{\mathfrak{K}_I := \sup_q \Big(|K_I^{(V)}(q)| + |K_I^{(G)}(q)| + |K_I^{(VG)}(q)| \Big), \qquad I \in \{O, D\}.} \tag{626}$$

2. Small-p expansion inputs

Write the projected momentum structure as

$$\mathcal{P}_I(p,q) = \mathcal{P}_I^{(0)}(q) + p_\mu \mathcal{P}_I^{(1)\mu}(q) + \frac{1}{2} p_\mu p_\nu \mathcal{P}_I^{(2)\mu\nu}(q) + O(|p|^3), \tag{627}$$

and the projected propagator as

$$G_I(p+q) = G_I(q) + p_\mu \partial_{q_\mu} G_I(q) + \frac{1}{2} p_\mu p_\nu \partial_{q_\mu} \partial_{q_\nu} G_I(q) + O(|p|^3). \tag{628}$$

Then the p^2-derivative of the self-energy at $p=0$ is generated by:

1. the second-order curvature of the projected vertex,
2. the second-order curvature of the projected propagator,
3. the mixed first-order vertex/propagator term.

3. Explicit kernel decomposition

Accordingly, for $I \in \{O, D\}$,

$$K_I^{(V)}(q) \sim \mathcal{P}_I^{(2)}(q)\, G_{\bar{I}}(q)\, G_I(q), \tag{629}$$

$$K_I^{(G)}(q) \sim \mathcal{P}_I^{(0)}(q)\, G_{\bar{I}}(q)\, \partial_q^2 G_I(q), \tag{630}$$

$$K_I^{(VG)}(q) \sim \mathcal{P}_I^{(1)}(q)\, G_{\bar{I}}(q)\, \partial_q G_I(q). \tag{631}$$

Here $\bar{I}$ denotes the opposite sector:

$$\bar{O} = D, \qquad \bar{D} = O.$$

4. Minimal derivative bounds on the propagators

Assume the projected spectra satisfy quadratic lower bounds

$$E_m^O(q) \geq \Delta_O + c_O|q|^2, \qquad E_n^D(q) \geq \Delta_D + c_D|q|^2, \qquad c_O, c_D > 0, \tag{632}$$

and define

$$G_O^{(m)}(q) = \frac{1}{E_m^O(q) + R_{O,k}(q)}, \qquad G_D^{(n)}(q) = \frac{1}{E_n^D(q) + R_{D,k}(q)}.$$

Then, schematically,

$$|\partial_q G_I(q)| \leq \frac{C_I^{(1)}(1+|q|)}{(\Delta_I + c_I|q|^2 + r_I^{\min})^2}, \tag{633}$$

$$|\partial_q^2 G_I(q)| \leq \frac{C_I^{(2)}(1+|q|^2)}{(\Delta_I + c_I|q|^2 + r_I^{\min})^3}, \tag{634}$$

for finite constants $C_I^{(1)}, C_I^{(2)}$ determined by the background Hessian smoothness and the regulator profile.
Thus define

$$\boxed{\mathfrak{D}_I^{(1)} := \sup_q |\partial_q G_I(q)|, \qquad \mathfrak{D}_I^{(2)} := \sup_q |\partial_q^2 G_I(q)|.} \tag{635}$$

5. Minimal derivative bounds on the projected vertices

Similarly define the projected momentum-structure bounds

$$\boxed{\mathfrak{P}_I^{(0)} := \sup_q |\mathcal{P}_I^{(0)}(q)|, \qquad \mathfrak{P}_I^{(1)} := \sup_q |\mathcal{P}_I^{(1)}(q)|, \qquad \mathfrak{P}_I^{(2)} := \sup_q |\mathcal{P}_I^{(2)}(q)|.} \tag{636}$$

These constants encode the background/projector dependence of the cubic vertices and are finite provided the projected transport kernel is smooth in the low-momentum region. This is consistent with the defect-projector transport reduction where the leading kernel is built from the smooth defect gradient $\theta'(\xi)\mathcal{M}^\mu(\theta)$. :contentReference[oaicite:2]index=2

6. Curvature bounds

Using the previous packages

$$\mathfrak{G}_O, \qquad \mathfrak{G}_D, \qquad \mathfrak{D}_O^{(1)}, \mathfrak{D}_D^{(1)}, \qquad \mathfrak{D}_O^{(2)}, \mathfrak{D}_D^{(2)},$$

we get the uniform bounds

$$|K_O^{(V)}(q)| \le \mathfrak{P}_O^{(2)}\, \mathfrak{G}_D\, \mathfrak{G}_O, \qquad |K_D^{(V)}(q)| \le \mathfrak{P}_D^{(2)}\, \mathfrak{G}_O\, \mathfrak{G}_D, \tag{637}$$

$$|K_O^{(G)}(q)| \le \mathfrak{P}_O^{(0)}\, \mathfrak{G}_D\, \mathfrak{D}_O^{(2)}, \qquad |K_D^{(G)}(q)| \le \mathfrak{P}_D^{(0)}\, \mathfrak{G}_O\, \mathfrak{D}_D^{(2)}, \tag{638}$$

$$|K_O^{(VG)}(q)| \le \mathfrak{P}_O^{(1)}\, \mathfrak{G}_D\, \mathfrak{D}_O^{(1)}, \qquad |K_D^{(VG)}(q)| \le \mathfrak{P}_D^{(1)}\, \mathfrak{G}_O\, \mathfrak{D}_D^{(1)}. \tag{639}$$

Therefore

$$\boxed{\mathfrak{K}_O \le \mathfrak{P}_O^{(2)}\mathfrak{G}_D\mathfrak{G}_O + \mathfrak{P}_O^{(0)}\mathfrak{G}_D\mathfrak{D}_O^{(2)} + \mathfrak{P}_O^{(1)}\mathfrak{G}_D\mathfrak{D}_O^{(1)},} \tag{640}$$

$$\boxed{\mathfrak{K}_D \le \mathfrak{P}_D^{(2)}\mathfrak{G}_O\mathfrak{G}_D + \mathfrak{P}_D^{(0)}\mathfrak{G}_O\mathfrak{D}_D^{(2)} + \mathfrak{P}_D^{(1)}\mathfrak{G}_O\mathfrak{D}_D^{(1)}.} \tag{641}$$

7. Conservative reduced form

If one does not want to keep all constants separate, define

$$\mathfrak{P}_*^{(a)} := \max\{\mathfrak{P}_O^{(a)}, \mathfrak{P}_D^{(a)}\}, \qquad \mathfrak{D}_*^{(a)} := \max\{\mathfrak{D}_O^{(a)}, \mathfrak{D}_D^{(a)}\}, \qquad \mathfrak{G}_* := \max\{\mathfrak{G}_O, \mathfrak{G}_D\}.$$

Then both sectors obey the conservative estimate

$$\boxed{\mathfrak{K}_O,\ \mathfrak{K}_D \le \mathfrak{G}_*^2\, \mathfrak{P}_*^{(2)} + \mathfrak{G}_*\, \mathfrak{D}_*^{(2)}\, \mathfrak{P}_*^{(0)} + \mathfrak{G}_*\, \mathfrak{D}_*^{(1)}\, \mathfrak{P}_*^{(1)}.} \tag{642}$$

8. Honest status

At this stage, the small-p curvature problem is fully reduced to:

1. spectral lower bounds Δ_O, Δ_D,
2. smoothness/derivative bounds on the projected propagators,
3. smoothness/derivative bounds on the projected momentum structures $\mathcal{P}_I$.

No further conceptual input is needed. What remains is explicit evaluation of these finitely many constants from the chosen background-adapted Hessian and regulator.

. Summary

$$\boxed{\mathfrak{K}_I := \sup_q \Big(|K_I^{(V)}(q)| + |K_I^{(G)}(q)| + |K_I^{(VG)}(q)| \Big), \qquad I \in \{O, D\}.}$$

With

$$\mathcal{P}_I(p, q) = \mathcal{P}_I^{(0)}(q) + p \cdot \mathcal{P}_I^{(1)}(q) + \frac{1}{2} p \cdot \mathcal{P}_I^{(2)}(q) \cdot p + \cdots ,$$

and

$$G_I(p+q) = G_I(q) + p \cdot \partial_q G_I(q) + \frac{1}{2} p \cdot \partial_q^2 G_I(q) \cdot p + \cdots ,$$

the three curvature kernels satisfy

$$|K_O^{(V)}| \le \mathfrak{P}_O^{(2)} \mathfrak{G}_D \mathfrak{G}_O, \qquad |K_D^{(V)}| \le \mathfrak{P}_D^{(2)} \mathfrak{G}_O \mathfrak{G}_D,$$

$$|K_O^{(G)}| \le \mathfrak{P}_O^{(0)} \mathfrak{G}_D \mathfrak{D}_O^{(2)}, \qquad |K_D^{(G)}| \le \mathfrak{P}_D^{(0)} \mathfrak{G}_O \mathfrak{D}_D^{(2)},$$

$$|K_O^{(VG)}| \le \mathfrak{P}_O^{(1)} \mathfrak{G}_D \mathfrak{D}_O^{(1)}, \qquad |K_D^{(VG)}| \le \mathfrak{P}_D^{(1)} \mathfrak{G}_O \mathfrak{D}_D^{(1)}.$$

Therefore

$$\boxed{\mathfrak{K}_O \le \mathfrak{P}_O^{(2)} \mathfrak{G}_D \mathfrak{G}_O + \mathfrak{P}_O^{(0)} \mathfrak{G}_D \mathfrak{D}_O^{(2)} + \mathfrak{P}_O^{(1)} \mathfrak{G}_D \mathfrak{D}_O^{(1)},}$$

$$\boxed{\mathfrak{K}_D \le \mathfrak{P}_D^{(2)} \mathfrak{G}_O \mathfrak{G}_D + \mathfrak{P}_D^{(0)} \mathfrak{G}_O \mathfrak{D}_D^{(2)} + \mathfrak{P}_D^{(1)} \mathfrak{G}_O \mathfrak{D}_D^{(1)}.}$$

A conservative unified bound is

$$\boxed{\mathfrak{K}_O, \mathfrak{K}_D \le \mathfrak{G}_*^2 \mathfrak{P}_*^{(2)} + \mathfrak{G}_* \mathfrak{D}_*^{(2)} \mathfrak{P}_*^{(0)} + \mathfrak{G}_* \mathfrak{D}_*^{(1)} \mathfrak{P}_*^{(1)}.}$$

So the curvature part of the OD-kernel package is now reduced to finitely many projected propagator- and vertex-derivative constants.

ANISOTROPY-RESPONSE DERIVATIVE BOUNDS FOR $\mathfrak{A}_{OD}$ AND $\mathfrak{A}_{DO}$

1. Definitions

The final two quantities in the minimal projected OD-kernel package are

$$\boxed{\mathfrak{A}_{OD} := \sup_{q,m,n} \left| \frac{\partial}{\partial \delta_D} \big(\mathcal{P}_O \, G_D \, G_O \, \partial_t R_k \big) \right|_{\delta=0},} \tag{643}$$

$$\boxed{\mathfrak{A}_{DO} := \sup_{q,m,n} \left| \frac{\partial}{\partial \delta_O} \big(\mathcal{P}_D \, G_O \, G_D \, \partial_t R_k \big) \right|_{\delta=0}.} \tag{644}$$

They control the off-diagonal entries of the mixed OD block:

$$(M_Z^{OD})_{OD}, \qquad (M_Z^{OD})_{DO}.$$

2. Product-rule expansion

Apply the product rule to (643):

$$\frac{\partial}{\partial\delta_D}\big(\mathcal{P}_O\, G_D\, G_O\, \partial_t R_k\big) = \left(\frac{\partial\mathcal{P}_O}{\partial\delta_D}\right) G_D G_O \partial_t R_k + \mathcal{P}_O \left(\frac{\partial G_D}{\partial\delta_D}\right) G_O \partial_t R_k$$
$$+ \mathcal{P}_O G_D \left(\frac{\partial G_O}{\partial\delta_D}\right) \partial_t R_k + \mathcal{P}_O G_D G_O \left(\frac{\partial(\partial_t R_k)}{\partial\delta_D}\right). \tag{645}$$

Similarly,

$$\frac{\partial}{\partial\delta_O}\big(\mathcal{P}_D\, G_O\, G_D\, \partial_t R_k\big) = \left(\frac{\partial\mathcal{P}_D}{\partial\delta_O}\right) G_O G_D \partial_t R_k + \mathcal{P}_D \left(\frac{\partial G_O}{\partial\delta_O}\right) G_D \partial_t R_k$$
$$+ \mathcal{P}_D G_O \left(\frac{\partial G_D}{\partial\delta_O}\right) \partial_t R_k + \mathcal{P}_D G_O G_D \left(\frac{\partial(\partial_t R_k)}{\partial\delta_O}\right). \tag{646}$$

Thus the anisotropy-response problem separates into four derivative packages:

1. projected-vertex anisotropy response,
2. D-sector propagator anisotropy response,
3. O-sector propagator anisotropy response,
4. regulator anisotropy response.

3. Resolvent identity for propagator derivatives

Write

$$G_I = (H_I + R_{I,k})^{-1}, \qquad I \in \{O, D\}.$$

Then the resolvent identity gives

$$\boxed{\frac{\partial G_I}{\partial\delta_J} = -\,G_I \left(\frac{\partial H_I}{\partial\delta_J} + \frac{\partial R_{I,k}}{\partial\delta_J}\right) G_I.} \tag{647}$$

Hence

$$\left|\frac{\partial G_I}{\partial\delta_J}\right| \le \|G_I\|^2 \left(\left\|\frac{\partial H_I}{\partial\delta_J}\right\| + \left\|\frac{\partial R_{I,k}}{\partial\delta_J}\right\|\right). \tag{648}$$

Define the Hessian-response constants

$$\boxed{\mathfrak{H}_{IJ} := \sup_q \left\|\frac{\partial H_I}{\partial\delta_J}\right\|, \qquad I, J \in \{O, D\},} \tag{649}$$

and regulator-response constants

$$\boxed{\mathfrak{R}^{(\delta)}_{IJ} := \sup_q \left|\frac{\partial R_{I,k}}{\partial\delta_J}\right|.} \tag{650}$$

Then

$$\boxed{\sup_q \left|\frac{\partial G_I}{\partial\delta_J}\right| \le \mathfrak{G}_I^2 \big(\mathfrak{H}_{IJ} + \mathfrak{R}^{(\delta)}_{IJ}\big).} \tag{651}$$

4. Projected-vertex anisotropy-response bounds

Define

$$\boxed{\mathfrak{P}^{(\delta)}_{O|D} := \sup_q \left|\frac{\partial \mathcal{P}_O}{\partial \delta_D}\right|, \qquad \mathfrak{P}^{(\delta)}_{D|O} := \sup_q \left|\frac{\partial \mathcal{P}_D}{\partial \delta_O}\right|.} \tag{652}$$

These encode how the projected cubic vertices deform when the sectoral stiffness mismatch varies.

Such terms are finite provided the projected momentum structures $\mathcal{P}_I$ are smooth functions of the sectoral wave-function factors near the isotropic point. This is exactly the type of finite Jacobian that the explicit M_Z program is supposed to compute. :contentReference[oaicite:2]index=2

5. Regulator anisotropy-response bounds

For a sectoral regulator of the form

$$R_{I,k}(q) = Z_{I,k} k^2 r(q^2/k^2),$$

the dependence on δ_J enters only through the sectoral wavefunction factors. Hence

$$\boxed{\mathfrak{R}^{(\delta)}_{k|D} := \sup_q \left|\frac{\partial(\partial_t R_k)}{\partial \delta_D}\right|, \qquad \mathfrak{R}^{(\delta)}_{k|O} := \sup_q \left|\frac{\partial(\partial_t R_k)}{\partial \delta_O}\right|.} \tag{653}$$

In the simplest sector-diagonal implementation, these are controlled by the same profile constants as $\mathfrak{R}_k$, up to one additional factor from $\partial Z_I/\partial \delta_J$.

6. Master bounds for $\mathfrak{A}_{OD}$ and $\mathfrak{A}_{DO}$

Using (645) together with the bounds already derived for

$$\mathfrak{G}_O, \qquad \mathfrak{G}_D, \qquad \mathfrak{R}_k,$$

and the new derivative packages, we obtain

$$\begin{aligned}\mathfrak{A}_{OD} \le{}& \mathfrak{P}^{(\delta)}_{O|D}\,\mathfrak{G}_D\,\mathfrak{G}_O\,\mathfrak{R}_k + \mathfrak{P}^{(0)}_O\,\mathfrak{G}_O\,\mathfrak{R}_k\,\mathfrak{G}_D^2(\mathfrak{H}_{DD} + \mathfrak{R}^{(\delta)}_{DD}) \\ &+ \mathfrak{P}^{(0)}_O\,\mathfrak{G}_D\,\mathfrak{R}_k\,\mathfrak{G}_O^2(\mathfrak{H}_{OD} + \mathfrak{R}^{(\delta)}_{OD}) + \mathfrak{P}^{(0)}_O\,\mathfrak{G}_D\,\mathfrak{G}_O\,\mathfrak{R}^{(\delta)}_{k|D}.\end{aligned} \tag{654}$$

Similarly,

$$\begin{aligned}\mathfrak{A}_{DO} \le{}& \mathfrak{P}^{(\delta)}_{D|O}\,\mathfrak{G}_O\,\mathfrak{G}_D\,\mathfrak{R}_k + \mathfrak{P}^{(0)}_D\,\mathfrak{G}_D\,\mathfrak{R}_k\,\mathfrak{G}_O^2(\mathfrak{H}_{OO} + \mathfrak{R}^{(\delta)}_{OO}) \\ &+ \mathfrak{P}^{(0)}_D\,\mathfrak{G}_O\,\mathfrak{R}_k\,\mathfrak{G}_D^2(\mathfrak{H}_{DO} + \mathfrak{R}^{(\delta)}_{DO}) + \mathfrak{P}^{(0)}_D\,\mathfrak{G}_O\,\mathfrak{G}_D\,\mathfrak{R}^{(\delta)}_{k|O}.\end{aligned} \tag{655}$$

These are the explicit anisotropy-response derivative bounds.

7. Conservative reduced form

Define the sector-unified constants

$$\mathfrak{G}_* := \max(\mathfrak{G}_O, \mathfrak{G}_D), \qquad \mathfrak{P}^{(0)}_* := \max(\mathfrak{P}^{(0)}_O, \mathfrak{P}^{(0)}_D),$$

$$\mathfrak{P}^{(\delta)}_* := \max(\mathfrak{P}^{(\delta)}_{O|D}, \mathfrak{P}^{(\delta)}_{D|O}), \qquad \mathfrak{H}_* := \max_{I,J} \mathfrak{H}_{IJ}, \qquad \mathfrak{R}^{(\delta)}_* := \max_{I,J} \mathfrak{R}^{(\delta)}_{IJ},$$

$$\mathfrak{R}^{(\delta)}_{k,*} := \max(\mathfrak{R}^{(\delta)}_{k|D}, \mathfrak{R}^{(\delta)}_{k|O}).$$

Then both $\mathfrak{A}_{OD}$ and $\mathfrak{A}_{DO}$ obey the conservative bound

$$\boxed{\mathfrak{A}_{OD},\ \mathfrak{A}_{DO} \le \mathfrak{P}^{(\delta)}_*\,\mathfrak{G}_*^2\,\mathfrak{R}_k + 2\,\mathfrak{P}^{(0)}_*\,\mathfrak{G}_*^3\,\mathfrak{R}_k\,(\mathfrak{H}_* + \mathfrak{R}^{(\delta)}_*) + \mathfrak{P}^{(0)}_*\,\mathfrak{G}_*^2\,\mathfrak{R}^{(\delta)}_{k,*}.} \tag{656}$$

8. Closure of the OD-mixed package

At this point, the entire minimal bound package for η_{OD} is explicit:

$$(\mathfrak{B}_\chi, \mathfrak{B}_n, \mathfrak{G}_O, \mathfrak{G}_D, \mathfrak{R}_k, \mathfrak{K}_O, \mathfrak{K}_D, \mathfrak{A}_{OD}, \mathfrak{A}_{DO}).$$

Therefore the master estimate

$$\eta_{OD} \lesssim \lambda_{OD}^2 \Big[\mathfrak{B}_\chi^2\, \mathfrak{G}_O \mathfrak{G}_D\, \mathfrak{R}_k\, \mathfrak{K}_O + \mathfrak{B}_n^2\, \mathfrak{G}_O \mathfrak{G}_D\, \mathfrak{R}_k\, \mathfrak{K}_D + \mathfrak{B}_\chi^2\, \mathfrak{A}_{OD} + \mathfrak{B}_n^2\, \mathfrak{A}_{DO}\Big]$$

is now fully reduced to finitely many explicit constants.

9. Honest status

No new conceptual assumption has been added here. The only new inputs are finite Jacobian bounds for:

- the background-dependent Hessian blocks,
- the projected cubic momentum structures,
- the sectoral regulator response.

This is exactly the level of explicitness at which the TECT M_Z frontier is supposed to close: no structural mystery remains, only finite derivative/kernel estimates. :contentReference[oaicite:3]index=3

. Summary

$$\boxed{\mathfrak{A}_{OD} := \sup \left| \frac{\partial}{\partial \delta_D} (\mathcal{P}_O G_D G_O \partial_t R_k) \right|, \qquad \mathfrak{A}_{DO} := \sup \left| \frac{\partial}{\partial \delta_O} (\mathcal{P}_D G_O G_D \partial_t R_k) \right|.}$$

Using the product rule and the resolvent identity

$$\boxed{\frac{\partial G_I}{\partial \delta_J} = -\, G_I \left(\frac{\partial H_I}{\partial \delta_J} + \frac{\partial R_{I,k}}{\partial \delta_J} \right) G_I,}$$

define

$$\mathfrak{H}_{IJ} := \sup_q \left\| \frac{\partial H_I}{\partial \delta_J} \right\|, \qquad \mathfrak{R}_{IJ}^{(\delta)} := \sup_q \left| \frac{\partial R_{I,k}}{\partial \delta_J} \right|,$$

$$\mathfrak{P}_{O|D}^{(\delta)} := \sup_q \left| \frac{\partial \mathcal{P}_O}{\partial \delta_D} \right|, \qquad \mathfrak{P}_{D|O}^{(\delta)} := \sup_q \left| \frac{\partial \mathcal{P}_D}{\partial \delta_O} \right|,$$

$$\mathfrak{R}_{k|D}^{(\delta)} := \sup_q \left| \frac{\partial (\partial_t R_k)}{\partial \delta_D} \right|, \qquad \mathfrak{R}_{k|O}^{(\delta)} := \sup_q \left| \frac{\partial (\partial_t R_k)}{\partial \delta_O} \right|.$$

Then

$$\boxed{\mathfrak{A}_{OD} \le \mathfrak{P}_{O|D}^{(\delta)}\, \mathfrak{G}_D\, \mathfrak{G}_O\, \mathfrak{R}_k + \mathfrak{P}_O^{(0)}\, \mathfrak{G}_O\, \mathfrak{R}_k\, \mathfrak{G}_D^2 (\mathfrak{H}_{DD} + \mathfrak{R}_{DD}^{(\delta)}) + \mathfrak{P}_O^{(0)}\, \mathfrak{G}_D\, \mathfrak{R}_k\, \mathfrak{G}_O^2 (\mathfrak{H}_{OD} + \mathfrak{R}_{OD}^{(\delta)}) + \mathfrak{P}_O^{(0)}\, \mathfrak{G}_D\, \mathfrak{G}_O\, \mathfrak{R}_{k|D}^{(\delta)}.}$$

$$\boxed{\mathfrak{A}_{DO} \le \mathfrak{P}_{D|O}^{(\delta)}\, \mathfrak{G}_O\, \mathfrak{G}_D\, \mathfrak{R}_k + \mathfrak{P}_D^{(0)}\, \mathfrak{G}_D\, \mathfrak{R}_k\, \mathfrak{G}_O^2 (\mathfrak{H}_{OO} + \mathfrak{R}_{OO}^{(\delta)}) + \mathfrak{P}_D^{(0)}\, \mathfrak{G}_O\, \mathfrak{R}_k\, \mathfrak{G}_D^2 (\mathfrak{H}_{DO} + \mathfrak{R}_{DO}^{(\delta)}) + \mathfrak{P}_D^{(0)}\, \mathfrak{G}_O\, \mathfrak{G}_D\, \mathfrak{R}_{k|O}^{(\delta)}.}$$

A conservative unified bound is

$$\boxed{\mathfrak{A}_{OD}, \mathfrak{A}_{DO} \le \mathfrak{P}_*^{(\delta)}\, \mathfrak{G}_*^2\, \mathfrak{R}_k + 2\, \mathfrak{P}_*^{(0)}\, \mathfrak{G}_*^3\, \mathfrak{R}_k\, (\mathfrak{H}_* + \mathfrak{R}_*^{(\delta)}) + \mathfrak{P}_*^{(0)}\, \mathfrak{G}_*^2\, \mathfrak{R}_{k,*}^{(\delta)}.}$$

Thus the entire minimal OD-mixed package is now reduced to finitely many explicit overlap, gap, regulator, curvature, and anisotropy-response constants.

MINIMAL BOUND PACKAGE FOR THE PROJECTED GAUGE/PROJECTOR KERNEL

1. Goal

The second remaining bottleneck in the M_Z-program is

$$\eta_{\mathrm{gf}} = \|M_Z^{(\mathrm{gf/proj})}|_{\mathcal{U}_{\mathrm{tr}}}\|.$$

The explicit sign-test formula has already been reduced to

$$\boxed{(M_Z^{\mathrm{gf/proj}})_{TJ} = \frac{\partial}{\partial \delta_J}\mathcal{K}_A(0;\bar{\Phi}_{\min})\bigg|_{\delta=0}, \qquad J \in \{T, O, D\},} \tag{657}$$

where the projected transverse polarization kernel is defined by

$$\Pi_T^{\mu\nu}(p)\,\partial_t \Gamma^{(2)}_{\mu\nu}(p;\bar{\Phi}_{\min}) = p^2\,\mathcal{K}_A(p^2;\bar{\Phi}_{\min}). \tag{658}$$

Thus the problem is to bound the anisotropy response of the projected small-p transverse polarization kernel.

2. Preconditions

This package is meaningful only under the gauge-consistent infrared conditions:

$$p^\mu \Gamma^{(2)}_{\mu\nu}(p) = 0, \qquad m^2_{A,k} \to 0, \qquad \text{gauge-preserving regulator}, \qquad \text{Class I/II operators only}. \tag{659}$$

These are the exact conditions under which the one-loop projected polarization remains transverse and no Proca mass is generated. :contentReference[oaicite:3]index=3 :contentReference[oaicite:4]index=4

3. Projected polarization decomposition

The projected one-loop polarization splits as

$$\mathcal{K}_A = \mathcal{K}_A^{(4)} + \mathcal{K}_A^{(33)}, \tag{660}$$

where:

- $\mathcal{K}_A^{(4)}$ is the projected tadpole/seagull contribution,
- $\mathcal{K}_A^{(33)}$ is the projected bubble/sunset contribution.

Hence

$$(M_Z^{\mathrm{gf/proj}})_{TJ} = \frac{\partial}{\partial \delta_J}\mathcal{K}_A^{(4)}(0)\bigg|_{\delta=0} + \frac{\partial}{\partial \delta_J}\mathcal{K}_A^{(33)}(0)\bigg|_{\delta=0}. \tag{661}$$

4. Package A: projected gauge propagator bound

Let

$$G_A^{\mathrm{phys}}(q) = \left(\Pi_A^{\mathrm{phys}}[\Gamma_A^{(2)} + R_{A,k}]\Pi_A^{\mathrm{phys}}\right)^{-1} \tag{662}$$

be the projected transverse gauge propagator.

Assume the projected gauge-sector spectrum has a nonnegative lower bound on the physical subspace,

$$E_A^{\mathrm{phys}}(q) + R_{A,k}(q) \geq \Delta_A^{\mathrm{phys}} \geq 0. \tag{663}$$

Then define

$$\boxed{\mathfrak{G}_A := \sup_q \|G_A^{\mathrm{phys}}(q)\| \leq (\Delta_A^{\mathrm{phys}})^{-1}} \tag{664}$$

whenever $\Delta_A^{\mathrm{phys}} > 0$, or the corresponding IR-regulated version if the physical transverse mode is strictly massless.

a. Remark. In the gauge-preserving Maxwell-type sector, the absence of a Proca mass means $m_{A,k}^2 = 0$, not that the projected propagator is uncontrolled; it means one must use the regulator-supported IR bound rather than a naive massive bound.

5. Package B: projected polarization vertex bounds

Let the projected cubic and quartic gauge vertices entering the transverse polarization be denoted schematically by

$$\Gamma^{(3),\mathrm{phys}}_{A\Phi\Phi}, \qquad \Gamma^{(4),\mathrm{phys}}_{AA\Phi\Phi},$$

where Φ denotes the coupled projected soft matter sectors.

Define the operator bounds

$$\boxed{\mathfrak{V}_A^{(3)} := \sup_{p,q} \big\| \Gamma^{(3),\mathrm{phys}}_{A\Phi\Phi}(p,q;\bar{\Phi}_{\min}) \big\|,} \tag{665}$$

$$\boxed{\mathfrak{V}_A^{(4)} := \sup_{p,q} \big\| \Gamma^{(4),\mathrm{phys}}_{AA\Phi\Phi}(p,q;\bar{\Phi}_{\min}) \big\|.} \tag{666}$$

These constants encode the size of the physically projected gauge-matter interaction vertices on the shell-/defect-adapted background.

6. Package C: transverse small-p curvature bounds

Expand the projected polarization kernel near $p = 0$:

$$\mathcal{K}_A(p^2) = \mathcal{K}_A^{(0)} + p^2 \mathcal{K}_A^{(1)} + O(p^4). \tag{667}$$

The anisotropy-flow contribution depends only on $\mathcal{K}_A(0)$, i.e. the coefficient of the transverse p^2-structure.

Define

$$\boxed{\mathfrak{K}_A^{(4)} := \sup_q |\mathcal{K}_A^{(4)}(0;q)|, \qquad \mathfrak{K}_A^{(33)} := \sup_q |\mathcal{K}_A^{(33)}(0;q)|.} \tag{668}$$

Equivalently, one may package them into

$$\boxed{\mathfrak{K}_A := \sup_q \big(|\mathcal{K}_A^{(4)}(0;q)| + |\mathcal{K}_A^{(33)}(0;q)| \big).} \tag{669}$$

7. Package D: anisotropy-response derivative bounds

The quantity η_{gf} requires derivatives with respect to δ_J. Thus define

$$\boxed{\mathfrak{A}_{A|J} := \sup_q \left| \frac{\partial}{\partial \delta_J} \mathcal{K}_A(0;q;\bar{\Phi}_{\min}) \right|_{\delta=0}, \qquad J \in \{T,O,D\}.} \tag{670}$$

These are the direct gauge/projector analogues of $\mathfrak{A}_{OD}, \mathfrak{A}_{DO}$ in the mixed matter-sector package.

8. Package E: projector-control constants

Because the A-sector is meaningful only after projection onto the transverse physical subspace, define

$$\boxed{\mathfrak{P}_A := \sup_p \|\Pi_A^{\mathrm{phys}}(p)\|,} \tag{671}$$

and the anisotropy-response of the projector itself

$$\boxed{\mathfrak{P}^{(\delta)}_{A|J} := \sup_p \left\| \frac{\partial \Pi^{\rm phys}_A(p)}{\partial \delta_J} \right\|, \qquad J \in \{T, O, D\}.} \tag{672}$$

For a standard transverse projector,

$$\Pi^{\mu\nu}_T(p) = \eta^{\mu\nu} - \frac{p^\mu p^\nu}{p^2},$$

one has $\mathfrak{P}_A \leq 1$ away from the trivial zero-mode singularity, and the remaining issue is the background/projector implementation in the physical A-sector.

9. Master bound for $\eta_{\rm gf}$

Collecting the packages above, one obtains the schematic estimate

$$\boxed{\eta_{\rm gf} \lesssim \mathfrak{P}^2_A \Big[\mathfrak{G}^2_A \, \mathfrak{V}^{(4)}_A \, \mathfrak{K}^{(4)}_A + \mathfrak{G}^3_A \, (\mathfrak{V}^{(3)}_A)^2 \, \mathfrak{K}^{(33)}_A \Big] + \sum_{J\in\{T,O,D\}} \mathfrak{P}^{(\delta)}_{A|J} \, \mathfrak{K}_A + \sum_{J\in\{T,O,D\}} \mathfrak{A}_{A|J}.} \tag{673}$$

A simpler conservative form is

$$\boxed{\eta_{\rm gf} \lesssim \mathfrak{P}^2_A \Big[\mathfrak{G}^2_A \, \mathfrak{V}^{(4)}_A \, \mathfrak{K}_A + \mathfrak{G}^3_A \, (\mathfrak{V}^{(3)}_A)^2 \, \mathfrak{K}_A \Big] + \mathfrak{P}^{(\delta)}_{A,*} \, \mathfrak{K}_A + \mathfrak{A}_{A,*},} \tag{674}$$

where

$$\mathfrak{P}^{(\delta)}_{A,*} := \max_J \mathfrak{P}^{(\delta)}_{A|J}, \qquad \mathfrak{A}_{A,*} := \max_J \mathfrak{A}_{A|J}.$$

10. Minimal package

Therefore the minimal bound package needed to obtain $\eta_{\rm gf}$ is exactly

$$\boxed{\big(\mathfrak{G}_A, \ \mathfrak{V}^{(3)}_A, \ \mathfrak{V}^{(4)}_A, \ \mathfrak{K}^{(4)}_A, \ \mathfrak{K}^{(33)}_A, \ \mathfrak{A}_{A|T}, \mathfrak{A}_{A|O}, \mathfrak{A}_{A|D}, \ \mathfrak{P}_A, \ \mathfrak{P}^{(\delta)}_{A|T}, \mathfrak{P}^{(\delta)}_{A|O}, \mathfrak{P}^{(\delta)}_{A|D}\big).} \tag{675}$$

Once these finitely many constants are bounded, $\eta_{\rm gf}$ follows immediately.

11. Honest status

This is the exact gauge/projector analogue of the already-closed OD-mixed package: no conceptual ambiguity remains, but the actual numerical or analytic values of the package constants have not yet been extracted from the full TECT background-adapted Hessian.

That is precisely the current frontier stated in the notes: the remaining task is the explicit evaluation of the full Hessian, threshold functions, anomalous dimensions, and projected flow kernels. :contentReference[oaicite:6]index=6

. Summary

$$\boxed{(M^{\rm gf/proj}_Z)_{TJ} = \frac{\partial}{\partial \delta_J} \mathcal{K}_A(0; \bar{\Phi}_{\rm min}) \Big|_{\delta=0}, \qquad J \in \{T, O, D\},}$$

with

$$\Pi_T^{\mu\nu}(p)\,\partial_t\Gamma^{(2)}_{\mu\nu}(p;\bar{\Phi}_{\min}) = p^2\,\mathcal{K}_A(p^2;\bar{\Phi}_{\min}).$$

The minimal bound package for

$$\eta_{\rm gf} = \|M_Z^{(\rm gf/proj)}|_{\mathcal{U}_{\rm tr}}\|$$

is:

$$\boxed{\mathfrak{G}_A = \sup_q \|G_A^{\rm phys}(q)\|,}$$

$$\boxed{\mathfrak{V}_A^{(3)} = \sup\|\Gamma^{(3),\rm phys}_{A\Phi\Phi}\|, \qquad \mathfrak{V}_A^{(4)} = \sup\|\Gamma^{(4),\rm phys}_{AA\Phi\Phi}\|,}$$

$$\boxed{\mathfrak{K}_A^{(4)} = \sup|\mathcal{K}_A^{(4)}(0;q)|, \qquad \mathfrak{K}_A^{(33)} = \sup|\mathcal{K}_A^{(33)}(0;q)|,}$$

$$\boxed{\mathfrak{A}_{A|J} = \sup\left|\frac{\partial}{\partial\delta_J}\mathcal{K}_A(0;q)\right|, \qquad J\in\{T,O,D\},}$$

$$\boxed{\mathfrak{P}_A = \sup\|\Pi_A^{\rm phys}\|, \qquad \mathfrak{P}^{(\delta)}_{A|J} = \sup\left\|\frac{\partial\Pi_A^{\rm phys}}{\partial\delta_J}\right\|.}$$

A master conservative estimate is

$$\boxed{\eta_{\rm gf} \lesssim \mathfrak{P}_A^2\Big[\mathfrak{G}_A^2\,\mathfrak{V}_A^{(4)}\,\mathfrak{K}_A + \mathfrak{G}_A^3\,(\mathfrak{V}_A^{(3)})^2\,\mathfrak{K}_A\Big] + \mathfrak{P}^{(\delta)}_{A,*}\,\mathfrak{K}_A + \mathfrak{A}_{A,*}.}$$

Thus the gauge/projector bottleneck is fully reduced to a finite package of projected propagator, vertex, curvature, anisotropy-response, and projector-control constants.

MINIMAL BOUND PACKAGE FOR THE COERCIVE NEGATIVITY CONSTANT $\mu_{\rm good}$

1. Definition of the good part

The physical anisotropy-flow matrix is decomposed as

$$M_Z = M_Z^{(\tau)} + M_Z^{(\nabla\rm ani)} + M_Z^{(\rm self)} + M_Z^{(OD)} + M_Z^{(\rm gf/proj)}.$$

Define the good part by

$$\boxed{M_Z^{\rm good} := M_Z^{(\tau)} + M_Z^{(\nabla\rm ani)} + M_Z^{(\rm self)}.} \tag{676}$$

Let

$$\mathcal{U}_{\rm tr} = \big\{u = (\delta_T,\delta_O,\delta_D)^T \in \mathbb{R}^3 : 2\delta_T + 2\delta_O + 2\delta_D = 0\big\}$$

be the physical traceless anisotropy subspace. The target is a coercive negativity estimate

$$\boxed{\langle u, M_Z^{\rm good}u\rangle \le -\mu_{\rm good}\|u\|^2 \qquad \forall u\in\mathcal{U}_{\rm tr}, \qquad \mu_{\rm good}>0.} \tag{677}$$

2. Splitting of the coercive constant

We split

$$\mu_{\rm good} = \mu_\tau + \mu_\nabla + \mu_{\rm self} - \varepsilon_{\rm cross}, \tag{678}$$

where:

- μ_τ is the negative contribution coming from the quartic anisotropy sector,
- μ_∇ is the negative contribution coming from the derivative-anisotropy sector,
- $\mu_{\rm self}$ is the diagonal stabilizing contribution from the self-sector,
- $\varepsilon_{\rm cross}$ is a finite correction accounting for noncommuting/projected-basis cross terms when the three pieces are represented in a common traceless basis.

Thus the minimal package for $\mu_{\rm good}$ is the package controlling

$$\mu_\tau, \qquad \mu_\nabla, \qquad \mu_{\rm self}, \qquad \varepsilon_{\rm cross}.$$

3. Package A: quartic-anisotropy negativity μ_τ

The quartic cubic-anisotropy sector has the explicit RG eigenvalue

$$\theta_\tau = \epsilon\frac{4 - N_{\rm eff}}{N_{\rm eff} + 8} = -\frac{\epsilon}{7} \qquad (N_{\rm eff} = 6). \tag{679}$$

This is already an explicit negative quantity in the current TECT truncation. :contentReference[oaicite:2]index=2
Therefore define

$$\boxed{\mu_\tau := \frac{\epsilon}{7}\, c_\tau, \qquad c_\tau > 0,} \tag{680}$$

where c_τ is the projection coefficient measuring how strongly the quartic-anisotropy eigen-direction overlaps the physical traceless derivative-anisotropy sector.

The minimal bound package for μ_τ is therefore:

$$\boxed{c_\tau := \inf_{\substack{u\in\mathcal{U}_{\rm tr}\\ \|u\|=1}} \langle u, P_\tau u\rangle,} \tag{681}$$

where P_τ denotes the effective projection of the quartic-anisotropy RG direction onto the traceless physical anisotropy sector.

Hence

$$\boxed{\mu_\tau \geq \frac{\epsilon}{7}\, c_\tau.} \tag{682}$$

4. Package B: derivative-anisotropy negativity μ_∇

The derivative-anisotropy sector is the direct driver of the physical anisotropy flow. At the current theorem level, the notes support the expectation that its RG direction is negative, but do not yet provide the final explicit M_Z eigenvalue proof.

Accordingly, define

$$\boxed{\mu_\nabla := \inf_{\substack{u\in\mathcal{U}_{\rm tr}\\ \|u\|=1}} \left(-\langle u, M_Z^{(\nabla{\rm ani})} u\rangle\right).} \tag{683}$$

The minimal package here is not a single number from symmetry, but an explicit lower bound extracted from the derivative-anisotropy sector:

$$\boxed{\mu_\nabla \geq c_\nabla > 0,} \tag{684}$$

where c_∇ is the finite lower coercivity constant of the derivative-anisotropy block on $\mathcal{U}_{\rm tr}$.

Operationally, this constant is what the explicit M_Z computation should eventually deliver in the good sector.

5. Package C: self-sector stabilizing contribution $\mu_{\rm self}$

The self-sector consists of the diagonal O- and D-stabilizing pieces generated by

$$\lambda_O \bar{n}_B, \qquad \lambda_D \bar{\chi}_B,$$

through the background-induced self-bubble contributions. These were previously identified as naturally diagonally stabilizing.

Define the self contribution on the traceless subspace by

$$\boxed{\mu_{\rm self} := \inf_{\substack{u\in\mathcal{U}_{\rm tr}\\ \|u\|=1}} \big(-\langle u, M_Z^{(\rm self)} u\rangle\big).} \tag{685}$$

The minimal package is a lower diagonal estimate:

$$\boxed{M_Z^{(\rm self)}|_{\mathcal{U}_{\rm tr}} \le -\mu_{\rm self}\,\mathbf{1}_{\mathcal{U}_{\rm tr}}, \qquad \mu_{\rm self} \ge 0.} \tag{686}$$

A sufficient way to produce $\mu_{\rm self}$ is to extract the minimum of the diagonal self-sector entries after projection to the traceless basis.

6. Package D: cross-term defect $\varepsilon_{\rm cross}$

The three good pieces need not be simultaneously diagonal in the same basis on $\mathcal{U}_{\rm tr}$. Therefore define the cross-term defect by

$$\boxed{\varepsilon_{\rm cross} := \sup_{\substack{u\in\mathcal{U}_{\rm tr}\\ \|u\|=1}} \Big|\langle u, \big(M_Z^{(\tau)} + M_Z^{(\nabla{\rm ani})} + M_Z^{(\rm self)}\big)u\rangle - \sum_{X\in\{\tau,\nabla,{\rm self}\}} \langle u, M_Z^{(X)} u\rangle\Big|.} \tag{687}$$

If the three sectors are represented in a common basis with no further mixing defect, then $\varepsilon_{\rm cross} = 0$. More generally one only needs a finite upper bound

$$\boxed{\varepsilon_{\rm cross} \le c_{\rm cross}.} \tag{688}$$

7. Master bound for $\mu_{\rm good}$

Combining the previous packages yields

$$\boxed{\mu_{\rm good} \ge \mu_\tau + \mu_\nabla + \mu_{\rm self} - \varepsilon_{\rm cross}.} \tag{689}$$

Hence a sufficient positivity criterion is

$$\boxed{\mu_\tau + \mu_\nabla + \mu_{\rm self} > \varepsilon_{\rm cross}.} \tag{690}$$

Then

$$\boxed{\mu_{\rm good} > 0.} \tag{691}$$

8. Minimal package summary

Therefore the minimal bound package for $\mu_{\rm good}$ is exactly

$$\boxed{\big(c_\tau,\ c_\nabla,\ \mu_{\rm self},\ c_{\rm cross}\big),} \tag{692}$$

or equivalently

$$\boxed{\big(\mu_\tau,\ \mu_\nabla,\ \mu_{\rm self},\ \varepsilon_{\rm cross}\big).} \tag{693}$$

Once these four quantities are known, one obtains $\mu_{\rm good}$ from (689).

9. Final sufficient negativity criterion revisited

Recall the bad-part bounds

$$\|M_Z^{(OD)}|_{\mathcal{U}_{\rm tr}}\| \le \eta_{OD}, \qquad \|M_Z^{(\rm gf/proj)}|_{\mathcal{U}_{\rm tr}}\| \le \eta_{\rm gf}.$$

Then the full sufficient negativity condition becomes

$$\boxed{\mu_\tau + \mu_\nabla + \mu_{\rm self} - \varepsilon_{\rm cross} > \eta_{OD} + \eta_{\rm gf}.} \tag{694}$$

This is the fully explicit master inequality for the negativity of

$$\mathrm{spec}(M_Z|_{\mathcal{U}_{\rm tr}}).$$

10. Honest status

At this point, the good-sector problem is reduced completely to finitely many lower-bound and mixing-defect constants. No new conceptual input is needed.

What remains open is only the explicit extraction of:

$$c_\tau, \qquad c_\nabla, \qquad \mu_{\rm self}, \qquad c_{\rm cross},$$

from the chosen TECT background-adapted FRG truncation. That is fully consistent with the current notes: quartic anisotropy is already explicitly irrelevant, whereas the derivative-anisotropy and velocity-locking problem still requires the explicit M_Z computation. :contentReference[oaicite:4]index=4 :contentReference[oaicite:5]index=5

. Summary

$$\boxed{M_Z^{\rm good} = M_Z^{(\tau)} + M_Z^{(\nabla\rm ani)} + M_Z^{(\rm self)}.}$$

The coercive negativity target is

$$\boxed{\langle u, M_Z^{\rm good} u\rangle \le -\mu_{\rm good}\|u\|^2 \qquad \forall u \in \mathcal{U}_{\rm tr}.}$$

A minimal decomposition is

$$\boxed{\mu_{\rm good} \ge \mu_\tau + \mu_\nabla + \mu_{\rm self} - \varepsilon_{\rm cross}.}$$

Here:

$$\boxed{\mu_\tau \ge \frac{\epsilon}{7}\, c_\tau \qquad (N_{\rm eff} = 6),}$$

coming from quartic anisotropy irrelevance,

$$\boxed{\mu_\nabla = \inf_{\|u\|=1,\ u\in\mathcal{U}_{\rm tr}} \big(-\langle u, M_Z^{(\nabla\rm ani)} u\rangle\big) \ge c_\nabla > 0,}$$

$$\boxed{\mu_{\rm self} = \inf_{\|u\|=1,\ u\in\mathcal{U}_{\rm tr}} \big(-\langle u, M_Z^{(\rm self)} u\rangle\big) \ge 0,}$$

$$\boxed{\varepsilon_{\mathrm{cross}} \le c_{\mathrm{cross}}.}$$

Thus the minimal package for μ_{good} is

$$\boxed{(c_\tau,\ c_\nabla,\ \mu_{\mathrm{self}},\ c_{\mathrm{cross}}).}$$

Together with the already-closed bad-part packages,

$$\|M_Z^{(OD)}|_{\mathcal{U}_{\mathrm{tr}}}\| \le \eta_{OD}, \qquad \|M_Z^{(\mathrm{gf/proj})}|_{\mathcal{U}_{\mathrm{tr}}}\| \le \eta_{\mathrm{gf}},$$

the final explicit sufficient negativity condition is

$$\boxed{\mu_\tau + \mu_\nabla + \mu_{\mathrm{self}} - \varepsilon_{\mathrm{cross}} > \eta_{OD} + \eta_{\mathrm{gf}}.}$$

If this holds, then

$$\boxed{\mathrm{spec}(M_Z|_{\mathcal{U}_{\mathrm{tr}}}) \subset (-\infty, 0).}$$

MASTER COMPUTATION CHECKLIST FOR CLOSING M_Z

1. Final target

The final target is

$$\boxed{\mathrm{spec}(M_Z|_{\mathcal{U}_{\mathrm{tr}}}) \subset (-\infty, 0),} \tag{695}$$

where

$$\mathcal{U}_{\mathrm{tr}} = \left\{u = (\delta_T, \delta_O, \delta_D)^T : 2\delta_T + 2\delta_O + 2\delta_D = 0\right\}.$$

The fully explicit sufficient criterion is

$$\boxed{\mu_\tau + \mu_\nabla + \mu_{\mathrm{self}} - \varepsilon_{\mathrm{cross}} > \eta_{OD} + \eta_{\mathrm{gf}}.} \tag{696}$$

Thus the problem is reduced to computing / bounding finitely many constants.

2. Checklist block I: OD-mixed bad-part bound

The first bad-sector quantity is

$$\eta_{OD} = \|M_Z^{(OD)}|_{\mathcal{U}_{\mathrm{tr}}}\|.$$

Its master bound is

$$\eta_{OD} \lesssim \lambda_{OD}^2 \Big[\mathfrak{B}_\chi^2\, \mathfrak{G}_O \mathfrak{G}_D\, \mathfrak{R}_k\, \mathfrak{K}_O + \mathfrak{B}_n^2\, \mathfrak{G}_O \mathfrak{G}_D\, \mathfrak{R}_k\, \mathfrak{K}_D + \mathfrak{B}_\chi^2\, \mathfrak{A}_{OD} + \mathfrak{B}_n^2\, \mathfrak{A}_{DO}\Big]. \tag{697}$$

Therefore the required OD-mixed checklist is:

a. (I-1) Background overlap bounds

$$\mathfrak{B}_\chi = \|P_O \bar{\chi}_B P_D\|_{\mathrm{HS}}, \qquad \mathfrak{B}_n = \|P_D \bar{n}_B P_O\|_{\mathrm{HS}}.$$

Minimal explicit bounds:

$$\mathfrak{B}_\chi \le \sqrt{N_O N_D}\, \|\bar{\chi}_B\|_{L^\infty_\xi}, \qquad \mathfrak{B}_n \le \sqrt{N_O N_D}\, \|\bar{n}_B\|_{L^\infty_\xi}.$$

b. (I-2) Projected propagator bounds

$$\mathfrak{G}_O = \sup_{m,q} |G_O^{(m)}(q)|, \qquad \mathfrak{G}_D = \sup_{n,q} |G_D^{(n)}(q)|.$$

Minimal explicit bounds:

$$\mathfrak{G}_O \le \frac{1}{\Delta_O + r_O^{\min}}, \qquad \mathfrak{G}_D \le \frac{1}{\Delta_D + r_D^{\min}}.$$

c. (I-3) Regulator derivative bound

$$\mathfrak{R}_k = \sup_q |\partial_t R_k(q)|.$$

Minimal explicit bound:

$$\mathfrak{R}_k \le k^2(\dot{Z}_{*,k} + 2Z_{*,k}),$$

or conservatively

$$\mathfrak{R}_k \lesssim Z_{*,k} k^2.$$

d. (I-4) Small-p curvature bounds

$$\mathfrak{K}_O, \qquad \mathfrak{K}_D.$$

Minimal explicit bounds:

$$\mathfrak{K}_O \le \mathfrak{P}_O^{(2)} \mathfrak{G}_D \mathfrak{G}_O + \mathfrak{P}_O^{(0)} \mathfrak{G}_D \mathfrak{D}_O^{(2)} + \mathfrak{P}_O^{(1)} \mathfrak{G}_D \mathfrak{D}_O^{(1)},$$

$$\mathfrak{K}_D \le \mathfrak{P}_D^{(2)} \mathfrak{G}_O \mathfrak{G}_D + \mathfrak{P}_D^{(0)} \mathfrak{G}_O \mathfrak{D}_D^{(2)} + \mathfrak{P}_D^{(1)} \mathfrak{G}_O \mathfrak{D}_D^{(1)}.$$

e. (I-5) Anisotropy-response derivative bounds

$$\mathfrak{A}_{OD}, \qquad \mathfrak{A}_{DO}.$$

Minimal explicit conservative form:

$$\mathfrak{A}_{OD}, \mathfrak{A}_{DO} \le \mathfrak{P}_*^{(\delta)} \mathfrak{G}_*^2 \mathfrak{R}_k + 2\mathfrak{P}_*^{(0)} \mathfrak{G}_*^3 \mathfrak{R}_k (\mathfrak{H}_* + \mathfrak{R}_*^{(\delta)}) + \mathfrak{P}_*^{(0)} \mathfrak{G}_*^2 \mathfrak{R}_{k,*}^{(\delta)}.$$

f. Deliverable of Block I Once $(\mathfrak{B}_\chi, \mathfrak{B}_n, \mathfrak{G}_O, \mathfrak{G}_D, \mathfrak{R}_k, \mathfrak{K}_O, \mathfrak{K}_D, \mathfrak{A}_{OD}, \mathfrak{A}_{DO})$ are estimated, η_{OD} is closed.

3. Checklist block II: gauge/projector bad-part bound

The second bad-sector quantity is

$$\eta_{\rm gf} = \| M_Z^{(\rm gf/proj)}|_{\mathcal{U}_{\rm tr}} \|.$$

Its conservative master bound is

$$\eta_{\rm gf} \lesssim \mathfrak{P}_A^2 \Big[\mathfrak{G}_A^2 \, \mathfrak{V}_A^{(4)} \, \mathfrak{K}_A + \mathfrak{G}_A^3 \, (\mathfrak{V}_A^{(3)})^2 \, \mathfrak{K}_A \Big] + \mathfrak{P}_{A,*}^{(\delta)} \, \mathfrak{K}_A + \mathfrak{A}_{A,*}. \tag{698}$$

Therefore the required gauge/projector checklist is:

g. (II-0) Preconditions Verify first:

$$p^\mu \Gamma_{\mu\nu}^{(2)} = 0, \qquad m_{A,k}^2 \to 0, \qquad \text{gauge-preserving regulator}, \qquad \text{Class I/II only}.$$

h. (II-1) Projected gauge propagator bound

$$\mathfrak{G}_A = \sup_q \| G_A^{\rm phys}(q) \|.$$

i. *(II-2) Projected gauge vertex bounds*

$$\mathfrak{V}_A^{(3)}, \qquad \mathfrak{V}_A^{(4)}.$$

j. *(II-3) Projected transverse curvature bounds*

$$\mathfrak{K}_A^{(4)}, \qquad \mathfrak{K}_A^{(33)}, \qquad \mathfrak{K}_A := \mathfrak{K}_A^{(4)} + \mathfrak{K}_A^{(33)}.$$

k. *(II-4) Gauge anisotropy-response bounds*

$$\mathfrak{A}_{A|T}, \qquad \mathfrak{A}_{A|O}, \qquad \mathfrak{A}_{A|D}.$$

l. *(II-5) Projector-control bounds*

$$\mathfrak{P}_A, \qquad \mathfrak{P}_{A|T}^{(\delta)}, \qquad \mathfrak{P}_{A|O}^{(\delta)}, \qquad \mathfrak{P}_{A|D}^{(\delta)}.$$

m. *Deliverable of Block II* Once

$$(\mathfrak{G}_A, \mathfrak{V}_A^{(3)}, \mathfrak{V}_A^{(4)}, \mathfrak{K}_A^{(4)}, \mathfrak{K}_A^{(33)}, \mathfrak{A}_{A|T}, \mathfrak{A}_{A|O}, \mathfrak{A}_{A|D}, \mathfrak{P}_A, \mathfrak{P}_{A|T}^{(\delta)}, \mathfrak{P}_{A|O}^{(\delta)}, \mathfrak{P}_{A|D}^{(\delta)})$$

are estimated, η_{gf} is closed.

4. Checklist block III: good-part coercive negativity

The good-sector quantity is

$$\mu_{\mathrm{good}}$$

with

$$M_Z^{\mathrm{good}} = M_Z^{(\tau)} + M_Z^{(\nabla\mathrm{ani})} + M_Z^{(\mathrm{self})}.$$

Its master lower bound is

$$\mu_{\mathrm{good}} \geq \mu_\tau + \mu_\nabla + \mu_{\mathrm{self}} - \varepsilon_{\mathrm{cross}}. \tag{699}$$

Therefore the required good-sector checklist is:

n. *(III-1) Quartic-anisotropy contribution* Use

$$\theta_\tau = -\frac{\epsilon}{7}$$

and extract the projection overlap coefficient

$$c_\tau = \inf_{\|u\|=1,\ u\in\mathcal{U}_{\mathrm{tr}}} \langle u, P_\tau u\rangle,$$

so that

$$\mu_\tau \geq \frac{\epsilon}{7} c_\tau.$$

o. *(III-2) Derivative-anisotropy coercive bound* Estimate

$$\mu_\nabla = \inf_{\|u\|=1,\ u\in\mathcal{U}_{\mathrm{tr}}} \big(-\langle u, M_Z^{(\nabla\mathrm{ani})} u\rangle\big) \geq c_\nabla > 0.$$

p. *(III-3) Self-sector stabilizing bound* Estimate

$$\mu_{\mathrm{self}} = \inf_{\|u\|=1,\ u\in\mathcal{U}_{\mathrm{tr}}} \big(-\langle u, M_Z^{(\mathrm{self})} u\rangle\big) \geq 0.$$

q. *(III-4) Cross-term defect* Estimate

$$\varepsilon_{\text{cross}} \le c_{\text{cross}}.$$

r. *Deliverable of Block III* Once

$$(c_{\mathcal{T}},\ c_{\nabla},\ \mu_{\text{self}},\ c_{\text{cross}})$$

are estimated, μ_{good} is closed.

5. Final execution test

After Blocks I–III are completed, perform the final comparison:

$$\boxed{\mu_{\text{good}} > \eta_{OD} + \eta_{\text{gf}}.} \tag{700}$$

If this holds, then

$$\boxed{\text{spec}(M_Z|_{\mathcal{U}_{\text{tr}}}) \subset (-\infty, 0).} \tag{701}$$

Hence

$$\delta_T, \delta_O, \delta_D \to 0,$$

and the soft infrared sector is velocity-locked and isotropic.

6. Minimal execution order

The most efficient order is:

$$\boxed{\textbf{Step 1: } \eta_{OD} \quad\Longrightarrow\quad \textbf{Step 2: } \eta_{\text{gf}} \quad\Longrightarrow\quad \textbf{Step 3: } \mu_{\text{good}} \quad\Longrightarrow\quad \textbf{Step 4: final inequality}.}$$

This order is optimal because:

- η_{OD} is the cleanest matter-sector bad-piece,
- η_{gf} is structurally more delicate but already fully packaged,
- μ_{good} is expected to be favorable and is best estimated once the bad-part scales are known.

7. Honest final status

At this point the M_Z-closure program is reduced completely to a finite checklist of explicit constants. No conceptual frontier remains in the M_Z setup itself. The only remaining work is the actual extraction of these constants from the chosen background-adapted FRG truncation.

. Summary

$$\boxed{\textbf{Master Computation Checklist for closing } M_Z}$$

The final sufficient condition is

$$\boxed{\mu_{\text{good}} > \eta_{OD} + \eta_{\text{gf}}.}$$

So the full execution checklist is:

$$\boxed{\textbf{Block I: close } \eta_{OD}}$$

by estimating

$$(\mathfrak{B}_\chi, \mathfrak{B}_n, \mathfrak{G}_O, \mathfrak{G}_D, \mathfrak{R}_k, \mathfrak{K}_O, \mathfrak{K}_D, \mathfrak{A}_{OD}, \mathfrak{A}_{DO}).$$

$$\boxed{\textbf{Block II: close } \eta_{\mathrm{gf}}}$$

by estimating

$$(\mathfrak{G}_A, \mathfrak{V}_A^{(3)}, \mathfrak{V}_A^{(4)}, \mathfrak{K}_A^{(4)}, \mathfrak{K}_A^{(33)}, \mathfrak{A}_{A|T}, \mathfrak{A}_{A|O}, \mathfrak{A}_{A|D}, \mathfrak{P}_A, \mathfrak{P}_{A|T}^{(\delta)}, \mathfrak{P}_{A|O}^{(\delta)}, \mathfrak{P}_{A|D}^{(\delta)}),$$

under the preconditions

$$p^\mu \Gamma^{(2)}_{\mu\nu} = 0, \qquad m_{A,k}^2 \to 0, \qquad \text{gauge-preserving regulator}, \qquad \text{Class I/II only}.$$

$$\boxed{\textbf{Block III: close } \mu_{\mathrm{good}}}$$

by estimating

$$(c_\tau,\ c_\nabla,\ \mu_{\mathrm{self}},\ c_{\mathrm{cross}}),$$

so that

$$\mu_{\mathrm{good}} \geq \mu_\tau + \mu_\nabla + \mu_{\mathrm{self}} - \varepsilon_{\mathrm{cross}}.$$

$$\boxed{\textbf{Block IV: final comparison}}$$

Check

$$\boxed{\mu_{\mathrm{good}} > \eta_{OD} + \eta_{\mathrm{gf}}.}$$

If true, then

$$\boxed{\mathrm{spec}(M_Z|_{\mathcal{U}_{\mathrm{tr}}}) \subset (-\infty, 0),}$$

hence

$$\boxed{\delta_T, \delta_O, \delta_D \to 0.}$$

Optimal execution order:

$$\boxed{\eta_{OD} \ \to\ \eta_{\mathrm{gf}} \ \to\ \mu_{\mathrm{good}} \ \to\ \mu_{\mathrm{good}} > \eta_{OD} + \eta_{\mathrm{gf}}.}$$

So the M_Z frontier is now a finite executable checklist, not an open structural problem.

STRONGEST HONEST CLOSURE THEOREM CURRENTLY AVAILABLE FOR TECT

A. Closed theorem-level inputs

At the theorem level, the following structures are already closed.

a. A1. Local Weyl/Dirac reduction. The local shell-reduced fermion problem is reduced to the finite witness data

$$W_\parallel, \qquad W_I, \qquad W_J,$$

or equivalently the reduced coefficient matrix M, such that local Weyl nondegeneracy is equivalent to

$$W_\parallel \neq 0, \qquad (W_I, W_J) \neq (0,0), \tag{702}$$

and opposite-valley doubling reduces to equality of the witness triples at $\pm G_*$.

b. A2. Stability of the local Dirac block. If the remote sector is spectrally gapped,

$$\mathrm{spec}(H_R) \cap (-\Delta, \Delta) = \varnothing, \qquad \Delta > 0,$$

and the mixing obeys

$$V(0) = 0, \qquad \|V(p)\| \leq c|p|,$$

then the Schur-complement correction is only

$$O(|p|^2/\Delta),$$

and does not alter the leading first-order Dirac block.

c. A3. Mass protection reduction. The constant-mass problem is reduced to a finite residual-symmetry algebra check: if no symmetry-allowed constant Hermitian 4×4 operator is a Dirac mass, then the reduced Dirac cone remains massless. :contentReference[oaicite:4]index=4

d. A4. Quartic anisotropy irrelevance. In the TECT infrared truncation with

$$\Phi = (A_1, A_2, n_1, n_2, \chi_1, \chi_2), \qquad N_{\mathrm{eff}} = 6,$$

the quartic cubic anisotropy eigenvalue is

$$\theta_\tau = \epsilon \frac{4 - N_{\mathrm{eff}}}{N_{\mathrm{eff}} + 8} = -\frac{\epsilon}{7} < 0. \tag{703}$$

Hence the leading quartic anisotropy is rigorously infrared-irrelevant at this schematic FRG level. :contentReference[oaicite:5]index=5

e. A5. Gauge-consistent transverse sector. The A-sector must satisfy

$$p^\mu \Gamma^{(2)}_{\mu\nu} = 0, \qquad m^2_{A,k} \to 0,$$

with a gauge-preserving regulator and no Class III gauge-breaking operators. Under these conditions, the projected polarization remains transverse and no spurious Proca mass is generated.

B. Explicit microscopic inputs still not yet verified

What is not yet theorem-level closed in the final microscopic realization is the following finite input list.

f. B1. Actual witness realization. The reduced witness theorem is closed, but the actual microscopic TECT/Class II realization still requires verification that the relevant transport tensor sector is genuinely present in the physical defect background.

g. B2. Opposite-valley realization. The formal pairing criterion is closed, but the actual equality of the witness triples at $\pm G_*$ still has to be checked microscopically. :contentReference[oaicite:8]index=8

h. B3. Remote gap. The remote-gap theorem is closed as a reduction theorem, but the actual positive value of

$$\Delta > 0$$

must still be computed from the relevant Bloch matrix / projected Hessian.

i. B4. Zero constant mixing. The operational test

$$\mathrm{Hom}_{H_*}(H_R, H_D) = 0$$

is explicit, but the actual shell-relevant irrep mismatch must still be checked in the final microscopic representation content. :contentReference[oaicite:10]index=10 *j. B5. Mass protection.* The reduction theorem is closed, but the actual residual constant algebra

$$A_{\mathrm{res}}$$

must still be computed in the final microscopic realization. :contentReference[oaicite:11]index=11

k. B6. Derivative-anisotropy sign. The physical derivative-anisotropy problem is reduced to the sign of the linearized kinetic-flow matrix

$$M_Z$$

on the traceless subspace

$$\mathcal{U}_{\mathrm{tr}} = \left\{ u = (\delta_T, \delta_O, \delta_D)^T : 2\delta_T + 2\delta_O + 2\delta_D = 0 \right\}.$$

The notes are explicit that this part remains open: the full Hessian, regulator, threshold functions, anomalous dimensions, and gauge constraints are still needed to determine the sign of the nontrivial eigenvalues of M_Z.

C. Strongest conditional closure theorem

Collect the finite microscopic inputs into the list:

1. **Class II / admissible carrier realization:** the actual microscopic tensor space contains a nontrivial symmetry-allowed Class II sector or equivalent admissible witness carrier;
2. **Weyl witness nonvanishing:**
$$W_\parallel^\pm \neq 0, \qquad (W_I^\pm, W_J^\pm) \neq (0,0);$$
3. **opposite-valley matching:**
$$(W_\parallel^-, W_I^-, W_J^-) = (W_\parallel^+, W_I^+, W_J^+);$$
4. **remote spectral gap:**
$$\Delta > 0;$$
5. **zero constant mixing:**
$$V(0) = 0, \qquad \|V(p)\| \leq c|p|;$$
6. **mass protection:** no symmetry-allowed constant Dirac mass exists in the actual residual microscopic algebra;
7. **gauge-consistent projected sector:**
$$p^\mu \Gamma^{(2)}_{\mu\nu} = 0, \qquad m^2_{A,k} \to 0, \qquad \text{gauge-preserving regulator}, \qquad \text{Class I/II only};$$
8. **IR-isotropy sign:**
$$\mathrm{spec}(M_Z|_{\mathcal{U}_{\mathrm{tr}}}) \subset (-\infty, 0).$$

Then the strongest currently honest conditional theorem is:

[Strongest honest conditional closure theorem for TECT] Assume items (1)–(8) above. Then the final microscopic TECT realization contains:

1. an actual pair of opposite local Weyl valleys at $\pm G_*$,
2. an actual local doubled Dirac block,
3. a spectrally isolated massless local Dirac cone,

4. an infrared soft sector whose sectoral kinetic anisotropies decay,

$$\delta_T, \delta_O, \delta_D \to 0,$$

so that the low-energy theory is velocity-locked and infrared-isotropic.

[Proof sketch] Items (2) and (3) imply actual Weyl valleys at $\pm G_*$ by the witness criterion. Item (4) implies these valleys pair into a local Dirac block. Items (5) and (6) imply the remote sector alters the local Dirac block only by higher-order terms and cannot generate a constant Dirac mass. Hence the microscopic theory contains a stable spectrally isolated massless local Dirac cone.

On the infrared bosonic side, item (7) ensures the A-sector contribution is physically meaningful. Item (8) implies that the physical derivative-anisotropy variables on the traceless sector decay under the RG flow. Combined with quartic anisotropy irrelevance (703), this yields infrared isotropy and velocity locking.

D. What is genuinely proved versus not yet proved

Therefore the exact logical status is:

l. What is already proved.

1. the theorem-level reduction of the Weyl/Dirac problem to finite witness/gap/mixing/mass conditions;
2. the theorem-level reduction of the isotropy problem to the sign of M_Z;
3. quartic anisotropy irrelevance for $N_{\text{eff}} = 6$;
4. the gauge-consistency preconditions for a physical transverse A-sector.

m. What is not yet proved.

1. that the final microscopic TECT action actually realizes the required witness carrier sector;
2. that the actual microscopic coefficients satisfy the witness conditions;
3. that the actual projected spectrum has the required positive gap Δ;
4. that the residual microscopic symmetry forbids constant mass terms;
5. that the physical eigenvalues of M_Z are in fact negative.

E. Honest final closure statement

Thus the strongest possible honest closure statement at the present stage is:

$$\boxed{\text{TECT is not yet fully unconditionally closed microscopically,}} \tag{704}$$

but

$$\boxed{\text{its remaining frontier has been reduced to a finite conditional verification theorem.}} \tag{705}$$

Equivalently:

no conceptual gap remains at the roadmap level,

and the only remaining work is the explicit microscopic verification of the finite inputs.

. Summary

$$\boxed{\textbf{Strongest honest closure available now}}$$

Already closed at theorem level:

$$\boxed{\text{(i) Weyl/Dirac reduction } \Longrightarrow \text{ finite witness/gap/mass tests,}}$$

$$\boxed{\text{(ii) quartic anisotropy irrelevance for } N_{\text{eff}} = 6,\ \theta_\tau = -\epsilon/7 < 0,}$$

$$\boxed{\text{(iii) gauge-consistency conditions for the physical transverse sector.}}$$

Still not yet verified microscopically:

$$\boxed{\text{actual witness nonvanishing, opposite-valley matching, remote gap } \Delta > 0,}$$

$$\boxed{\text{zero constant mixing, mass protection, and } \operatorname{spec}(M_Z|_{\mathcal{U}_{\text{tr}}}) < 0.}$$

Hence the strongest current conditional theorem is:

$$\boxed{\begin{array}{l}\text{If the final microscopic TECT realization satisfies:}\\ \quad \text{(1) actual admissible witness carrier sector,}\\ \quad \text{(2) actual Weyl witness nonvanishing at both valleys,}\\ \quad \text{(3) opposite-valley witness matching,}\\ \quad \text{(4) remote gap } \Delta > 0,\\ \quad \text{(5) zero constant mixing,}\\ \quad \text{(6) mass protection,}\\ \quad \text{(7) gauge-consistent projected } A\text{-sector,}\\ \quad \text{(8) } \operatorname{spec}(M_Z|_{\mathcal{U}_{\text{tr}}}) \subset (-\infty, 0),\end{array}}$$

then

$$\boxed{\text{TECT contains an actual stable massless local Dirac cone}}$$

and

$$\boxed{\delta_T, \delta_O, \delta_D \to 0,}$$

so the infrared soft sector is velocity-locked and isotropic.

Therefore the current exact status is:

$$\boxed{\text{not fully unconditionally closed yet,}}$$

but

$$\boxed{\text{fully reduced to a finite conditional verification theorem.}}$$

DIRAC STEP 1: EXPLICIT WITNESS COMPUTATION INSIDE THE CONCRETE CLASS II ANSATZ

1. Goal

The local shell-reduced Weyl/Dirac problem is already reduced to the explicit witness triple

$$(W_{\parallel}, W_I, W_J),$$

or equivalently to the reduced mixed symbol coefficients

$$(\lambda_{\parallel}, \alpha, \beta).$$

For the natural local $S2$ branch, the mixed symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1\sigma_1 + p_2\sigma_2) + \beta(-p_2\sigma_1 + p_1\sigma_2). \tag{706}$$

The exact local nondegeneracy condition is

$$\boxed{\lambda_{\parallel} \neq 0, \qquad (\alpha, \beta) \neq (0,0).} \tag{707}$$

This is explicitly stated in the local CP^2-defect transport notes. :contentReference[oaicite:4]index=4 :contentReference[oaicite:5]index=5

2. Concrete Class II transport ansatz

The concrete minimal microscopic transport action is

$$\mathcal{L}_{\text{trans}}^{\min} = -i\,\eta^{\dagger}\Big[g_c\,\theta'(\xi)\,(\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\text{tr}}\,n^{\mu} + g_E\,\theta'(\xi)\,E^{\mu}{}_a\Sigma_a + g_O\,\theta'(\xi)\,J^{\mu}{}_a\Sigma_a\Big]\partial_{y^{\mu}}\eta. \tag{708}$$

Projecting this transport operator onto the graded low-mode sector gives the explicit coefficients

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512}\,g_c\,\theta_0^3\kappa^3, \tag{709}$$

$$\alpha = -\frac{\sqrt{3}\pi}{16}\,g_E\,\theta_0\kappa, \qquad \beta = -\frac{\sqrt{3}\pi}{16}\,g_O\,\theta_0\kappa. \tag{710}$$

Equivalently, in the notation of the same derivation,

$$\lambda_{\parallel} = \frac{g_c}{2} I_{\theta'^3}, \qquad \alpha = g_E I_{\theta}, \qquad \beta = g_O I_{\theta},$$

with the explicit defect integrals already computed. :contentReference[oaicite:6]index=6

Therefore

$$\boxed{g_c \neq 0 \Longleftrightarrow \lambda_{\parallel} \neq 0, \qquad (g_E, g_O) \neq (0,0) \Longleftrightarrow (\alpha, \beta) \neq (0,0).} \tag{711}$$

So Step 1 reduces to the finite coefficient extraction problem for

$$g_c, \qquad g_E, \qquad g_O.$$

3. Minimal Class II seed parameterization

Now choose the shell-adapted Class II microscopic tensor basis in the axial frame $(\hat{n}, e_1, e_2)$:

$$m^a(P_0) = m\,\delta^{a3}, \tag{712}$$

$$C^{33}_{ij} = c_\parallel\, \hat n_i \hat n_j, \tag{713}$$

$$C^{31}_{ij} = c_I\, \hat n_i e_{1j} + c_J\, \hat n_i e_{2j}, \qquad C^{32}_{ij} = -\, c_J\, \hat n_i e_{1j} + c_I\, \hat n_i e_{2j}. \tag{714}$$

This is exactly the minimal shell-relevant Class II seed family appearing in the Class II completion notes. :contentReference[oaicite:7]index=7

Using the shell-frame decomposition of the induced tensor, the Class II correction has the shell-relevant coefficients

$$\delta A \propto c_\parallel, \qquad \delta B \propto c_I, \qquad \delta C \propto c_J,$$

and therefore the corresponding witness shifts satisfy

$$g_c \propto c_\parallel, \qquad g_E \propto c_I, \qquad g_O \propto c_J. \tag{715}$$

Equivalently, the induced witness corrections are

$$\delta W_\parallel \propto A, \qquad \delta W_I \propto B, \qquad \delta W_J \propto C.$$

:contentReference[oaicite:8]index=8 :contentReference[oaicite:9]index=9

4. Explicit witness seeds

The seed directions are now immediate.

a. Longitudinal seed. Take

$$c_\parallel \neq 0, \qquad c_I = c_J = 0.$$

Then

$$g_c \neq 0, \qquad g_E = g_O = 0, \tag{716}$$

hence

$$\lambda_\parallel \neq 0.$$

b. Even transverse seed. Take

$$c_\parallel = 0, \qquad c_I \neq 0, \qquad c_J = 0.$$

Then

$$g_E \neq 0, \qquad g_c = g_O = 0, \tag{717}$$

hence

$$\alpha \neq 0.$$

c. Odd transverse seed. Take

$$c_\parallel = 0, \qquad c_I = 0, \qquad c_J \neq 0.$$

Then

$$g_O \neq 0, \qquad g_c = g_E = 0, \tag{718}$$

hence

$$\beta \neq 0.$$

Therefore the chosen Class II parameterization already contains:

1. an explicit longitudinal witness seed;
2. an explicit transverse witness seed.

This is exactly the finite-witness closure theorem proved in the notes. :contentReference[oaicite:10]index=10

5. Determinant and local Weyl criterion

For the $S2$ symbol (706),

$$\det A_1(p) = \lambda_{\|} p_{\|} \Big(\alpha^2 (p_1^2 + p_2^2) + \beta^2 (p_1^2 + p_2^2) \Big)$$

up to the standard shell normalization factor. Equivalently, in the shell witness notation,

$$\det M = (\kappa_* |G_*|)^3 W_{\|} \big(W_I^2 + W_J^2 \big). \tag{719}$$

Hence local Weyl nondegeneracy is exactly

$$\boxed{W_{\|} \neq 0, \qquad (W_I, W_J) \neq (0,0),} \tag{720}$$

or equivalently

$$\boxed{g_c \neq 0, \qquad (g_E, g_O) \neq (0,0).} \tag{721}$$

:contentReference[oaicite:11]index=11

6. What is actually closed in Step 1

At this point, the following is rigorously closed inside the Class II truncation:

1. the local $S2$ symbol is explicitly written;
2. the witness coefficients are explicitly matched to (g_c, g_E, g_O);
3. the minimal Class II tensor family contains explicit longitudinal and transverse witness seeds;
4. therefore the witness problem is generically closed on the allowed finite-dimensional Class II tensor space.

In particular, once one longitudinal seed and one transverse seed exist, finite-dimensional genericity implies:

$$\boxed{g_c(N) \neq 0 \quad \text{and} \quad (g_E(N), g_O(N)) \neq (0,0)} \tag{722}$$

for an open dense full-measure subset of the allowed Class II tensor space V_{II}. :contentReference[oaicite:12]index=12

7. Honest remaining open point

What is *not* yet closed is stronger than this Step 1 result:

$$\boxed{\text{the true complete TECT action must still be shown to realize a nontrivial Class II sector in the physically relevant defect backg}} \tag{723}$$

So the exact honest conclusion is:

$$\boxed{\text{inside the concrete Class II truncation, Step 1 is closed;}} \tag{724}$$

but

$$\boxed{\text{the full microscopic TECT realization of that Class II sector remains to be verified.}} \tag{725}$$

This is stated explicitly in the notes. :contentReference[oaicite:13]index=13 :contentReference[oaicite:14]index=14

. Summary

$$\boxed{A_1(p) = \lambda_\parallel p_\parallel \sigma_3 + \alpha(p_1\sigma_1 + p_2\sigma_2) + \beta(-p_2\sigma_1 + p_1\sigma_2).}$$

For the concrete Class II transport ansatz,

$$\boxed{\lambda_\parallel = -\frac{3\sqrt{3}\pi}{512}\, g_c\, \theta_0^3 \kappa^3, \qquad \alpha = -\frac{\sqrt{3}\pi}{16}\, g_E\, \theta_0 \kappa, \qquad \beta = -\frac{\sqrt{3}\pi}{16}\, g_O\, \theta_0 \kappa.}$$

Therefore the exact local nondegeneracy condition is

$$\boxed{g_c \neq 0, \qquad (g_E, g_O) \neq (0,0).}$$

For the minimal shell-adapted Class II seed family

$$m^a(P_0) = m\delta^{a3}, \qquad C^{33}_{ij} = c_\parallel \hat{n}_i \hat{n}_j,$$

$$C^{31}_{ij} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \qquad C^{32}_{ij} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j},$$

one has

$$\boxed{g_c \propto c_\parallel, \qquad g_E \propto c_I, \qquad g_O \propto c_J.}$$

Hence explicit witness seeds already exist:

$$c_\parallel \neq 0,\ c_I = c_J = 0 \Longrightarrow g_c \neq 0,$$

$$c_\parallel = 0,\ c_I \neq 0,\ c_J = 0 \Longrightarrow g_E \neq 0,$$

$$c_\parallel = 0,\ c_I = 0,\ c_J \neq 0 \Longrightarrow g_O \neq 0.$$

Thus:

$$\boxed{\text{inside the concrete Class II truncation, the Step 1 witness problem is closed.}}$$

What remains open is only:

$$\boxed{\text{whether the full complete TECT action actually realizes a nontrivial Class II sector in the relevant defect background.}}$$

DIRAC STEP 2: OPPOSITE-VALLEY WITNESS MATCHING IN THE CONCRETE CLASS II ANSATZ

1. Goal

The local shell-reduced Dirac criterion is not merely:

$$\text{Weyl at } +G_*, \qquad \text{Weyl at } -G_*.$$

One also needs opposite-valley pairing in the matched frame, equivalently

$$\boxed{(W^-_\parallel, W^-_I, W^-_J) = (W^+_\parallel, W^+_I, W^+_J).} \tag{726}$$

This is exactly the finite witness formulation of local Dirac doubling. :contentReference[oaicite:8]index=8 :contentReference[oaicite:9]index=9

2. Positive-valley witness data

At the positive valley $+G_*$, with adapted frame

$$\{\hat{n}^+, e_1^+, e_2^+\}, \qquad \hat{n}^+ = \hat{n} = \frac{G_*}{|G_*|},$$

the reduced shell-enriched operator is

$$\delta L^{(1)}_{\rm enr} = g_\parallel O_\parallel + g_I O_I + g_J O_J,$$

with

$$O_\parallel = p_\parallel \sigma_3, \qquad O_I = p_1\sigma_1 + p_2\sigma_2, \qquad O_J = -p_2\sigma_1 + p_1\sigma_2.$$

The corresponding witness functionals satisfy

$$W_\parallel^+ = g_\parallel^+, \qquad W_I^+ = g_I^+, \qquad W_J^+ = g_J^+. \tag{727}$$

:contentReference[oaicite:10]index=10

Inside the concrete Class II transport ansatz, these coefficients are read as

$$g_\parallel^+ \leftrightarrow g_c^+, \qquad g_I^+ \leftrightarrow g_E^+, \qquad g_J^+ \leftrightarrow g_O^+.$$

Thus Step 1 already gives

$$W_\parallel^+ = g_c^+, \qquad W_I^+ = g_E^+, \qquad W_J^+ = g_O^+.$$

3. Negative-valley frame and matching convention

At the opposite valley $-G_*$, one does *not* compare coefficients in an arbitrary raw coordinate frame. One first chooses the matched opposite-valley frame

$$\{\hat{n}^-, e_1^-, e_2^-\}$$

so that the opposite-valley Hamiltonian is compared in the standard doubled form

$$h^-(p) = -h^+(-p)$$

to leading order. This is the exact witness formulation of opposite-valley pairing in the notes. :contentReference[oaicite:11]index=11 :contentReference[oaicite:12]index=12

Therefore the correct matching problem is not "do the raw coefficients literally coincide in the same laboratory basis," but rather:

$$\boxed{\text{after the matched opposite-valley frame choice, do the two witness triples coincide?}}$$

4. Class II tensor data at the two valleys

Inside the concrete minimal Class II family,

$$m^a(P_0) = m\,\delta^{a3}, \qquad C^{33}_{ij} = c_\parallel \hat{n}_i\hat{n}_j, \qquad C^{31}_{ij} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \qquad C^{32}_{ij} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}. \tag{728}$$

On this family, the shell/defect dictionary gives

$$g_c \propto c_\parallel, \qquad g_E \propto c_I, \qquad g_O \propto c_J. \tag{729}$$

:contentReference[oaicite:13]index=13 :contentReference[oaicite:14]index=14

Now pass from $+G_*$ to $-G_*$. Then

$$\hat{n}^- = -\hat{n}^+.$$

If one simultaneously uses the matched transverse basis

$$e_1^- = -e_1^+, \qquad e_2^- = e_2^+$$

or any equivalent right-handed matched choice, then the shell-relevant tensor structures are read in the same oriented local coordinates. Under this matched-frame prescription, the coefficients extracted by the witness functionals are the same scalar amplitudes:

$$c_\parallel^- = c_\parallel^+, \qquad c_I^- = c_I^+, \qquad c_J^- = c_J^+. \tag{730}$$

Hence

$$g_c^- = g_c^+, \qquad g_E^- = g_E^+, \qquad g_O^- = g_O^+. \tag{731}$$

5. Witness matching

By the witness dictionary,

$$W_\parallel^\pm = g_c^\pm, \qquad W_I^\pm = g_E^\pm, \qquad W_J^\pm = g_O^\pm.$$

Therefore (731) implies

$$\boxed{W_\parallel^- = W_\parallel^+, \qquad W_I^- = W_I^+, \qquad W_J^- = W_J^+.} \tag{732}$$

This is exactly the opposite-valley witness matching criterion required for local Dirac doubling. :contentReference[oaicite:15]index=15

6. Consequence

Combining Step 1 and Step 2:

1. Step 1 gives local Weyl nondegeneracy at one valley:

$$g_c \neq 0, \qquad (g_E, g_O) \neq (0,0).$$

2. The same concrete Class II seed family exists at the opposite valley.
3. In the matched opposite-valley frame, the witness triples coincide:

$$(W_\parallel^-, W_I^-, W_J^-) = (W_\parallel^+, W_I^+, W_J^+).$$

Therefore, inside the concrete Class II truncation,

$$\boxed{\text{opposite-valley pairing is closed at the witness level.}} \tag{733}$$

7. Honest remaining open point

What remains open is stronger than the statement above:

$$\boxed{\text{the full microscopic TECT action must still be shown to realize the required opposite shell pair } \pm G_* \text{ as actual low-energy vall}} \tag{734}$$

This is explicitly listed in the notes as one of the remaining microscopic tasks. :contentReference[oaicite:16]index=16 :contentReference[oaicite:17]index=17

So the exact honest conclusion is:

$$\boxed{\text{inside the concrete Class II witness truncation, Step 2 is closed;}}$$

but

$$\boxed{\text{the full microscopic existence of the opposite shell-pair valleys remains to be verified.}}$$

. Summary

$$\boxed{\textbf{Dirac Step 2: opposite-valley witness matching}}$$

Local Dirac doubling requires

$$\boxed{(W_{\parallel}^{-}, W_{I}^{-}, W_{J}^{-}) = (W_{\parallel}^{+}, W_{I}^{+}, W_{J}^{+}).}$$

Inside the concrete Class II ansatz,

$$\boxed{g_c \propto c_{\parallel}, \qquad g_E \propto c_I, \qquad g_O \propto c_J,}$$

and the witness dictionary is

$$\boxed{W_{\parallel} = g_c, \qquad W_I = g_E, \qquad W_J = g_O.}$$

Using the matched opposite-valley frame, the same seed amplitudes are read at the two valleys:

$$\boxed{c_{\parallel}^{-} = c_{\parallel}^{+}, \qquad c_{I}^{-} = c_{I}^{+}, \qquad c_{J}^{-} = c_{J}^{+},}$$

hence

$$\boxed{g_c^{-} = g_c^{+}, \qquad g_E^{-} = g_E^{+}, \qquad g_O^{-} = g_O^{+}.}$$

Therefore

$$\boxed{W_{\parallel}^{-} = W_{\parallel}^{+}, \qquad W_{I}^{-} = W_{I}^{+}, \qquad W_{J}^{-} = W_{J}^{+}.}$$

So:

$$\boxed{\text{inside the concrete Class II witness truncation, opposite-valley pairing is closed at the witness level.}}$$

What remains open is only:

$$\boxed{\text{whether the full microscopic TECT action actually realizes the shell pair } \pm G_* \text{ as genuine low-energy valleys.}}$$

DIRAC STEP 3: ZERO CONSTANT MIXING $V(0) = 0$ IN THE CONCRETE CLASS II SETTING

1. Goal

After Step 1 (actual Weyl witness inside the Class II truncation) and Step 2 (opposite-valley witness matching), the next stability condition for the local Dirac block is

$$\boxed{V(0) = 0.} \tag{735}$$

Here $V(p)$ is the off-diagonal mixing block between the local doubled Dirac sector and the remote sector:

$$H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^{\dagger} & H_R(p) \end{pmatrix}. \tag{736}$$

The symbol $V(0) = 0$ means absence of *constant hybridization* at the crossing, not vanishing vacuum energy. :contentReference[oaicite:5]index=5 :contentReference[oaicite:6]index=6

2. The theorem-level zero-mixing statement

Let H_* be the little group at the selected shell crossing $p = 0$, and assume the full Hilbert space decomposes as

$$\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R, \tag{737}$$

where

- $\mathcal{H}_D$ is the isolated doubled shell block carrying the local Dirac sector,
- $\mathcal{H}_R$ is the orthogonal remote sector.

Assume further that the decomposition is H_*-invariant and that the microscopic Hamiltonian is H_*-equivariant at $p = 0$. Then the constant mixing block obeys

$$V(0) \in \mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D). \tag{738}$$

Hence

$$\boxed{\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \Longrightarrow V(0) = 0.} \tag{739}$$

This is already theorem-level closed. :contentReference[oaicite:7]index=7 :contentReference[oaicite:8]index=8

3. Practical sectorwise audit

The theorem is turned into a finite practical audit by decomposing the remote sector into H_*-invariant pieces

$$\mathcal{H}_R = \bigoplus_a K_a. \tag{740}$$

If every sector K_a differs from $\mathcal{H}_D$ in at least one shell-relevant label, then

$$\mathrm{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0 \qquad \forall a, \tag{741}$$

hence

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \qquad \Longrightarrow \qquad V(0) = 0. \tag{742}$$

The practical checklist given in the notes is:
a remote sector is excluded from constant mixing if it differs from the local Dirac block in at least one of

1. the selected shell-pair label $\pm G_*$,
2. transverse versus longitudinal/volumetric character,
3. presence of the enriched shell-relevant internal doublet,
4. residual grading / parity / chirality,
5. any extra conserved microscopic label.

:contentReference[oaicite:9]index=9 :contentReference[oaicite:10]index=10

4. Minimal truncated remote sector

The smallest dangerous remote completion used in the notes is

$$\mathcal{H}^{\text{tr}}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}}. \tag{743}$$

Here:

- $\mathcal{H}_D \cong \mathbb{C}^2_{\text{pair}} \otimes \mathbb{C}^2_{\text{int}}$ is the local doubled Dirac carrier,

- $\mathcal{H}_{\rm vol}$ is the volumetric/longitudinal sector,
- $\mathcal{H}_{\rm high}$ is the higher-shell sector not belonging to the selected shell pair $\pm G_*$,
- $\mathcal{H}_{\rm non\text{-}enr}$ is the same-shell but non-enriched sector,
- $\mathcal{H}_{\rm wrong\text{-}int}$ contains modes with internal content not matching the selected enriched shell doublet.

This is explicitly introduced as the smallest truncation containing all previously identified dangerous remote channels. :contentReference[oaicite:11]index=11 :contentReference[oaicite:12]index=12

5. Why the concrete Class II ansatz is compatible with $V(0) = 0$

The canonical minimal Class II representative is

$$m^a(P_0) = m\delta^{a3}, \qquad C^{33}_{ij} = c_\parallel \hat{n}_i\hat{n}_j, \qquad C^{31}_{ij} = c_I\hat{n}_i e_{1j} + c_J\hat{n}_i e_{2j}, \qquad C^{32}_{ij} = -c_J\hat{n}_i e_{1j} + c_I\hat{n}_i e_{2j}. \tag{744}$$

Its effect is to renormalize the shell-level first-order coefficients

$$A_{\rm eff}, \qquad B_{\rm eff}, \qquad C_{\rm eff},$$

or equivalently the Weyl witness triple. It does *not* by construction insert an independent constant 4×4 Dirac mass term, and it does not by itself define a constant Dirac–remote intertwiner. Therefore the exact logical status is:

$$\boxed{\text{Class II is compatible with } V(0) = 0, \text{ but does not itself prove } V(0) = 0.} \tag{745}$$

This is explicitly stated in the compatibility theorem for the canonical minimal Class II choice. :contentReference[oaicite:13]index=13 :contentReference[oaicite:14]index=14

6. Concrete closure of Step 3

What can now be honestly closed is the following theorem.
[Step 3 closure in the concrete Class II setting] Assume:

1. the concrete Class II witness truncation of Steps 1–2 is realized on the local doubled shell block $\mathcal{H}_D$;
2. the full microscopic crossing is H_*-equivariant at $p = 0$;
3. the remote sector admits an H_*-invariant decomposition
$$\mathcal{H}_R = \mathcal{H}_{\rm vol} \oplus \mathcal{H}_{\rm high} \oplus \mathcal{H}_{\rm non\text{-}enr} \oplus \mathcal{H}_{\rm wrong\text{-}int};$$
4. each remote summand differs from $\mathcal{H}_D$ in at least one shell-relevant label.

Then

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0, \qquad V(0) = 0. \tag{746}$$

By H_*-equivariance and invariant block decomposition,

$$V(0) \in \mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D).$$

Because

$$\mathcal{H}_R = \bigoplus_a K_a,$$

one has

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) \cong \bigoplus_a \mathrm{Hom}_{H_*}(K_a, \mathcal{H}_D).$$

By assumption, every K_a mismatches $\mathcal{H}_D$ in at least one shell-relevant label, hence

$$\mathrm{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0 \qquad \forall a.$$

Therefore

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0,$$

and thus $V(0) = 0$.

7. Honest remaining open point

The theorem above closes Step 3 only *conditionally on the actual remote-sector label audit.* Therefore the exact honest status is:

$$\boxed{\text{Step 3 is reduced completely to a finite remote-sector irrep/label audit.}} \tag{747}$$

What is still not yet fully closed microscopically is:

$$\boxed{\text{the actual final TECT truncation must still be audited to prove that no remote sector shares the same shell-relevant irrep as } \mathcal{H}} \tag{748}$$

This is exactly how the notes summarize the present status of Task V. :contentReference[oaicite:15]index=15 :contentReference[oaicite:16]index=16 :contentReference[oaicite:17]index=17

8. Strongest honest conclusion of Step 3

Thus the strongest currently honest conclusion is:

$$\boxed{\text{inside the concrete Class II ansatz, } V(0) = 0 \text{ is not automatic,}} \tag{749}$$

but

$$\boxed{\text{its verification is fully reduced to a finite little-group mismatch audit of the remote sector.}} \tag{750}$$

So Step 3 is closed at the reduction-theorem level.

. Summary

$$\boxed{H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^\dagger & H_R(p) \end{pmatrix}, \qquad V(0) \in \mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D).}$$

Hence the theorem-level zero-mixing statement is

$$\boxed{\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \Longrightarrow V(0) = 0.}$$

The practical finite audit is: a remote sector is excluded from constant mixing if it differs from the local Dirac block in at least one of

$$\boxed{\text{shell-pair label,} \quad \text{transverse vs volumetric character,} \quad \text{presence of enriched internal doublet,} \quad \text{grading/parity/chirality,} \quad \text{extra}}$$

In the minimal dangerous truncation,

$$\boxed{\mathcal{H}_{\text{full}}^{\text{tr}} = \mathcal{H}_D \oplus \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}}.}$$

The canonical minimal Class II representative

$$\boxed{m^a(P_0) = m\delta^{a3}, \; C_{ij}^{33} = c_\parallel \hat{n}_i \hat{n}_j, \; C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \; C_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}}$$

renormalizes the shell-level derivative witness coefficients but does not by itself prove $V(0) = 0$.
Therefore:

$$\boxed{\text{Class II is compatible with } V(0) = 0, \text{ but } V(0) = 0 \text{ still requires an independent little-group mismatch audit.}}$$

So the strongest honest Step 3 conclusion is:

$$\boxed{\text{zero constant mixing is fully reduced to a finite remote-sector irrep/label audit.}}$$

DIRAC STEP 4: REMOTE SPECTRAL GAP $\Delta > 0$ IN THE CONCRETE CLASS II SETTING

1. Goal

After Step 3, the next stability condition for the local doubled Dirac block is

$$\boxed{\mathrm{spec}(H_R(p)) \cap (-\Delta, \Delta) = \varnothing \qquad \text{for small } p, \qquad \Delta > 0.} \tag{751}$$

This means that the remote sector remains spectrally separated from the local Dirac crossing.

The notes are explicit that the remote-gap theorem itself is already closed, but the actual value of the gap must still be checked in the final microscopic truncation. :contentReference[oaicite:8]index=8 :contentReference[oaicite:9]index=9

2. Pointwise and uniform spectral lower-bound theorem

Let the remote block be written as

$$H_R(p) = D_R(p) + E_R(p),$$

where:

- $D_R(p)$ is the remote diagonal / benchmark part,
- $E_R(p)$ is the off-diagonal / correction part.

Define

$$\delta_0(p) := \min_a |(D_R)_{aa}(p)|, \qquad \eta_R(p) := \|E_R(p)\|. \tag{752}$$

Then the pointwise remote-gap theorem states:

$$\boxed{\delta_0(p) > \eta_R(p) \Longrightarrow \mathrm{spec}(H_R(p)) \cap (-\Delta(p), \Delta(p)) = \varnothing, \qquad \Delta(p) \geq \delta_0(p) - \eta_R(p) > 0.} \tag{753}$$

:contentReference[oaicite:10]index=10

Fix a neighborhood

$$N_\rho := \{p : |p| \leq \rho\},$$

and define

$$\delta_{0,\rho} := \inf_{|p|\leq\rho} \delta_0(p), \qquad \eta_{R,\rho} := \sup_{|p|\leq\rho} \eta_R(p). \tag{754}$$

Then the uniform theorem is

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho} \Longrightarrow \mathrm{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \varnothing \quad \forall |p| \leq \rho,} \tag{755}$$

with

$$\boxed{\Delta_\rho := \delta_{0,\rho} - \eta_{R,\rho} > 0.} \tag{756}$$

:contentReference[oaicite:11]index=11

Thus the remote-gap problem is already reduced to a finite Bloch-spectral inequality.

3. Perturbative gap-stability theorem

Assume there exists an unperturbed microscopic realization

$$H_R^{(0)}(p) \quad \text{with} \quad \operatorname{spec}(H_R^{(0)}(p)) \cap (-\Delta_0, \Delta_0) = \varnothing \qquad (\Delta_0 > 0), \tag{757}$$

for all sufficiently small p. Now perturb by

$$H_R(p) = H_R^{(0)}(p) + \delta H_R(p), \qquad \sup_{|p| \le p_*} \|\delta H_R(p)\| \le \varepsilon. \tag{758}$$

Then the bounded-perturbation remote-gap theorem gives

$$\boxed{\varepsilon < \Delta_0 \Longrightarrow \operatorname{spec}(H_R(p)) \cap (-\Delta, \Delta) = \varnothing, \qquad \Delta := \Delta_0 - \varepsilon > 0.} \tag{759}$$

Moreover, under the stronger bound

$$\varepsilon < \frac{\Delta_0}{2},$$

one gets

$$\Delta > \frac{\Delta_0}{2} > 0.$$

:contentReference[oaicite:12]index=12 :contentReference[oaicite:13]index=13

This is precisely the theorem-level meaning of the statement that a sufficiently small Class II correction is gap-compatible if the starting realization is already gapped.

4. Resolvent bound and Schur-complement consequence

If $|E| < \Delta/2$, then

$$\|(H_R(p) - E)^{-1}\| \le \frac{2}{\Delta}. \tag{760}$$

Therefore, if in addition

$$V(0) = 0, \qquad \|V(p)\| \le c|p|,$$

the Schur complement obeys

$$\|\Sigma(E, p)\| \le \frac{2c^2}{\Delta}|p|^2, \tag{761}$$

so the remote sector modifies the local Dirac Hamiltonian only at order

$$O(|p|^2/\Delta).$$

This is the exact theorem-level mechanism by which the remote gap enters spectral isolation. :contentReference[oaicite:14]index=14 :contentReference[oaicite:15]index=15

5. Concrete Class II compatibility statement

The canonical minimal Class II representative modifies only the shell-relevant derivative tensor coefficients:

$$A_{\text{eff}} = A_0 - \mu c_{\|}, \qquad B_{\text{eff}} = B_0 - \mu c_I, \qquad C_{\text{eff}} = C_0 - \mu c_J,$$

with

$$\mu = \frac{\lambda g_X}{M_X^2} m.$$

It does not by itself prove a remote gap. The exact honest theorem in the Class II notes is:

$$\boxed{\text{Class II is compatible with } \Delta > 0, \text{ but } \Delta > 0 \text{ still requires an independent microscopic gap check,}} \tag{762}$$

or, alternatively, perturbative stability from an already-gapped realization:

$$\boxed{\|\delta H_{\text{Class II}}\| < \frac{\Delta_0}{2} \Longrightarrow \Delta \geq \Delta_0 - \|\delta H_{\text{Class II}}\| > \frac{\Delta_0}{2} > 0.} \tag{763}$$

:contentReference[oaicite:16]index=16 :contentReference[oaicite:17]index=17

6. Strongest honest closure of Step 4

Thus the strongest theorem-level closure available now is the following.
[Step 4 closure in the concrete Class II setting] Assume:

1. the final microscopic TECT truncation admits a remote block decomposition $H_R(p)$;
2. either:
   (a) the direct Bloch-spectral inequality holds on a small neighborhood N_ρ,
   $$\delta_{0,\rho} > \eta_{R,\rho},$$
   or
   (b) there exists an already-gapped unperturbed realization with gap $\Delta_0 > 0$, and the Class II correction obeys
   $$\sup_{|p| \leq p_*} \|\delta H_R(p)\| < \Delta_0;$$
3. the local Dirac block satisfies Step 3:
   $$V(0) = 0, \qquad \|V(p)\| \leq c|p|.$$

Then the remote sector has a positive gap

$$\Delta > 0,$$

and the Schur-complement correction to the local Dirac block is only

$$O(|p|^2/\Delta).$$

In case (2a), apply the uniform spectral lower-bound theorem (755)–(756). In case (2b), apply the bounded perturbation theorem (759). Then use the resolvent estimate (760) together with $\|V(p)\| \leq c|p|$ to obtain (761). This proves that the remote sector remains spectrally separated and contributes only subleading curvature corrections.

7. Honest remaining open point

What remains open is not the theorem but the actual microscopic input:

$$\boxed{\text{the actual numerical or analytic verification of } \delta_{0,\rho} > \eta_{R,\rho} \text{ (or of a concrete already-gapped realization with small perturbation)}} \tag{764}$$

This is explicitly listed in the notes as one of the remaining finite microscopic tasks. :contentReference[oaicite:18]index=18 :contentReference[oaicite:19]index=19

8. Strongest honest conclusion of Step 4

Therefore the strongest honest Step 4 conclusion is:

$$\boxed{\text{the remote-gap problem is completely reduced to a finite Bloch-spectral inequality (or bounded-perturbation gap audit);}} \tag{765}$$

and

$$\boxed{\text{the concrete Class II ansatz is gap-compatible, but does not by itself prove } \Delta > 0.} \tag{766}$$

So Step 4 is closed at the reduction-theorem level.

. Summary

$$\boxed{\textbf{Dirac Step 4: remote gap } \Delta > 0}$$

The theorem-level remote-gap criterion is:

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho} \Longrightarrow \operatorname{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \varnothing \quad (|p| \le \rho),}$$

with

$$\boxed{\Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0.}$$

An equivalent perturbative stability route is:

$$\boxed{\operatorname{spec}(H_R^{(0)}(p)) \cap (-\Delta_0, \Delta_0) = \varnothing, \qquad \sup_{|p| \le p_*} \|\delta H_R(p)\| < \Delta_0}$$

implies

$$\boxed{\Delta = \Delta_0 - \sup \|\delta H_R\| > 0.}$$

Then, if

$$\boxed{V(0) = 0, \qquad \|V(p)\| \le c|p|,}$$

the resolvent obeys

$$\boxed{\|(H_R(p) - E)^{-1}\| \le \frac{2}{\Delta} \qquad (|E| < \Delta/2),}$$

and the Schur-complement correction is only

$$\boxed{\|\Sigma(E,p)\| \le \frac{2c^2}{\Delta}|p|^2.}$$

For the concrete Class II ansatz:

$$\boxed{\text{Class II is compatible with } \Delta > 0,}$$

but

$$\boxed{\Delta > 0 \text{ still requires an independent microscopic gap check,}}$$

or a sufficiently small bounded perturbation of an already-gapped realization.
So the strongest honest Step 4 conclusion is:

$$\boxed{\text{the remote-gap problem is fully reduced to a finite Bloch-spectral inequality / bounded-perturbation audit.}}$$

DIRAC STEP 5: MASS PROTECTION IN THE CONCRETE CLASS II SETTING

1. Goal

After:

- Step 1: actual local Weyl witness,
- Step 2: opposite-valley witness matching,
- Step 3: zero constant mixing $V(0) = 0$,
- Step 4: remote spectral gap $\Delta > 0$,

the final local Dirac-stability condition is:

$$\boxed{\text{no symmetry-allowed constant Dirac mass exists on the isolated doubled shell block.}} \tag{767}$$

The notes are explicit that this question is no longer an infinite-dimensional PDE problem. It is exactly a finite algebraic classification problem on 4×4 Hermitian matrices. :contentReference[oaicite:11]index=11

2. Minimal doubled Dirac block and mass space

Assume the matched local doubled Dirac block has the form

$$H_D(q) = \rho_3 \otimes (q_i \sigma_i), \qquad i = 1, 2, 3. \tag{768}$$

The constant Dirac mass space is defined by

$$\mathcal{M} := \{K \in \mathrm{Herm}(4) : \{K, \rho_3 \otimes \sigma_i\} = 0 \ \forall i = 1, 2, 3\}. \tag{769}$$

The exact classification theorem gives

$$\boxed{\mathcal{M} = \mathrm{span}_{\mathbb{R}}\{\rho_1 \otimes \mathbf{1}_2,\ \rho_2 \otimes \mathbf{1}_2\}.} \tag{770}$$

Thus the constant mass space is exactly two-dimensional. :contentReference[oaicite:12]index=12 :contentReference[oaicite:13]index=13

Without any further symmetry, these masses are allowed, and the node is gapped:

$$H_D^{(m)}(q) = \rho_3 \otimes (q_i \sigma_i) + m_1 \rho_1 \otimes \mathbf{1}_2 + m_2 \rho_2 \otimes \mathbf{1}_2,$$

with spectrum

$$E_\pm(q) = \pm\sqrt{|q|^2 + m_1^2 + m_2^2}.$$

Therefore:

$$\boxed{\text{the minimal doubled Dirac algebra alone does not protect the node.}} \tag{771}$$

:contentReference[oaicite:14]index=14

3. Finite mass-protection criterion

Let G_{res} be the actual residual low-energy symmetry group acting on the doubled shell block, and let

$$A_{\mathrm{res}} := \{K \in \mathrm{Herm}(4) : U(g) K U(g)^{-1} = K \ \forall g \in G_{\mathrm{res}}\} \tag{772}$$

be the corresponding allowed constant operator algebra.

Then the exact finite criterion is:

$$\boxed{A_{\rm res} \cap \mathcal{M} = \{0\} \Longrightarrow \text{mass protection.}} \tag{773}$$

Equivalently, one may solve:

$$U(g)KU(g)^{-1} = K, \qquad \{K, \Gamma_i\} = 0,$$

and check whether the only solution is $K = 0$. :contentReference[oaicite:15]index=15

Thus Step 5 is already theorem-level reduced to a finite 4×4 linear algebra problem.

4. Explicit sufficient mechanism: residual valley $U(1)_V$

Define the valley-charge generator

$$Q_V := \rho_3 \otimes \mathbf{1}_2, \qquad U_V(\theta) = e^{i\theta Q_V}. \tag{774}$$

The kinetic Dirac block is invariant because

$$[Q_V, \rho_3 \otimes \sigma_i] = 0.$$

Under this symmetry the two mass generators rotate as

$$U_V(\theta)(\rho_1 \otimes \mathbf{1}_2)U_V(\theta)^{-1} = (\cos 2\theta)\rho_1 \otimes \mathbf{1}_2 + (\sin 2\theta)\rho_2 \otimes \mathbf{1}_2, \tag{775}$$

$$U_V(\theta)(\rho_2 \otimes \mathbf{1}_2)U_V(\theta)^{-1} = -(\sin 2\theta)\rho_1 \otimes \mathbf{1}_2 + (\cos 2\theta)\rho_2 \otimes \mathbf{1}_2. \tag{776}$$

Hence the mass space transforms as a nontrivial doublet and no nonzero mass element is invariant under all θ. Therefore

$$\boxed{A_{U(1)_V} \cap \mathcal{M} = \{0\}.} \tag{777}$$

So exact residual valley $U(1)_V$ is a sufficient mechanism for full constant mass protection. :contentReference[oaicite:16]index=16 :contentReference[oaicite:17]index=17

5. Weaker sufficient mechanism: residual $\mathbb{Z}_2^V$

The full continuous $U(1)_V$ is stronger than necessary. The discrete remnant

$$S_V := \rho_3 \otimes \mathbf{1}_2, \qquad S_V^2 = \mathbf{1}_4 \tag{778}$$

already suffices, because

$$S_V(\rho_1 \otimes \mathbf{1}_2)S_V^{-1} = -(\rho_1 \otimes \mathbf{1}_2), \qquad S_V(\rho_2 \otimes \mathbf{1}_2)S_V^{-1} = -(\rho_2 \otimes \mathbf{1}_2). \tag{779}$$

Thus an exact residual $\mathbb{Z}_2^V$ also forbids all constant Dirac masses:

$$\boxed{A_{\mathbb{Z}_2^V} \cap \mathcal{M} = \{0\}.} \tag{780}$$

:contentReference[oaicite:18]index=18

6. Stability under remote-sector elimination

If the residual valley symmetry is exact in the full microscopic theory and the remote-sector elimination preserves that symmetry, then no symmetry-preserving self-energy correction can generate a constant Dirac mass. In the little-group scalar sector the self-energy at $(\omega, p) = (0, 0)$ has the restricted form

$$\Sigma(0,0) = a_0\, \mathbf{1}_4 + a_3\, \tau_z \otimes \mathbf{1}_2,$$

so no term proportional to

$$\tau_x \otimes \mathbf{1}_2 \quad \text{or} \quad \tau_y \otimes \mathbf{1}_2$$

can be generated. Hence mass protection is radiatively stable, provided the residual valley symmetry is exact. :contentReference[oaicite:19]index=19

7. Compatibility with the concrete Class II ansatz

The canonical minimal Class II representative is

$$m^a(P_0) = m\delta^{a3}, \qquad C^{33}_{ij} = c_{\parallel}\hat{n}_i\hat{n}_j, \qquad C^{31}_{ij} = c_I\hat{n}_i e_{1j} + c_J\hat{n}_i e_{2j}, \qquad C^{32}_{ij} = -c_J\hat{n}_i e_{1j} + c_I\hat{n}_i e_{2j}. \tag{781}$$

Its role is to renormalize the derivative shell tensor coefficients A, B, C, equivalently the first-order local Weyl/Dirac witnesses. It does not by construction insert an independent constant 4×4 Dirac mass operator.

Therefore the exact honest compatibility statement is:

$$\boxed{\text{Class II is mass-protection compatible, but does not by itself prove mass protection.}} \tag{782}$$

The actual proof of mass protection still depends on the residual symmetry algebra of the isolated doubled shell block.

8. Strongest honest closure of Step 5

We can now state the strongest theorem-level closure available in the concrete Class II setting.

[Step 5 closure in the concrete Class II setting] Assume:

1. Steps 1–4 already hold, so that an isolated doubled local Dirac block

   $$H_D(q) = \rho_3 \otimes (q_i\sigma_i) + O(|q|^2/\Delta)$$

   exists;

2. the actual residual low-energy symmetry algebra on the isolated doubled shell block contains an exact valley symmetry $U(1)_V$ generated by $Q_V = \rho_3 \otimes \mathbf{1}_2$, or at least its $\mathbb{Z}_2^V$ remnant generated by $S_V = \rho_3 \otimes \mathbf{1}_2$;

3. remote-sector elimination preserves that residual symmetry.

Then

$$A_{\text{res}} \cap \mathcal{M} = \{0\}, \qquad \mathcal{M} = \operatorname{span}_{\mathbb{R}}\{\rho_1 \otimes \mathbf{1}_2, \rho_2 \otimes \mathbf{1}_2\}, \tag{783}$$

hence no symmetry-allowed constant Dirac mass exists, and the isolated local Dirac cone is mass-protected.

By the explicit mass-space classification, every constant Dirac mass belongs to

$$\mathcal{M} = \operatorname{span}_{\mathbb{R}}\{\rho_1 \otimes \mathbf{1}_2, \rho_2 \otimes \mathbf{1}_2\}.$$

Under exact $U(1)_V$ these two generators rotate as a nontrivial doublet, so none is invariant unless its coefficient vanishes. Hence

$$A_{U(1)_V} \cap \mathcal{M} = \{0\}.$$

The same conclusion follows already from the $\mathbb{Z}_2^V$ remnant, since both mass generators are odd under S_V. Because remote-sector elimination preserves the symmetry, no radiative constant mass term can be generated. Therefore all symmetry-allowed constant Dirac masses are forbidden.

9. Honest remaining open point

What is still open microscopically is:

$$\boxed{\text{whether the actual full TECT/Class II realization indeed preserves exact residual } U(1)_V \text{ (or at least } \mathbb{Z}_2^V\text{).}} \tag{784}$$

The notes are explicit that this is still a separate microscopic check. :contentReference[oaicite:21]index=21 :contentReference[oaicite:22]index=22

10. Strongest honest conclusion of Step 5

Therefore the strongest honest Step 5 conclusion is:

$$\boxed{\text{mass protection is fully reduced to a finite residual-symmetry exclusion test on the isolated doubled shell block;}} \tag{785}$$

more concretely,

$$\boxed{A_{\text{res}} \cap \operatorname{span}_{\mathbb{R}}\{\rho_1 \otimes \mathbf{1}_2, \rho_2 \otimes \mathbf{1}_2\} = \{0\}.} \tag{786}$$

And an exact residual valley $U(1)_V$ or $\mathbb{Z}_2^V$ is an explicit sufficient mechanism. So Step 5 is closed at the reduction-theorem level.

. Summary

$$\boxed{\textbf{Dirac Step 5: mass protection}}$$

For the isolated doubled local Dirac block

$$H_D(q) = \rho_3 \otimes (q_i \sigma_i),$$

the constant Dirac mass space is exactly

$$\boxed{\mathcal{M} = \operatorname{span}_{\mathbb{R}}\{\rho_1 \otimes \mathbf{1}_2,\ \rho_2 \otimes \mathbf{1}_2\}.}$$

Thus the finite mass-protection criterion is

$$\boxed{A_{\text{res}} \cap \mathcal{M} = \{0\} \Longrightarrow \text{mass protection.}}$$

An explicit sufficient mechanism is exact residual valley symmetry

$$\boxed{Q_V = \rho_3 \otimes \mathbf{1}_2, \qquad U_V(\theta) = e^{i\theta Q_V},}$$

because under it

$$\boxed{U_V(\theta)(\rho_1 \otimes \mathbf{1}_2)U_V(\theta)^{-1} = (\cos 2\theta)\rho_1 \otimes \mathbf{1}_2 + (\sin 2\theta)\rho_2 \otimes \mathbf{1}_2,}$$

$$\boxed{U_V(\theta)(\rho_2 \otimes \mathbf{1}_2)U_V(\theta)^{-1} = -(\sin 2\theta)\rho_1 \otimes \mathbf{1}_2 + (\cos 2\theta)\rho_2 \otimes \mathbf{1}_2.}$$

Hence

$$\boxed{A_{U(1)_V} \cap \mathcal{M} = \{0\}.}$$

A weaker but still sufficient mechanism is the residual

$$\boxed{\mathbb{Z}_2^V : \quad S_V = \rho_3 \otimes \mathbf{1}_2,}$$

under which both mass generators are odd, so again

$$\boxed{A_{\mathbb{Z}_2^V} \cap \mathcal{M} = \{0\}.}$$

For the concrete Class II ansatz:

$$\boxed{\text{Class II is mass-protection compatible,}}$$

but

$$\boxed{\text{mass protection itself still requires an independent residual-symmetry check.}}$$

So the strongest honest Step 5 conclusion is:

$$\boxed{\text{mass protection is fully reduced to a finite residual-symmetry exclusion test on } \mathcal{M} = \operatorname{span}_{\mathbb{R}}\{\rho_1 \otimes \mathbf{1}_2, \rho_2 \otimes \mathbf{1}_2\}.}$$

LOCAL DIRAC CLOSURE THEOREM IN THE CONCRETE CLASS II SETTING

A. Setting

We consider the shell-reduced local fermion problem near a selected shell pair

$$\pm G_*,$$

with the concrete Class II microscopic witness truncation acting on the low-energy doubled shell block. The local doubled shell carrier is denoted by

$$\mathcal{H}_D,$$

and the remote sector by

$$\mathcal{H}_R, \qquad \mathcal{H}_{\mathrm{full}} = \mathcal{H}_D \oplus \mathcal{H}_R.$$

The goal is to determine when the full microscopic TECT linearization contains a stable, spectrally isolated, massless local Dirac cone.

B. Step 1: actual local Weyl witness inside the concrete Class II truncation

On the natural local $S2$ branch, the reduced mixed symbol is

$$A_1(p) = \lambda_\parallel p_\parallel \sigma_3 + \alpha(p_1\sigma_1 + p_2\sigma_2) + \beta(-p_2\sigma_1 + p_1\sigma_2), \tag{787}$$

with

$$\lambda_\parallel = -\frac{3\sqrt{3}\pi}{512}\, g_c\, \theta_0^3 \kappa^3, \qquad \alpha = -\frac{\sqrt{3}\pi}{16}\, g_E\, \theta_0 \kappa, \qquad \beta = -\frac{\sqrt{3}\pi}{16}\, g_O\, \theta_0 \kappa. \tag{788}$$

Therefore local Weyl nondegeneracy is exactly

$$\boxed{g_c \neq 0, \qquad (g_E, g_O) \neq (0,0).} \tag{789}$$

Inside the minimal shell-adapted Class II tensor family,

$$g_c \propto c_\parallel, \qquad g_E \propto c_I, \qquad g_O \propto c_J,$$

so explicit longitudinal and transverse witness seeds exist. Hence the witness problem is closed inside the concrete Class II truncation: an open dense full-measure subset of the allowed Class II tensor space produces a nondegenerate local Weyl symbol. :contentReference[oaicite:4]index=4 :contentReference[oaicite:5]index=5 :contentReference[oaicite:6]index=6

C. Step 2: opposite-valley witness matching

Let

$$(W_\parallel^\pm, W_I^\pm, W_J^\pm)$$

denote the witness triples at $\pm G_*$. Local Dirac doubling requires not only Weyl nondegeneracy at each valley, but also matched opposite-valley data:

$$\boxed{(W_\parallel^-, W_I^-, W_J^-) = (W_\parallel^+, W_I^+, W_J^+).} \tag{790}$$

In the matched opposite-valley frame, the same Class II seed amplitudes are read at the two valleys:

$$c_\parallel^- = c_\parallel^+, \qquad c_I^- = c_I^+, \qquad c_J^- = c_J^+,$$

hence

$$g_c^- = g_c^+, \qquad g_E^- = g_E^+, \qquad g_O^- = g_O^+,$$

and therefore

$$\boxed{W_{\parallel}^- = W_{\parallel}^+, \qquad W_I^- = W_I^+, \qquad W_J^- = W_J^+.} \tag{791}$$

Thus opposite-valley pairing is closed at the witness level inside the concrete Class II truncation. :contentReference[oaicite:7]index=7 :contentReference[oaicite:8]index=8

D. Step 3: zero constant mixing

Write the full block Hamiltonian as

$$H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^\dagger & H_R(p) \end{pmatrix}. \tag{792}$$

At the crossing $p = 0$, let H_* be the corresponding little group. Then the theorem-level statement is

$$V(0) \in \mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D), \tag{793}$$

hence

$$\boxed{\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \Longrightarrow V(0) = 0.} \tag{794}$$

The practical remote-sector audit is finite: if every remote sector differs from $\mathcal{H}_D$ in at least one shell-relevant label (shell-pair, transverse/volumetric character, enriched internal doublet, grading/parity/chirality, extra conserved label), then no constant intertwiner exists. Thus zero constant mixing is fully reduced to a finite remote-sector irrep/label mismatch audit. The concrete Class II ansatz is compatible with this condition, but does not by itself prove it. :contentReference[oaicite:9]index=9 :contentReference[oaicite:10]index=10 :contentReference[oaicite:11]index=11

E. Step 4: remote spectral gap

Let the remote block admit the decomposition

$$H_R(p) = D_R(p) + E_R(p),$$

and define

$$\delta_0(p) := \min_a |(D_R)_{aa}(p)|, \qquad \eta_R(p) := \|E_R(p)\|.$$

Then the pointwise and uniform remote-gap theorems imply:

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho} \Longrightarrow \mathrm{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \varnothing \quad \forall |p| \le \rho,} \tag{795}$$

with

$$\boxed{\Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0.} \tag{796}$$

Alternatively, if an unperturbed realization has gap $\Delta_0 > 0$ and the Class II correction obeys

$$\sup_{|p| \le p_*} \|\delta H_R(p)\| < \Delta_0,$$

then

$$\boxed{\Delta = \Delta_0 - \sup \|\delta H_R\| > 0.} \tag{797}$$

If in addition Step 3 holds, i.e.

$$V(0) = 0, \qquad \|V(p)\| \leq c|p|,$$

then the Schur complement correction is

$$\Sigma(E, p) = O(|p|^2/\Delta),$$

so the remote sector affects the local Dirac block only at subleading order. Thus the remote-gap problem is fully reduced to a finite Bloch-spectral inequality / bounded perturbation audit. Again, the concrete Class II ansatz is gap-compatible, but does not by itself prove $\Delta > 0$. :contentReference[oaicite:12]index=12 :contentReference[oaicite:13]index=13 :contentReference[oaicite:14]index=14

F. Step 5: mass protection

For the isolated doubled local Dirac block

$$H_D(q) = \rho_3 \otimes (q_i \sigma_i), \qquad i = 1, 2, 3, \tag{798}$$

the constant mass space is exactly

$$\boxed{\mathcal{M} = \mathrm{span}_{\mathbb{R}}\{\rho_1 \otimes \mathbf{1}_2,\ \rho_2 \otimes \mathbf{1}_2\}.} \tag{799}$$

Thus the finite mass-protection criterion is

$$\boxed{A_{\mathrm{res}} \cap \mathcal{M} = \{0\} \Longrightarrow \text{mass protection.}} \tag{800}$$

A sufficient mechanism is exact residual valley symmetry

$$Q_V = \rho_3 \otimes \mathbf{1}_2, \qquad U_V(\theta) = e^{i\theta Q_V},$$

or already its $\mathbb{Z}_2^V$ remnant generated by

$$S_V = \rho_3 \otimes \mathbf{1}_2.$$

Under either symmetry, both mass generators in $\mathcal{M}$ are forbidden, hence

$$A_{\mathrm{res}} \cap \mathcal{M} = \{0\}.$$

The concrete Class II ansatz is compatible with this, but does not itself prove that the full microscopic realization preserves the needed residual symmetry. :contentReference[oaicite:15]index=15 :contentReference[oaicite:16]index=16 :contentReference[oaicite:17]index=17

G. Local Dirac Closure Theorem

We now combine Steps 1–5.
[Local Dirac Closure Theorem in the concrete Class II setting] Assume:

1. **Class II witness realization:** the concrete shell-adapted Class II tensor family is realized on the selected local shell pair, with

$$g_c \neq 0, \qquad (g_E, g_O) \neq (0, 0);$$

2. **opposite-valley witness matching:** in the matched opposite-valley frame,

$$(W_\parallel^-, W_I^-, W_J^-) = (W_\parallel^+, W_I^+, W_J^+);$$

3. **zero constant mixing:**

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0, \qquad V(0) = 0;$$

4. **remote spectral gap:** either

$$\delta_{0,\rho} > \eta_{R,\rho}$$

for some shell neighborhood, or there exists an already-gapped realization with sufficiently small Class II correction, so that

$$\Delta > 0;$$

5. **mass protection:**

$$A_{\text{res}} \cap \operatorname{span}_{\mathbb{R}}\{\rho_1 \otimes \mathbf{1}_2, \rho_2 \otimes \mathbf{1}_2\} = \{0\}.$$

Then the full microscopic TECT truncation contains an isolated local massless Dirac block

$$H_{\text{eff}}(q) = \rho_3 \otimes (q_i \sigma_i) + O(|q|^2/\Delta),$$

i.e. an actual stable local Dirac cone.

[Proof sketch] By Step 1, the local Class II truncation produces a nondegenerate Weyl symbol at one valley. By Step 2, the opposite valley carries the same witness triple in the matched frame, so the two valleys pair into a local doubled Dirac block. By Step 3, no constant hybridization with the remote sector is allowed at the crossing. By Step 4, the remote block remains spectrally separated by a positive gap $\Delta > 0$, so the Schur-complement correction is only $O(|q|^2/\Delta)$. By Step 5, no symmetry-allowed constant Dirac mass exists on the isolated doubled shell block. Therefore the resulting local Dirac cone is both spectrally isolated and massless.

H. Honest final status

The theorem above is the strongest currently honest closure statement for the local Dirac sector.

What is already closed:

1. the reduction of the full local fermion problem to the finite five-step audit above;
2. the full theorem-level logic of Steps 1–5 inside the concrete Class II framework.

What is not yet unconditionally closed:

1. that the *actual* complete microscopic TECT action realizes the required Class II witness carrier sector;
2. that the *actual* remote microscopic sector satisfies the irrep mismatch needed for $V(0) = 0$;
3. that the *actual* remote block satisfies the Bloch-spectral gap inequality;
4. that the *actual* residual symmetry algebra forbids all constant masses.

Thus the precise final status is:

$$\boxed{\text{the local Dirac problem is fully reduced to a finite conditional closure theorem in the concrete Class II setting.}} \tag{801}$$

It is not yet a fully unconditional microscopic theorem for the complete TECT action.

. Summary

$$\boxed{\textbf{Local Dirac Closure Theorem in the concrete Class II setting}}$$

Step 1:

$$\boxed{g_c \neq 0, \qquad (g_E, g_O) \neq (0,0)}$$

gives actual local Weyl nondegeneracy.

Step 2:

$$\boxed{(W_{\parallel}^{-}, W_{I}^{-}, W_{J}^{-}) = (W_{\parallel}^{+}, W_{I}^{+}, W_{J}^{+})}$$

gives opposite-valley pairing into a local doubled Dirac block.

Step 3:

$$\boxed{\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \Longrightarrow V(0) = 0}$$

removes constant hybridization with the remote sector.

Step 4:

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho} \Longrightarrow \Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0}$$

(or perturbative stability from an already-gapped realization) gives remote spectral isolation and

$$\boxed{\Sigma(E, p) = O(|p|^2/\Delta).}$$

Step 5:

$$\boxed{\mathcal{M} = \mathrm{span}_{\mathbb{R}}\{\rho_1 \otimes \mathbf{1}_2, \rho_2 \otimes \mathbf{1}_2\}, \qquad A_{\mathrm{res}} \cap \mathcal{M} = \{0\}}$$

gives mass protection.

Therefore, if the concrete Class II setting satisfies the five finite inputs above, then the full microscopic truncation contains an isolated local massless Dirac cone:

$$\boxed{H_{\mathrm{eff}}(q) = \rho_3 \otimes (q_i \sigma_i) + O(|q|^2/\Delta).}$$

Honest final status:

$$\boxed{\text{the local Dirac problem is fully reduced to a finite conditional closure theorem,}}$$

but

$$\boxed{\text{the complete microscopic TECT realization of the required Class II carrier, gap, and residual symmetry is still an independent}}$$

TECT PARTIAL CLOSURE THEOREM: LOCAL DIRAC CLOSURE + IR ISOTROPY CONDITIONAL CLOSURE

A. Global setting

The present TECT program has two logically distinct but coupled low-energy frontiers:

1. the *local fermionic frontier*: existence of an actual stable massless local Dirac cone;
2. the *infrared bosonic frontier*: decay of sectoral derivative anisotropies and infrared velocity locking.

The current theorem-level status is that both frontiers have been reduced to finite conditional verification problems.

B. Local Dirac closure input

Let $\mathcal{H}_D$ denote the isolated doubled shell block, and $\mathcal{H}_R$ the remote sector. The local Dirac closure in the concrete Class II setting is reduced to the following finite input list:

1. **actual Class II witness carrier realization:** the microscopic TECT defect/background realizes a nontrivial shell-relevant Class II sector;

2. **local Weyl witness nonvanishing:**

$$g_c \neq 0, \qquad (g_E, g_O) \neq (0,0); \tag{802}$$

3. **opposite-valley witness matching:**

$$(W_\parallel^-, W_I^-, W_J^-) = (W_\parallel^+, W_I^+, W_J^+); \tag{803}$$

4. **zero constant mixing:**

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \Longrightarrow V(0) = 0; \tag{804}$$

5. **remote spectral gap:**

$$\delta_{0,\rho} > \eta_{R,\rho} \Longrightarrow \Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0; \tag{805}$$

6. **mass protection:**

$$A_{\mathrm{res}} \cap \mathrm{span}_{\mathbb{R}}\{\rho_1 \otimes \mathbf{1}_2, \rho_2 \otimes \mathbf{1}_2\} = \{0\}. \tag{806}$$

If these conditions hold, then the local fermionic sector contains an isolated massless local Dirac cone

$$H_{\mathrm{eff}}(q) = \rho_3 \otimes (q_i \sigma_i) + O(|q|^2/\Delta). \tag{807}$$

This is exactly the strongest currently honest local Dirac closure theorem in the concrete Class II setting.

C. Infrared isotropy closure input

Let

$$u = (\delta_T, \delta_O, \delta_D)^T$$

denote the physical derivative-anisotropy variables constrained by

$$2\delta_T + 2\delta_O + 2\delta_D = 0. \tag{808}$$

Let

$$\mathcal{U}_{\mathrm{tr}} = \{u \in \mathbb{R}^3 : \ 2\delta_T + 2\delta_O + 2\delta_D = 0\}$$

be the traceless physical anisotropy subspace.

The loop-level flow is

$$\partial_t u = M_Z u + O(u^2). \tag{809}$$

Decompose

$$M_Z = M_Z^{(\tau)} + M_Z^{(\nabla\mathrm{ani})} + M_Z^{(\mathrm{self})} + M_Z^{(OD)} + M_Z^{(\mathrm{gf/proj})}. \tag{810}$$

Define

$$M_Z^{\mathrm{good}} = M_Z^{(\tau)} + M_Z^{(\nabla\mathrm{ani})} + M_Z^{(\mathrm{self})}, \qquad M_Z^{\mathrm{bad}} = M_Z^{(OD)} + M_Z^{(\mathrm{gf/proj})}. \tag{811}$$

Assume the following explicit bounds:

a. Good-part coercive negativity

$$\langle u, M_Z^{\mathrm{good}} u\rangle \leq -\mu_{\mathrm{good}} \|u\|^2 \qquad \forall u \in \mathcal{U}_{\mathrm{tr}}, \qquad \mu_{\mathrm{good}} > 0. \tag{812}$$

b. Bad-part norm bounds

$$\|M_Z^{(OD)}|_{\mathcal{U}_{\rm tr}}\| \le \eta_{OD}, \qquad \|M_Z^{(\rm gf/proj)}|_{\mathcal{U}_{\rm tr}}\| \le \eta_{\rm gf}. \tag{813}$$

Then the explicit sufficient IR-isotropy condition is

$$\boxed{\mu_{\rm good} > \eta_{OD} + \eta_{\rm gf}.} \tag{814}$$

Indeed, for all $u \in \mathcal{U}_{\rm tr}$,

$$\begin{aligned}\langle u, M_Z u\rangle &= \langle u, M_Z^{\rm good} u\rangle + \langle u, M_Z^{\rm bad} u\rangle \\ &\le -\mu_{\rm good}\|u\|^2 + (\eta_{OD} + \eta_{\rm gf})\|u\|^2 \\ &= -(\mu_{\rm good} - \eta_{OD} - \eta_{\rm gf})\|u\|^2.\end{aligned} \tag{815}$$

Hence (814) implies

$$\boxed{\mathrm{spec}(M_Z|_{\mathcal{U}_{\rm tr}}) \subset (-\infty, 0),} \tag{816}$$

and therefore

$$\boxed{\delta_T, \delta_O, \delta_D \to 0 \quad \text{in the infrared.}} \tag{817}$$

D. TECT Partial Closure Theorem

We can now combine the local Dirac closure and the IR-isotropy closure.
[TECT Partial Closure Theorem] Assume:

1. **local Dirac closure inputs:** the finite local Dirac conditions (802)–(806) hold in the concrete Class II setting;
2. **gauge-consistent soft sector:** the projected A-sector satisfies

$$p^\mu \Gamma^{(2)}_{\mu\nu} = 0, \qquad m^2_{A,k} \to 0, \qquad \text{gauge-preserving regulator}, \qquad \text{Class I/II only;} \tag{818}$$

3. **IR-isotropy input:** the explicit inequality (814) holds:

$$\mu_{\rm good} > \eta_{OD} + \eta_{\rm gf}.$$

Then:

1. the microscopic TECT truncation contains an isolated local massless Dirac cone;
2. the physical derivative-anisotropy variables decay under the infrared flow:

$$\delta_T, \delta_O, \delta_D \to 0;$$

3. the low-energy soft sector is infrared-isotropic and velocity-locked.

[Proof sketch] By the local Dirac closure theorem, the finite witness/pairing/mixing/gap/mass-protection conditions imply the existence of an isolated massless local Dirac cone. By the gauge-consistency assumptions, the projected A-sector contribution to the flow is physically meaningful. By the explicit good-vs-bad inequality (814), the linearized anisotropy matrix M_Z has strictly negative spectrum on the physical traceless subspace. Hence the derivative anisotropies decay in the infrared. Therefore TECT admits both an isolated local massless Dirac cone and an infrared-isotropic soft sector.

E. Exact logical status

The theorem above is the strongest currently honest whole-system closure statement.

c. What is already theorem-level closed.

1. the reduction of the local Dirac problem to a finite five-step audit;
2. quartic anisotropy irrelevance for $N_{\text{eff}} = 6$:

$$\theta_\tau = -\frac{\epsilon}{7} < 0;$$

3. the reduction of the derivative-anisotropy problem to the sign of M_Z;
4. the explicit sufficient IR-isotropy criterion

$$\mu_{\text{good}} > \eta_{OD} + \eta_{\text{gf}}.$$

d. What is not yet unconditionally closed.

1. that the actual complete microscopic TECT action realizes the required Class II witness carrier;
2. that the actual microscopic remote sector satisfies the label/irrep audit for $V(0) = 0$;
3. that the actual microscopic remote block satisfies the gap inequality;
4. that the actual residual symmetry excludes constant masses;
5. that the actual background-adapted FRG data satisfy

$$\mu_{\text{good}} > \eta_{OD} + \eta_{\text{gf}}.$$

Thus the exact current status is

$$\boxed{\text{TECT is not yet fully unconditionally closed,}} \tag{819}$$

but

$$\boxed{\text{it is fully reduced to a finite partial closure theorem whose remaining inputs are explicit microscopic verification tasks.}} \tag{820}$$

. Summary

$$\boxed{\textbf{TECT Partial Closure Theorem}}$$

Local Dirac side: if the concrete Class II setting satisfies

$$\boxed{g_c \neq 0, \qquad (g_E, g_O) \neq (0, 0),}$$

$$\boxed{(W_\parallel^-, W_I^-, W_J^-) = (W_\parallel^+, W_I^+, W_J^+),}$$

$$\boxed{\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0, \qquad \Delta > 0,}$$

$$\boxed{A_{\text{res}} \cap \mathrm{span}_{\mathbb{R}}\{\rho_1 \otimes \mathbf{1}_2, \rho_2 \otimes \mathbf{1}_2\} = \{0\},}$$

then TECT contains an isolated local massless Dirac cone:

$$\boxed{H_{\text{eff}}(q) = \rho_3 \otimes (q_i \sigma_i) + O(|q|^2/\Delta).}$$

IR isotropy side: if the projected soft sector satisfies

$$\boxed{p^\mu \Gamma^{(2)}_{\mu\nu} = 0, \qquad m^2_{A,k} \to 0, \qquad \text{gauge-preserving regulator}, \qquad \text{Class I/II only},}$$

and if

$$\boxed{\mu_{\text{good}} > \eta_{OD} + \eta_{\text{gf}},}$$

then

$$\boxed{\operatorname{spec}(M_Z|_{\mathcal{U}_{\text{tr}}}) \subset (-\infty, 0),}$$

hence

$$\boxed{\delta_T, \delta_O, \delta_D \to 0.}$$

Therefore, under the finite microscopic input list above,

$$\boxed{\text{TECT contains an isolated local massless Dirac cone and an infrared-isotropic velocity-locked soft sector.}}$$

Honest final status:

$$\boxed{\text{not yet a fully unconditional microscopic theorem,}}$$

but

$$\boxed{\text{already a finite partial closure theorem.}}$$

ACTUAL MICROSCOPIC WITNESS CARRIER AUDIT

1. Goal

The remaining microscopic question after the concrete Class II witness theorem is:

$$\boxed{\text{Does the true complete TECT action actually realize a nontrivial Class II sector in the physically relevant defect background?}} \tag{821}$$

Inside the concrete Class II truncation, the witness problem is already closed: once one longitudinal seed and one transverse seed exist, local Weyl nondegeneracy is generic in the allowed Class II tensor space. :contentReference[oaicite:5]index=5 :contentReference[oaicite:6]index=6

Therefore the remaining task is not to re-prove the witness theorem, but to audit whether the full microscopic TECT linearization actually contains the required carrier sector.

2. What the witness actually needs

The shell-level Weyl/Dirac witness is controlled by the enriched microscopic tensor

$$N^a_{ij},$$

through the shell-adapted contractions

$$\lambda_\parallel, \qquad \alpha, \qquad \beta, \qquad \det M = (\kappa_* |G_*|)^3 A(B^2 + C^2).$$

The minimal shell-adapted tensor family is

$$N^3_{ij} = A\,\hat{n}_i \hat{n}_j, \qquad N^1_{ij} = B\,\hat{n}_i e_{1j} + C\,\hat{n}_i e_{2j}, \qquad N^2_{ij} = -C\,\hat{n}_i e_{1j} + B\,\hat{n}_i e_{2j}. \tag{822}$$

Hence the witness requires a microscopic object carrying simultaneously:

1. one spatial derivative index i or j,
2. one reduced internal / shell-tangent index a.

This is explicitly identified in the embedding/classification discussion. :contentReference[oaicite:7]index=7 :contentReference[oaicite:8]index=8

3. Minimal UV classification

The microscopic classification theorem states that any minimal local UV extension producing a nonzero shell/projector mixing source must effectively introduce an (i, a)-carrier. At lowest derivative order, the minimal viable classes are:

1. Class I: direct shell–projector derivative couplings,
2. Class II: heavy auxiliary internal vector mediator X_i^a,
3. Class III: connection-like mediator A_i^a.

Among them, the most natural minimal UV extension is Class II. :contentReference[oaicite:9]index=9

Therefore the actual carrier audit reduces to:

$$\boxed{\text{Does the microscopic TECT linearization contain an effective shell-relevant } (i, a)\text{-carrier channel?}} \tag{823}$$

4. Explicit Class II mechanism

The explicit Class II quadratic source is generated by

$$L_{\mathrm{UV}}^{(II)} = \frac{M_X^2}{2} X_i^a X_i^a + \alpha X_i^a (\partial_i \Psi^\dagger T^a \Psi + \text{h.c.}) + \beta X_i^a \partial_i \delta P^a \, \psi. \tag{824}$$

Around the locked background, eliminating X_i^a produces schematically

$$L_{\mathrm{mix}}^{(2)} \sim -\frac{\alpha\beta}{M_X^2} (\partial_i \phi_0) \, C^a[\psi] \, \partial_i \xi^a, \tag{825}$$

which after shell reduction gives precisely the target quadratic mixing form

$$(\partial_i \phi_0)(\partial_j \psi) \, \xi^a,$$

hence a nonzero induced correction

$$\delta N_{ij}^a \neq 0.$$

This is the first mathematically consistent route to a derived nonzero shell-enriched tensor correction from projector-sector physics. :contentReference[oaicite:10]index=10

So the audit target is no longer vague: one must verify that the actual microscopic TECT linearization contains or effectively induces this same (i, a)-carrier channel.

5. Embedding criterion for the explicit seed subspace

The explicit witness seed subspace

$$V_{\mathrm{seed}} = \mathrm{span}\{N_{\mathrm{seed}}^{(\|)}, N_{\mathrm{seed}}^{(\perp,E)}, N_{\mathrm{seed}}^{(\perp,O)}\}$$

is not an arbitrary effective ansatz. The embedding statement is:

$$\boxed{\text{If the true microscopic transport space contains an allowed } (i, a)\text{-carrier sector, then the explicit seed subspace can be embedde}} \tag{826}$$

Equivalently, the carrier audit is successful if the symmetry-allowed microscopic transport space contains at least:

1. one longitudinal shell-adapted seed direction;
2. one transverse shell-adapted seed direction.

Then the concrete Class II witness theorem applies and local Weyl nondegeneracy is generic in that sector. :contentReference[oaicite:11]index=11 :contentReference[oaicite:12]index=12

6. Operational audit checklist

The actual microscopic carrier audit is therefore:

a. *(A) Linearize the microscopic TECT equation* around the physically relevant defect background and isolate all terms linear in fluctuations and first order in worldvolume derivatives.

b. *(B) Extract the shell-relevant transport operator* in the form

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}.$$

The odd transport kernel generated by the defect background is already known to arise naturally from the microscopic projector-gradient term:

$$V^\mu(\xi) \sim (\partial_\xi P_B)T^\mu, \qquad V^\mu(-\xi) = -V^\mu(\xi).$$

:contentReference[oaicite:13]index=13

c. *(C) Determine whether the projected overlap matrices*

$$(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle$$

span a nontrivial traceless Hermitian sector. At the effective level, this is exactly what generates the mixed Dirac symbol. :contentReference[oaicite:14]index=14

d. *(D) Match the induced shell tensor* to the shell-adapted axial basis

$$(A, B, C) \leftrightarrow (\lambda_\|, \alpha, \beta).$$

If at least one longitudinal and one transverse component survive, then the seed criterion is satisfied.

e. *(E) Compute the projection test*

$$P_V T^{(\alpha)} \neq 0$$

or equivalently produce one explicit admissible seed tensor for each required channel. This is the finite operational completion criterion already stated in the roadmap. :contentReference[oaicite:15]index=15 :contentReference[oaicite:16]index=16

7. Strongest honest conclusion

Thus the actual microscopic witness carrier audit is now completely reduced to the following finite statement:

$$\boxed{\text{the carrier audit succeeds iff the full microscopic TECT linearization contains a shell-relevant nontrivial } (i,a)\text{-carrier sector wh}} \tag{827}$$

This is not yet an unconditional theorem for the complete TECT action, because the actual projection test has not yet been carried out in the final microscopic linearization. But it is no longer conceptually open: it is an explicit finite projection / seed-embedding audit.

. Summary

$$\boxed{\textbf{Actual microscopic witness carrier audit}}$$

The witness needs a shell-relevant enriched tensor

$$N_{ij}^a,$$

so the microscopic theory must effectively contain an object with both:

$$\boxed{\text{one spatial derivative index } (i \text{ or } j), \qquad \text{one reduced internal index } a.}$$

The minimal shell-adapted tensor family is

$$\boxed{N^3_{ij} = A\,\hat{n}_i\hat{n}_j, \qquad N^1_{ij} = B\,\hat{n}_i e_{1j} + C\,\hat{n}_i e_{2j}, \qquad N^2_{ij} = -C\,\hat{n}_i e_{1j} + B\,\hat{n}_i e_{2j}.}$$

The minimal UV classes capable of producing such a carrier are:

$$\boxed{\text{Class I, Class II, Class III,}}$$

with Class II the most natural minimal mediated completion.

The explicit Class II route introduces a heavy auxiliary internal vector mediator

$$\boxed{X^a_i,}$$

whose elimination produces

$$\boxed{L^{(2)}_{\rm mix} \sim (\partial_i\phi_0)(\partial_j\psi)\,\xi^a \Longrightarrow \delta N^a_{ij} \neq 0.}$$

Therefore the real microscopic audit is:

$$\boxed{\text{Does the full microscopic TECT linearization contain an effective shell-relevant nontrivial } (i,a)\text{-carrier sector?}}$$

Operationally, this reduces to:

1. linearize the microscopic TECT equation around the relevant defect background;
2. extract the first-order transport kernel

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu};$$

3. project it to the low-mode sector

$$(M^\mu)_{ab} = \langle u^+_a, V^\mu u^-_b\rangle;$$

4. match the induced shell tensor to the axial basis

$$(A, B, C) \leftrightarrow (\lambda_\parallel, \alpha, \beta);$$

5. verify that at least one longitudinal and one transverse seed survive.

So the strongest honest conclusion is:

$$\boxed{\text{the actual microscopic witness carrier problem is fully reduced to a finite projection / seed-embedding audit.}}$$

PROJECTION TEST $P_V T^{(\alpha)} \neq 0$ IN CONCRETE CARRIER-AUDIT LANGUAGE

1. Abstract tensor-space formulation

Let

$$W = \mathbb{R}^3_{\rm out} \otimes \mathbb{R}^3_{\rm grad} \otimes U_{\rm int}$$

be the ambient shell-enriched tensor space, and let

$$V \subset W$$

be the finite-dimensional real vector space of all symmetry-allowed microscopic enriched tensors surviving the microscopic TECT linearization around the condensed defect background.

For each shell-Dirac channel $\alpha = 1, 2$, define the shell contraction

$$L_\alpha(N) = \langle T^{(\alpha)}, N\rangle, \tag{828}$$

where $T^{(\alpha)}$ is the target covector built from the selected shell direction, transverse frame, and internal reduced mode. The exact dichotomy is:

$$\boxed{L_\alpha|_V \equiv 0 \iff P_V T^{(\alpha)} = 0.} \tag{829}$$

Therefore:

$$\boxed{P_V T^{(\alpha)} = 0 \Longrightarrow \text{the channel is symmetry-forbidden,}} \tag{830}$$

$$\boxed{P_V T^{(\alpha)} \neq 0 \Longrightarrow \text{the channel is generically nonvanishing on an open dense full-measure subset of } V.} \tag{831}$$

This is the exact practical criterion stated in the finite witness reduction. :contentReference[oaicite:5]index=5

2. Carrier-audit reinterpretation

The abstract condition $P_V T^{(\alpha)} \neq 0$ has the following concrete meaning:

$$\boxed{\text{the actual microscopic TECT linearization contains at least one shell-relevant admissible transport direction whose projection o}} \tag{832}$$

Equivalently, the carrier audit succeeds in channel α iff the symmetry-allowed microscopic transport space V contains at least one tensor

$$N^{(\alpha)} \in V$$

such that

$$L_\alpha(N^{(\alpha)}) \neq 0.$$

Thus $P_V T^{(\alpha)} \neq 0$ is not merely a formal projection statement: it is the statement that the true microscopic theory actually realizes a nontrivial seed in the corresponding shell-relevant carrier direction.

3. Stronger practical sufficient condition: explicit seed test

A stronger practical sufficient condition is:

$$\boxed{\exists\ N^{(\alpha)}_{\text{seed}} \in V \quad \text{such that} \quad \langle T^{(\alpha)}, N^{(\alpha)}_{\text{seed}} \rangle \neq 0.} \tag{833}$$

Then automatically

$$\boxed{P_V T^{(\alpha)} \neq 0.} \tag{834}$$

This is exactly the seed/witness test stated in the witness-reduction notes. :contentReference[oaicite:6]index=6

So the actual microscopic carrier audit has two equivalent operational routes:

a. Route A: direct projection route

1. determine the residual symmetry group H;
2. determine the allowed tensor space $V \subset W$;
3. compute the orthogonal projection $P_V T^{(\alpha)}$.

b. Route B: explicit seed route

1. determine the residual symmetry group H;
2. determine the allowed tensor space $V \subset W$;
3. construct one explicit admissible tensor $N^{(\alpha)}_{\text{seed}} \in V$;
4. check that $\langle T^{(\alpha)}, N^{(\alpha)}_{\text{seed}} \rangle \neq 0$.

4. Concrete Class II seed subspace

Inside the concrete Class II ansatz, the explicit seed directions are already known:

$$N^{(\parallel)}_{\text{seed}}: \quad V^{\mu}_{(\parallel)}(\xi) = \frac{1}{2}\,\theta'(\xi)^3\,n^{\mu}\sigma_3, \tag{835}$$

$$N^{(\perp,E)}_{\text{seed}}: \quad V^{\mu}_{(\perp,E)}(\xi) = \theta'(\xi)\,E^{\mu}{}_{a}\sigma_a, \tag{836}$$

$$N^{(\perp,O)}_{\text{seed}}: \quad V^{\mu}_{(\perp,O)}(\xi) = \theta'(\xi)\,J^{\mu}{}_{a}\sigma_a. \tag{837}$$

For

$$N = c_{\parallel}N^{(\parallel)}_{\text{seed}} + c_E N^{(\perp,E)}_{\text{seed}} + c_O N^{(\perp,O)}_{\text{seed}},$$

the reduced witnesses are exactly

$$\lambda_{\parallel} = \frac{1}{2}c_{\parallel}I_{\theta'^3}, \qquad \alpha = c_E I_{\theta}, \qquad \beta = c_O I_{\theta}. \tag{838}$$

Since

$$I_{\theta} \neq 0, \qquad I_{\theta'^3} \neq 0,$$

the symbol is nondegenerate iff

$$\boxed{c_{\parallel} \neq 0, \qquad (c_E, c_O) \neq (0,0).} \tag{839}$$

Thus inside the concrete Class II ansatz space, the finite witness problem is already exactly closed. :contentReference[oaicite:7]index=7 :contentReference[oaicite:8]index=8

5. What remains open microscopically

Because the Class II seed subspace is explicit, the remaining question is no longer "do such seeds exist abstractly?" Instead it is:

$$\boxed{\text{Does the complete microscopic TECT transport space actually contain } V_{\text{seed}} = \operatorname{span}\{N^{(\parallel)}_{\text{seed}}, N^{(\perp,E)}_{\text{seed}}, N^{(\perp,O)}_{\text{seed}}\}\text{, or at least its requ}} \tag{840}$$

This is explicitly identified as the remaining frontier. :contentReference[oaicite:9]index=9

6. Sharp carrier-audit completion criterion

Therefore the actual microscopic carrier audit can now be stated sharply:

[Carrier-audit completion criterion] Let $V \subset W$ be the symmetry-allowed microscopic transport space obtained from the full TECT linearization around the relevant defect background. Then the witness carrier audit succeeds iff either of the following equivalent conditions holds:

1. for each required shell channel α,

$$P_V T^{(\alpha)} \neq 0;$$

2. V contains at least one admissible longitudinal seed and at least one admissible transverse seed with nonzero overlap against the corresponding target covectors.

In particular, because the explicit Class II seed subspace already realizes the required longitudinal and transverse overlaps, the remaining microscopic task is exactly:

$$\boxed{\text{prove that the full microscopic TECT transport space contains } V_{\text{seed}}\text{, or at least one longitudinal and one transverse seed direc}} \tag{841}$$

7. Strongest honest conclusion

Thus the strongest honest conclusion is:

$$\boxed{P_V T^{(\alpha)} \neq 0} \tag{842}$$

means:

$$\boxed{\text{the actual microscopic TECT linearization contains a shell-relevant nontrivial carrier direction in channel } \alpha.} \tag{843}$$

So the remaining carrier problem is no longer an infinite-dimensional PDE mystery. It is an explicit finite projection / seed-embedding audit on the symmetry-allowed microscopic transport space.

. Summary

$$\boxed{L_\alpha(N) = \langle T^{(\alpha)}, N\rangle, \qquad L_\alpha|_V \equiv 0 \iff P_V T^{(\alpha)} = 0.}$$

Hence:

$$\boxed{P_V T^{(\alpha)} = 0 \Longrightarrow \text{channel } \alpha \text{ is symmetry-forbidden,}}$$

$$\boxed{P_V T^{(\alpha)} \neq 0 \Longrightarrow \text{channel } \alpha \text{ is generically nonvanishing on } V.}$$

In concrete carrier-audit language,

$$\boxed{P_V T^{(\alpha)} \neq 0 \iff \text{the full microscopic TECT transport space contains at least one shell-relevant admissible carrier direction wi}$$

A stronger practical sufficient condition is:

$$\boxed{\exists N^{(\alpha)}_{\mathrm{seed}} \in V \text{ such that } \langle T^{(\alpha)}, N^{(\alpha)}_{\mathrm{seed}}\rangle \neq 0 \Longrightarrow P_V T^{(\alpha)} \neq 0.}$$

Inside the concrete Class II ansatz, explicit seeds already exist:

$$\boxed{N^{(\|)}_{\mathrm{seed}}, \qquad N^{(\perp,E)}_{\mathrm{seed}}, \qquad N^{(\perp,O)}_{\mathrm{seed}}.}$$

They satisfy

$$\boxed{\lambda_\| = \frac{1}{2} c_\| I_{\theta'^3}, \qquad \alpha = c_E I_\theta, \qquad \beta = c_O I_\theta,}$$

so the witness problem is already exactly closed inside that ansatz.

Therefore the actual remaining microscopic question is:

$$\boxed{\text{does the complete microscopic TECT transport space contain } V_{\mathrm{seed}}, \text{ or at least one longitudinal and one transverse seed directi}$$

So the carrier audit is fully reduced to a finite projection / seed-embedding test.

MINIMAL PRACTICAL SUFFICIENT CONDITION FOR ACTUAL MICROSCOPIC CARRIER SURVIVAL

1. Ambient formulation

Let

$$W = \mathbb{R}^3_{\text{out}} \otimes \mathbb{R}^3_{\text{grad}} \otimes U_{\text{int}}$$

be the ambient enriched tensor space, and let

$$V_{\text{micro}} \subset W$$

be the actual symmetry-allowed transport tensor space extracted from the full microscopic TECT linearization around the relevant defect background.

For each shell channel α, define the shell witness functional

$$L_\alpha(N) = \langle T^{(\alpha)}, N \rangle.$$

Then the exact dichotomy is:

$$P_{V_{\text{micro}}} T^{(\alpha)} = 0 \iff L_\alpha|_{V_{\text{micro}}} \equiv 0,$$

and

$$P_{V_{\text{micro}}} T^{(\alpha)} \neq 0 \iff L_\alpha \text{ is generically nonzero on } V_{\text{micro}}.$$

2. Strong practical sufficient condition

A stronger sufficient condition is the existence of one explicit admissible seed:

$$\exists\, N^{(\alpha)}_{\text{seed}} \in V_{\text{micro}} \quad \text{such that} \quad \langle T^{(\alpha)}, N^{(\alpha)}_{\text{seed}} \rangle \neq 0.$$

Then automatically

$$P_{V_{\text{micro}}} T^{(\alpha)} \neq 0.$$

Therefore the actual carrier audit succeeds if one can exhibit:

1. one longitudinal seed $N_\parallel \in V_{\text{micro}}$ with

$$\langle T^{(\parallel)}, N_\parallel \rangle \neq 0;$$

2. one transverse seed $N_\perp \in V_{\text{micro}}$ with

$$\langle T^{(\perp)}, N_\perp \rangle \neq 0.$$

3. Minimal practical sufficient condition

Hence the minimal practical sufficient condition for actual microscopic carrier survival is:

$$\boxed{\exists\, N_\parallel, N_\perp \in V_{\text{micro}} \text{ such that } \langle T^{(\parallel)}, N_\parallel \rangle \neq 0, \qquad \langle T^{(\perp)}, N_\perp \rangle \neq 0.} \tag{844}$$

This immediately implies

$$P_{V_{\text{micro}}} T^{(\parallel)} \neq 0, \qquad P_{V_{\text{micro}}} T^{(\perp)} \neq 0,$$

hence the witness channels survive generically.

4. Relation to the concrete Class II seed space

Inside the concrete Class II ansatz, the explicit seed subspace is already known:

$$V_{\text{seed}} = \text{span}\{N^{(\|)}_{\text{seed}}, N^{(\perp,E)}_{\text{seed}}, N^{(\perp,O)}_{\text{seed}}\}.$$

Therefore a still sharper practical sufficient condition is:

$$\boxed{V_{\text{seed}} \cap V_{\text{micro}} \text{ contains at least one nonzero longitudinal seed and at least one nonzero transverse seed.}} \tag{845}$$

Equivalently, it is enough to prove

$$N^{(\|)}_{\text{seed}} \in V_{\text{micro}}, \qquad N^{(\perp,E)}_{\text{seed}} \in V_{\text{micro}}$$

or

$$N^{(\|)}_{\text{seed}} \in V_{\text{micro}}, \qquad N^{(\perp,O)}_{\text{seed}} \in V_{\text{micro}}.$$

5. Strongest honest practical conclusion

Thus the remaining microscopic carrier problem does not require proving the full Class II completion first. A much weaker and sufficient condition already closes the witness-carrier question:

$$\boxed{\text{actual microscopic carrier survival is proved once one longitudinal and one transverse admissible seed survive in the true micros}} \tag{846}$$

. Summary

Minimal practical sufficient condition for actual microscopic carrier survival

Let V_{micro} be the true symmetry-allowed microscopic transport space.
It is sufficient to find:

$$\boxed{\exists\, N_\| \in V_{\text{micro}} \text{ such that } \langle T^{(\|)}, N_\| \rangle \neq 0,}$$

and

$$\boxed{\exists\, N_\perp \in V_{\text{micro}} \text{ such that } \langle T^{(\perp)}, N_\perp \rangle \neq 0.}$$

Then

$$\boxed{P_{V_{\text{micro}}} T^{(\|)} \neq 0, \qquad P_{V_{\text{micro}}} T^{(\perp)} \neq 0,}$$

so the longitudinal and transverse witness channels both survive generically.
Since the concrete Class II ansatz already provides explicit seeds

$$\boxed{N^{(\|)}_{\text{seed}}, \qquad N^{(\perp,E)}_{\text{seed}}, \qquad N^{(\perp,O)}_{\text{seed}},}$$

an even sharper sufficient condition is simply:

$$\boxed{V_{\text{micro}} \text{ contains at least one longitudinal seed direction and at least one transverse seed direction from } V_{\text{seed}}.}$$

So the practical carrier audit reduces to a finite seed-embedding check, not a full PDE proof.

MICROSCOPIC EXTRACTION PROTOCOL FOR V_{micro}

1. Purpose

The remaining microscopic witness-carrier problem is no longer:

“solve the full PDE exactly”.

It is the finite extraction problem:

determine the actual symmetry-allowed microscopic transport space V_{micro}

from the full TECT linearization around the relevant defect background, and test whether it contains one longitudinal and one transverse seed direction. :contentReference[oaicite:6]index=6 :contentReference[oaicite:7]index=7

2. Step 0: choose the defect background

Fix a physically relevant static defect projector background

$$P_B(\xi), \qquad P_B(-\xi) = P_B(\xi), \tag{847}$$

depending only on the transverse coordinate ξ.

Its tangent bundle is

$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^{\perp}, \qquad \dim_{\mathbb{C}} Q_B = 2. \tag{848}$$

Thus the fluctuation field on the defect core is a complex doublet. This is the natural microscopic origin of the low-mode internal 2-component sector. :contentReference[oaicite:8]index=8

3. Step 1: linearize the microscopic TECT equation

Start from the microscopic condensate equation linearized around the condensed background,

$$i\partial_t \Phi = L(-i\nabla)\Phi. \tag{849}$$

Equivalently, in projector form one has the linearized fluctuation equation

$$\rho\partial_t^2 \delta P - C_{ij}\partial_i\partial_j \delta P + H_B^{\mathrm{stat}}\delta P + T(\delta P) = 0, \tag{850}$$

where the first-order transport term comes from derivatives acting on the background projector as well as on the fluctuation. :contentReference[oaicite:9]index=9

4. Step 2: isolate the first-order transport kernel

Expanding derivatives on $P_B + \delta P$ yields the first-order transport term

$$T(\delta P) = -C_{ij}\Big[(\partial_i P_B)\partial_j \delta P + (\partial_j P_B)\partial_i \delta P\Big]. \tag{851}$$

Writing derivatives along the defect worldvolume as

$$y^{\mu} = (t, x_1, x_2),$$

this takes the form

$$\boxed{L_1 = -iV^{\mu}(\xi)\partial_{y^{\mu}},} \tag{852}$$

with

$$V^{\mu}(\xi) \sim (\partial_{\xi} P_B)T^{\mu}. \tag{853}$$

Because $P_B(-\xi) = P_B(\xi)$, one has

$$\boxed{V^{\mu}(-\xi) = -V^{\mu}(\xi).} \tag{854}$$

Thus the microscopic TECT linearization naturally supplies the odd transport kernel required in the reduced Dirac framework. :contentReference[oaicite:10]index=10

5. Step 3: construct the low-mode sector

Solve the static transverse operator on the defect core

$$H_B = -\partial_\xi^2 + U_B(\xi) \tag{855}$$

on the rank-two bundle Q_B, and extract its low even/odd modes

$$u_a^+(\xi), \qquad u_b^-(\xi), \qquad a, b = 1, 2.$$

This is the isolated graded $2+2$ sector required by the reduced Dirac theorem. :contentReference[oaicite:11]index=11

6. Step 4: project the transport kernel

Compute the projected overlap matrices

$$\boxed{(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle = \int d\xi\, (u_a^+)^\dagger V^\mu(\xi) u_b^-.} \tag{856}$$

Since

$$u_a^+(\xi) \text{ is even}, \qquad u_b^-(\xi) \text{ is odd}, \qquad V^\mu(\xi) \text{ is odd},$$

the integrand is even, so the overlap is generically nonzero. This is the direct microscopic origin of the mixed first-order symbol. :contentReference[oaicite:12]index=12

At this point, one may already regard the span of the projected matrices M^μ as the effective extracted transport space.

7. Step 5: choose the selected shell pair and project to the valley block

Fix a selected shell vector $G_* \in S_{\mathrm{BCC}}$ and write

$$k = G_* + p, \qquad p = p_\parallel \hat{n} + p_1 e_1 + p_2 e_2, \qquad \hat{n} = \frac{G_*}{|G_*|}.$$

Assume the microscopic symbol admits the Taylor expansion

$$L(G_* + p) = L_* + p_i K_i + O(|p|^2), \qquad K_i = \partial_{k_i} L(k)\big|_{k=G_*}. \tag{857}$$

Let P_* denote the orthogonal projector onto the selected low-energy 2-dimensional valley block H_*. Then the reduced first-order Hamiltonian is

$$h^+_{\mathrm{lin}}(p) = P_* \,(p_i K_i)\, P_*\big|_{H_*}. \tag{858}$$

Since $h^+_{\mathrm{lin}}(p)$ is Hermitian 2×2, define the Pauli-resolved coefficient matrix

$$M_{\mu i} = \frac{1}{2} \mathrm{Tr}_{H_*}\big(\sigma_\mu P_* K_i P_*\big), \qquad \mu = 1, 2, 3. \tag{859}$$

Then

$$h^+_{\mathrm{lin}}(p) = M_{\mu i} p_i \sigma_\mu.$$

Thus the local Weyl data are determined directly by the projected first derivatives K_i. :contentReference[oaicite:13]index=13

8. Step 6: extract the witness coefficients

Choose the adapted shell frame $(\hat{n}, e_1, e_2)$. Then the concrete shell witnesses are read as the coefficients of

$$H_{\rm wit}(p) = \lambda_{\parallel}(\hat{n}\cdot p)\sigma_3 + \alpha(p_1\sigma_1 + p_2\sigma_2) + \beta(-p_2\sigma_1 + p_1\sigma_2). \tag{860}$$

Equivalently:

$$\lambda_{\parallel} = M_{3,\parallel}, \qquad \alpha = \text{coefficient of } (p_1\sigma_1 + p_2\sigma_2), \qquad \beta = \text{coefficient of } (-p_2\sigma_1 + p_1\sigma_2). \tag{861}$$

Now define the actual microscopic transport tensor space $V_{\rm micro}$ to be the finite-dimensional real span of all shell-relevant microscopic tensors contributing to these projected coefficients.

Then the witness carrier audit reduces to the finite test:

$$\boxed{\text{find one admissible longitudinal seed with } \lambda_{\parallel} \neq 0, \quad \text{and one admissible transverse seed with } (\alpha,\beta) \neq (0,0).} \tag{862}$$

By the finite witness theorem, this is enough to guarantee generic nondegeneracy on an open dense full-measure subset of $V_{\rm micro}$. :contentReference[oaicite:14]index=14 :contentReference[oaicite:15]index=15

9. Practical yes/no outputs of the protocol

The extraction protocol produces exactly four logically distinct outputs:

a. *(O1) Carrier survival* Do we obtain at least one longitudinal and one transverse seed direction? If yes, the microscopic witness carrier survives.

b. *(O2) Valley witness* Do the resulting coefficients satisfy

$$\lambda_{\parallel} \neq 0, \qquad (\alpha,\beta) \neq (0,0)?$$

If yes, the selected valley is Weyl.

c. *(O3) Opposite-valley matching* Repeat the same extraction at $-G_*$ and check matched-frame equality of the witness triple.

d. *(O4) Data for later audits* The same extraction provides the actual low-energy block and projected operators needed for:

$$V(0) = 0, \qquad \Delta > 0, \qquad \text{mass protection}, \qquad M_Z.$$

10. Strongest honest conclusion

Thus the microscopic extraction protocol for $V_{\rm micro}$ is now completely explicit:

$$\boxed{P_B(\xi) \implies V^{\mu}(\xi) \implies u_a^{\pm} \implies (M^{\mu})_{ab} \implies P_* K_i P_* \implies (\lambda_{\parallel}, \alpha, \beta) \implies \text{actual witness carrier audit.}} \tag{863}$$

No further conceptual ambiguity remains. What remains is the actual evaluation of these projections in the chosen microscopic TECT linearization.

. Summary

$$\boxed{\textbf{Microscopic extraction protocol for } V_{\rm micro}}$$

1. Fix a defect background

$$\boxed{P_B(\xi), \qquad P_B(-\xi) = P_B(\xi).}$$

2. Linearize the microscopic TECT equation around it:

$$\boxed{i\partial_t \Phi = L(-i\nabla)\Phi.}$$

3. Extract the first-order transport term

$$\boxed{L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \qquad V^\mu(\xi) \sim (\partial_\xi P_B)T^\mu, \qquad V^\mu(-\xi) = -V^\mu(\xi).}$$

4. Solve the static transverse defect operator and obtain the even/odd low modes

$$\boxed{u_a^+, \qquad u_b^-.}$$

5. Compute the projected overlap matrices

$$\boxed{(M^\mu)_{ab} = \langle u_a^+, V^\mu u_b^- \rangle.}$$

6. At a selected shell pair $\pm G_*$, compute the projected first derivatives

$$\boxed{M_{\mu i} = \frac{1}{2}\mathrm{Tr}_{H_*}\big(\sigma_\mu P_* K_i P_*\big).}$$

7. Read off the witness coefficients

$$\boxed{\lambda_\parallel, \qquad \alpha, \qquad \beta}$$

from the adapted shell Hamiltonian

$$\boxed{H_{\mathrm{wit}}(p) = \lambda_\parallel(\hat{n}\cdot p)\sigma_3 + \alpha(p_1\sigma_1 + p_2\sigma_2) + \beta(-p_2\sigma_1 + p_1\sigma_2).}$$

Then the actual microscopic carrier audit is reduced to the finite test:

$$\boxed{\text{find one admissible longitudinal seed with } \lambda_\parallel \neq 0, \quad \text{and one admissible transverse seed with } (\alpha,\beta) \neq (0,0).}$$

So:

$$\boxed{P_B \to V^\mu \to u^\pm \to (M^\mu)_{ab} \to P_* K_i P_* \to (\lambda_\parallel, \alpha, \beta)}$$

is the full microscopic extraction chain for the actual witness carrier problem.

MINIMAL EXPLICIT TOY EXTRACTION OF THE MICROSCOPIC CARRIER

1. Reflection-symmetric defect background

Take the explicit CP^2-type defect representative

$$z_B(\xi) = \begin{pmatrix} \cos\theta(\xi) \\ 0 \\ \sin\theta(\xi) \end{pmatrix}, \qquad \theta(-\xi) = \theta(\xi), \qquad \theta(\xi) = \theta_0\,\mathrm{sech}(\kappa\xi). \tag{864}$$

Then

$$P_B(\xi) = z_B(\xi)z_B(\xi)^\dagger, \qquad P_B(-\xi) = P_B(\xi), \tag{865}$$

and

$$\partial_\xi z_B(\xi) = \theta'(\xi)e_1(\xi), \qquad \partial_\xi P_B(\xi) = \theta'(\xi)\big(e_1 z_B^\dagger + z_B e_1^\dagger\big). \tag{866}$$

Hence every leading transport kernel linear in the defect gradient factors through $\theta'(\xi)$. :contentReference[oaicite:11]index=11

2. Explicit Q_B-frame

A convenient even orthonormal frame of

$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^{\perp}$$

is

$$e_1(\xi) = \begin{pmatrix} -\sin\theta(\xi) \\ 0 \\ \cos\theta(\xi) \end{pmatrix}, \qquad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}. \tag{867}$$

Both are reflection-even because $\theta(\xi)$ is even. :contentReference[oaicite:12]index=12

3. Explicit even/odd low modes

Choose the concrete even/odd defect profiles

$$f_+(\xi) = \sqrt{\frac{\kappa}{2}}\,\mathrm{sech}(\kappa\xi), \qquad f_-(\xi) = \sqrt{\frac{3\kappa}{2}}\,\tanh(\kappa\xi)\,\mathrm{sech}(\kappa\xi). \tag{868}$$

Then define

$$u_a^+(\xi) = f_+(\xi)e_a(\xi), \qquad u_a^-(\xi) = f_-(\xi)e_a(\xi), \qquad a = 1, 2. \tag{869}$$

These are not merely trial functions: they are exact zero modes of the explicit graded defect operator

$$L_0 = \begin{pmatrix} H_+ & 0 \\ 0 & H_- \end{pmatrix} \otimes \mathbf{1}_2,$$

with

$$H_+ = -\partial_\xi^2 + \kappa^2 - 2\kappa^2\mathrm{sech}^2(\kappa\xi), \qquad H_- = -\partial_\xi^2 + \kappa^2 - 6\kappa^2\mathrm{sech}^2(\kappa\xi).$$

Thus the 2+2 graded low-energy carrier is explicit. :contentReference[oaicite:13]index=13 :contentReference[oaicite:14]index=14

4. Odd transport kernel

Take the leading worldvolume transport kernel in factorized form

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \qquad V^\mu(\xi) = \theta'(\xi)\,\Sigma^\mu, \tag{870}$$

where Σ^μ is a constant Hermitian 2×2 matrix acting on the local Q_B-doublet. Because $\theta(\xi)$ is even,

$$\theta'(-\xi) = -\theta'(\xi), \qquad V^\mu(-\xi) = -V^\mu(\xi). \tag{871}$$

Hence same-parity overlaps vanish, while mixed overlaps are allowed. :contentReference[oaicite:15]index=15 :contentReference[oaicite:16]index=16

5. Mixed overlap integral

The projected mixed overlap matrices are

$$(M^\mu)_{ab} = \int_{-\infty}^{\infty} d\xi\,(u_a^+)^\dagger V^\mu(\xi)u_b^-. \tag{872}$$

Substituting (869) and (870),

$$(M^\mu)_{ab} = \int d\xi\, f_+(\xi)f_-(\xi)\, e_a^\dagger(\xi)\,\theta'(\xi)\Sigma^\mu\, e_b(\xi). \tag{873}$$

If Σ^μ is constant in the chosen Q_B-frame, then

$$M^\mu = I_\theta\,\Sigma^\mu, \qquad I_\theta := \int_{-\infty}^{\infty} d\xi\, f_+(\xi)f_-(\xi)\theta'(\xi). \tag{874}$$

6. Explicit evaluation of I_θ

Using

$$\theta(\xi) = \theta_0 \operatorname{sech}(\kappa\xi), \qquad \theta'(\xi) = -\theta_0\kappa \operatorname{sech}(\kappa\xi)\tanh(\kappa\xi),$$

together with (868), we have

$$\begin{aligned} f_+(\xi)f_-(\xi)\theta'(\xi) &= \sqrt{\frac{\kappa}{2}}\sqrt{\frac{3\kappa}{2}}\operatorname{sech}(\kappa\xi)\ \tanh(\kappa\xi)\operatorname{sech}(\kappa\xi)\ \big(-\theta_0\kappa\operatorname{sech}(\kappa\xi)\tanh(\kappa\xi)\big) \\ &= -\frac{\sqrt{3}\,\kappa}{2}\,\theta_0\kappa\operatorname{sech}^3(\kappa\xi)\tanh^2(\kappa\xi). \end{aligned} \tag{875}$$

Thus

$$I_\theta = -\frac{\sqrt{3}\,\theta_0\kappa^2}{2}\int_{-\infty}^{\infty} d\xi \operatorname{sech}^3(\kappa\xi)\tanh^2(\kappa\xi). \tag{876}$$

Set $x = \kappa\xi$. Then

$$d\xi = \frac{dx}{\kappa},$$

so

$$I_\theta = -\frac{\sqrt{3}\,\theta_0\kappa}{2}\int_{-\infty}^{\infty} dx \operatorname{sech}^3 x\ \tanh^2 x. \tag{877}$$

Using

$$\tanh^2 x = 1 - \operatorname{sech}^2 x,$$

we get

$$\int_{-\infty}^{\infty}\operatorname{sech}^3 x\ \tanh^2 x\,dx = \int_{-\infty}^{\infty}\operatorname{sech}^3 x\,dx - \int_{-\infty}^{\infty}\operatorname{sech}^5 x\,dx = \frac{\pi}{2} - \frac{3\pi}{8} = \frac{\pi}{8}. \tag{878}$$

Therefore

$$\boxed{I_\theta = -\frac{\sqrt{3}\pi}{16}\,\theta_0\kappa.} \tag{879}$$

This matches the explicit formula already recorded in the notes. :contentReference[oaicite:17]index=17
Hence

$$\boxed{\theta_0\kappa \neq 0 \Longrightarrow I_\theta \neq 0.} \tag{880}$$

7. Explicit projected symbol

It follows that

$$\boxed{M^\mu = I_\theta\,\Sigma^\mu.} \tag{881}$$

Hence the reduced principal symbol is

$$A_1(p) = p_\mu M^\mu = I_\theta\,p_\mu\Sigma^\mu. \tag{882}$$

If one chooses

$$\Sigma^1 = v_1\sigma_1, \qquad \Sigma^2 = v_2\sigma_2, \qquad \Sigma^3 = v_3\sigma_3, \qquad v_i \neq 0, \tag{883}$$

then

$$A_1(p) = I_\theta\,(v_1p_1\sigma_1 + v_2p_2\sigma_2 + v_3p_3\sigma_3), \tag{884}$$

and therefore

$$\det A_1(p) = -\,I_\theta^2\,\big(v_1^2p_1^2 + v_2^2p_2^2 + v_3^2p_3^2\big). \tag{885}$$

So for generic $p \neq 0$,

$$\boxed{\det A_1(p) \neq 0 \qquad \text{provided} \qquad I_\theta \neq 0, \quad v_1v_2v_3 \neq 0.} \tag{886}$$

This is exactly the explicit minimal effective Dirac realization described in the notes. :contentReference[oaicite:18]index=18 :contentReference[oaicite:19]index=19

8. Carrier-audit interpretation

The toy extraction proves the following.

$$\boxed{\text{the microscopic extraction chain } P_B \to V^\mu \to u^\pm \to M^\mu \to A_1(p) \text{ can be carried out explicitly in a concrete defect model.}} \tag{887}$$

What it proves:

1. an explicit reflection-symmetric defect background exists;
2. an explicit $2+2$ graded low-mode carrier exists;
3. the odd transport kernel arises naturally from $\partial_\xi P_B$;
4. the mixed overlap integral is explicitly nonzero:

$$I_\theta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa \neq 0;$$

5. with a full-rank Σ^μ, the projected symbol is Dirac.

What it does *not* prove:

$$\boxed{\text{that the full complete microscopic TECT defect linearization is literally identical to this toy model.}} \tag{888}$$

This is exactly the honest limitation stated in the notes. :contentReference[oaicite:20]index=20 :contentReference[oaicite:21]index=21

. Summary

$$\boxed{\textbf{Minimal explicit toy extraction}}$$

Take the reflection-symmetric defect background

$$\boxed{z_B(\xi) = \begin{pmatrix} \cos\theta(\xi) \\ 0 \\ \sin\theta(\xi) \end{pmatrix}, \qquad \theta(\xi) = \theta_0\,\mathrm{sech}(\kappa\xi),}$$

and the even Q_B-frame

$$\boxed{e_1(\xi) = \begin{pmatrix} -\sin\theta(\xi) \\ 0 \\ \cos\theta(\xi) \end{pmatrix}, \qquad e_2(\xi) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.}$$

Choose explicit even/odd low modes

$$\boxed{f_+(\xi) = \sqrt{\frac{\kappa}{2}}\,\mathrm{sech}(\kappa\xi), \qquad f_-(\xi) = \sqrt{\frac{3\kappa}{2}}\,\tanh(\kappa\xi)\mathrm{sech}(\kappa\xi),}$$

$$\boxed{u_a^{\pm}(\xi) = f_{\pm}(\xi)e_a(\xi).}$$

Take the odd transport kernel

$$\boxed{V^\mu(\xi) = \theta'(\xi)\Sigma^\mu, \qquad V^\mu(-\xi) = -V^\mu(\xi).}$$

Then the projected mixed overlap is

$$\boxed{(M^\mu)_{ab} = \int d\xi\, f_+(\xi)f_-(\xi)e_a^\dagger(\xi)V^\mu(\xi)e_b(\xi) = I_\theta\,(\Sigma^\mu)_{ab},}$$

with

$$\boxed{I_\theta = \int d\xi\, f_+ f_- \theta' = -\frac{\sqrt{3}\pi}{16}\,\theta_0\kappa.}$$

Hence

$$\boxed{\theta_0\kappa \neq 0 \Longrightarrow I_\theta \neq 0.}$$

Therefore the reduced symbol is

$$\boxed{A_1(p) = p_\mu M^\mu = I_\theta\, p_\mu \Sigma^\mu.}$$

If

$$\boxed{\Sigma^1 = v_1\sigma_1, \qquad \Sigma^2 = v_2\sigma_2, \qquad \Sigma^3 = v_3\sigma_3, \qquad v_1v_2v_3 \neq 0,}$$

then

$$\boxed{\det A_1(p) = -\,I_\theta^2\left(v_1^2p_1^2 + v_2^2p_2^2 + v_3^2p_3^2\right) \neq 0 \quad \text{for generic } p \neq 0.}$$

So the toy model explicitly realizes:

$$\boxed{P_B \to V^\mu \to u^\pm \to M^\mu \to A_1(p) \sim p_i\sigma_i.}$$

This proves explicit realizability of the projected Dirac mechanism, but not yet its necessity in the full microscopic TECT PDE.

DIRECT MATCHING OF THE TOY EXTRACTION TO THE S2 WITNESS BRANCH

1. S2 witness branch

The natural local symmetry-compatible branch is

$$A_1(p) = \lambda_\| p_\| \sigma_3 + \alpha(p_1\sigma_1 + p_2\sigma_2) + \beta(-p_2\sigma_1 + p_1\sigma_2), \tag{889}$$

with

$$\Lambda = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_\| \end{pmatrix}, \qquad \det\Lambda = \lambda_\|(\alpha^2+\beta^2). \tag{890}$$

Thus S2 nondegeneracy is exactly

$$\boxed{\lambda_\| \neq 0, \qquad (\alpha,\beta)\neq(0,0).} \tag{891}$$

This is the exact local branch already derived in the witness analysis. :contentReference[oaicite:3]index=3

2. Toy extracted symbol

From the explicit toy extraction one obtains

$$(M^\mu)_{ab} = I_\theta(\Sigma^\mu)_{ab}, \qquad I_\theta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa. \tag{892}$$

Hence the projected first-order symbol is

$$A_1^{\mathrm{toy}}(p) = p_\mu M^\mu = I_\theta\, p_\mu \Sigma^\mu. \tag{893}$$

To compare with the S2 branch, choose the adapted shell/worldvolume frame

$$p = (p_1, p_2, p_\|),$$

and write the constant 2×2 matrices in the form

$$\Sigma^1 = g_E\sigma_1 - g_O\sigma_2, \qquad \Sigma^2 = g_O\sigma_1 + g_E\sigma_2, \qquad \Sigma^\| = g_c^{(\mathrm{toy})}\sigma_3. \tag{894}$$

Then

$$\begin{aligned} A_1^{\mathrm{toy}}(p) &= I_\theta\Big[p_1(g_E\sigma_1 - g_O\sigma_2) + p_2(g_O\sigma_1 + g_E\sigma_2) + p_\| g_c^{(\mathrm{toy})}\sigma_3\Big] \\ &= I_\theta\Big[g_c^{(\mathrm{toy})}p_\|\sigma_3 + g_E(p_1\sigma_1 + p_2\sigma_2) + g_O(-p_2\sigma_1 + p_1\sigma_2)\Big]. \end{aligned} \tag{895}$$

Therefore the toy-extracted S2 coefficients are

$$\boxed{\lambda_\|^{\mathrm{toy}} = I_\theta\, g_c^{(\mathrm{toy})}, \qquad \alpha^{\mathrm{toy}} = I_\theta\, g_E, \qquad \beta^{\mathrm{toy}} = I_\theta\, g_O.} \tag{896}$$

3. Comparison with the concrete Class II local formula

For the concrete local Class II transport ansatz, the exact projected witnesses are

$$\boxed{\lambda_\| = \frac{g_c}{2} I_{\theta'^3}, \qquad \alpha = g_E I_\theta, \qquad \beta = g_O I_\theta,} \tag{897}$$

with

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256}\,\theta_0^3\kappa^3, \qquad I_\theta = -\frac{\sqrt{3}\pi}{16}\,\theta_0\kappa. \tag{898}$$

:contentReference[oaicite:4]index=4

Comparing (896) with (897), one finds:

a. Transverse match. The transverse coefficients agree exactly:

$$\boxed{\alpha^{\mathrm{toy}} = \alpha = g_E I_\theta, \qquad \beta^{\mathrm{toy}} = \beta = g_O I_\theta.} \tag{899}$$

b. Longitudinal mismatch in origin. The longitudinal coefficient in the exact Class II transport is not of I_θ-type. Instead it is generated by the primitive commutator channel

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} = \frac{1}{2}\,\theta'(\xi)^2 \sigma_3,$$

which produces the transport seed

$$V^\mu_{(\|)}(\xi) = \frac{1}{2}\,\theta'(\xi)^3\, n^\mu \sigma_3,$$

hence the witness

$$\lambda_\| = \frac{g_c}{2} I_{\theta'^3}.$$

So the exact longitudinal sector lives in a distinct primitive channel.

4. Interpretation

The toy extraction therefore proves the following.

$$\boxed{\text{The odd-gradient toy kernel already realizes the full transverse S2 witness sector } (\alpha, \beta).} \tag{900}$$

However:

$$\boxed{\text{the longitudinal witness } \lambda_\| \text{ requires an additional primitive } \sigma_3\text{-type channel,}} \tag{901}$$

namely the commutator-square channel of the concrete Class II transport action.

Thus the toy model is not the full local Class II transport kernel. It is precisely the transverse part of it.

5. Strongest honest consequence

Combining the toy extraction and the exact local Class II formulas, the strongest honest conclusion is:

1. The transverse witness carrier is explicitly realized:

$$(\alpha, \beta) = (g_E I_\theta, g_O I_\theta), \qquad I_\theta \neq 0.$$

2. The longitudinal witness requires a stronger primitive shell channel:

$$\lambda_\| = \frac{g_c}{2} I_{\theta'^3}, \qquad I_{\theta'^3} \neq 0.$$

3. Therefore the full S2 nondegeneracy condition is exactly

$$\boxed{g_c \neq 0, \qquad (g_E, g_O) \neq (0, 0),} \tag{902}$$

but the toy extraction directly proves only the second part, i.e. the survival of the transverse sector.

6. Practical consequence for the carrier audit

The actual microscopic carrier audit is now sharply split into two subaudits:

c. (A) Transverse carrier audit. Verify that the full microscopic transport kernel contains the odd-gradient channel whose low-mode projection produces

$$I_\theta \neq 0, \qquad (\alpha, \beta) \neq (0, 0).$$

d. (B) Longitudinal carrier audit. Verify that the full microscopic transport kernel contains the primitive commutator-square channel whose low-mode projection produces

$$I_{\theta'^3} \neq 0, \qquad \lambda_\parallel \neq 0.$$

Thus the toy model already solves half of the microscopic witness-carrier problem explicitly: the transverse half. What remains is the explicit microscopic realization of the longitudinal primitive channel.

. Summary

$$\boxed{\textbf{Direct matching of toy extraction to the S2 witness branch}}$$

The S2 branch is

$$\boxed{A_1(p) = \lambda_\parallel p_\parallel \sigma_3 + \alpha(p_1\sigma_1 + p_2\sigma_2) + \beta(-p_2\sigma_1 + p_1\sigma_2),}$$

with

$$\boxed{\det \Lambda = \lambda_\parallel(\alpha^2 + \beta^2).}$$

From the toy extraction,

$$\boxed{M^\mu = I_\theta\, \Sigma^\mu, \qquad I_\theta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa.}$$

Choosing

$$\boxed{\Sigma^1 = g_E\sigma_1 - g_O\sigma_2, \qquad \Sigma^2 = g_O\sigma_1 + g_E\sigma_2, \qquad \Sigma^\parallel = g_c^{(\mathrm{toy})}\sigma_3,}$$

gives

$$\boxed{A_1^{\mathrm{toy}}(p) = I_\theta\Big[g_c^{(\mathrm{toy})} p_\parallel \sigma_3 + g_E(p_1\sigma_1 + p_2\sigma_2) + g_O(-p_2\sigma_1 + p_1\sigma_2)\Big].}$$

Hence the toy coefficients are

$$\boxed{\lambda_\parallel^{\mathrm{toy}} = I_\theta g_c^{(\mathrm{toy})}, \qquad \alpha^{\mathrm{toy}} = I_\theta g_E, \qquad \beta^{\mathrm{toy}} = I_\theta g_O.}$$

The exact local Class II coefficients are

$$\boxed{\lambda_\parallel = \frac{g_c}{2} I_{\theta'^3}, \qquad \alpha = g_E I_\theta, \qquad \beta = g_O I_\theta.}$$

Therefore:

$$\boxed{\alpha, \beta \text{ match exactly between the toy extraction and the exact local Class II formula.}}$$

But:

$$\boxed{\lambda_\parallel \text{ does } \textit{not} \text{ come from the toy } I_\theta\text{-channel; it comes from the distinct primitive } I_{\theta'^3}\text{-channel.}}$$

So the toy extraction explicitly proves the transverse S2 witness sector, while the longitudinal witness still requires the primitive commutator-square transport channel.

ACTUAL MICROSCOPIC CARRIER AUDIT: TRANSVERSE SOLVED, LONGITUDINAL REMAINING

1. The minimal witness criterion

The full local Weyl/Dirac witness problem is reduced to the existence of:

1. one admissible longitudinal witness,
2. one admissible transverse witness.

Equivalently, in the shell-adapted S2 branch,

$$\boxed{\lambda_{\parallel} \neq 0, \qquad (\alpha, \beta) \neq (0,0).} \tag{903}$$

This is the exact finite witness reduction of the local Weyl problem. :contentReference[oaicite:5]index=5 :contentReference[oaicite:6]index=6

2. Concrete explicit seed space

Inside the concrete Class II seed subspace

$$V_{\text{seed}} = \text{span}\Big\{ N_{\text{seed}}^{(\|)}, N_{\text{seed}}^{(\perp,E)}, N_{\text{seed}}^{(\perp,O)} \Big\},$$

the projected witnesses are exactly

$$\boxed{\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \qquad \alpha = c_E I_{\theta}, \qquad \beta = c_O I_{\theta}.} \tag{904}$$

Therefore the reduced S2 symbol is nondegenerate iff

$$\boxed{c_{\parallel} \neq 0, \qquad (c_E, c_O) \neq (0,0).} \tag{905}$$

This means that inside the concrete ansatz space itself, the finite witness problem is already exactly closed. :contentReference[oaicite:7]index=7 :contentReference[oaicite:8]index=8

3. Explicit transverse solution

The explicit transverse-even seed is

$$V^{\mu}_{(\perp,E)}(\xi) = \theta'(\xi)\, E^{\mu}{}_{a} \sigma_a, \tag{906}$$

for which

$$\lambda_{\parallel} = 0, \qquad \alpha = I_{\theta}, \qquad \beta = 0. \tag{907}$$

Hence

$$T_{\perp} = \begin{pmatrix} I_{\theta} & 0 \\ 0 & I_{\theta} \end{pmatrix}, \qquad \det T_{\perp} = I_{\theta}^2 > 0. \tag{908}$$

So an explicit transverse witness already exists. :contentReference[oaicite:9]index=9

Likewise, the optional transverse-odd seed

$$V^{\mu}_{(\perp,O)}(\xi) = \theta'(\xi)\, J^{\mu}{}_{a} \sigma_a$$

gives

$$\lambda_{\parallel} = 0, \qquad \alpha = 0, \qquad \beta = I_{\theta} \neq 0,$$

so the transverse channel is doubly realizable. :contentReference[oaicite:10]index=10

Thus the transverse half of the carrier audit is already explicitly solved:

$$\boxed{(\alpha, \beta) \neq (0,0) \quad \text{is explicitly realizable.}} \tag{909}$$

4. Explicit longitudinal requirement

The explicit longitudinal seed is

$$V^{\mu}_{(\parallel)}(\xi) = \frac{1}{2}\,\theta'(\xi)^3\, n^{\mu}\sigma_3, \tag{910}$$

for which

$$\lambda_{\parallel} = \frac{1}{2} I_{\theta'^3}, \qquad \alpha = 0, \qquad \beta = 0. \tag{911}$$

Since $I_{\theta'^3} \neq 0$, an explicit longitudinal witness exists *inside the concrete seed space.* :contentReference[oaicite:11]index=11

However, unlike the transverse toy extraction built directly from the odd-gradient channel $\theta'(\xi)\Sigma^{\mu}$, the longitudinal witness requires the distinct primitive channel

$$\theta'(\xi)^3\,\sigma_3.$$

Therefore the actual remaining microscopic question is no longer whether a longitudinal witness can exist in principle, but whether the full microscopic TECT linearization actually generates this primitive channel in the physically relevant defect background.

Equivalently:

$$\boxed{\text{the unresolved microscopic carrier question is precisely the survival of the longitudinal primitive channel.}} \tag{912}$$

5. Embedding statement

The embedding theorem already proves that if the true microscopic TECT transport sector contains one of the minimal viable UV classes (Class I or Class II), then there exists a three-dimensional microscopic tensor subspace

$$V^{(3)}_{\mathrm{mic}} \subset V_{\mathrm{mic}}$$

whose reduced defect/projector image is isomorphic to

$$V_{\mathrm{seed}}.$$

Equivalently, the longitudinal seed witness and the transverse seed doublet can both be realized by admissible microscopic transport tensors. :contentReference[oaicite:12]index=12

Therefore the remaining question is not abstract compatibility. It is the concrete yes/no question:

$$\boxed{\text{does the full complete microscopic TECT action actually realize Class I or Class II in the relevant defect background?}} \tag{913}$$

6. Strongest honest carrier-audit theorem

We can now state the sharpest honest result.

[Actual microscopic carrier audit: transverse solved, longitudinal remaining] Assume the relevant defect background admits the explicit odd-gradient transport extraction described earlier, so that the transverse seed channel produces a nonzero overlap

$$I_{\theta} \neq 0.$$

Then:

1. the transverse witness carrier survives explicitly in the true microscopic transport extraction;
2. hence the transverse nondegeneracy condition

   $$(\alpha, \beta) \neq (0, 0)$$

   is no longer the open part of the carrier problem;

3. the only remaining microscopic carrier question is whether the full TECT linearization contains the primitive longitudinal seed channel producing

$$\lambda_{\|} = \frac{1}{2} c_{\|} I_{\theta'^3} \neq 0.$$

Therefore the full witness-carrier audit is reduced to one residual yes/no question:

$$\boxed{\text{does the physically relevant microscopic TECT linearization realize a nonzero longitudinal primitive channel?}}$$

7. Honest final status

Thus the exact logical status is:

$$\boxed{\text{the carrier problem is no longer “three channels open”; it is “transverse solved, longitudinal remaining”.}} \tag{914}$$

This is a substantial strengthening of the previous frontier statement, because the surviving open part is now only one primitive shell-relevant channel rather than the whole witness tensor problem.

. Summary

$$\boxed{\textbf{Actual microscopic carrier audit: transverse solved, longitudinal remaining}}$$

The full local witness criterion is

$$\boxed{\lambda_{\|} \neq 0, \qquad (\alpha, \beta) \neq (0,0).}$$

Inside the explicit seed space,

$$\boxed{\lambda_{\|} = \frac{1}{2} c_{\|} I_{\theta'^3}, \qquad \alpha = c_E I_\theta, \qquad \beta = c_O I_\theta.}$$

Explicit transverse seed:

$$\boxed{V^\mu_{(\perp,E)}(\xi) = \theta'(\xi)\, E^\mu{}_a \sigma_a}$$

gives

$$\boxed{\lambda_{\|} = 0, \qquad \alpha = I_\theta, \qquad \beta = 0, \qquad \det T_\perp = I_\theta^2 > 0.}$$

Explicit longitudinal seed:

$$\boxed{V^\mu_{(\|)}(\xi) = \frac{1}{2}\, \theta'(\xi)^3\, n^\mu \sigma_3}$$

gives

$$\boxed{\lambda_{\|} = \frac{1}{2} I_{\theta'^3} \neq 0, \qquad \alpha = \beta = 0.}$$

Therefore:

$$\boxed{\text{the transverse witness carrier is explicitly solved,}}$$

because the odd-gradient extraction already realizes $I_\theta \neq 0$;
while

$$\boxed{\text{the only remaining microscopic carrier question is the survival of the longitudinal primitive channel } \theta'^3\sigma_3.}$$

So the witness-carrier frontier is now reduced to one residual yes/no question:

$$\boxed{\text{does the full microscopic TECT linearization realize a nonzero longitudinal primitive channel?}}$$

MICROSCOPIC ORIGIN OF THE LONGITUDINAL PRIMITIVE CHANNEL $\theta'^3\sigma_3$

1. What still remains open

The current carrier-audit status is:

$$\text{transverse solved, longitudinal remaining.}$$

Inside the explicit seed space one already has

$$\lambda_\parallel = \frac{1}{2}c_\parallel I_{\theta'^3}, \qquad \alpha = c_E I_\theta, \qquad \beta = c_O I_\theta, \tag{915}$$

with

$$I_\theta \neq 0, \qquad I_{\theta'^3} \neq 0.$$

So the unresolved microscopic question is not whether a longitudinal witness can exist in principle, but:

$$\boxed{\text{what is the natural microscopic origin of the primitive longitudinal seed } V^\mu_{(\parallel)}(\xi) = \frac{1}{2}\,\theta'(\xi)^3\,n^\mu\sigma_3?} \tag{916}$$

2. Primitive longitudinal seed already identified in the explicit seed space

The explicit longitudinal seed recorded in the concrete Class II construction is

$$\boxed{V^\mu_{(\parallel)}(\xi) := \theta'(\xi)\,(\Pi_Q\mathcal{T}_\xi^2\Pi_Q)_{\mathrm{tr}}\,n^\mu = \frac{1}{2}\,\theta'(\xi)^3\,n^\mu\sigma_3.} \tag{917}$$

Its projected witnesses are

$$\lambda_\parallel = \frac{1}{2}I_{\theta'^3} \neq 0, \qquad \alpha = 0, \qquad \beta = 0. \tag{918}$$

So the only remaining issue is microscopic survival of the operator

$$(\Pi_Q\mathcal{T}_\xi^2\Pi_Q)_{\mathrm{tr}}.$$

:contentReference[oaicite:2]index=2

3. Geometric interpretation of the channel

Let $P_B(\xi)$ be the reflection-symmetric defect projector background, and

$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^\perp$$

its rank-two tangent bundle. The first-order transport kernel is already known to arise from the defect gradient:

$$L_1 = -iV^\mu(\xi)\partial_{y^\mu}, \qquad V^\mu(\xi) \sim (\partial_\xi P_B)\, T^\mu, \qquad V^\mu(-\xi) = -V^\mu(\xi).$$

This generates the transverse odd-gradient channel. :contentReference[oaicite:3]index=3

To obtain a longitudinal witness, one needs a σ_3-type traceless operator on the internal Q_B-doublet. The natural source is the *quadratic transverse response* of the defect geometry in the Q_B-sector:

$$\mathcal{T}_\xi : \Gamma(Q_B) \to \Gamma(Q_B), \qquad \mathcal{T}_\xi^2 : \Gamma(Q_B) \to \Gamma(Q_B). \tag{919}$$

Projecting to Q_B and removing the scalar part gives

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} = \Pi_Q \mathcal{T}_\xi^2 \Pi_Q - \frac{1}{2}\mathrm{Tr}_{Q_B}\big(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q\big)\mathbf{1}_2. \tag{920}$$

This is the unique lowest-order internal traceless scalar compatible with the shell-adapted longitudinal channel.

4. Why σ_3 appears naturally

In the explicit defect frame

$$Q_B = \mathrm{span}_{\mathbb{C}}\{e_1, e_2\},$$

the two tangent directions are not geometrically equivalent:

- e_1 is the direction directly tied to the defect bending through $\partial_\xi z_B = \theta' e_1$,
- e_2 is the orthogonal internal spectator direction.

Therefore the quadratic response operator $\mathcal{T}_\xi^2$ need not act isotropically on (e_1, e_2). Its traceless part naturally distinguishes the two directions, and in the adapted frame this distinction is represented by

$$\boxed{(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} \propto \sigma_3.} \tag{921}$$

For the explicit seed construction, the proportionality is

$$\boxed{(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} = \frac{1}{2}\,\theta'(\xi)^2 \sigma_3.} \tag{922}$$

Then multiplying by the already-present odd defect gradient factor gives

$$\boxed{\theta'(\xi)\,(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} = \frac{1}{2}\,\theta'(\xi)^3\,\sigma_3.} \tag{923}$$

This is exactly the primitive longitudinal seed. :contentReference[oaicite:4]index=4

5. Parity and why this channel is allowed

For a reflection-symmetric defect,

$$\theta(-\xi) = \theta(\xi), \qquad \theta'(-\xi) = -\theta'(\xi).$$

Hence

$$\theta'(\xi)^2 \quad \text{is even}, \qquad \theta'(\xi)^3 \quad \text{is odd}.$$

Therefore:

- $(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} \sim \theta'^2 \sigma_3$ is an even internal operator,
- multiplying by $\theta'(\xi)$ makes the full transport kernel odd.

Thus the mixed overlap with even/odd low modes is parity-allowed:

$$u_a^+(\xi) \text{ even}, \qquad u_b^-(\xi) \text{ odd}, \qquad V^\mu_{(\parallel)}(\xi) \text{ odd} \Longrightarrow (u_a^+)^\dagger V^\mu_{(\parallel)} u_b^- \text{ is even}. \tag{924}$$

So the primitive longitudinal channel is not only symmetry-compatible; it is exactly of the parity type required to survive the low-mode projection.

6. Projected witness coefficient

With

$$V^\mu_{(\|)}(\xi) = \frac{1}{2}\,\theta'(\xi)^3\, n^\mu \sigma_3,$$

the projected mixed symbol produces

$$\lambda_\| = \frac{1}{2}\int d\xi\, f_+(\xi) f_-(\xi)\theta'(\xi)^3 = \frac{1}{2} I_{\theta'^3}. \tag{925}$$

The explicit defect calculation gives

$$\boxed{I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256}\,\theta_0^3\kappa^3 \neq 0} \tag{926}$$

for nontrivial defect amplitude $\theta_0\kappa \neq 0$. :contentReference[oaicite:5]index=5 Hence

$$\boxed{\lambda_\| \neq 0} \tag{927}$$

provided the primitive channel actually survives in the full microscopic transport extraction.

7. Strongest honest microscopic statement

Therefore the sharpest honest microscopic statement is:

$$\boxed{\text{the longitudinal primitive channel is geometrically natural if the defect } Q_B\text{-sector has a non-isotropic quadratic response whose}} \tag{928}$$

In that case, the odd defect gradient automatically upgrades it to the transport seed

$$\frac{1}{2}\,\theta'^3 n^\mu \sigma_3,$$

which yields $\lambda_\| \neq 0$.

So the remaining carrier question is not "invent a mysterious extra term," but the much sharper audit:

$$\boxed{\text{does the full microscopic TECT linearization produce a nonzero traceless quadratic } Q_B\text{-response channel?}} \tag{929}$$

8. Honest final status

What is already effectively established:

1. the transverse odd-gradient channel is explicitly realized;
2. the longitudinal primitive seed has an explicit geometrically natural form;
3. its projected coefficient is explicitly nonzero once the channel survives.

What remains open:

$$\boxed{\text{the actual survival of } (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} \text{ in the full microscopic TECT transport extraction.}} \tag{930}$$

Thus the entire microscopic witness-carrier frontier is now reduced to one residual yes/no question about one primitive longitudinal channel.

. Summary

$$\boxed{\textbf{Microscopic origin of the longitudinal primitive channel}}$$

The explicit longitudinal seed is

$$\boxed{V^{\mu}_{(\parallel)}(\xi) = \theta'(\xi)\,(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}}\, n^\mu = \frac{1}{2}\,\theta'(\xi)^3\, n^\mu \sigma_3.}$$

So the primitive microscopic ingredient is

$$\boxed{(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}}.}$$

In the adapted Q_B-frame, the defect-bending direction e_1 and the spectator direction e_2 are not equivalent, so the quadratic Q_B-response need not be isotropic. Its traceless part naturally takes the form

$$\boxed{(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} \propto \sigma_3.}$$

For the explicit seed construction,

$$\boxed{(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} = \frac{1}{2}\,\theta'(\xi)^2 \sigma_3,}$$

hence

$$\boxed{\theta'(\xi)\,(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} = \frac{1}{2}\,\theta'(\xi)^3\, \sigma_3.}$$

This produces the longitudinal witness

$$\boxed{\lambda_\parallel = \frac{1}{2} I_{\theta'^3}, \qquad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256}\,\theta_0^3 \kappa^3 \neq 0.}$$

Therefore the remaining microscopic question is sharply reduced to:

$$\boxed{\text{does the full microscopic TECT linearization produce a nonzero traceless quadratic } Q_B\text{-response channel } (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}}?}$$

So the carrier frontier is now:

$$\boxed{\text{transverse explicit, longitudinal reduced to one primitive yes/no channel.}}$$

MINIMAL SUFFICIENT CONDITION FOR SURVIVAL OF THE LONGITUDINAL PRIMITIVE CHANNEL

1. The remaining microscopic question

The carrier-audit status is now:

$$\text{transverse solved, longitudinal remaining.}$$

Inside the explicit seed space one has

$$\lambda_\parallel = \frac{1}{2} c_\parallel I_{\theta'^3}, \qquad \alpha = c_E I_\theta, \qquad \beta = c_O I_\theta. \tag{931}$$

Hence the only residual microscopic question is whether the full microscopic TECT linearization actually realizes the longitudinal seed direction.

2. Primitive longitudinal seed

The explicit longitudinal seed is

$$\boxed{V^{\mu}_{(\|)}(\xi) = \theta'(\xi)\, (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\rm tr}\, n^{\mu}.} \tag{932}$$

In the explicit representative used in the seed construction,

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\rm tr} = \frac{1}{2}\, \theta'(\xi)^2\, \sigma_3, \tag{933}$$

so

$$V^{\mu}_{(\|)}(\xi) = \frac{1}{2}\, \theta'(\xi)^3\, n^{\mu} \sigma_3. \tag{934}$$

Projecting to the even/odd low modes gives

$$\lambda_{\|} = \frac{1}{2} \int d\xi\, f_+(\xi) f_-(\xi) \theta'(\xi)^3 = \frac{1}{2} I_{\theta'^3}. \tag{935}$$

Since

$$I_{\theta'^3} \neq 0 \qquad (\theta_0 \kappa \neq 0), \tag{936}$$

the longitudinal witness is nonzero whenever the primitive channel survives. :contentReference[oaicite:3]index=3

3. Minimal sufficient condition: operator form

Therefore the weakest natural sufficient condition is:

$$\boxed{(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\rm tr} \not\equiv 0 \quad \text{and} \quad \int d\xi\, f_+(\xi) f_-(\xi)\, \theta'(\xi)\, (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\rm tr} \neq 0.} \tag{937}$$

This implies that the microscopic transport extraction contains a nonzero longitudinal primitive seed, hence

$$\boxed{\lambda_{\|} \neq 0.} \tag{938}$$

4. Geometric sufficient criterion

A more geometric sufficient condition is:

$$\boxed{\text{the quadratic } Q_B\text{-response is anisotropic in the adapted defect frame } (e_1, e_2),} \tag{939}$$

meaning that the two internal directions are not eigen-equivalent at quadratic order, so that

$$\Pi_Q \mathcal{T}_\xi^2 \Pi_Q \neq \frac{1}{2} \mathrm{Tr}_{Q_B} (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)\, \mathbf{1}_2.$$

Equivalently,

$$\boxed{(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\rm tr} \neq 0.} \tag{940}$$

If this anisotropic traceless quadratic response is aligned with the shell axis through the factor n^{μ}, then the odd defect gradient upgrades it to the first-order transport kernel (932), and the parity structure guarantees that the even/odd low-mode overlap is allowed.

So the geometric content of the sufficient condition is simply:

$$\boxed{\text{non-isotropic quadratic } Q_B\text{-response} + \text{odd defect gradient} + \text{nonzero low-mode overlap} \Longrightarrow \lambda_{\|} \neq 0.} \tag{941}$$

5. Practical finite test

Hence the actual microscopic longitudinal audit can be implemented as the finite three-step test:

a. *(L1)* Compute the projected quadratic response

$$\Pi_Q \mathcal{T}_\xi^2 \Pi_Q.$$

b. *(L2)* Remove the scalar part and check whether the traceless part vanishes:

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\rm tr} \stackrel{?}{=} 0.$$

c. *(L3)* If it is nonzero, compute the one-dimensional overlap

$$\boxed{J_\| := \int d\xi\, f_+(\xi) f_-(\xi)\, \theta'(\xi)\, (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\rm tr}.}$$

If

$$\boxed{J_\| \neq 0,} \tag{942}$$

then the longitudinal primitive channel survives and

$$\lambda_\| \neq 0.$$

6. Strongest honest conclusion

Therefore the sharpest minimal sufficient condition for longitudinal primitive survival is:

$$\boxed{(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\rm tr} \not\equiv 0 \quad \text{and} \quad J_\| \neq 0.} \tag{943}$$

This is much weaker than proving the full Class II completion. It is only a finite operator-and-overlap check. Inside the explicit seed representative, this condition is satisfied automatically because

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\rm tr} = \frac{1}{2}\theta'(\xi)^2 \sigma_3, \qquad J_\| = \frac{1}{2} I_{\theta'^3} \neq 0.$$

So the open frontier is not conceptual. It is the verification that the full microscopic TECT linearization obeys this same finite condition.

. Summary

Minimal sufficient condition for survival of the longitudinal primitive channel

The primitive longitudinal seed is

$$\boxed{V^\mu_{(\|)}(\xi) = \theta'(\xi)\, (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\rm tr}\, n^\mu.}$$

Therefore a minimal sufficient condition for longitudinal survival is:

$$\boxed{(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\rm tr} \not\equiv 0}$$

and

$$\boxed{J_\| := \int d\xi\, f_+(\xi) f_-(\xi)\, \theta'(\xi)\, (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\rm tr} \neq 0.}$$

Then

$$\boxed{\lambda_\parallel \neq 0.}$$

In the explicit seed representative,

$$\boxed{(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\rm tr} = \frac{1}{2}\,\theta'(\xi)^2\,\sigma_3,}$$

so

$$\boxed{V^\mu_{(\parallel)}(\xi) = \frac{1}{2}\,\theta'(\xi)^3\,n^\mu\sigma_3, \qquad J_\parallel = \frac{1}{2} I_{\theta'^3} \neq 0.}$$

Hence the remaining microscopic frontier is reduced to one finite test:

$$\boxed{\text{does the full microscopic TECT linearization produce a nonzero traceless quadratic } Q_B\text{-response whose low-mode overlap } J_\parallel \text{ is}}$$

This is the weakest practical sufficient condition for closing the longitudinal half of the witness-carrier audit.

DIRAC CLOSURE STATUS UPDATE: TRANSVERSE FIXED, LONGITUDINAL REDUCED TO ONE FINITE TEST

1. Previous frontier formulation

Previously, the local Dirac closure theorem required the following carrier-level microscopic input:

$$\boxed{\text{the full microscopic TECT transport space must contain one admissible longitudinal witness direction and one admissible trans}} \tag{944}$$

Equivalently, the local S2 witness criterion required

$$\boxed{\lambda_\parallel \neq 0, \qquad (\alpha,\beta) \neq (0,0).} \tag{945}$$

At that stage, both the longitudinal and transverse halves of the witness-carrier problem appeared open.

2. Explicit transverse resolution

Inside the explicit seed space, one has

$$\lambda_\parallel = \frac{1}{2} c_\parallel I_{\theta'^3}, \qquad \alpha = c_E I_\theta, \qquad \beta = c_O I_\theta. \tag{946}$$

An explicit transverse-even seed exists:

$$V^\mu_{(\perp,E)}(\xi) = \theta'(\xi) E^\mu{}_a \sigma_a, \tag{947}$$

for which

$$\lambda_\parallel = 0, \qquad \alpha = I_\theta, \qquad \beta = 0.$$

Likewise an explicit transverse-odd seed exists:

$$V^\mu_{(\perp,O)}(\xi) = \theta'(\xi) J^\mu{}_a \sigma_a, \qquad \lambda_\parallel = 0, \qquad \alpha = 0, \qquad \beta = I_\theta.$$

Moreover, the explicit toy extraction evaluates

$$\boxed{I_\theta = -\frac{\sqrt{3}\pi}{16}\,\theta_0\kappa \neq 0 \qquad (\theta_0\kappa \neq 0).} \tag{948}$$

Hence the transverse witness channel is no longer hypothetical: it is explicitly realized.
Therefore:

$$\boxed{(\alpha,\beta) \neq (0,0) \quad \text{is already explicitly solved at the carrier level.}} \tag{949}$$

3. Longitudinal primitive channel

The explicit longitudinal seed is

$$\boxed{V^{\mu}_{(\|)}(\xi) = \theta'(\xi)\,(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\rm tr}\, n^{\mu}.} \tag{950}$$

In the explicit seed representative,

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\rm tr} = \frac{1}{2}\,\theta'(\xi)^2 \sigma_3, \tag{951}$$

so

$$V^{\mu}_{(\|)}(\xi) = \frac{1}{2}\,\theta'(\xi)^3\, n^{\mu} \sigma_3. \tag{952}$$

Projecting to the graded low modes gives

$$\boxed{\lambda_{\|} = \frac{1}{2} I_{\theta'^3}, \qquad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256}\,\theta_0^3 \kappa^3 \neq 0.} \tag{953}$$

Thus the longitudinal channel is also explicitly realizable *inside the seed representative.* What remains open is more specific: whether the full microscopic TECT transport extraction actually preserves this primitive channel.

4. Minimal remaining longitudinal test

The weakest practical sufficient condition for longitudinal survival is:

$$\boxed{(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\rm tr} \not\equiv 0 \quad \text{and} \quad J_{\|} \neq 0,} \tag{954}$$

where

$$\boxed{J_{\|} := \int d\xi\, f_+(\xi) f_-(\xi)\, \theta'(\xi)\, (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\rm tr}.} \tag{955}$$

If (954) holds, then

$$\boxed{\lambda_{\|} \neq 0.} \tag{956}$$

So the longitudinal half of the witness-carrier problem has been reduced to one primitive finite operator-and-overlap test.

5. Updated carrier-audit status

Combining Sections 2–4 gives the sharpest updated statement:

$$\boxed{\text{the carrier audit is no longer “transverse + longitudinal open”.}} \tag{957}$$

Instead:

$$\boxed{\text{transverse carrier survival is explicitly solved,}} \tag{958}$$

and

$$\boxed{\text{the only remaining carrier question is the survival of the primitive longitudinal channel.}} \tag{959}$$

Equivalently, the entire microscopic carrier frontier is reduced to the single finite yes/no test (954).

6. Updated local Dirac closure theorem

The local Dirac closure theorem may therefore be sharpened as follows:
[Updated local Dirac closure status] At the carrier level, the concrete Class II local Dirac closure problem is reduced to:

1. **already fixed:** transverse witness survival,
$$(\alpha, \beta) \neq (0, 0),$$
via the explicit odd-gradient extraction with $I_\theta \neq 0$;
2. **remaining finite test:** longitudinal primitive survival,
$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} \not\equiv 0 \quad \text{and} \quad J_\parallel \neq 0 \Longrightarrow \lambda_\parallel \neq 0.$$

Hence the carrier portion of the local Dirac theorem is now reduced to one residual primitive yes/no test rather than a full open tensor-space audit.

7. Honest final status

Thus the strongest honest status update is:

$$\boxed{\text{Local Dirac closure is substantially sharper than before:}} \tag{960}$$

$$\boxed{\text{carrier problem } = \text{transverse fixed} + \text{one remaining longitudinal primitive test.}} \tag{961}$$

This is a real closure improvement, because one entire half of the previous carrier frontier has already been explicitly settled.

. Summary

$$\boxed{\textbf{Dirac closure status update}}$$

Previously, the witness-carrier frontier looked like:

$$\boxed{\lambda_\parallel \neq 0 \quad \text{and} \quad (\alpha, \beta) \neq (0, 0)}$$

with both halves apparently open.
Now the status is sharper.
Inside the explicit seed space:

$$\boxed{\lambda_\parallel = \frac{1}{2} c_\parallel I_{\theta'^3}, \qquad \alpha = c_E I_\theta, \qquad \beta = c_O I_\theta.}$$

The transverse half is explicitly solved because:

$$\boxed{I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \neq 0,}$$

so explicit transverse seeds already realize

$$\boxed{(\alpha, \beta) \neq (0, 0).}$$

The only remaining microscopic carrier question is the longitudinal primitive channel:

$$\boxed{V^{\mu}_{(\|)}(\xi) = \theta'(\xi)(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} n^{\mu}.}$$

Its weakest practical sufficient survival test is:

$$\boxed{(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} \not\equiv 0 \quad \text{and} \quad J_{\|} = \int d\xi\, f_+ f_-\, \theta'(\xi)\, (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} \neq 0.}$$

Then

$$\boxed{\lambda_{\|} \neq 0.}$$

So the updated carrier-audit status is:

$$\boxed{\text{transverse fixed, longitudinal reduced to one finite primitive test.}}$$

UPDATED LOCAL DIRAC CLOSURE THEOREM: TRANSVERSE FIXED, LONGITUDINAL ONE-TEST REMAINING

1. Original local Dirac closure structure

The local Dirac closure problem was reduced to five finite ingredients:

1. actual microscopic witness-carrier realization,
2. opposite-valley witness matching,
3. zero constant mixing $V(0) = 0$,
4. remote spectral gap $\Delta > 0$,
5. mass protection.

Previously, Step 1 was still phrased as the existence of both:

$$\lambda_{\|} \neq 0, \qquad (\alpha, \beta) \neq (0, 0).$$

So the carrier problem looked like a two-sided open problem.

2. Updated carrier status

Inside the explicit seed space, the S2 witness coefficients are

$$\lambda_{\|} = \frac{1}{2} c_{\|} I_{\theta'^3}, \qquad \alpha = c_E I_\theta, \qquad \beta = c_O I_\theta. \tag{962}$$

The transverse channel is already explicitly realized by the odd-gradient extraction, since

$$I_\theta \neq 0 \tag{963}$$

for a nontrivial defect profile. Therefore explicit admissible transverse seeds exist with

$$(\alpha, \beta) \neq (0, 0). \tag{964}$$

Hence the transverse half of the carrier problem is no longer open.
The only remaining microscopic carrier issue is the longitudinal primitive channel

$$V^{\mu}_{(\|)}(\xi) = \theta'(\xi)\, (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}}\, n^{\mu}, \tag{965}$$

together with the overlap test

$$J_{\parallel} := \int d\xi \, f_{+}(\xi) f_{-}(\xi) \, \theta'(\xi) \, (\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\mathrm{tr}}. \tag{966}$$

A minimal sufficient condition is

$$(\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\mathrm{tr}} \not\equiv 0, \qquad J_{\parallel} \neq 0, \tag{967}$$

which implies

$$\lambda_{\parallel} \neq 0. \tag{968}$$

Therefore Step 1 is now sharpened from

two-channel open carrier problem

to

transverse fixed + one remaining longitudinal primitive test.

3. Updated theorem statement

We now restate the local Dirac closure theorem in its sharpened form.
[Updated Local Dirac Closure Theorem] Assume the following.
a. *(S1) Carrier input: updated form.* The transverse witness carrier is already present, i.e.

$$(\alpha, \beta) \neq (0, 0) \tag{969}$$

is explicitly realized by the microscopic odd-gradient transport extraction.
In addition, the longitudinal primitive test succeeds:

$$(\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\mathrm{tr}} \not\equiv 0, \qquad J_{\parallel} \neq 0, \tag{970}$$

hence

$$\lambda_{\parallel} \neq 0. \tag{971}$$

b. *(S2) Opposite-valley witness matching.* In the matched opposite-valley frame,

$$(W_{\parallel}^{-}, W_{I}^{-}, W_{J}^{-}) = (W_{\parallel}^{+}, W_{I}^{+}, W_{J}^{+}). \tag{972}$$

c. *(S3) Zero constant mixing.*

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \Longrightarrow V(0) = 0. \tag{973}$$

d. *(S4) Remote spectral gap.* There exists $\Delta > 0$ such that

$$\mathrm{spec}(H_R(p)) \cap (-\Delta, \Delta) = \varnothing \tag{974}$$

for the relevant low-momentum neighborhood, equivalently by a direct Bloch inequality or bounded-perturbation gap stability.
e. *(S5) Mass protection.*

$$A_{\mathrm{res}} \cap \mathrm{span}_{\mathbb{R}}\{\rho_1 \otimes \mathbf{1}_2, \rho_2 \otimes \mathbf{1}_2\} = \{0\}. \tag{975}$$

Then the full microscopic truncation contains an isolated local massless Dirac cone

$$H_{\mathrm{eff}}(q) = \rho_3 \otimes (q_i \sigma_i) + O(|q|^2/\Delta). \tag{976}$$

4. Proof sketch

f. Step A: Weyl nondegeneracy. By (969) and (971), both parts of the S2 nondegeneracy condition hold:

$$\lambda_\parallel \neq 0, \qquad (\alpha, \beta) \neq (0, 0).$$

Hence the selected valley carries a nondegenerate local Weyl symbol.

g. Step B: doubling into Dirac. By (972), the opposite valley carries the matched witness triple, so the two valleys pair into a local doubled Dirac block.

h. Step C: elimination of remote constant hybridization. By (973), no constant Dirac-remote intertwiner exists, so $V(0) = 0$.

i. Step D: spectral isolation. By (974), the remote sector remains gapped by $\Delta > 0$, and the Schur complement only contributes $O(|q|^2/\Delta)$.

j. Step E: masslessness. By (975), no symmetry-allowed constant Dirac mass is present on the isolated doubled block.

Hence the resulting local Dirac cone is spectrally isolated and massless.

5. Strongest honest status

The sharpened theorem shows that the carrier frontier has improved substantially:

$$\boxed{\text{old status: carrier problem} = \text{longitudinal} + \text{transverse both open;}} \tag{977}$$

$$\boxed{\text{new status: carrier problem} = \text{transverse fixed} + \text{one remaining longitudinal primitive test.}} \tag{978}$$

So the full local Dirac closure frontier is no longer spread across a broad witness-carrier ambiguity. It is concentrated in one residual microscopic test:

$$\boxed{(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} \not\equiv 0 \quad \text{and} \quad J_\parallel \neq 0.} \tag{979}$$

6. Exact logical status

What is already effectively closed:

1. transverse witness survival,
2. opposite-valley witness matching criterion,
3. zero-mixing criterion,
4. remote-gap criterion,
5. mass-protection criterion.

What remains open microscopically:

1. actual survival of the longitudinal primitive channel,
2. the final microscopic audits behind Steps 3–5 in the full TECT realization.

Therefore the most accurate current statement is:

$$\boxed{\text{the local Dirac problem is now reduced to a sharpened finite conditional theorem with only one residual primitive carrier test o}} \tag{980}$$

. Summary

$$\boxed{\textbf{Updated Local Dirac Closure Theorem}}$$

The carrier part is no longer fully open.
Inside the explicit seed structure:

$$\boxed{\lambda_{\parallel} = \frac{1}{2} c_{\parallel} I_{\theta'^3}, \qquad \alpha = c_E I_\theta, \qquad \beta = c_O I_\theta.}$$

Now:

$$\boxed{(\alpha, \beta) \neq (0,0) \text{ is already explicitly fixed by the transverse odd-gradient extraction,}}$$

so the transverse half is solved.
The only remaining carrier test is:

$$\boxed{(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} \not\equiv 0 \quad \text{and} \quad J_{\parallel} = \int d\xi \, f_+ f_- \, \theta'(\xi) \, (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} \neq 0,}$$

which implies

$$\boxed{\lambda_{\parallel} \neq 0.}$$

Therefore the updated local Dirac closure inputs are:

$$\boxed{(1) \text{ transverse fixed}, \qquad (2) \text{ one remaining longitudinal primitive test},}$$

together with

$$\boxed{(3) \text{ opposite-valley matching}, \quad (4) \text{ zero constant mixing}, \quad (5) \text{ remote gap}, \quad (6) \text{ mass protection}.}$$

If all of these hold, then

$$\boxed{H_{\mathrm{eff}}(q) = \rho_3 \otimes (q_i \sigma_i) + O(|q|^2/\Delta)}$$

is an isolated local massless Dirac cone.
So the new status is:

$$\boxed{\text{transverse fixed, longitudinal one-test remaining.}}$$

EXPLICIT CALCULATION PROTOCOL FOR $(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}}$ AND $J_{\parallel}$

1. Goal

The only remaining primitive carrier test on the witness side is:

$$\boxed{(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} \not\equiv 0 \quad \text{and} \quad J_{\parallel} \neq 0.} \tag{981}$$

Here

$$J_{\parallel} := \int d\xi \, f_+(\xi) f_-(\xi) \, \theta'(\xi) \, (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}}. \tag{982}$$

If (1002) holds, then the longitudinal primitive channel survives and

$$\lambda_{\parallel} \neq 0.$$

2. Input data

Fix:

1. a reflection-symmetric defect background $P_B(\xi)$;
2. the corresponding rank-two tangent bundle
$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^{\perp};$$
3. a local orthonormal Q_B-frame
$$e_1(\xi),\ e_2(\xi);$$
4. the graded low-mode profiles
$$u_a^+(\xi) = f_+(\xi)e_a(\xi), \qquad u_a^-(\xi) = f_-(\xi)e_a(\xi), \qquad a = 1, 2.$$

The concrete toy representative already provides a natural choice of $f_\pm$ and e_a, but the protocol below is independent of the exact choice.

3. Step 1: define the microscopic transverse-response operator

Let $\mathcal{T}_\xi$ be the microscopic transverse-response operator induced by the linearized TECT dynamics on sections of Q_B:

$$\mathcal{T}_\xi : \Gamma(Q_B) \to \Gamma(Q_B). \tag{983}$$

Operationally, $\mathcal{T}_\xi$ is the ξ-dependent Q_B-sector operator obtained after:

1. linearizing the microscopic TECT equation around $P_B(\xi)$,
2. isolating the transverse/internal doublet sector,
3. restricting the resulting operator to the rank-two bundle Q_B.

4. Step 2: compute the 2×2 matrix of $\mathcal{T}_\xi$

In the local orthonormal frame $\{e_1, e_2\}$, define

$$(T_\xi)_{ab} := \langle e_a(\xi), \mathcal{T}_\xi e_b(\xi)\rangle, \qquad a, b = 1, 2. \tag{984}$$

Thus T_ξ is an explicit 2×2 Hermitian matrix-valued function of ξ.
This is the first concrete output of the protocol.

5. Step 3: compute the quadratic response

Now square the operator in the Q_B-frame:

$$T_\xi^{(2)} := T_\xi^2. \tag{985}$$

Then extract the traceless part

$$\boxed{S_\xi := (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} = T_\xi^2 - \frac{1}{2}\mathrm{Tr}(T_\xi^2)\mathbf{1}_2.} \tag{986}$$

The first yes/no test is now:

$$\boxed{S_\xi \overset{?}{\equiv} 0.} \tag{987}$$

If $S_\xi \equiv 0$, then the longitudinal primitive channel is absent. If $S_\xi \not\equiv 0$, continue.

6. Step 4: Pauli decomposition of the traceless part

Any traceless Hermitian 2×2 matrix can be written as

$$S_\xi = s_1(\xi)\sigma_1 + s_2(\xi)\sigma_2 + s_3(\xi)\sigma_3. \tag{988}$$

The longitudinal witness channel specifically cares about the σ_3-type component in the adapted frame, because the explicit primitive representative is

$$S_\xi^{\mathrm{exp}} = \frac{1}{2}\theta'(\xi)^2\sigma_3.$$

Therefore the second useful diagnostic is:

$$\boxed{s_3(\xi) \overset{?}{\not\equiv} 0.} \tag{989}$$

A sufficient condition for longitudinal survival is:

$$\boxed{s_3(\xi) \not\equiv 0.} \tag{990}$$

7. Step 5: form the primitive transport kernel

Given S_ξ, define the longitudinal primitive transport kernel

$$\boxed{V^\mu_{(\parallel)}(\xi) = \theta'(\xi)\, S_\xi\, n^\mu.} \tag{991}$$

Since $\theta'(\xi)$ is odd for a reflection-symmetric defect and S_ξ is even whenever it is built from quadratic response data, $V^\mu_{(\parallel)}(\xi)$ is odd, exactly as required for an even/odd low-mode coupling.

8. Step 6: compute the projected overlap

The low-mode projected longitudinal overlap is

$$(M^\mu_{(\parallel)})_{ab} = \int d\xi\, (u_a^+)^\dagger V^\mu_{(\parallel)}(\xi) u_b^-. \tag{992}$$

Using

$$u_a^\pm = f_\pm e_a,$$

this becomes

$$(M^\mu_{(\parallel)})_{ab} = n^\mu \int d\xi\, f_+(\xi) f_-(\xi)\, \theta'(\xi)\, S_{\xi,ab}. \tag{993}$$

The scalar channel of interest is therefore

$$\boxed{J_\parallel := \int d\xi\, f_+(\xi) f_-(\xi)\, \theta'(\xi)\, S_\xi.} \tag{994}$$

If one projects further onto the σ_3 component, write

$$\boxed{J_\parallel^{(3)} := \int d\xi\, f_+(\xi) f_-(\xi)\, \theta'(\xi)\, s_3(\xi).} \tag{995}$$

Then a sharp practical sufficient condition is:

$$\boxed{J_\parallel^{(3)} \neq 0.} \tag{996}$$

9. Step 7: extract $\lambda_\parallel$

The longitudinal witness coefficient is read off from the projected symbol:

$$A_{1,\parallel}(p) = \lambda_\parallel p_\parallel \sigma_3. \tag{997}$$

In the explicit representative one has

$$S_\xi = \frac{1}{2}\theta'(\xi)^2\sigma_3,$$

so

$$J_\parallel^{(3)} = \frac{1}{2}\int d\xi\, f_+ f_- \theta'^3 = \frac{1}{2} I_{\theta'^3},$$

and therefore

$$\boxed{\lambda_\parallel = J_\parallel^{(3)} = \frac{1}{2} I_{\theta'^3}.} \tag{998}$$

More generally, the protocol says:

$$\boxed{\lambda_\parallel \neq 0 \quad \Longleftarrow \quad J_\parallel^{(3)} \neq 0.} \tag{999}$$

10. Minimal sufficient condition in algorithmic form

The whole longitudinal audit reduces to the following finite algorithm.

a. Input

$$P_B(\xi), \quad e_a(\xi), \quad f_\pm(\xi), \quad \mathcal{T}_\xi.$$

b. Algorithm

1. Compute T_ξ from (984).
2. Compute S_ξ from (986).
3. Check whether $S_\xi \equiv 0$.
4. Decompose $S_\xi = s_1\sigma_1 + s_2\sigma_2 + s_3\sigma_3$.
5. Compute $J_\parallel^{(3)}$ from (995).

c. Output

$$\boxed{J_\parallel^{(3)} \neq 0 \Longrightarrow \lambda_\parallel \neq 0 \Longrightarrow \text{longitudinal primitive carrier survives.}} \tag{1000}$$

11. Strongest honest conclusion

Thus the explicit calculation protocol is now fully fixed. The remaining microscopic longitudinal question is no longer vague at all:

$$\boxed{\text{compute the } 2\times 2\ Q_B\text{-sector matrix } T_\xi,\ \text{form } S_\xi,\ \text{and test whether } J_\parallel^{(3)} \neq 0.} \tag{1001}$$

This is a finite operator-and-overlap calculation, not an open-ended conceptual problem.

. Summary

$$\boxed{\textbf{Explicit calculation protocol for } (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} \text{ and } J_\parallel}$$

Given

$$\boxed{P_B(\xi), \quad Q_B(\xi), \quad e_1(\xi), e_2(\xi), \quad f_\pm(\xi), \quad \mathcal{T}_\xi,}$$

do the following:

1. Compute the 2×2 Q_B-sector matrix

$$\boxed{(T_\xi)_{ab} = \langle e_a, \mathcal{T}_\xi e_b \rangle.}$$

2. Square it and remove the scalar part:

$$\boxed{S_\xi = T_\xi^2 - \frac{1}{2}\mathrm{Tr}(T_\xi^2)\mathbf{1}_2 = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}}.}$$

3. Decompose

$$\boxed{S_\xi = s_1(\xi)\sigma_1 + s_2(\xi)\sigma_2 + s_3(\xi)\sigma_3.}$$

4. Compute the longitudinal overlap

$$\boxed{J_\parallel^{(3)} = \int d\xi \, f_+(\xi) f_-(\xi) \, \theta'(\xi) \, s_3(\xi).}$$

Then

$$\boxed{J_\parallel^{(3)} \neq 0 \Longrightarrow \lambda_\parallel \neq 0.}$$

So the remaining longitudinal primitive audit is reduced to one finite calculation:

$$\boxed{T_\xi \to S_\xi \to s_3(\xi) \to J_\parallel^{(3)}.}$$

If $J_\parallel^{(3)} \neq 0$, the longitudinal primitive carrier survives.

TOY EVALUATION OF THE LONGITUDINAL PRIMITIVE PROTOCOL

1. Goal

We apply the previously derived longitudinal primitive protocol to the explicit toy defect representative. The protocol was:

$$T_\xi \;\longrightarrow\; S_\xi = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} \;\longrightarrow\; s_3(\xi) \;\longrightarrow\; J_\parallel^{(3)}.$$

The aim is to verify explicitly that

$$S_\xi \not\equiv 0, \qquad s_3(\xi) \not\equiv 0, \qquad J_\parallel^{(3)} \neq 0.$$

2. Defect data

Use the explicit reflection-symmetric defect profile

$$\theta(\xi) = \theta_0 \operatorname{sech}(\kappa\xi), \qquad \theta(-\xi) = \theta(\xi),$$

so that

$$\theta'(\xi) = -\theta_0 \kappa \operatorname{sech}(\kappa\xi) \tanh(\kappa\xi).$$

Take the even/odd low-mode profiles

$$f_+(\xi) = \sqrt{\frac{\kappa}{2}} \operatorname{sech}(\kappa\xi), \qquad f_-(\xi) = \sqrt{\frac{3\kappa}{2}} \tanh(\kappa\xi) \operatorname{sech}(\kappa\xi).$$

These are the same explicit low-mode profiles already used in the seed construction and toy extraction.

3. Correct longitudinal representative

A naive attempt would be to model T_ξ itself as proportional to $\theta'(\xi)\sigma_3$. But then

$$T_\xi^2 \propto \theta'(\xi)^2 \mathbf{1}_2,$$

so its traceless part would vanish. Therefore the correct toy representative is not a model for T_ξ itself, but directly for the quadratic traceless response:

$$\boxed{S_\xi := (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} = \frac{1}{2}\, \theta'(\xi)^2\, \sigma_3.}$$

This is exactly the explicit longitudinal primitive seed already identified in the seed-space construction. :contentReference[oaicite:2]index=2

4. Pauli decomposition

Write

$$S_\xi = s_1(\xi)\sigma_1 + s_2(\xi)\sigma_2 + s_3(\xi)\sigma_3.$$

For the explicit representative above,

$$\boxed{s_1(\xi) = 0, \qquad s_2(\xi) = 0, \qquad s_3(\xi) = \frac{1}{2}\, \theta'(\xi)^2.}$$

Hence

$$s_3(\xi) \not\equiv 0$$

whenever the defect is nontrivial, i.e. $\theta_0 \kappa \neq 0$.

5. Longitudinal overlap

By definition,

$$J_\parallel^{(3)} := \int d\xi\, f_+(\xi) f_-(\xi)\, \theta'(\xi)\, s_3(\xi).$$

Substituting $s_3(\xi) = \frac{1}{2}\theta'(\xi)^2$, we get

$$J_{\parallel}^{(3)} = \frac{1}{2}\int d\xi\, f_+(\xi)f_-(\xi)\,\theta'(\xi)^3 = \frac{1}{2}I_{\theta'^3}.$$

From the explicit evaluation already recorded in the notes,

$$\boxed{I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256}\,\theta_0^3\kappa^3.}$$

Therefore

$$\boxed{J_{\parallel}^{(3)} = -\frac{3\sqrt{3}\pi}{512}\,\theta_0^3\kappa^3.}$$

So for a nontrivial defect profile,

$$\boxed{J_{\parallel}^{(3)} \neq 0.}$$

6. Consequence for the longitudinal witness

The longitudinal projected symbol is

$$A_{1,\parallel}(p) = \lambda_{\parallel}p_{\parallel}\sigma_3.$$

In this toy representative the protocol gives

$$\lambda_{\parallel} = J_{\parallel}^{(3)} = \frac{1}{2}I_{\theta'^3}.$$

Hence

$$\boxed{\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512}\,\theta_0^3\kappa^3 \neq 0}$$

for $\theta_0\kappa \neq 0$.

Thus the longitudinal primitive channel survives in the toy representative.

7. Interpretation

The toy evaluation shows:

1. the longitudinal primitive protocol is computationally concrete;
2. the correct object to test is the traceless quadratic response S_ξ, not a guessed linear T_ξ;
3. in the explicit representative,

$$S_\xi = \frac{1}{2}\theta'(\xi)^2\sigma_3$$

   is nonzero;
4. the weighted low-mode overlap

$$J_{\parallel}^{(3)} = \frac{1}{2}I_{\theta'^3}$$

   is nonzero;
5. therefore the remaining longitudinal witness condition is satisfied in the explicit seed representative.

8. Honest limitation

This does *not* prove that the full microscopic TECT linearization produces the same S_ξ automatically. What it proves is:

$$\boxed{\text{the residual longitudinal primitive test is realizable, finite, and explicitly passes in the concrete representative.}}$$

So the remaining microscopic frontier is only the matching problem: does the actual full TECT Q_B-sector quadratic response reduce to this same nonzero traceless structure?

. Summary

$$\boxed{\textbf{Toy evaluation of the longitudinal primitive protocol}}$$

Use the explicit representative

$$\boxed{S_\xi := (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} = \frac{1}{2}\,\theta'(\xi)^2\,\sigma_3.}$$

Then its Pauli decomposition is

$$\boxed{s_1(\xi) = 0, \qquad s_2(\xi) = 0, \qquad s_3(\xi) = \frac{1}{2}\,\theta'(\xi)^2.}$$

Hence the longitudinal overlap is

$$\boxed{J_\parallel^{(3)} = \int d\xi\, f_+(\xi) f_-(\xi)\,\theta'(\xi)\,s_3(\xi) = \frac{1}{2}\int d\xi\, f_+ f_-\,\theta'^3 = \frac{1}{2} I_{\theta'^3}.}$$

Using

$$\boxed{I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256}\,\theta_0^3\kappa^3,}$$

we get

$$\boxed{J_\parallel^{(3)} = -\frac{3\sqrt{3}\pi}{512}\,\theta_0^3\kappa^3 \neq 0 \qquad (\theta_0\kappa \neq 0).}$$

Therefore

$$\boxed{\lambda_\parallel = J_\parallel^{(3)} \neq 0.}$$

So in the explicit toy representative, the longitudinal primitive protocol succeeds completely:

$$\boxed{S_\xi \neq 0, \qquad s_3(\xi) \neq 0, \qquad J_\parallel^{(3)} \neq 0.}$$

This proves explicit realizability of the remaining primitive longitudinal test, though not yet its automatic survival in the full microscopic TECT linearization.

EXPLICIT CALCULATION PROTOCOL FOR $(\Pi_Q\mathcal{T}_\xi^2\Pi_Q)_{\mathrm{tr}}$ AND $J_\parallel$

1. Goal

The remaining primitive carrier test is:

$$\boxed{(\Pi_Q\mathcal{T}_\xi^2\Pi_Q)_{\mathrm{tr}} \not\equiv 0 \quad \text{and} \quad J_\parallel^{(3)} \neq 0.} \tag{1002}$$

If this holds, then the longitudinal witness survives:

$$\boxed{\lambda_\parallel \neq 0.} \tag{1003}$$

2. Input data

Fix the following data:

1. a reflection-symmetric defect background
$$P_B(\xi), \qquad P_B(-\xi) = P_B(\xi);$$
2. the corresponding rank-two tangent bundle
$$Q_B(\xi) = (\mathbb{C}z_B(\xi))^\perp;$$
3. a local orthonormal frame of Q_B,
$$e_1(\xi),\ e_2(\xi);$$
4. the graded low-mode profiles
$$u_a^+(\xi) = f_+(\xi)e_a(\xi), \qquad u_a^-(\xi) = f_-(\xi)e_a(\xi), \qquad a = 1, 2.$$

The explicit seed construction already shows that longitudinal and transverse seed representatives exist in the concrete ansatz space.

3. Step 1: define the microscopic transverse-response operator

Let

$$\mathcal{T}_\xi : \Gamma(Q_B) \to \Gamma(Q_B) \tag{1004}$$

be the microscopic Q_B-sector transverse-response operator obtained from the linearized TECT dynamics around the chosen defect background.

Operationally, $\mathcal{T}_\xi$ is extracted after:

1. linearizing the microscopic TECT equation around $P_B(\xi)$,
2. restricting to the defect-localized low-energy internal doublet sector,
3. expressing the result as an operator on Q_B.

4. Step 2: compute the 2×2 matrix of $\mathcal{T}_\xi$

In the orthonormal frame $\{e_1, e_2\}$, define

$$\boxed{(T_\xi)_{ab} = \langle e_a(\xi), \mathcal{T}_\xi e_b(\xi)\rangle, \qquad a, b = 1, 2.} \tag{1005}$$

This produces an explicit 2×2 Hermitian matrix-valued function T_ξ.

5. Step 3: compute the quadratic traceless response

Square the matrix:

$$T_\xi^{(2)} := T_\xi^2. \tag{1006}$$

Then remove the scalar part:

$$\boxed{S_\xi := (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\rm tr} = T_\xi^2 - \frac{1}{2}\mathrm{Tr}(T_\xi^2)\mathbf{1}_2.} \tag{1007}$$

The first yes/no test is:

$$\boxed{S_\xi \stackrel{?}{\equiv} 0.} \tag{1008}$$

If $S_\xi \equiv 0$, the longitudinal primitive channel is absent. If $S_\xi \not\equiv 0$, continue.

6. Step 4: Pauli decomposition

Any traceless Hermitian 2×2 matrix admits the decomposition

$$\boxed{S_\xi = s_1(\xi)\sigma_1 + s_2(\xi)\sigma_2 + s_3(\xi)\sigma_3.} \tag{1009}$$

The longitudinal primitive channel specifically cares about the σ_3-component.
Thus the second diagnostic is:

$$\boxed{s_3(\xi) \stackrel{?}{\not\equiv} 0.} \tag{1010}$$

A useful sufficient condition is simply:

$$\boxed{s_3(\xi) \not\equiv 0.} \tag{1011}$$

7. Step 5: form the longitudinal primitive overlap

Define the primitive transport kernel

$$\boxed{V^\mu_{(\parallel)}(\xi) = \theta'(\xi)\, S_\xi\, n^\mu.} \tag{1012}$$

For a reflection-symmetric defect, $\theta'(\xi)$ is odd, while S_ξ is even if it comes from quadratic response data. Hence $V^\mu_{(\parallel)}$ is odd, so it can couple even and odd low modes.
The projected overlap is

$$(M^\mu_{(\parallel)})_{ab} = \int d\xi\, (u_a^+)^\dagger V^\mu_{(\parallel)}(\xi) u_b^-. \tag{1013}$$

Using $u_a^\pm = f_\pm e_a$, this becomes

$$(M^\mu_{(\parallel)})_{ab} = n^\mu \int d\xi\, f_+(\xi) f_-(\xi)\, \theta'(\xi)\, S_{\xi,ab}. \tag{1014}$$

Projecting further to the σ_3 component gives the scalar diagnostic:

$$\boxed{J_\parallel^{(3)} := \int d\xi\, f_+(\xi) f_-(\xi)\, \theta'(\xi)\, s_3(\xi).} \tag{1015}$$

Then the sharp practical sufficient condition is:

$$\boxed{J_\parallel^{(3)} \neq 0 \Longrightarrow \lambda_\parallel \neq 0.} \tag{1016}$$

8. Consistency with the explicit seed representative

In the explicit seed representative,

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\rm tr} = \frac{1}{2}\,\theta'(\xi)^2 \sigma_3, \tag{1017}$$

so

$$s_1(\xi) = 0, \qquad s_2(\xi) = 0, \qquad s_3(\xi) = \frac{1}{2}\,\theta'(\xi)^2.$$

Therefore

$$J_\parallel^{(3)} = \frac{1}{2} \int d\xi\, f_+(\xi) f_-(\xi) \theta'(\xi)^3 = \frac{1}{2} I_{\theta'^3}. \tag{1018}$$

Since $I_{\theta'^3} \neq 0$, this gives $\lambda_\parallel \neq 0$.

9. Algorithmic summary

The whole longitudinal primitive audit is now a finite algorithm:
a. Input

$$P_B(\xi), \quad e_a(\xi), \quad f_\pm(\xi), \quad \mathcal{T}_\xi.$$

b. Algorithm

1. compute T_ξ from (1005);
2. compute S_ξ from (1007);
3. check whether $S_\xi \equiv 0$;
4. decompose S_ξ as in (1009);
5. compute $J_\parallel^{(3)}$ from (1015).

c. Output

$$\boxed{J_\parallel^{(3)} \neq 0 \Longrightarrow \lambda_\parallel \neq 0 \Longrightarrow \text{longitudinal primitive carrier survives.}} \tag{1019}$$

10. Strongest honest conclusion

Thus the remaining longitudinal primitive question is no longer vague:

$$\boxed{\text{compute } T_\xi,\ \text{form } S_\xi,\ \text{extract } s_3(\xi),\ \text{and test } J_\parallel^{(3)} \neq 0.} \tag{1020}$$

This is a finite operator-and-overlap calculation, not an open-ended conceptual problem.

. Summary

$$\boxed{\textbf{Explicit calculation protocol for } (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\rm tr} \text{ and } J_\parallel}$$

Given

$$\boxed{P_B(\xi), \quad Q_B(\xi), \quad e_1(\xi), e_2(\xi), \quad f_\pm(\xi), \quad \mathcal{T}_\xi,}$$

do the following:

1. Compute the 2×2 Q_B-sector matrix

$$\boxed{(T_\xi)_{ab} = \langle e_a, \mathcal{T}_\xi e_b \rangle.}$$

2. Square it and remove the scalar part:

$$\boxed{S_\xi = T_\xi^2 - \frac{1}{2}\mathrm{Tr}(T_\xi^2)\mathbf{1}_2 = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}}.}$$

3. Decompose

$$\boxed{S_\xi = s_1(\xi)\sigma_1 + s_2(\xi)\sigma_2 + s_3(\xi)\sigma_3.}$$

4. Compute the longitudinal overlap

$$\boxed{J_\parallel^{(3)} = \int d\xi\, f_+(\xi) f_-(\xi)\, \theta'(\xi)\, s_3(\xi).}$$

Then

$$\boxed{J_\parallel^{(3)} \neq 0 \Longrightarrow \lambda_\parallel \neq 0.}$$

So the remaining longitudinal primitive audit is reduced to one finite chain:

$$\boxed{T_\xi \to S_\xi \to s_3(\xi) \to J_\parallel^{(3)}.}$$

TOY EVALUATION OF THE LONGITUDINAL PRIMITIVE PROTOCOL

1. Goal

We apply the previously derived longitudinal primitive protocol to the explicit toy defect representative. The protocol is:

$$T_\xi \;\longrightarrow\; S_\xi = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} \;\longrightarrow\; s_3(\xi) \;\longrightarrow\; J_\parallel^{(3)}.$$

The objective is to verify explicitly that

$$S_\xi \not\equiv 0, \qquad s_3(\xi) \not\equiv 0, \qquad J_\parallel^{(3)} \neq 0.$$

2. Explicit defect data

Take the reflection-symmetric defect profile

$$\theta(\xi) = \theta_0 \,\mathrm{sech}(\kappa\xi), \qquad \theta(-\xi) = \theta(\xi),$$

so that

$$\theta'(\xi) = -\theta_0 \kappa \,\mathrm{sech}(\kappa\xi) \tanh(\kappa\xi).$$

Take the explicit even/odd low-mode profiles

$$f_+(\xi) = \sqrt{\frac{\kappa}{2}}\,\mathrm{sech}(\kappa\xi), \qquad f_-(\xi) = \sqrt{\frac{3\kappa}{2}}\,\tanh(\kappa\xi)\,\mathrm{sech}(\kappa\xi).$$

These are the same explicit profiles used in the seed construction.

3. Correct toy representative for the quadratic traceless response

The longitudinal primitive protocol is controlled by

$$S_\xi = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}},$$

not by an arbitrary guess for T_ξ itself.

In the explicit seed representative, the correct toy choice is

$$\boxed{S_\xi = \frac{1}{2}\,\theta'(\xi)^2\,\sigma_3.} \tag{1021}$$

This is exactly the explicit traceless quadratic Q_B-response underlying the longitudinal seed.

4. Pauli decomposition

Write

$$S_\xi = s_1(\xi)\sigma_1 + s_2(\xi)\sigma_2 + s_3(\xi)\sigma_3.$$

From (1021), we read off

$$\boxed{s_1(\xi) = 0, \qquad s_2(\xi) = 0, \qquad s_3(\xi) = \frac{1}{2}\,\theta'(\xi)^2.} \tag{1022}$$

Hence

$$s_3(\xi) \not\equiv 0$$

for any nontrivial defect profile $\theta_0\kappa \neq 0$.

5. Longitudinal overlap

The primitive longitudinal overlap is

$$J_\parallel^{(3)} := \int d\xi\, f_+(\xi) f_-(\xi)\,\theta'(\xi)\,s_3(\xi).$$

Using (1022),

$$\begin{aligned} J_\parallel^{(3)} &= \int d\xi\, f_+(\xi) f_-(\xi)\,\theta'(\xi)\,\frac{1}{2}\,\theta'(\xi)^2 \\ &= \frac{1}{2}\int d\xi\, f_+(\xi) f_-(\xi)\,\theta'(\xi)^3 \\ &= \frac{1}{2} I_{\theta'^3}. \end{aligned} \tag{1023}$$

For the explicit defect profile,

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256}\,\theta_0^3\kappa^3. \tag{1024}$$

Therefore

$$\boxed{J_\parallel^{(3)} = -\frac{3\sqrt{3}\pi}{512}\,\theta_0^3\kappa^3.} \tag{1025}$$

Hence, for $\theta_0\kappa \neq 0$,

$$\boxed{J_\parallel^{(3)} \neq 0.} \tag{1026}$$

6. Longitudinal witness coefficient

The longitudinal part of the reduced symbol is

$$A_{1,\parallel}(p) = \lambda_\parallel p_\parallel \sigma_3.$$

By the explicit seed formula,

$$\lambda_\parallel = \frac{1}{2} I_{\theta'^3}.$$

Combining with (1023), we obtain

$$\boxed{\lambda_\parallel = J_\parallel^{(3)}.} \tag{1027}$$

Thus

$$\boxed{\lambda_\parallel = -\frac{3\sqrt{3}\pi}{512}\,\theta_0^3\kappa^3 \neq 0.} \tag{1028}$$

7. Consequence

Therefore the longitudinal primitive protocol succeeds completely in the toy representative:

$$S_\xi \neq 0, \qquad s_3(\xi) \neq 0, \qquad J_\parallel^{(3)} \neq 0, \qquad \lambda_\parallel \neq 0.$$

So the residual primitive longitudinal test is not merely formal: it is explicitly realizable and explicitly passes in the concrete representative.

8. Honest limitation

What this does *not* prove is that the full microscopic TECT linearization automatically produces the same

$$S_\xi = \frac{1}{2}\theta'(\xi)^2\sigma_3.$$

What it *does* prove is:

$$\boxed{\text{the residual longitudinal primitive test is finite, explicit, and passes in the concrete representative.}} \tag{1029}$$

Thus the remaining microscopic frontier is only the final matching question: does the actual full TECT Q_B-sector quadratic response reduce to the same nonzero traceless structure?

. Summary

Toy evaluation of the longitudinal primitive protocol

Use the explicit representative

$$\boxed{S_\xi = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} = \frac{1}{2}\,\theta'(\xi)^2\,\sigma_3.}$$

Then the Pauli decomposition is

$$\boxed{s_1(\xi) = 0, \qquad s_2(\xi) = 0, \qquad s_3(\xi) = \frac{1}{2}\,\theta'(\xi)^2.}$$

Therefore

$$\boxed{J_\parallel^{(3)} = \int d\xi\, f_+(\xi) f_-(\xi)\, \theta'(\xi)\, s_3(\xi) = \frac{1}{2}\int d\xi\, f_+ f_-\, \theta'^3 = \frac{1}{2} I_{\theta'^3}.}$$

Using

$$\boxed{I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256}\,\theta_0^3\kappa^3,}$$

we get

$$\boxed{J_\parallel^{(3)} = -\frac{3\sqrt{3}\pi}{512}\,\theta_0^3\kappa^3 \neq 0 \qquad (\theta_0\kappa \neq 0).}$$

Hence

$$\boxed{\lambda_\parallel = J_\parallel^{(3)} \neq 0.}$$

So in the explicit toy representative,

$$\boxed{S_\xi \neq 0, \qquad s_3(\xi) \neq 0, \qquad J_\parallel^{(3)} \neq 0,}$$

and the longitudinal primitive channel survives completely.

NEAR-CLOSURE STATEMENT FOR THE LOCAL DIRAC SECTOR

1. Logical position of the local Dirac problem

The local Dirac sector was originally reduced to five finite ingredients:

1. witness-carrier realization,
2. opposite-valley witness matching,
3. zero constant mixing $V(0) = 0$,
4. remote spectral gap $\Delta > 0$,
5. mass protection.

At an earlier stage, the carrier problem itself still looked genuinely open because both the transverse and longitudinal witness channels had to be justified microscopically.
The updated situation is much sharper.

2. Witness sector: now effectively explicit

Inside the explicit seed subspace, the S2 witness coefficients are

$$\lambda_\parallel = \frac{1}{2} c_\parallel I_{\theta'^3}, \qquad \alpha = c_E I_\theta, \qquad \beta = c_O I_\theta. \tag{1030}$$

Hence the S2 nondegeneracy condition

$$\boxed{\lambda_{\parallel} \neq 0, \qquad (\alpha, \beta) \neq (0,0)} \tag{1031}$$

is equivalent to

$$\boxed{c_{\parallel} \neq 0, \qquad (c_E, c_O) \neq (0,0).} \tag{1032}$$

This is already exact inside the explicit seed ansatz. :contentReference[oaicite:5]index=5

a. Transverse channel. The transverse odd-gradient extraction gives

$$I_\theta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa \neq 0, \tag{1033}$$

so explicit transverse seed directions already realize

$$(\alpha, \beta) \neq (0,0).$$

Hence the transverse witness half is fixed.

b. Longitudinal channel. The explicit primitive longitudinal representative is

$$V^{\mu}_{(\parallel)}(\xi) = \theta'(\xi)(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}}\, n^\mu = \frac{1}{2}\,\theta'(\xi)^3\, n^\mu \sigma_3. \tag{1034}$$

Its projected witness is

$$\lambda_{\parallel} = \frac{1}{2} I_{\theta'^3}, \qquad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256}\theta_0^3\kappa^3 \neq 0. \tag{1035}$$

Equivalently, in the protocol language,

$$S_\xi = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} = \frac{1}{2}\theta'(\xi)^2\sigma_3, \qquad J_{\parallel}^{(3)} = \frac{1}{2} I_{\theta'^3} \neq 0. \tag{1036}$$

So the longitudinal primitive channel also explicitly survives in the representative model.

Therefore, at the witness level,

$$\boxed{\text{both the transverse and longitudinal channels are explicitly realized in the seed/toy representative.}} \tag{1037}$$

3. Opposite-valley structure

The opposite-valley pairing condition was reduced to the equality of the witness triples

$$\boxed{(W_{\parallel}^-, W_I^-, W_J^-) = (W_{\parallel}^+, W_I^+, W_J^+)} \tag{1038}$$

in the matched opposite-valley frame. Inside the concrete Class II seed family, this matching is already closed at the witness level. Thus the transition

$$\text{Weyl at one valley} \Longrightarrow \text{local doubled Dirac block}$$

is structurally closed once the same seed sector is realized at both valleys.

4. Remote-sector audits already reduced to finite tests

The three remaining stability conditions were already reduced to finite audits:

c. Zero constant mixing.

$$\boxed{\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \Longrightarrow V(0) = 0.} \tag{1039}$$

So constant hybridization is reduced to a finite little-group irrep mismatch audit.

d. Remote spectral gap.

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho} \Longrightarrow \Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0,} \tag{1040}$$

or equivalently bounded-perturbation stability from an already-gapped realization. Hence the remote-gap issue is reduced to a finite Bloch-spectral inequality.

e. Mass protection. For the isolated doubled local Dirac block,

$$\mathcal{M} = \operatorname{span}_{\mathbb{R}}\{\rho_1 \otimes \mathbf{1}_2, \rho_2 \otimes \mathbf{1}_2\}, \tag{1041}$$

and mass protection reduces to

$$\boxed{A_{\text{res}} \cap \mathcal{M} = \{0\}.} \tag{1042}$$

This is a finite residual-symmetry exclusion test. Exact valley $U(1)_V$ or its $\mathbb{Z}_2^V$ remnant is already a sufficient mechanism.

5. What near-closure means here

The explicit representative now shows that the previously open carrier part is not merely abstractly possible, but explicitly realized in a concrete defect model. Therefore the local Dirac frontier is no longer:

"Does any complete carrier realization exist at all?"

Instead it is the much sharper matching problem:

$$\boxed{\text{Does the full microscopic TECT linearization match the explicit seed/toy representative in the required shell-relevant channels?}} \tag{1043}$$

This matching question has three components:

1. **witness matching to the representative:** does the full microscopic transport extraction contain the same longitudinal and transverse seed directions?
2. **remote-sector matching:** does the full microscopic remote sector satisfy the irrep mismatch and gap inequalities?
3. **residual-symmetry matching:** does the full microscopic low-energy algebra preserve the symmetry needed to forbid constant masses?

6. Near-closure theorem

We may therefore state the strongest honest near-closure theorem for the local Dirac sector.

[Near-closure statement for the local Dirac sector] Assume that the full microscopic TECT linearization around the physically relevant defect background matches the explicit seed/toy representative in the following sense:

1. the shell-relevant transport extraction contains the explicit transverse seed directions and a longitudinal primitive channel matching

   $$S_\xi = (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\text{tr}} = \frac{1}{2}\theta'(\xi)^2\sigma_3$$

   up to nonzero scalar deformation, so that

   $$\lambda_\| \neq 0, \qquad (\alpha, \beta) \neq (0, 0);$$

2. the opposite-valley matched witness triples coincide;
3. the remote sector satisfies the finite zero-mixing and gap audits;
4. the residual low-energy symmetry satisfies the finite mass-protection exclusion test.

Then the local low-energy fermionic sector contains an isolated massless Dirac cone

$$H_{\text{eff}}(q) = \rho_3 \otimes (q_i\sigma_i) + O(|q|^2/\Delta).$$

7. Honest final status

The significance of the theorem is not that the local Dirac sector is fully unconditionally proved already. Rather, it says that the remaining frontier is now only a final microscopic matching problem.

Therefore the exact current status is:

$$\boxed{\text{the local Dirac sector is in a near-closure state:}} \tag{1044}$$

$$\boxed{\text{an explicit representative already realizes all required channels, and only full microscopic matching remains.}} \tag{1045}$$

This is stronger than a generic conditional theorem, because the explicit seed/toy representative already demonstrates concrete realizability of every required witness component.

. Summary

Near-closure statement for the local Dirac sector

At the witness level, the explicit seed/toy representative already realizes both halves of the S2 nondegeneracy condition:

$$\boxed{(\alpha,\beta)\neq(0,0)}$$

via the explicit transverse odd-gradient channel, and

$$\boxed{\lambda_{\parallel}\neq 0}$$

via the explicit longitudinal primitive channel

$$\boxed{V^{\mu}_{(\|)}(\xi)=\theta'(\xi)(\Pi_Q\mathcal{T}_\xi^2\Pi_Q)_{\mathrm{tr}}\,n^\mu=\frac{1}{2}\theta'(\xi)^3n^\mu\sigma_3.}$$

So the witness side is no longer abstractly open; it is explicitly realizable.

The remaining ingredients are already finite audits:

$$\boxed{(W_\|^-,W_I^-,W_J^-)=(W_\|^+,W_I^+,W_J^+)}$$

for opposite-valley pairing,

$$\boxed{\mathrm{Hom}_{H_*}(\mathcal{H}_R,\mathcal{H}_D)=0}$$

for zero constant mixing,

$$\boxed{\delta_{0,\rho}>\eta_{R,\rho}\Longrightarrow\Delta_\rho>0}$$

for remote spectral isolation,

and

$$\boxed{A_{\mathrm{res}}\cap\mathrm{span}_{\mathbb{R}}\{\rho_1\otimes\mathbf{1}_2,\rho_2\otimes\mathbf{1}_2\}=\{0\}}$$

for mass protection.

Therefore the exact current status is:

$$\boxed{\text{the local Dirac sector is in a near-closure state:}}$$

$$\boxed{\text{all required channels are explicitly realized in the seed/toy representative,}}$$

and

$$\boxed{\text{only the final matching to the full microscopic TECT linearization remains.}}$$

FINAL MICROSCOPIC MATCHING CHECKLIST FOR THE LOCAL DIRAC SECTOR

Final target

The final target is to show that the full microscopic TECT linearization matches the explicit seed/toy representative strongly enough to produce an isolated local massless Dirac cone:

$$H_{\mathrm{eff}}(q) = \rho_3 \otimes (q_i \sigma_i) + O(|q|^2/\Delta).$$

At this stage, the local Dirac problem is no longer an open structural problem. It is a finite matching checklist.

Checklist 1: witness-side microscopic matching

Extract the actual microscopic projected first-order symbol at the selected valley $+G_*$, and read off the S2 coefficients

$$A_1(p) = \lambda_\parallel p_\parallel \sigma_3 + \alpha(p_1\sigma_1 + p_2\sigma_2) + \beta(-p_2\sigma_1 + p_1\sigma_2).$$

Check:

$$\boxed{\lambda_\parallel \neq 0, \qquad (\alpha, \beta) \neq (0, 0).} \tag{1046}$$

This splits into:

a. (1a) transverse matching Verify that the microscopic odd-gradient transport extraction reproduces the nonzero transverse witness channel, i.e.

$$I_\theta \neq 0 \quad \Longrightarrow \quad (\alpha, \beta) \neq (0, 0).$$

b. (1b) longitudinal matching Verify that the microscopic Q_B-sector quadratic response reproduces the primitive longitudinal channel:

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} \not\equiv 0, \qquad J_\parallel^{(3)} \neq 0 \quad \Longrightarrow \quad \lambda_\parallel \neq 0.$$

Checklist 2: opposite-valley matching

Repeat the same witness extraction at the opposite shell point $-G_*$, and compare in the matched opposite-valley frame.

Check:

$$\boxed{(W_\parallel^-, W_I^-, W_J^-) = (W_\parallel^+, W_I^+, W_J^+).} \tag{1047}$$

If true, the two local Weyl valleys pair into a doubled local Dirac block.

Checklist 3: zero constant mixing audit

Let

$$\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R$$

be the microscopic decomposition into the local doubled shell block and the remote sector. At the crossing point, identify the little group H_*.

Check:

$$\boxed{\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0.} \tag{1048}$$

Equivalently, verify that every remote sector differs from $\mathcal{H}_D$ in at least one shell-relevant label. Then

$$V(0) = 0.$$

Checklist 4: remote spectral gap audit

For the remote block $H_R(p)$, define

$$\delta_{0,\rho} := \inf_{|p|\le\rho} \delta_0(p), \qquad \eta_{R,\rho} := \sup_{|p|\le\rho} \eta_R(p).$$

Check either:

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho},} \tag{1049}$$

which gives

$$\Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0,$$

or, alternatively, check bounded perturbation stability from an already-gapped realization:

$$\boxed{\sup_{|p|\le p_*} \|\delta H_R(p)\| < \Delta_0.} \tag{1050}$$

If either holds, then the remote sector is spectrally isolated by a positive gap $\Delta > 0$.

Checklist 5: mass-protection audit

For the isolated doubled local Dirac block, the constant mass space is

$$\mathcal{M} = \mathrm{span}_{\mathbb{R}}\{\rho_1 \otimes \mathbf{1}_2, \rho_2 \otimes \mathbf{1}_2\}.$$

Let A_{res} be the actual residual constant symmetry-allowed operator algebra on the low-energy doubled shell block.

Check:

$$\boxed{A_{\text{res}} \cap \mathcal{M} = \{0\}.} \tag{1051}$$

A sufficient mechanism is exact residual valley $U(1)_V$ or its $\mathbb{Z}_2^V$ remnant.

Pass condition

If all five checks pass, then the full microscopic TECT linearization matches the explicit representative strongly enough to yield an isolated local massless Dirac cone:

$$\boxed{H_{\text{eff}}(q) = \rho_3 \otimes (q_i\sigma_i) + O(|q|^2/\Delta).} \tag{1052}$$

Most compressed version

The entire local Dirac matching problem is therefore compressed to:

$$\boxed{\textbf{(C1)}\ \lambda_\parallel \neq 0,\ (\alpha,\beta)\neq(0,0),} \tag{1053}$$

$$\boxed{\textbf{(C2)}\ (W_\parallel^-, W_I^-, W_J^-) = (W_\parallel^+, W_I^+, W_J^+),} \tag{1054}$$

$$\boxed{\textbf{(C3)}\ \mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0,} \tag{1055}$$

$$\boxed{\textbf{(C4)}\ \Delta > 0,} \tag{1056}$$

$$\boxed{\textbf{(C5)}\ A_{\mathrm{res}} \cap \mathcal{M} = \{0\}.} \tag{1057}$$

If **(C1)**–**(C5)** hold, the local Dirac sector is closed.

. Summary

Final microscopic matching checklist for the local Dirac sector

Check the following five items:

$$\boxed{\textbf{(C1)}\ \lambda_\parallel \neq 0, \qquad (\alpha,\beta)\neq(0,0)}$$

from the actual microscopic witness extraction;

$$\boxed{\textbf{(C2)}\ (W_\parallel^-, W_I^-, W_J^-) = (W_\parallel^+, W_I^+, W_J^+)}$$

in the matched opposite-valley frame;

$$\boxed{\textbf{(C3)}\ \mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0}$$

so that $V(0) = 0$;

$$\boxed{\textbf{(C4)}\ \Delta > 0}$$

via either $\delta_{0,\rho} > \eta_{R,\rho}$ or bounded perturbation gap stability;

$$\boxed{\textbf{(C5)}\ A_{\mathrm{res}} \cap \mathrm{span}_{\mathbb{R}}\{\rho_1 \otimes \mathbf{1}_2, \rho_2 \otimes \mathbf{1}_2\} = \{0\}}$$

for mass protection.

If all five checks pass, then

$$\boxed{H_{\mathrm{eff}}(q) = \rho_3 \otimes (q_i \sigma_i) + O(|q|^2/\Delta)}$$

is an isolated local massless Dirac cone.

So the local Dirac problem is now a finite microscopic matching checklist, not an open structural problem.

FINAL MICROSCOPIC MATCHING CHECKLIST FOR THE LOCAL DIRAC SECTOR

Final target

The final target is to show that the full microscopic TECT linearization matches the explicit seed/representative strongly enough to produce an isolated local massless Dirac cone:

$$H_{\text{eff}}(q) = \rho_3 \otimes (q_i\sigma_i) + O(|q|^2/\Delta).$$

At this stage, the local Dirac problem is no longer an open structural problem. It is a finite matching checklist.

Checklist 1: witness-side microscopic matching

Extract the actual microscopic projected first-order symbol at the selected valley $+G_*$, and read off the S2 coefficients

$$A_1(p) = \lambda_\| p_\| \sigma_3 + \alpha(p_1\sigma_1 + p_2\sigma_2) + \beta(-p_2\sigma_1 + p_1\sigma_2).$$

Check:

$$\boxed{\lambda_\| \neq 0, \qquad (\alpha, \beta) \neq (0,0).} \tag{1058}$$

Inside the explicit seed space one already has

$$\lambda_\| = \frac{1}{2}c_\| I_{\theta'^3}, \qquad \alpha = c_E I_\theta, \qquad \beta = c_O I_\theta,$$

so the exact seed-space nondegeneracy condition is

$$c_\| \neq 0, \qquad (c_E, c_O) \neq (0,0).$$

Checklist 2: opposite-valley matching

Repeat the same witness extraction at $-G_*$, and compare in the matched opposite-valley frame.
Check:

$$\boxed{(W_\|^-, W_I^-, W_J^-) = (W_\|^+, W_I^+, W_J^+).} \tag{1059}$$

If true, the two local Weyl valleys pair into a doubled local Dirac block.

Checklist 3: zero constant mixing audit

Let

$$\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R$$

be the microscopic decomposition into the local doubled shell block and the remote sector. At the crossing point, identify the little group H_*.
Check:

$$\boxed{\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0.} \tag{1060}$$

Equivalently, verify that every remote sector differs from $\mathcal{H}_D$ in at least one shell-relevant label. Then

$$V(0) = 0.$$

Checklist 4: remote spectral gap audit

For the remote block $H_R(p)$, define

$$\delta_{0,\rho} := \inf_{|p|\le\rho} \delta_0(p), \qquad \eta_{R,\rho} := \sup_{|p|\le\rho} \eta_R(p).$$

Check either:

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho},} \tag{1061}$$

which gives

$$\Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0,$$

or, alternatively, check bounded perturbation stability from an already-gapped realization:

$$\boxed{\sup_{|p|\le p_*} \|\delta H_R(p)\| < \Delta_0.} \tag{1062}$$

If either holds, then the remote sector is spectrally isolated by a positive gap $\Delta > 0$.

Checklist 5: mass-protection audit

For the isolated doubled local Dirac block, the constant mass space is

$$\mathcal{M} = \mathrm{span}_{\mathbb{R}}\{\rho_1 \otimes \mathbf{1}_2, \rho_2 \otimes \mathbf{1}_2\}.$$

Let $A_{\rm res}$ be the actual residual constant symmetry-allowed operator algebra on the low-energy doubled shell block. Check:

$$\boxed{A_{\rm res} \cap \mathcal{M} = \{0\}.} \tag{1063}$$

A sufficient mechanism is exact residual valley $U(1)_V$ or its $\mathbb{Z}_2^V$ remnant.

Pass condition

If all five checks pass, then the full microscopic TECT linearization yields an isolated local massless Dirac cone:

$$\boxed{H_{\rm eff}(q) = \rho_3 \otimes (q_i\sigma_i) + O(|q|^2/\Delta).} \tag{1064}$$

Most compressed version

The entire local Dirac matching problem is therefore compressed to:

$$\boxed{\textbf{(C1)}\ \lambda_\parallel \neq 0,\ (\alpha,\beta) \neq (0,0),} \tag{1065}$$

$$\boxed{\textbf{(C2)}\ (W_\parallel^-, W_I^-, W_J^-) = (W_\parallel^+, W_I^+, W_J^+),} \tag{1066}$$

$$\boxed{\textbf{(C3)}\ \mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0,} \tag{1067}$$

$$\boxed{\textbf{(C4)}\ \Delta > 0,} \tag{1068}$$

$$\boxed{\textbf{(C5)}\ A_{\rm res} \cap \mathcal{M} = \{0\}.} \tag{1069}$$

If **(C1)**–**(C5)** hold, the local Dirac sector is closed.

. Summary

$$\boxed{\textbf{Final microscopic matching checklist for the local Dirac sector}}$$

Check the following five items:

$$\boxed{\textbf{(C1)}\ \lambda_{\parallel} \neq 0, \qquad (\alpha, \beta) \neq (0,0)}$$

from the actual microscopic witness extraction;

$$\boxed{\textbf{(C2)}\ (W_{\parallel}^{-}, W_{I}^{-}, W_{J}^{-}) = (W_{\parallel}^{+}, W_{I}^{+}, W_{J}^{+})}$$

in the matched opposite-valley frame;

$$\boxed{\textbf{(C3)}\ \mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0}$$

so that $V(0) = 0$;

$$\boxed{\textbf{(C4)}\ \Delta > 0}$$

via either $\delta_{0,\rho} > \eta_{R,\rho}$ or bounded perturbation gap stability;

$$\boxed{\textbf{(C5)}\ A_{\mathrm{res}} \cap \mathrm{span}_{\mathbb{R}}\{\rho_1 \otimes \mathbf{1}_2, \rho_2 \otimes \mathbf{1}_2\} = \{0\}}$$

for mass protection.

If all five checks pass, then

$$\boxed{H_{\mathrm{eff}}(q) = \rho_3 \otimes (q_i \sigma_i) + O(|q|^2/\Delta)}$$

is an isolated local massless Dirac cone.

LONGITUDINAL SEED EMBEDDING TEST

1. Goal

The only remaining unresolved part of C1 is whether the full microscopic transport space

$$V_{\mathrm{micro}}$$

actually contains the explicit longitudinal seed direction

$$N_{\mathrm{seed}}^{(\parallel)} \sim V_{(\parallel)}^{\mu}(\xi) = \theta'(\xi)\, (\Pi_Q \mathcal{T}_{\xi}^{2} \Pi_Q)_{\mathrm{tr}}\, n^{\mu}.$$

Inside the explicit representative this becomes

$$\boxed{V_{(\parallel)}^{\mu}(\xi) = \frac{1}{2}\, \theta'(\xi)^3\, n^{\mu} \sigma_3.} \tag{1070}$$

So the embedding problem is:

$$\boxed{V_{\mathrm{micro}} \ni N_{\mathrm{seed}}^{(\parallel)}\ ?} \tag{1071}$$

2. Minimal tensor structure that must appear

A longitudinal seed requires the simultaneous presence of the following three ingredients in the microscopic transport kernel:

a. (L1) odd defect-gradient factor An odd scalar profile proportional to

$$\theta'(\xi).$$

b. (L2) traceless internal Q_B-scalar A 2×2 Hermitian traceless operator on the Q_B-doublet, i.e.

$$(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} = s_1(\xi)\sigma_1 + s_2(\xi)\sigma_2 + s_3(\xi)\sigma_3.$$

c. (L3) shell-axis worldvolume factor A contraction with the distinguished shell/worldvolume direction n^μ.

Thus the minimal longitudinal transport structure is

$$\boxed{V^\mu_{\mathrm{micro}}(\xi) \supset \theta'(\xi)\Big[s_1(\xi)\sigma_1 + s_2(\xi)\sigma_2 + s_3(\xi)\sigma_3\Big]\, n^\mu.} \tag{1072}$$

For the explicit representative, only the σ_3-part survives:

$$s_1 = s_2 = 0, \qquad s_3 = \frac{1}{2}\,\theta'(\xi)^2.$$

3. Embedding criterion in Pauli language

Let the actual microscopic first-order transport kernel projected to the Q_B-doublet be

$$V^\mu_{\mathrm{micro}}(\xi) = v^\mu_0(\xi)\mathbf{1}_2 + v^\mu_1(\xi)\sigma_1 + v^\mu_2(\xi)\sigma_2 + v^\mu_3(\xi)\sigma_3. \tag{1073}$$

The longitudinal seed embedding succeeds if the σ_3-component contains a nonzero shell-axis odd piece:

$$\boxed{v^\mu_3(\xi) \supset n^\mu\,\theta'(\xi)\,q(\xi), \qquad q(\xi) \not\equiv 0,} \tag{1074}$$

with $q(\xi)$ even. In the explicit representative,

$$q(\xi) = \frac{1}{2}\,\theta'(\xi)^2.$$

So a sharp sufficient criterion is:

$$\boxed{\text{the projected microscopic kernel has a nonzero odd } n^\mu\sigma_3\text{-component.}} \tag{1075}$$

4. Equivalent overlap test

Instead of reading the Pauli coefficient directly, one may test the overlap against the explicit longitudinal seed profile. Define

$$\boxed{\mathcal{E}_\parallel := \int d\xi\,\theta'(\xi)\,\mathrm{Tr}_{Q_B}\Big[\sigma_3\, n_\mu\, V^\mu_{\mathrm{micro}}(\xi)\Big].} \tag{1076}$$

Then a sufficient condition for longitudinal embedding is

$$\boxed{\mathcal{E}_\parallel \neq 0.} \tag{1077}$$

If one also includes the graded low-mode weights, the strongest practical version is

$$\boxed{\mathcal{J}_\parallel := \int d\xi\, f_+(\xi) f_-(\xi)\,\mathrm{Tr}_{Q_B}\Big[\sigma_3\, n_\mu\, V^\mu_{\mathrm{micro}}(\xi)\Big] \neq 0.} \tag{1078}$$

This is the direct embedding analogue of the representative identity

$$J^{(3)}_\parallel = \frac{1}{2} I_{\theta'^3} \neq 0.$$

5. Relation to the representative

In the explicit representative,

$$V^\mu_{\rm rep}(\xi) = \frac{1}{2}\,\theta'(\xi)^3\,n^\mu\sigma_3.$$

Then

$$\mathrm{Tr}_{Q_B}\Big[\sigma_3\,n_\mu\,V^\mu_{\rm rep}(\xi)\Big] = \theta'(\xi)^3$$

up to normalization, and

$$\mathcal{J}_\parallel \propto \int d\xi\, f_+ f_-\,\theta'^3 = I_{\theta'^3} \neq 0.$$

So the representative passes the embedding test exactly.

6. Final practical algorithm

The longitudinal seed embedding test is therefore:

d. *Step A* Compute the projected microscopic kernel $V^\mu_{\rm micro}(\xi)$ on the Q_B-doublet.
e. *Step B* Pauli-decompose it as in (1073).
f. *Step C* Check whether $v^\mu_3(\xi)$ contains a nonzero odd n^μ-aligned piece.
g. *Step D* Compute the weighted overlap $\mathcal{J}_\parallel$ in (1078).
h. *Pass condition*

$$\boxed{\mathcal{J}_\parallel \neq 0 \Longrightarrow N^{(\parallel)}_{\rm seed}\ \text{is embedded in}\ V_{\rm micro} \Longrightarrow \lambda_\parallel \neq 0.} \tag{1079}$$

7. Strongest honest conclusion

Thus the last unresolved part of C1 is now reduced to a single finite matching test:

$$\boxed{\text{Does the full microscopic projected transport kernel contain a nonzero odd } n^\mu\sigma_3\text{-component with nonvanishing graded overlap?}} \tag{1080}$$

This is strictly sharper than the earlier carrier-audit phrasing. It is not a broad search over tensor spaces anymore; it is one explicit coefficient/overlap test.

. Summary

$$\boxed{\textbf{Longitudinal seed embedding test}}$$

The explicit longitudinal representative is

$$\boxed{V^\mu_{(\parallel)}(\xi) = \frac{1}{2}\,\theta'(\xi)^3\,n^\mu\sigma_3.}$$

So in the full microscopic projected kernel

$$\boxed{V^\mu_{\rm micro}(\xi) = v^\mu_0(\xi)\mathbf{1}_2 + v^\mu_1(\xi)\sigma_1 + v^\mu_2(\xi)\sigma_2 + v^\mu_3(\xi)\sigma_3,}$$

the embedding succeeds if the σ_3-component contains a nonzero odd shell-axis piece:

$$\boxed{v^\mu_3(\xi) \supset n^\mu\,\theta'(\xi)\,q(\xi), \qquad q(\xi) \not\equiv 0.}$$

A direct weighted overlap test is

$$\boxed{\mathcal{J}_{\parallel} := \int d\xi\, f_{+}(\xi) f_{-}(\xi)\, \mathrm{Tr}_{Q_B}\Big[\sigma_3\, n_\mu\, V^{\mu}_{\mathrm{micro}}(\xi)\Big].}$$

If

$$\boxed{\mathcal{J}_{\parallel} \neq 0,}$$

then

$$\boxed{N^{(\parallel)}_{\mathrm{seed}} \subset V_{\mathrm{micro}}, \qquad \lambda_{\parallel} \neq 0.}$$

So the last unresolved part of C1 is now one explicit finite test:

$$\boxed{\text{nonzero odd } n^{\mu}\sigma_3\text{-component} + \text{nonvanishing graded overlap.}}$$

C2: OPPOSITE-VALLEY MATCHING

1. Goal

After C1, the selected valley $+G_*$ carries a nondegenerate local Weyl symbol. To obtain a local doubled Dirac block, one must verify that the opposite valley $-G_*$ carries the matched witness data.

The exact C2 condition is

$$\boxed{(W^{-}_{\parallel}, W^{-}_{I}, W^{-}_{J}) = (W^{+}_{\parallel}, W^{+}_{I}, W^{+}_{J})} \tag{1081}$$

in the matched opposite-valley frame. This is the finite witness formulation of local Dirac doubling. :contentReference[oaicite:2]index=2

2. What must be compared

At $+G_*$, extract the projected first-order symbol

$$A^{+}_{1}(p) = \lambda^{+}_{\parallel} p_{\parallel}\sigma_3 + \alpha^{+}(p_1\sigma_1 + p_2\sigma_2) + \beta^{+}(-p_2\sigma_1 + p_1\sigma_2),$$

and define

$$W^{+}_{\parallel} = \lambda^{+}_{\parallel}, \qquad W^{+}_{I} = \alpha^{+}, \qquad W^{+}_{J} = \beta^{+}.$$

At $-G_*$, do the same:

$$A^{-}_{1}(p) = \lambda^{-}_{\parallel} p_{\parallel}\sigma_3 + \alpha^{-}(p_1\sigma_1 + p_2\sigma_2) + \beta^{-}(-p_2\sigma_1 + p_1\sigma_2),$$

with

$$W^{-}_{\parallel} = \lambda^{-}_{\parallel}, \qquad W^{-}_{I} = \alpha^{-}, \qquad W^{-}_{J} = \beta^{-}.$$

Then C2 is exactly (1081).

3. Why raw comparison is not the right comparison

One must not compare the coefficients in an arbitrary laboratory basis. The correct comparison is done after choosing the matched opposite-valley frame

$$\{\hat{n}^{+}, e^{+}_{1}, e^{+}_{2}\}, \qquad \{\hat{n}^{-}, e^{-}_{1}, e^{-}_{2}\},$$

so that the opposite-valley Hamiltonian is brought to the standard doubled form.

Thus the real C2 question is:

$$\boxed{\text{after matched-frame identification, do the two witness triples coincide?}}$$

This is exactly how the witness-level opposite-valley theorem is formulated.

4. Concrete Class II seed family

Inside the concrete Class II family, the same seed amplitudes

$$c_{\parallel}, \qquad c_I, \qquad c_J$$

determine the witness data at both valleys. In the matched opposite-valley frame, they are read identically:

$$c_{\parallel}^{-} = c_{\parallel}^{+}, \qquad c_I^{-} = c_I^{+}, \qquad c_J^{-} = c_J^{+}.$$

Hence

$$W_{\parallel}^{-} = W_{\parallel}^{+}, \qquad W_I^{-} = W_I^{+}, \qquad W_J^{-} = W_J^{+}.$$

Therefore, inside the explicit seed/representative family, C2 is already closed at the witness level. :contentReference[oaicite:4]index=4

5. Final microscopic matching test for C2

So the only remaining microscopic task is not to invent a new criterion, but to verify that the full microscopic extraction produces the same matched seed sector at both valleys.

Operationally:

a. *(C2-a)* Extract

$$(W_{\parallel}^{+}, W_I^{+}, W_J^{+})$$

from the projected microscopic symbol at $+G_*$.

b. *(C2-b)* Extract

$$(W_{\parallel}^{-}, W_I^{-}, W_J^{-})$$

from the projected microscopic symbol at $-G_*$.

c. *(C2-c)* Pass to the matched opposite-valley frame.

d. *(C2-d)* Check

$$(W_{\parallel}^{-}, W_I^{-}, W_J^{-}) = (W_{\parallel}^{+}, W_I^{+}, W_J^{+}).$$

If true, C2 passes.

6. Strongest honest conclusion

Thus C2 is already structurally closed. The remaining part is only a final microscopic matching check:

$$\boxed{\text{does the full microscopic TECT extraction yield the same witness triple at } \pm G_* \text{ in the matched frame?}} \tag{1082}$$

This is a finite comparison problem, not an open structural problem.

. Summary

$$\boxed{\textbf{C2: opposite-valley matching}}$$

The exact condition is

$$\boxed{(W_{\parallel}^{-}, W_I^{-}, W_J^{-}) = (W_{\parallel}^{+}, W_I^{+}, W_J^{+})}$$

in the matched opposite-valley frame.

Inside the explicit Class II seed/representative family, this is already closed, because the same seed amplitudes

$$\boxed{c_\parallel, \qquad c_I, \qquad c_J}$$

are read identically at the two valleys after matched-frame identification.

So the remaining microscopic C2 task is only:

$$\boxed{\text{extract the witness triple at } +G_* \text{ and } -G_*, \text{ pass to the matched frame, and compare them.}}$$

If they are equal, then the two Weyl valleys pair into a local doubled Dirac block.

C3: ZERO CONSTANT MIXING

1. Goal

Write the full microscopic Hamiltonian near the selected crossing as

$$H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^\dagger & H_R(p) \end{pmatrix},$$

where $\mathcal{H}_D$ is the local doubled shell Dirac block and $\mathcal{H}_R$ is the remote sector.

The target is

$$\boxed{V(0) = 0.}$$

2. Theorem-level criterion

Let H_* be the little group / residual symmetry at the crossing $p = 0$. Assume the Hamiltonian is H_*-equivariant and the decomposition

$$\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R$$

is H_*-invariant.

Then

$$V(0) \in \operatorname{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D).$$

Hence

$$\boxed{\operatorname{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \Longrightarrow V(0) = 0.} \tag{1083}$$

3. Sectorwise reduction

Decompose the remote sector into H_*-invariant pieces

$$\mathcal{H}_R = \bigoplus_a K_a.$$

Then

$$\operatorname{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) \cong \bigoplus_a \operatorname{Hom}_{H_*}(K_a, \mathcal{H}_D).$$

So it is enough to prove

$$\operatorname{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0 \qquad \forall a.$$

4. Practical label audit

A remote sector K_a is excluded from constant mixing if it differs from $\mathcal{H}_D$ in at least one shell-relevant label:

1. selected shell-pair label $\pm G_*$,
2. transverse vs longitudinal/volumetric character,
3. enriched internal doublet label,
4. residual parity / grading / chirality,
5. any extra conserved microscopic label.

Therefore the practical test is:

$$\boxed{\text{for every } K_a \subset \mathcal{H}_R, \text{ find at least one mismatch label relative to } \mathcal{H}_D.}$$

Then

$$\mathrm{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0$$

for all a, hence

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0$$

and therefore $V(0) = 0$.

5. Minimal truncation checklist

In the minimal dangerous truncation

$$\mathcal{H}_R = \mathcal{H}_{\mathrm{vol}} \oplus \mathcal{H}_{\mathrm{high}} \oplus \mathcal{H}_{\mathrm{non-enr}} \oplus \mathcal{H}_{\mathrm{wrong-int}} \oplus \mathcal{H}_{\mathrm{extra}},$$

the default eliminations are:

$$\mathcal{H}_{\mathrm{vol}}: \text{ volumetric mismatch,}$$

$$\mathcal{H}_{\mathrm{high}}: \text{ different shell orbit mismatch,}$$

$$\mathcal{H}_{\mathrm{non-enr}}: \text{ no enriched internal doublet,}$$

$$\mathcal{H}_{\mathrm{wrong-int}}: \text{ wrong internal representation,}$$

while $\mathcal{H}_{\mathrm{extra}}$ must be checked case by case.

6. Completion criterion

Thus C3 is complete once one proves

$$\boxed{\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0.}$$

Equivalently:

$$\boxed{\text{the local Dirac block and the remote sector share no common shell-relevant irreducible representation.}}$$

Then

$$\boxed{V(0) = 0.}$$

. Summary

$$\boxed{\textbf{C3: zero constant mixing}}$$

The exact theorem-level criterion is

$$\boxed{V(0) \in \mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D), \qquad \mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \Longrightarrow V(0) = 0.}$$

So C3 reduces to a finite sectorwise label audit.
Decompose

$$\boxed{\mathcal{H}_R = \bigoplus_a K_a.}$$

A sector K_a is excluded from constant mixing if it differs from $\mathcal{H}_D$ in at least one of:

$$\boxed{\pm G_*, \quad \text{transverse vs volumetric character}, \quad \text{enriched internal doublet}, \quad \text{parity/grading/chirality}, \quad \text{extra conserved label.}}$$

Hence the completion criterion is

$$\boxed{\forall a,\ \mathrm{Hom}_{H_*}(K_a, \mathcal{H}_D) = 0 \Longrightarrow \mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \Longrightarrow V(0) = 0.}$$

So C3 is a finite remote-sector label audit, not an open structural problem.

C4: REMOTE GAP $\Delta > 0$

1. Goal

Let the microscopic Hamiltonian near the selected crossing be written as

$$H_{\mathrm{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^\dagger & H_R(p) \end{pmatrix},$$

where H_D acts on the local doubled shell Dirac block $\mathcal{H}_D$ and H_R acts on the remote sector $\mathcal{H}_R$. The C4 target is:

$$\boxed{\mathrm{spec}(H_R(p)) \cap (-\Delta, \Delta) = \varnothing \qquad \text{for small } p, \quad \Delta > 0.} \tag{1084}$$

2. Route A: direct Bloch/spectral lower-bound test

Write the remote block as

$$H_R(p) = D_R(p) + E_R(p),$$

where $D_R(p)$ is a diagonal or benchmark part and $E_R(p)$ is the remaining correction.
Define

$$\delta_0(p) := \min_a |(D_R)_{aa}(p)|, \qquad \eta_R(p) := \|E_R(p)\|. \tag{1085}$$

For a low-momentum neighborhood

$$N_\rho = \{p : |p| \le \rho\},$$

define

$$\delta_{0,\rho} := \inf_{|p|\le\rho} \delta_0(p), \qquad \eta_{R,\rho} := \sup_{|p|\le\rho} \eta_R(p). \tag{1086}$$

Then the direct remote-gap criterion is:

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho} \Longrightarrow \Delta_\rho := \delta_{0,\rho} - \eta_{R,\rho} > 0,} \tag{1087}$$

and hence

$$\boxed{\mathrm{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \varnothing \qquad \forall |p| \le \rho.} \tag{1088}$$

This is the finite Bloch/spectral lower-bound route.

3. Route B: bounded perturbation stability from an already-gapped realization

Suppose there exists a reference remote block $H_R^{(0)}(p)$ satisfying

$$\boxed{\mathrm{spec}(H_R^{(0)}(p)) \cap (-\Delta_0, \Delta_0) = \varnothing \qquad (\Delta_0 > 0)} \tag{1089}$$

for all $|p| \le p_*$, and write

$$H_R(p) = H_R^{(0)}(p) + \delta H_R(p). \tag{1090}$$

If

$$\boxed{\sup_{|p|\le p_*} \|\delta H_R(p)\| \le \varepsilon < \Delta_0,} \tag{1091}$$

then the perturbed remote block still has a positive gap

$$\boxed{\Delta = \Delta_0 - \varepsilon > 0, \qquad \mathrm{spec}(H_R(p)) \cap (-\Delta, \Delta) = \varnothing.} \tag{1092}$$

A common strong form is

$$\|\delta H_R\| < \Delta_0/2,$$

which implies

$$\Delta > \Delta_0/2 > 0.$$

4. Consequence for the resolvent and Schur complement

Once C4 holds, for $|E| < \Delta/2$,

$$\boxed{\|(H_R(p) - E)^{-1}\| \le \frac{2}{\Delta}.} \tag{1093}$$

If C3 also holds, i.e.

$$V(0) = 0, \qquad \|V(p)\| \le c|p|,$$

then the Schur-complement self-energy obeys

$$\boxed{\Sigma(E,p) = V(p)(H_R(p) - E)^{-1}V(p)^\dagger = O\left(\frac{|p|^2}{\Delta}\right).} \tag{1094}$$

Hence the remote sector changes the local Dirac block only at quadratic order.

5. Minimal practical checklist for C4

To complete C4 in practice, do one of the following.

a. Direct route.

1. Build the reduced remote block $H_R(p)$.
2. Choose a benchmark diagonal part $D_R(p)$.
3. Compute $\delta_{0,\rho}$.
4. Estimate $\eta_{R,\rho}$.
5. Check whether $\delta_{0,\rho} > \eta_{R,\rho}$.

b. Perturbative route.

1. Choose an already-gapped reference model $H_R^{(0)}(p)$ with gap Δ_0.
2. Compute the correction $\delta H_R(p)$.
3. Estimate $\sup \|\delta H_R\|$.
4. Check whether $\sup \|\delta H_R\| < \Delta_0$.

6. Strongest honest conclusion

Therefore C4 is not an open structural problem. It is a finite spectral check:

$$\boxed{\text{either prove } \delta_{0,\rho} > \eta_{R,\rho} \quad \text{or prove } \sup \|\delta H_R\| < \Delta_0.} \tag{1095}$$

If either succeeds, then $\Delta > 0$ is established.

. Summary

$$\boxed{\textbf{C4: remote gap } \Delta > 0}$$

There are two standard ways to prove C4.

a. Route A: direct lower-bound test Write

$$\boxed{H_R(p) = D_R(p) + E_R(p),}$$

define

$$\boxed{\delta_{0,\rho} := \inf_{|p|\le\rho} \min_a |(D_R)_{aa}(p)|, \qquad \eta_{R,\rho} := \sup_{|p|\le\rho} \|E_R(p)\|,}$$

and check

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho}.}$$

Then

$$\boxed{\Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0.}$$

b. Route B: bounded perturbation stability If

$$\boxed{\operatorname{spec}(H_R^{(0)}(p)) \cap (-\Delta_0, \Delta_0) = \varnothing}$$

and

$$\boxed{\sup_{|p| \le p_*} \|\delta H_R(p)\| < \Delta_0,}$$

then

$$\boxed{\Delta = \Delta_0 - \sup \|\delta H_R\| > 0.}$$

Once C4 holds,

$$\boxed{\|(H_R(p) - E)^{-1}\| \le \frac{2}{\Delta} \qquad (|E| < \Delta/2),}$$

and together with C3,

$$\boxed{\Sigma(E, p) = O\left(\frac{|p|^2}{\Delta}\right).}$$

So C4 is a finite Bloch/spectral lower-bound check, not an open structural problem.

ACTUAL C3 AUDIT IN THE MINIMAL TRUNCATION

1. Target

We want to prove, not merely restate, that in the minimal shell-relevant truncation

$$V(0) = 0.$$

The theorem-level criterion is already known:

$$V(0) \in \operatorname{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D), \qquad \operatorname{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \Longrightarrow V(0) = 0.$$

So the task is a sectorwise actual mismatch audit.

2. Minimal truncated Hilbert space

Use the minimal truncation

$$\mathcal{H}_{\rm full}^{\rm tr} = \mathcal{H}_D \oplus \mathcal{H}_{\rm vol} \oplus \mathcal{H}_{\rm high} \oplus \mathcal{H}_{\rm non\text{-}enr} \oplus \mathcal{H}_{\rm wrong\text{-}int}. \tag{1096}$$

Here

$$\mathcal{H}_D \cong \mathbb{C}^2_{\rm pair} \otimes \mathbb{C}^2_{\rm int}$$

is the local doubled Dirac carrier, while the four remote sectors are explicitly identified as the nearest potentially dangerous competitors. This is exactly the first nontrivial finite truncation introduced for Tasks III and V. :contentReference[oaicite:3]index=3

3. Sectorwise audit

We now verify, one by one, that each remote sector differs from $\mathcal{H}_D$ in at least one shell-relevant label.

c. 3.1. Volumetric sector

$$\mathcal{H}_{\text{vol}}$$

is the volumetric/longitudinal sector, whereas $\mathcal{H}_D$ is the transverse shell-relevant doubled Dirac carrier. Hence there is an immediate mismatch in the label

transverse vs longitudinal/volumetric character.

Therefore

$$\boxed{\mathrm{Hom}_{H_*}(\mathcal{H}_{\text{vol}}, \mathcal{H}_D) = 0.} \tag{1097}$$

d. 3.2. Higher-shell sector

$$\mathcal{H}_{\text{high}}$$

contains modes not belonging to the selected shell pair $\pm G_*$, while $\mathcal{H}_D$ is built precisely on that selected shell pair. Hence there is an immediate mismatch in the label

selected shell-pair / shell orbit.

Therefore

$$\boxed{\mathrm{Hom}_{H_*}(\mathcal{H}_{\text{high}}, \mathcal{H}_D) = 0.} \tag{1098}$$

e. 3.3. Same-shell but non-enriched sector

$$\mathcal{H}_{\text{non-enr}}$$

may lie on the same shell, but by definition does not carry the shell-relevant enriched internal doublet. By contrast,

$$\mathcal{H}_D \cong \mathbb{C}^2_{\text{pair}} \otimes \mathbb{C}^2_{\text{int}}$$

does carry exactly that doubled internal structure. Hence there is an immediate mismatch in the label

presence of enriched internal doublet.

Therefore

$$\boxed{\mathrm{Hom}_{H_*}(\mathcal{H}_{\text{non-enr}}, \mathcal{H}_D) = 0.} \tag{1099}$$

f. 3.4. Wrong-internal sector

$$\mathcal{H}_{\text{wrong-int}}$$

contains modes whose internal content does not match the selected shell doublet. Hence there is an immediate mismatch in the label

internal representation.

Therefore

$$\boxed{\mathrm{Hom}_{H_*}(\mathcal{H}_{\text{wrong-int}}, \mathcal{H}_D) = 0.} \tag{1100}$$

4. Assemble the result

Since

$$\mathcal{H}_R^{\text{tr}} = \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}},$$

we have

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R^{\mathrm{tr}}, \mathcal{H}_D) \cong \mathrm{Hom}_{H_*}(\mathcal{H}_{\mathrm{vol}}, \mathcal{H}_D) \oplus \mathrm{Hom}_{H_*}(\mathcal{H}_{\mathrm{high}}, \mathcal{H}_D) \oplus \mathrm{Hom}_{H_*}(\mathcal{H}_{\text{non-enr}}, \mathcal{H}_D) \oplus \mathrm{Hom}_{H_*}(\mathcal{H}_{\text{wrong-int}}, \mathcal{H}_D).$$

Using (1097)–(1100), every summand vanishes. Hence

$$\boxed{\mathrm{Hom}_{H_*}(\mathcal{H}_R^{\mathrm{tr}}, \mathcal{H}_D) = 0.} \tag{1101}$$

By H_*-equivariance of the truncated Hamiltonian at $p = 0$,

$$V(0) \in \mathrm{Hom}_{H_*}(\mathcal{H}_R^{\mathrm{tr}}, \mathcal{H}_D).$$

Therefore

$$\boxed{V(0) = 0.} \tag{1102}$$

5. Exact status

Thus C3 is actually completed in the minimal truncation:

$$\boxed{\text{for } \mathcal{H}_R^{\mathrm{tr}} = \mathcal{H}_{\mathrm{vol}} \oplus \mathcal{H}_{\mathrm{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}}, \quad V(0) = 0 \text{ is proved.}} \tag{1103}$$

What would remain open only concerns any additional remote sectors beyond the minimal truncation, if such sectors are later added. But for the stated minimal truncation, C3 is done.

. Summary

$$\boxed{\textbf{Actual C3 result in the minimal truncation}}$$

Use

$$\boxed{\mathcal{H}_{\mathrm{full}}^{\mathrm{tr}} = \mathcal{H}_D \oplus \mathcal{H}_{\mathrm{vol}} \oplus \mathcal{H}_{\mathrm{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}}.}$$

Then:

$$\boxed{\mathcal{H}_{\mathrm{vol}} \text{ mismatches } \mathcal{H}_D \text{ by transverse vs volumetric character} \Longrightarrow \mathrm{Hom}_{H_*}(\mathcal{H}_{\mathrm{vol}}, \mathcal{H}_D) = 0,}$$

$$\boxed{\mathcal{H}_{\mathrm{high}} \text{ mismatches } \mathcal{H}_D \text{ by shell orbit} \Longrightarrow \mathrm{Hom}_{H_*}(\mathcal{H}_{\mathrm{high}}, \mathcal{H}_D) = 0,}$$

$$\boxed{\mathcal{H}_{\text{non-enr}} \text{ mismatches } \mathcal{H}_D \text{ by lack of enriched internal doublet} \Longrightarrow \mathrm{Hom}_{H_*}(\mathcal{H}_{\text{non-enr}}, \mathcal{H}_D) = 0,}$$

$$\boxed{\mathcal{H}_{\text{wrong-int}} \text{ mismatches } \mathcal{H}_D \text{ by wrong internal representation} \Longrightarrow \mathrm{Hom}_{H_*}(\mathcal{H}_{\text{wrong-int}}, \mathcal{H}_D) = 0.}$$

Hence

$$\boxed{\mathrm{Hom}_{H_*}(\mathcal{H}_R^{\mathrm{tr}}, \mathcal{H}_D) = 0 \Longrightarrow V(0) = 0.}$$

So, in the minimal truncation, **C3 is completed**.

ACTUAL C4 STATUS IN THE MINIMAL TRUNCATION

1. Minimal truncated remote block

In the minimal shell-relevant truncation, the full Hilbert space is

$$\mathcal{H}^{\mathrm{tr}}_{\mathrm{full}} = \mathcal{H}_D \oplus \mathcal{H}_{\mathrm{vol}} \oplus \mathcal{H}_{\mathrm{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}}.$$

The remote sector is therefore

$$\mathcal{H}^{\mathrm{tr}}_{R} = \mathcal{H}_{\mathrm{vol}} \oplus \mathcal{H}_{\mathrm{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}}.$$

Write the truncated Hamiltonian as

$$H^{\mathrm{tr}}_{\mathrm{full}}(p) = \begin{pmatrix} H_D(p) & V_{DR}(p) \\ V_{DR}(p)^{\dagger} & H_R(p) \end{pmatrix}, \qquad H_R(p) = D_R(p) + R_R(p).$$

Here $D_R(p)$ is the diagonal sector-center part and $R_R(p)$ is the remote-remote mixing remainder.

2. Exact lower-bound reduction

Define

$$\delta_0(p) = \min\{|\mu_{\mathrm{vol}}(p)|, |\mu_{\mathrm{high}}(p)|, |\mu_{\text{non-enr}}(p)|, |\mu_{\text{wrong-int}}(p)|\},$$

$$\eta_R(p) = \|R_R(p)\|.$$

Then the minimal truncation remote-gap theorem gives the pointwise operator bound

$$\boxed{\|H_R(p)\psi\| \geq (\delta_0(p) - \eta_R(p))\|\psi\|.}$$

Hence, if

$$\delta_0(p) > \eta_R(p),$$

then the remote sector has a pointwise gap

$$\mathrm{spec}(H_R(p)) \cap (-\Delta(p), \Delta(p)) = \varnothing, \qquad \Delta(p) \geq \delta_0(p) - \eta_R(p) > 0.$$

On a neighborhood

$$N_\rho = \{p : |p| \leq \rho\},$$

define

$$\delta_{0,\rho} = \inf_{|p|\leq\rho} \delta_0(p), \qquad \eta_{R,\rho} = \sup_{|p|\leq\rho} \eta_R(p).$$

Then

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho} \Longrightarrow \mathrm{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \varnothing \quad (|p| \leq \rho),}$$

with

$$\boxed{\Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0.}$$

Equivalently, one may use the Gershgorin lower bound

$$\Delta_{G,\rho} := \inf_{|p|\leq\rho} \min_a \Big(|(H_R)_{aa}(p)| - \sum_{b\neq a} |(H_R)_{ab}(p)| \Big) > 0.$$

3. What is actually completed

Therefore, at the theorem level, C4 is already completely reduced to the finite inequality

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho} \quad \text{or equivalently} \quad \Delta_{G,\rho} > 0.}$$

This means that the abstract remote-gap question has been converted into a finite-dimensional lower-bound problem on the minimal truncated remote matrix.

4. What is not yet numerically/analytically completed

To actually conclude

$$\Delta_\rho > 0,$$

one still needs the explicit lower bounds for the four sector-center energies:

$$\mu_{\text{vol}}(p), \qquad \mu_{\text{high}}(p), \qquad \mu_{\text{non-enr}}(p), \qquad \mu_{\text{wrong-int}}(p),$$

and an explicit upper bound for

$$\eta_R(p) = \|R_R(p)\|.$$

At the present stage, the uploaded material does not yet provide these four actual center-energy formulas in a fully evaluated form for the final microscopic TECT realization. Hence the final sign of

$$\delta_{0,\rho} - \eta_{R,\rho}$$

has not yet been numerically or analytically verified.

5. Exact honest status of C4

So the honest status is:

$$\boxed{\text{C4 is closed as a finite lower-bound theorem in the minimal truncation,}}$$

but

$$\boxed{\text{the actual positive value of } \Delta_\rho \text{ still requires explicit evaluation of } \mu_{\text{vol}}, \mu_{\text{high}}, \mu_{\text{non-enr}}, \mu_{\text{wrong-int}}, \eta_{R,\rho}.}$$

Thus C4 is not structurally open anymore, but it is not yet numerically completed either.

. Summary

$$\boxed{\textbf{Actual C4 status}}$$

In the minimal truncation,

$$\mathcal{H}_R^{\text{tr}} = \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non-enr}} \oplus \mathcal{H}_{\text{wrong-int}}, \qquad H_R(p) = D_R(p) + R_R(p).$$

Define

$$\delta_0(p) = \min\{|\mu_{\text{vol}}(p)|, |\mu_{\text{high}}(p)|, |\mu_{\text{non-enr}}(p)|, |\mu_{\text{wrong-int}}(p)|\}, \qquad \eta_R(p) = \|R_R(p)\|.$$

Then

$$\boxed{\|H_R(p)\psi\| \geq (\delta_0(p) - \eta_R(p))\|\psi\|.}$$

Hence, on

$$N_\rho = \{p : |p| \leq \rho\},$$

if

$$\boxed{\delta_{0,\rho} > \eta_{R,\rho},}$$

then

$$\boxed{\Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0, \qquad \operatorname{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \varnothing.}$$

So C4 is already reduced to a finite matrix lower-bound problem.
What is still not completed is the actual evaluation of

$$\boxed{\mu_{\text{vol}},\ \mu_{\text{high}},\ \mu_{\text{non-enr}},\ \mu_{\text{wrong-int}},\ \eta_{R,\rho}.}$$

Therefore:

$$\boxed{\text{C4 is theorem-level closed in the minimal truncation, but not yet numerically/analytically fully verified.}}$$

ACTUAL C5 AUDIT: MASS PROTECTION

1. Goal

We want to determine whether the isolated doubled local Dirac block admits any symmetry-allowed constant mass term.
The low-energy block is

$$H_D(q) = \rho_3 \otimes (q_i\sigma_i), \qquad i = 1, 2, 3. \tag{1104}$$

The C5 target is

$$\boxed{A_{\text{res}} \cap \mathcal{M} = \{0\},} \tag{1105}$$

where $\mathcal{M}$ is the constant Dirac mass space and A_{res} is the residual symmetry-allowed constant operator algebra.

2. Actual mass-space classification

The constant mass space is defined by

$$\mathcal{M} := \{K \in \operatorname{Herm}(4) : \{K, \rho_3 \otimes \sigma_i\} = 0 \ \forall i = 1, 2, 3\}. \tag{1106}$$

The exact classification theorem gives

$$\boxed{\mathcal{M} = \operatorname{span}_{\mathbb{R}}\{\rho_1 \otimes \mathbf{1}_2,\ \rho_2 \otimes \mathbf{1}_2\}.} \tag{1107}$$

So there are exactly two independent constant Dirac masses. :contentReference[oaicite:6]index=6
Therefore any possible constant mass has the form

$$K_m = m_1\, \rho_1 \otimes \mathbf{1}_2 + m_2\, \rho_2 \otimes \mathbf{1}_2. \tag{1108}$$

3. Valley symmetry test

Define the residual valley-charge generator

$$Q_V := \rho_3 \otimes \mathbf{1}_2, \qquad U_V(\theta) = e^{i\theta Q_V}. \tag{1109}$$

Since

$$[Q_V, \rho_3 \otimes \sigma_i] = 0,$$

the kinetic Dirac block is valley symmetric.

Now act on the two mass generators:

$$U_V(\theta)(\rho_1 \otimes \mathbf{1}_2)U_V(\theta)^{-1} = (\cos 2\theta)\rho_1 \otimes \mathbf{1}_2 + (\sin 2\theta)\rho_2 \otimes \mathbf{1}_2, \tag{1110}$$

$$U_V(\theta)(\rho_2 \otimes \mathbf{1}_2)U_V(\theta)^{-1} = -(\sin 2\theta)\rho_1 \otimes \mathbf{1}_2 + (\cos 2\theta)\rho_2 \otimes \mathbf{1}_2. \tag{1111}$$

Thus the mass space $\mathcal{M}$ transforms as a nontrivial doublet under $U(1)_V$. Hence no nonzero mass element is $U(1)_V$-invariant:

$$\boxed{A_{U(1)_V} \cap \mathcal{M} = \{0\}.} \tag{1112}$$

So exact residual valley $U(1)_V$ implies mass protection.

4. Weaker discrete symmetry is already sufficient

Even the discrete remnant

$$S_V := \rho_3 \otimes \mathbf{1}_2, \qquad S_V^2 = \mathbf{1}_4 \tag{1113}$$

is enough. Indeed,

$$S_V(\rho_1 \otimes \mathbf{1}_2)S_V^{-1} = -(\rho_1 \otimes \mathbf{1}_2), \qquad S_V(\rho_2 \otimes \mathbf{1}_2)S_V^{-1} = -(\rho_2 \otimes \mathbf{1}_2). \tag{1114}$$

Hence both mass generators are odd under S_V, so again

$$\boxed{A_{\mathbb{Z}_2^V} \cap \mathcal{M} = \{0\}.} \tag{1115}$$

Thus an exact residual $\mathbb{Z}_2^V$ is already sufficient for mass protection. :contentReference[oaicite:8]index=8

5. Connection with C3

C3 already showed, in the minimal truncation, that

$$V(0) = 0$$

because

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R^{\mathrm{tr}}, \mathcal{H}_D) = 0.$$

This means constant Dirac–remote hybridization is absent at the crossing. Therefore the remaining source of a constant gap is not remote hybridization, but an allowed constant operator living directly in the isolated doubled block.

But the previous sections show that any such constant operator must lie in $\mathcal{M}$, and if residual valley symmetry is exact, then no nonzero element of $\mathcal{M}$ is allowed.

So C3 + residual valley symmetry implies:

$$\boxed{\text{no constant mass is generated at the local Dirac point.}} \tag{1116}$$

6. Actual C5 result

Therefore the actual C5 audit gives the following:

$$\boxed{\text{At the level of the isolated doubled Dirac block, the complete mass classification is done.}} \tag{1117}$$

$$\boxed{\text{If the residual microscopic symmetry contains } U(1)_V \text{ or at least } \mathbb{Z}_2^V, \text{ then C5 is proved.}} \tag{1118}$$

Equivalently:

$$\boxed{A_{\text{res}} \cap \text{span}_{\mathbb{R}}\{\rho_1 \otimes \mathbf{1}_2, \rho_2 \otimes \mathbf{1}_2\} = \{0\}.} \tag{1119}$$

7. Exact honest status

What is actually completed:

1. the full constant mass-space classification;
2. the proof that residual $U(1)_V$ forbids all masses;
3. the proof that residual $\mathbb{Z}_2^V$ already suffices.

What still remains open microscopically:

$$\boxed{\text{whether the full microscopic TECT truncation preserves exact residual } U(1)_V \text{ or at least } \mathbb{Z}_2^V.} \tag{1120}$$

So C5 is not structurally open. It is reduced to one residual symmetry-preservation matching question.

. Summary

$$\boxed{\textbf{Actual C5 result}}$$

For the isolated doubled local Dirac block

$$\boxed{H_D(q) = \rho_3 \otimes (q_i\sigma_i),}$$

the full constant mass space is exactly

$$\boxed{\mathcal{M} = \text{span}_{\mathbb{R}}\{\rho_1 \otimes \mathbf{1}_2, \rho_2 \otimes \mathbf{1}_2\}.}$$

If the residual microscopic symmetry contains

$$\boxed{Q_V = \rho_3 \otimes \mathbf{1}_2}$$

or already its discrete remnant

$$\boxed{S_V = \rho_3 \otimes \mathbf{1}_2,}$$

then both mass generators are forbidden, so

$$\boxed{A_{\text{res}} \cap \mathcal{M} = \{0\}.}$$

Hence:

$$\boxed{\text{C5 is completed once residual valley } U(1)_V \text{ or } \mathbb{Z}_2^V \text{ is actually preserved in the microscopic truncation.}}$$

So C5 is reduced to one final residual-symmetry preservation check.

C1 LAST UNRESOLVED PART: ACTUAL LONGITUDINAL SEED EMBEDDING

1. What remains

Inside the explicit seed/representative family, the longitudinal witness is already realized:

$$V^{\mu}_{(\|)}(\xi) = \theta'(\xi)(\Pi_Q \mathcal{T}_{\xi}^2 \Pi_Q)_{\mathrm{tr}}\, n^{\mu} = \frac{1}{2}\,\theta'(\xi)^3\, n^{\mu}\sigma_3,$$

and hence

$$\lambda_{\|} = \frac{1}{2} I_{\theta'^3} \neq 0.$$

So the remaining C1 question is not whether a longitudinal seed can exist in principle. It is only:

$$\boxed{\text{Does the full microscopic projected transport kernel actually contain a nonzero } n^{\mu}\sigma_3 \text{ component?}} \tag{1121}$$

2. Microscopic projected kernel

Let the full microscopic first-order transport kernel projected to the Q_B-doublet be

$$V^{\mu}_{\mathrm{micro}}(\xi) = v^{\mu}_0(\xi)\mathbf{1}_2 + v^{\mu}_1(\xi)\sigma_1 + v^{\mu}_2(\xi)\sigma_2 + v^{\mu}_3(\xi)\sigma_3. \tag{1122}$$

The longitudinal seed embedding is controlled only by the σ_3 channel.

3. Shell-axis projection

Let n^{μ} be the distinguished shell/worldvolume axis. Project the σ_3 coefficient along n^{μ}:

$$\boxed{q_{\|}(\xi) := n_{\mu}\, v^{\mu}_3(\xi).} \tag{1123}$$

Then the microscopic kernel contains the desired longitudinal structure iff

$$\boxed{q_{\|}(\xi) \not\equiv 0.} \tag{1124}$$

4. Graded low-mode overlap

Let

$$u^{+}_a(\xi) = f_{+}(\xi)e_a(\xi), \qquad u^{-}_a(\xi) = f_{-}(\xi)e_a(\xi)$$

be the even/odd low modes. The actual projected longitudinal overlap is

$$\mathcal{J}_{\|} := \int d\xi\, f_{+}(\xi)f_{-}(\xi)\, q_{\|}(\xi). \tag{1125}$$

This is the full microscopic analogue of the representative quantity

$$\frac{1}{2} I_{\theta'^3}.$$

5. Pass/fail criterion

The actual longitudinal seed embedding passes iff

$$\boxed{\mathcal{J}_{\parallel} \neq 0.} \tag{1126}$$

If $\mathcal{J}_{\parallel} \neq 0$, then the microscopic transport space contains the longitudinal seed direction, hence

$$\boxed{\lambda_{\parallel} \neq 0.} \tag{1127}$$

If $\mathcal{J}_{\parallel} = 0$, then the present microscopic truncation does not realize the longitudinal witness.

6. Final reduction of C1

Therefore the last unresolved part of C1 is completely reduced to the single scalar test:

$$\boxed{\mathcal{J}_{\parallel} = \int d\xi\, f_{+}(\xi) f_{-}(\xi)\, n_{\mu} v_{3}^{\mu}(\xi) \overset{?}{\neq} 0.} \tag{1128}$$

So C1 is now split as:

$$\text{transverse part: fixed}, \qquad \text{longitudinal part: one scalar overlap test.}$$

. Summary

$$\boxed{\textbf{C1 final unresolved test}}$$

Write the full microscopic projected transport kernel as

$$\boxed{V^{\mu}_{\text{micro}}(\xi) = v_{0}^{\mu}(\xi)\mathbf{1}_{2} + v_{1}^{\mu}(\xi)\sigma_{1} + v_{2}^{\mu}(\xi)\sigma_{2} + v_{3}^{\mu}(\xi)\sigma_{3}.}$$

Project the σ_3 channel along the shell axis:

$$\boxed{q_{\parallel}(\xi) := n_{\mu} v_{3}^{\mu}(\xi).}$$

Then define the graded overlap

$$\boxed{\mathcal{J}_{\parallel} := \int d\xi\, f_{+}(\xi) f_{-}(\xi)\, q_{\parallel}(\xi).}$$

If

$$\boxed{\mathcal{J}_{\parallel} \neq 0,}$$

then

$$\boxed{\lambda_{\parallel} \neq 0,}$$

so the longitudinal seed is embedded in the full microscopic transport space.
Therefore the last unresolved part of C1 is reduced to one scalar coefficient/overlap test:

$$\boxed{\int d\xi\, f_{+}(\xi) f_{-}(\xi)\, n_{\mu} v_{3}^{\mu}(\xi) \neq 0.}$$

ACTUAL RESIDUAL SYMMETRY PRESERVATION CHECK FOR C5

1. Goal

The final unresolved part of C5 is not the mass classification itself. That is already closed:

$$\mathcal{M} = \operatorname{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2,\ \tau_y \otimes \mathbf{1}_2\}.$$

The remaining question is whether the corrected microscopic shell block actually preserves a residual symmetry that excludes all elements of $\mathcal{M}$.

2. Corrected shell block

For the canonical minimal Class II correction, the shell-level coefficients are renormalized as

$$A_{\text{eff}} = A_0 - \mu c_{\parallel}, \qquad B_{\text{eff}} = B_0 - \mu c_I, \qquad C_{\text{eff}} = C_0 - \mu c_J.$$

Thus the corrected local doubled shell block remains of the form

$$\boxed{H_D(p) = \tau_z \otimes \Big(B_{\text{eff}} p_1 \sigma_1 + C_{\text{eff}} p_2 \sigma_2 + A_{\text{eff}} p_{\parallel} \sigma_3\Big).} \tag{1129}$$

So the Class II correction changes only the derivative shell tensor coefficients. It does not insert an independent constant shell-level mass operator.

3. Residual valley-charge generator

Define

$$\boxed{Q_V = \tau_z \otimes \mathbf{1}_2.} \tag{1130}$$

Then for each kinetic generator,

$$[Q_V, \tau_z \otimes \sigma_i] = 0, \qquad i = 1, 2, 3.$$

Hence

$$\boxed{[Q_V, H_D(p)] = 0.} \tag{1131}$$

So the corrected shell block preserves the residual valley $U(1)_V$ symmetry.

4. Exclusion of constant masses

Any constant Dirac mass lies in

$$\mathcal{M} = \operatorname{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2,\ \tau_y \otimes \mathbf{1}_2\}.$$

Under

$$U_V(\theta) = e^{i\theta Q_V},$$

these generators transform as

$$U_V(\theta)(\tau_x \otimes \mathbf{1}_2)U_V(\theta)^{-1} = (\cos 2\theta)\tau_x \otimes \mathbf{1}_2 + (\sin 2\theta)\tau_y \otimes \mathbf{1}_2,$$

$$U_V(\theta)(\tau_y \otimes \mathbf{1}_2)U_V(\theta)^{-1} = -(\sin 2\theta)\tau_x \otimes \mathbf{1}_2 + (\cos 2\theta)\tau_y \otimes \mathbf{1}_2.$$

Thus no nonzero element of $\mathcal{M}$ is invariant under the residual $U(1)_V$. Therefore

$$\boxed{A_{U(1)_V} \cap \mathcal{M} = \{0\}.} \tag{1132}$$

So shell-level mass protection holds.

5. Actual shell-level result

Combining (1129), (1131), and (1132), we obtain:

$$\boxed{\text{the canonical minimal Class II corrected shell block preserves residual valley symmetry and forbids all constant Dirac masses.}} \tag{1133}$$

6. Honest remaining microscopic issue

What still remains open is the stronger full-microscopic statement: does the complete TECT Hamiltonian, including the remote sector and its elimination, preserve the same symmetry?

If the full microscopic theory admits a symmetry action

$$(U_\theta \oplus U_{R,\theta})H_{\text{full}}(p)(U_\theta \oplus U_{R,\theta})^{-1} = H_{\text{full}}(p),$$

then the Schur-complement effective Hamiltonian inherits it, and in particular the zero-momentum self-energy satisfies

$$[Q_V, \Sigma(0,0)] = 0.$$

Under little-group scalarity this implies

$$\Sigma(0,0) = a_0\mathbf{1}_4 + a_3\tau_z \otimes \mathbf{1}_2,$$

so no radiatively generated term proportional to

$$\tau_x \otimes \mathbf{1}_2 \quad \text{or} \quad \tau_y \otimes \mathbf{1}_2$$

can appear. :contentReference[oaicite:8]index=8

Thus the only residual open part of C5 is:

$$\boxed{\text{full microscopic symmetry inheritance under remote-sector elimination.}} \tag{1134}$$

. Summary

$$\boxed{\textbf{Actual C5 shell-level result}}$$

For the canonical minimal Class II correction, the corrected shell block remains

$$\boxed{H_D(p) = \tau_z \otimes \Big(B_{\text{eff}}p_1\sigma_1 + C_{\text{eff}}p_2\sigma_2 + A_{\text{eff}}p_\parallel\sigma_3\Big).}$$

Hence the residual valley-charge generator

$$\boxed{Q_V = \tau_z \otimes \mathbf{1}_2}$$

still satisfies

$$\boxed{[Q_V, H_D(p)] = 0.}$$

Since the full constant mass space is

$$\boxed{\mathcal{M} = \operatorname{span}_{\mathbb{R}}\{\tau_x \otimes \mathbf{1}_2, \tau_y \otimes \mathbf{1}_2\},}$$

and these two generators transform nontrivially under $U(1)_V$,

$$\boxed{A_{U(1)_V} \cap \mathcal{M} = \{0\}.}$$

Therefore:

$$\boxed{\text{shell-level C5 is passed: the corrected local shell block preserves residual valley symmetry and forbids all constant Dirac masses}}$$

The only remaining microscopic open part is:

$$\boxed{\text{does the full microscopic TECT theory, including remote-sector elimination, preserve the same residual symmetry?}}$$

$$\boxed{\Delta_{\rm min} := \min\{\Delta_{\rm vol}, \Delta_{\rm high}, \Delta_{\text{non-enr}}, \Delta_{\text{wrong-int}}\}, \qquad \eta_{R,\rho} < \Delta_{\rm min}}$$

$$\Longrightarrow \boxed{\Delta_\rho \geq \Delta_{\rm min} - \eta_{R,\rho} > 0.}$$

$$\boxed{H^{(0)}_{\rm vol}(\omega, k) = M_\sigma^2 + c_\sigma^2 k^2 - \omega^2.}$$

In the low-energy regime $|\omega|, |k| \ll M_\sigma$, after restricting to a sufficiently small neighborhood,

$$\boxed{H^{(0)}_{\rm vol}(\omega, k) \geq \frac{M_\sigma^2}{2} > 0.}$$

Therefore

$$\boxed{\operatorname{dist}\big(\operatorname{spec}(H^{(0)}_{\rm vol}), 0\big) \geq \frac{M_\sigma^2}{2},}$$

and equivalently, at the Hamiltonian level,

$$\boxed{\Delta^{(0)}_{\rm vol} \geq c_{\rm vol} M_\sigma > 0.}$$

ACTUAL COMPUTATION OF THE HIGHER-SHELL LOWER BOUND $\Delta^{(0)}_{\rm high}$

1. Goal

We want to prove an explicit positive lower bound for the higher-shell part of the remote spectrum. The relevant diagonal kernel is of Brazovskii type:

$$\Lambda(Q) = r_* + K(|Q|^2 - q_*^2)^2, \qquad K > 0. \tag{1135}$$

For the higher-shell sector, define the radial shell-separation parameter

$$\delta_{\rm shell} = \inf_{Q \in \mathcal{Q}_{\rm high}} \big| |Q|^2 - q_*^2 \big|. \tag{1136}$$

Then the theorem-level sufficient bound is

$$\boxed{K\delta_{\rm shell}^2 > |r_*| \Longrightarrow \Delta^{(0)}_{\rm high} \geq K\delta_{\rm shell}^2 - |r_*| > 0.} \tag{1137}$$

This is explicitly stated in the notes. :contentReference[oaicite:4]index=4

2. BCC reciprocal-shell arithmetic

For the selected BCC first shell, the integer reciprocal representatives are

$$\widetilde{S}_{\mathrm{BCC}} = \{(\pm1,\pm1,0),(\pm1,0,\pm1),(0,\pm1,\pm1)\},$$

and the compatible reciprocal lattice is

$$L^*_{\mathrm{BCC}} = \{(h,k,\ell)\in\mathbb{Z}^3 : h+k+\ell\equiv 0 \pmod 2\}.$$

The notes compute the first radial separation above the selected shell and conclude:

$$\boxed{\delta_{\mathrm{shell}} = 2} \tag{1138}$$

in the integer reciprocal convention. :contentReference[oaicite:5]index=5

3. Insert the explicit BCC value

Substituting (1138) into (1137), we obtain

$$\boxed{\Delta^{(0)}_{\mathrm{high}} \geq K\delta^2_{\mathrm{shell}} - |r_*| = 4K - |r_*|.} \tag{1139}$$

Therefore, a completely explicit sufficient positivity condition is

$$\boxed{4K > |r_*| \Longrightarrow \Delta^{(0)}_{\mathrm{high}} > 0.} \tag{1140}$$

Equivalently, in unit-shell normalization the same statement is sometimes written as

$$\widehat{\Delta}^{(0)}_{\mathrm{high}} \geq K - |r_*|.$$

:contentReference[oaicite:6]index=6

4. Interpretation

This is an actual completed sectorwise result, not merely a structural reduction:

1. the higher-shell radial separation is no longer abstract;
2. the BCC arithmetic fixes it explicitly to $\delta_{\mathrm{shell}} = 2$;
3. therefore the higher-shell diagonal remote lower bound is explicitly

$$\Delta^{(0)}_{\mathrm{high}} \geq 4K - |r_*|.$$

So the higher-shell part of C4 is genuinely closed once the standard low-shell regime

$$4K > |r_*|$$

is imposed.

5. Position inside the tail-gap theorem

The tail-gap decomposition theorem collects the explicit sectorwise bounds:

$$\Delta^{(0)}_{\text{high}}, \qquad \Delta^{(0)}_{\text{vol}}, \qquad \Delta^{(0)}_{\text{wrong-int}}, \qquad \Delta^{(0)}_{\text{extra}}.$$

With

$$\Delta^{\text{tail}}_{\text{diag},\rho} = \min\{\Delta^{(0)}_{\text{high}}, \Delta^{(0)}_{\text{vol}}, \Delta^{(0)}_{\text{wrong-int}}, \Delta^{(0)}_{\text{extra}}\},$$

if

$$\eta_{\text{tail},\rho} < \Delta^{\text{tail}}_{\text{diag},\rho},$$

then the full tail block is uniformly gapped:

$$\Delta_{\text{tail},\rho} \geq \Delta^{\text{tail}}_{\text{diag},\rho} - \eta_{\text{tail},\rho} > 0.$$

Thus, after the present computation, the higher-shell contribution is no longer part of the open C4 frontier.

. Summary

$$\boxed{\textbf{Actual higher-shell result}}$$

For the Brazovskii-type diagonal kernel

$$\boxed{\Lambda(Q) = r_* + K(|Q|^2 - q_*^2)^2,}$$

the higher-shell lower bound is

$$\boxed{\Delta^{(0)}_{\text{high}} \geq K\delta^2_{\text{shell}} - |r_*|.}$$

For the selected BCC first shell, the reciprocal-shell arithmetic gives

$$\boxed{\delta_{\text{shell}} = 2.}$$

Therefore

$$\boxed{\Delta^{(0)}_{\text{high}} \geq 4K - |r_*|.}$$

Hence the explicit sufficient condition is

$$\boxed{4K > |r_*| \Longrightarrow \Delta^{(0)}_{\text{high}} > 0.}$$

So the higher-shell piece of C4 is actually closed.

ACTUAL COMPUTATION OF THE WRONG-INTERNAL LOWER BOUND $\Delta^{(0)}_{\text{wrong-int}}$

1. Goal

We want an explicit positive lower bound for the wrong-internal contribution to the remote gap.
Let U_D denote the selected internal shell-point doublet space with energy E_*, and let

$$U_{\text{wrong}} = Q(\mathbb{C}^3)$$

denote the complementary wrong-internal sector, where Q is the orthogonal projector onto the complement of the selected doublet.
Let I_0 be the unperturbed internal shell-point operator.

2. Exact identification of the wrong-internal gap

The wrong-internal diagonal contribution is identified as

$$\boxed{\Delta^{(0)}_{\text{wrong-int}} = m^{(0)}_{\text{int}} = \operatorname{dist}\bigl(E_*, \operatorname{spec}(QI_0Q)\bigr).} \tag{1141}$$

This is stated explicitly in the notes. :contentReference[oaicite:4]index=4

So the wrong-internal gap is simply the spectral separation between the selected internal doublet energy and the complement internal spectrum.

3. Minimal $\mathbb{C}^3$ carrier

In the minimal $\mathbb{C}^3$ shell-point carrier, the selected internal sector is a transverse doublet with energy E_*, while the complement is the one-dimensional longitudinal direction $\hat{n}$, satisfying

$$I_0\hat{n} = E_L\hat{n}. \tag{1142}$$

Hence

$$\operatorname{spec}(QI_0Q) = \{E_L\}.$$

Therefore

$$\boxed{\Delta^{(0)}_{\text{wrong-int}} = m^{(0)}_{\text{int}} = |E_L - E_*|.} \tag{1143}$$

This is exactly the minimal-carrier wrong-internal gap corollary. :contentReference[oaicite:5]index=5

Consequently,

$$\boxed{E_L \neq E_* \Longrightarrow \Delta^{(0)}_{\text{wrong-int}} > 0.} \tag{1144}$$

4. Stability under bounded internal perturbations

Now allow a bounded perturbation of the internal shell-point operator:

$$I = I_0 + \delta I, \qquad \|Q\delta IQ\| \leq \varepsilon_{\text{int}}.$$

Then the perturbed complement spectrum satisfies

$$\boxed{m_{\text{int}} := \operatorname{dist}\bigl(E_*, \operatorname{spec}(QIQ)\bigr) \geq m^{(0)}_{\text{int}} - \varepsilon_{\text{int}}.} \tag{1145}$$

Hence, if

$$\boxed{\varepsilon_{\text{int}} < m^{(0)}_{\text{int}},} \tag{1146}$$

then

$$\boxed{m_{\text{int}} > 0.} \tag{1147}$$

So the wrong-internal sector gap is stable under sufficiently small internal perturbations. :contentReference[oaicite:6]index=6

5. Consequence for the tail-gap theorem

The tail-gap decomposition theorem uses

$$\Delta^{\text{tail}}_{\text{diag},\rho} = \min\{\Delta^{(0)}_{\text{high}}, \Delta^{(0)}_{\text{vol}}, \Delta^{(0)}_{\text{wrong-int}}, \Delta^{(0)}_{\text{extra}}\}.$$

The present computation gives the explicit wrong-internal contribution:

$$\boxed{\Delta^{(0)}_{\text{wrong-int}} = |E_L - E_*|}$$

in the minimal carrier, and more generally

$$\boxed{\Delta^{(0)}_{\text{wrong-int}} = \operatorname{dist}\big(E_*, \operatorname{spec}(QI_0Q)\big).}$$

Thus the wrong-internal piece of C4 is no longer structurally open. It is reduced to one actual internal spectral-separation number.

6. Strongest honest conclusion

Therefore the wrong-internal sector of C4 is actually closed up to one explicit internal spectral input:

$$\boxed{E_L \neq E_* \Longrightarrow \Delta^{(0)}_{\text{wrong-int}} = |E_L - E_*| > 0.} \tag{1148}$$

Moreover, this positivity is stable under bounded internal perturbations.

. Summary

$$\boxed{\textbf{Actual wrong-internal result}}$$

The wrong-internal lower bound is

$$\boxed{\Delta^{(0)}_{\text{wrong-int}} = m^{(0)}_{\text{int}} = \operatorname{dist}\big(E_*, \operatorname{spec}(QI_0Q)\big).}$$

In the minimal $\mathbb{C}^3$ shell-point carrier,

$$\boxed{I_0\hat{n} = E_L\hat{n}, \qquad \Delta^{(0)}_{\text{wrong-int}} = |E_L - E_*|.}$$

Hence

$$\boxed{E_L \neq E_* \Longrightarrow \Delta^{(0)}_{\text{wrong-int}} > 0.}$$

Under a bounded internal perturbation

$$\boxed{\|Q\delta IQ\| \le \varepsilon_{\text{int}},}$$

one still has

$$\boxed{m_{\text{int}} \ge m^{(0)}_{\text{int}} - \varepsilon_{\text{int}},}$$

so if

$$\boxed{\varepsilon_{\text{int}} < m^{(0)}_{\text{int}},}$$

then

$$\boxed{m_{\text{int}} > 0.}$$

So the wrong-internal piece of C4 is actually closed once the selected internal doublet is spectrally separated from the longitudinal internal mode.

ACTUAL ESTIMATE OF THE TAIL/REMOTE MIXING NORM $\eta_{\mathrm{rem},\rho}$

1. Goal

The last unresolved part of C4 is no longer the diagonal remote gaps themselves, but the total remote mixing budget

$$\eta_{\mathrm{rem},\rho}.$$

The full remote block is written as

$$H_R^{\mathrm{full}}(p) = D_{\mathrm{rem}}(p) + R_{\mathrm{rem}}(p), \tag{1149}$$

where $D_{\mathrm{rem}}(p)$ is the diagonal benchmark operator and $R_{\mathrm{rem}}(p)$ collects all remote–remote off-diagonal mixing, tail couplings, and bounded residual corrections.

Define

$$\boxed{\eta_{\mathrm{rem},\rho} := \sup_{|p|\le\rho} \|R_{\mathrm{rem}}(p)\|.} \tag{1150}$$

The C4 objective is to show

$$\Delta_{\mathrm{bench},\rho} > \eta_{\mathrm{rem},\rho},$$

which then implies a positive full remote gap.

2. Explicit master bound

The notes give the following explicit estimate:

$$\boxed{\eta_{\mathrm{rem},\rho} \le \sum_{a\in\mathcal{A}_{\mathrm{act}}} K_a \rho^{r_a} + \sum_{a\in\mathcal{A}_{\mathrm{act}}^{(H)}} \frac{\widetilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a}, \qquad r_a \ge 1.} \tag{1151}$$

Here:

- $\mathcal{A}_{\mathrm{act}}$ denotes active low-order remote couplings,
- $\mathcal{A}_{\mathrm{act}}^{(H)}$ denotes heavy-mediated contributions,
- $K_a, \widetilde{K}_a > 0$ are operator-norm constants,
- $r_a \ge 1$ is the order at which the corresponding coupling begins in momentum,
- $M_a^{\sigma_a}$ encodes heavy-mass suppression.

This is the exact quantitative form of the tail-coupling estimate theorem. :contentReference[oaicite:3]index=3

3. Immediate consequence: infrared smallness

Since every exponent satisfies $r_a \ge 1$, one immediately gets

$$\boxed{\eta_{\mathrm{rem},\rho} \xrightarrow[\rho\to 0]{} 0.} \tag{1152}$$

More concretely, if

$$r_{\min} := \min_a r_a \ge 1,$$

then for sufficiently small ρ,

$$\eta_{\mathrm{rem},\rho} = O(\rho^{r_{\min}}). \tag{1153}$$

Thus the total remote mixing is an infrared-small quantity.

4. Combination with the diagonal benchmark

The full remote-gap theorem uses the benchmark inequality

$$\boxed{\Delta_{\mathrm{bench},\rho} > \eta_{\mathrm{rem},\rho}.} \tag{1154}$$

If this holds, then the full remote block has a uniform positive gap:

$$\boxed{\Delta_{\mathrm{full},\rho} \ge \Delta_{\mathrm{bench},\rho} - \eta_{\mathrm{rem},\rho} > 0.} \tag{1155}$$

This is the master explicit spectral inequality theorem.

5. Explicit sufficient benchmark form

The notes also give the stronger explicit sufficient inequality

$$\boxed{\min\left\{4K - |r_*|,\ c_{\mathrm{vol}} M_\sigma,\ q_*^2 \left|(\lambda+\mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2}\right|,\ \Delta^{(0)}_{\mathrm{extra},\rho}\right\} > \eta_{\mathrm{rem},\rho}.} \tag{1156}$$

This packages the higher-shell, volumetric, wrong-internal/effective-internal, and extra-sector diagonal lower bounds into one benchmark condition. :contentReference[oaicite:5]index=5

6. What is actually closed now

At this stage, the structural problem is finished:

1. $\eta_{\mathrm{rem},\rho}$ has an explicit operator-norm upper bound;
2. that upper bound tends to zero as $\rho \to 0$;
3. therefore, once the diagonal benchmark is positive, there exists a sufficiently small ρ such that

$$\Delta_{\mathrm{bench},\rho} > \eta_{\mathrm{rem},\rho}.$$

So the remaining practical work is only to estimate the constants

$$K_a, \qquad \widetilde{K}_a, \qquad r_a, \qquad M_a,$$

for the actual microscopic truncation.

7. Strongest honest conclusion

Therefore the last missing part of C4 is no longer conceptual. It is reduced to a finite small-coupling estimate:

$$\boxed{\eta_{\mathrm{rem},\rho} \le \sum_{a\in\mathcal{A}_{\mathrm{act}}} K_a \rho^{r_a} + \sum_{a\in\mathcal{A}^{(H)}_{\mathrm{act}}} \frac{\widetilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a}, \qquad r_a \ge 1,} \tag{1157}$$

together with the explicit benchmark inequality

$$\boxed{\Delta_{\mathrm{bench},\rho} > \eta_{\mathrm{rem},\rho} \Longrightarrow \Delta_{\mathrm{full},\rho} > 0.} \tag{1158}$$

In particular, infrared smallness of the tail/remote couplings is already theorem-level established.

. Summary

$$\boxed{\textbf{Actual result for the last C4 bottleneck}}$$

The full remote block is written as

$$\boxed{H_R^{\mathrm{full}}(p) = D_{\mathrm{rem}}(p) + R_{\mathrm{rem}}(p), \qquad \eta_{\mathrm{rem},\rho} := \sup_{|p|\le\rho} \|R_{\mathrm{rem}}(p)\|.}$$

The explicit master bound is

$$\boxed{\eta_{\mathrm{rem},\rho} \le \sum_{a\in\mathcal{A}_{\mathrm{act}}} K_a \rho^{r_a} + \sum_{a\in\mathcal{A}_{\mathrm{act}}^{(H)}} \frac{\widetilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a}, \qquad r_a \ge 1.}$$

Hence

$$\boxed{\eta_{\mathrm{rem},\rho} \to 0 \qquad (\rho \to 0).}$$

Therefore, if the diagonal benchmark gap is positive and satisfies

$$\boxed{\Delta_{\mathrm{bench},\rho} > \eta_{\mathrm{rem},\rho},}$$

then the full remote block has a uniform positive gap

$$\boxed{\Delta_{\mathrm{full},\rho} \ge \Delta_{\mathrm{bench},\rho} - \eta_{\mathrm{rem},\rho} > 0.}$$

A strong explicit sufficient form is

$$\boxed{\min\left\{4K - |r_*|,\ c_{\mathrm{vol}} M_\sigma,\ q_*^2 \left|(\lambda+\mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2}\right|,\ \Delta_{\mathrm{extra},\rho}^{(0)}\right\} > \eta_{\mathrm{rem},\rho}.}$$

So the last C4 bottleneck is reduced to estimating finitely many constants $K_a, \widetilde{K}_a, r_a, M_a$, not to any new structural problem.

$$\boxed{\text{If the full microscopic TECT theory preserves exact residual valley symmetry,}}$$

$$\boxed{(U_\theta \oplus U_{R,\theta}) H_{\mathrm{full}}(p) (U_\theta \oplus U_{R,\theta})^{-1} = H_{\mathrm{full}}(p),}$$

then the Schur-complement effective shell Hamiltonian inherits the same symmetry:

$$\boxed{U_\theta H_{\mathrm{eff}}(p,\omega) U_\theta^{-1} = H_{\mathrm{eff}}(p,\omega).}$$

Hence

$$\boxed{[Q_V, \Sigma(0,0)] = 0.}$$

If little-group scalarity also holds, then

$$\boxed{\Sigma(0,0) = a_0\, \mathbf{1}_4 + a_3\, \tau_z \otimes \mathbf{1}_2,}$$

so no term proportional to

$$\boxed{\tau_x \otimes \mathbf{1}_2 \quad \text{or} \quad \tau_y \otimes \mathbf{1}_2}$$

can be generated. Therefore radiative constant Dirac masses are forbidden.

REDUCTION OF THE LAST C1 BOTTLENECK TO THE COEFFICIENT A

1. Remaining issue

The last unresolved part of C1 had been written as the longitudinal overlap test

$$\mathcal{J}_{\parallel} = \int d\xi \, f_+(\xi) f_-(\xi) \, n_\mu v_3^\mu(\xi).$$

The question is whether this quantity is nonzero in the full microscopic TECT linearization.

2. Extraction theorem

Let the microscopic linearized operator near a selected shell point G_* admit

$$L(G_* + p) = L_* + p_i K_i + O(|p|^2),$$

and let P_* be the projector onto the two-dimensional low-energy valley space H_*. Then the reduced first-order Hamiltonian is

$$h^+_{\text{lin}}(p) = P_*(p_i K_i) P_*|_{H_*},$$

and the Pauli-resolved coefficients are

$$M_{\mu i} = \frac{1}{2} \operatorname{Tr}_{H_*}(\sigma_\mu P_* K_i P_*).$$

Thus the local Weyl data are read directly from the projected first derivatives K_i. :contentReference[oaicite:3]index=3

3. Minimal actual operator ansatz

For the minimal actual symmetry-compatible operator,

$$L_{\text{act}}(G_* + p) = E_* \mathbf{1}_4 + A(\hat{n} \cdot p) \, \tau_3 \otimes \mathbf{1}_2 + B(p_1 \tau_1 \otimes s_1 + p_2 \tau_1 \otimes s_2) + C(-p_2 \tau_1 \otimes s_1 + p_1 \tau_1 \otimes s_2) + O(|p|^2),$$

the projected two-band valley Hamiltonian is

$$h_+(p) = A p_\parallel \sigma_3 + B(p_1 \sigma_1 + p_2 \sigma_2) + C(-p_2 \sigma_1 + p_1 \sigma_2) + O(|p|^2).$$

Therefore

$$\boxed{\lambda_\parallel = A, \qquad \alpha = B, \qquad \beta = C.} \tag{1159}$$

:contentReference[oaicite:4]index=4

4. Microscopic tensor form

Equivalently, the shell-compatible enriched tensors are

$$N^3_{ij} = \frac{A}{\kappa_* |G_*|} \hat{n}_i \hat{n}_j, \qquad N^1_{ij} = \frac{1}{\kappa_* |G_*|} (B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}), \qquad N^2_{ij} = \frac{1}{\kappa_* |G_*|} (-C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}).$$

So the longitudinal projector channel is exactly the coefficient A.

5. Final reduction

Hence the last unresolved longitudinal question is not a separate mysterious overlap problem. It is exactly the microscopic coefficient test

$$\boxed{A \neq 0.} \tag{1160}$$

Equivalently,

$$\mathcal{J}_{\parallel} \neq 0 \iff \lambda_{\parallel} \neq 0 \iff A \neq 0,$$

up to the normalization conventions of the chosen projected basis.

6. Strongest honest conclusion

So the last unresolved part of C1 is now compressed to:

$$\boxed{\text{Extract the coefficient } A \text{ from the full microscopic projected first-order operator and check whether } A \neq 0.} \tag{1161}$$

No further structural reduction is needed.

. Summary

$$\boxed{\textbf{Final reduction of the last C1 bottleneck}}$$

The longitudinal overlap test is not independent. By the extraction theorem, the projected first-order microscopic operator directly yields the Weyl coefficients

$$\boxed{\lambda_{\parallel},\ \alpha,\ \beta.}$$

For the minimal actual operator ansatz,

$$\boxed{\lambda_{\parallel} = A, \qquad \alpha = B, \qquad \beta = C.}$$

Equivalently, the longitudinal projector channel is

$$\boxed{N^3_{ij} = \frac{A}{\kappa_* |G_*|} \hat{n}_i \hat{n}_j.}$$

Therefore the last unresolved part of C1 is simply:

$$\boxed{A \neq 0.}$$

So the remaining local microscopic task is no longer "compute a mysterious overlap," but

$$\boxed{\text{extract } A \text{ from the full microscopic projected first-order operator and verify } A \neq 0.}$$

NEXT TASK

Compute the projected first-order microscopic operator near the selected shell point G_*:

$$L(G_* + p) = L_* + p_i K_i + O(|p|^2).$$

Let P_* be the projector onto the low-energy valley space H_*. Then

$$h^+_{\text{lin}}(p) = P_*(p_i K_i)P_*|_{H_*}.$$

In the adapted frame $(\hat{n}, e_1, e_2)$, decompose

$$h^+_{\text{lin}}(p) = Ap_\parallel \sigma_3 + B(p_1\sigma_1 + p_2\sigma_2) + C(-p_2\sigma_1 + p_1\sigma_2).$$

The immediate goal is to verify

$$\boxed{A \neq 0.}$$

This is the last unresolved microscopic part of C1.

ACTUAL EXTRACTION FORMULA FOR THE LAST C1 CONDITION: $A \neq 0$

1. Microscopic linearization near the selected shell point

Let the full microscopic linearized operator admit the expansion near the selected shell point G_*:

$$L(G_* + p) = L_* + p_i K_i + O(|p|^2), \qquad K_i := \partial_{k_i} L(k)\big|_{k=G_*}. \tag{1162}$$

Let P_* be the orthogonal projector onto the two-dimensional low-energy valley space $H_* \cong \mathbb{C}^2$. Then the reduced first-order valley Hamiltonian is

$$h^+_{\text{lin}}(p) = P_*(p_i K_i)P_*|_{H_*}. \tag{1163}$$

By Pauli decomposition,

$$h^+_{\text{lin}}(p) = a_0(p)\mathbf{1}_2 + a_\mu(p)\sigma_\mu, \qquad a_\mu(p) = M_{\mu i} p_i, \tag{1164}$$

with

$$M_{\mu i} = \frac{1}{2}\,\mathrm{Tr}_{H_*}\big(\sigma_\mu P_* K_i P_*\big). \tag{1165}$$

Thus all local Weyl data are determined directly by the projected first derivatives K_i. :contentReference[oaicite:4]index=4

2. Adapted shell frame and longitudinal coefficient

Choose the adapted frame

$$p = p_\parallel \hat{n} + p_1 e_1 + p_2 e_2, \qquad \hat{n} = \frac{G_*}{|G_*|}.$$

In the minimal actual symmetry-compatible ansatz, the reduced Hamiltonian is

$$h_+(p) = Ap_\parallel \sigma_3 + B(p_1\sigma_1 + p_2\sigma_2) + C(-p_2\sigma_1 + p_1\sigma_2) + O(|p|^2). \tag{1166}$$

Hence

$$\boxed{\lambda_\parallel = A, \qquad \alpha = B, \qquad \beta = C.} \tag{1167}$$

:contentReference[oaicite:5]index=5

Therefore the last unresolved part of C1 is simply the longitudinal coefficient test

$$\boxed{A \neq 0.} \tag{1168}$$

3. Direct projector formula for A

Let

$$K_\parallel := \hat{n}_i K_i.$$

Then the coefficient of $p_\parallel \sigma_3$ is

$$\boxed{A = \frac{1}{2}\,\mathrm{Tr}_{H_*}\big(\sigma_3 P_* K_\parallel P_*\big).} \tag{1169}$$

So the full microscopic C1 question has now become the single finite projector-contraction test

$$\boxed{\frac{1}{2}\,\mathrm{Tr}_{H_*}\big(\sigma_3 P_* K_\parallel P_*\big) \overset{?}{\neq} 0.} \tag{1170}$$

4. Equivalent tensor-contraction formula

In microscopic tensor language, the minimal enriched tensors realizing the actual operator are

$$N^3_{ij} = \frac{A}{\kappa_*|G_*|}\hat{n}_i\hat{n}_j, \qquad N^1_{ij} = \frac{1}{\kappa_*|G_*|}(B\hat{n}_i e_{1j} + C\hat{n}_i e_{2j}), \qquad N^2_{ij} = \frac{1}{\kappa_*|G_*|}(-C\hat{n}_i e_{1j} + B\hat{n}_i e_{2j}). \tag{1171}$$

Equivalently,

$$\lambda_\parallel = \kappa_*|G_*|A, \qquad \alpha = \kappa_*|G_*|B, \qquad \beta = \kappa_*|G_*|C. \tag{1172}$$

:contentReference[oaicite:6]index=6

Thus the longitudinal channel is equivalently the microscopic tensor contraction

$$\boxed{\hat{n}_i\hat{n}_j N^3_{ij} \neq 0 \Longleftrightarrow A \neq 0.} \tag{1173}$$

This is the tensor-form version of the same test.

5. Final reduction of C1

Therefore the last unresolved microscopic content of C1 is no longer a vague overlap problem. It is exactly one of the following equivalent tests:

$$\boxed{A \neq 0,} \tag{1174}$$

or equivalently

$$\boxed{\frac{1}{2}\,\mathrm{Tr}_{H_*}\big(\sigma_3 P_* K_\parallel P_*\big) \neq 0,} \tag{1175}$$

or equivalently

$$\boxed{\hat{n}_i\hat{n}_j N^3_{ij} \neq 0.} \tag{1176}$$

No further structural reduction is needed.

. Summary

$$\boxed{\textbf{Final actual form of the last C1 test}}$$

Near the selected shell point G_*,

$$\boxed{L(G_* + p) = L_* + p_i K_i + O(|p|^2), \qquad h^+_{\mathrm{lin}}(p) = P_*(p_i K_i) P_*|_{H_*}.}$$

In the adapted shell frame,

$$\boxed{h_+(p) = A p_\parallel \sigma_3 + B(p_1\sigma_1 + p_2\sigma_2) + C(-p_2\sigma_1 + p_1\sigma_2) + O(|p|^2),}$$

so

$$\boxed{\lambda_\parallel = A, \qquad \alpha = B, \qquad \beta = C.}$$

Therefore the last unresolved part of C1 is simply

$$\boxed{A \neq 0.}$$

Equivalently,

$$\boxed{A = \frac{1}{2}\,\mathrm{Tr}_{H_*}\big(\sigma_3 P_* K_\parallel P_*\big), \qquad K_\parallel = \hat{n}_i K_i,}$$

so the test is

$$\boxed{\frac{1}{2}\,\mathrm{Tr}_{H_*}\big(\sigma_3 P_* K_\parallel P_*\big) \neq 0.}$$

In microscopic tensor language, this is equivalently

$$\boxed{N^3_{ij} = \frac{A}{\kappa_* |G_*|}\hat{n}_i \hat{n}_j \quad \Longrightarrow \quad \hat{n}_i \hat{n}_j N^3_{ij} \neq 0 \iff A \neq 0.}$$

METHOD A: DIRECT PROJECTOR-DERIVATIVE EXTRACTION OF A

1. Starting point

Near the selected shell point G_*, write the microscopic linearized operator as

$$L(G_* + p) = L_* + p_i K_i + O(|p|^2), \qquad K_i := \partial_{k_i} L(k)\big|_{k=G_*}. \tag{1177}$$

Let P_* be the orthogonal projector onto the two-dimensional low-energy valley space H_*.
Then the reduced first-order valley Hamiltonian is

$$h^+_{\mathrm{lin}}(p) = P_*(p_i K_i) P_*|_{H_*}. \tag{1178}$$

2. Pauli-resolved coefficient matrix

Since $h^+_{\mathrm{lin}}(p)$ is Hermitian on $H_* \cong \mathbb{C}^2$, it decomposes as

$$h^+_{\mathrm{lin}}(p) = a_0(p)\mathbf{1}_2 + a_\mu(p)\sigma_\mu, \qquad a_\mu(p) = M_{\mu i} p_i, \tag{1179}$$

with

$$M_{\mu i} = \frac{1}{2}\,\mathrm{Tr}_{H_*}\big(\sigma_\mu P_* K_i P_*\big). \tag{1180}$$

Thus all local Weyl data are extracted directly from the projected first derivatives K_i. :contentReference[oaicite:4]index=4

3. Adapted shell frame

Choose the adapted frame

$$p = p_{\parallel}\hat{n} + p_1 e_1 + p_2 e_2, \qquad \hat{n} = \frac{G_*}{|G_*|}.$$

Then define

$$K_{\parallel} := \hat{n}_i K_i, \qquad K_1 := e_{1i} K_i, \qquad K_2 := e_{2i} K_i. \tag{1181}$$

In the minimal symmetry-compatible Weyl form,

$$h_+(p) = A p_{\parallel}\sigma_3 + B(p_1\sigma_1 + p_2\sigma_2) + C(-p_2\sigma_1 + p_1\sigma_2) + O(|p|^2). \tag{1182}$$

Therefore

$$\boxed{A = \frac{1}{2}\operatorname{Tr}_{H_*}\big(\sigma_3 P_* K_{\parallel} P_*\big).} \tag{1183}$$

Likewise, B and C are extracted from the σ_1, σ_2 components of $P_* K_1 P_*, P_* K_2 P_*$. :contentReference[oaicite:5]index=5 :contentReference[oaicite:6]index=6

4. What Method A reduces the problem to

Hence the final C1 condition is

$$\boxed{A \neq 0 \iff \frac{1}{2}\operatorname{Tr}_{H_*}\big(\sigma_3 P_* K_{\parallel} P_*\big) \neq 0.} \tag{1184}$$

So Method A completely removes the need for any additional abstract reduction. The only remaining task is to compute the microscopic K_i from the true TECT linearized operator and insert them into (1183). :contentReference[oaicite:7]index=7

5. Cross-check with the explicit local representative

In the explicit actual-local representative, the projected overlap formulas give

$$\lambda_{\parallel} = \int d\xi\, f_+(\xi) f_-(\xi) \theta'(\xi)\, h_{\parallel}(\theta(\xi)), \tag{1185}$$

and for the explicit canonical representative,

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \qquad \alpha = I_\theta \Lambda_E, \qquad \beta = I_\theta \Lambda_O. \tag{1186}$$

Thus in the representative model,

$$A = \lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3},$$

so Method A indeed returns a nonzero longitudinal coefficient whenever $\lambda_c \neq 0$.

6. Honest status

What is rigorously closed:

1. the exact projector-trace formula for A,

2. the equivalence $A \neq 0 \iff \lambda_{\parallel} \neq 0$,
3. the explicit nonzero value of A in the representative model.

What remains open:

$$\boxed{\text{the actual microscopic computation of } K_i = \partial_{k_i} L(k)\big|_{k=G_*} \text{ for the full TECT linearized operator.}} \tag{1187}$$

So Method A converts the final C1 bottleneck into one explicit coefficient-extraction calculation from the true microscopic PDE.

. Summary

$$\boxed{\textbf{Method A reduces the last C1 problem to one projector trace}}$$

Near the selected shell point G_*,

$$\boxed{L(G_* + p) = L_* + p_i K_i + O(|p|^2), \qquad h^+_{\text{lin}}(p) = P_*(p_i K_i) P_*|_{H_*}.}$$

In the adapted shell frame,

$$\boxed{h_+(p) = A p_{\parallel} \sigma_3 + B(p_1\sigma_1 + p_2\sigma_2) + C(-p_2\sigma_1 + p_1\sigma_2) + O(|p|^2),}$$

so

$$\boxed{A = \frac{1}{2} \operatorname{Tr}_{H_*}\big(\sigma_3 P_* K_{\parallel} P_*\big), \qquad K_{\parallel} = \hat{n}_i K_i.}$$

Therefore the final unresolved C1 condition is simply

$$\boxed{A \neq 0 \iff \frac{1}{2} \operatorname{Tr}_{H_*}\big(\sigma_3 P_* K_{\parallel} P_*\big) \neq 0.}$$

In the explicit local representative,

$$\boxed{A = \lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3},}$$

so Method A already gives a nonzero longitudinal coefficient there.

Thus Method A turns the last C1 bottleneck into one explicit coefficient-extraction calculation from the full microscopic TECT linearized operator.

METHOD A PUSHED ONE STEP FURTHER: EXPLICIT FORMULA FOR A

1. Projector-derivative extraction formula

Near the selected shell point G_*,

$$L(G_* + p) = L_* + p_i K_i + O(|p|^2), \qquad K_i = \partial_{k_i} L(k)\big|_{k=G_*}.$$

Let P_* be the projector onto the two-dimensional low-energy valley space H_*. Then

$$h^+_{\text{lin}}(p) = P_*(p_i K_i) P_*|_{H_*}.$$

In the adapted shell frame,

$$h_+(p) = A p_{\parallel} \sigma_3 + B(p_1\sigma_1 + p_2\sigma_2) + C(-p_2\sigma_1 + p_1\sigma_2) + O(|p|^2),$$

so

$$A = \lambda_{\parallel}.$$

2. Explicit projected symbol in the actual local projector-defect operator class

For the concrete minimal microscopic transport term already extracted from the local projector-defect geometry, the projected mixed symbol is

$$M^\mu = \frac{\lambda_c}{2} I_{\theta'^3}\, n^\mu \sigma_3 + I_\theta \Big[\Lambda_E E^\mu{}_a \sigma_a + \Lambda_O J^\mu{}_a \sigma_a\Big].$$

Hence the reduced S2 symbol is

$$A_1(p) = \lambda_\| p_\| \sigma_3 + \alpha(p_1\sigma_1 + p_2\sigma_2) + \beta(-p_2\sigma_1 + p_1\sigma_2),$$

with

$$\boxed{\lambda_\| = \frac{\lambda_c}{2} I_{\theta'^3}, \qquad \alpha = I_\theta \Lambda_E, \qquad \beta = I_\theta \Lambda_O.} \tag{1188}$$

Therefore

$$\boxed{A = \lambda_\| = \frac{\lambda_c}{2} I_{\theta'^3}.} \tag{1189}$$

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3. Explicit defect integral

For the explicit reflection-symmetric defect profile,

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256}\, \theta_0^3 \kappa^3.$$

Substituting into (1189),

$$\boxed{A = -\frac{3\sqrt{3}\pi}{512}\, \lambda_c\, \theta_0^3 \kappa^3.} \tag{1190}$$

Therefore, for a nontrivial defect profile $\theta_0 \kappa \neq 0$,

$$\boxed{\lambda_c \neq 0 \Longrightarrow A \neq 0.} \tag{1191}$$

4. Final nondegeneracy condition in this explicit operator class

Since also

$$I_\theta \neq 0,$$

the transverse coefficients satisfy

$$\alpha = I_\theta \Lambda_E, \qquad \beta = I_\theta \Lambda_O.$$

Thus the full S2 nondegeneracy condition becomes

$$\boxed{\lambda_c \neq 0, \qquad (\Lambda_E, \Lambda_O) \neq (0,0).} \tag{1192}$$

So within the explicit actual-local projector-defect operator class, Method A completely reduces the last C1 issue to checking the scalar coefficient λ_c.

. Summary

$$\boxed{\textbf{Method A actual result in the explicit local operator class}}$$

The longitudinal coefficient is

$$\boxed{A = \lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}.}$$

For the explicit defect profile,

$$\boxed{I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256}\,\theta_0^3 \kappa^3.}$$

Hence

$$\boxed{A = -\frac{3\sqrt{3}\pi}{512}\,\lambda_c\,\theta_0^3 \kappa^3.}$$

Therefore

$$\boxed{\lambda_c \neq 0 \Longrightarrow A \neq 0.}$$

Likewise,

$$\boxed{\alpha = I_\theta \Lambda_E, \qquad \beta = I_\theta \Lambda_O,}$$

so the full explicit S2 nondegeneracy condition is

$$\boxed{\lambda_c \neq 0, \qquad (\Lambda_E, \Lambda_O) \neq (0,0).}$$

NEXT ACTUAL TASK AFTER METHOD A: MICROSCOPIC IDENTIFICATION OF λ_c

1. What Method A already achieved

Method A reduced the last unresolved part of C1 to the longitudinal coefficient

$$A = \lambda_\parallel.$$

In the explicit local projector-defect operator class one has

$$\boxed{\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}, \qquad \alpha = I_\theta \Lambda_E, \qquad \beta = I_\theta \Lambda_O.} \tag{1193}$$

Since $I_{\theta'^3} \neq 0$ for a nontrivial defect profile, this implies

$$\boxed{\lambda_c \neq 0 \Longrightarrow A \neq 0.} \tag{1194}$$

:contentReference[oaicite:5]index=5

Therefore the actual remaining C1 problem is no longer $A \neq 0$ directly, but the microscopic identification of the coefficient λ_c.

2. Microscopic origin of the longitudinal channel

The key internal channel is

$$\mathcal{T}_\xi = i[P_B, \partial_\xi P_B]. \tag{1195}$$

Direct restriction gives

$$\Pi_Q \mathcal{T}_\xi \Pi_Q = 0,$$

but its quadratic channel is nontrivial:

$$\boxed{\Pi_Q \mathcal{T}_\xi^2 \Pi_Q = \frac{\theta'(\xi)^2}{2}\big(\mathbf{1}_2 + \sigma_3\big).} \tag{1196}$$

A naive singlet source proportional to $\theta'^2\sigma_3$ fails by parity because

$$f_+ \text{ is even}, \qquad f_- \text{ is odd}, \qquad \theta'^2 \text{ is even},$$

so

$$\int d\xi\, f_+ f_-\, \theta'^2 = 0.$$

Therefore the viable longitudinal first-order channel must contain one additional odd factor:

$$\boxed{V_\parallel^\mu(\xi) = \lambda_c\, \theta'(\xi)\, \Pi_Q \mathcal{T}_\xi^2 \Pi_Q\, n^\mu.} \tag{1197}$$

Taking the traceless part gives

$$\boxed{V_{\parallel,\mathrm{tr}}^\mu(\xi) = \frac{\lambda_c}{2}\, \theta'(\xi)^3\, n^\mu \sigma_3.} \tag{1198}$$

This is exactly the longitudinal witness channel.

3. Minimal composite transport operator

Combining the longitudinal commutator-induced singlet with the transverse doublet source, the minimal composite local transport operator is

$$\boxed{V_{\mathrm{comp}}^\mu(\xi) = \frac{\lambda_c}{2}\, \theta'(\xi)^3\, n^\mu \sigma_3 + \theta'(\xi)\Big[\Lambda_E\, E^\mu{}_a \sigma_a + \Lambda_O\, J^\mu{}_a \sigma_a\Big].} \tag{1199}$$

This produces the full S2 witness triplet explicitly, with

$$\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}, \qquad \alpha = I_\theta \Lambda_E, \qquad \beta = I_\theta \Lambda_O.$$

4. Final microscopic coefficient problem

Therefore the strongest honest current reduction is:

$$\boxed{\text{the complete microscopic local Dirac closure is reduced to extracting the finite coefficients } (\lambda_c, \Lambda_E, \Lambda_O) \text{ from the true TECT lin}} \tag{1200}$$

In particular, the last longitudinal issue is:

$$\boxed{\text{Does the full microscopic TECT linearization generate the commutator-induced longitudinal coupling } \lambda_c \neq 0?} \tag{1201}$$

5. Strongest honest conclusion

So the next actual task is not another structural reduction. It is the coefficient-identification step:

$$\boxed{\text{start from the true local TECT projector/condensate transport term, project it to the } Q_B\text{-doublet, and read off the coefficient}} \quad (1202)$$

If that coefficient is nonzero, then $A \neq 0$ follows immediately.

. Summary

$$\boxed{\textbf{Next actual task after Method A}}$$

Method A already reduced the longitudinal condition to

$$\boxed{A = \lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}.}$$

Since

$$\boxed{I_{\theta'^3} \neq 0}$$

for a nontrivial defect,

$$\boxed{\lambda_c \neq 0 \Longrightarrow A \neq 0.}$$

The microscopic origin of λ_c is the commutator-induced channel

$$\boxed{\mathcal{T}_\xi = i[P_B, \partial_\xi P_B], \qquad \Pi_Q \mathcal{T}_\xi^2 \Pi_Q = \frac{\theta'(\xi)^2}{2}(\mathbf{1}_2 + \sigma_3).}$$

Hence the viable longitudinal transport term is

$$\boxed{V_{\parallel}^{\mu}(\xi) = \lambda_c\, \theta'(\xi)\, \Pi_Q \mathcal{T}_\xi^2 \Pi_Q\, n^{\mu},}$$

whose traceless part is

$$\boxed{V_{\parallel,\mathrm{tr}}^{\mu}(\xi) = \frac{\lambda_c}{2}\, \theta'(\xi)^3\, n^{\mu} \sigma_3.}$$

Therefore the next real microscopic task is:

$$\boxed{\text{derive the true local TECT transport term and extract the coefficient multiplying } \theta'(\xi)(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} n^{\mu}.}$$

If that coefficient is nonzero, then

$$\boxed{A \neq 0.}$$

METHOD B: MICROSCOPIC ORIGIN OF THE LONGITUDINAL COEFFICIENT λ_c

1. Why the primitive channel fails

A natural primitive traceless internal tensor is

$$T(P) = 3P - \mathbf{1}_3. \quad (1203)$$

However, on the defect tangent bundle

$$Q_B = (\mathbb{C}z_B)^{\perp},$$

its restriction is

$$\boxed{T(P_B)\big|_{Q_B} = -\mathbf{1}_{Q_B}.} \tag{1204}$$

So the primitive channel produces only an identity-valued transport operator on Q_B, not a traceless Pauli triplet, and in particular not a distinguished longitudinal σ_3-splitting. :contentReference[oaicite:4]index=4

Therefore the primitive channel is too coarse to produce the full S2 witness structure.

2. The first genuinely gradient-sensitive channel

The first nontrivial gradient-sensitive refinement is

$$\boxed{\mathcal{T}_\xi := i[P_B, \partial_\xi P_B].} \tag{1205}$$

For the explicit background,

$$\mathcal{T}_\xi = i\theta'(\xi)\big(z_B e_1^\dagger - e_1 z_B^\dagger\big). \tag{1206}$$

Its direct restriction vanishes:

$$\boxed{\Pi_Q \mathcal{T}_\xi \Pi_Q = 0.} \tag{1207}$$

But its quadratic channel is nontrivial:

$$\boxed{\Pi_Q \mathcal{T}_\xi^2 \Pi_Q = \frac{\theta'(\xi)^2}{2}\big(\mathbf{1}_2 + \sigma_3\big).} \tag{1208}$$

Hence the traceless part is

$$\boxed{(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}} = \frac{\theta'(\xi)^2}{2}\sigma_3.} \tag{1209}$$

This is the first natural longitudinal σ_3-type splitting channel.

3. Parity obstruction and the need for one more odd factor

The graded low modes satisfy

$$u_a^+(\xi) = f_+(\xi)e_a(\xi), \qquad u_a^-(\xi) = f_-(\xi)e_a(\xi),$$

with

$$f_+ \text{ even}, \qquad f_- \text{ odd}.$$

Since $\theta'(\xi)^2$ is even, the overlap

$$\int d\xi\, f_+(\xi) f_-(\xi) \theta'(\xi)^2$$

vanishes by parity. Therefore the quadratic channel (1209) cannot by itself contribute to the first-order mixed symbol.

Hence one additional odd factor is required, and the minimal viable longitudinal source is

$$\boxed{V_\parallel^\mu(\xi) = \lambda_c\, \theta'(\xi)\, (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\mathrm{tr}}\, n^\mu.} \tag{1210}$$

Using (1209), this becomes

$$\boxed{V_\parallel^\mu(\xi) = \frac{\lambda_c}{2}\, \theta'(\xi)^3\, n^\mu \sigma_3.} \tag{1211}$$

This is exactly the longitudinal witness channel. :contentReference[oaicite:6]index=6

4. Mixed-symbol projection

Projecting to the graded low modes gives

$$(M^{\mu})_{ab} = \int d\xi\, f_{+}(\xi) f_{-}(\xi)\, e_a^{\dagger}(\xi) V_{\parallel}^{\mu}(\xi) e_b(\xi). \tag{1212}$$

Substituting (1211),

$$M_{\parallel}^{\mu} = \frac{\lambda_c}{2}\left(\int d\xi\, f_{+}(\xi) f_{-}(\xi) \theta'(\xi)^3\right) n^{\mu}\sigma_3 = \frac{\lambda_c}{2} I_{\theta'^3}\, n^{\mu}\sigma_3. \tag{1213}$$

Thus the longitudinal Weyl coefficient is

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}.} \tag{1214}$$

Since $I_{\theta'^3} \neq 0$ for a nontrivial defect profile,

$$\boxed{\lambda_c \neq 0 \Longrightarrow \lambda_{\parallel} \neq 0 \Longrightarrow A \neq 0.} \tag{1215}$$

5. The minimal composite transport operator

Combining the longitudinal commutator-induced singlet with the transverse doublet channels yields the minimal composite transport operator

$$\boxed{V_{\mathrm{comp}}^{\mu}(\xi) = \frac{\lambda_c}{2}\,\theta'(\xi)^3\, n^{\mu}\sigma_3 + \theta'(\xi)\Big[\Lambda_E\, E^{\mu}{}_a\sigma_a + \Lambda_O\, J^{\mu}{}_a\sigma_a\Big].} \tag{1216}$$

Its projected witnesses are

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \qquad \alpha = I_{\theta}\Lambda_E, \qquad \beta = I_{\theta}\Lambda_O.} \tag{1217}$$

Therefore the full S2 nondegeneracy condition is exactly

$$\boxed{\lambda_c \neq 0, \qquad (\Lambda_E, \Lambda_O) \neq (0,0).} \tag{1218}$$

6. Strongest honest conclusion

Thus Method B proves the following. The longitudinal coefficient λ_c is not an arbitrary free parameter introduced by hand. It is precisely the coefficient of the first viable commutator-induced longitudinal transport channel

$$\theta'(\xi)\,(\Pi_Q \mathcal{T}_{\xi}^{2} \Pi_Q)_{\mathrm{tr}}\, n^{\mu}.$$

Therefore the last unresolved microscopic content of C1 is:

$$\boxed{\text{Does the full microscopic TECT linearization generate this commutator-induced longitudinal channel with nonzero coefficient } \lambda} \tag{1219}$$

This is the sharpest currently honest formulation of the remaining longitudinal witness problem.

. Summary

Method B result

The primitive channel

$$\boxed{T(P) = 3P - \mathbf{1}_3}$$

is too coarse on Q_B, because

$$\boxed{T(P_B)\big|_{Q_B} = -\mathbf{1}_{Q_B}.}$$

The first genuinely gradient-sensitive longitudinal source is instead

$$\boxed{\mathcal{T}_\xi = i[P_B, \partial_\xi P_B].}$$

Although

$$\boxed{\Pi_Q \mathcal{T}_\xi \Pi_Q = 0,}$$

its quadratic channel is nontrivial:

$$\boxed{\Pi_Q \mathcal{T}_\xi^2 \Pi_Q = \frac{\theta'(\xi)^2}{2}\big(\mathbf{1}_2 + \sigma_3\big), \qquad (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\rm tr} = \frac{\theta'(\xi)^2}{2}\sigma_3.}$$

Because θ'^2 is even, one more odd factor is required. Hence the minimal viable longitudinal source is

$$\boxed{V_\parallel^\mu(\xi) = \lambda_c\, \theta'(\xi)\, (\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\rm tr}\, n^\mu = \frac{\lambda_c}{2}\, \theta'(\xi)^3\, n^\mu \sigma_3.}$$

Projecting to the graded low modes yields

$$\boxed{\lambda_\parallel = \frac{\lambda_c}{2} I_{\theta'^3}.}$$

Therefore

$$\boxed{\lambda_c \neq 0 \Longrightarrow \lambda_\parallel \neq 0 \Longrightarrow A \neq 0.}$$

So Method B reduces the remaining longitudinal witness problem to one sharp microscopic question:

$$\boxed{\text{Does the full microscopic TECT linearization generate the commutator-induced channel } \theta'(\xi)(\Pi_Q \mathcal{T}_\xi^2 \Pi_Q)_{\rm tr} n^\mu \text{ with nonzero coef}}$$

$$\boxed{\lambda_\parallel = \kappa_* |G_*|\, \hat{n}_i \hat{n}_j N_{ij}^3.}$$

For the axial shell-compatible family

$$N_{ij}^3 = A\, \hat{n}_i \hat{n}_j, \qquad N_{ij}^1 = B\, \hat{n}_i e_{1j} + C\, \hat{n}_i e_{2j}, \qquad N_{ij}^2 = -C\, \hat{n}_i e_{1j} + B\, \hat{n}_i e_{2j},$$

one gets

$$\boxed{\lambda_\parallel = \kappa_* |G_*| A, \qquad \alpha = \kappa_* |G_*| B, \qquad \beta = \kappa_* |G_*| C.}$$

Hence

$$\boxed{A \neq 0 \iff \hat{n}_i \hat{n}_j N_{ij}^3 \neq 0.}$$

METHOD B APPLIED TO THE MINIMAL LOCAL PDE ANSATZ

Start from the shell-enriched first-order coupling

$$\delta L_{\mathrm{enr}} = N^a_{ij}(\partial_i\phi_0)(-i\partial_j)\otimes s^a.$$

For the minimal projector/enriched completion,

$$\delta L_{\mathrm{enr}} = g_\parallel(\partial_i\phi_0)\hat{n}_i\big(\hat{n}\cdot(-i\nabla)\big)\otimes s_3 + g_I(\partial_i\phi_0)\hat{n}_i\Big[(e_1\cdot(-i\nabla))\otimes s_1 + (e_2\cdot(-i\nabla))\otimes s_2\Big] + g_J(\partial_i\phi_0)\hat{n}_i\Big[-(e_2\cdot(-i\nabla))\otimes s_1 + (e_1\cdot(-i\nabla))\otimes s_2\Big].$$

Comparing term by term gives

$$\boxed{N^3_{ij} = g_\parallel\hat{n}_i\hat{n}_j, \qquad N^1_{ij} = g_I\hat{n}_ie_{1j} - g_J\hat{n}_ie_{2j}, \qquad N^2_{ij} = g_I\hat{n}_ie_{2j} + g_J\hat{n}_ie_{1j}.}$$

Hence

$$\boxed{\hat{n}_i\hat{n}_jN^3_{ij} = g_\parallel,}$$

and therefore

$$\boxed{A \neq 0 \iff \hat{n}_i\hat{n}_jN^3_{ij} \neq 0 \iff g_\parallel \neq 0.}$$

Using the shell-matching formula,

$$\boxed{\lambda_\parallel = \kappa_*|G_*|\,g_\parallel, \qquad \alpha = \kappa_*|G_*|\,g_I, \qquad \beta = -\kappa_*|G_*|\,g_J.}$$

So the local Weyl condition becomes

$$\boxed{g_\parallel \neq 0, \qquad (g_I, g_J) \neq (0,0).}$$

For the Class II auxiliary-mediator realization,

$$\delta N^b_{ij} = -\frac{\lambda g_X}{M_X^2}\,m^aC^{ab}_{ij}.$$

If the shell-relevant decomposition of $m^aC^{ab}_{ij}$ contains

$$A\,\hat{n}_i\hat{n}_j\delta^{b3},$$

then

$$\boxed{\delta N^3_{ij} = -\frac{\lambda g_X}{M_X^2}A\,\hat{n}_i\hat{n}_j + \cdots, \qquad \delta W_\parallel = -\frac{\lambda g_X}{M_X^2}A \neq 0.}$$

Thus Method B reduces the microscopic longitudinal question to a single shell-relevant tensor coefficient:

$$\boxed{\hat{n}_i\hat{n}_jN^3_{ij} \neq 0.}$$

$$\delta L_{\mathrm{enr}} = N^a_{ij}(\partial_i\phi_0)(-i\partial_j)\otimes s^a$$

$$N^3_{ij} = g_\parallel\hat{n}_i\hat{n}_j, \qquad N^1_{ij} = g_I\hat{n}_ie_{1j} - g_J\hat{n}_ie_{2j}, \qquad N^2_{ij} = g_I\hat{n}_ie_{2j} + g_J\hat{n}_ie_{1j}$$

$$\hat{n}_i\hat{n}_jN^3_{ij} = g_\parallel$$

$$\lambda_\parallel = \kappa_*|G_*|\,g_\parallel$$

$$A \neq 0 \iff g_\parallel \neq 0.$$

METHOD B PUSHED TO THE FINAL MICROSCOPIC COEFFICIENT FORMULA

1. Tensor-contraction extraction of the longitudinal coefficient

For the shell-enriched first-order coupling

$$\delta L_{\mathrm{enr}} = N^a_{ij}(\partial_i\phi_0)(-i\partial_j)\otimes s^a,$$

the longitudinal Weyl coefficient is extracted by

$$\lambda_\parallel = \kappa_*|G_*|\,\hat n_i\hat n_j N^3_{ij}.$$

In the minimal shell-compatible ansatz

$$N^3_{ij} = g_\parallel \hat n_i\hat n_j,$$

hence

$$\boxed{g_\parallel = \hat n_i\hat n_j N^3_{ij}.}$$

Therefore

$$g_\parallel \neq 0 \Longrightarrow \lambda_\parallel \neq 0 \Longrightarrow A \neq 0.$$

2. Class II induced enriched tensor

For the Class II heavy auxiliary mediator completion, the induced enriched tensor is

$$\boxed{\delta N^b_{ij} = -\frac{\lambda g_X}{M_X^2}\, m^a C^{ab}_{ij}.}$$

After shell-adapted projection, the longitudinal axial coefficient is obtained by contracting the $b=3$ component along $\hat n_i\hat n_j$:

$$\delta g_\parallel = \hat n_i\hat n_j\,\delta N^3_{ij}.$$

Substituting the Class II formula gives

$$\boxed{\delta g_\parallel = -\frac{\lambda g_X}{M_X^2}\, m^a\,\hat n_i\hat n_j\,C^{a3}_{ij}.}$$

Thus the last microscopic longitudinal test is precisely

$$\boxed{m^a\,\hat n_i\hat n_j\,C^{a3}_{ij} \neq 0.}$$

3. Minimal seed realization

For the minimal seed choice

$$m^a = m\,\delta^{a3}, \qquad C^{33}_{ij} = c_\parallel \hat n_i\hat n_j,$$

one gets

$$m^a\,\hat n_i\hat n_j\,C^{a3}_{ij} = m\,\hat n_i\hat n_j\,C^{33}_{ij} = m\,c_\parallel.$$

Therefore

$$\boxed{\delta g_\parallel = -\frac{\lambda g_X}{M_X^2}\, m\,c_\parallel.}$$

Hence

$$\boxed{m\,c_\parallel \neq 0 \Longrightarrow g_\parallel \neq 0 \Longrightarrow A \neq 0.}$$

4. Final reduction

So Method B reduces the final longitudinal witness problem to one explicit shell-relevant tensor coefficient:

$$\boxed{\text{compute } m^a \hat{n}_i \hat{n}_j C^{a3}_{ij}.}$$

If it is nonzero, then the longitudinal coefficient is nonzero.

$$\delta L_{\text{enr}} = N^a_{ij}(\partial_i \phi_0)(-i\partial_j) \otimes s^a$$

For the minimal enriched local ansatz,

$$N^3_{ij} = g_\parallel \hat{n}_i \hat{n}_j, \qquad N^1_{ij} = g_I \hat{n}_i e_{1j} - g_J \hat{n}_i e_{2j}, \qquad N^2_{ij} = g_I \hat{n}_i e_{2j} + g_J \hat{n}_i e_{1j}.$$

Hence

$$\lambda_\parallel = \kappa_* |G_*| \, \hat{n}_i \hat{n}_j N^3_{ij} = \kappa_* |G_*| \, g_\parallel.$$

Therefore, within this minimal enriched local ansatz,

$$g_\parallel \neq 0 \Longrightarrow \lambda_\parallel \neq 0.$$

METHOD B: CORRECTED FINAL VERSION

1. Scope

We work inside the *minimal enriched local ansatz*, not yet the full microscopic TECT linearization. Let

$$\delta L_{\text{enr}} = N^a_{ij}(\partial_i \phi_0)(-i\partial_j) \otimes s^a, \qquad a = 1, 2, 3.$$

Fix a selected shell pair $\pm G_*$, define

$$\hat{n} = \frac{G_*}{|G_*|},$$

and choose an adapted transverse frame (e_1, e_2).

2. Exact tensor-contraction formulas

The shell-level Weyl coefficients are read from the microscopic enriched tensor by

$$\boxed{\lambda_\parallel = \kappa_* |G_*| \, \hat{n}_i \hat{n}_j N^3_{ij},} \tag{1220}$$

$$\boxed{\alpha = \frac{\kappa_* |G_*|}{2} \left(e_{1j} \hat{n}_i N^1_{ij} + e_{2j} \hat{n}_i N^2_{ij} \right),} \tag{1221}$$

$$\boxed{\beta = \frac{\kappa_* |G_*|}{2} \left(e_{2j} \hat{n}_i N^1_{ij} - e_{1j} \hat{n}_i N^2_{ij} \right),} \tag{1222}$$

where the overall sign of β is convention-dependent under reorientation of the transverse frame. :contentReference[oaicite:2]index=2

3. Minimal shell-compatible enriched ansatz

For the minimal enriched local ansatz,

$$\boxed{N^3_{ij} = g_\parallel \hat{n}_i \hat{n}_j, \qquad N^1_{ij} = g_I \hat{n}_i e_{1j} - g_J \hat{n}_i e_{2j}, \qquad N^2_{ij} = g_I \hat{n}_i e_{2j} + g_J \hat{n}_i e_{1j}.} \tag{1223}$$

Then

$$\hat{n}_i \hat{n}_j N^3_{ij} = g_\parallel (\hat{n} \cdot \hat{n})^2 = g_\parallel, \tag{1224}$$

so

$$\boxed{\lambda_\parallel = \kappa_* |G_*|\, g_\parallel.} \tag{1225}$$

Likewise,

$$\boxed{\alpha = \kappa_* |G_*|\, g_I, \qquad \beta = \pm\, \kappa_* |G_*|\, g_J,} \tag{1226}$$

where the sign of β depends on the transverse orientation convention.

4. Weyl criterion

Therefore the shell-level Weyl determinant is

$$\det M = (\kappa_* |G_*|)^3\, g_\parallel\, (g_I^2 + g_J^2), \tag{1227}$$

and the exact local Weyl criterion inside this ansatz is

$$\boxed{g_\parallel \neq 0, \qquad (g_I, g_J) \neq (0,0).} \tag{1228}$$

5. Exact honest status

What is rigorously established is:

1. inside the minimal enriched local ansatz, the longitudinal coefficient is exactly controlled by

$$\hat{n}_i \hat{n}_j N^3_{ij};$$

2. the shell-level Weyl condition is exactly

$$g_\parallel \neq 0, \qquad (g_I, g_J) \neq (0,0);$$

3. no further structural reduction is needed at this ansatz level.

What remains open is:

$$\boxed{\text{whether the true full microscopic TECT linearization actually generates the same nonzero enriched tensor coefficients } N^a_{ij}.} \tag{1229}$$

. Summary

$\boxed{\textbf{Open problems on the full PDE side}}$

1. The final full microscopic TECT condensate PDE is not yet uniquely closed from first principles.
2. The actual linearized microscopic operator $L(k)$ has not yet been derived in final form.
3. Hence the true coefficients $K_i = \partial_{k_i} L(k)|_{k=G_*}$ are still not explicitly computed.
4. The actual shell/valley crossing $\pm G_*$, spectral isolation, chiral grading,
   and Clifford isotropy are not yet verified directly in the full TECT dynamics.
5. Equivalently, the remaining microscopic Dirac task is the explicit extraction of g_c, g_E, g_O
   (or N^a_{ij}, or $\lambda_\parallel, \alpha, \beta$) from the true full PDE.

$\boxed{\textbf{Open problems on the FRG side}}$

1. Quartic cubic anisotropy is largely under control, but derivative anisotropy remains open.
2. The kinetic-flow matrix M_Z has not yet been fully computed.
3. Therefore the signs of the derivative-anisotropy eigenvalues are not yet known.
4. Sectoral wavefunction flows and velocity-locking flows are not yet fully evaluated.
5. Hence full IR isotropy / emergent Lorentz invariance is not yet completely closed.